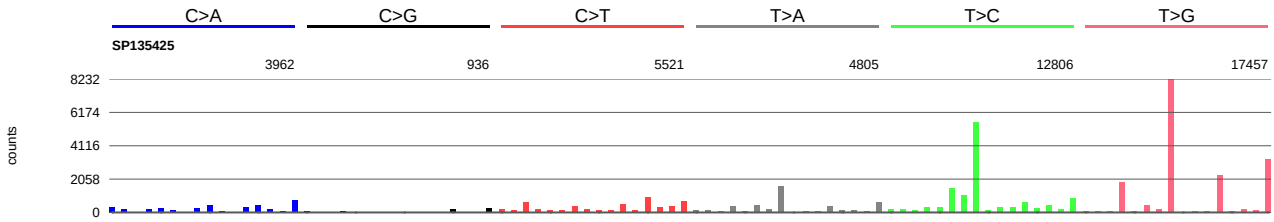
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8232
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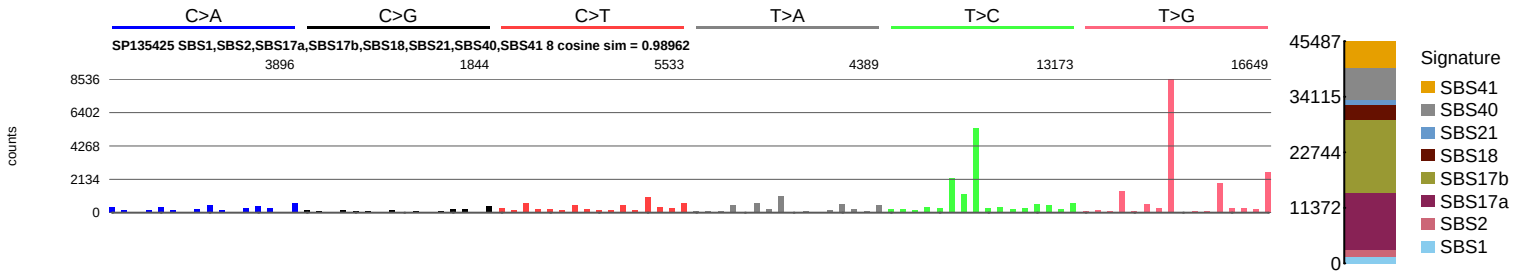
counts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SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98962
3896
1844
5533
4389
13173
16649
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
0
Signature
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SBS40
SBS21
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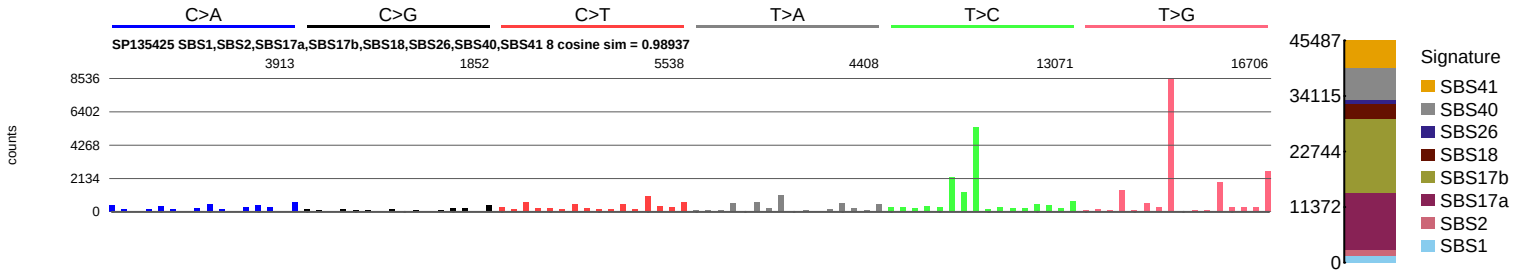
SBS17b
SBS17a
SBS2
SBS1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SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98937
3913
1852
5538
4408
13071
16706
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
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Signature
SBS41
SBS40
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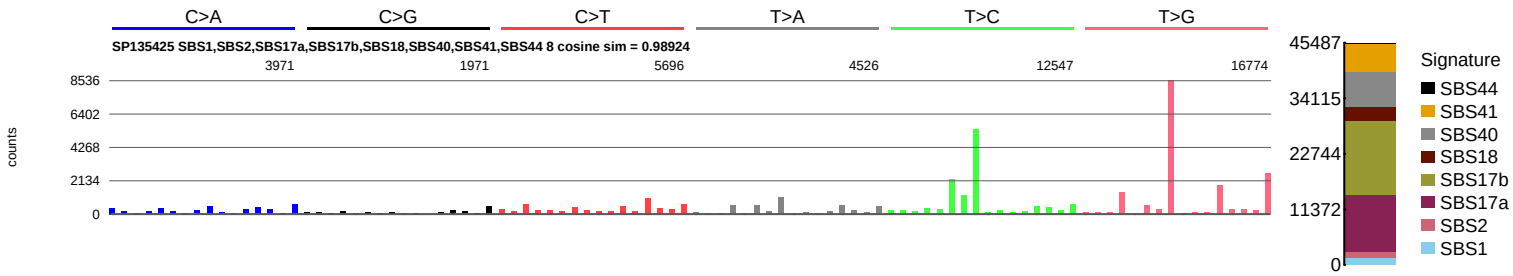
SBS18
SBS17b
SBS17a
SBS2
SBS1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1971
5696
4526
12547
16774
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
0
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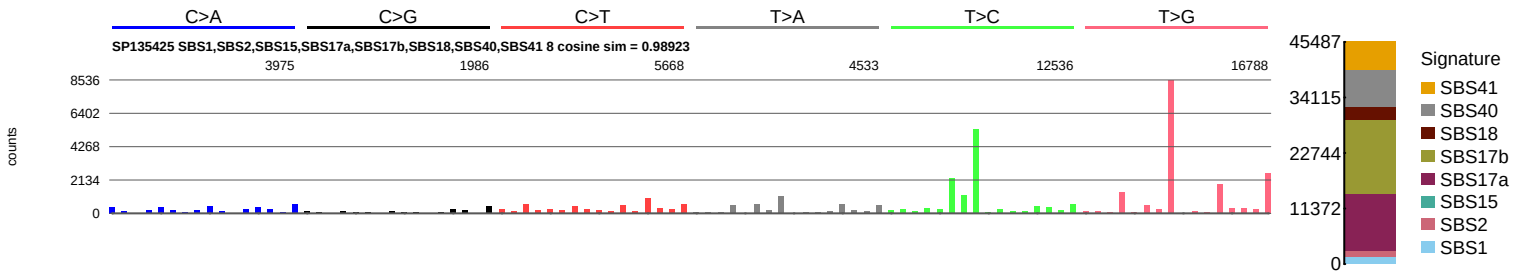
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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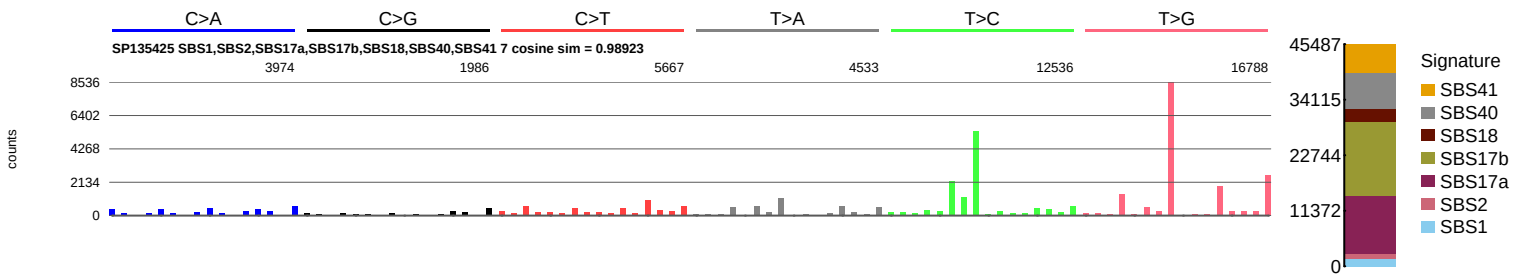
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98923
3975
1986
5668
4533
12536
16788
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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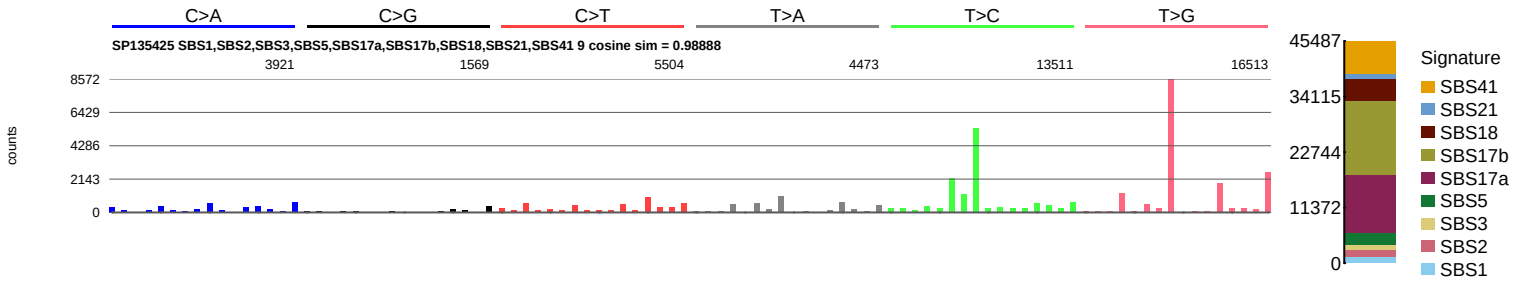
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98923
3974
1986
5667
4533
12536
16788
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98888
3921
1569
5504
4473
13511
16513
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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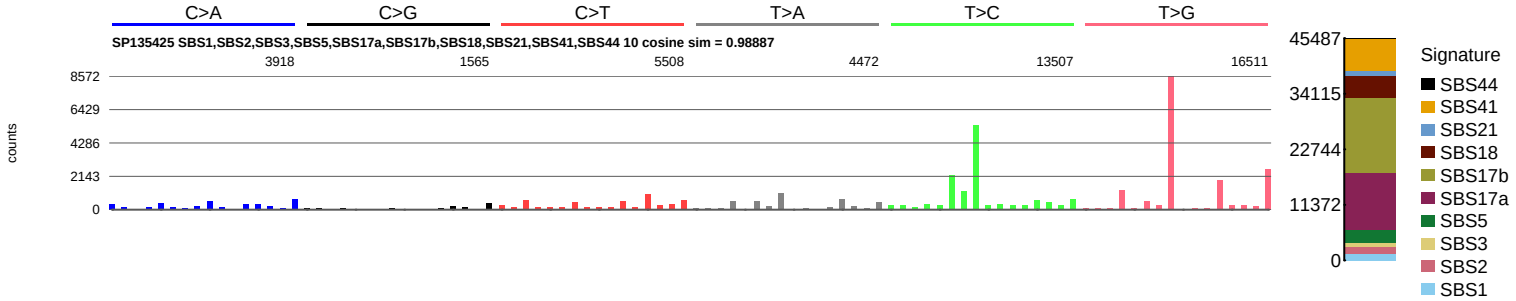
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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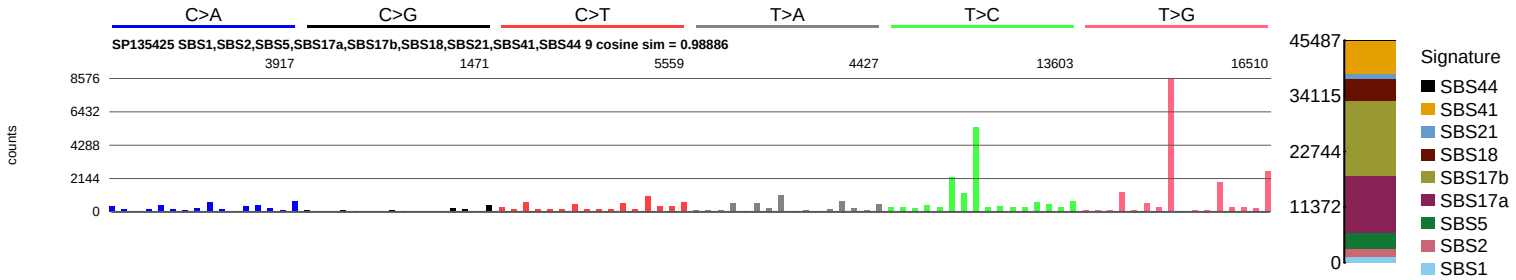
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98887
3918
1565
5508
4472
13507
16511
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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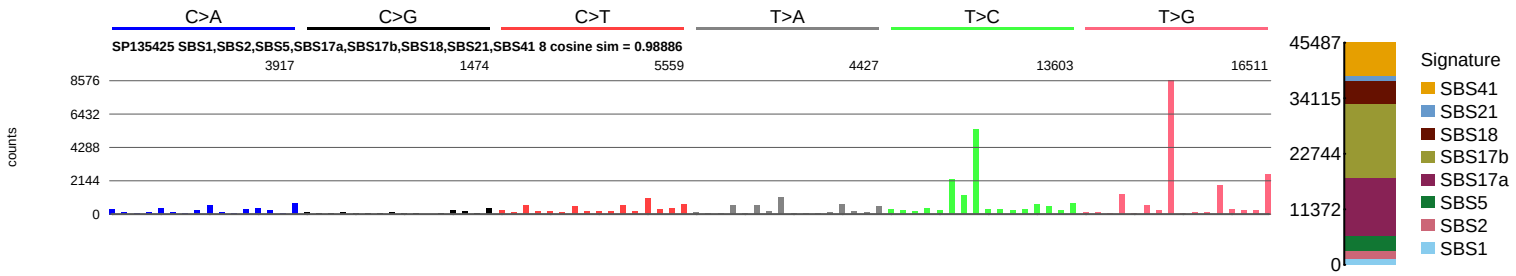
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98886
3917
1471
5559
4427
13603
16510
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98886
3917
1474
5559
4427
13603
16511
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
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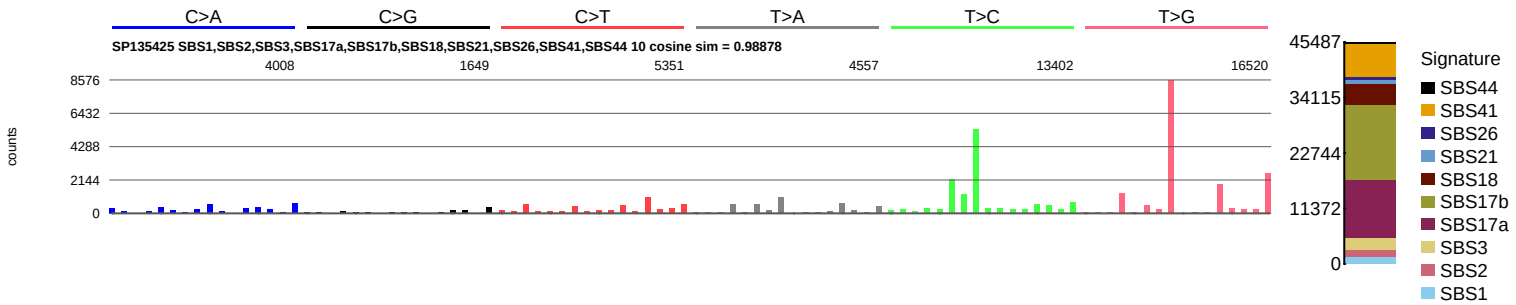
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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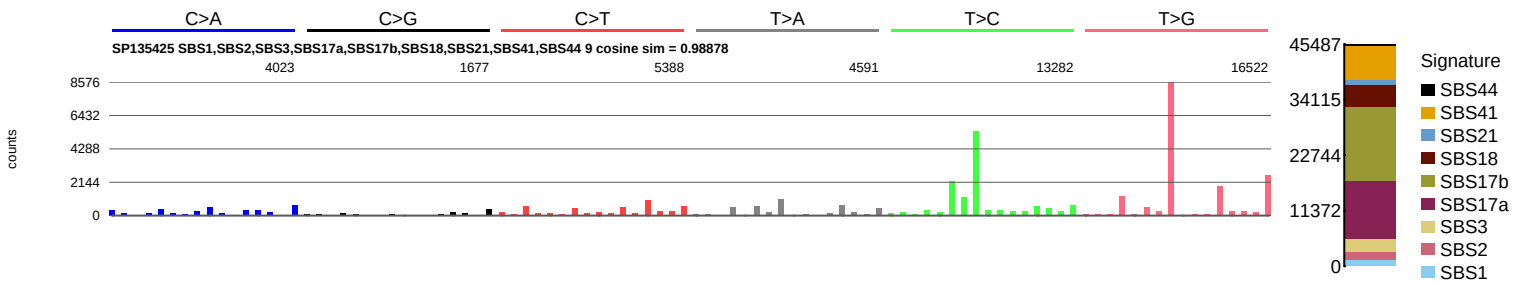
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98878
4008
1649
5351
4557
13402
16520
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98878
4023
1677
5388
4591
13282
16522
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
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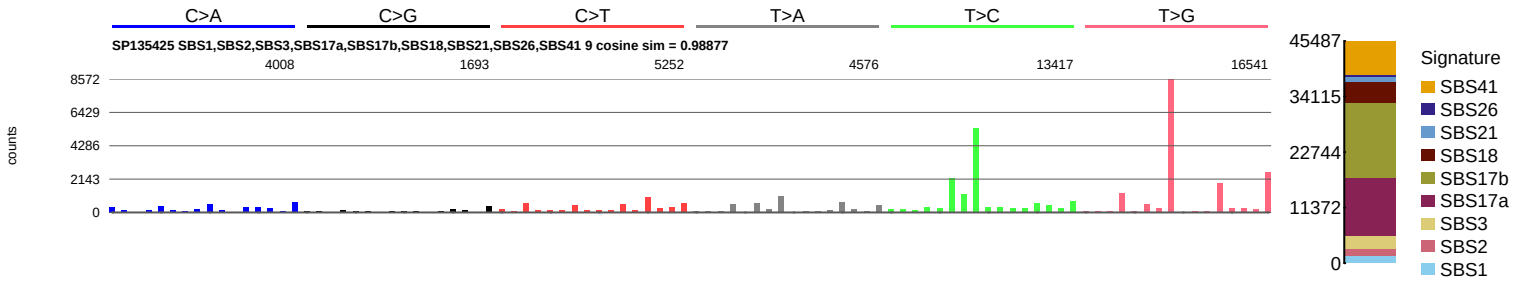
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98877
4008
1693
5252
4576
13417
16541
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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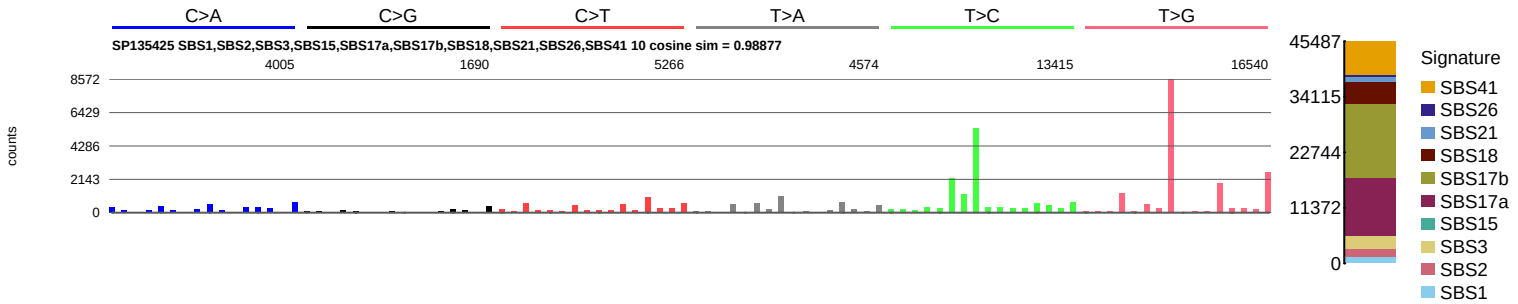
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98877
4005
1690
5266
4574
13415
16540
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
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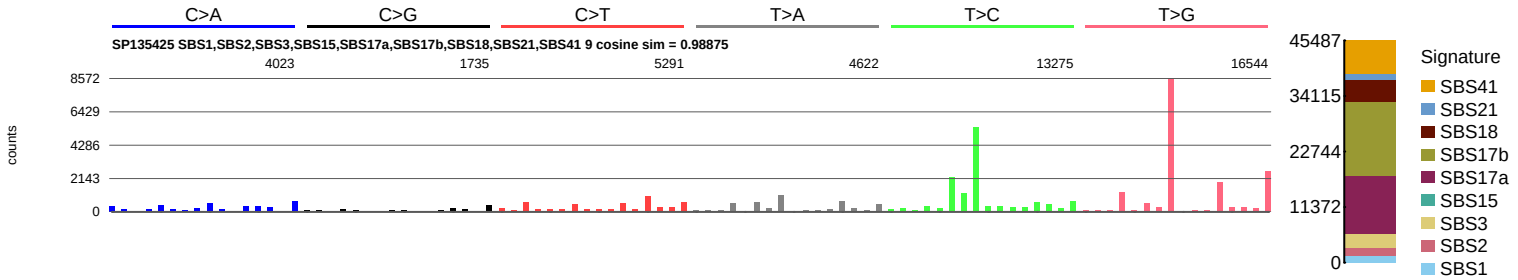
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98875
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1735
5291
4622
13275
16544
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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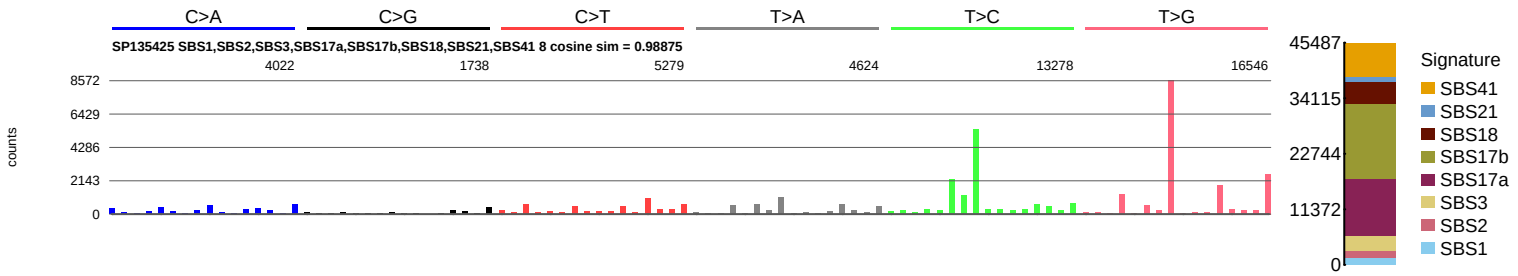
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98875
4022
1738
5279
4624
13278
16546
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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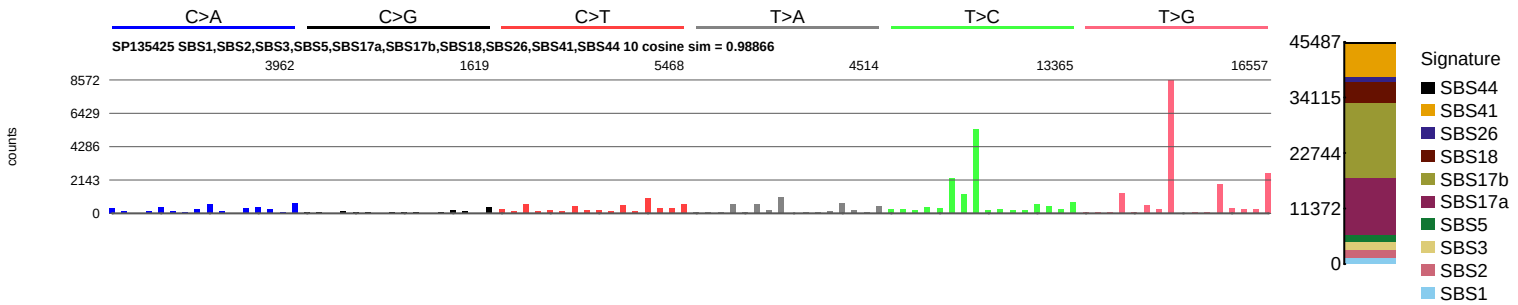
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98866
3962
1619
5468
4514
13365
16557
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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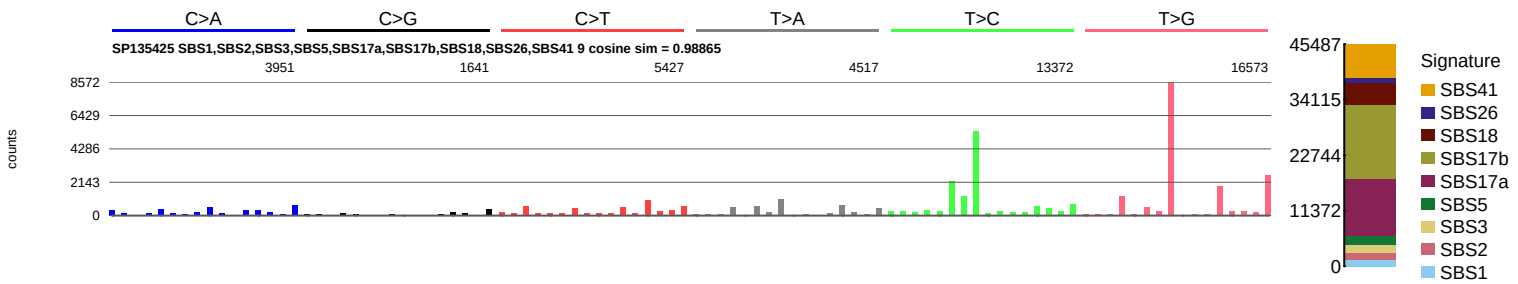
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98865
3951
1641
5427
4517
13372
16573
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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Signature
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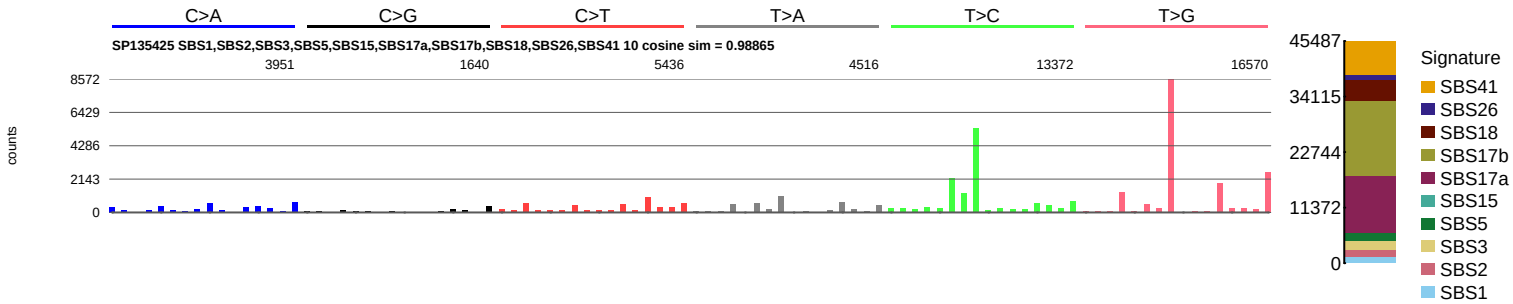
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98865
3951
1640
5436
4516
13372
16570
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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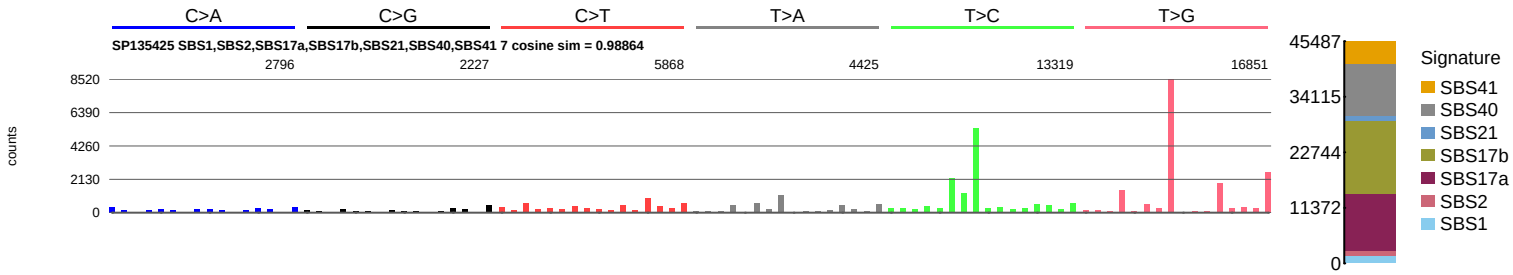
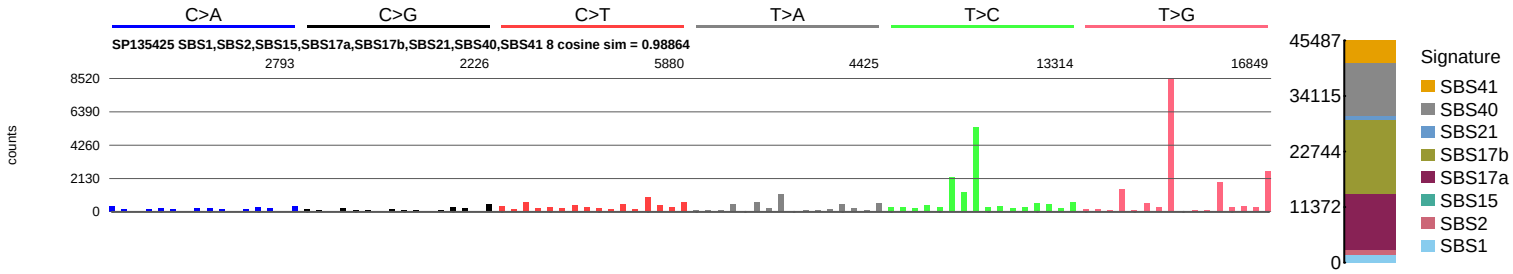
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98864
2796
2227
5868
4425
13319
16851
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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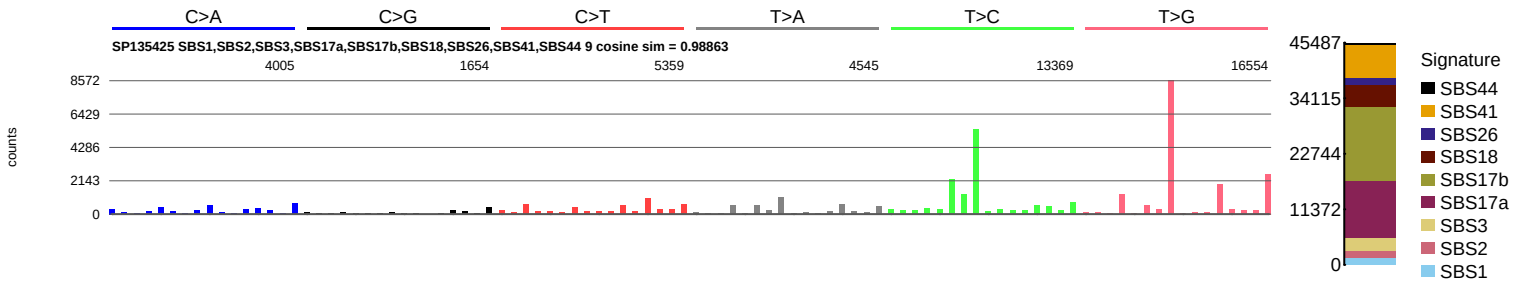
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98864
2793
2226
5880
4425
13314
16849
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98863
4005
1654
5359
4545
13369
16554
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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Signature
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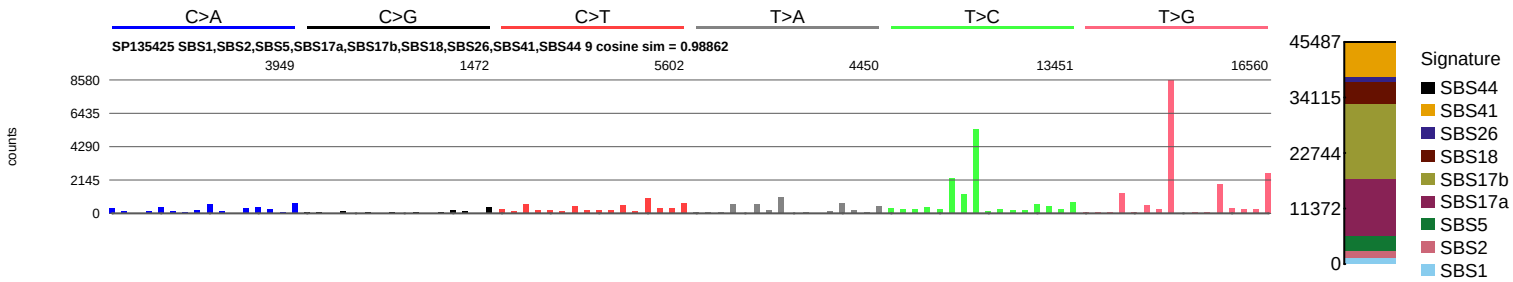
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98862
3949
1472
5602
4450
13451
16560
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
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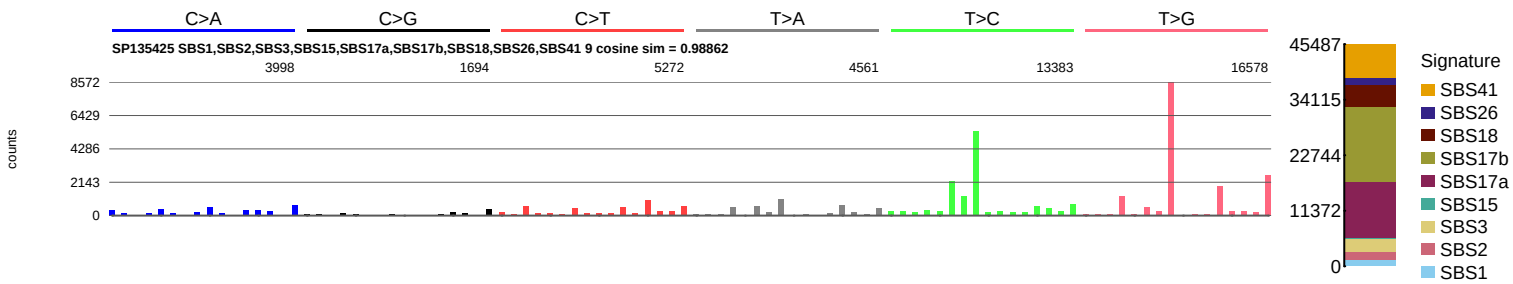
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98862
3998
1694
5272
4561
13383
16578
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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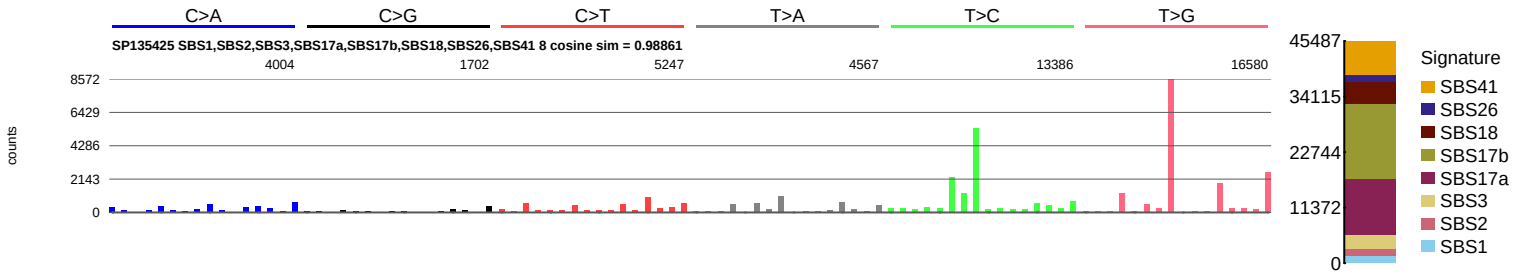
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98861
4004
1702
5247
4567
13386
16580
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
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SBS41
SBS26
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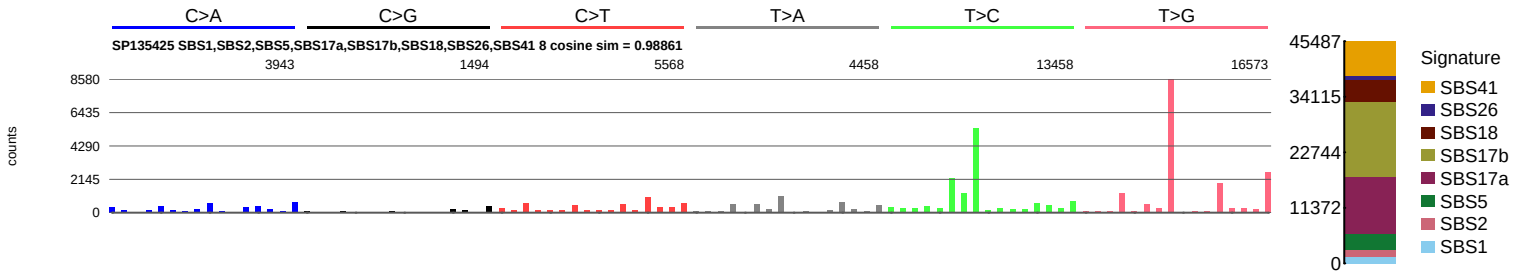
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98861
3943
1494
5568
4458
13458
16573
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
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Signature
SBS41
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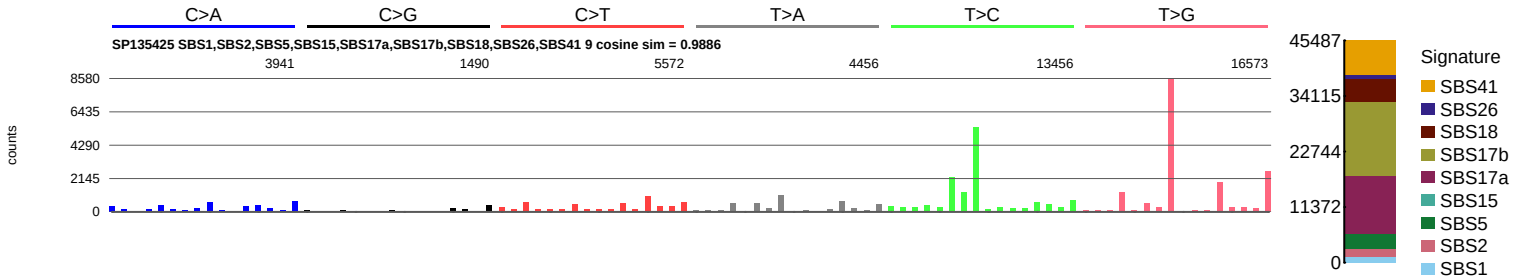
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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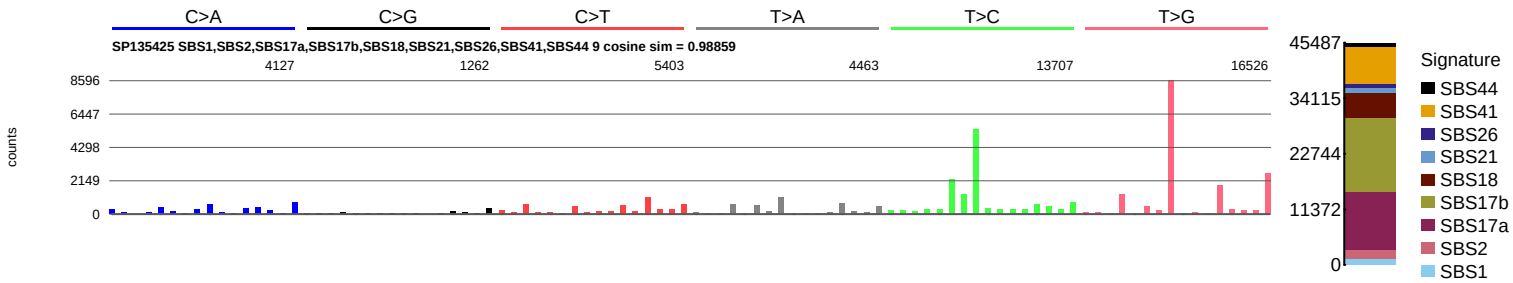
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.9886
3941
1490
5572
4456
13456
16573
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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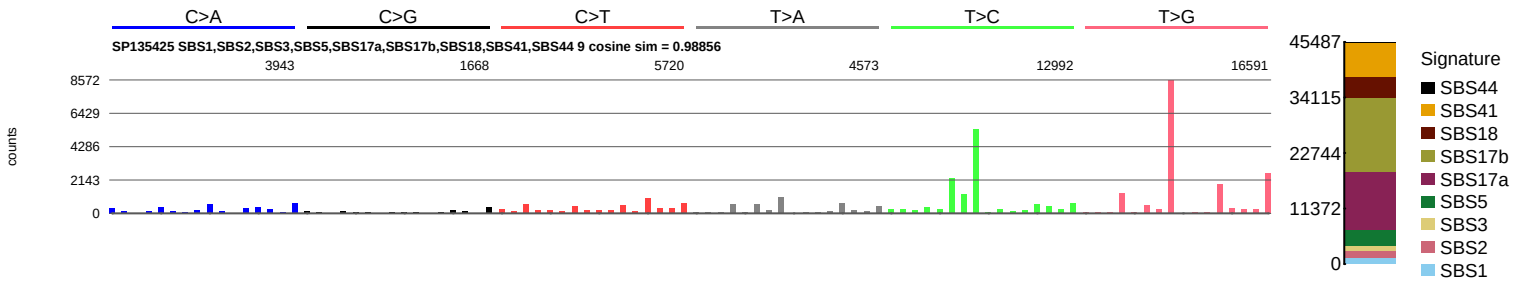
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98859
4127
1262
5403
4463
13707
16526
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98856
3943
1668
5720
4573
12992
16591
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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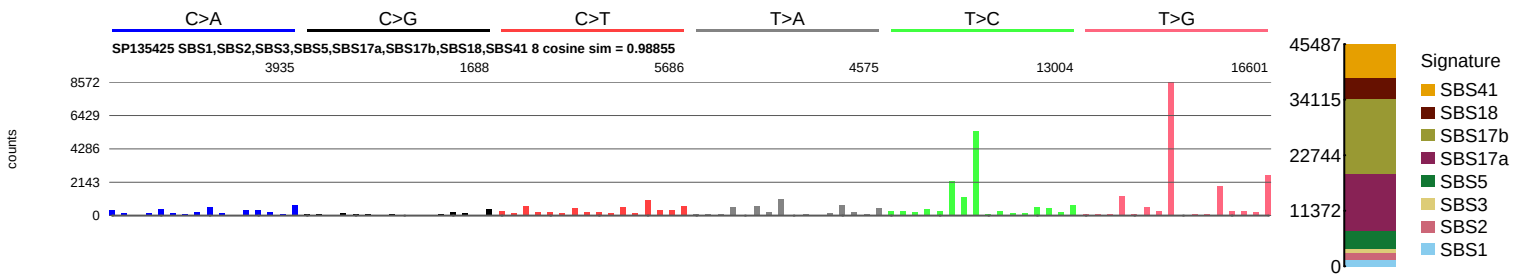
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98855
3935
1688
5686
4575
13004
16601
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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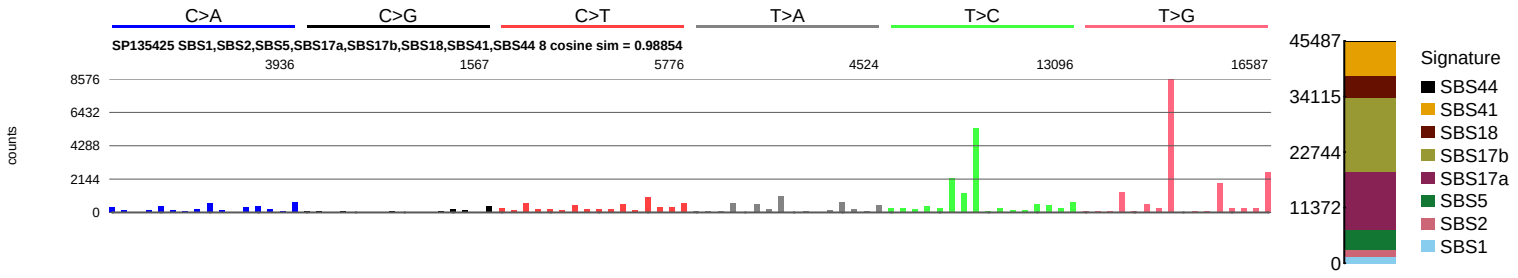
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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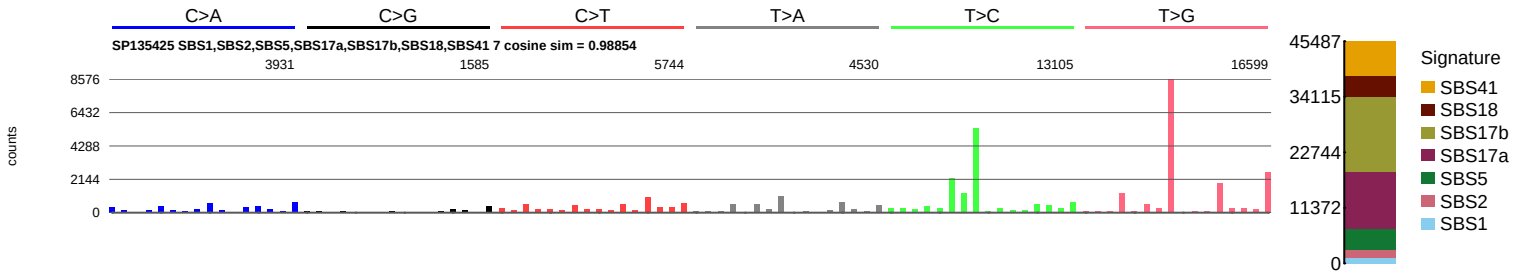
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98854
3936
1567
5776
4524
13096
16587
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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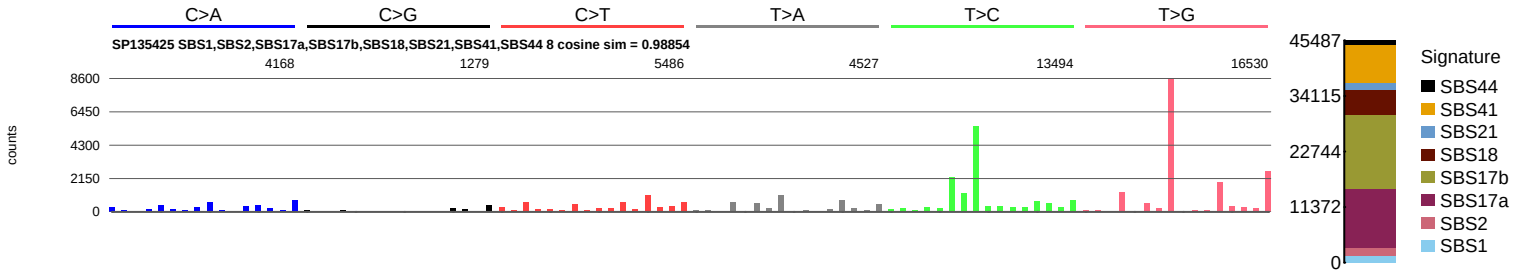
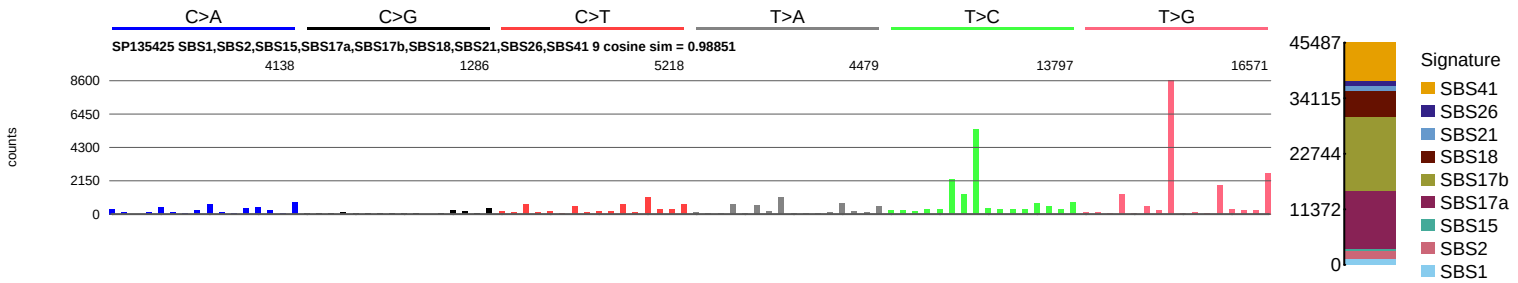
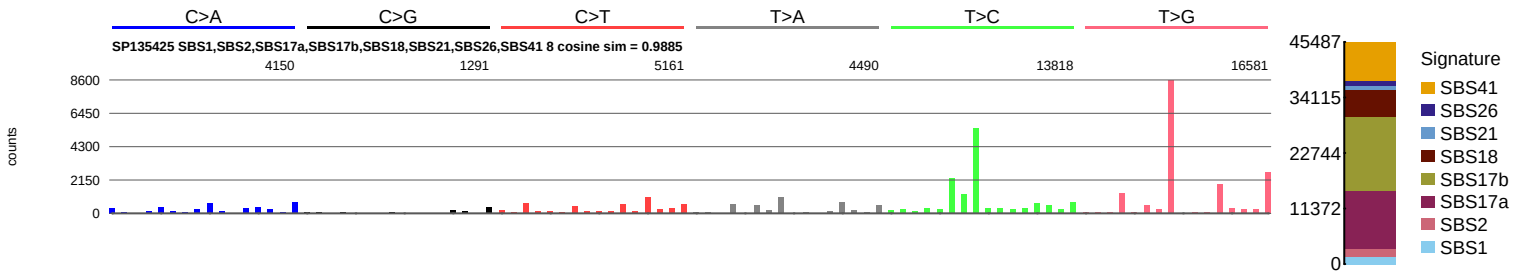
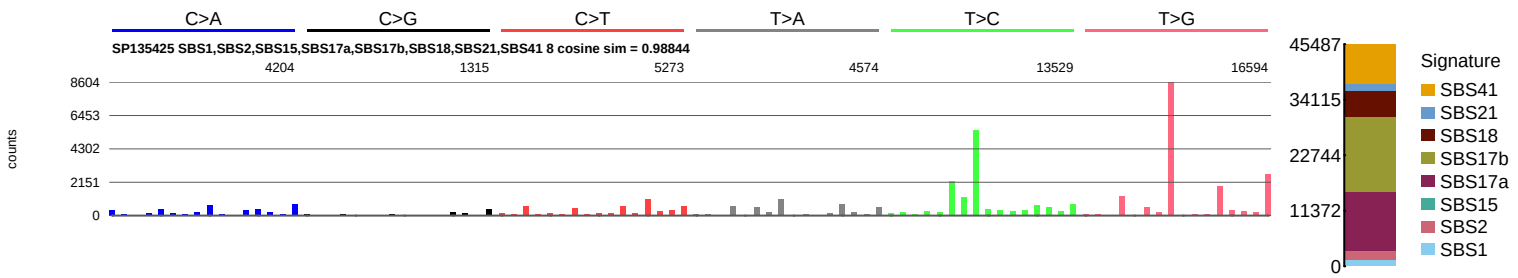
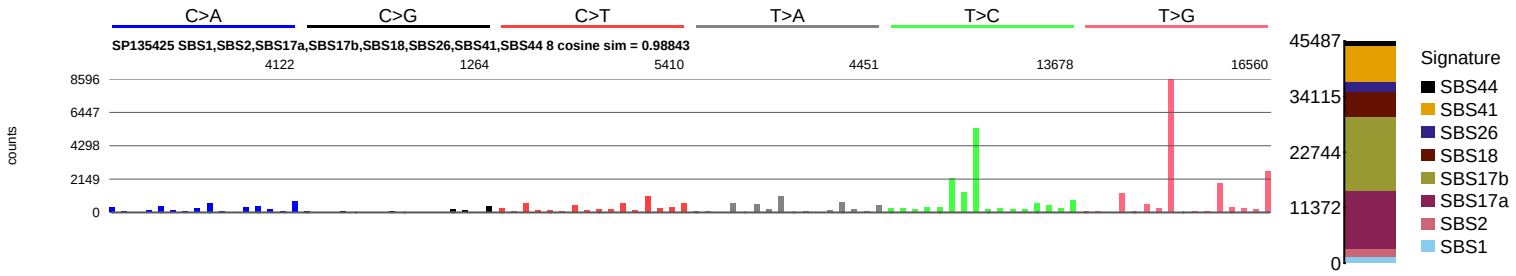
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98854
3931
1585
5744
4530
13105
16599
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

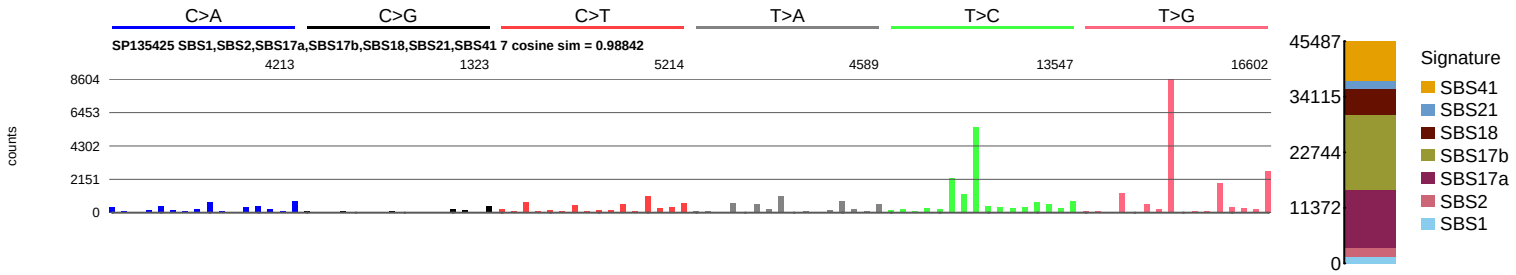

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98854
4168
1279
5486
4527
13494
16530
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

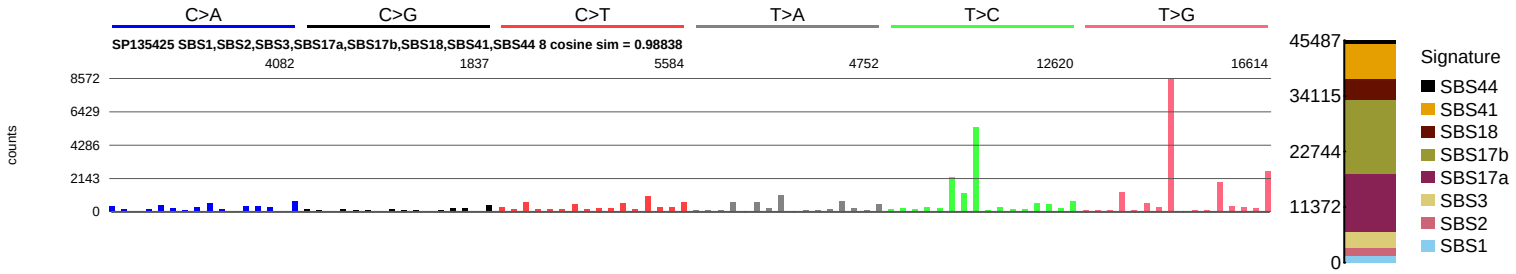

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98851
4138
1286
5218
4479
13797
16571
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

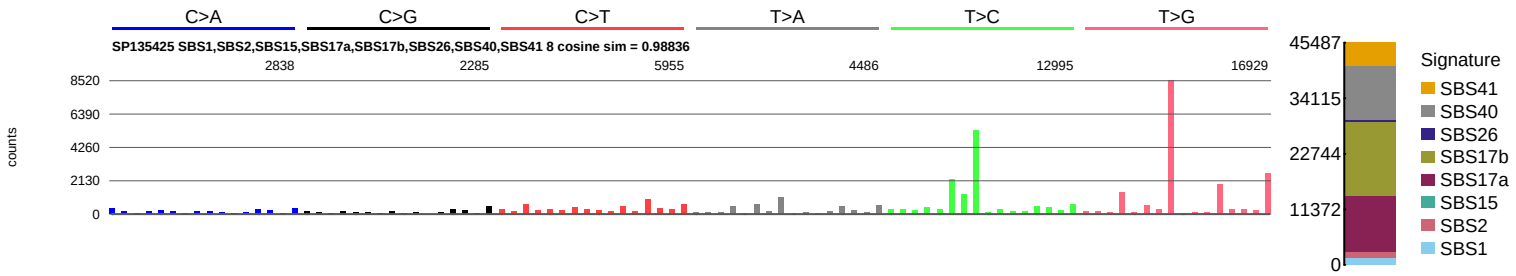

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.9885
4150
1291
5161
4490
13818
16581
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98844
4204
1315
5273
4574
13529
16594
8604
6453
4302
2151
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98843
4122
1264
5410
4451
13678
16560
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98842
4213
1323
5214
4589
13547
16602
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98838
4082
1837
5584
4752
12620
16614
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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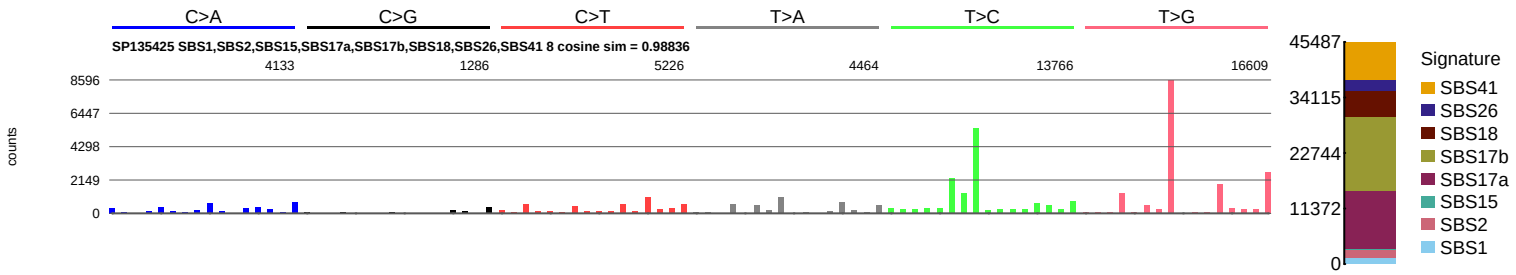
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98836
2838
2285
5955
4486
12995
16929
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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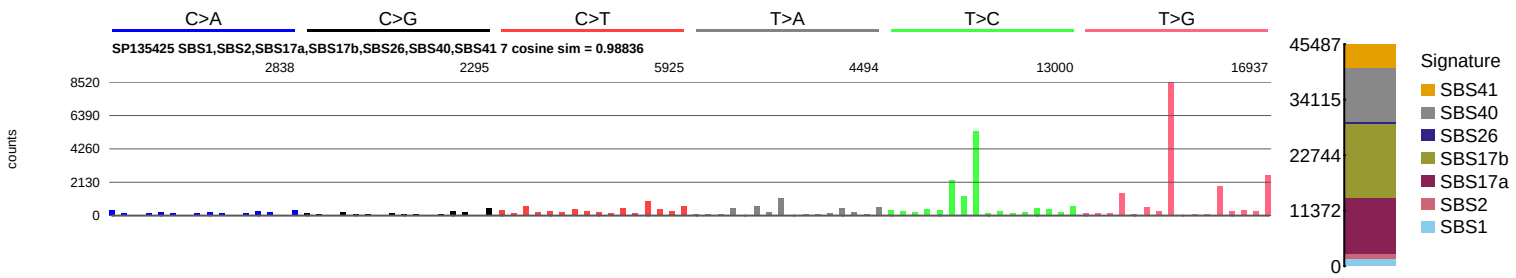
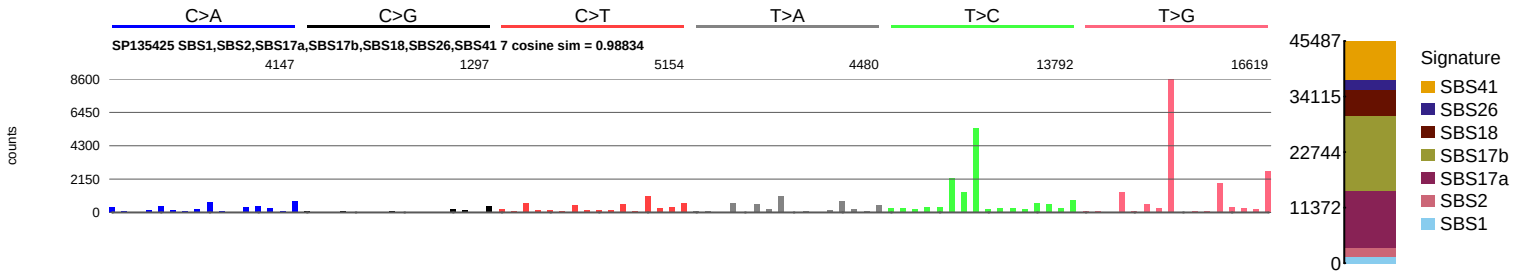
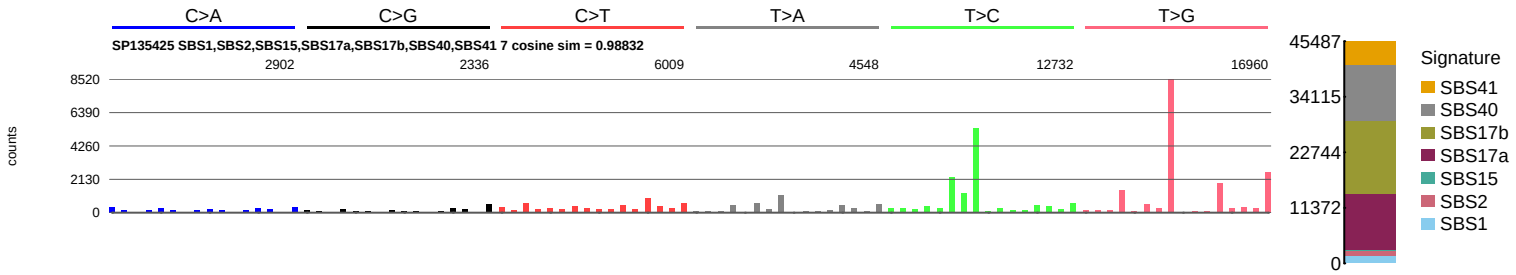
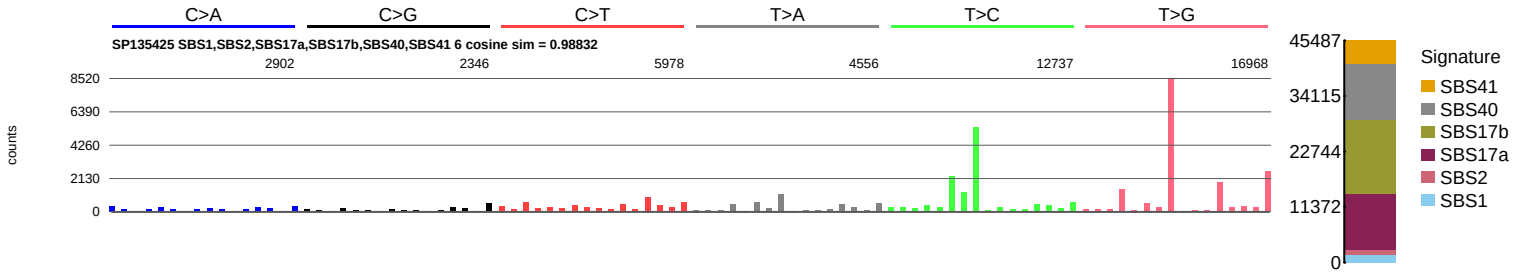
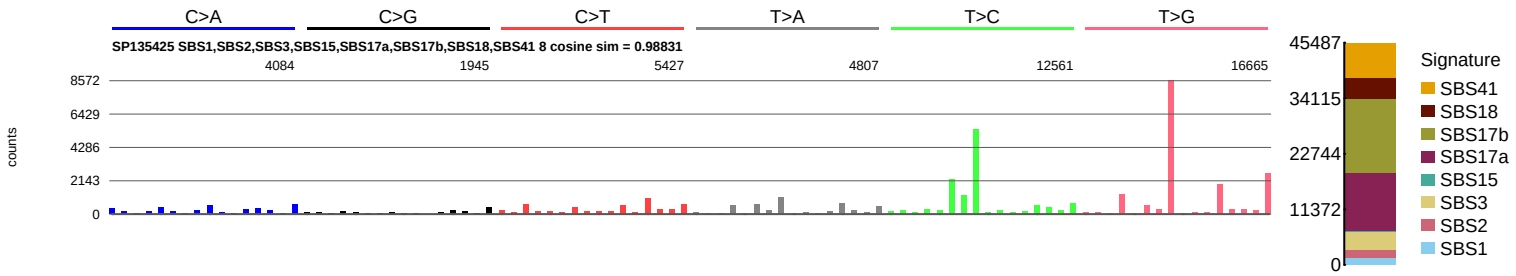
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98836
4133
1286
5226
4464
13766
16609
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

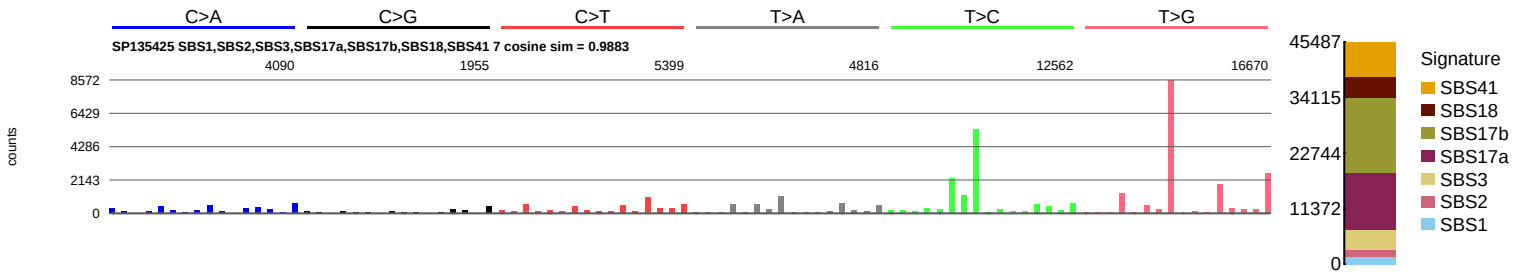

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98836
2838
2295
5925
4494
13000
16937
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

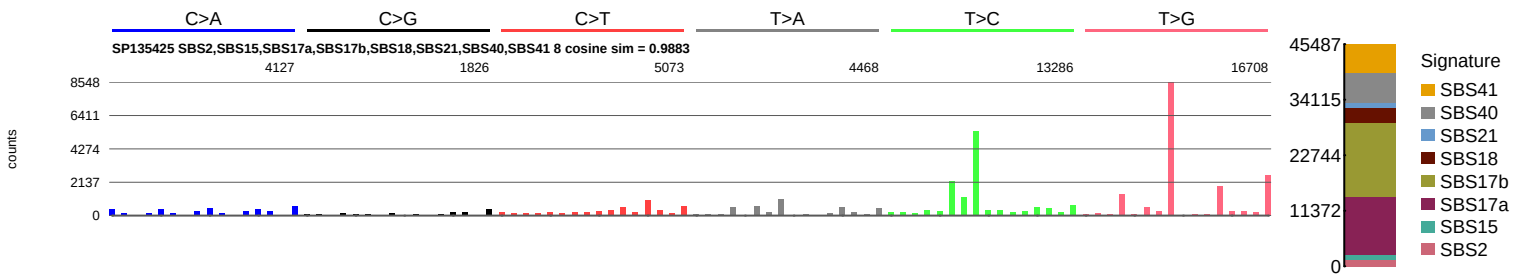

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98834
4147
1297
5154
4480
13792
16619
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98832
2902
2336
6009
4548
12732
16960
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98832
2902
2346
5978
4556
12737
16968
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98831
4084
1945
5427
4807
12561
16665
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9883
4090
1955
5399
4816
12562
16670
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.9883
4127
1826
5073
4468
13286
16708
counts
8548
6411
4274
2137
0
45487
34115
22744
11372
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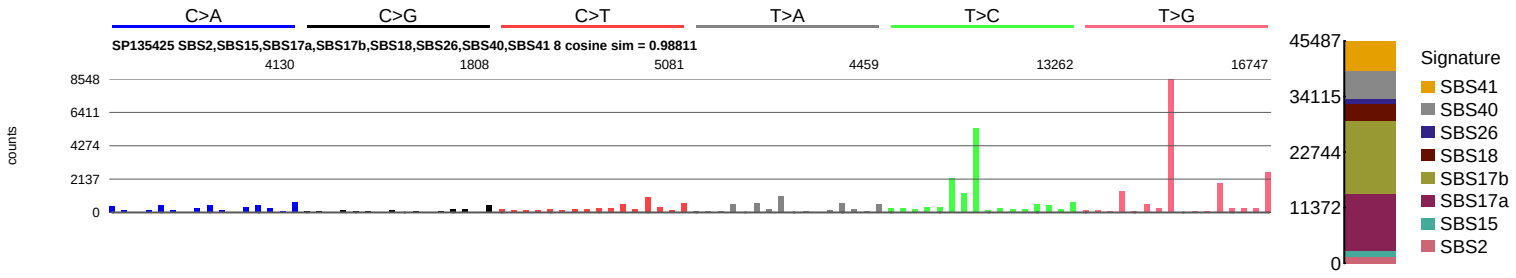
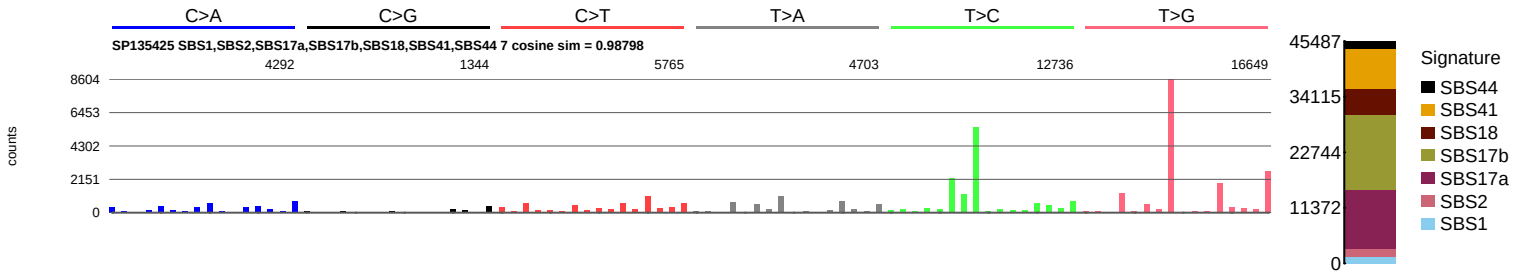
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98811
4130
1808
5081
4459
13262
16747
8548
6411
4274
2137
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
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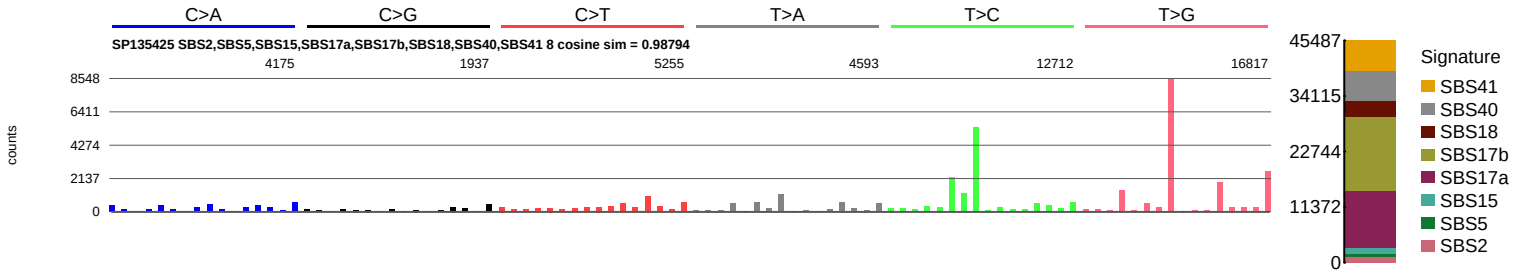
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98798
4292
1344
5765
4703
12736
16649
8604
6453
4302
2151
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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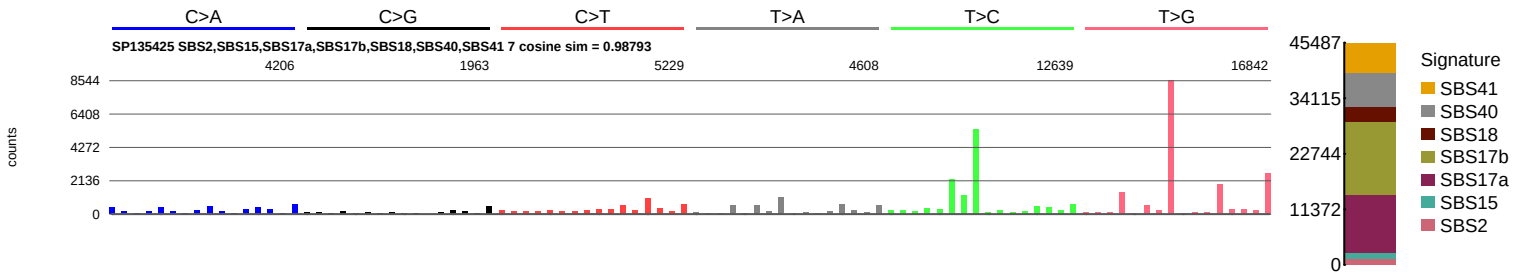
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98794
4175
1937
5255
4593
12712
16817
8548
6411
4274
2137
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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C>G
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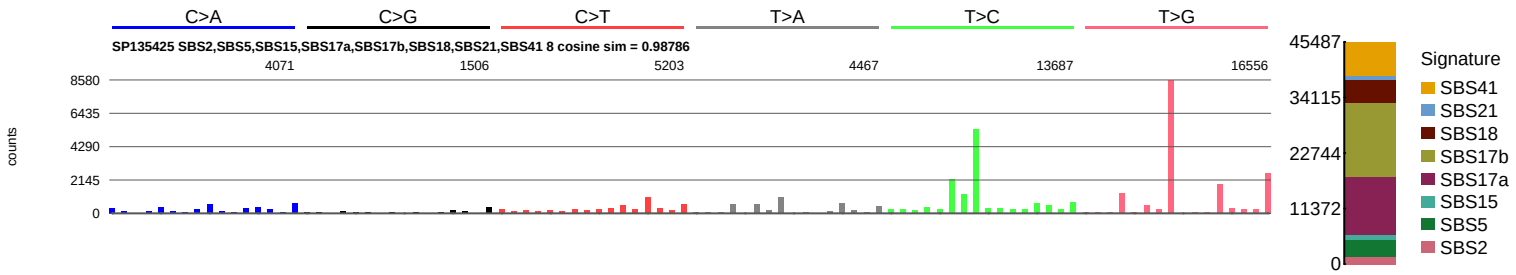
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98793
4206
1963
5229
4608
12639
16842
8544
6408
4272
2136
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2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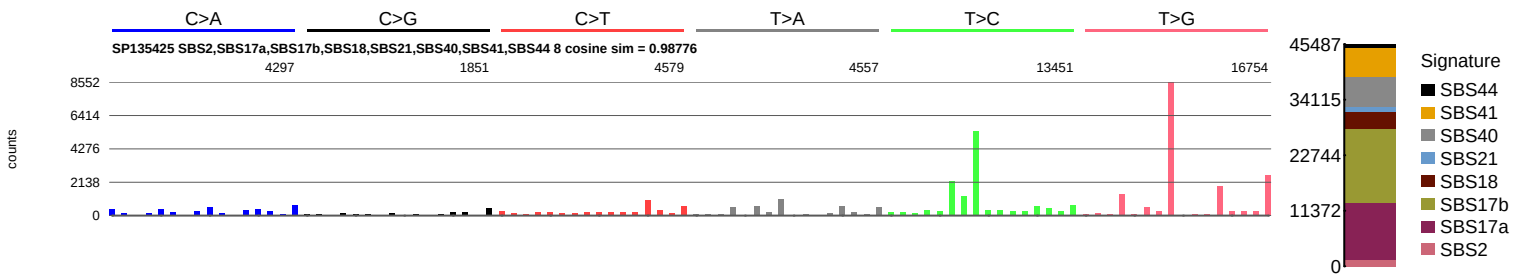
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98786
4071
1506
5203
4467
13687
16556
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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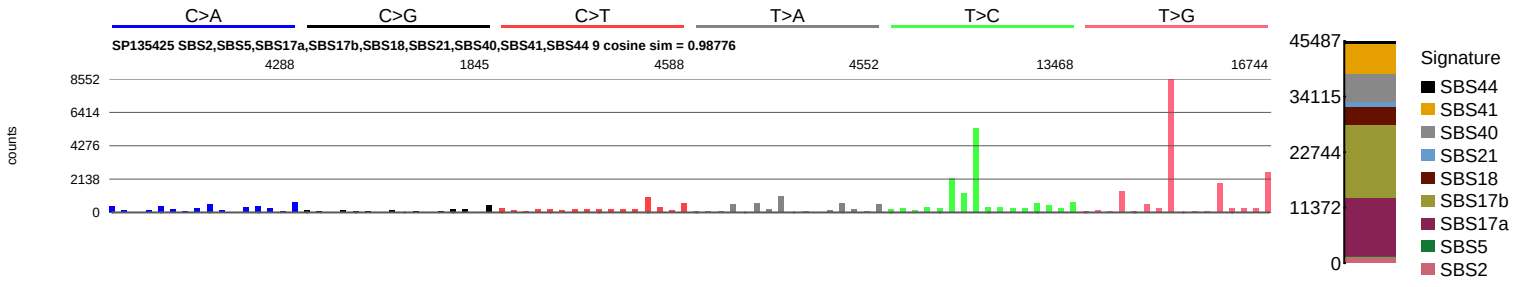
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98776
4297
1851
4579
4557
13451
16754
8552
6414
4276
2138
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2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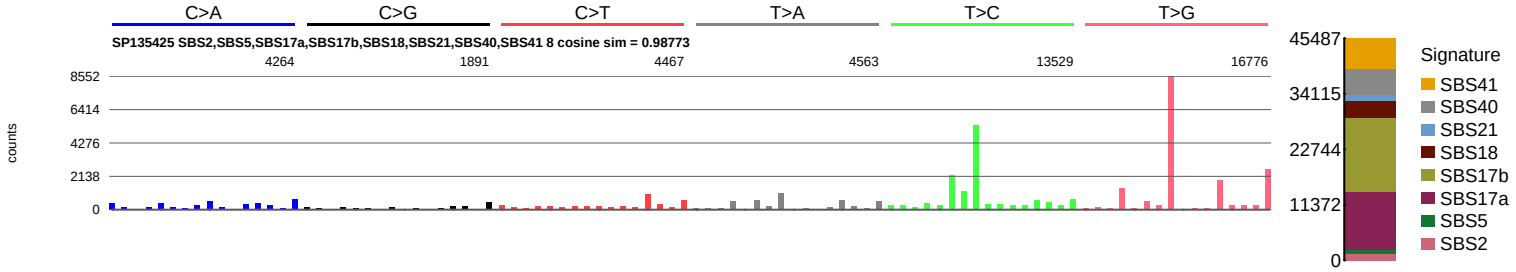
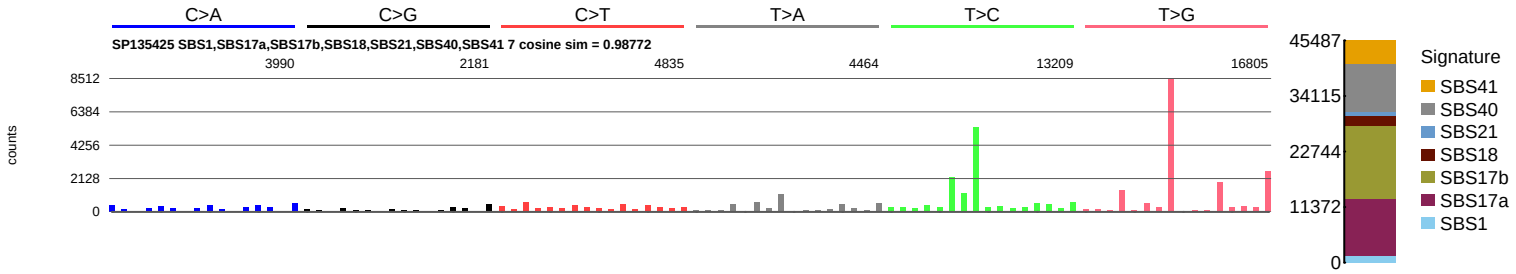
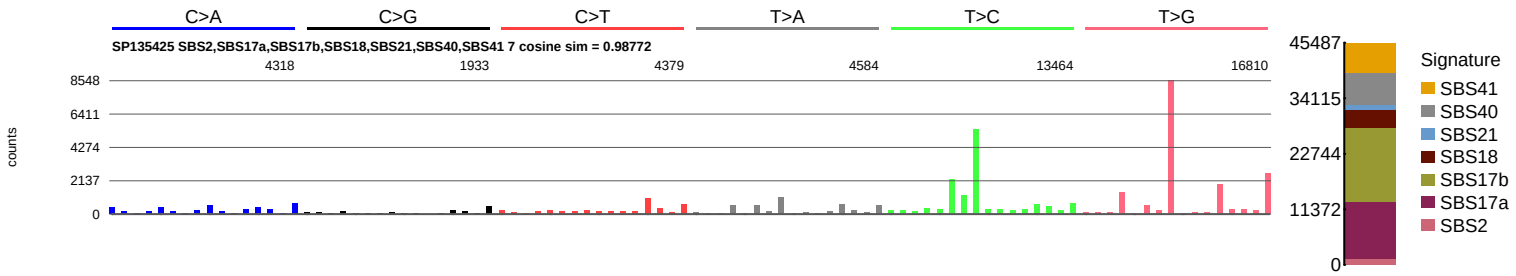
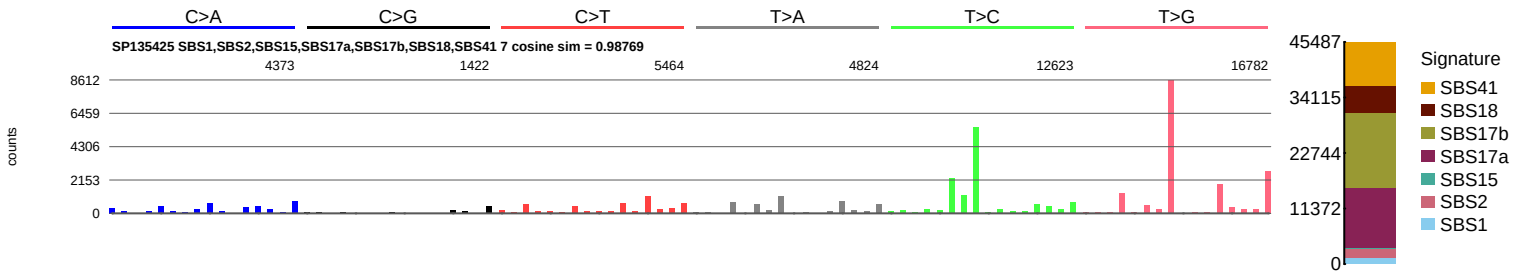
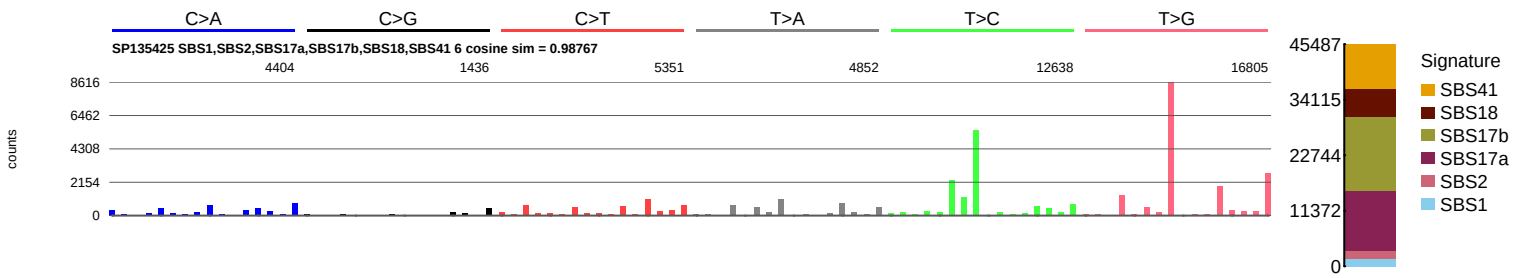
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98776
4288
1845
4588
4552
13468
16744
8552
6414
4276
2138
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

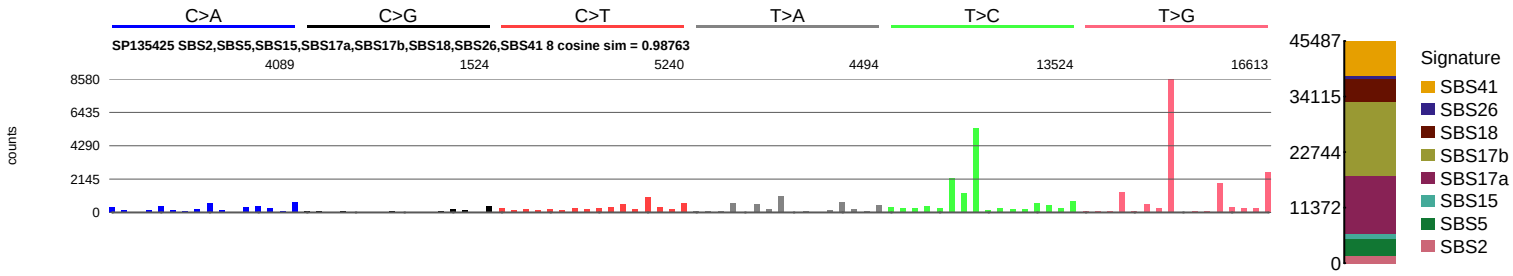

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98773
4264
1891
4467
4563
13529
16776
8552
6414
4276
2138
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

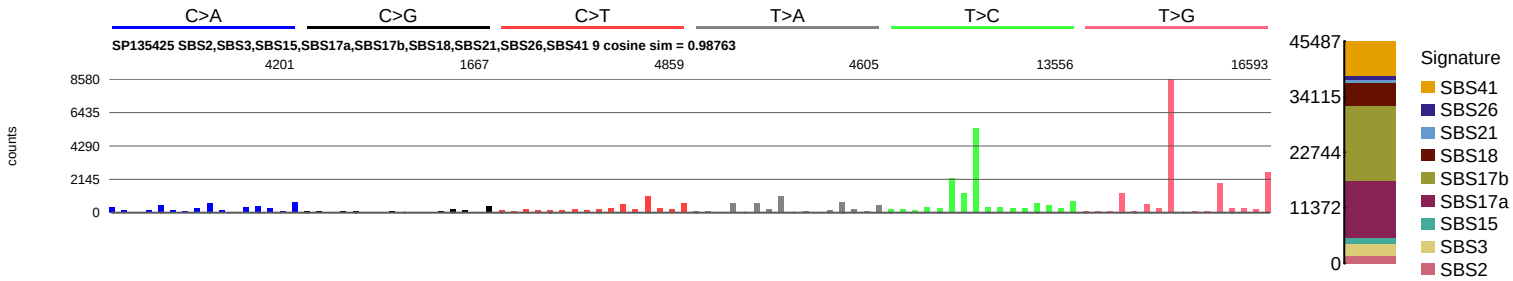

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98772
3990
2181
4835
4464
13209
16805
8512
6384
4256
2128
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

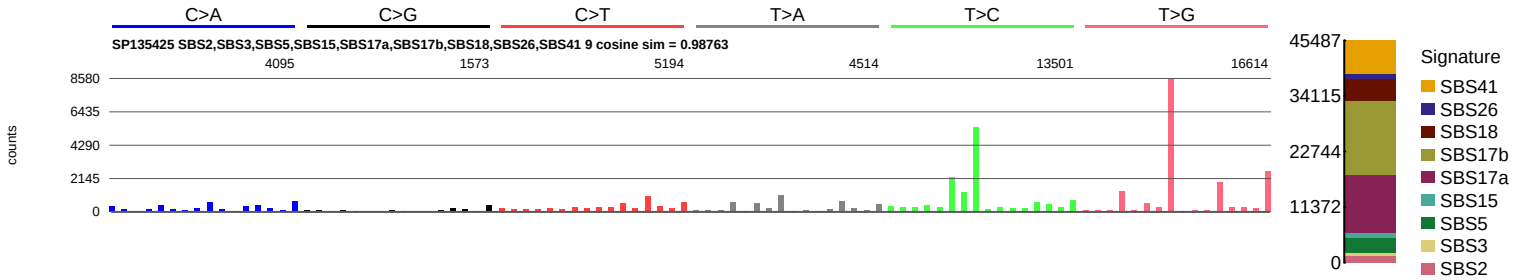

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98772
4318
1933
4379
4584
13464
16810
8548
6411
4274
2137
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2

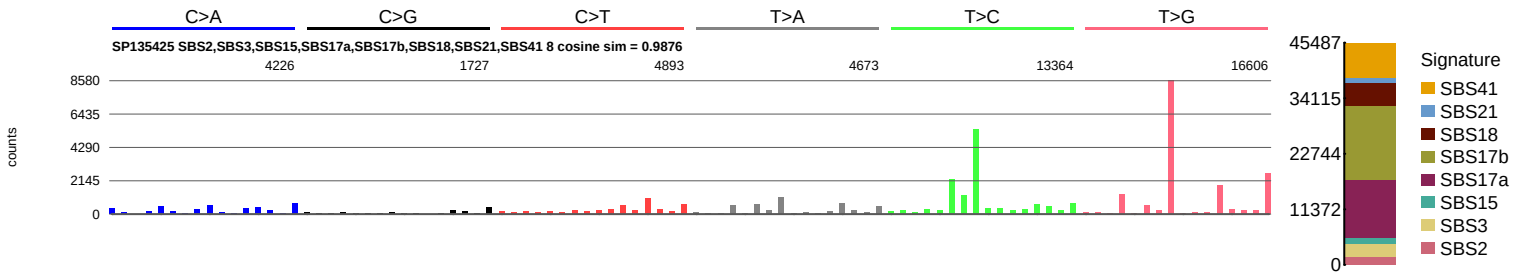

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98769
4373
1422
5464
4824
12623
16782
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

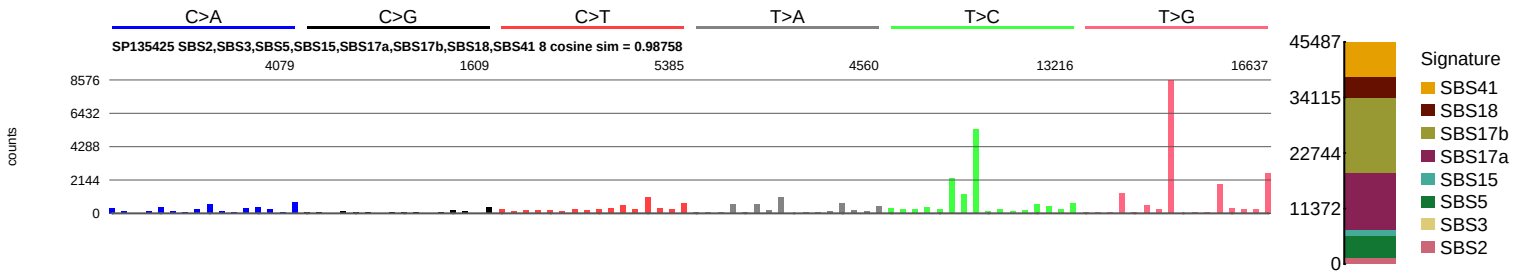

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98767
4404
1436
5351
4852
12638
16805
8616
6462
4308
2154
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98763
4089
1524
5240
4494
13524
16613
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98763
4201
1667
4859
4605
13556
16593
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98763
4095
1573
5194
4514
13501
16614
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.9876
4226
1727
4893
4673
13364
16606
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98758
4079
1609
5385
4560
13216
16637
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
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Signature
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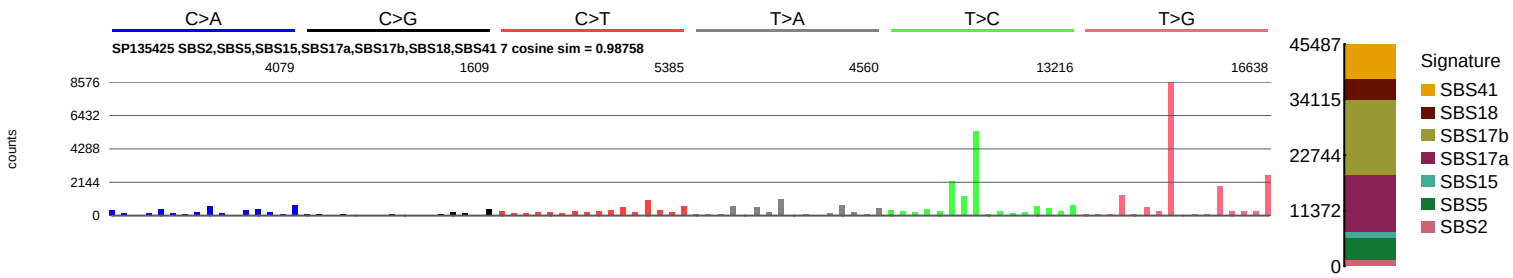
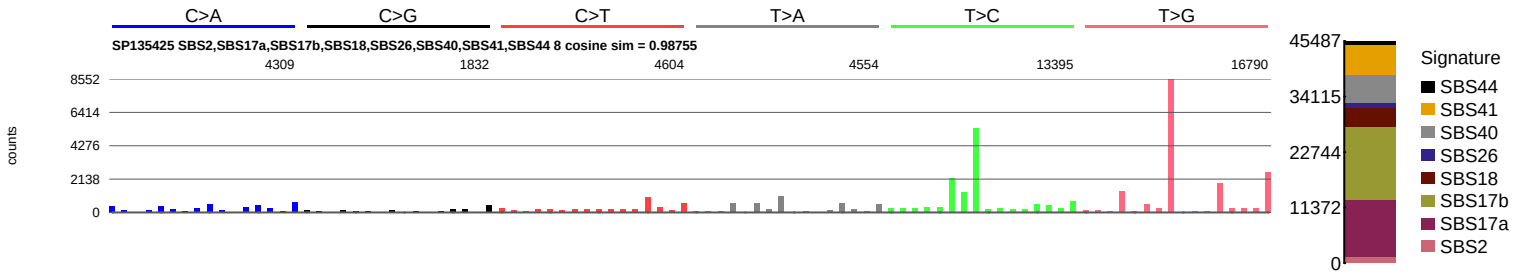
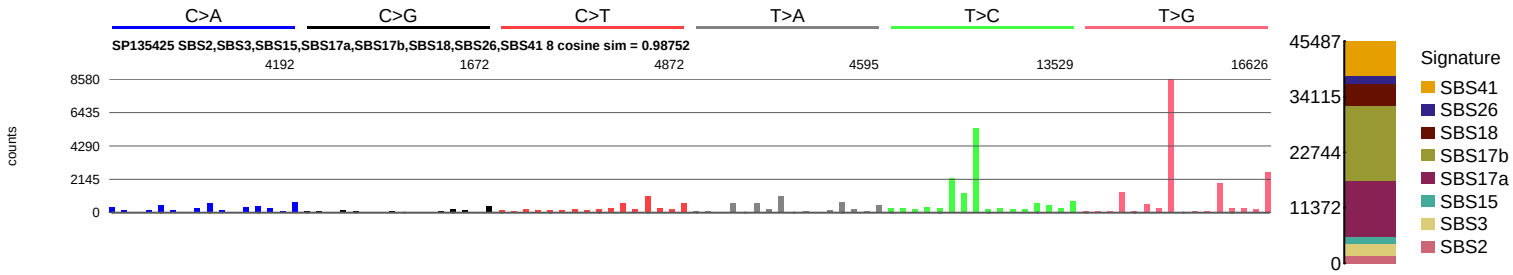
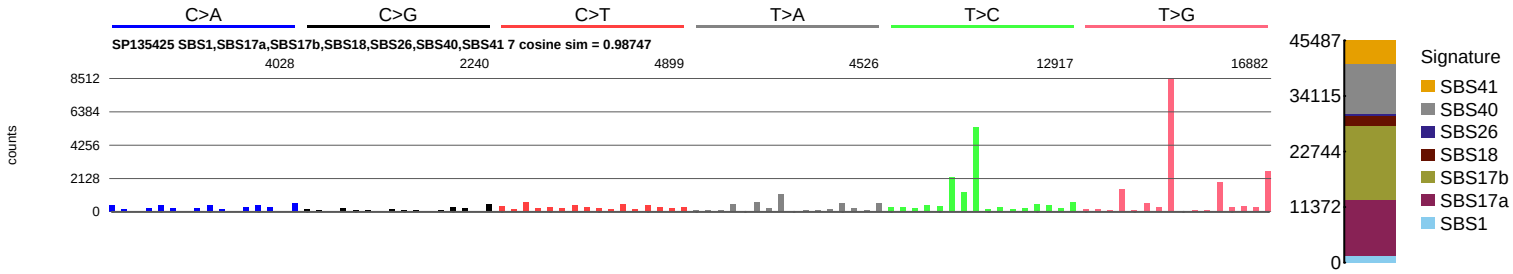
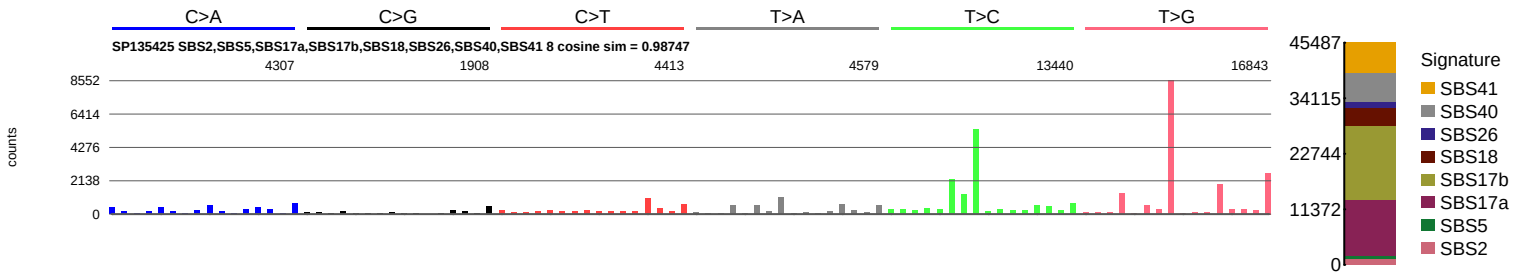
SBS5
SBS3
SBS2

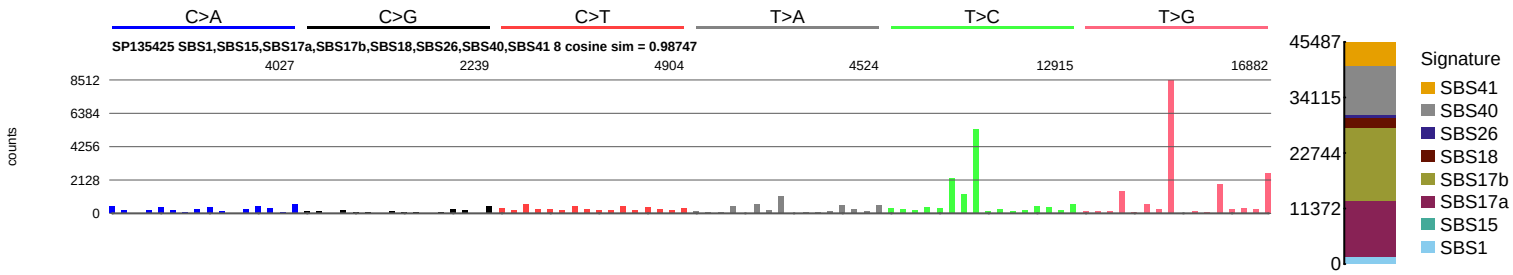

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98758
4079
1609
5385
4560
13216
16638
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

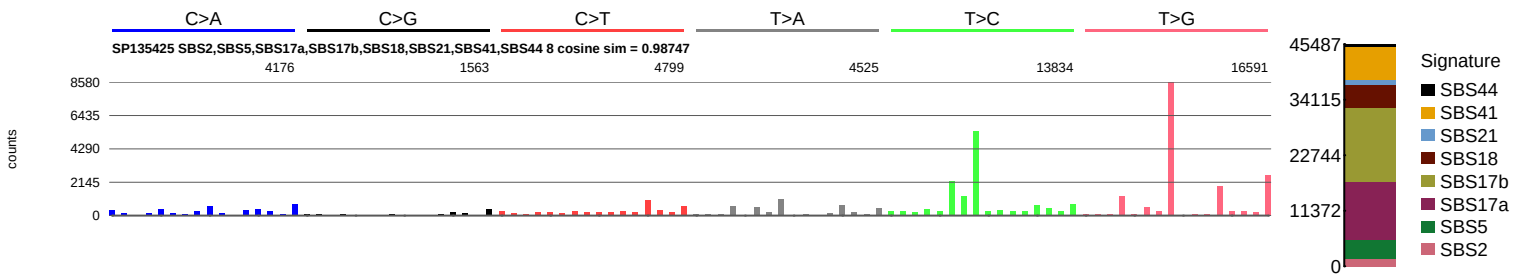

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98755
4309
1832
4604
4554
13395
16790
counts
8552
6414
4276
2138
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2

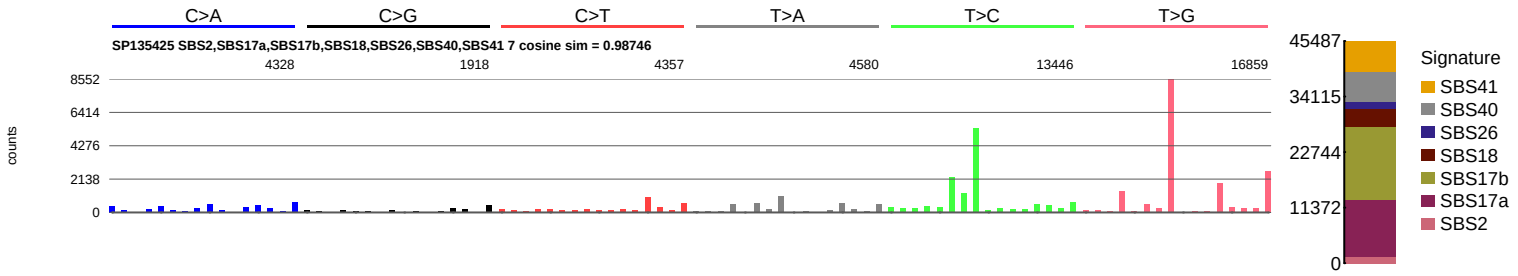

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98752
4192
1672
4872
4595
13529
16626
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

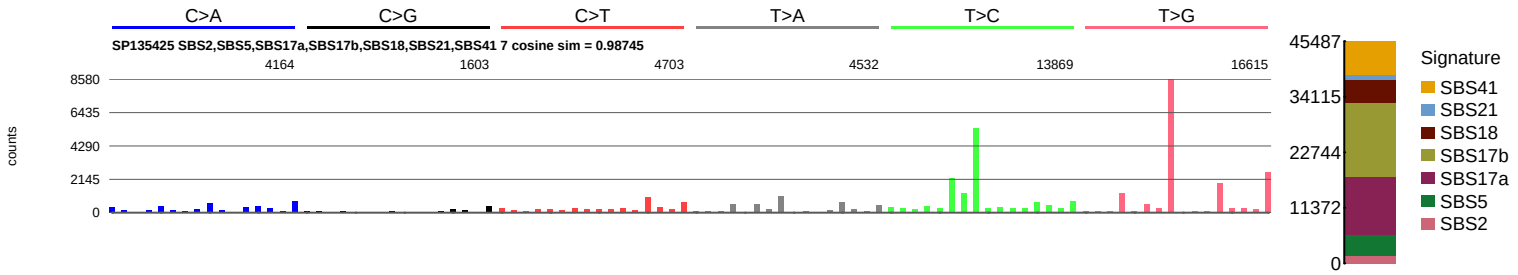

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98747
4028
2240
4899
4526
12917
16882
counts
8512
6384
4256
2128
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98747
4307
1908
4413
4579
13440
16843
counts
8552
6414
4276
2138
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98747
4027
2239
4904
4524
12915
16882
counts
8512
6384
4256
2128
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98747
4176
1563
4799
4525
13834
16591
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98746
4328
1918
4357
4580
13446
16859
8552
6414
4276
2138
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98745
4164
1603
4703
4532
13869
16615
8580
6435
4290
2145
0
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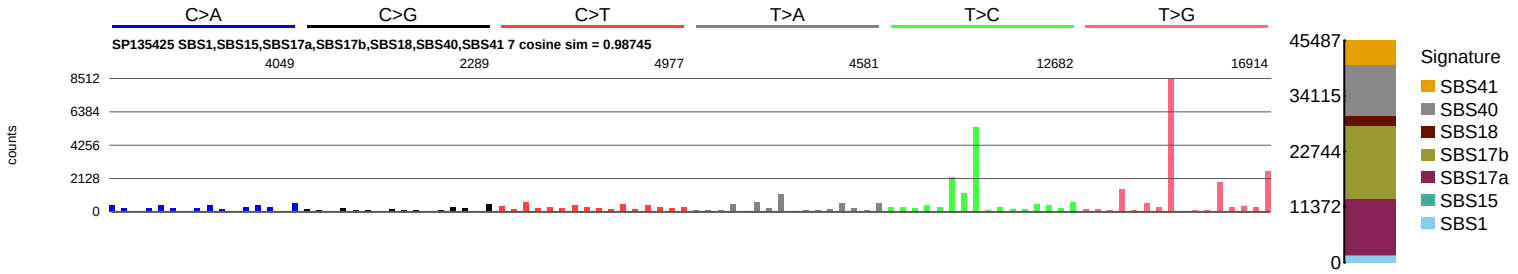
SBS5
SBS2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4049
2289
4977
4581
12682
16914
8512
6384
4256
2128
0
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22744
11372
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SBS17b
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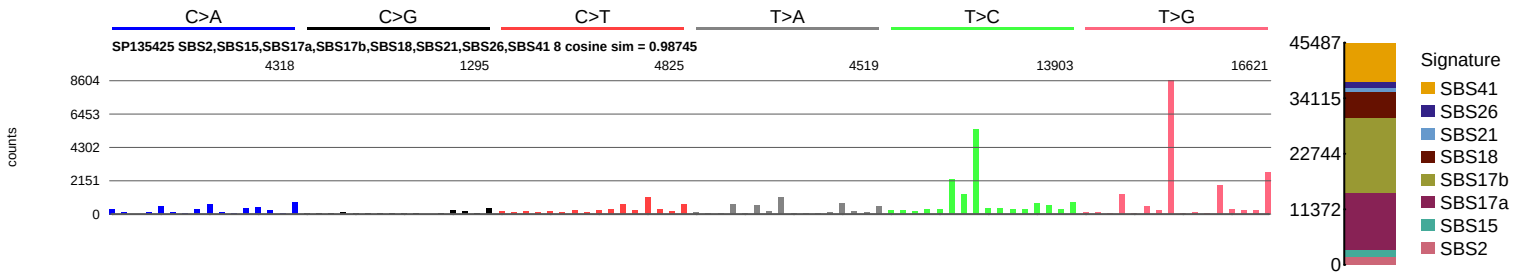
SBS15
SBS1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SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98745
4318
1295
4825
4519
13903
16621
8604
6453
4302
2151
0
counts
45487
34115
22744
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Signature
SBS41
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SBS21
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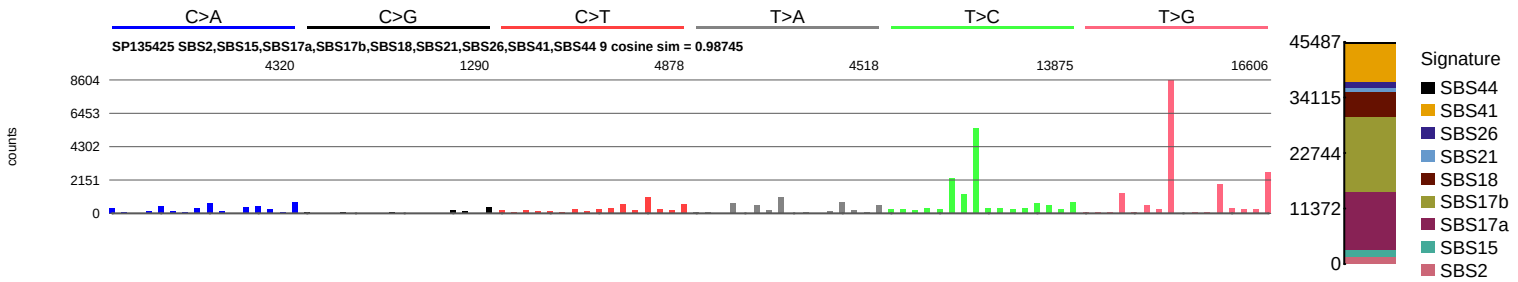
SBS17b
SBS17a
SBS15
SBS2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SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98745
4320
1290
4878
4518
13875
16606
8604
6453
4302
2151
0
counts
45487
34115
22744
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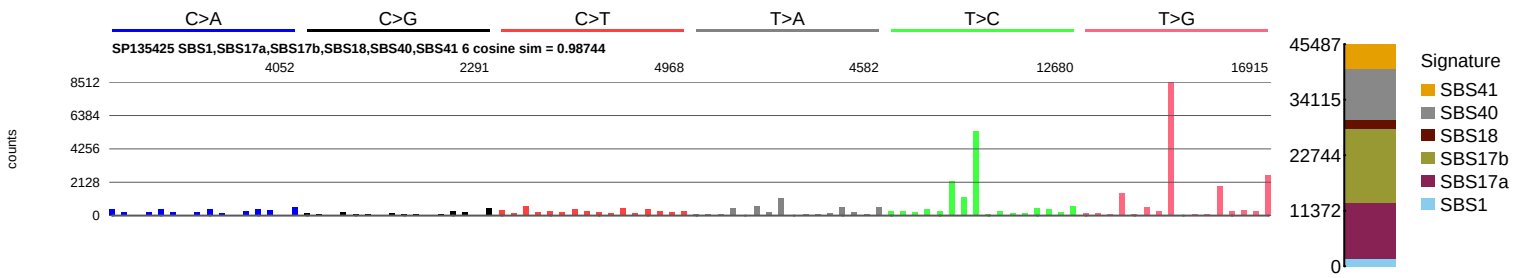
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98744
4052
2291
4968
4582
12680
16915
8512
6384
4256
2128
0
counts
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34115
22744
11372
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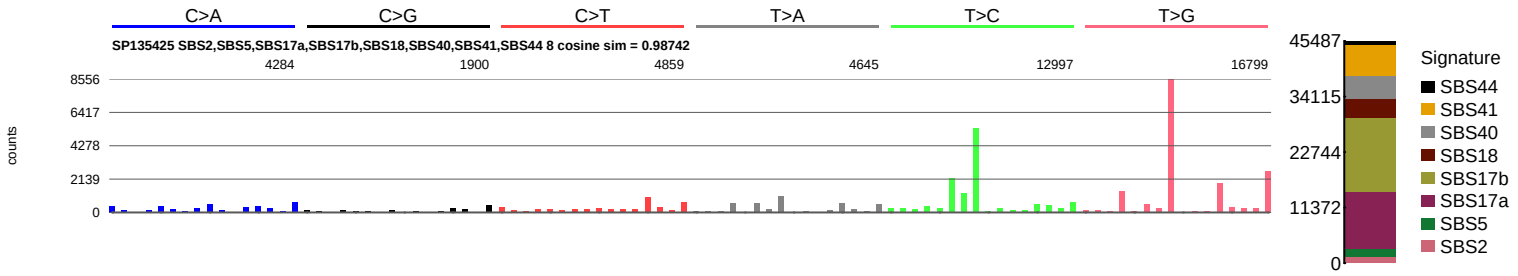
SBS18
SBS17b
SBS17a
SBS1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SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98742
4284
1900
4859
4645
12997
16799
8556
6417
4278
2139
0
counts
45487
34115
22744
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SBS44
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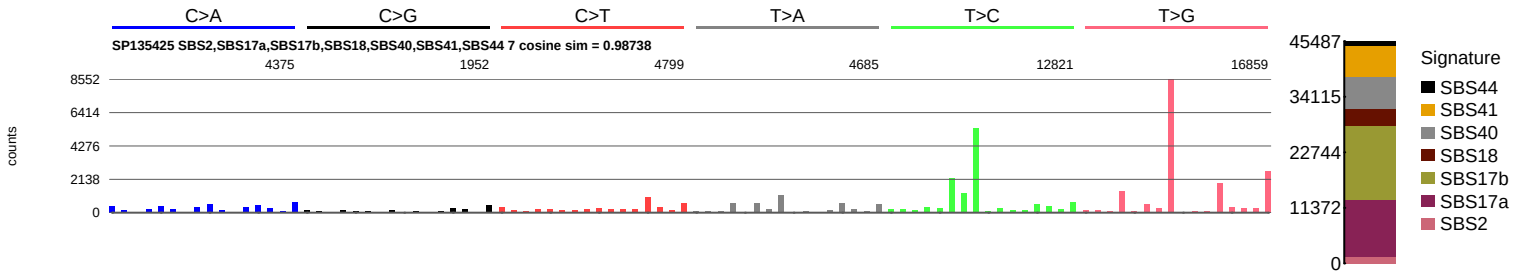
SBS17b
SBS17a
SBS5
SBS2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SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98738
4375
1952
4799
4685
12821
16859
8552
6414
4276
2138
0
counts
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34115
22744
11372
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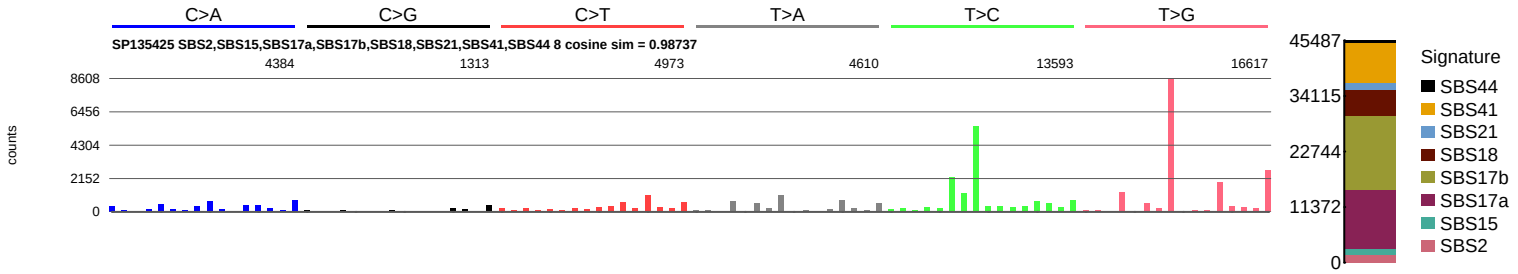
SBS40
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98737
4384
1313
4973
4610
13593
16617
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
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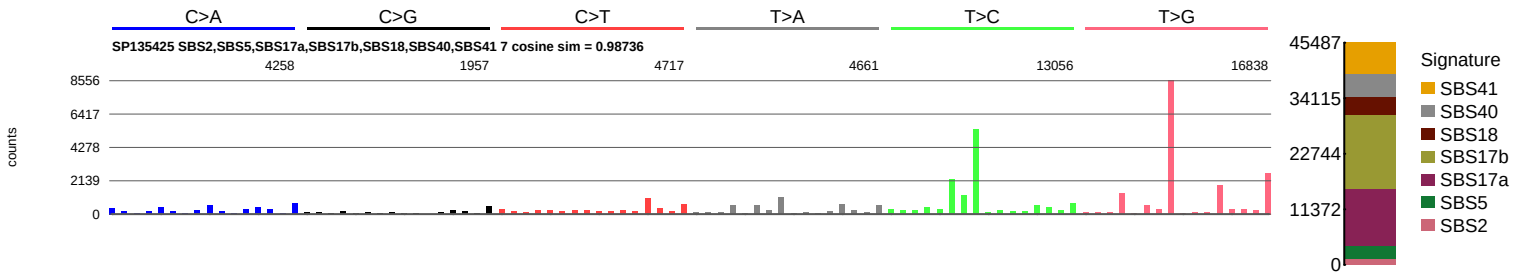
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
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T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98736
4258
1957
4717
4661
13056
16838
8556
6417
4278
2139
0
counts
45487
34115
22744
11372
0
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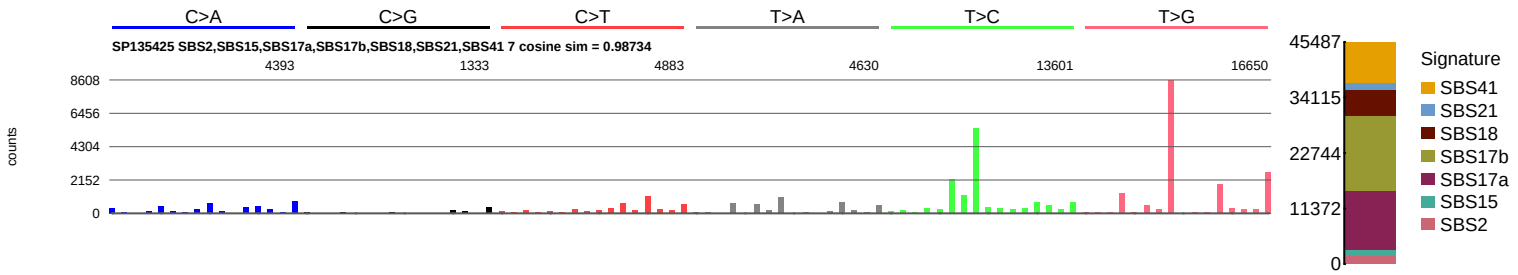
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98734
4393
1333
4883
4630
13601
16650
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
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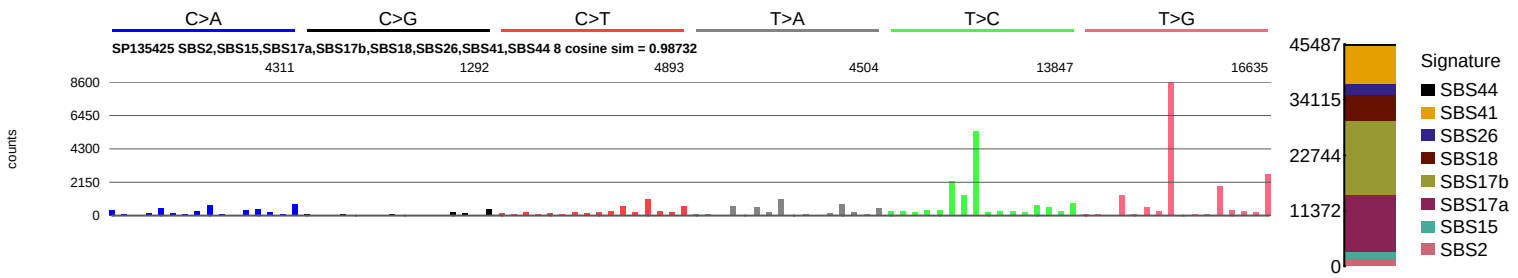
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2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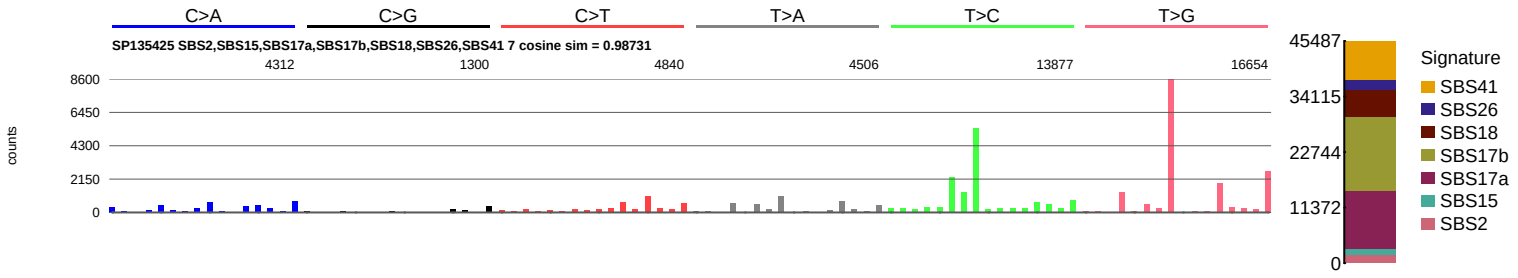
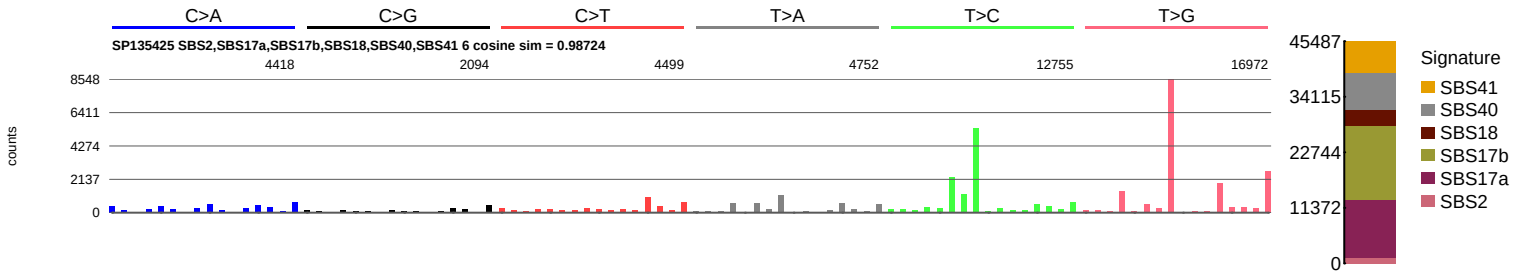
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98732
4311
1292
4893
4504
13847
16635
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98731
4312
1300
4840
4506
13877
16654
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2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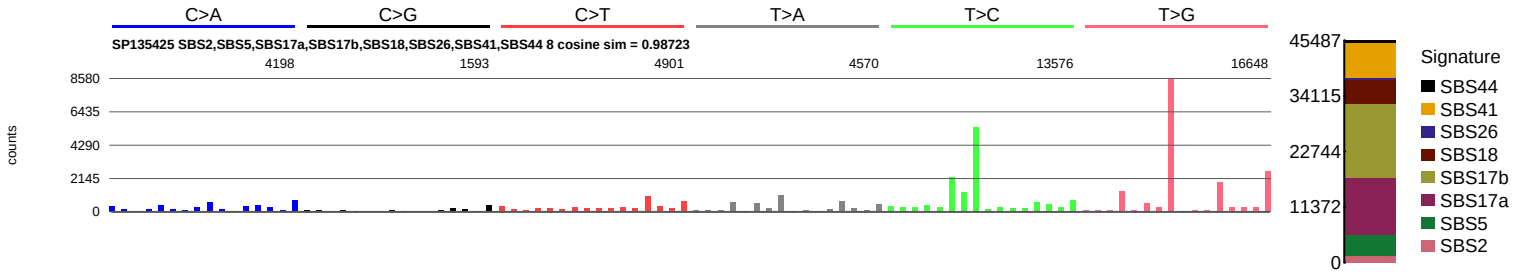
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98724
4418
2094
4499
4752
12755
16972
counts
8548
6411
4274
2137
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98723
4198
1593
4901
4570
13576
16648
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
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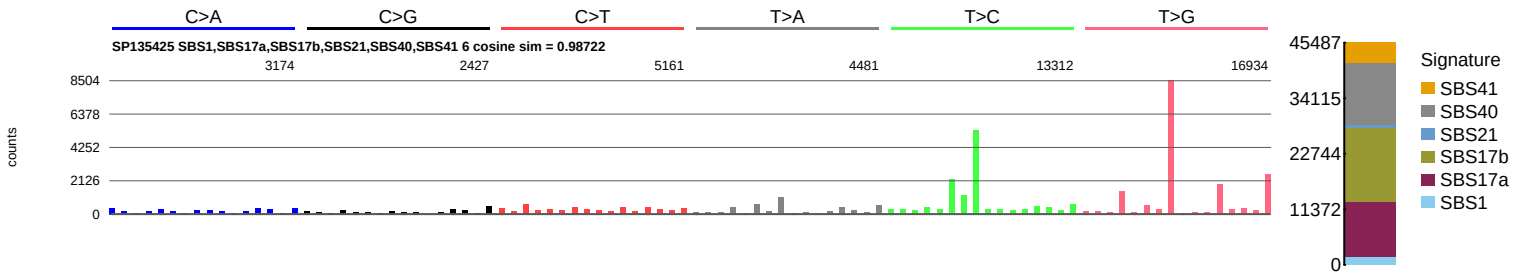
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2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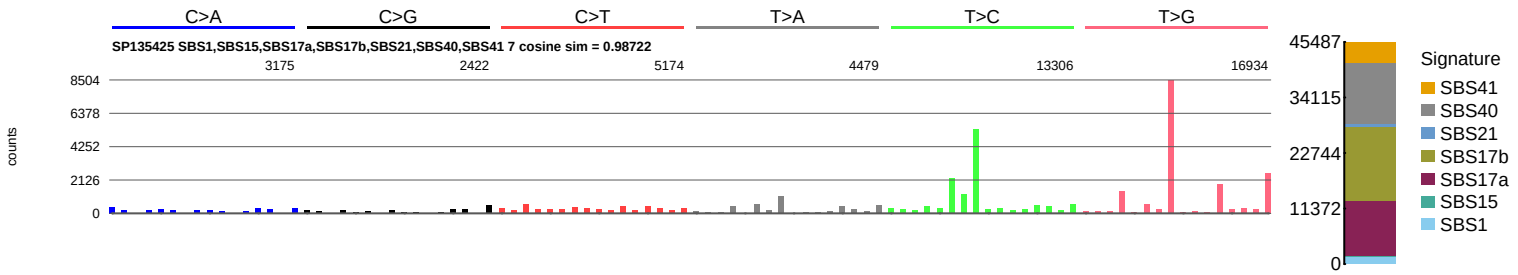
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98722
3174
2427
5161
4481
13312
16934
counts
8504
6378
4252
2126
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98722
3175
2422
5174
4479
13306
16934
counts
8504
6378
4252
2126
0
45487
34115
22744
11372
0
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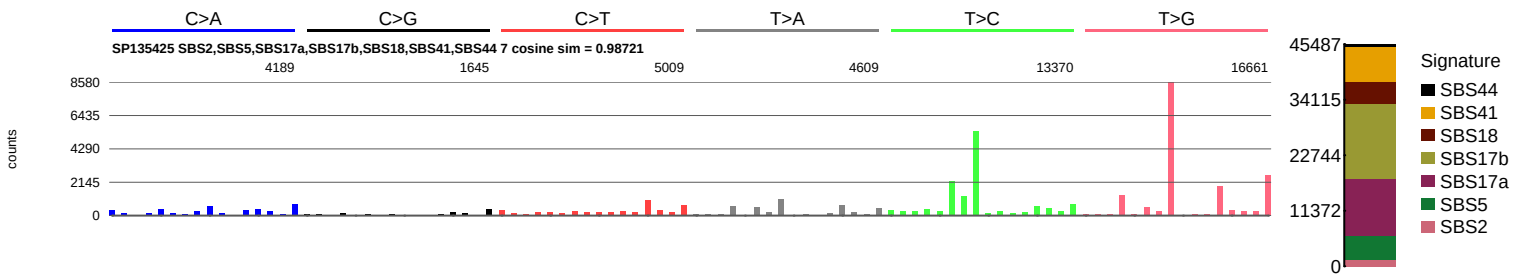
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98721
4189
1645
5009
4609
13370
16661
counts
8580
6435
4290
2145
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45487
34115
22744
11372
0
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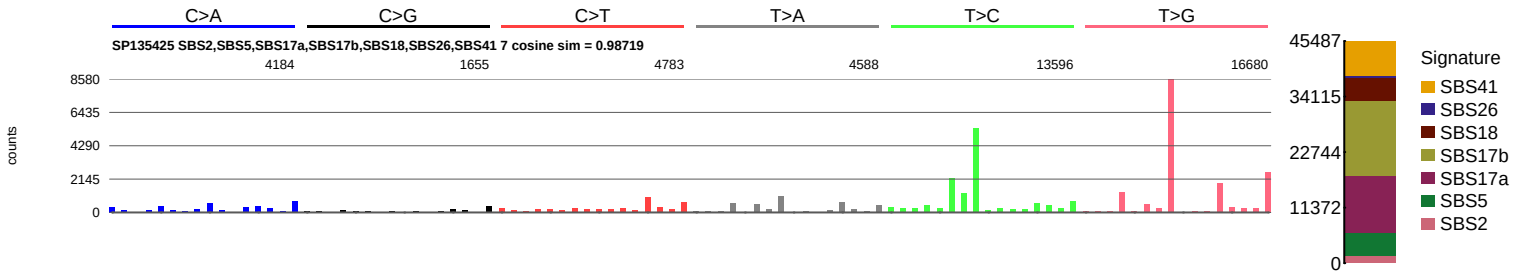
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98719
4184
1655
4783
4588
13596
16680
8580
6435
4290
2145
0
counts
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34115
22744
11372
0
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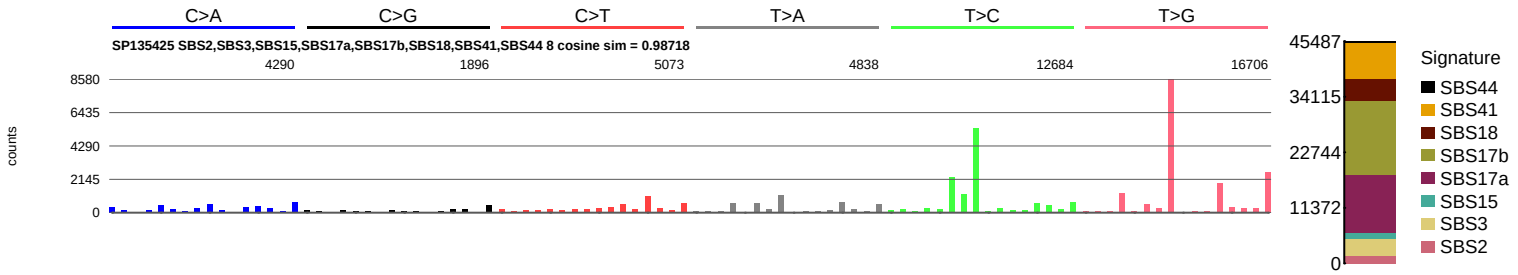
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
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T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98718
4290
1896
5073
4838
12684
16706
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
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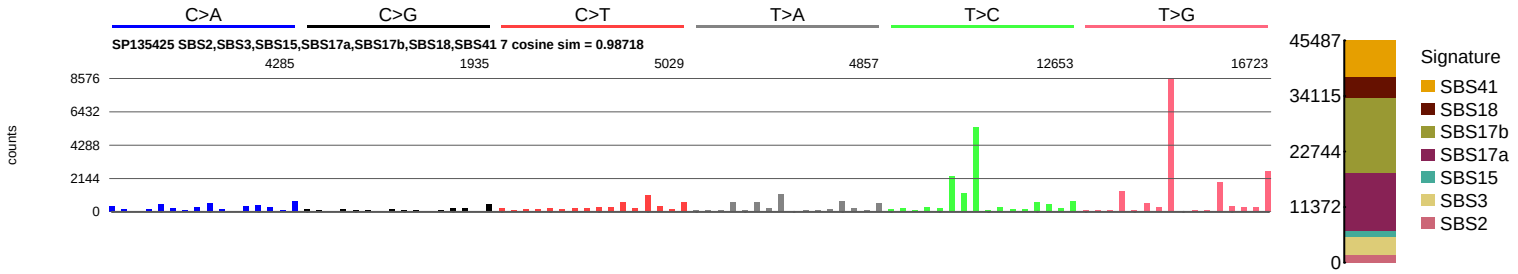
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98718
4285
1935
5029
4857
12653
16723
8576
6432
4288
2144
0
counts
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34115
22744
11372
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SBS18
SBS17b
SBS17a
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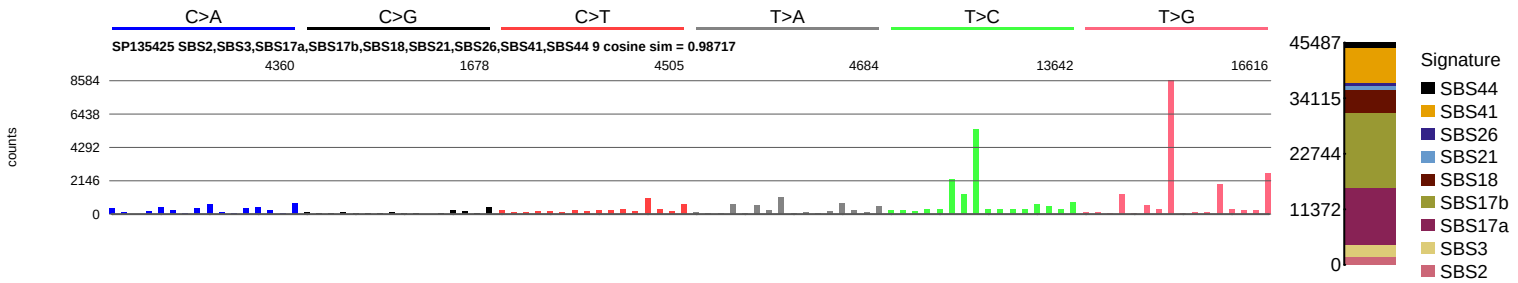
SBS3
SBS2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C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98717
4360
1678
4505
4684
13642
16616
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
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SBS21
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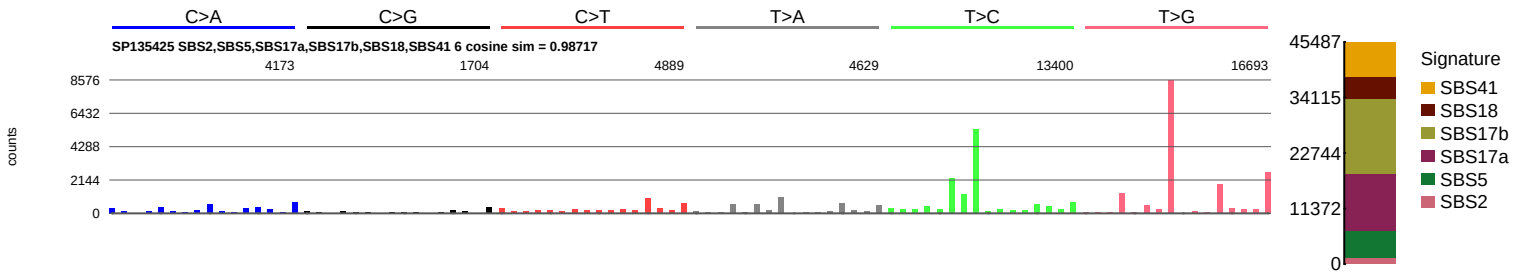
SBS3
SBS2


C>A
C>G
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T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98717
4173
1704
4889
4629
13400
16693
8576
6432
4288
2144
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counts
45487
34115
22744
11372
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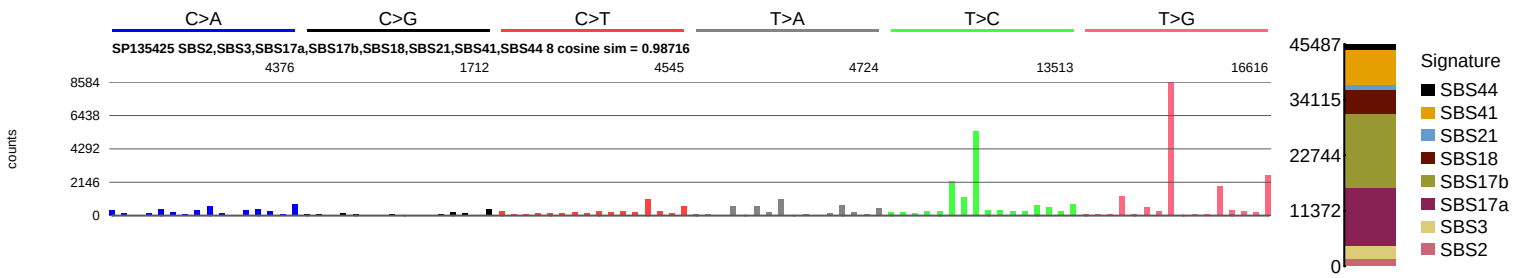
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98716
4376
1712
4545
4724
13513
16616
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
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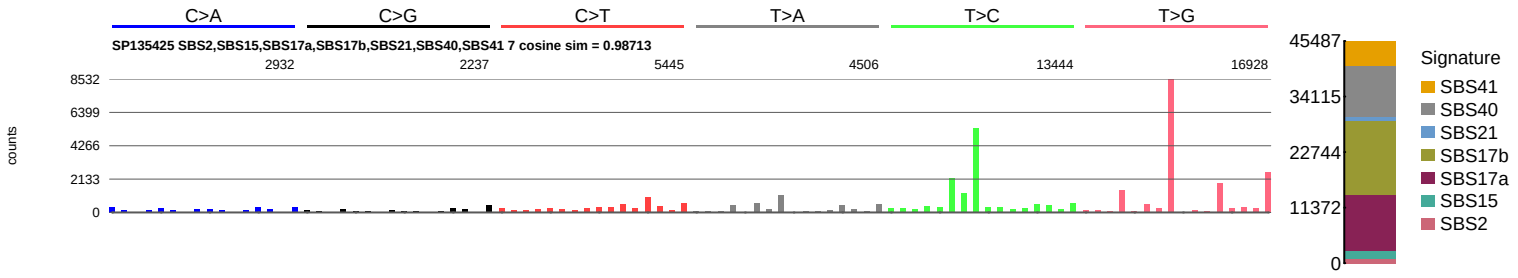
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98713
2932
2237
5445
4506
13444
16928
8532
6399
4266
2133
0
counts
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34115
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11372
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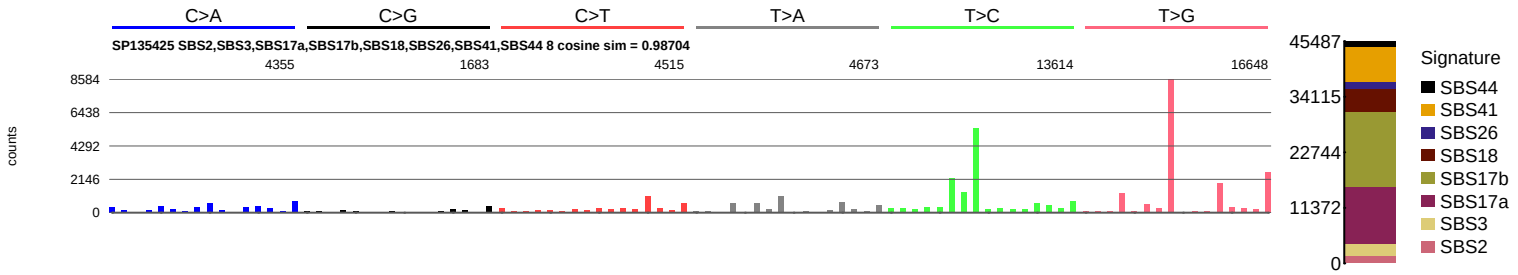
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98704
4355
1683
4515
4673
13614
16648
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
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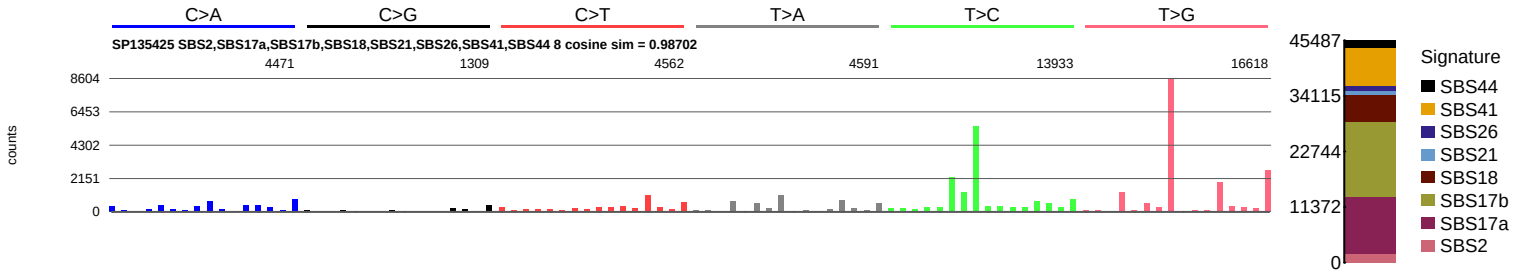
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98702
4471
1309
4562
4591
13933
16618
8604
6453
4302
2151
0
counts
45487
34115
22744
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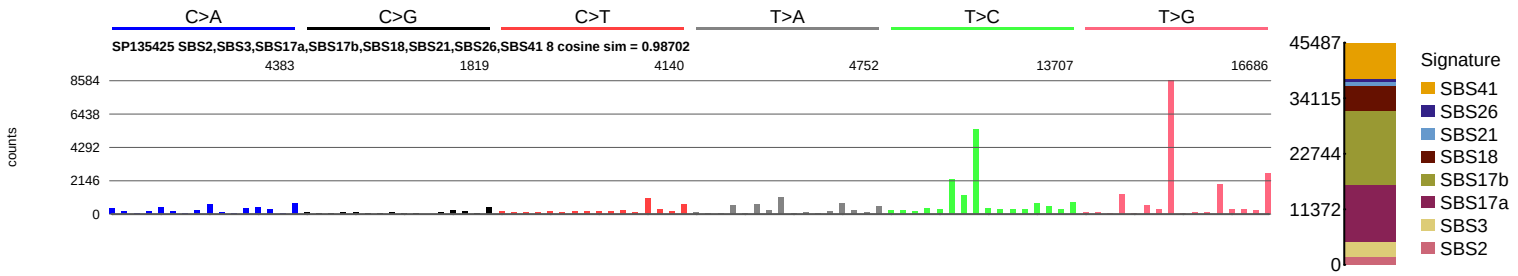
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SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98702
4383
1819
4140
4752
13707
16686
8584
6438
4292
2146
0
counts
45487
34115
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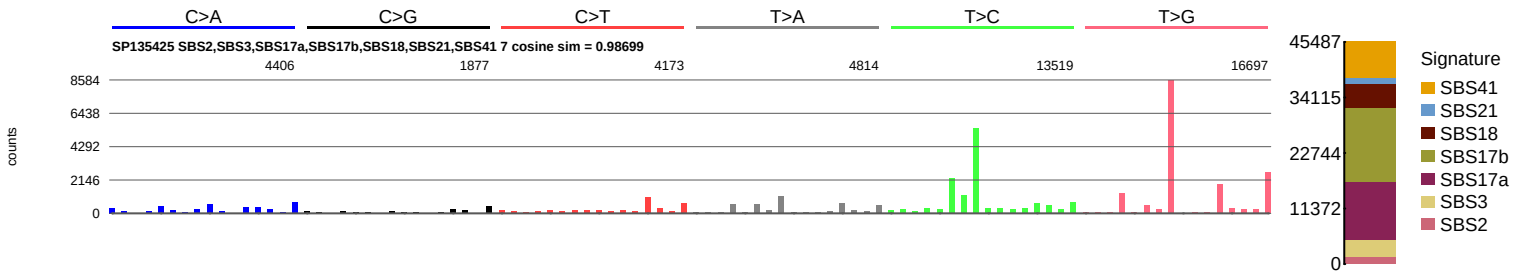
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SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98699
4406
1877
4173
4814
13519
16697
8584
6438
4292
2146
0
counts
45487
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22744
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Signature
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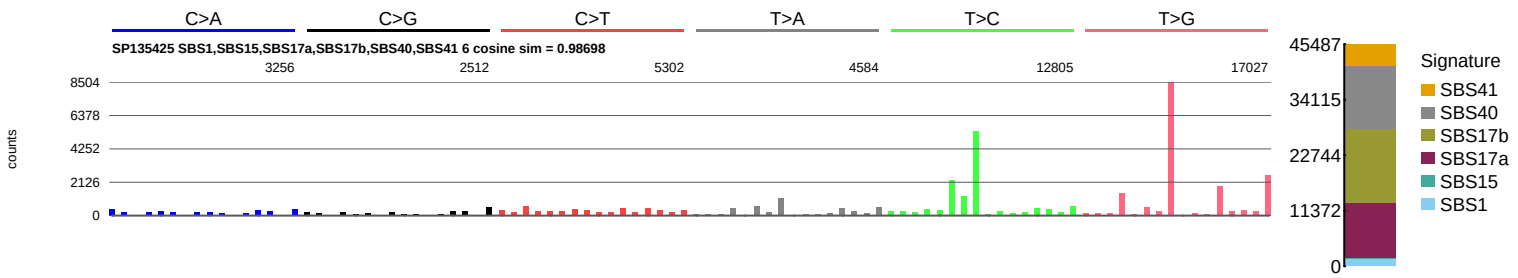
SBS17b
SBS17a
SBS3
SBS2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C>T
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T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98698
3256
2512
5302
4584
12805
17027
8504
6378
4252
2126
0
counts
45487
34115
22744
11372
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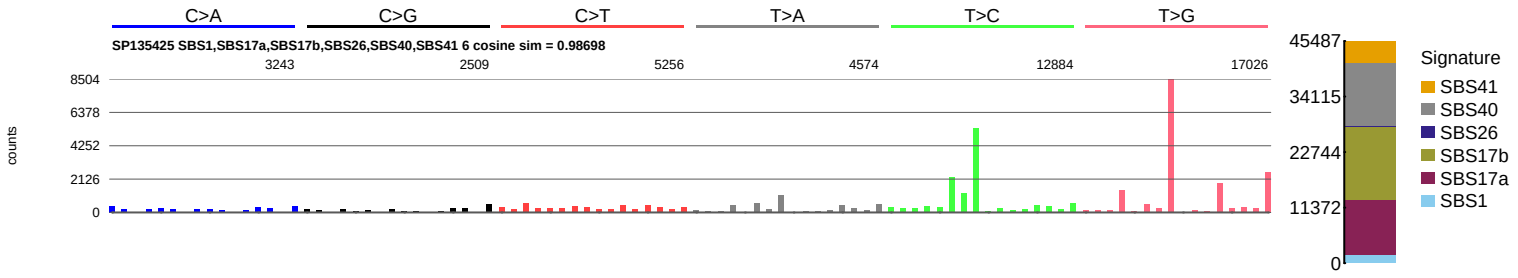
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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C>G
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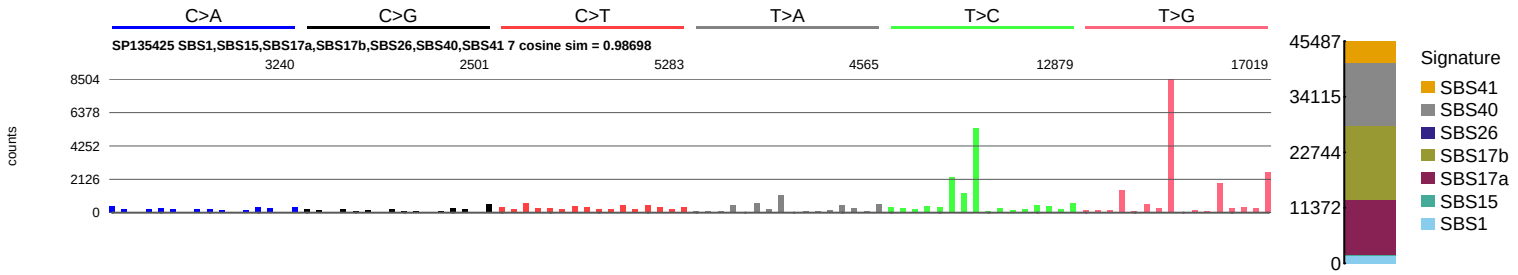
SP135425 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98698
3243
2509
5256
4574
12884
17026
counts
8504
6378
4252
2126
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98698
3240
2501
5283
4565
12879
17019
counts
8504
6378
4252
2126
0
45487
34115
22744
11372
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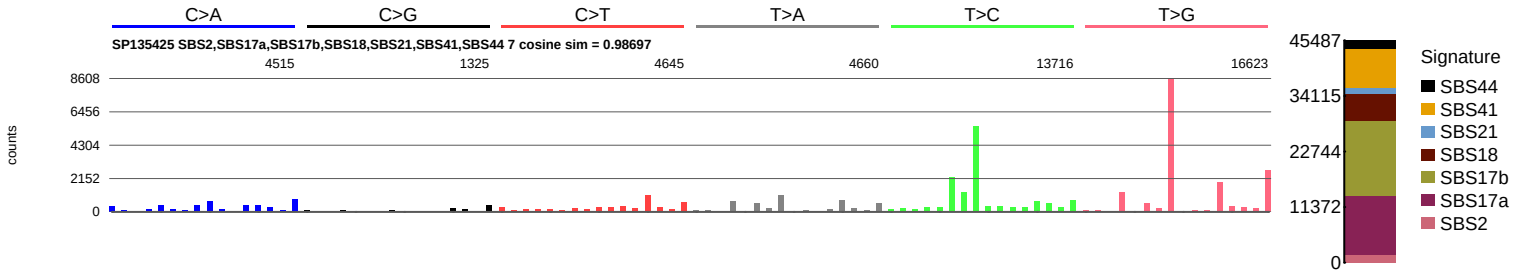
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98697
4515
1325
4645
4660
13716
16623
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
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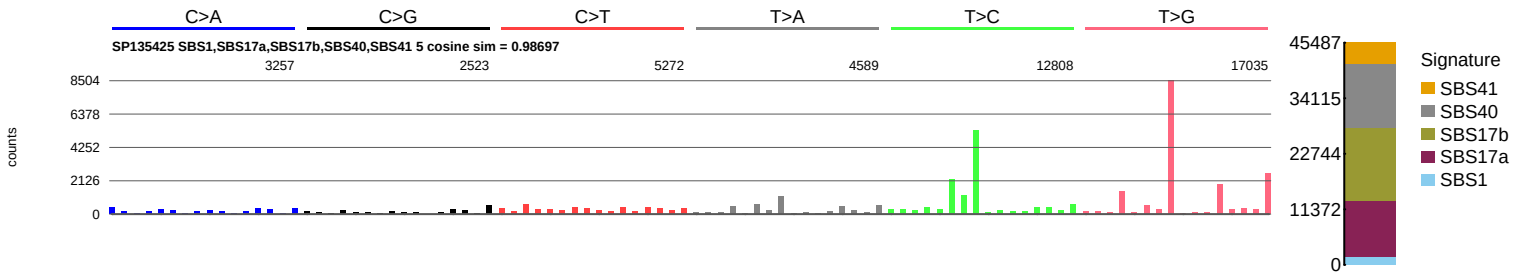
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2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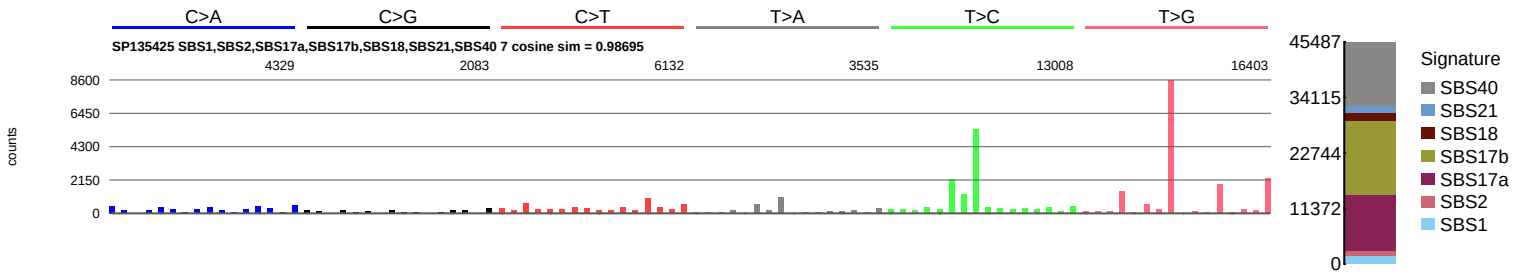
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98697
3257
2523
5272
4589
12808
17035
counts
8504
6378
4252
2126
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98695
4329
2083
6132
3535
13008
16403
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
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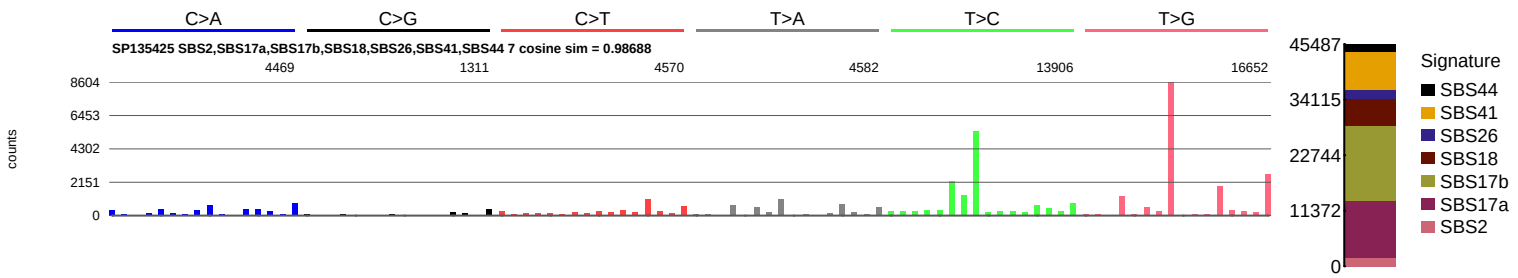
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98688
4469
1311
4570
4582
13906
16652
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
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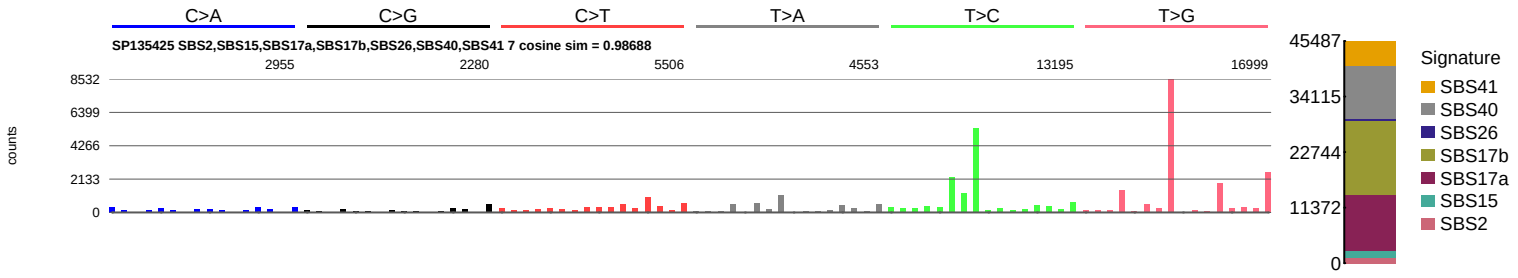
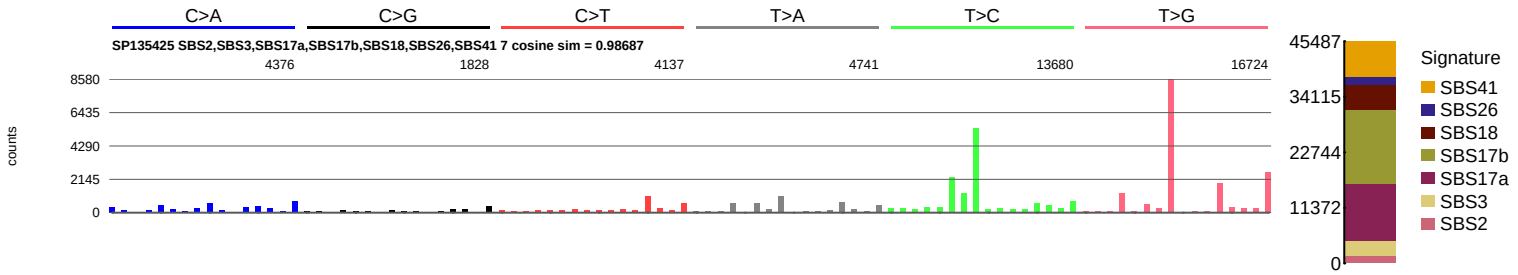
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98688
2955
2280
5506
4553
13195
16999
8532
6399
4266
2133
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98687
4376
1828
4137
4741
13680
16724
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
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Signature
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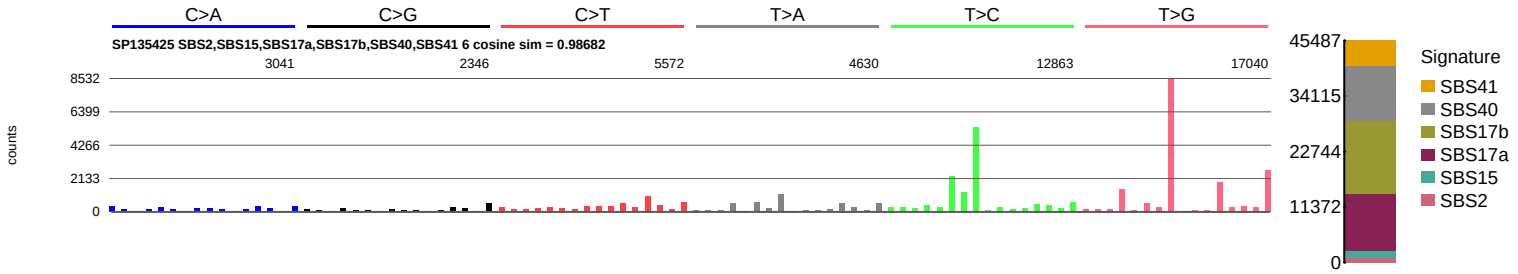
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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C>G
C>T
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T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98682
3041
2346
5572
4630
12863
17040
8532
6399
4266
2133
0
counts
45487
34115
22744
11372
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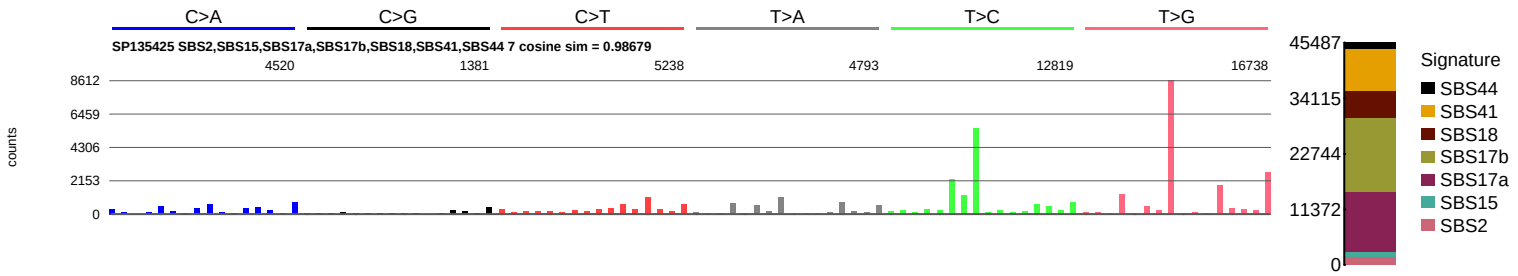
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98679
4520
1381
5238
4793
12819
16738
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
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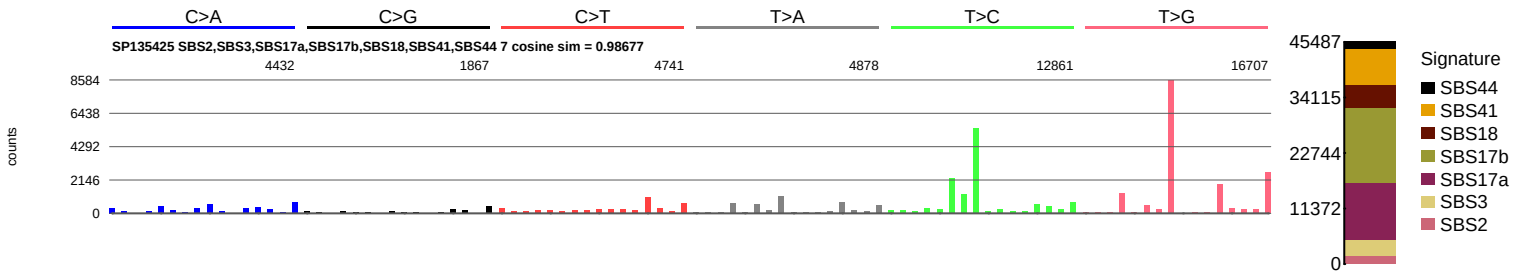
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98677
4432
1867
4741
4878
12861
16707
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
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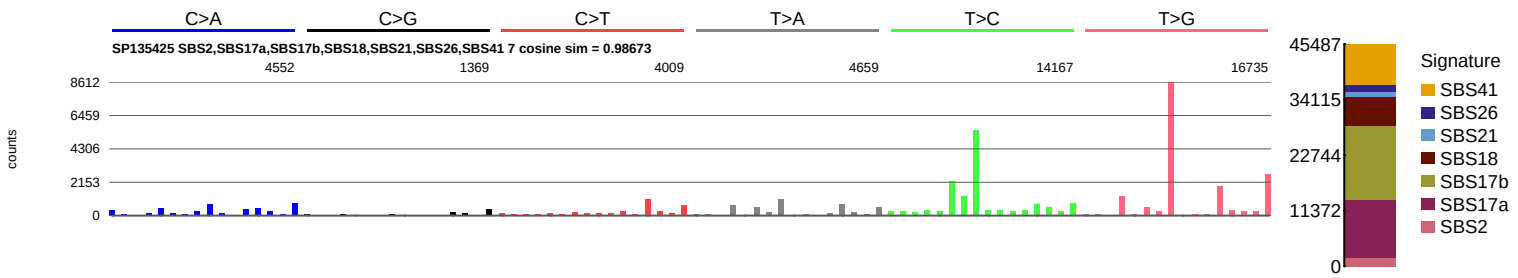
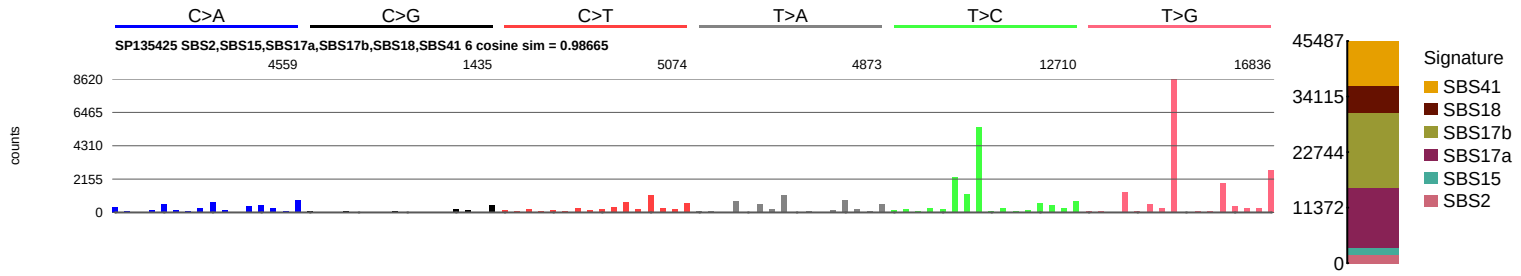
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98673
4552
1369
4009
4659
14167
16735
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98665
4559
1435
5074
4873
12710
16836
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
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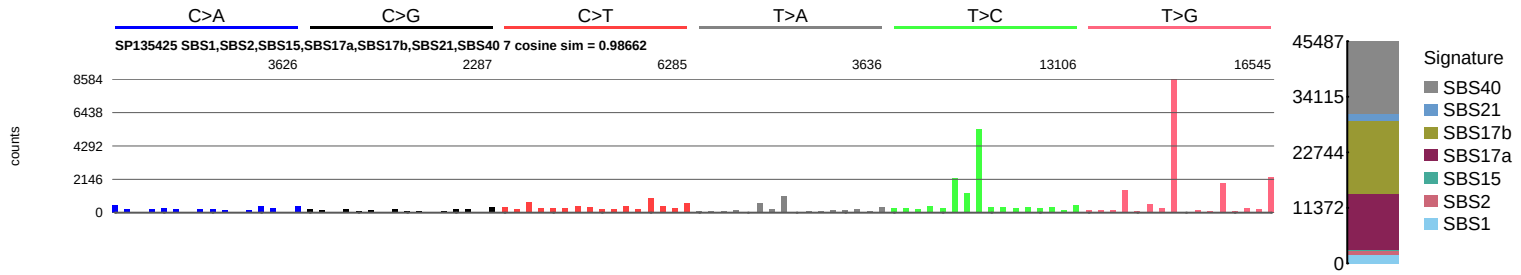
SBS18
SBS17b
SBS17a
SBS15
SBS2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C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98662
3626
2287
6285
3636
13106
16545
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
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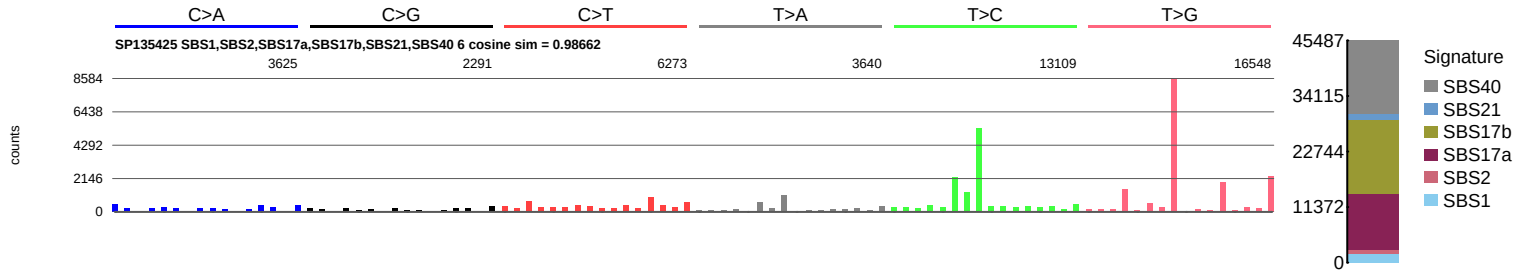
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98662
3625
2291
6273
3640
13109
16548
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
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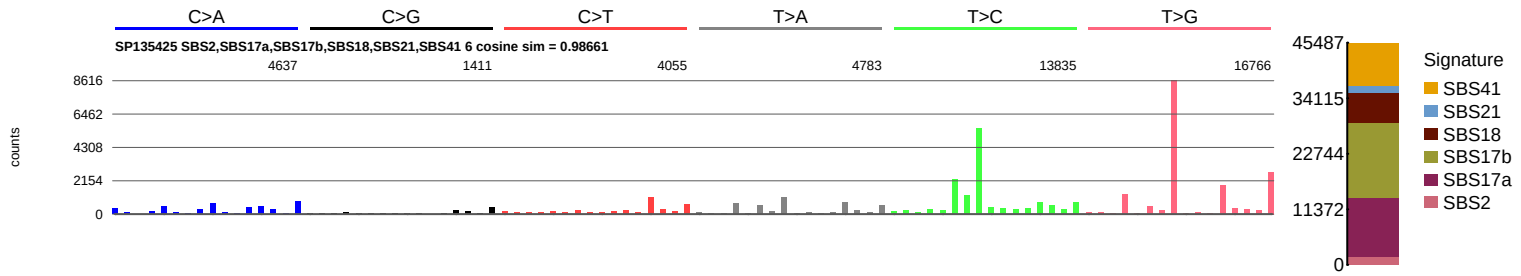
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98661
4637
1411
4055
4783
13835
16766
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
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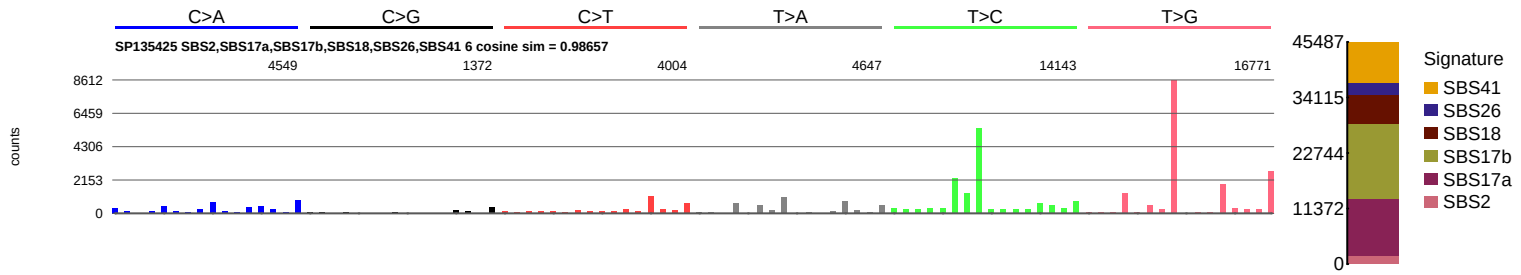
SBS21
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98657
4549
1372
4004
4647
14143
16771
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
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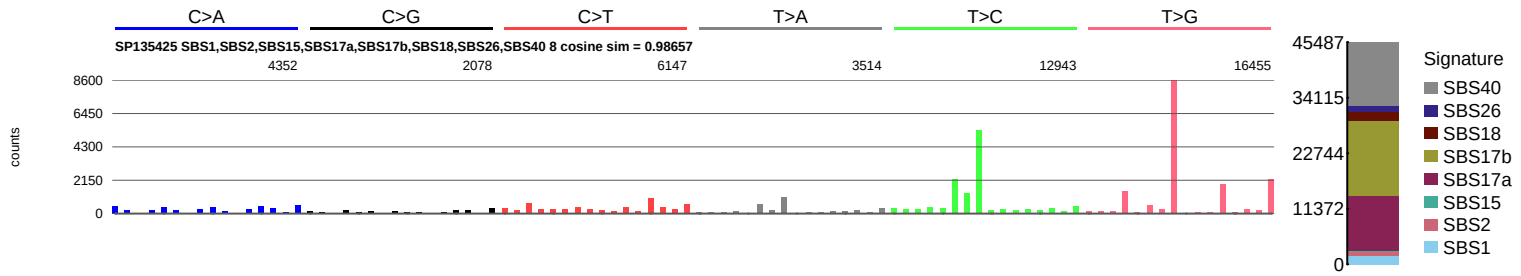
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98657
4352
2078
6147
3514
12943
16455
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
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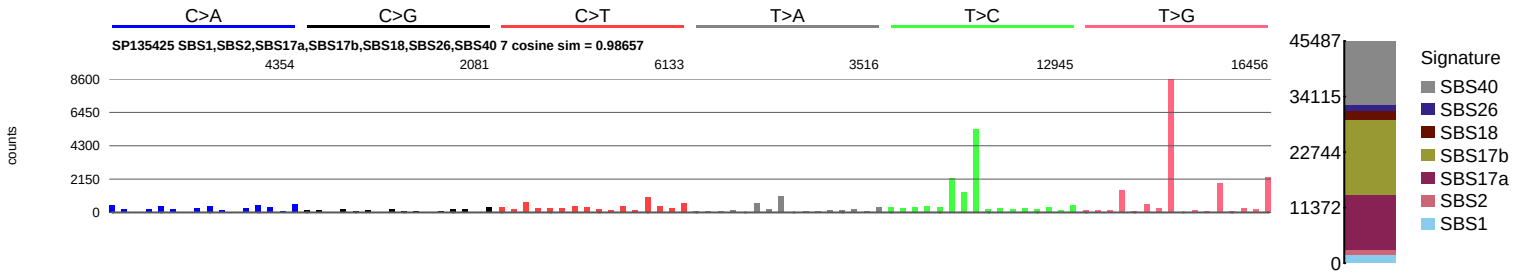
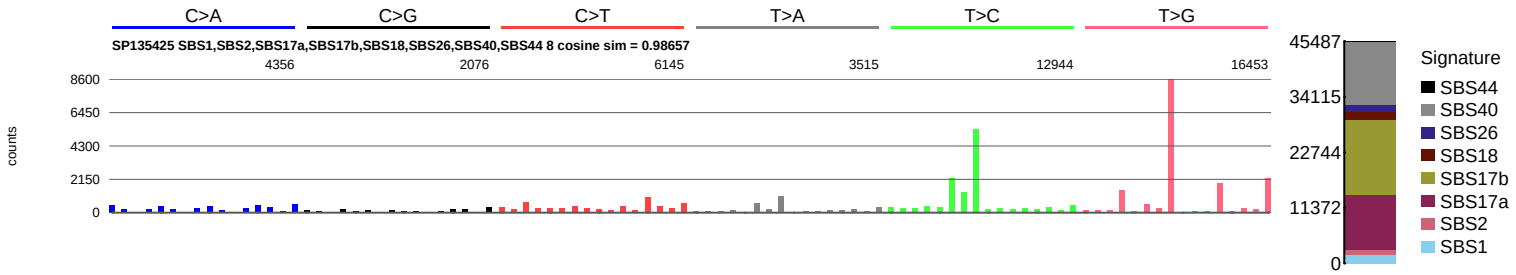
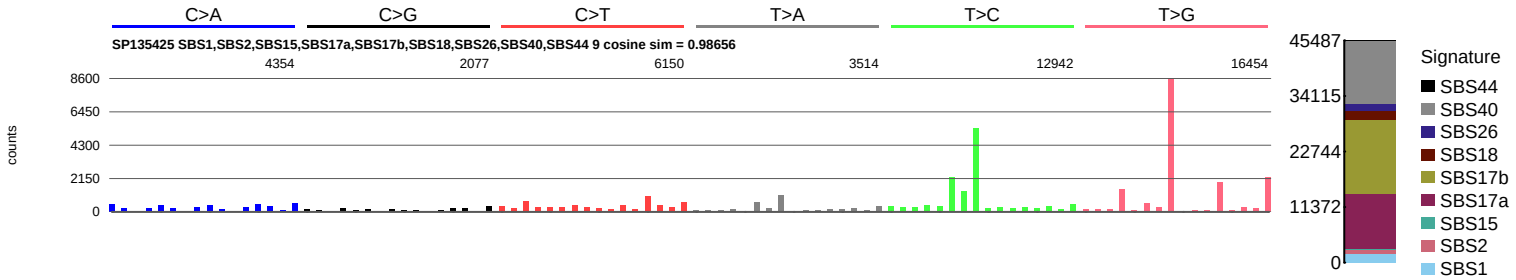
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98657
4354
2081
6133
3516
12945
16456
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98657
4356
2076
6145
3515
12944
16453
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98656
4354
2077
6150
3514
12942
16454
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
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Signature
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SBS40
SBS26
SBS18
SBS17b
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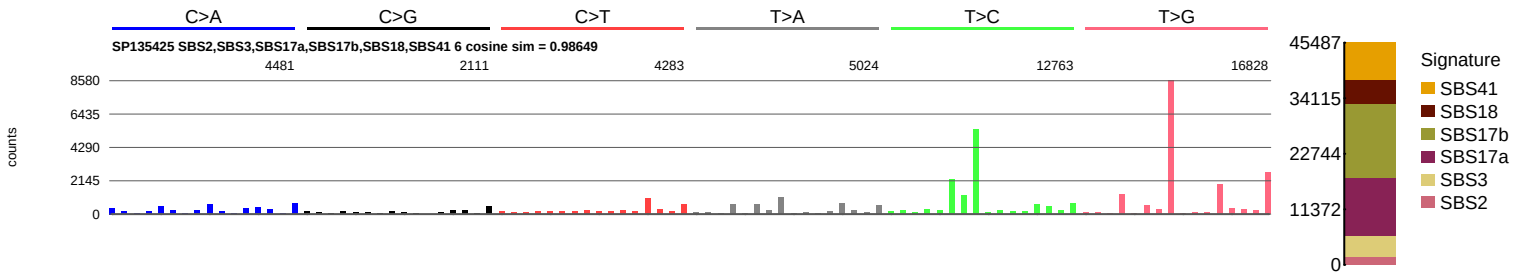
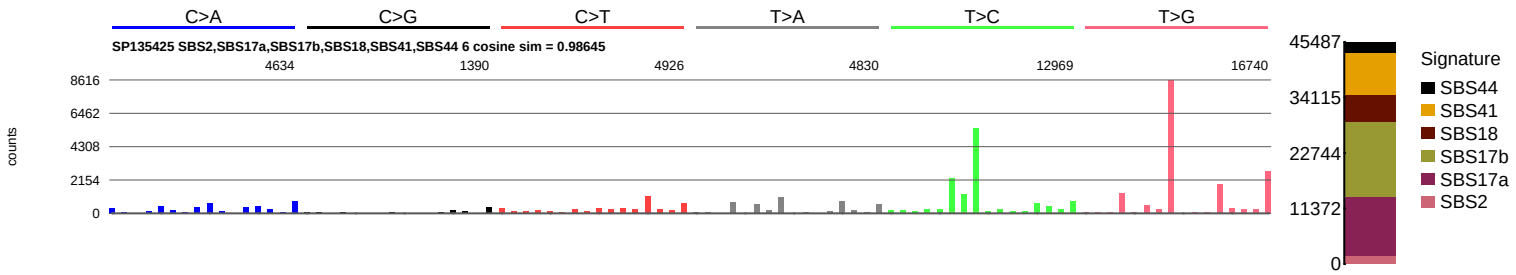
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98649
4481
2111
4283
5024
12763
16828
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2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SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98645
4634
1390
4926
4830
12969
16740
8616
6462
4308
2154
0
counts
45487
34115
22744
11372
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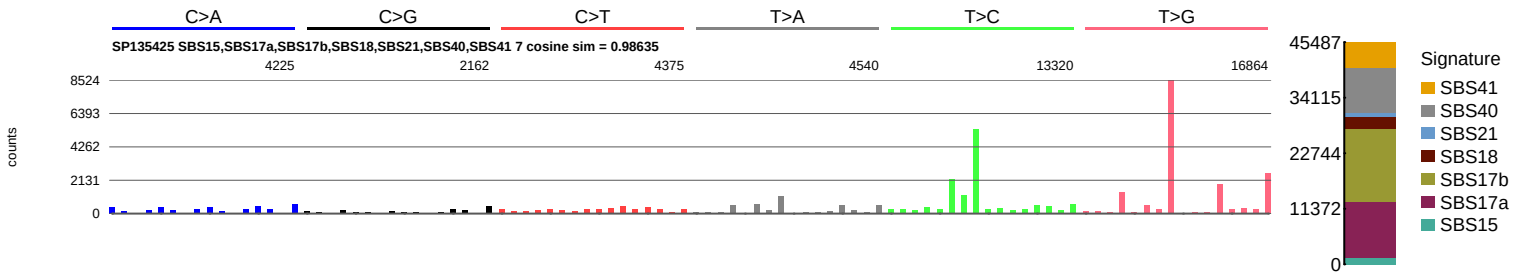
SBS18
SBS17b
SBS17a
SBS2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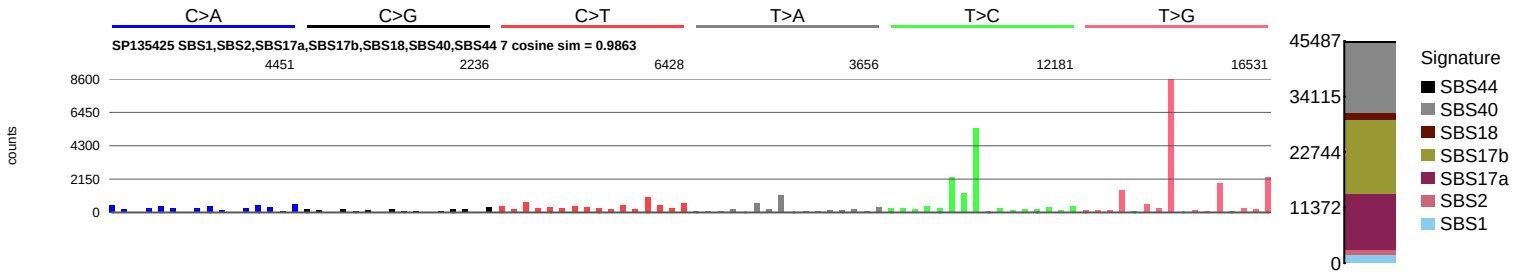
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98635
4225
2162
4375
4540
13320
16864
8524
6393
4262
2131
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15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SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9863
4451
2236
6428
3656
12181
16531
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
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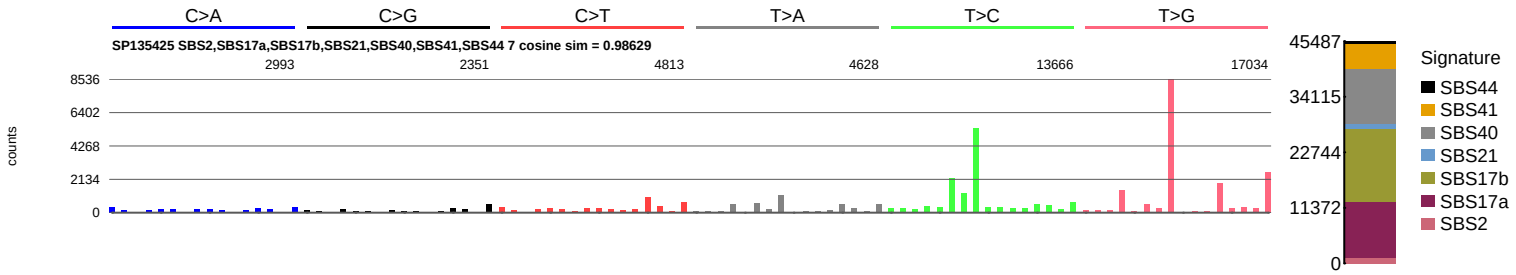
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98629
2993
2351
4813
4628
13666
17034
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
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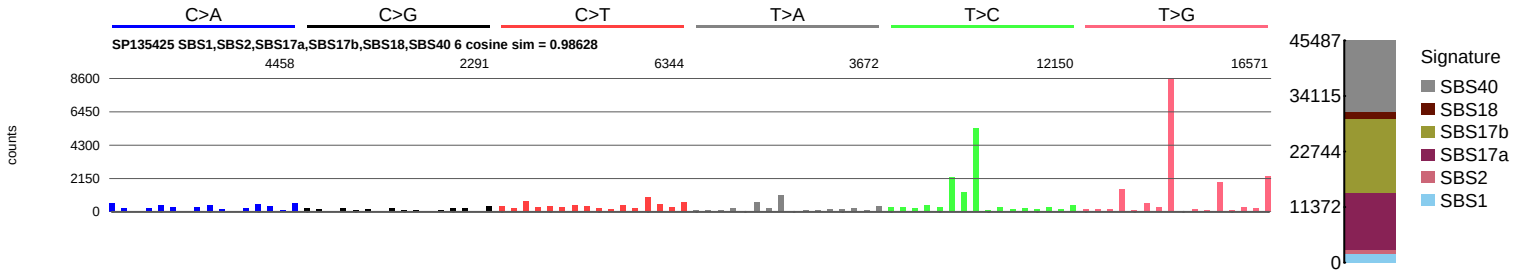
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2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SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98628
4458
2291
6344
3672
12150
16571
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
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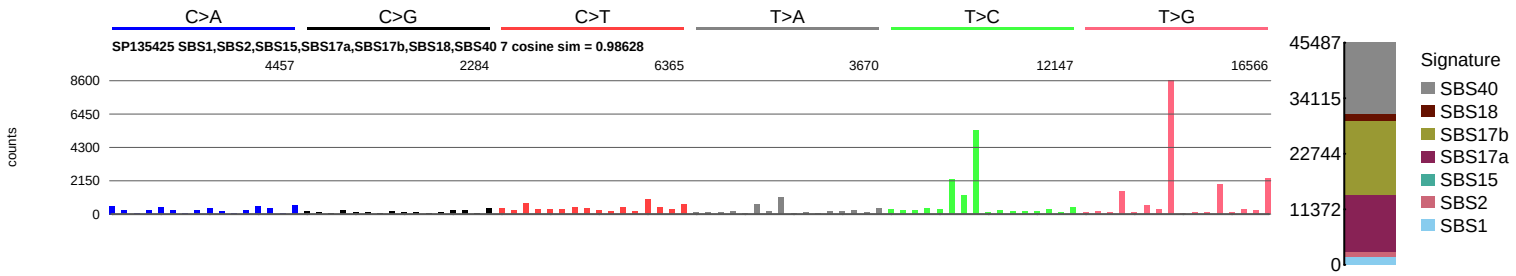
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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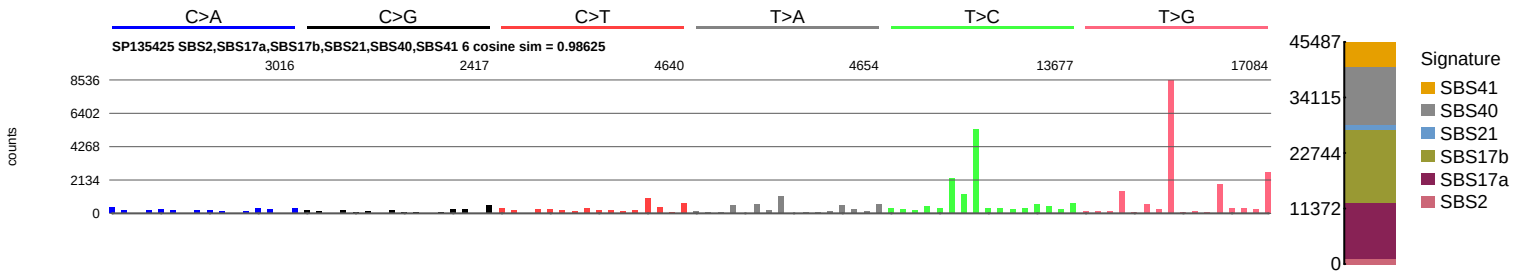
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98628
4457
2284
6365
3670
12147
16566
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98625
3016
2417
4640
4654
13677
17084
8536
6402
4268
2134
0
counts
45487
34115
22744
11372
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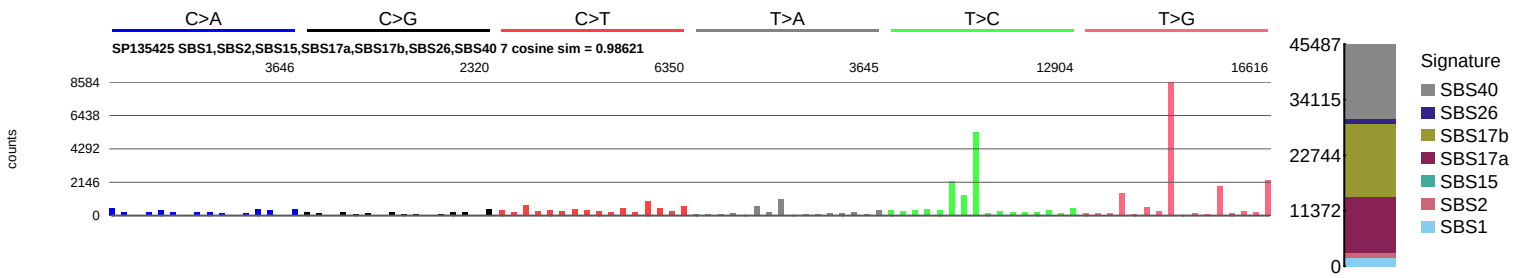
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98621
3646
2320
6350
3645
12904
16616
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
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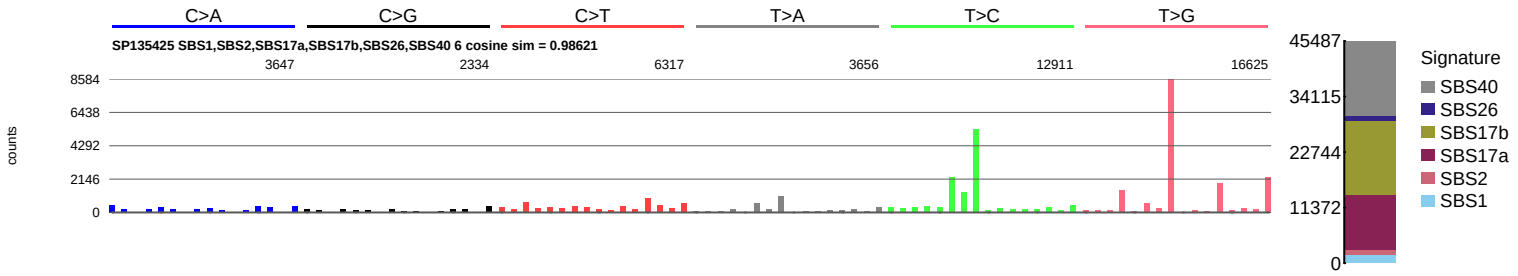
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98621
3647
2334
6317
3656
12911
16625
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
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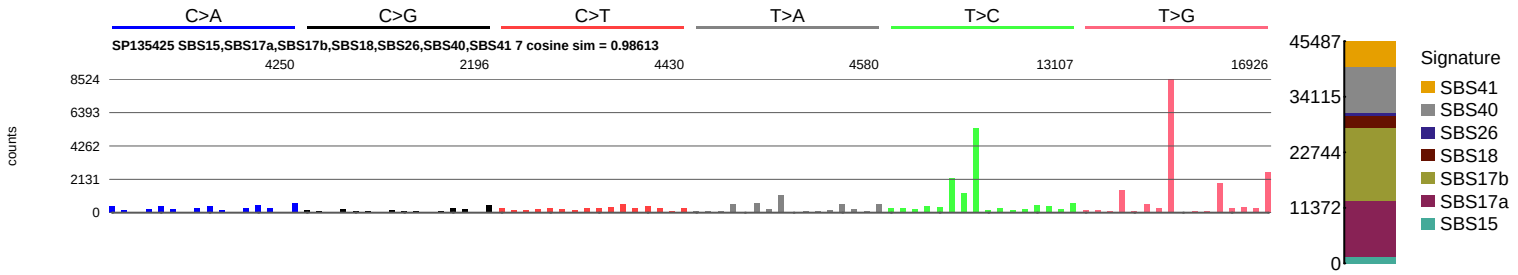
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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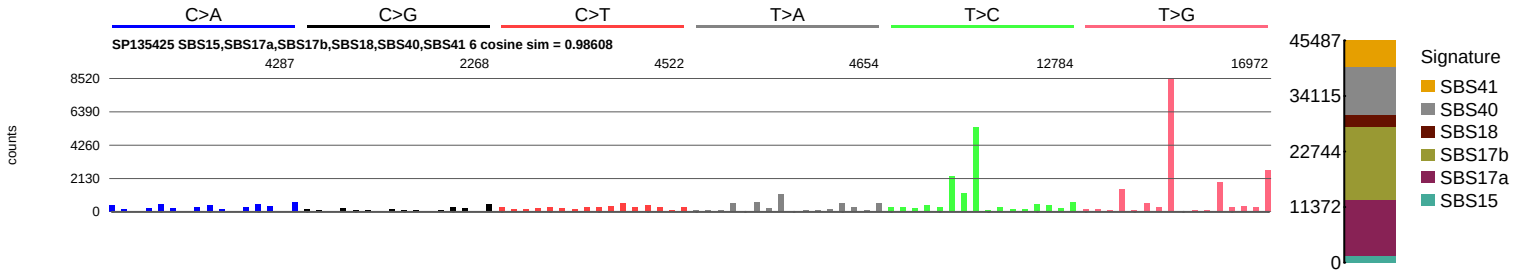
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98613
4250
2196
4430
4580
13107
16926
counts
8524
6393
4262
2131
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15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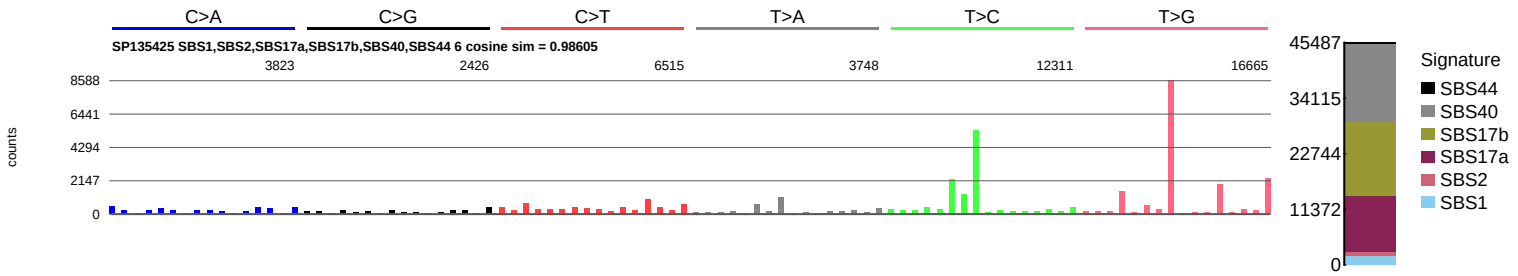
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98608
4287
2268
4522
4654
12784
16972
counts
8520
6390
4260
2130
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15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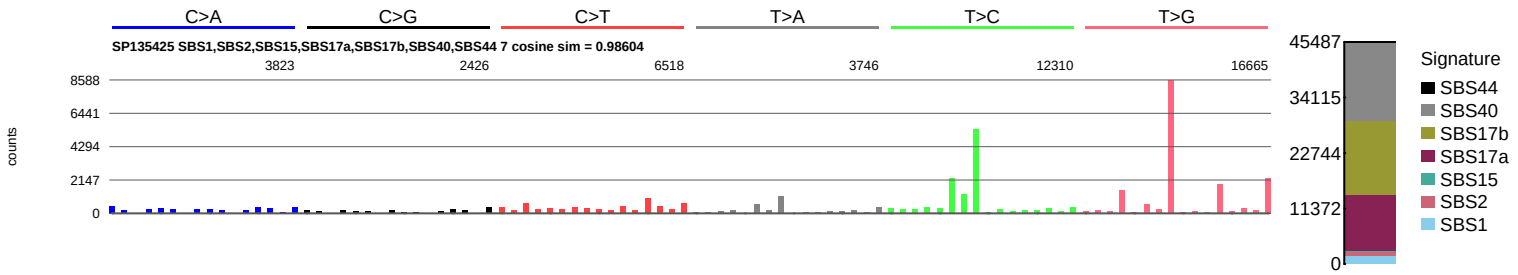
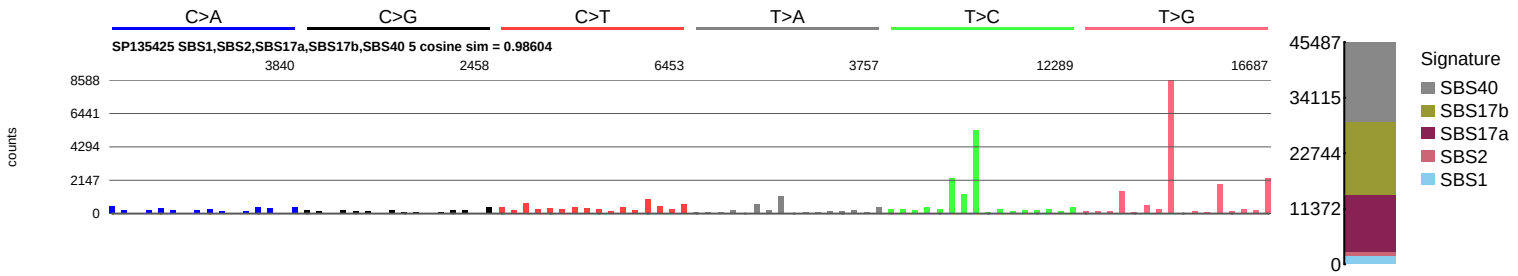
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98605
3823
2426
6515
3748
12311
16665
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98604
3823
2426
6518
3746
12310
16665
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98604
3840
2458
6453
3757
12289
16687
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
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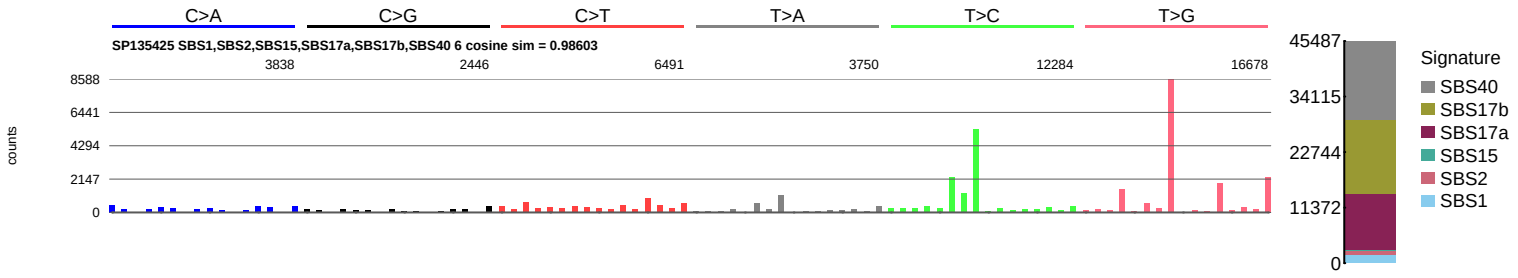
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98603
3838
2446
6491
3750
12284
16678
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
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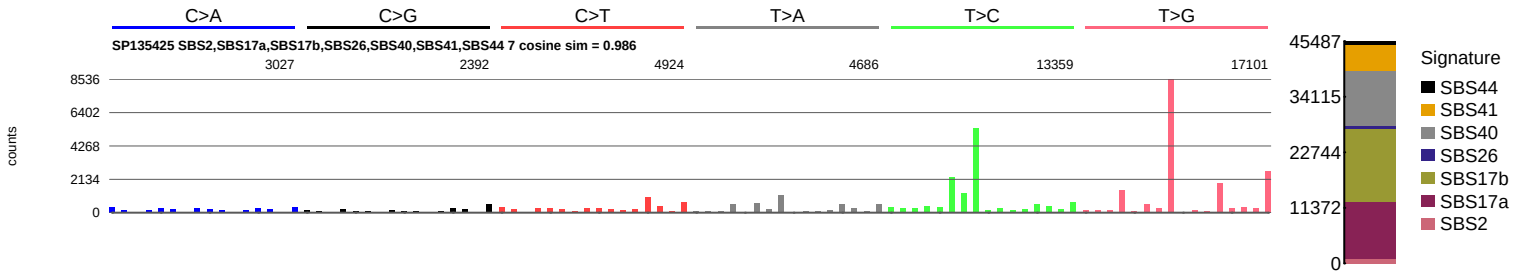
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.986
3027
2392
4924
4686
13359
17101
counts
8536
6402
4268
2134
0
45487
34115
22744
11372
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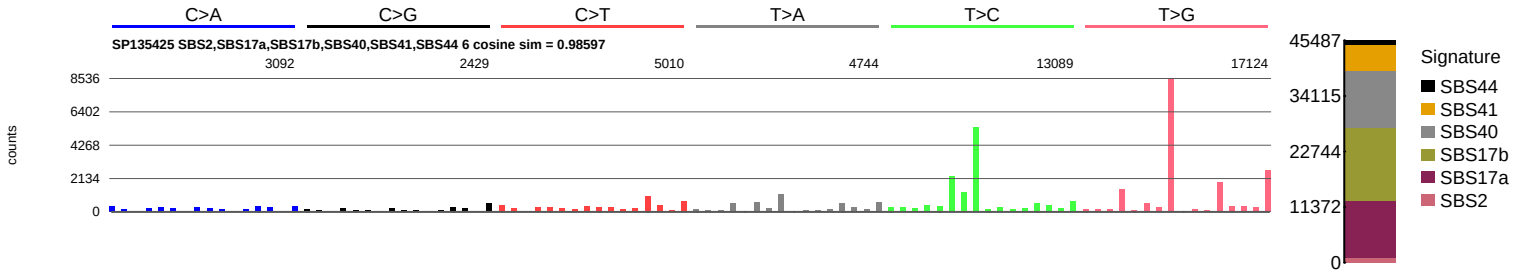
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2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C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98597
3092
2429
5010
4744
13089
17124
counts
8536
6402
4268
2134
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45487
34115
22744
11372
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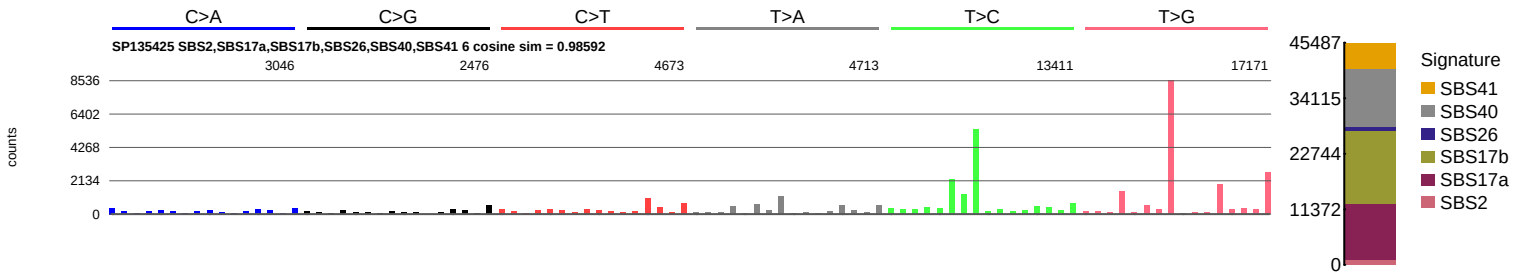
SBS41
SBS40
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98592
3046
2476
4673
4713
13411
17171
counts
8536
6402
4268
2134
0
45487
34115
22744
11372
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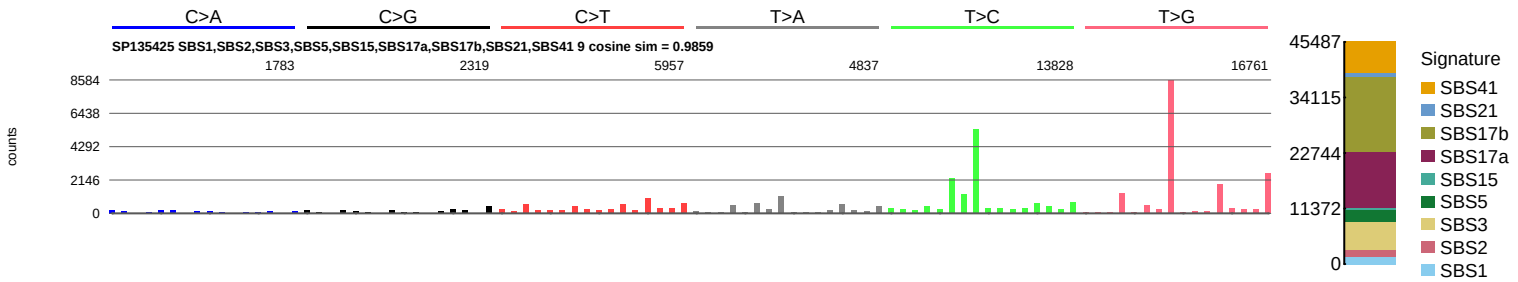
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2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T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9859
1783
2319
5957
4837
13828
16761
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
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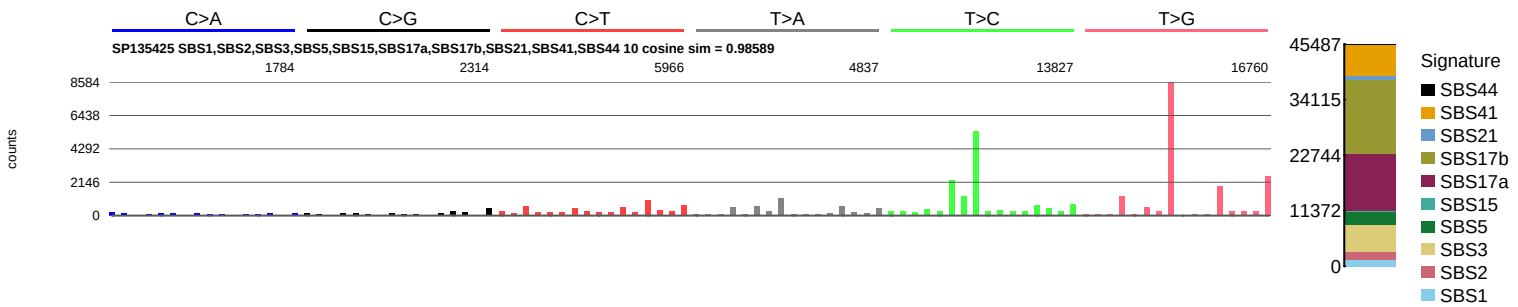
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98589
1784
2314
5966
4837
13827
16760
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
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SBS17b
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SBS15
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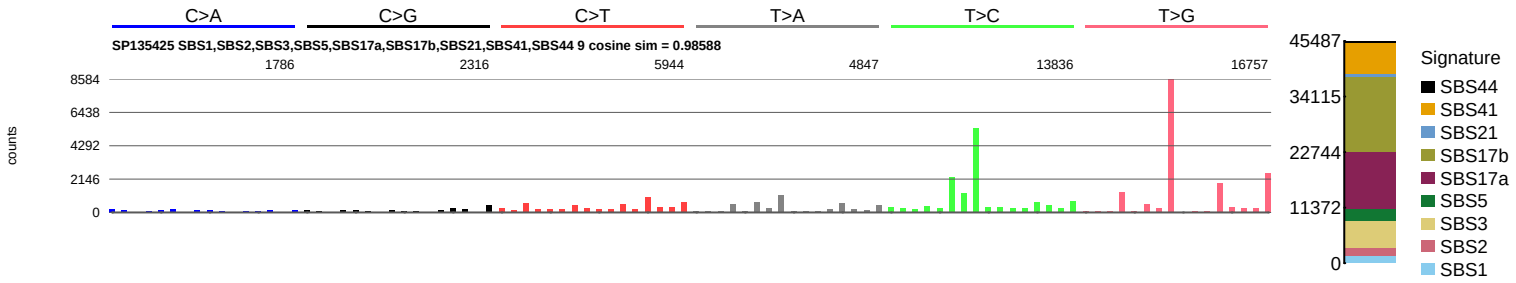
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98588
1786
2316
5944
4847
13836
16757
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
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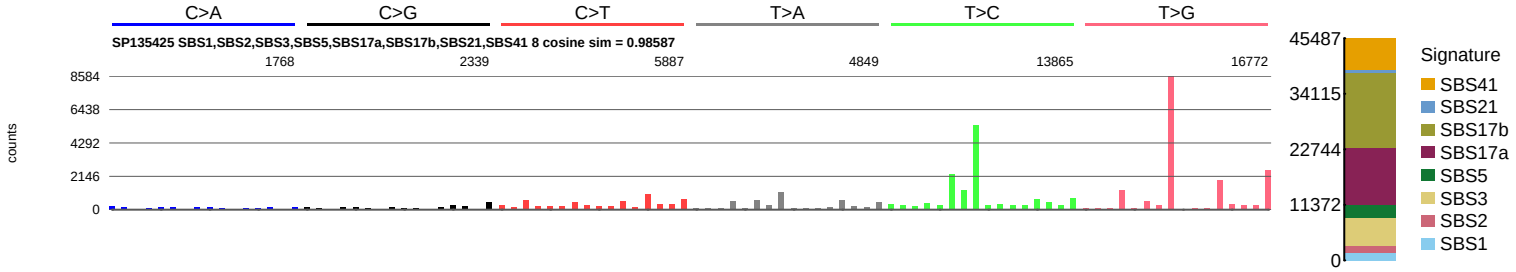
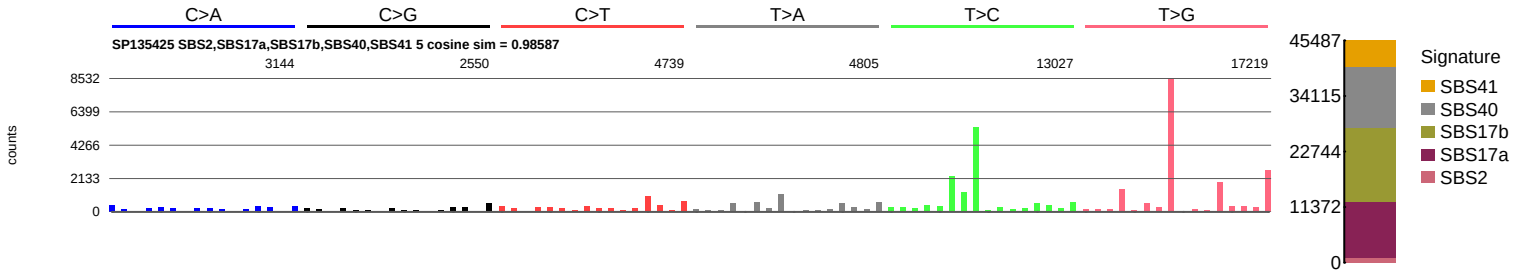
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98587
1768
2339
5887
4849
13865
16772
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98587
3144
2550
4739
4805
13027
17219
counts
8532
6399
4266
2133
0
45487
34115
22744
11372
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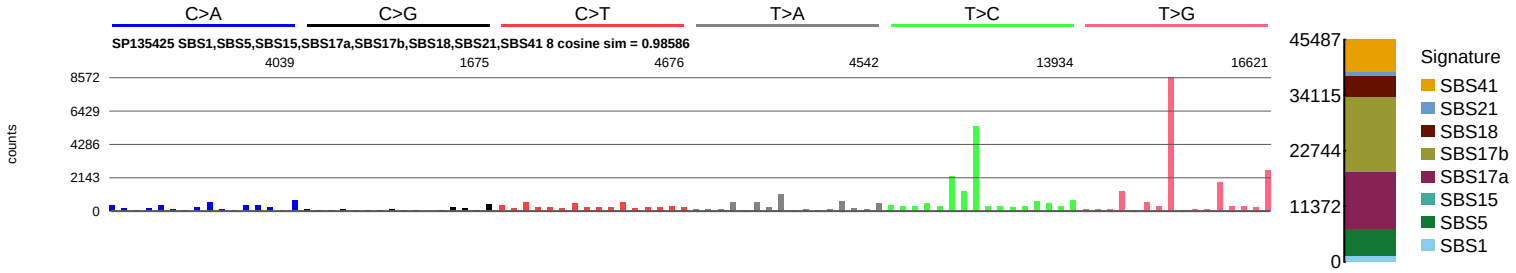
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98586
4039
1675
4676
4542
13934
16621
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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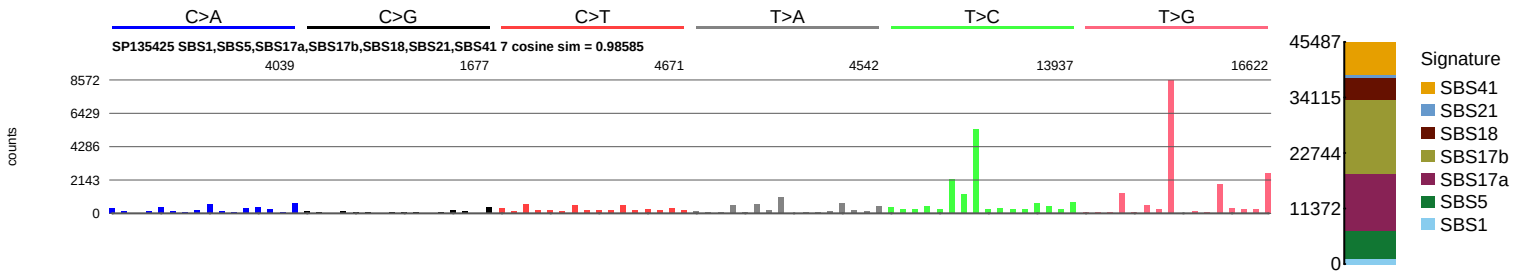
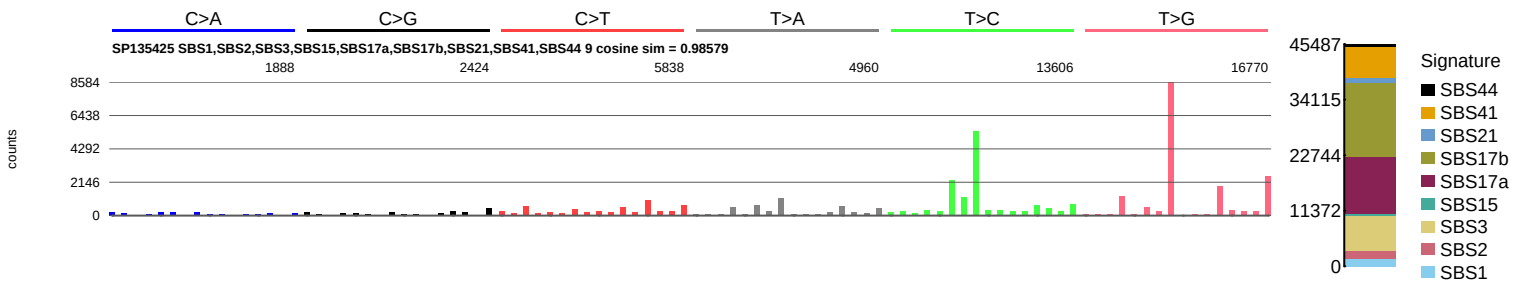
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98585
4039
1677
4671
4542
13937
16622
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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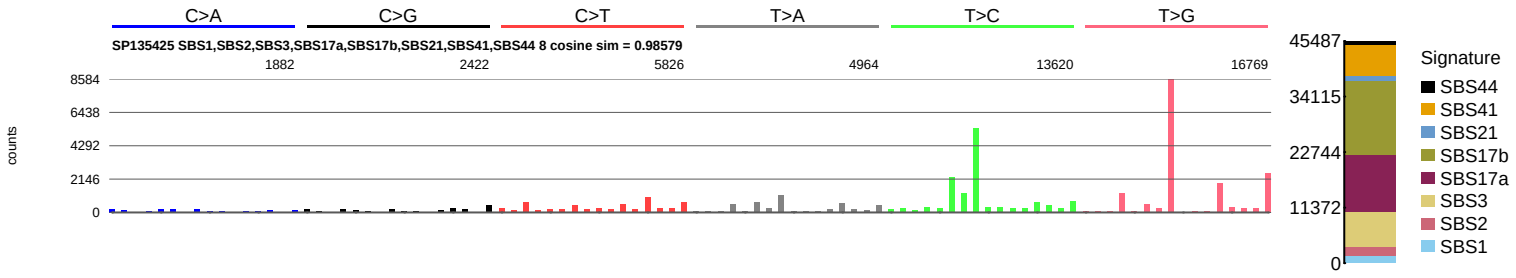
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98579
1888
2424
5838
4960
13606
16770
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
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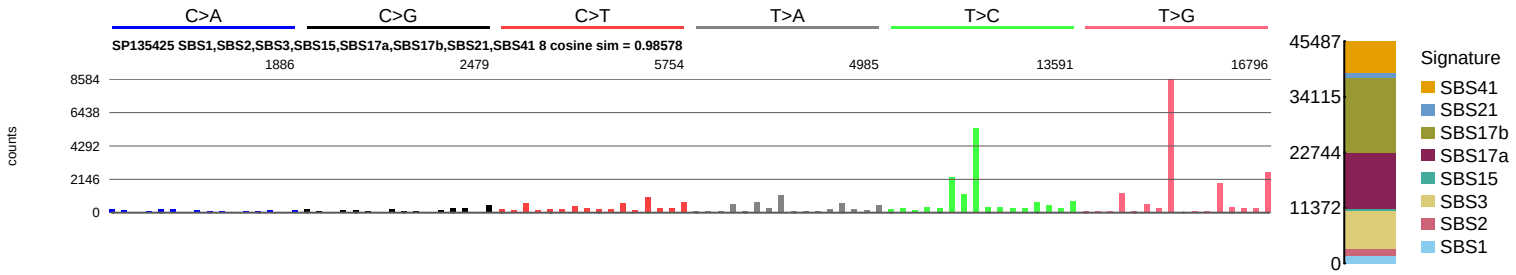
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98579
1882
2422
5826
4964
13620
16769
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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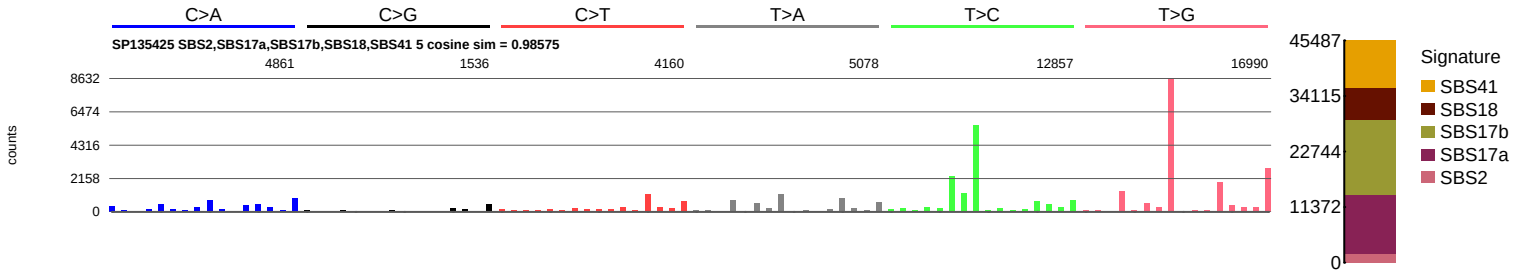
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98578
1886
2479
5754
4985
13591
16796
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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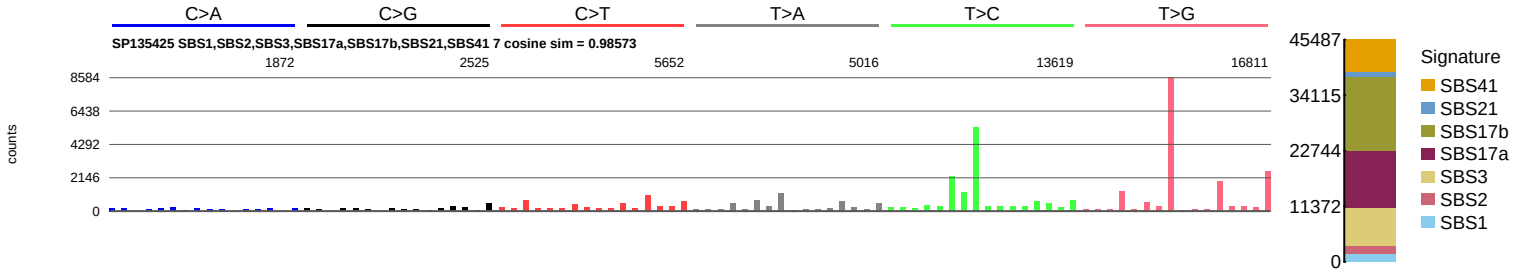
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98575
4861
1536
4160
5078
12857
16990
8632
6474
4316
2158
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
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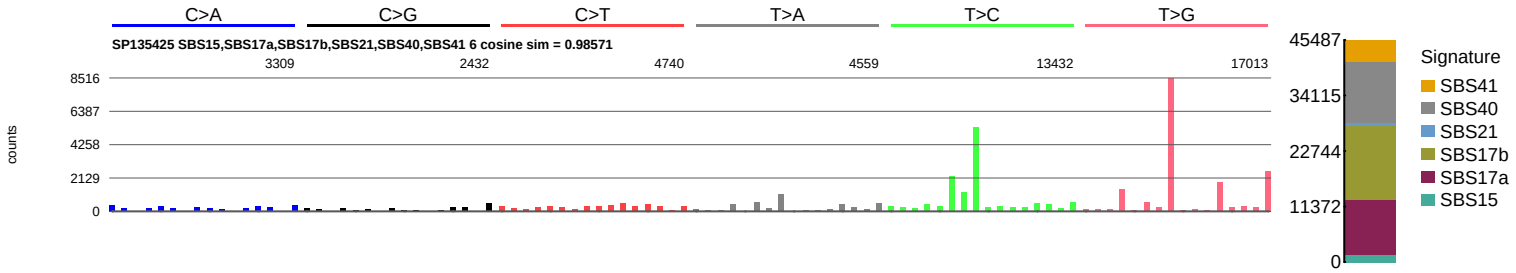
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98573
1872
2525
5652
5016
13619
16811
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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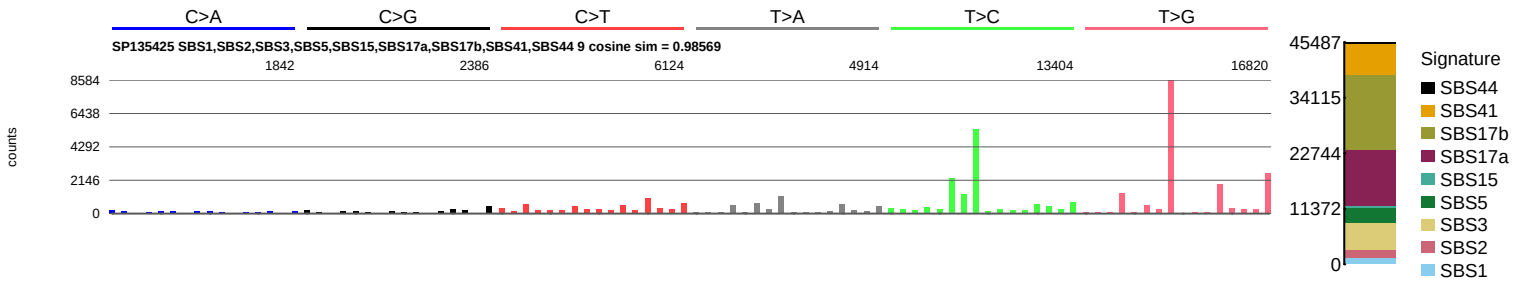
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98571
3309
2432
4740
4559
13432
17013
8516
6387
4258
2129
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15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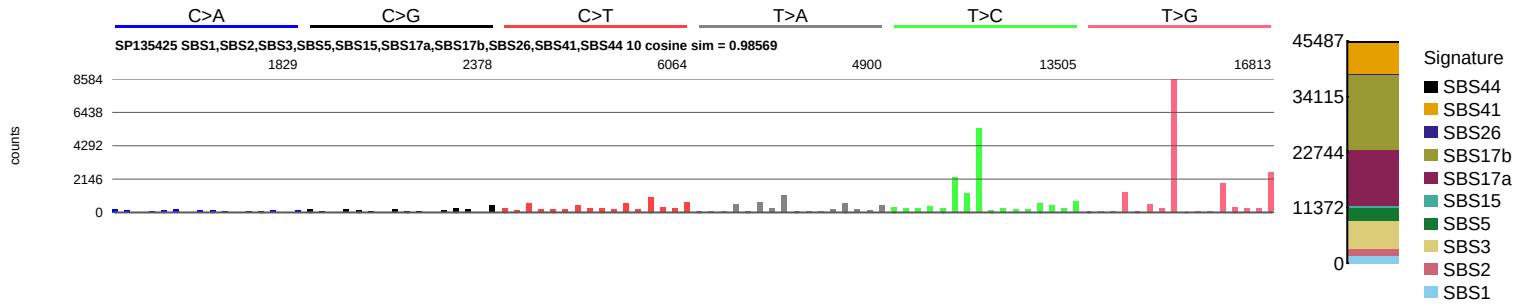
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98569
1842
2386
6124
4914
13404
16820
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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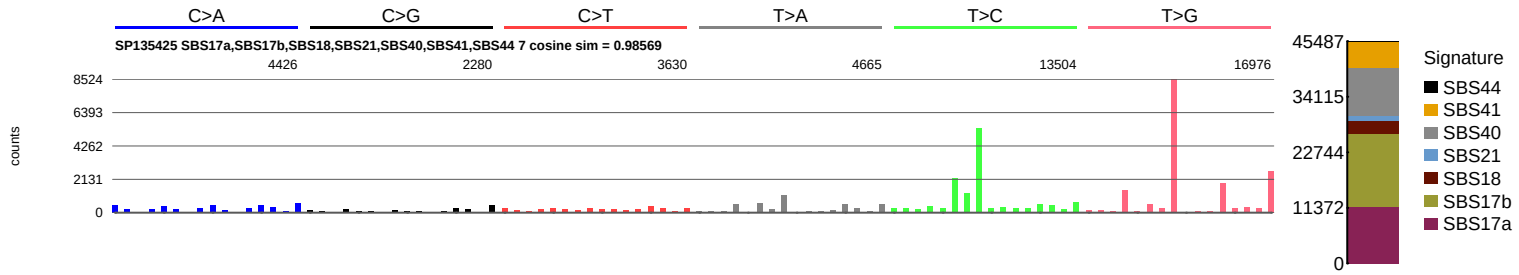
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 10 cosine sim = 0.98569
1829
2378
6064
4900
13505
16813
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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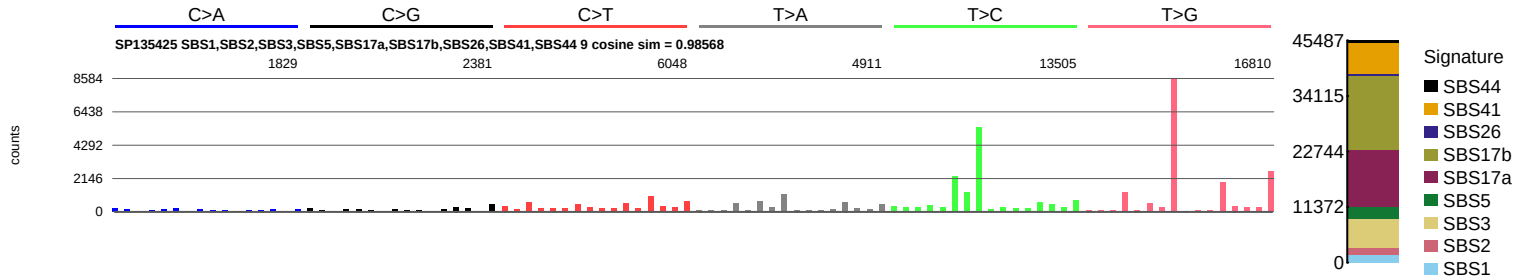
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.98569
4426
2280
3630
4665
13504
16976
8524
6393
4262
2131
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a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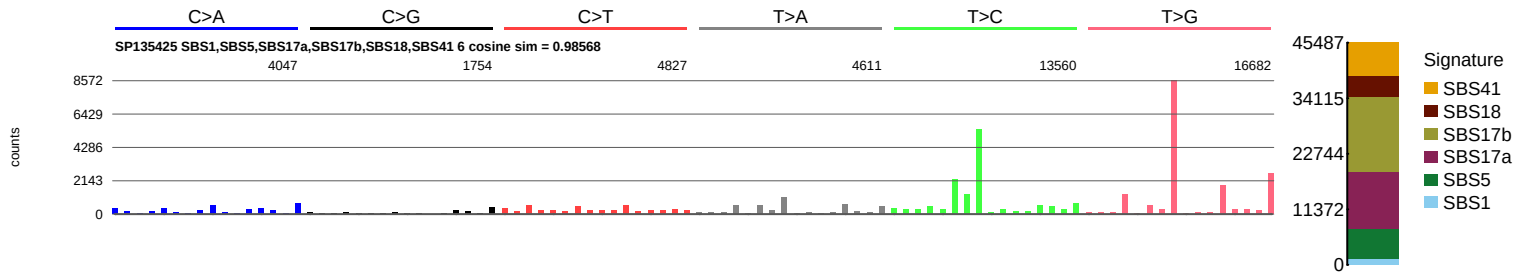
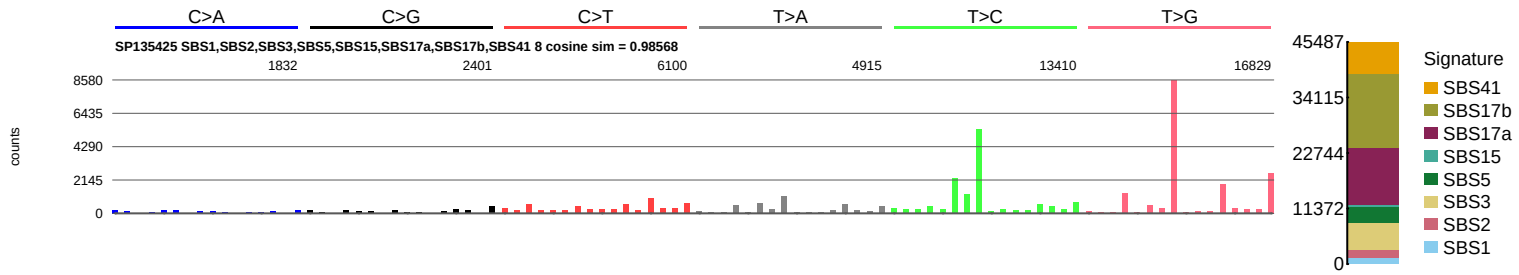
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98568
1829
2381
6048
4911
13505
16810
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98568
4047
1754
4827
4611
13560
16682
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1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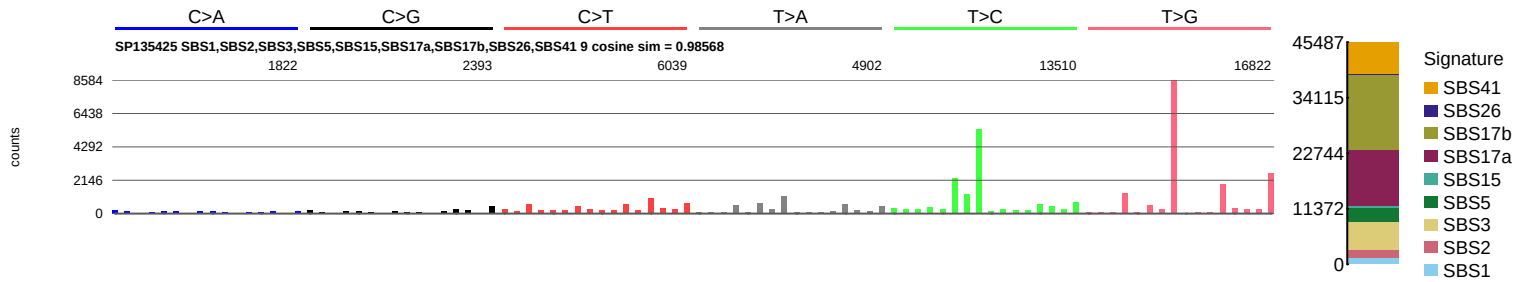
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98568
1832
2401
6100
4915
13410
16829
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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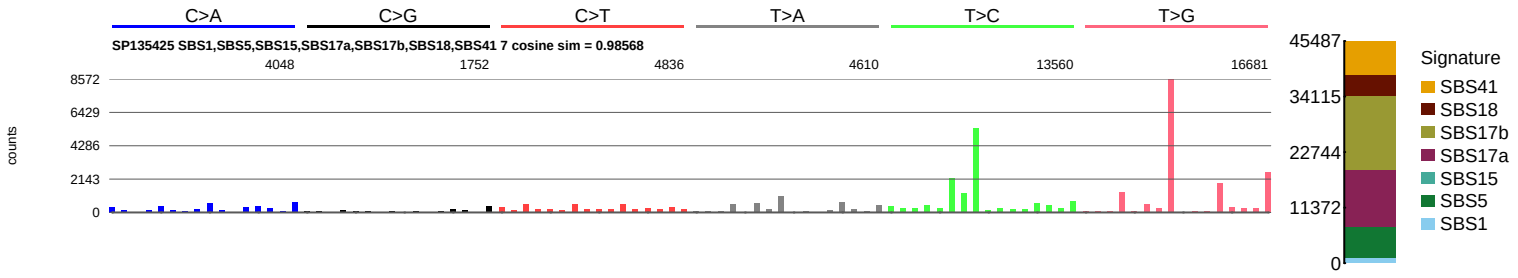
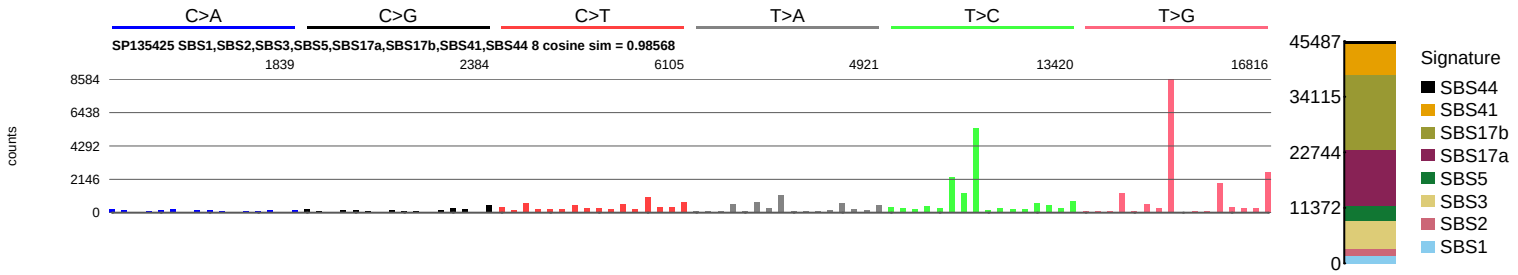
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS41 9 cosine sim = 0.98568
1822
2393
6039
4902
13510
16822
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98568
4048
1752
4836
4610
13560
16681
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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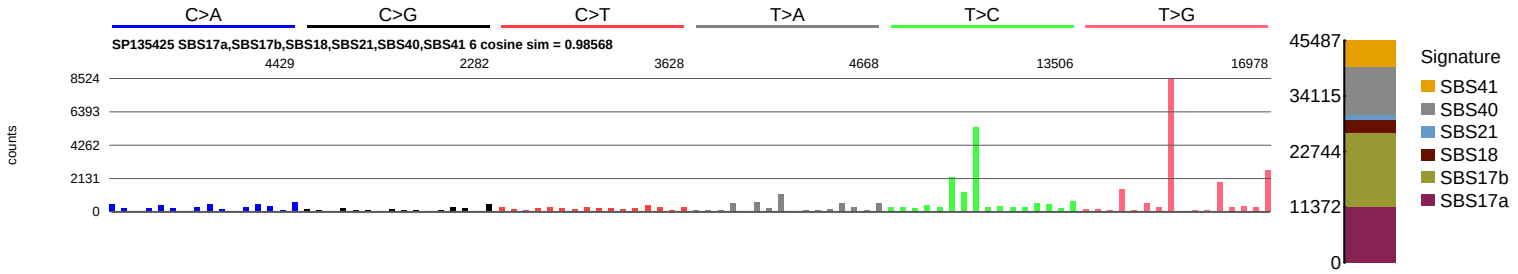
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98568
1839
2384
6105
4921
13420
16816
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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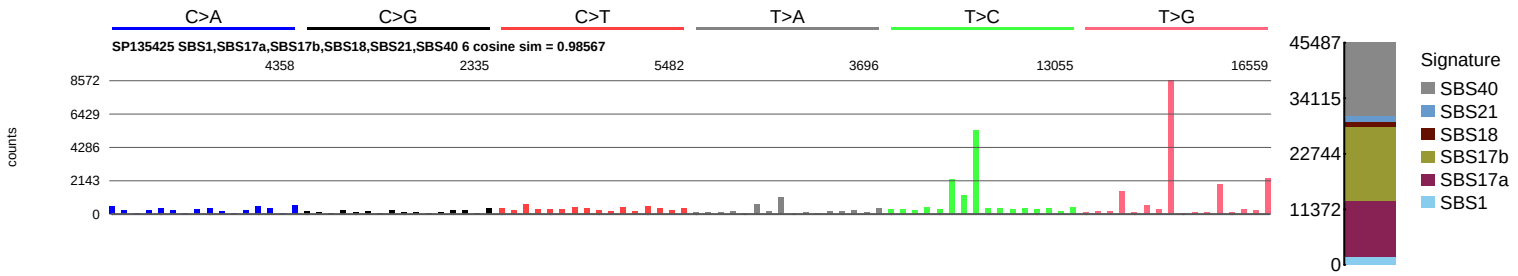
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 6 cosine sim = 0.98568
4429
2282
3628
4668
13506
16978
8524
6393
4262
2131
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a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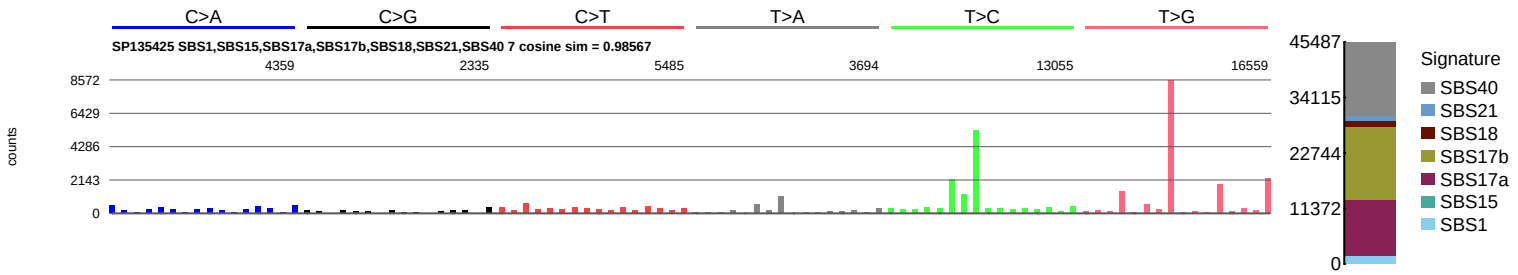
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98567
4358
2335
5482
3696
13055
16559
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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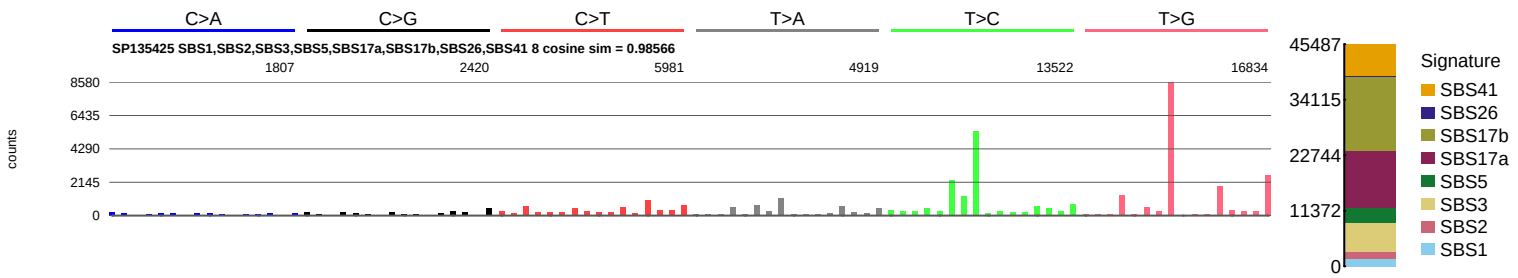
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98567
4359
2335
5485
3694
13055
16559
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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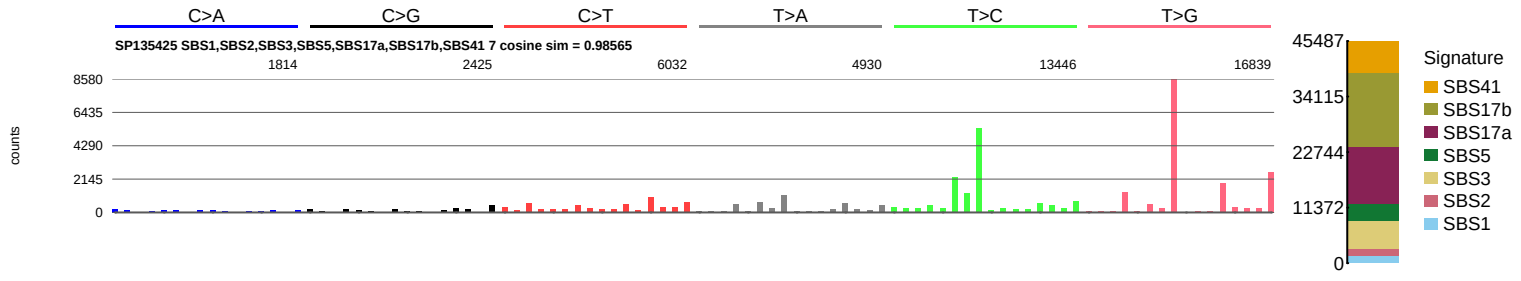
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98566
1807
2420
5981
4919
13522
16834
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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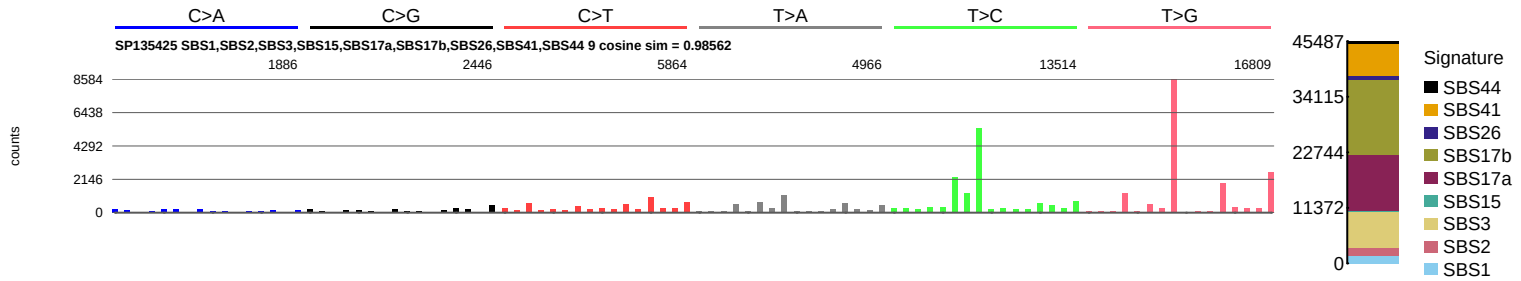
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98565
1814
2425
6032
4930
13446
16839
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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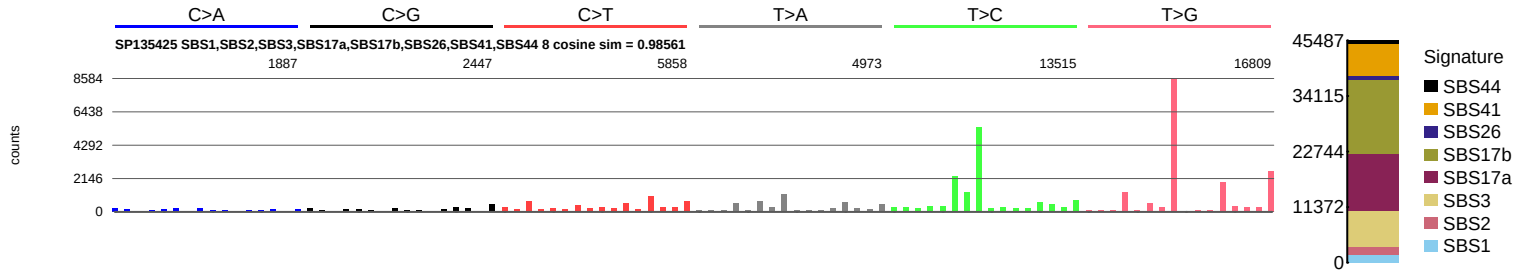
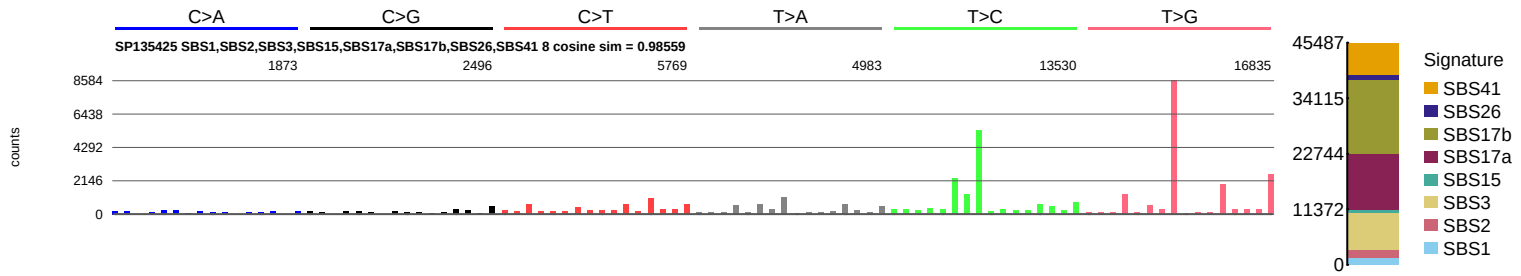
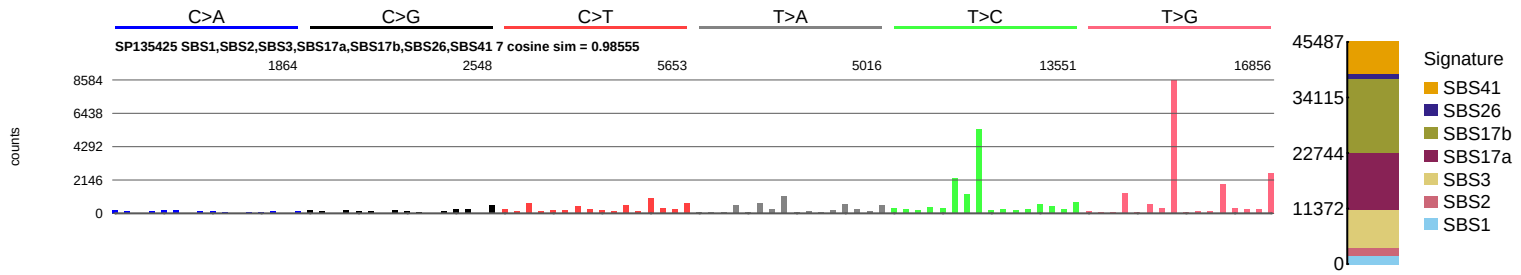
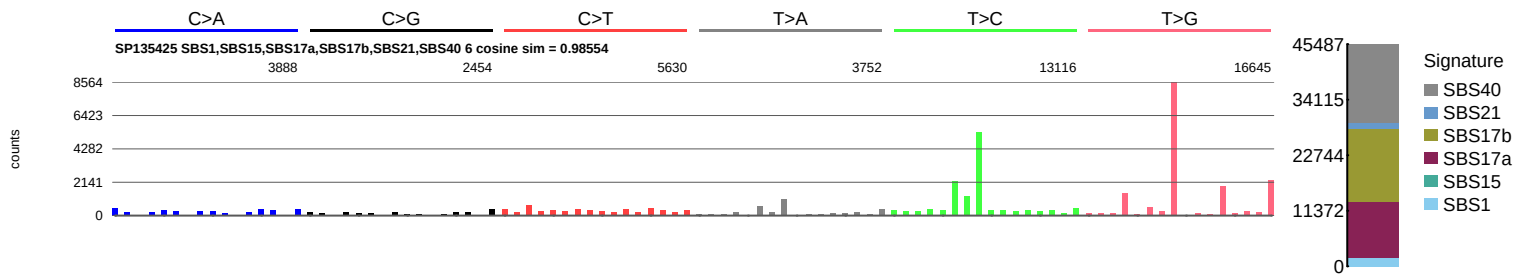
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98562
1886
2446
5864
4966
13514
16809
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98561
1887
2447
5858
4973
13515
16809
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98559
1873
2496
5769
4983
13530
16835
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98555
1864
2548
5653
5016
13551
16856
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98554
3888
2454
5630
3752
13116
16645
8564
6423
4282
2141
0
counts
45487
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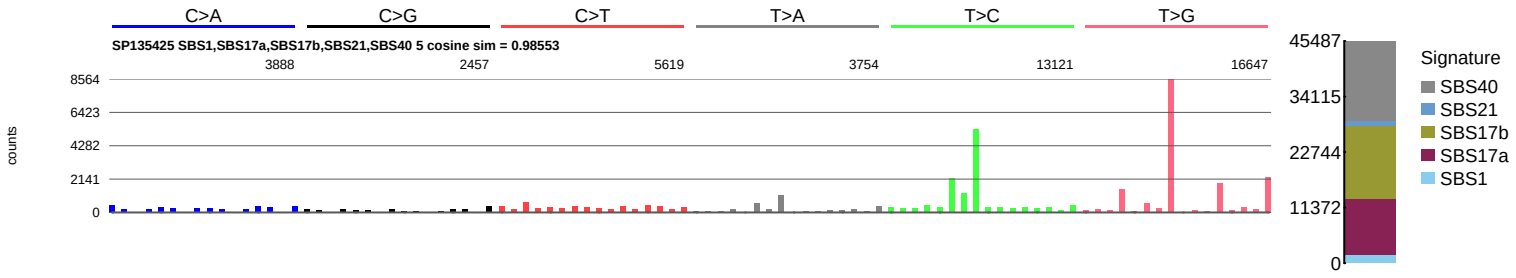
SBS1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SP135425 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98553
3888
2457
5619
3754
13121
16647
8564
6423
4282
2141
0
counts
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34115
22744
11372
0
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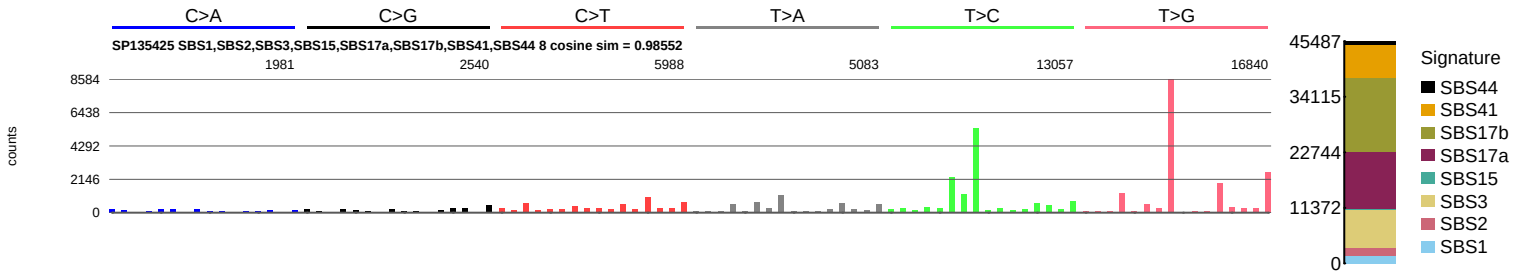
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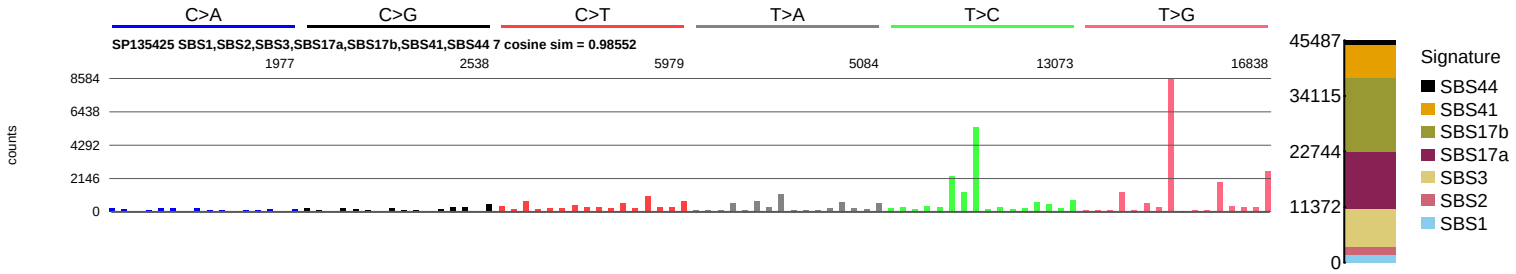
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98552
1981
2540
5988
5083
13057
16840
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98552
1977
2538
5979
5084
13073
16838
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
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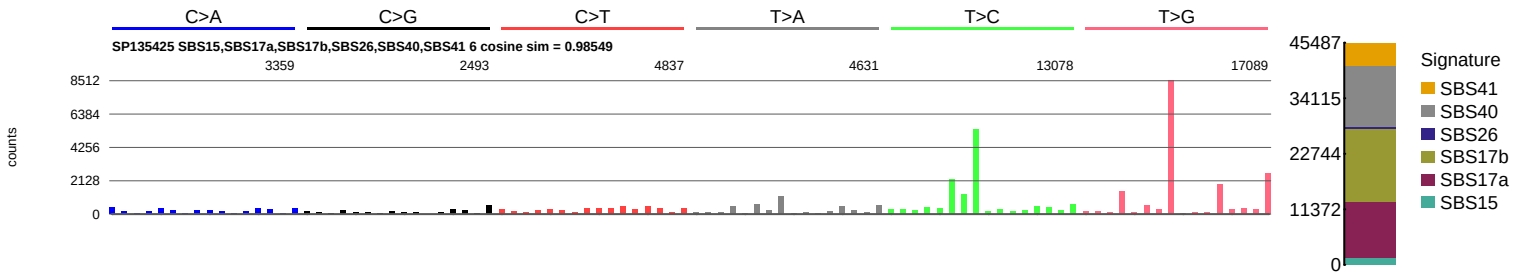
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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SP135425 SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98549
3359
2493
4837
4631
13078
17089
8512
6384
4256
2128
0
counts
45487
34115
22744
11372
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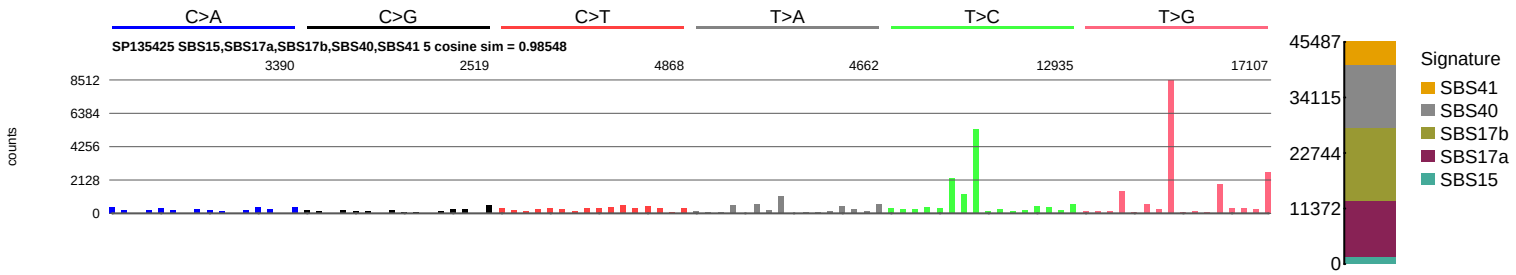
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15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SP135425 SBS15,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98548
3390
2519
4868
4662
12935
17107
8512
6384
4256
2128
0
counts
45487
34115
22744
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SBS40
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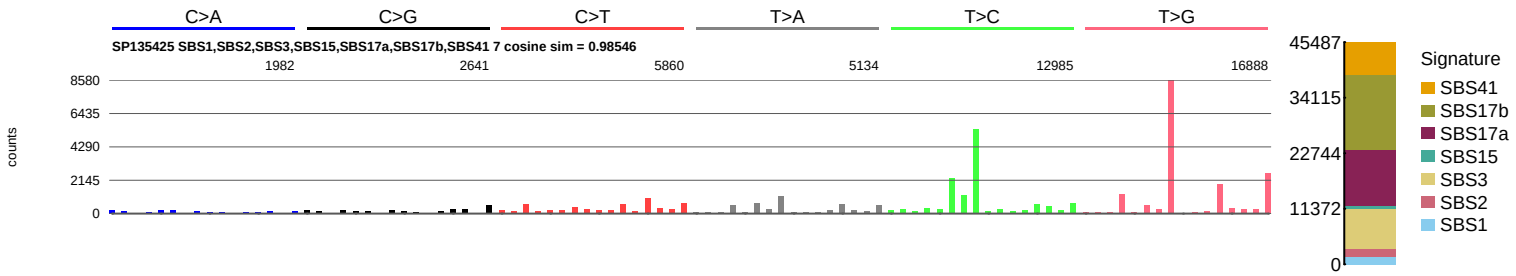
SBS17a
SBS15


C>A
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C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98546
1982
2641
5860
5134
12985
16888
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
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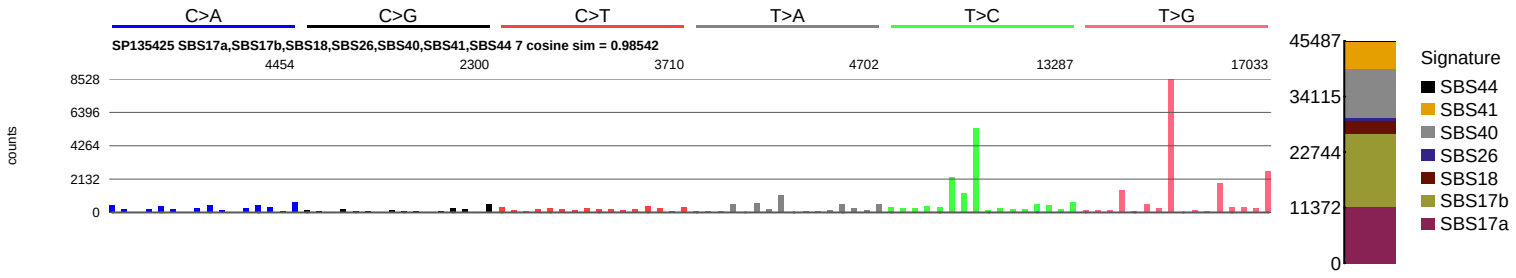
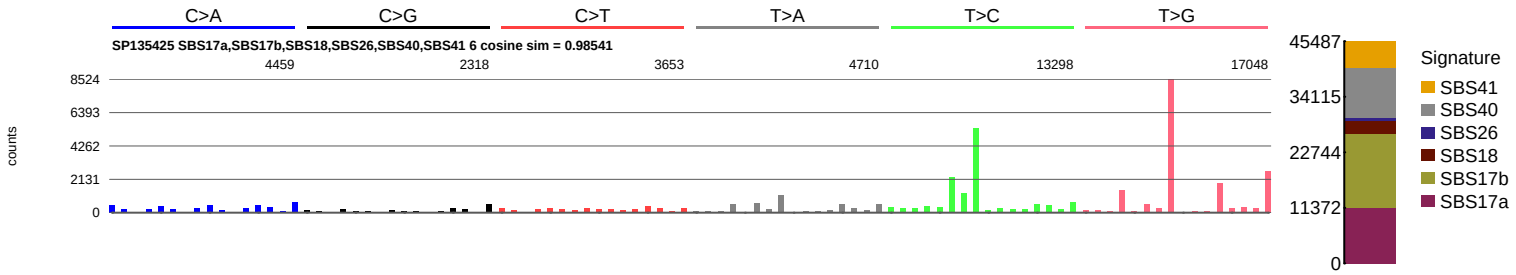
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98542
4454
2300
3710
4702
13287
17033
counts
8528
6396
4264
2132
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 6 cosine sim = 0.98541
4459
2318
3653
4710
13298
17048
counts
8524
6393
4262
2131
0
45487
34115
22744
11372
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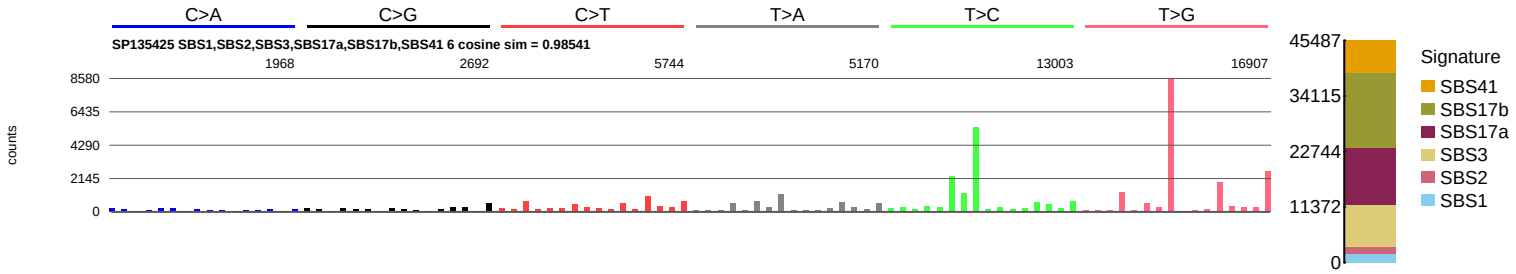
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98541
1968
2692
5744
5170
13003
16907
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
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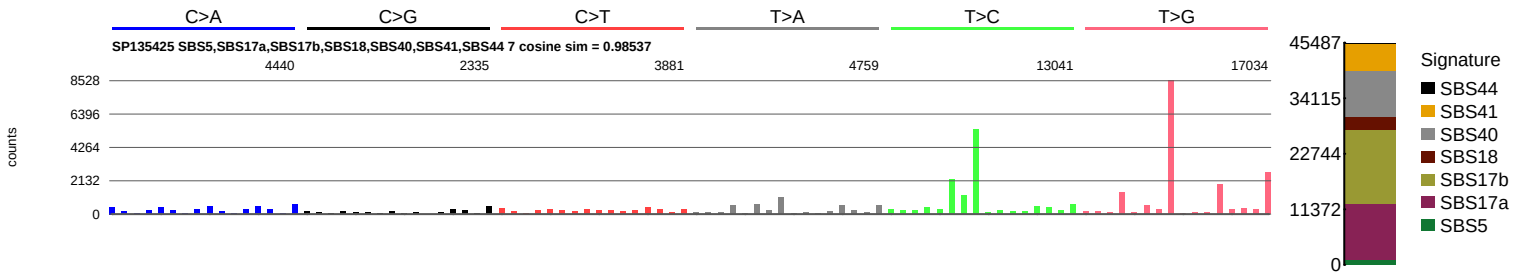
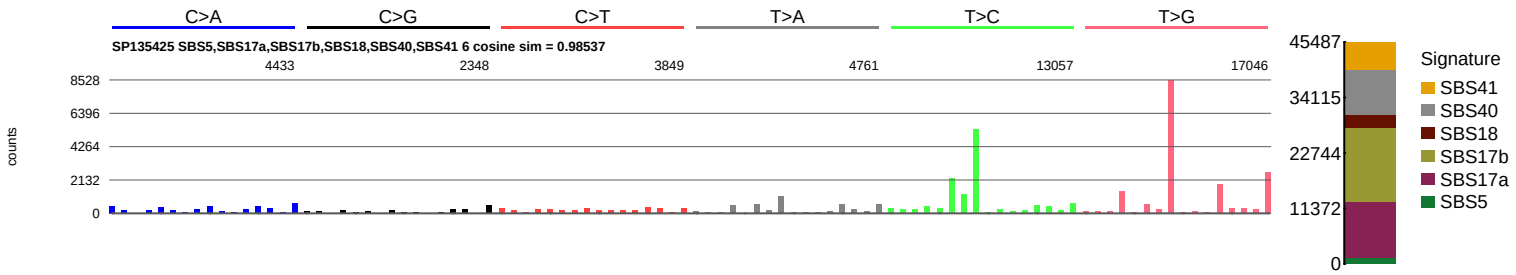
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98537
4440
2335
3881
4759
13041
17034
counts
8528
6396
4264
2132
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5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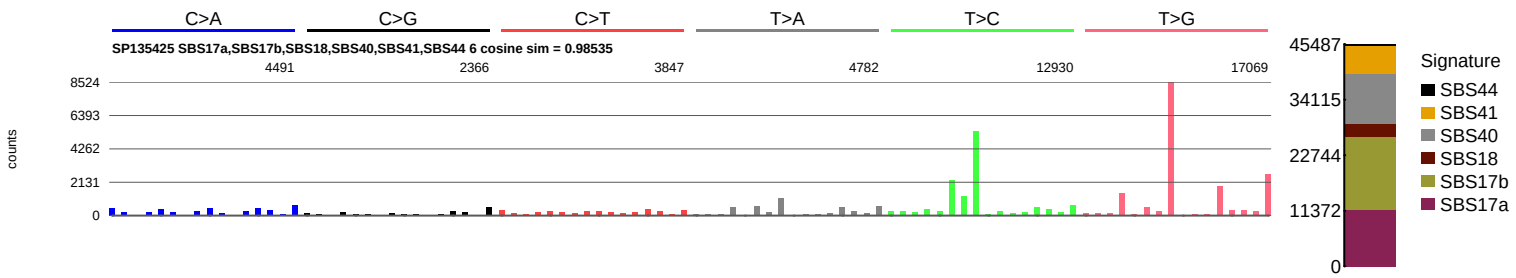
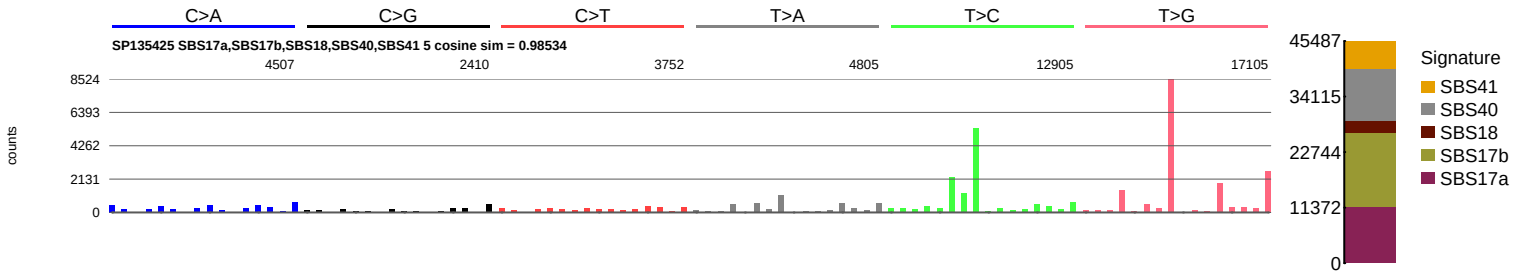
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98537
4433
2348
3849
4761
13057
17046
counts
8528
6396
4264
2132
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 6 cosine sim = 0.98535
4491
2366
3847
4782
12930
17069
counts
8524
6393
4262
2131
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS40,SBS41 5 cosine sim = 0.98534
4507
2410
3752
4805
12905
17105
8524
6393
4262
2131
0
counts
45487
34115
22744
11372
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Signature
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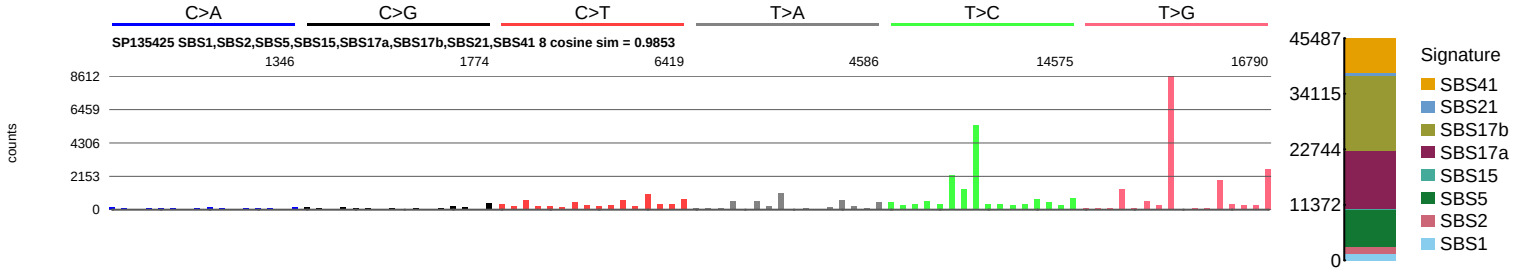
SBS40
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9853
1346
1774
6419
4586
14575
16790
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
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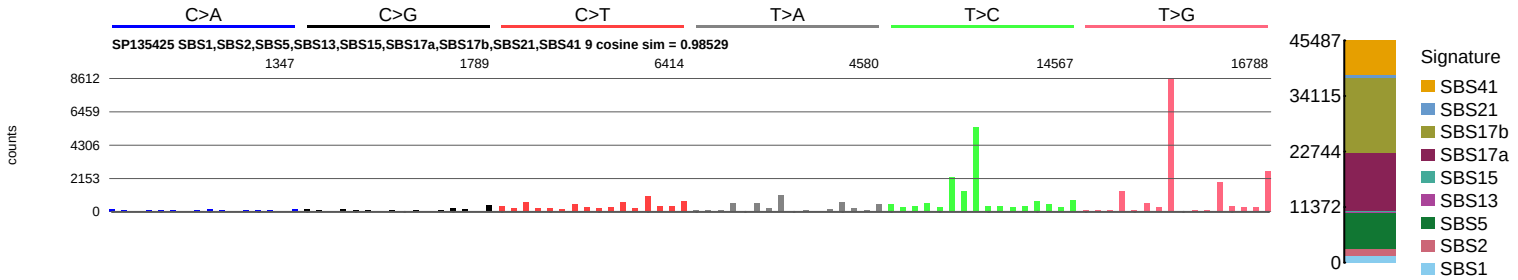
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98529
1347
1789
6414
4580
14567
16788
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
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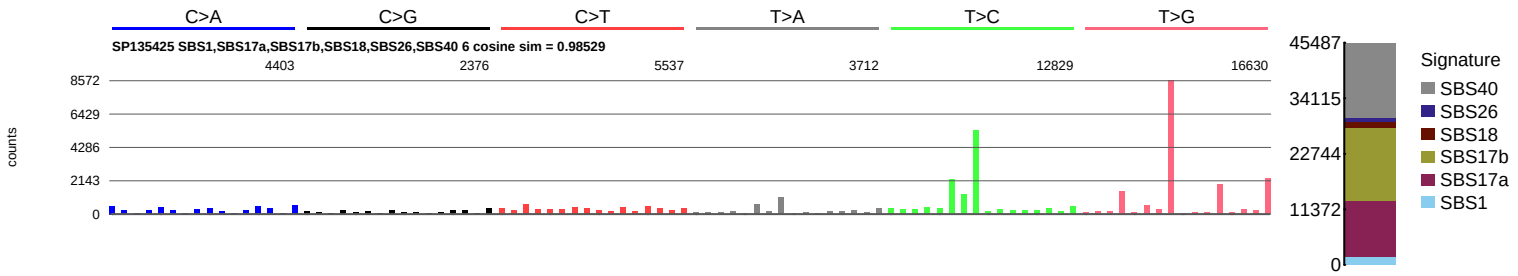
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98529
4403
2376
5537
3712
12829
16630
8572
6429
4286
2143
0
counts
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34115
22744
11372
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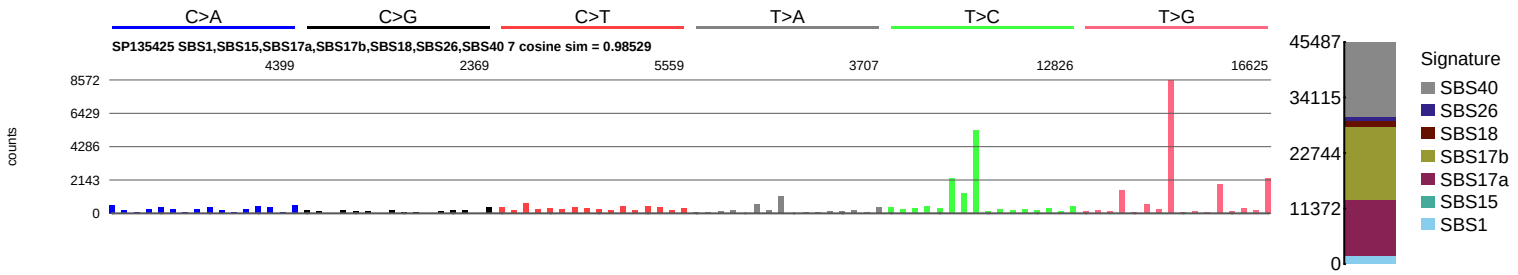
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98529
4399
2369
5559
3707
12826
16625
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
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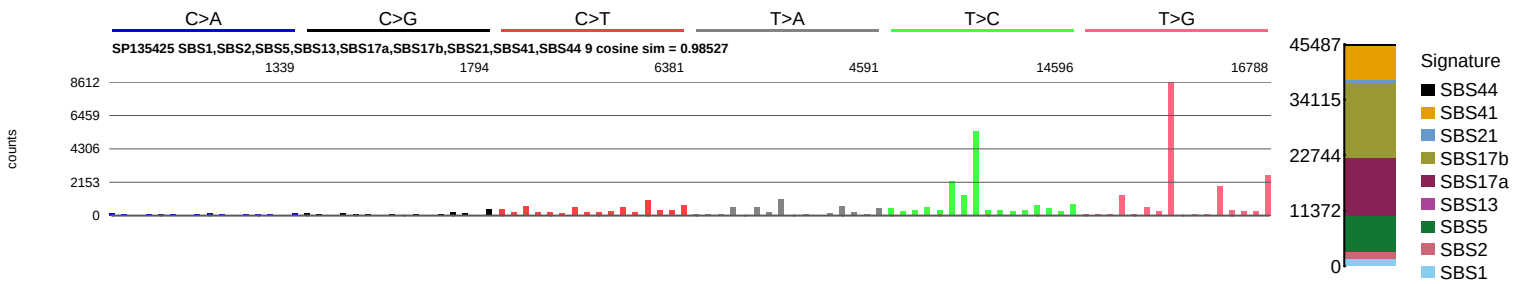
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98527
1339
1794
6381
4591
14596
16788
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
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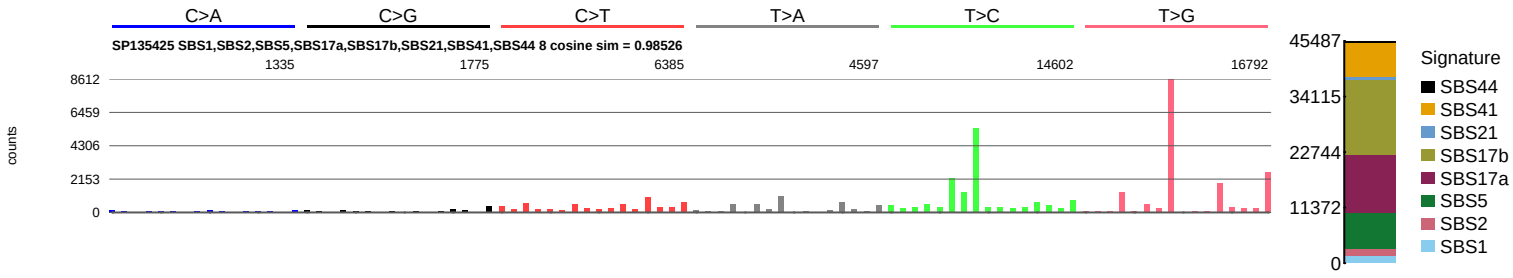
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98526
1335
1775
6385
4597
14602
16792
counts
8612
6459
4306
2153
0
45487
34115
22744
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SBS17b
SBS17a
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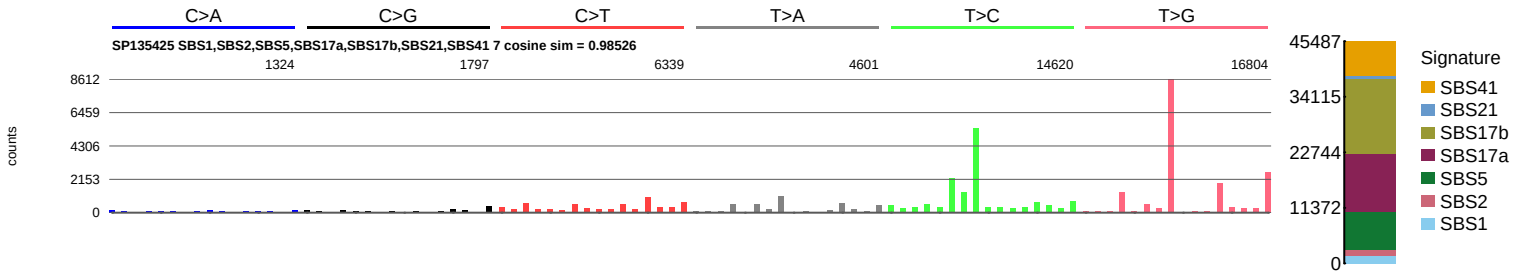
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98526
1324
1797
6339
4601
14620
16804
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
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SBS17b
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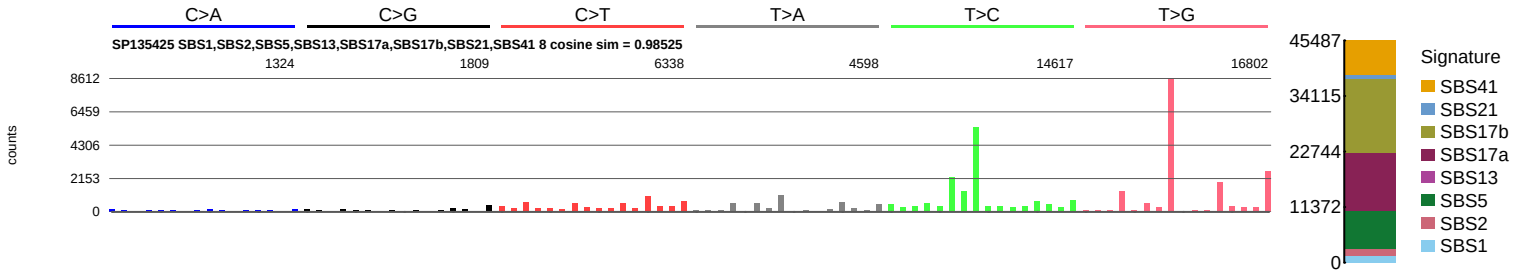
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98525
1324
1809
6338
4598
14617
16802
counts
8612
6459
4306
2153
0
45487
34115
22744
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SBS17b
SBS17a
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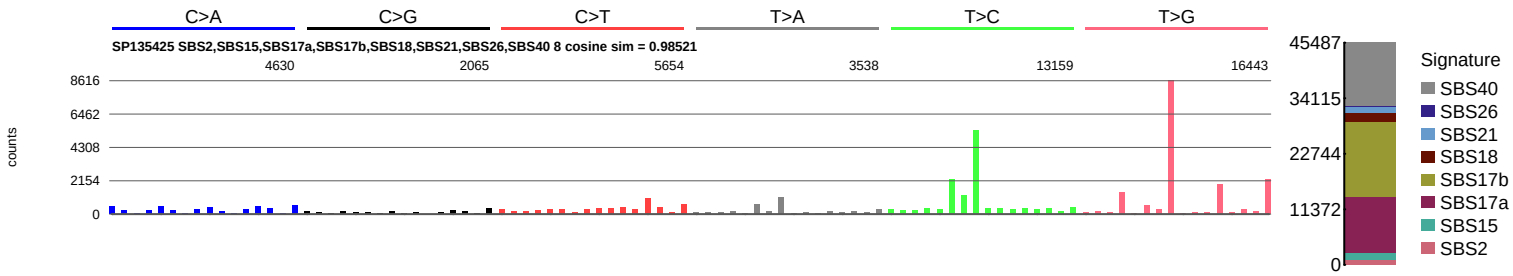
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98521
4630
2065
5654
3538
13159
16443
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
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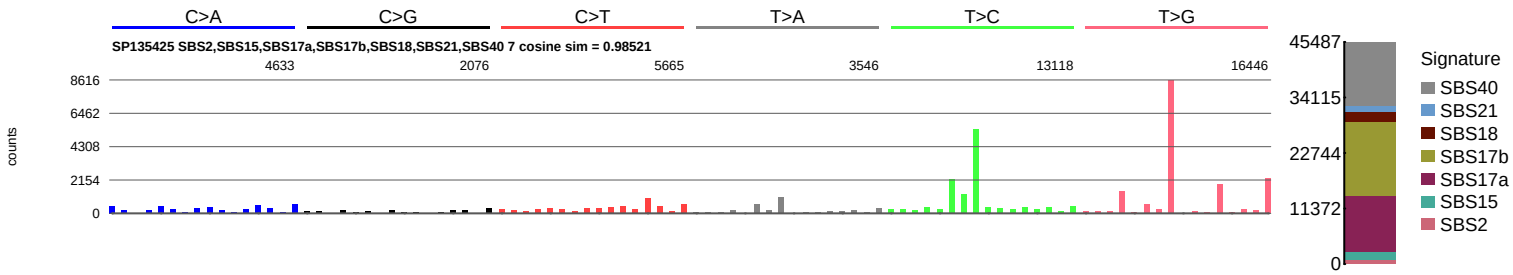
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98521
4633
2076
5665
3546
13118
16446
counts
8616
6462
4308
2154
0
45487
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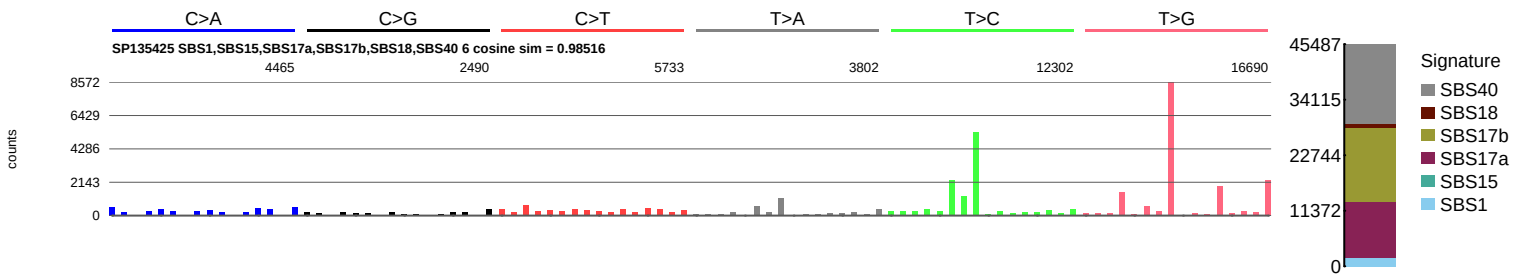
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98516
4465
2490
5733
3802
12302
16690
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
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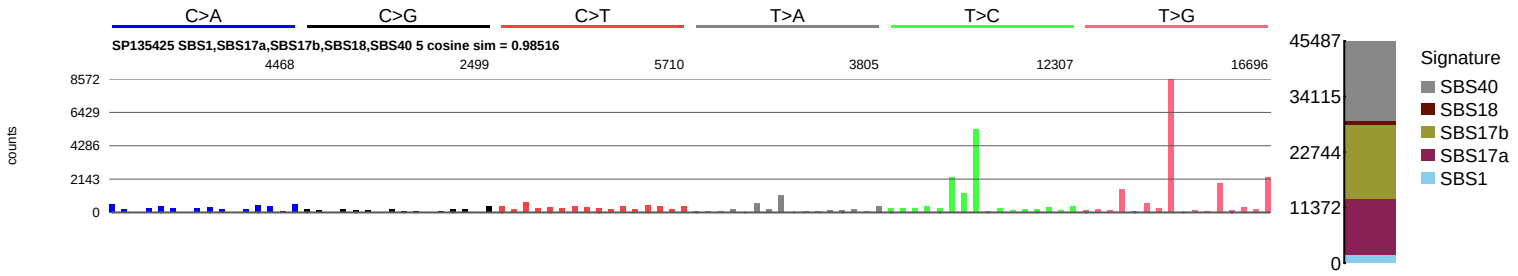
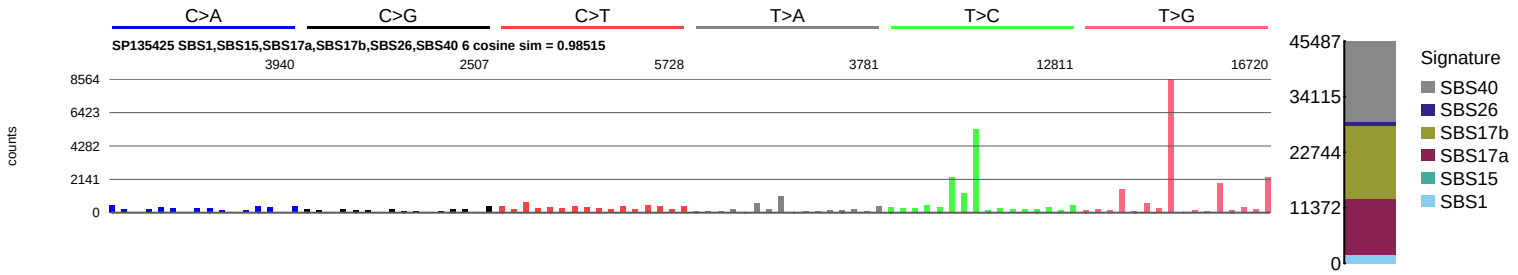
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98516
4468
2499
5710
3805
12307
16696
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1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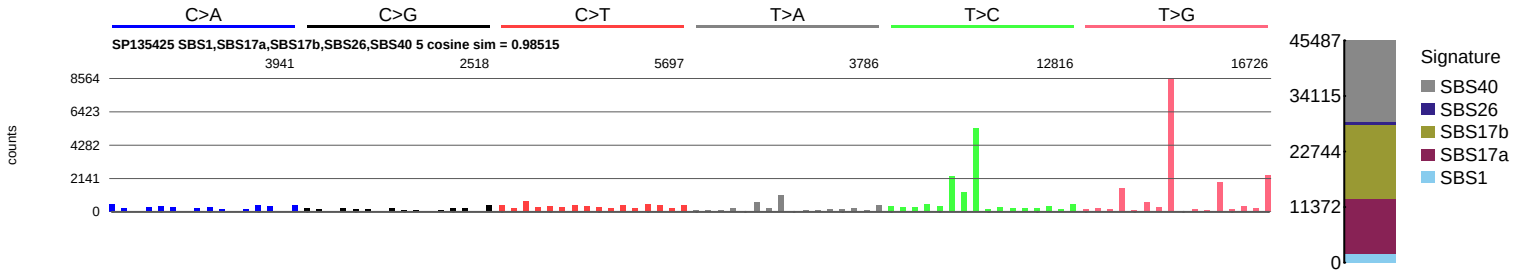
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98515
3940
2507
5728
3781
12811
16720
8564
6423
4282
2141
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1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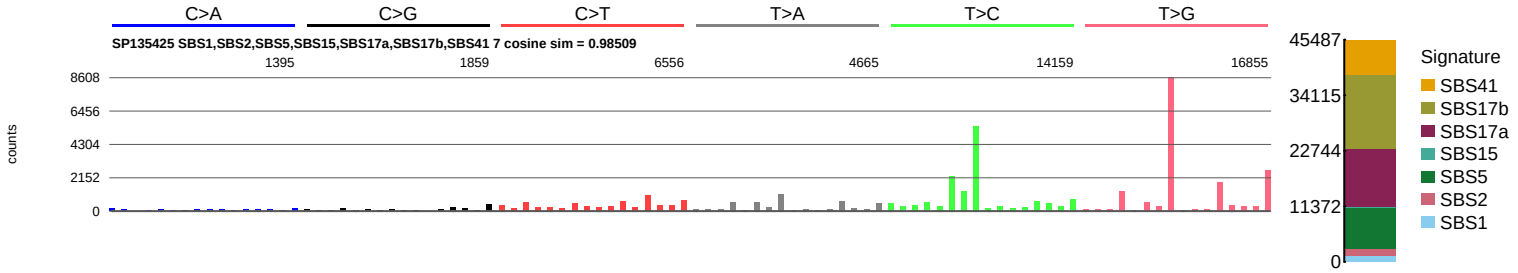
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98515
3941
2518
5697
3786
12816
16726
8564
6423
4282
2141
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1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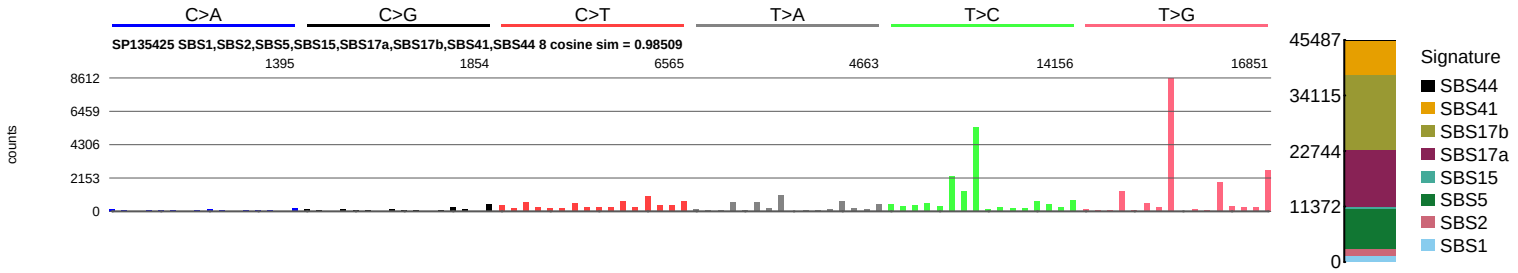
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98509
1395
1859
6556
4665
14159
16855
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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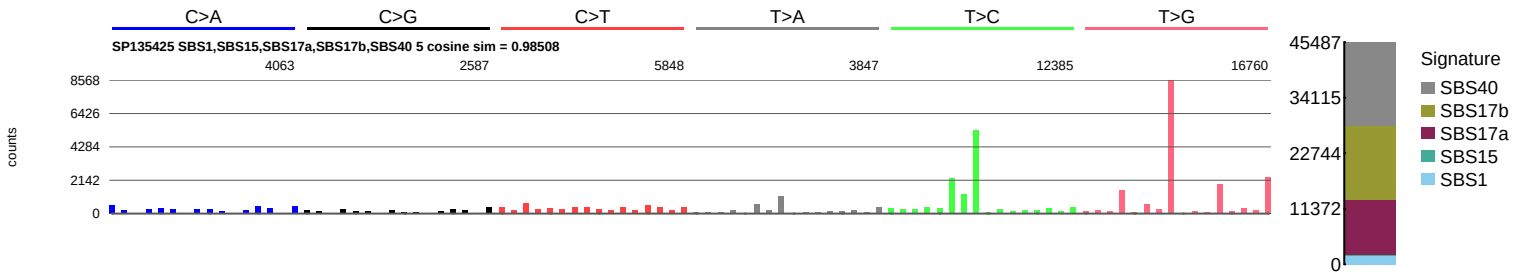
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98509
1395
1854
6565
4663
14156
16851
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
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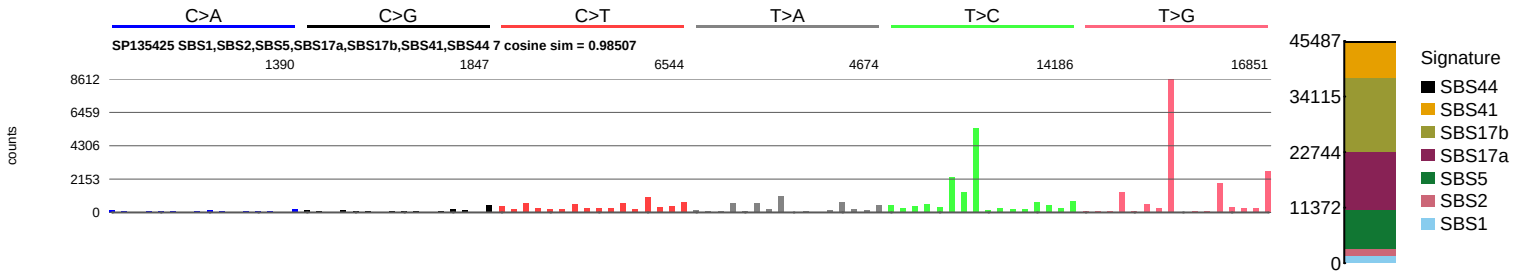
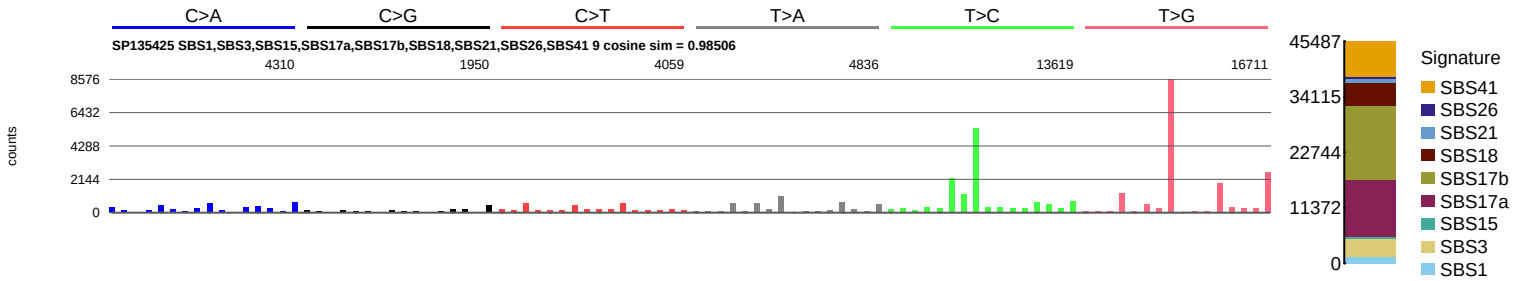
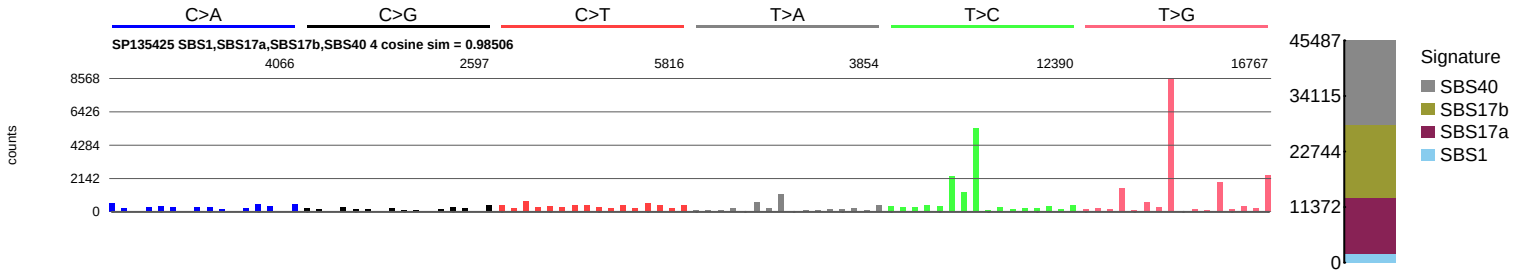
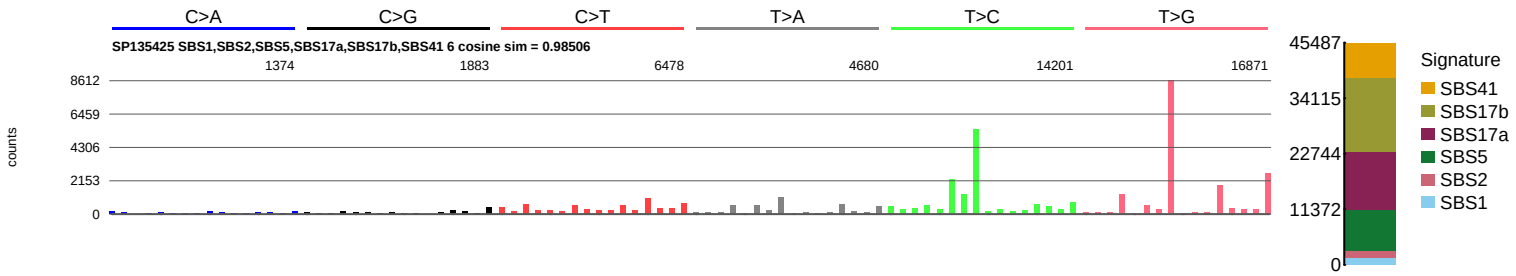
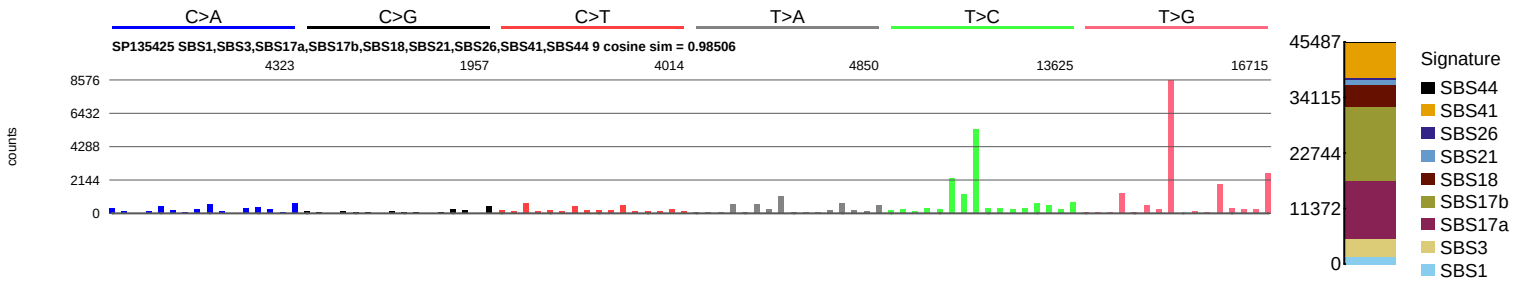
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98508
4063
2587
5848
3847
12385
16760
8568
6426
4284
2142
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98507
1390
1847
6544
4674
14186
16851
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98506
4310
1950
4059
4836
13619
16711
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98506
4066
2597
5816
3854
12390
16767
counts
8568
6426
4284
2142
0
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98506
1374
1883
6478
4680
14201
16871
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1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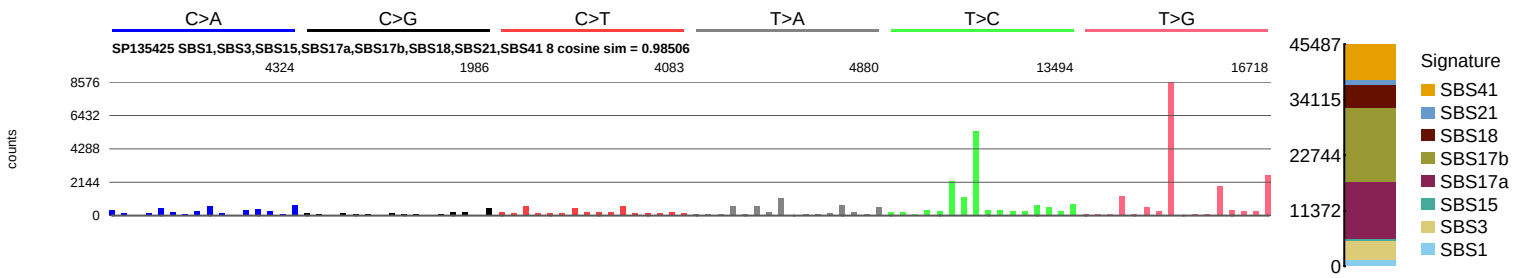
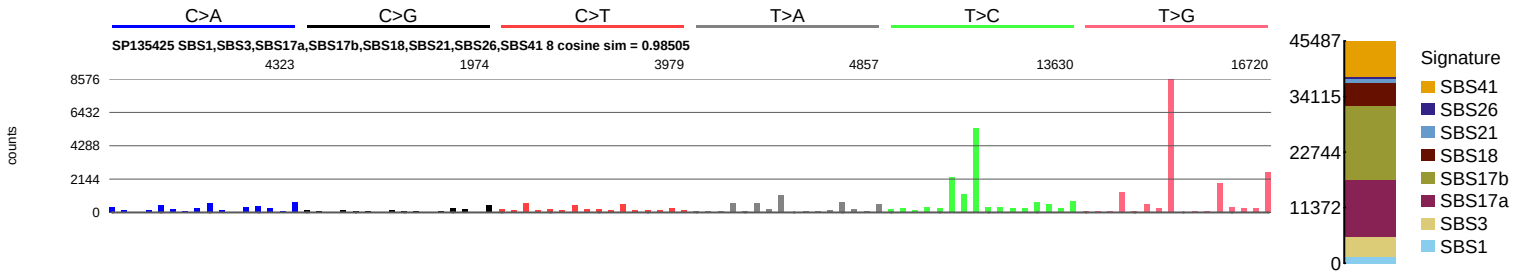
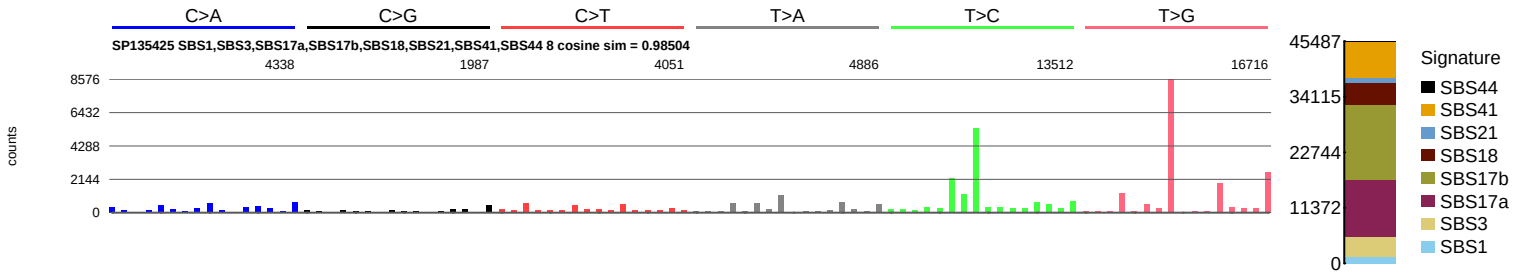
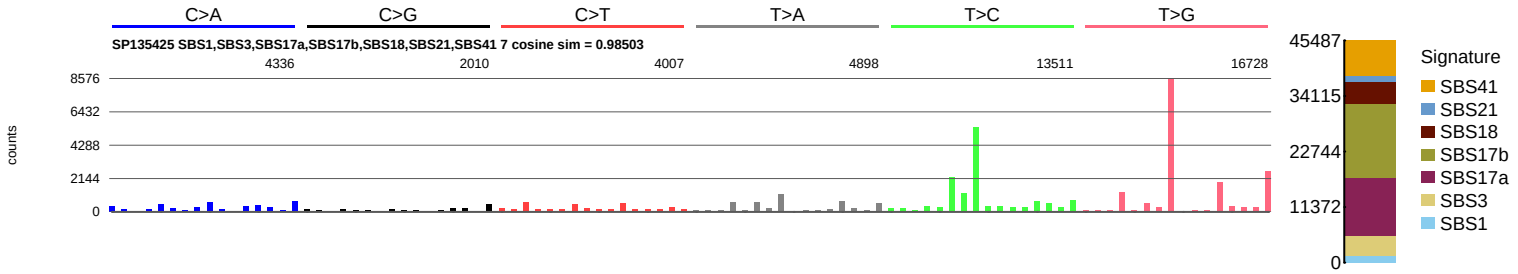
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98506
4323
1957
4014
4850
13625
16715
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98506
4324
1986
4083
4880
13494
16718
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98505
4323
1974
3979
4857
13630
16720
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98504
4338
1987
4051
4886
13512
16716
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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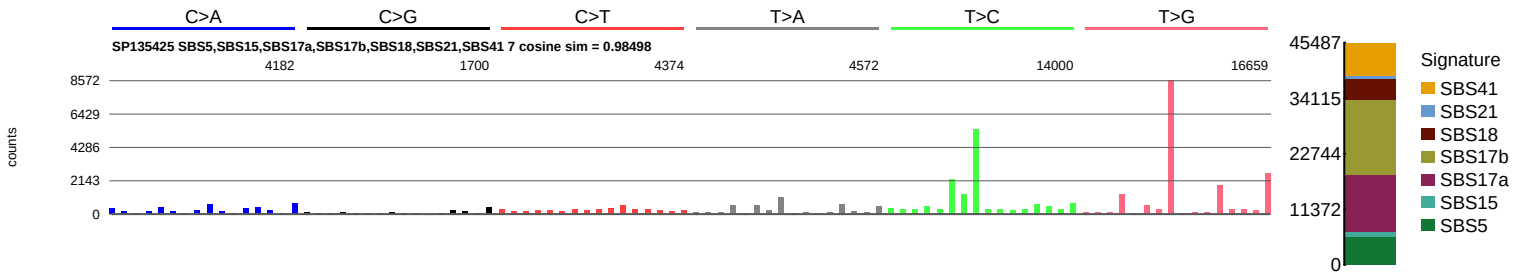
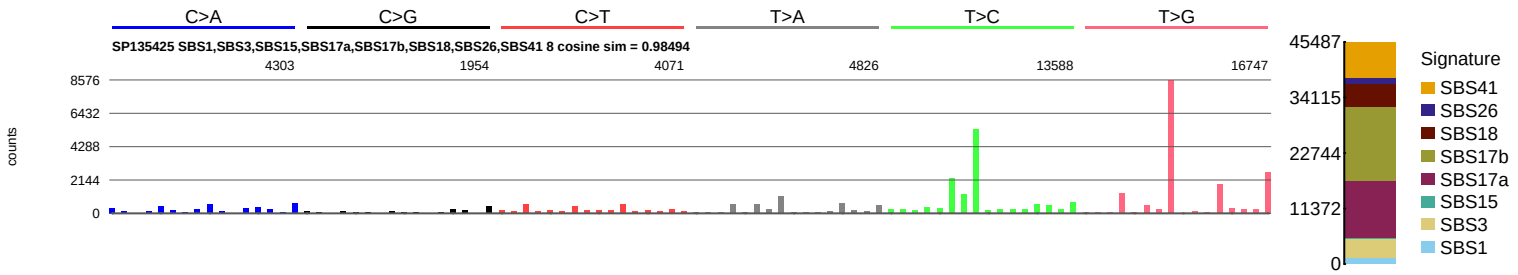
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98503
4336
2010
4007
4898
13511
16728
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98498
4182
1700
4374
4572
14000
16659
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5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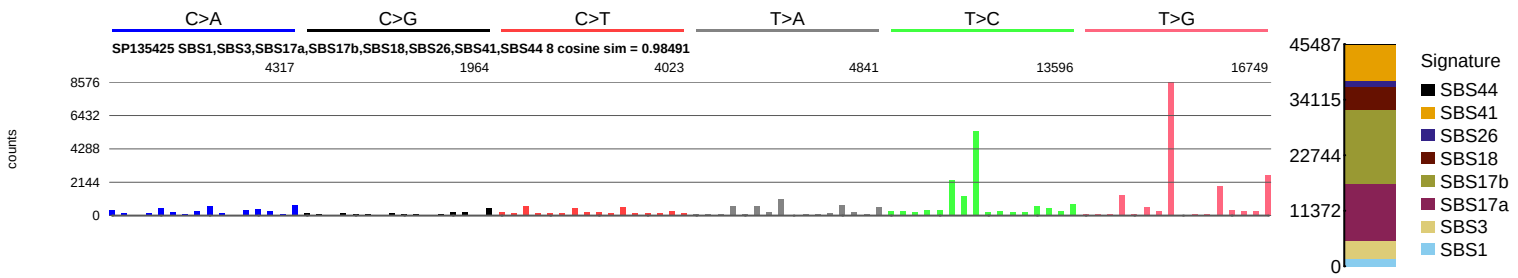
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98494
4303
1954
4071
4826
13588
16747
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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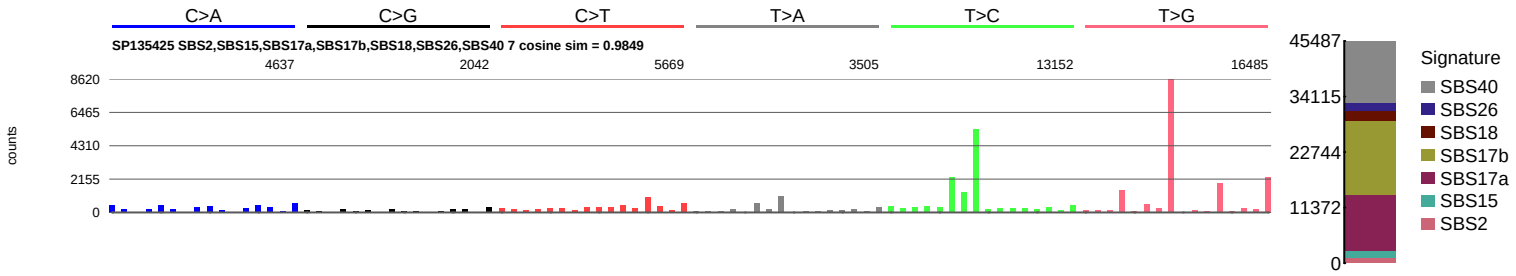
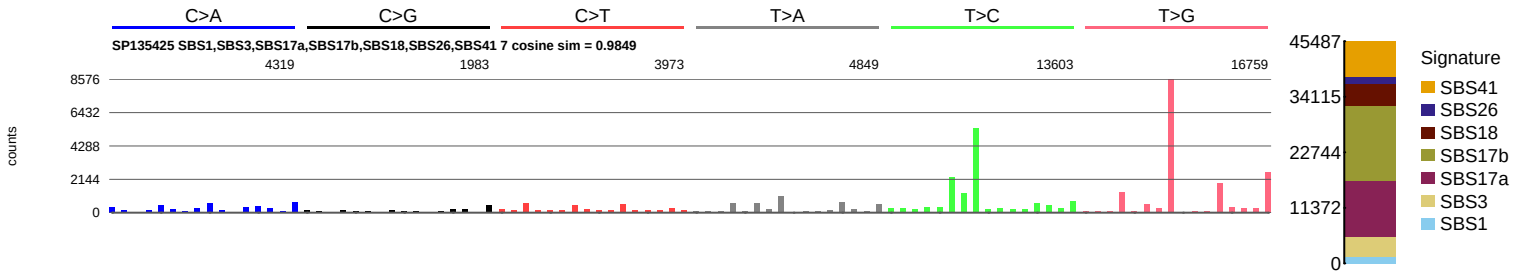
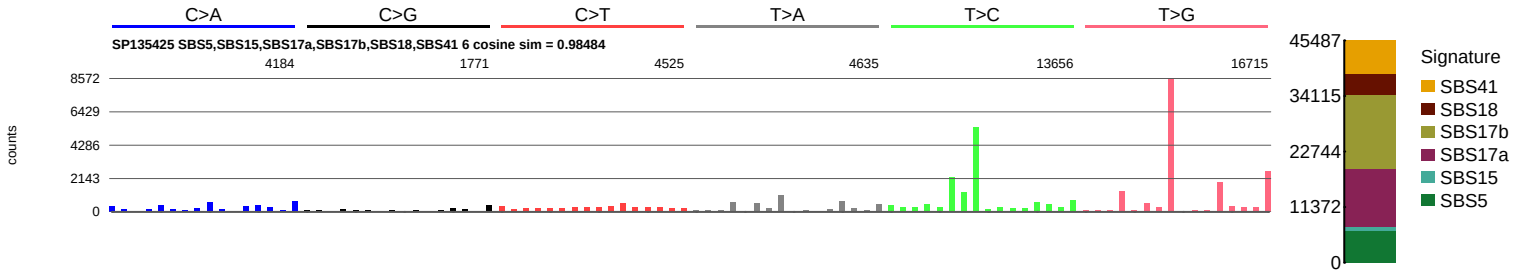
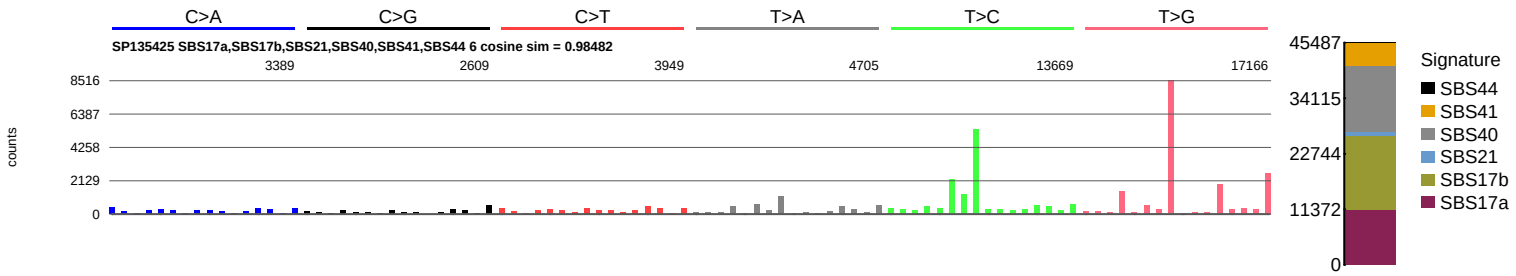
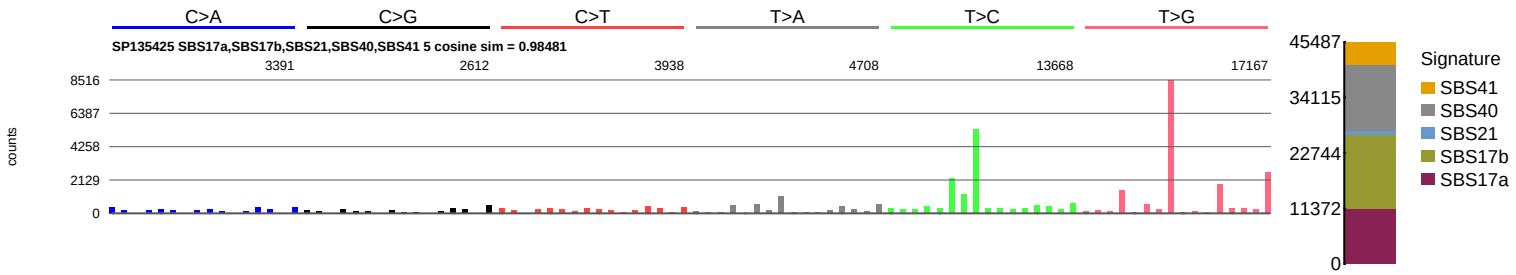
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98491
4317
1964
4023
4841
13596
16749
counts
8576
6432
4288
2144
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

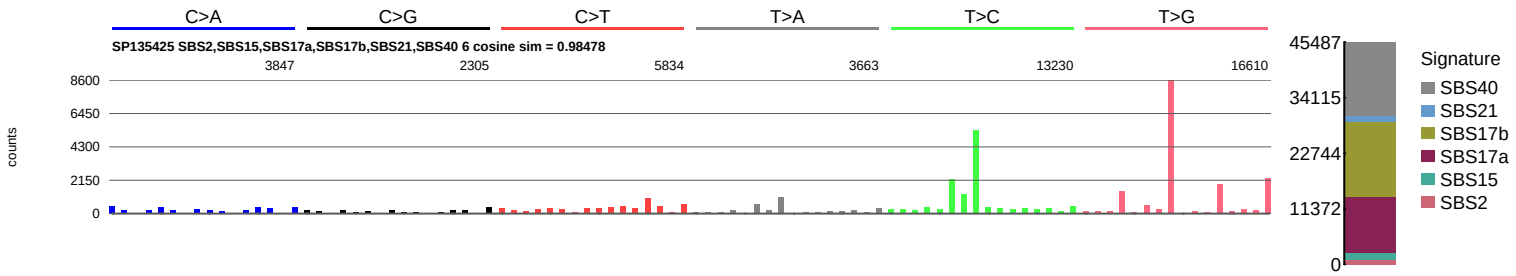

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.9849
4637
2042
5669
3505
13152
16485
8620
6465
4310
2155
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2

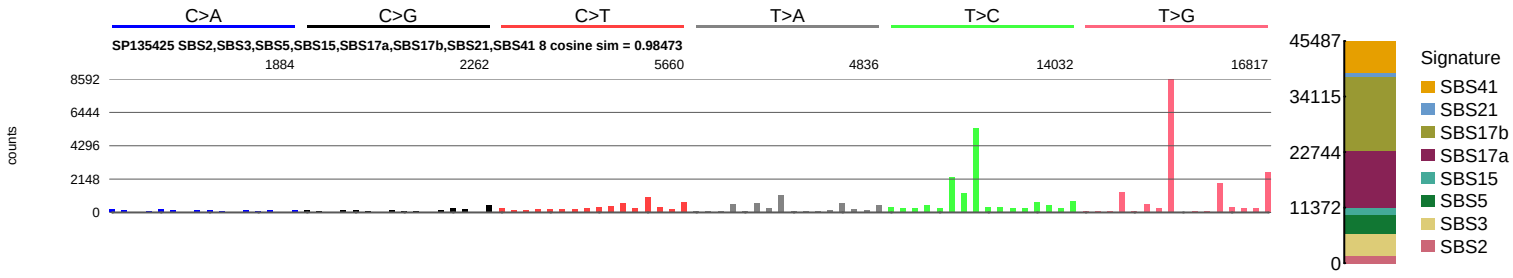

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9849
4319
1983
3973
4849
13603
16759
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1

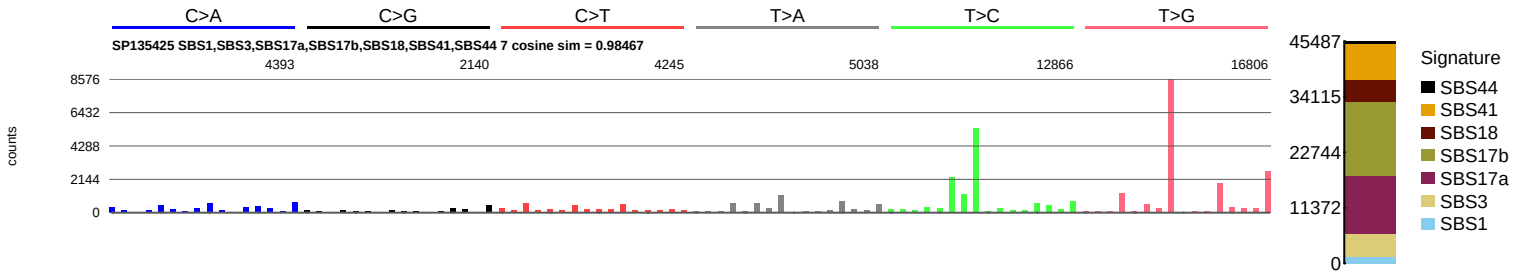

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98484
4184
1771
4525
4635
13656
16715
8572
6429
4286
2143
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5

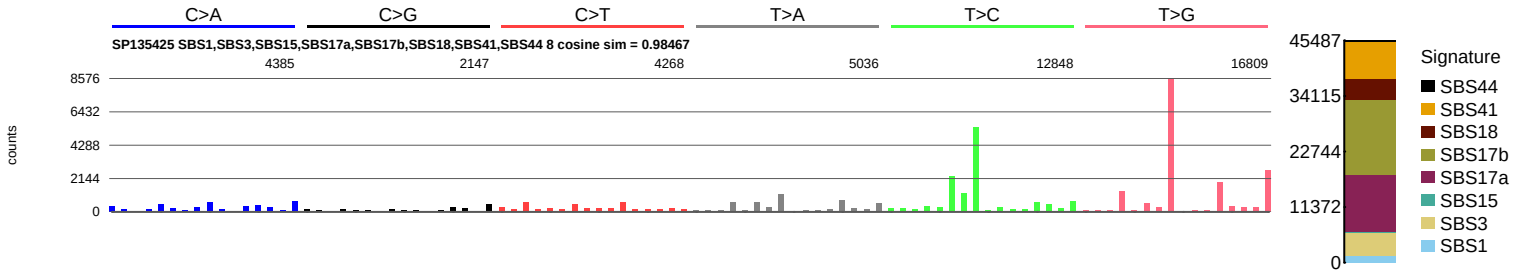

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 6 cosine sim = 0.98482
3389
2609
3949
4705
13669
17166
8516
6387
4258
2129
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS21,SBS40,SBS41 5 cosine sim = 0.98481
3391
2612
3938
4708
13668
17167
8516
6387
4258
2129
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98478
3847
2305
5834
3663
13230
16610
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98473
1884
2262
5660
4836
14032
16817
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98467
4393
2140
4245
5038
12866
16806
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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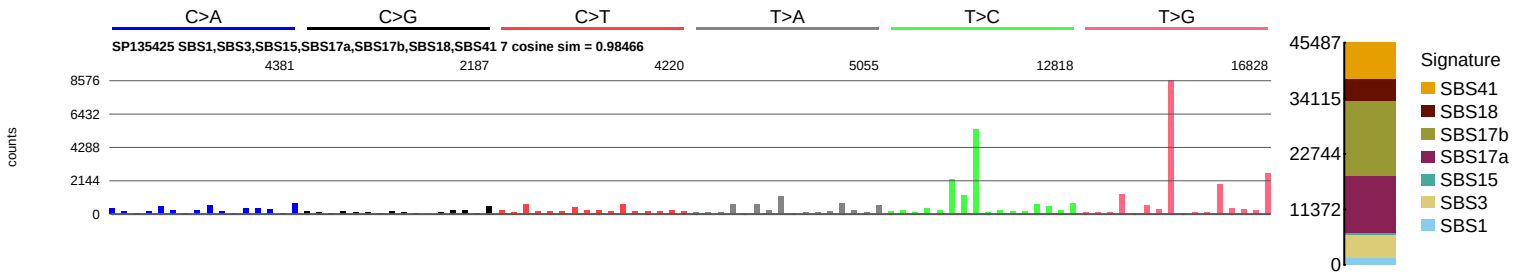
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98467
4385
2147
4268
5036
12848
16809
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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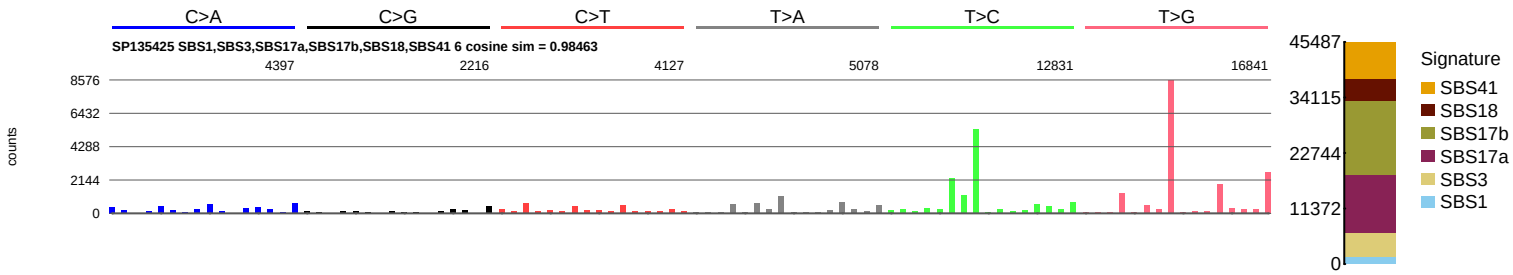
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98466
4381
2187
4220
5055
12818
16828
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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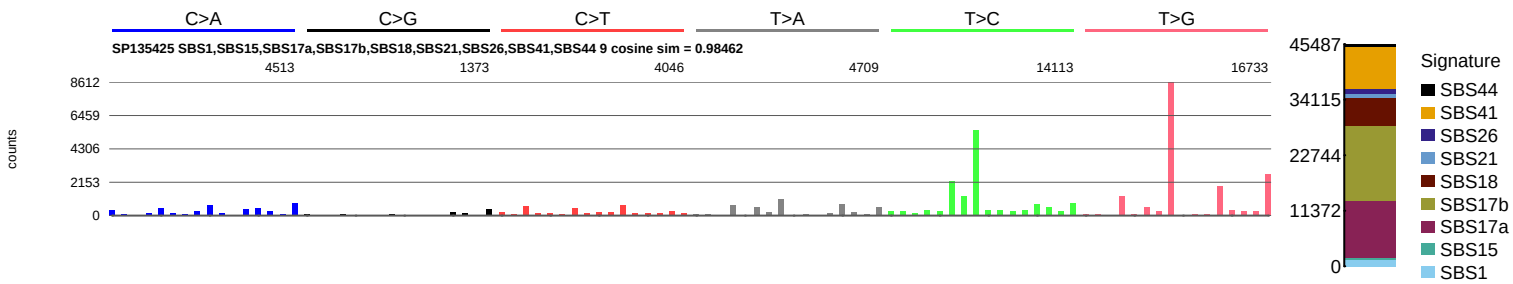
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98463
4397
2216
4127
5078
12831
16841
8576
6432
4288
2144
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1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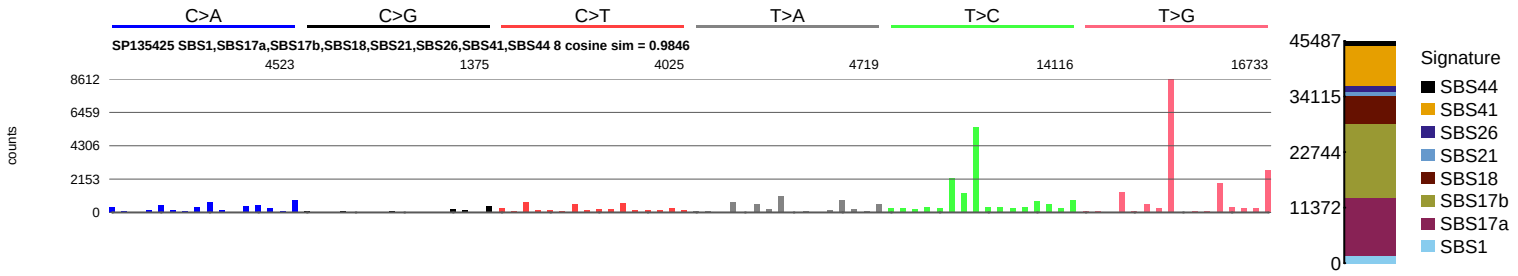
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98462
4513
1373
4046
4709
14113
16733
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>T
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T>C
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SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.9846
4523
1375
4025
4719
14116
16733
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
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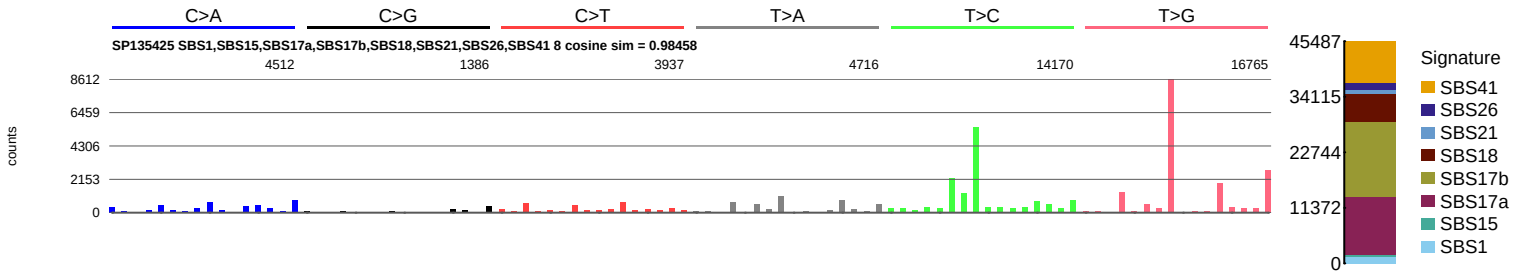
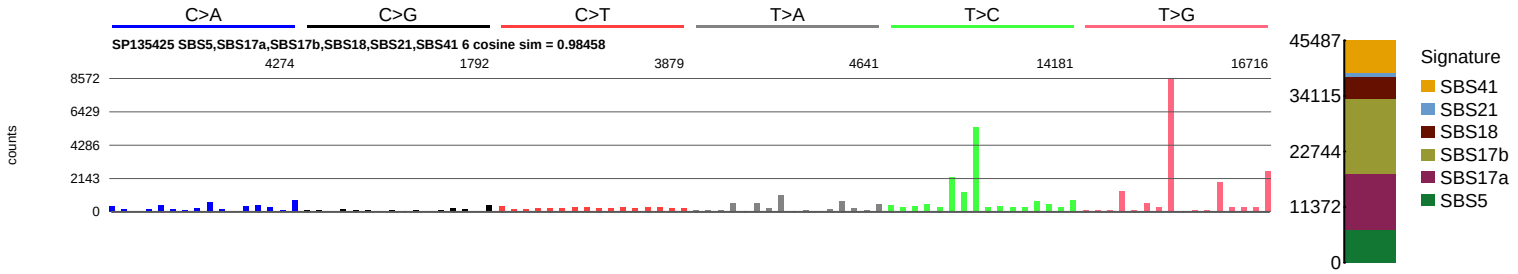
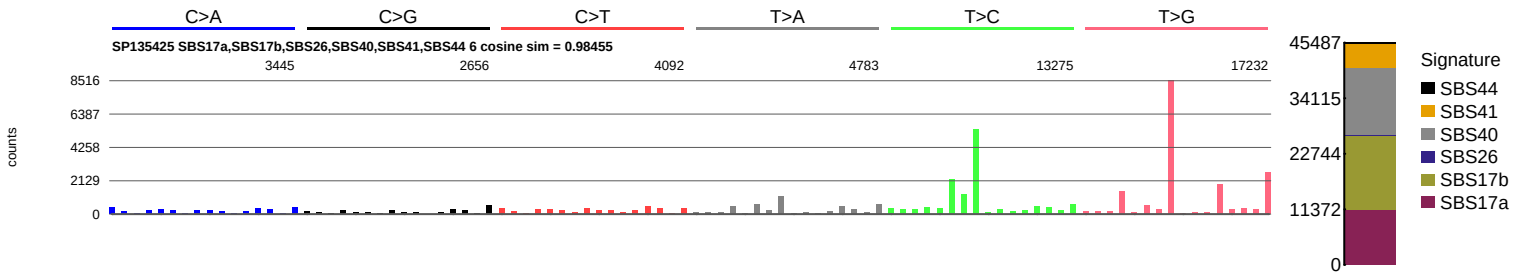
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98458
4512
1386
3937
4716
14170
16765
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98458
4274
1792
3879
4641
14181
16716
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5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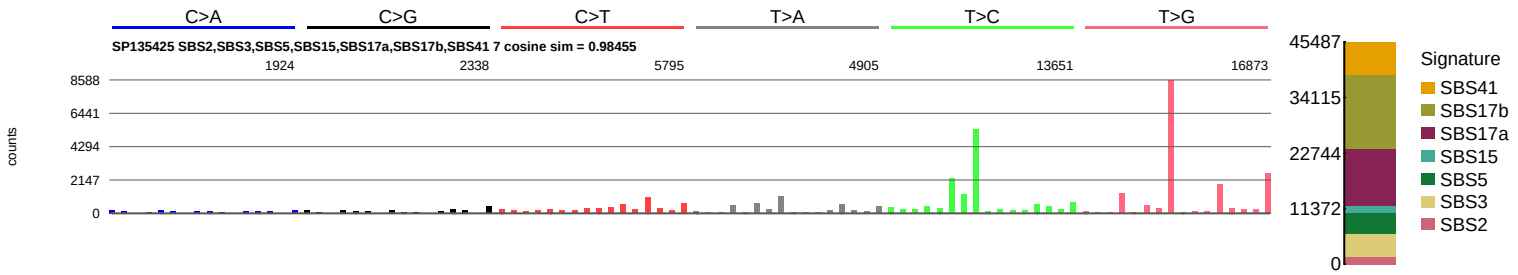
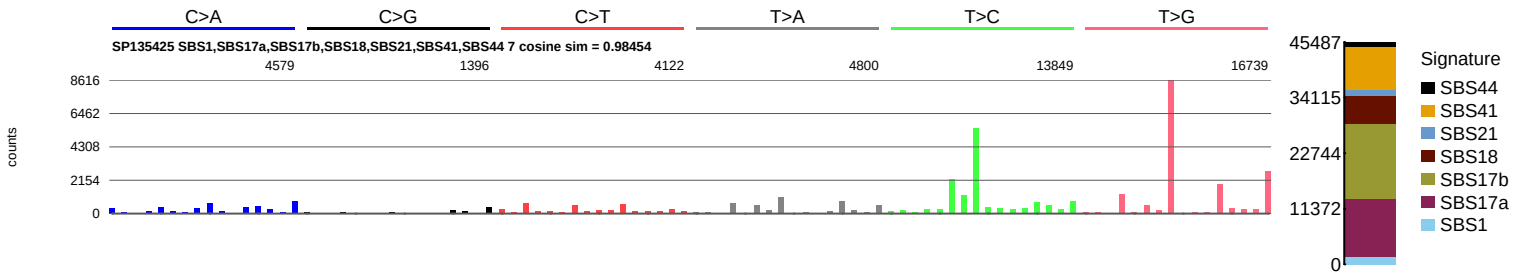
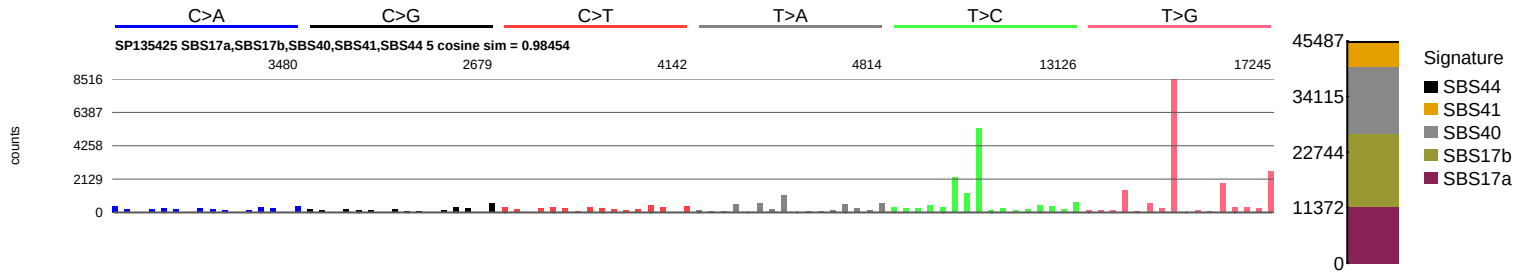
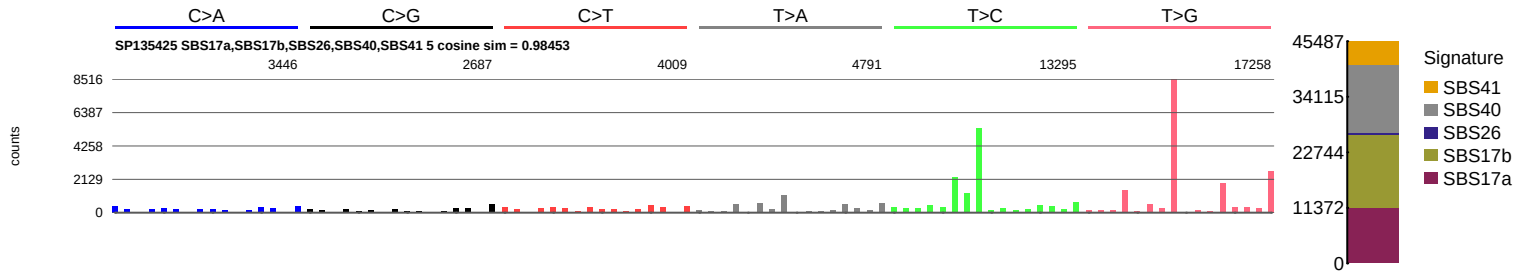
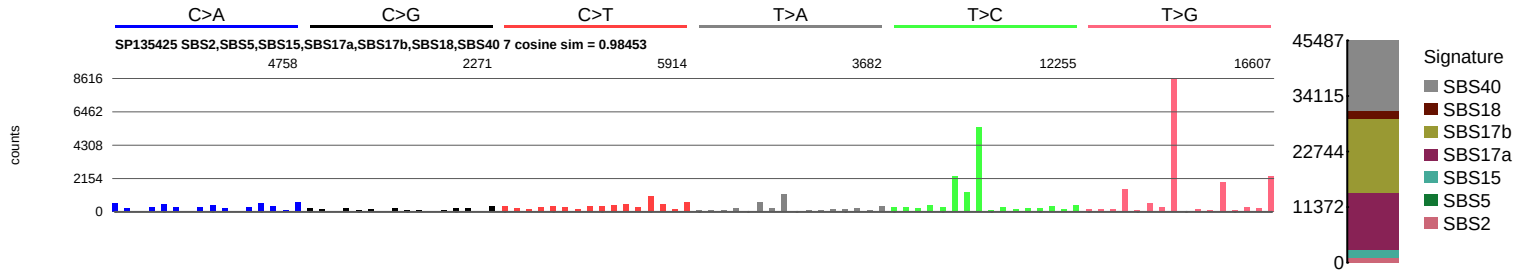
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 6 cosine sim = 0.98455
3445
2656
4092
4783
13275
17232
counts
8516
6387
4258
2129
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a

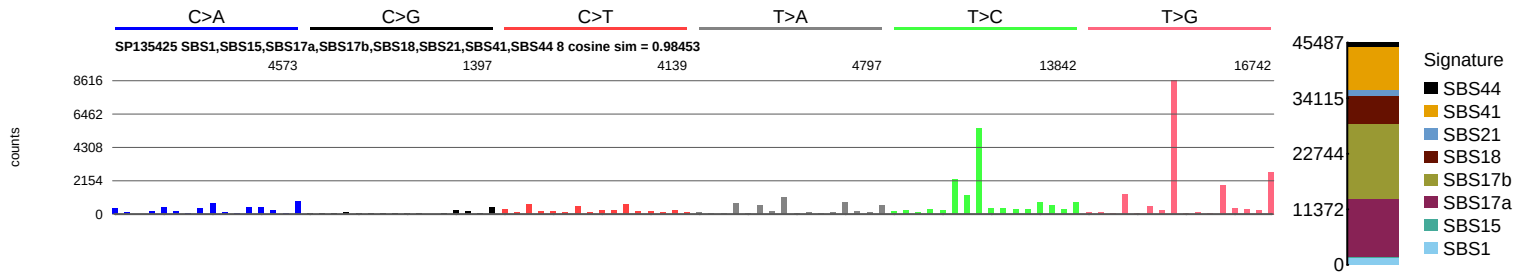

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98455
1924
2338
5795
4905
13651
16873
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98454
4579
1396
4122
4800
13849
16739
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS40,SBS41,SBS44 5 cosine sim = 0.98454
3480
2679
4142
4814
13126
17245
counts
8516
6387
4258
2129
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS26,SBS40,SBS41 5 cosine sim = 0.98453
3446
2687
4009
4791
13295
17258
counts
8516
6387
4258
2129
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98453
4758
2271
5914
3682
12255
16607
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2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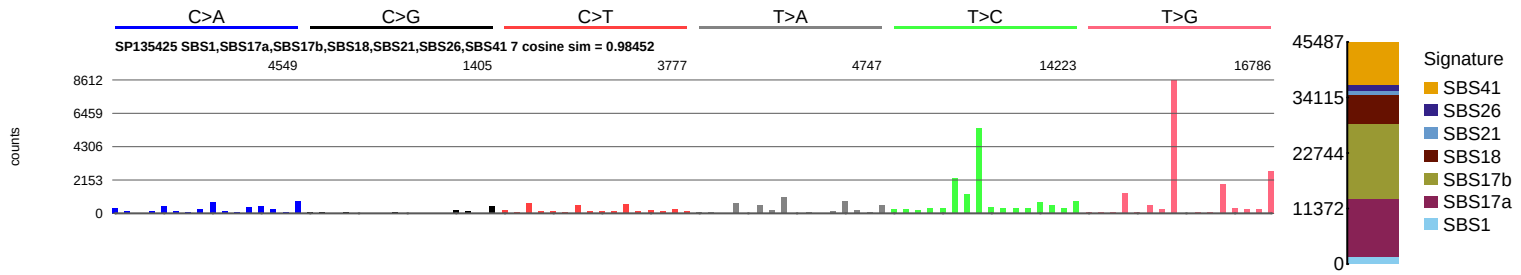
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98453
4573
1397
4139
4797
13842
16742
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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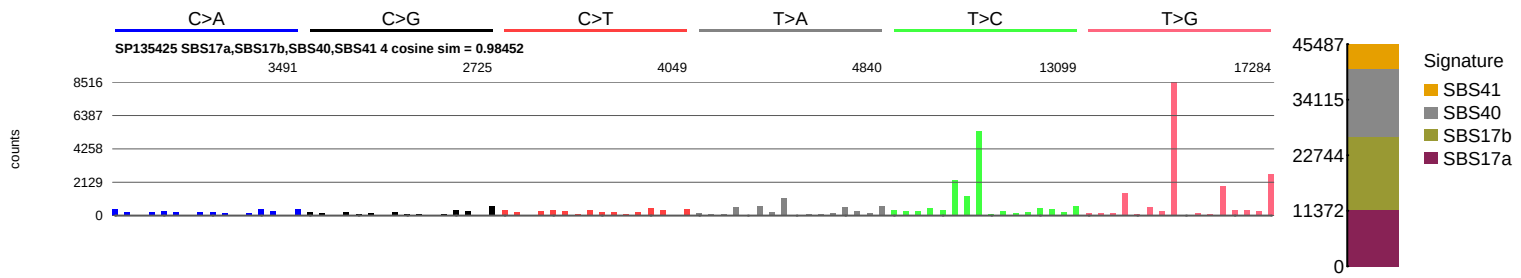
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98452
4549
1405
3777
4747
14223
16786
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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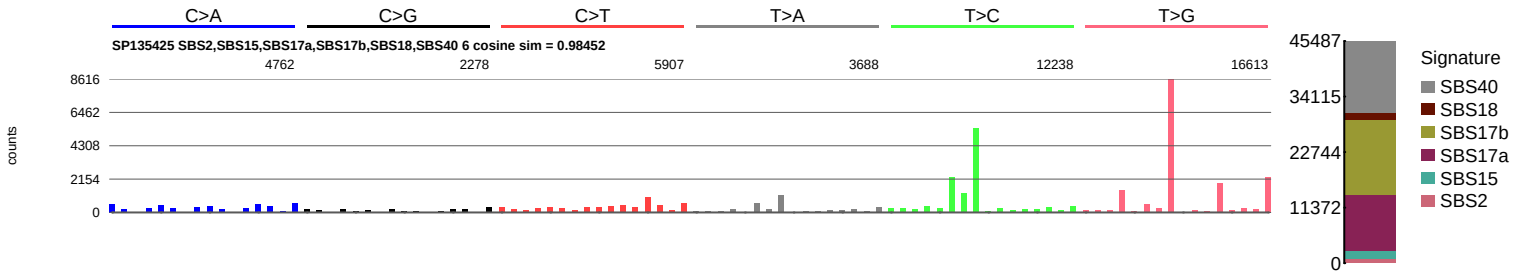
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS40,SBS41 4 cosine sim = 0.98452
3491
2725
4049
4840
13099
17284
counts
8516
6387
4258
2129
0
45487
34115
22744
11372
0
Signature
SBS41
SBS40
SBS17b
SBS17a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C>G
C>T
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T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98452
4762
2278
5907
3688
12238
16613
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
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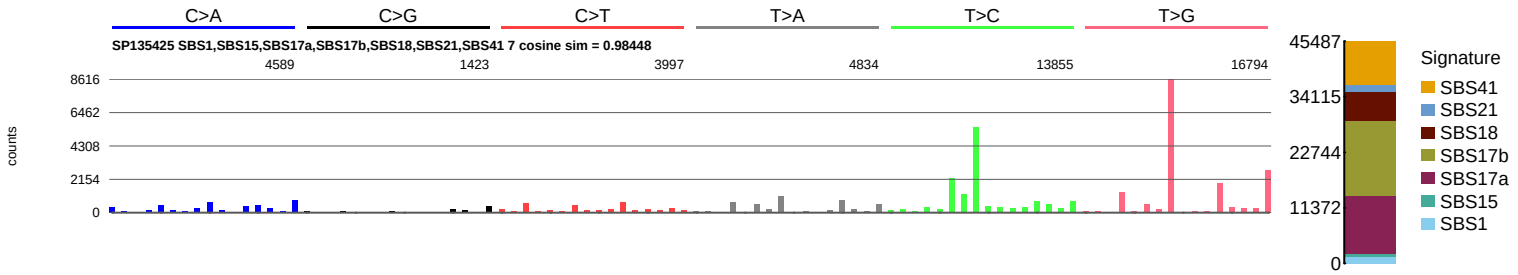
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98448
4589
1423
3997
4834
13855
16794
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
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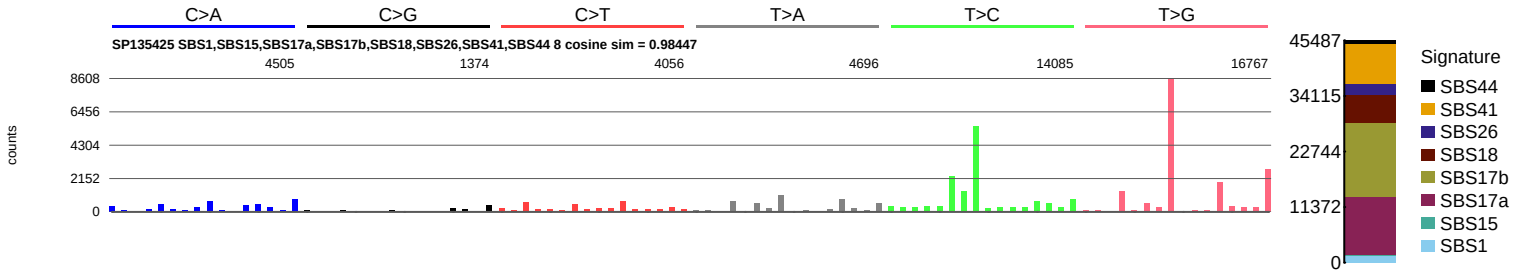
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98447
4505
1374
4056
4696
14085
16767
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
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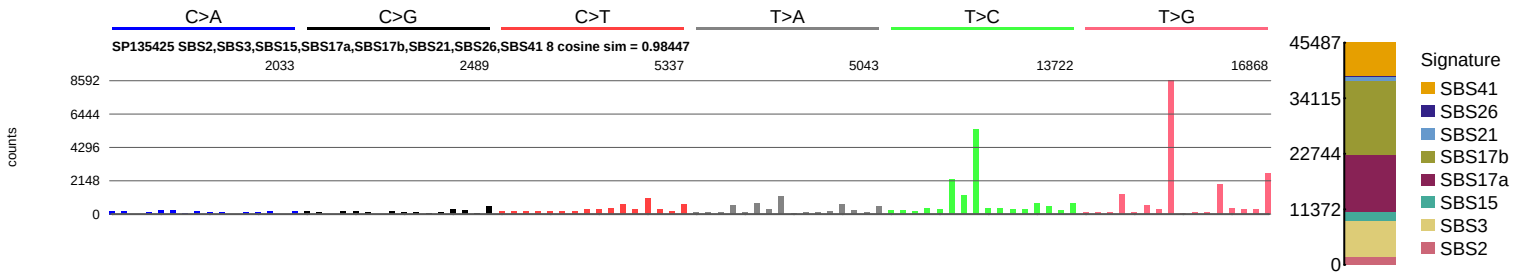
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98447
2033
2489
5337
5043
13722
16868
counts
8592
6444
4296
2148
0
45487
34115
22744
11372
0
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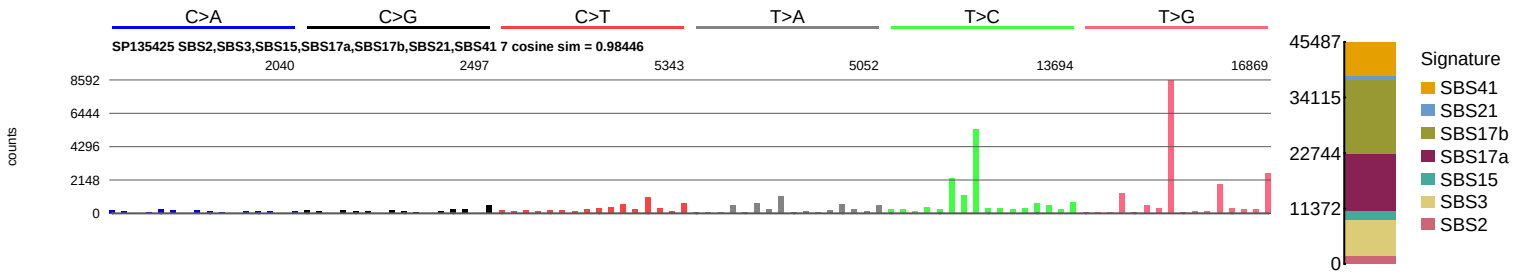
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98446
2040
2497
5343
5052
13694
16869
counts
8592
6444
4296
2148
0
45487
34115
22744
11372
0
Signature
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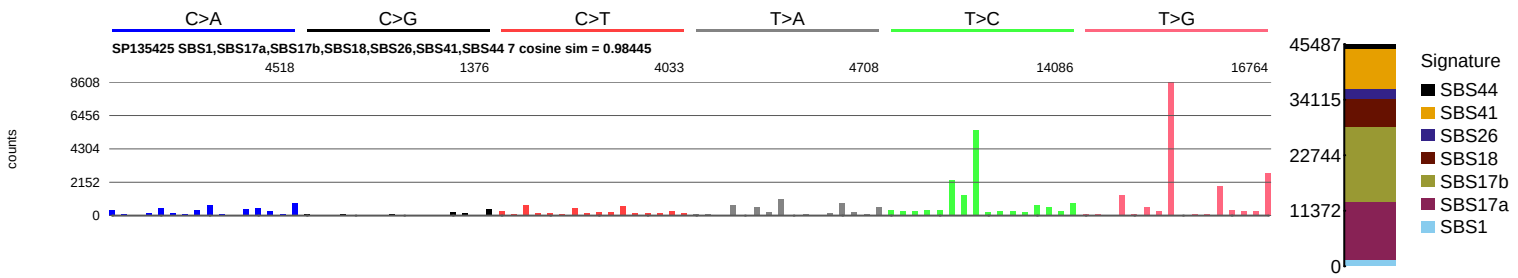
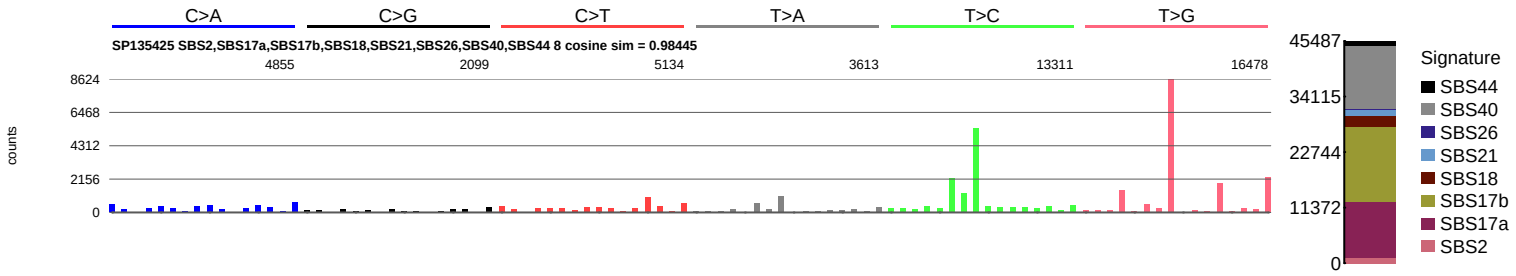
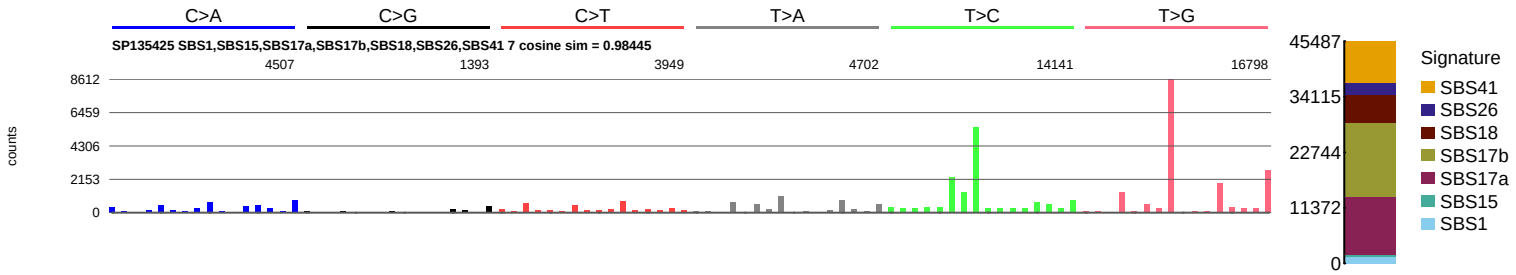
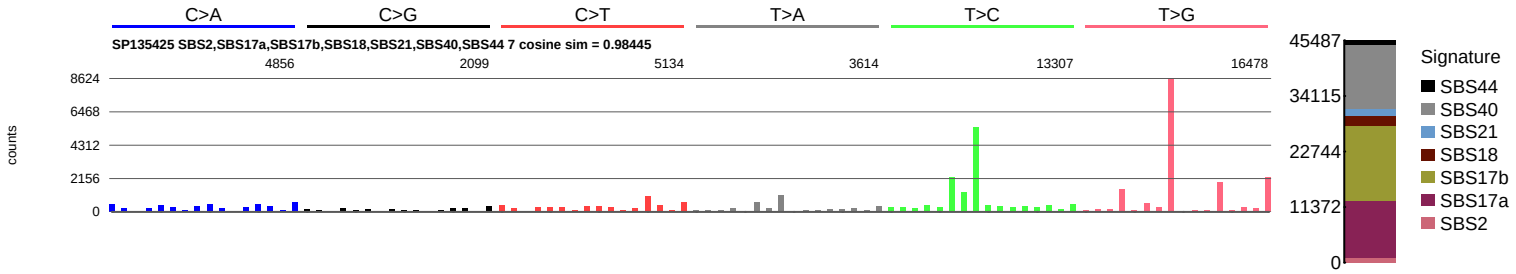
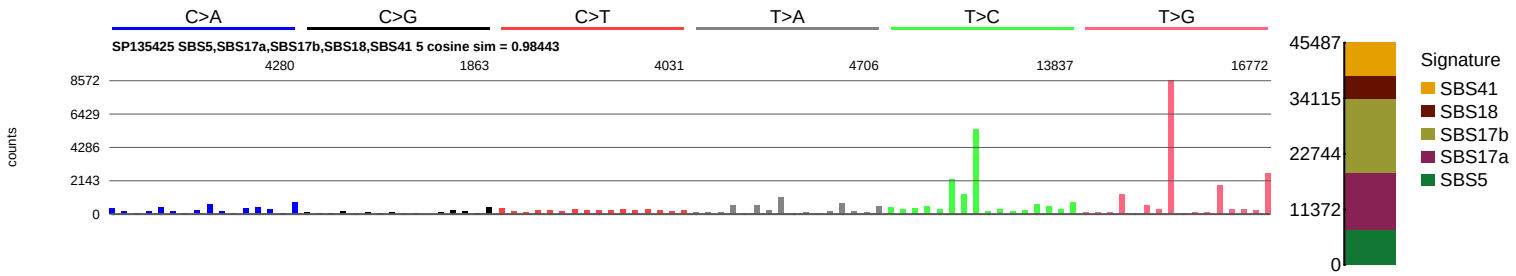
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98445
4518
1376
4033
4708
14086
16764
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98445
4855
2099
5134
3613
13311
16478
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98445
4507
1393
3949
4702
14141
16798
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98445
4856
2099
5134
3614
13307
16478
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
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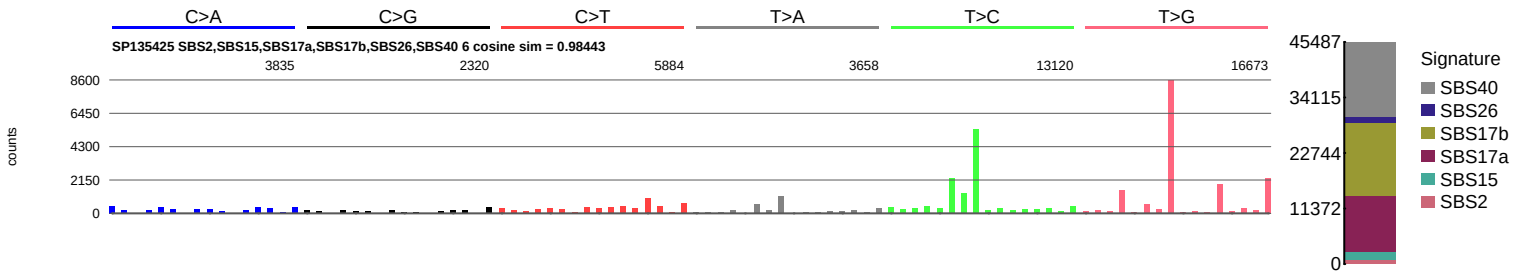
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98443
4280
1863
4031
4706
13837
16772
counts
8572
6429
4286
2143
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5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C>G
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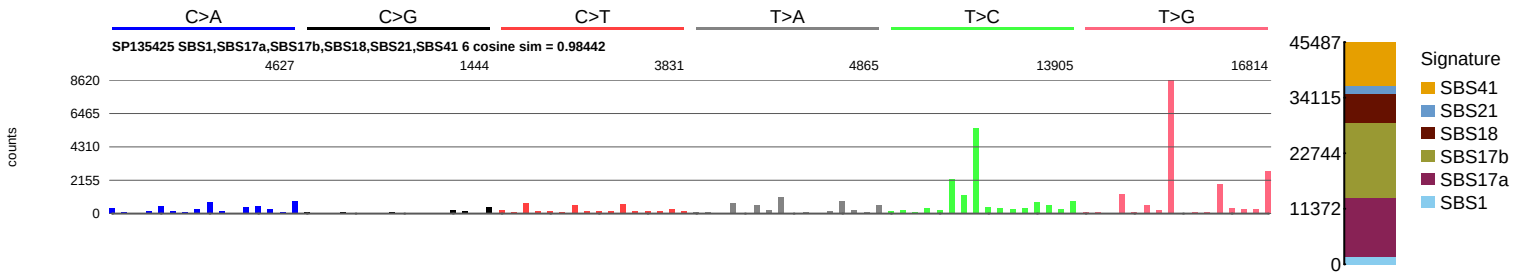
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98443
3835
2320
5884
3658
13120
16673
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2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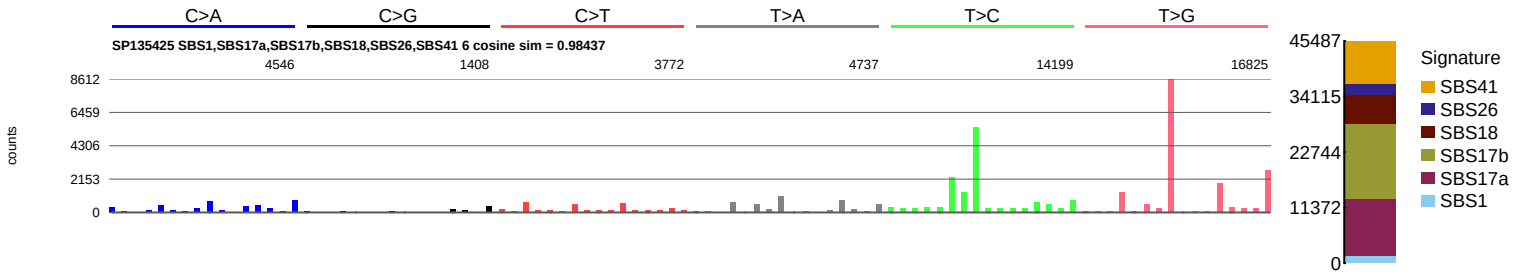
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98442
4627
1444
3831
4865
13905
16814
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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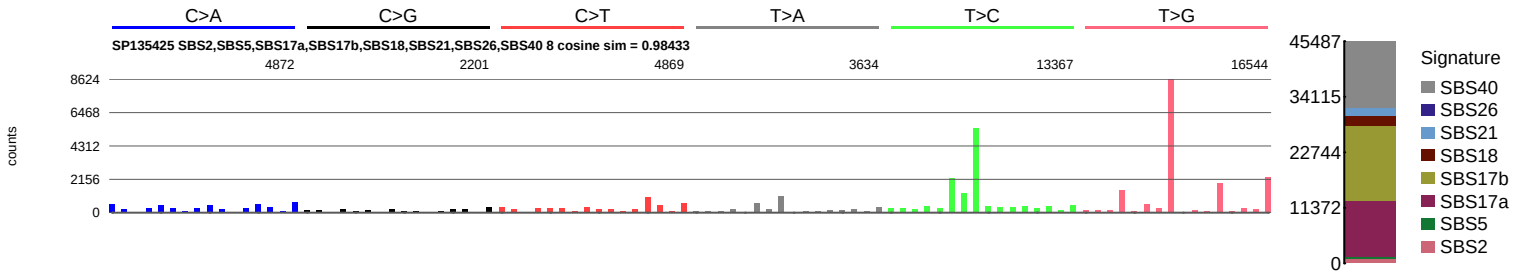
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98437
4546
1408
3772
4737
14199
16825
counts
8612
6459
4306
2153
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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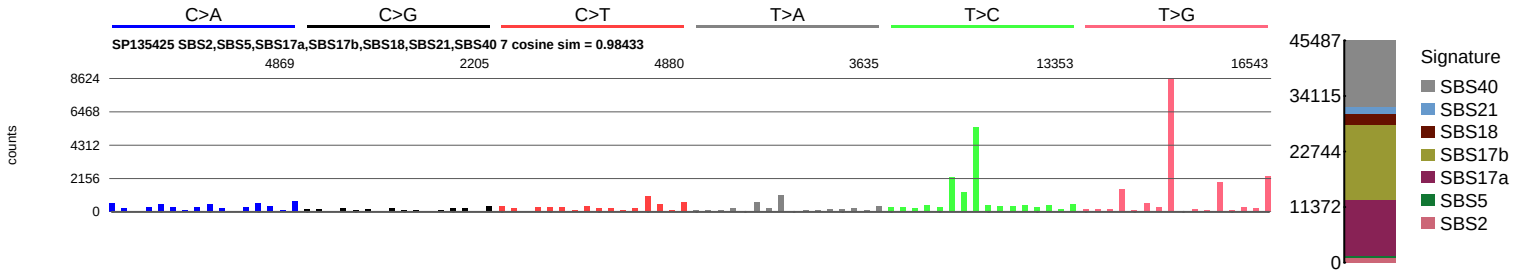
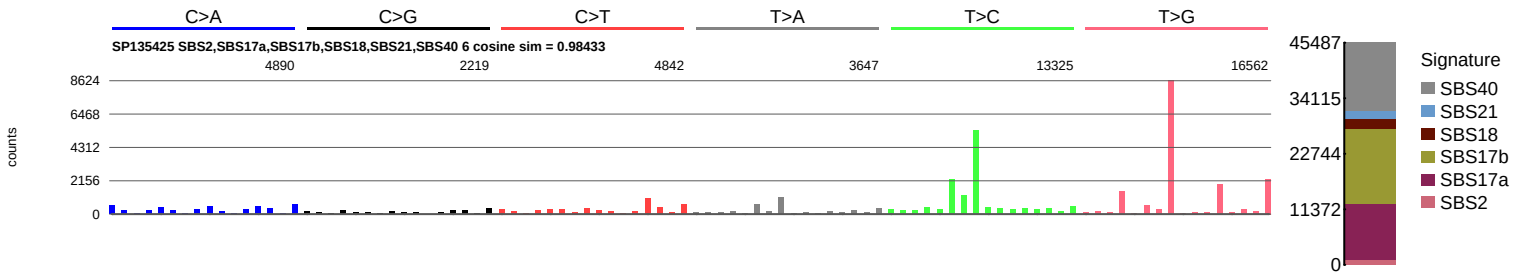
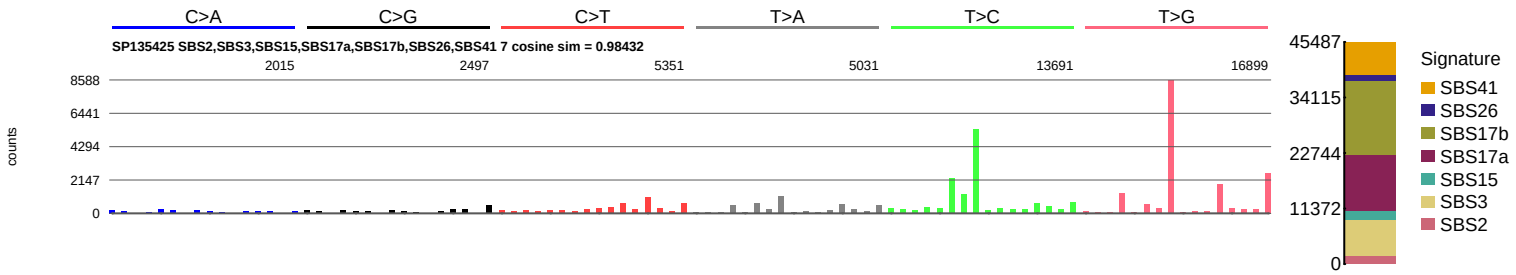
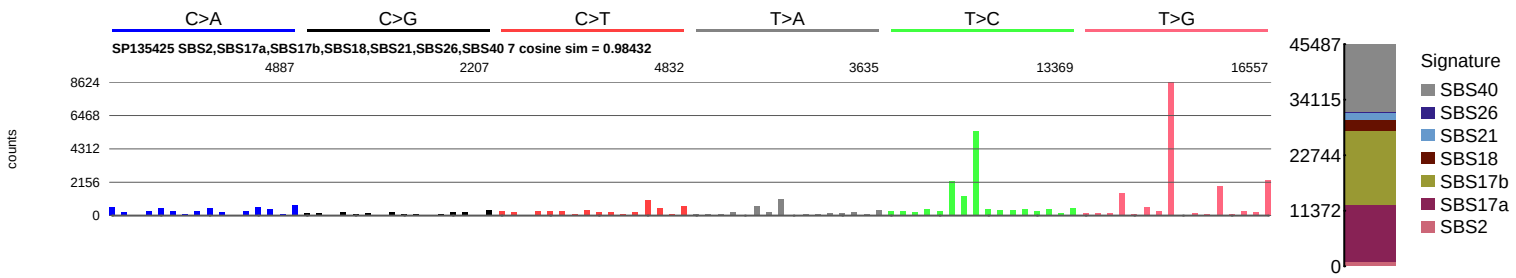
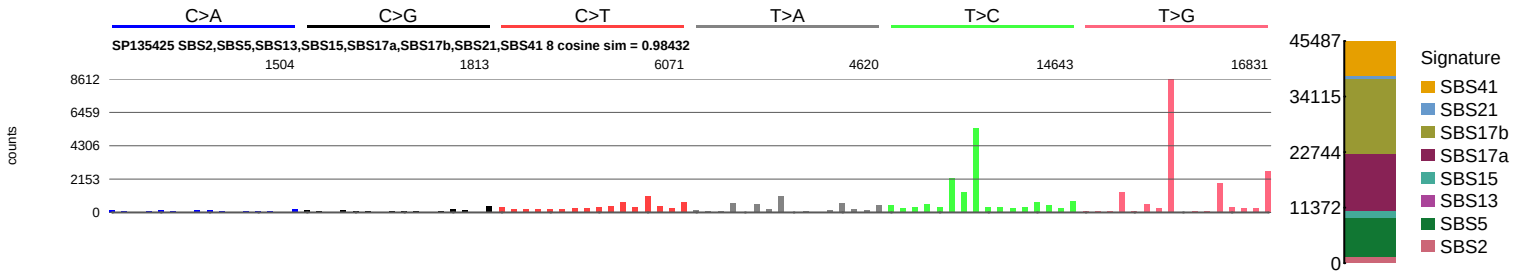
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98433
4872
2201
4869
3634
13367
16544
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

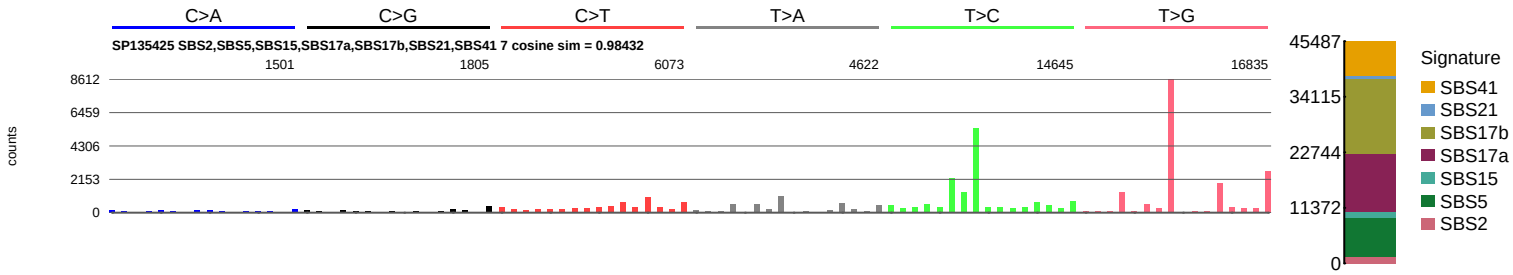

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98433
4869
2205
4880
3635
13353
16543
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2

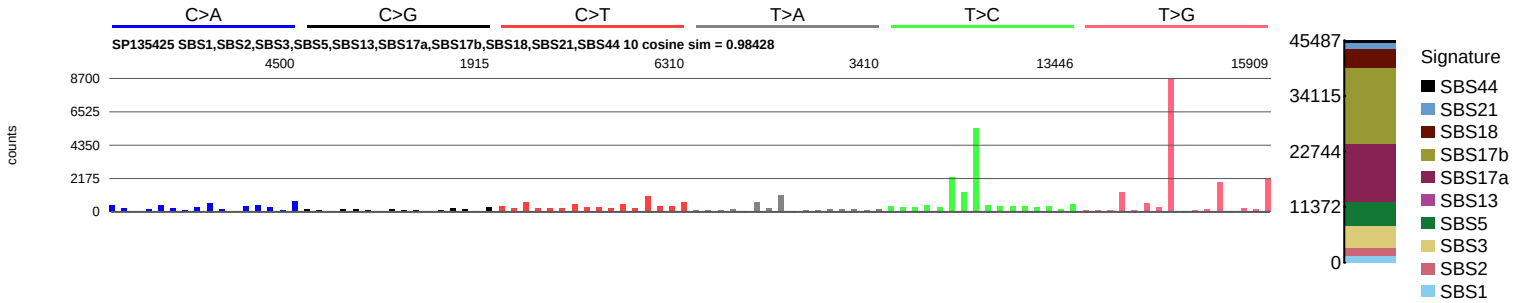

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98433
4890
2219
4842
3647
13325
16562
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2

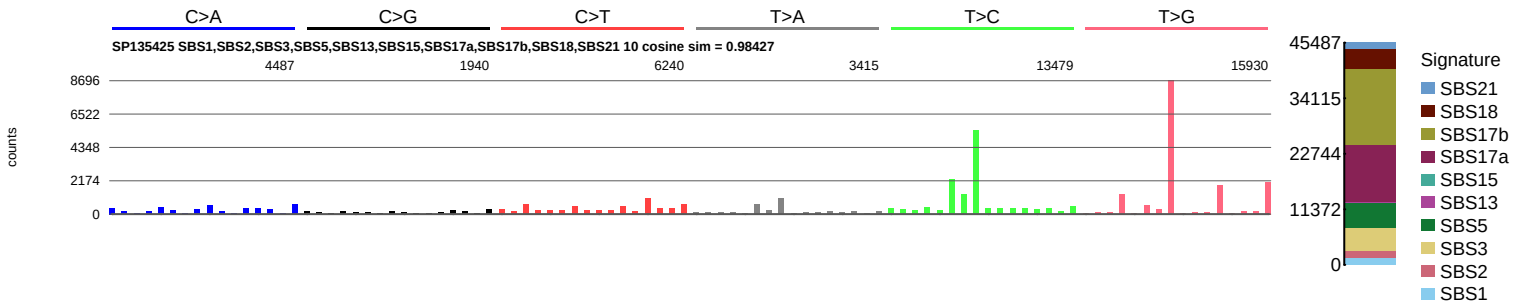

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98432
2015
2497
5351
5031
13691
16899
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2

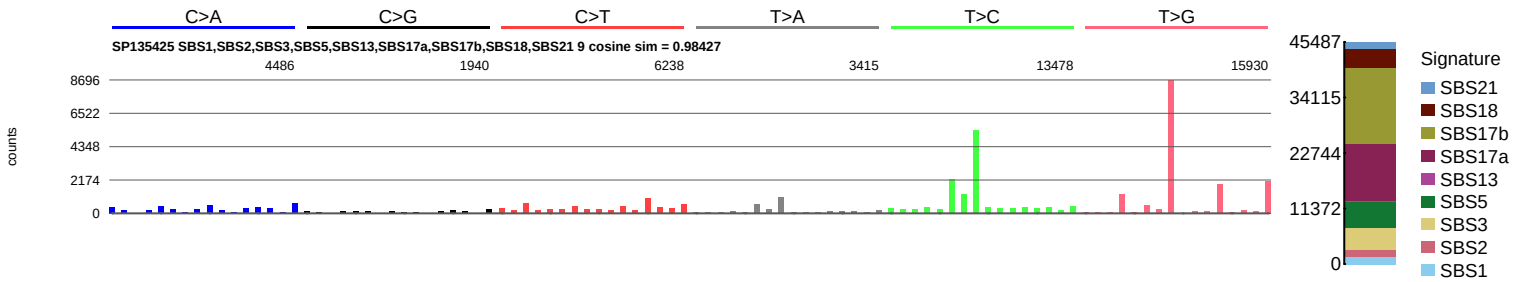

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98432
4887
2207
4832
3635
13369
16557
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2

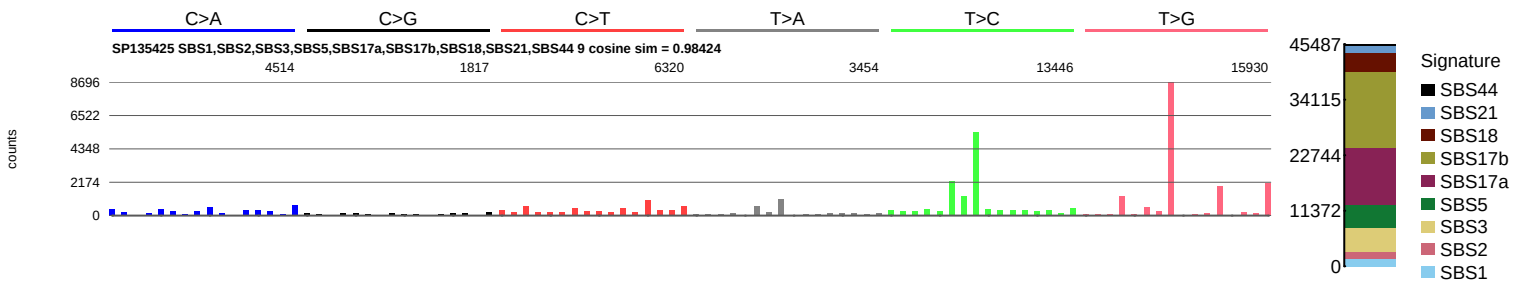

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98432
1504
1813
6071
4620
14643
16831
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

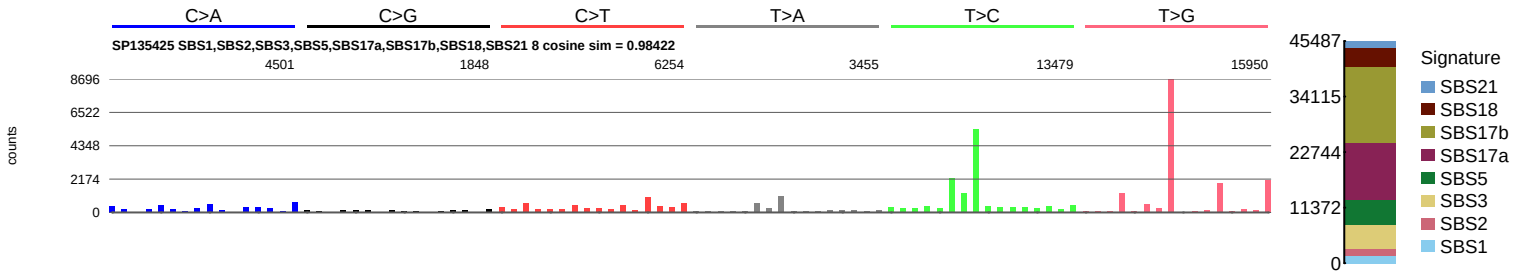

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98432
1501
1805
6073
4622
14645
16835
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98428
4500
1915
6310
3410
13446
15909
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98427
4487
1940
6240
3415
13479
15930
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98427
4486
1940
6238
3415
13478
15930
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98424
4514
1817
6320
3454
13446
15930
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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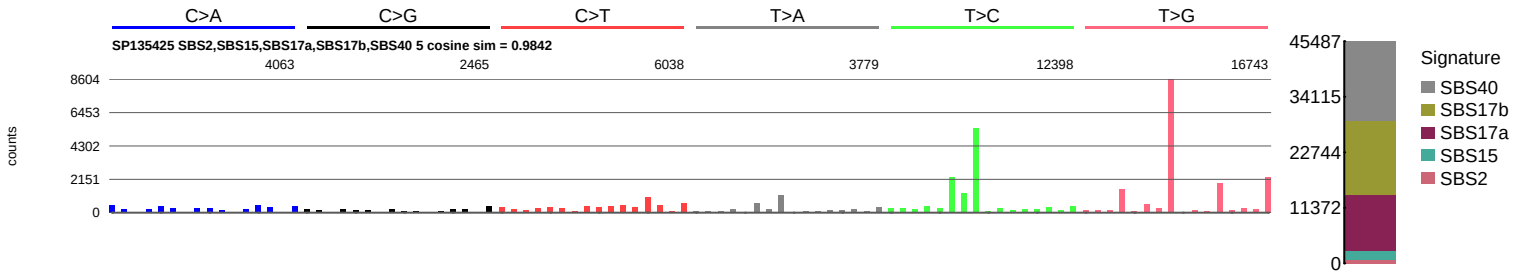
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98422
4501
1848
6254
3455
13479
15950
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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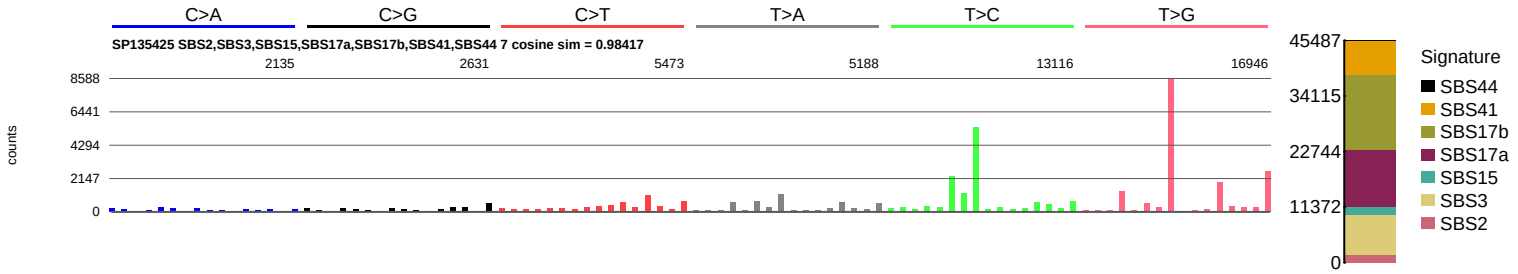
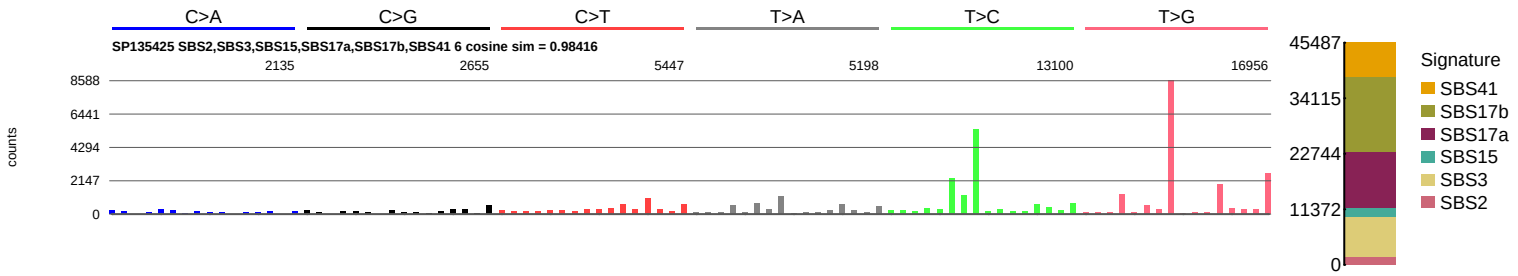
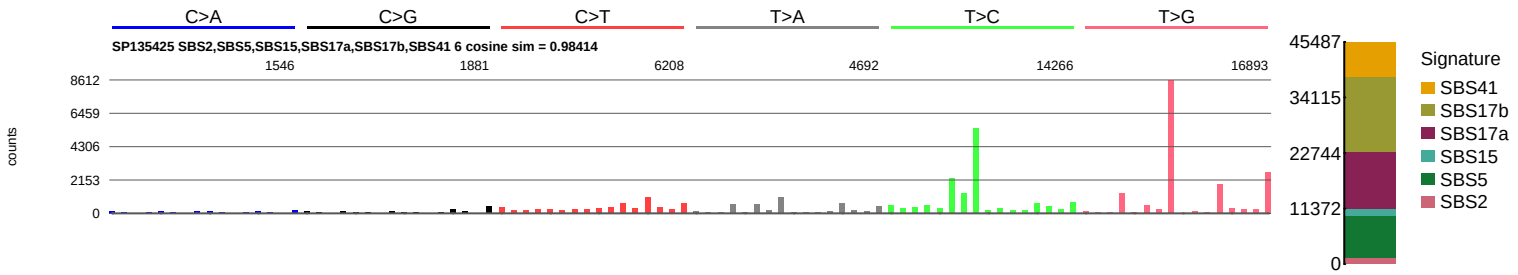
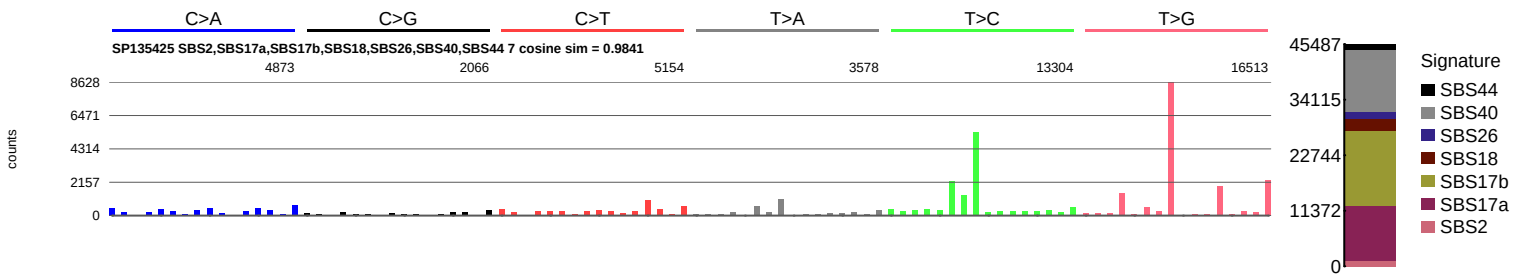
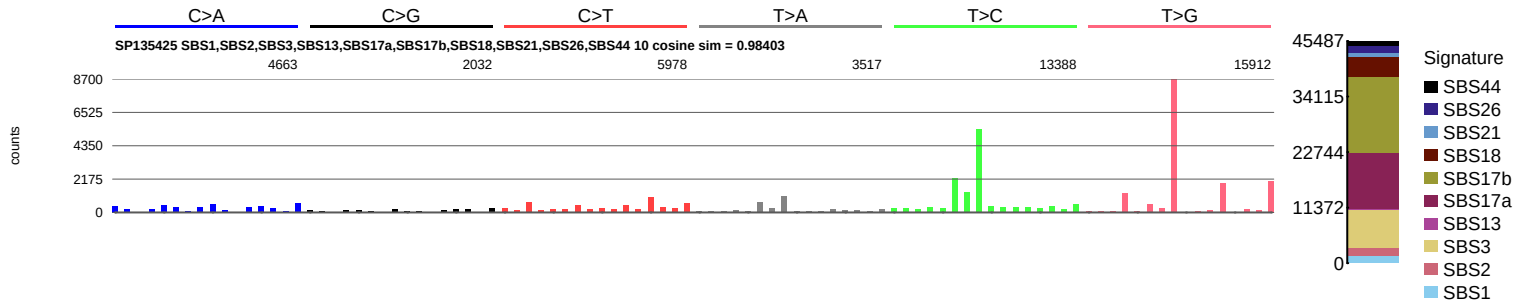
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.9842
4063
2465
6038
3779
12398
16743
8604
6453
4302
2151
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2

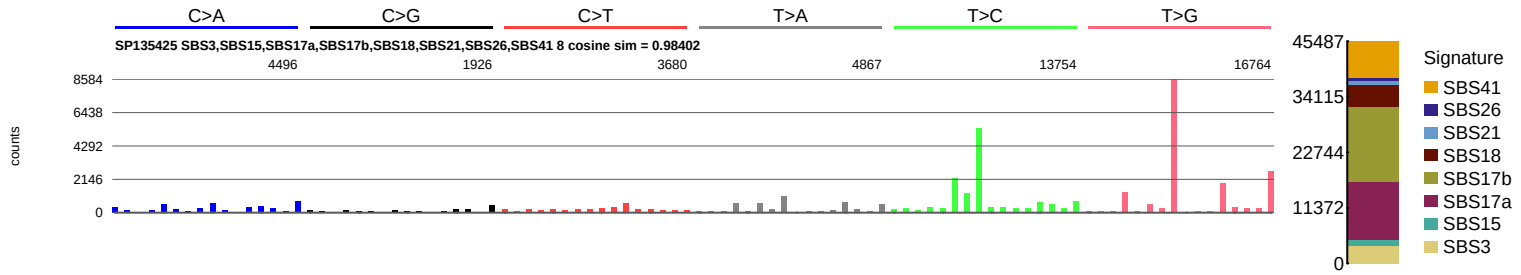

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98417
2135
2631
5473
5188
13116
16946
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2

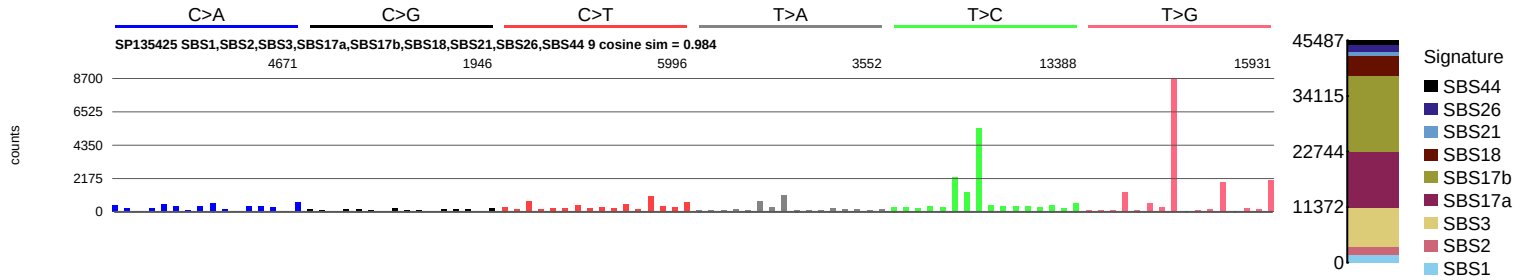

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98416
2135
2655
5447
5198
13100
16956
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2

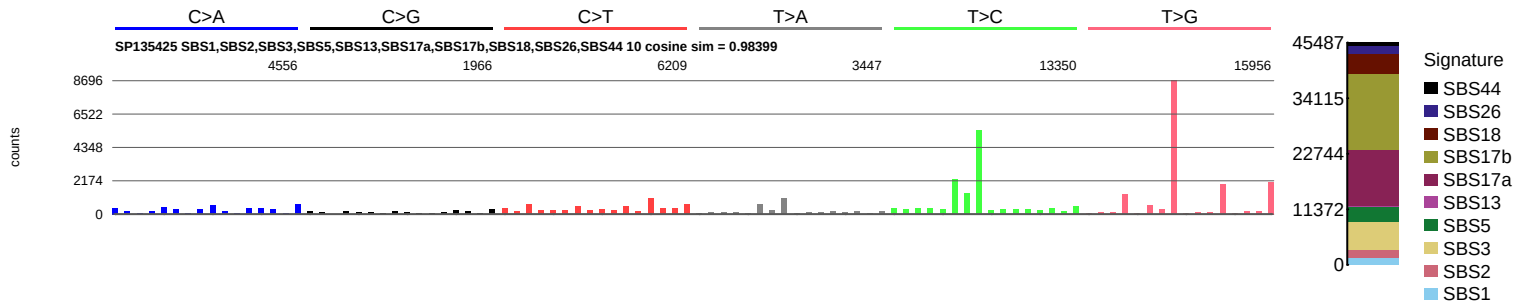

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98414
1546
1881
6208
4692
14266
16893
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.9841
4873
2066
5154
3578
13304
16513
8628
6471
4314
2157
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98403
4663
2032
5978
3517
13388
15912
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98402
4496
1926
3680
4867
13754
16764
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.984
4671
1946
5996
3552
13388
15931
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98399
4556
1966
6209
3447
13350
15956
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS44
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SBS18
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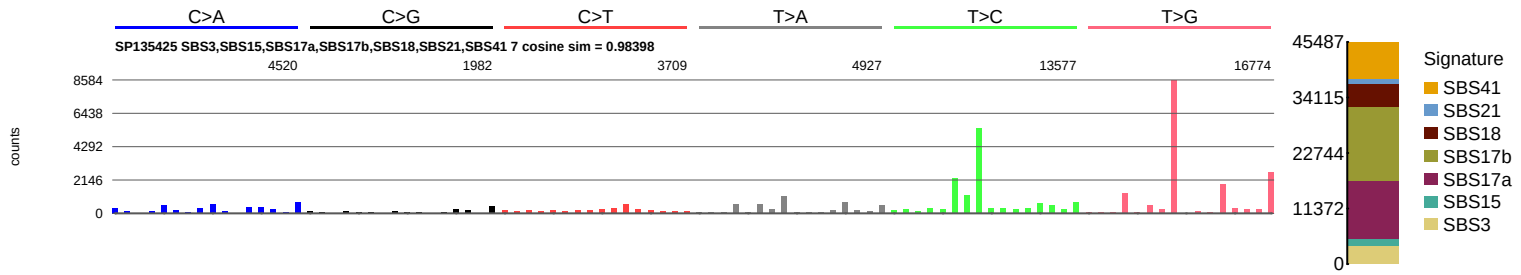
SBS1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C>G
C>T
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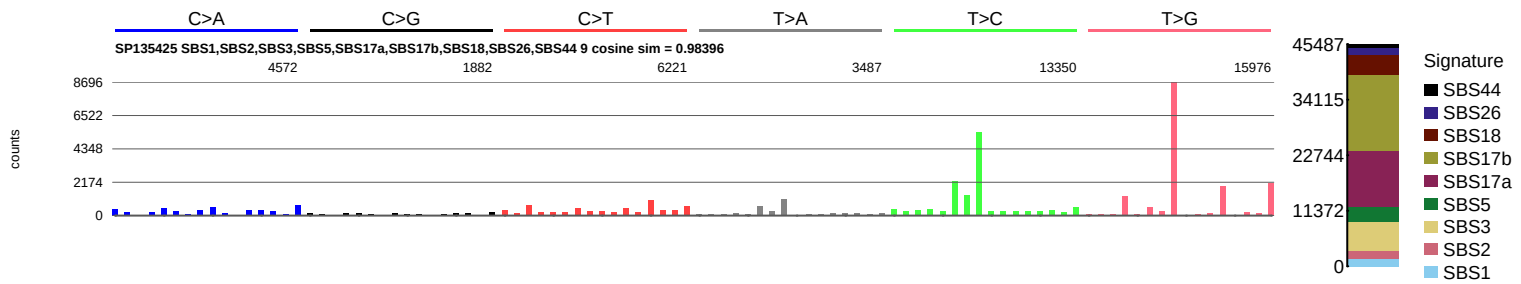
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98398
4520
1982
3709
4927
13577
16774
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3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C>T
T>A
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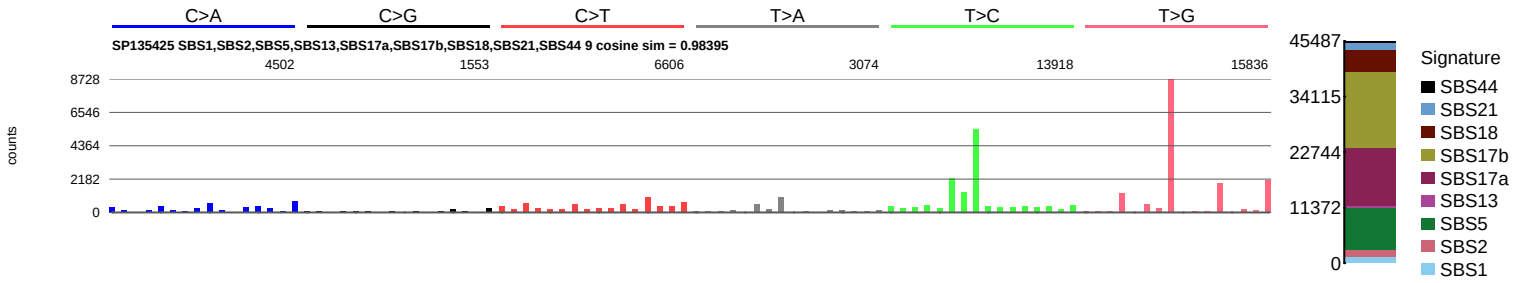
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98396
4572
1882
6221
3487
13350
15976
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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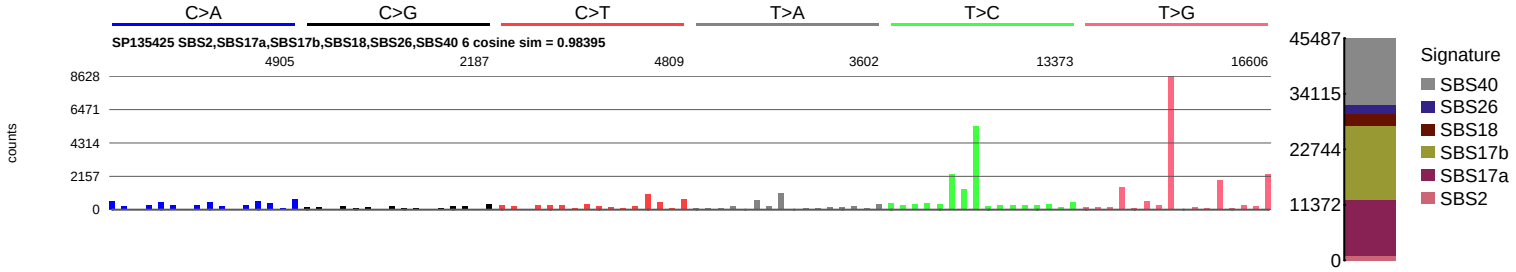
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98395
4502
1553
6606
3074
13918
15836
counts
8728
6546
4364
2182
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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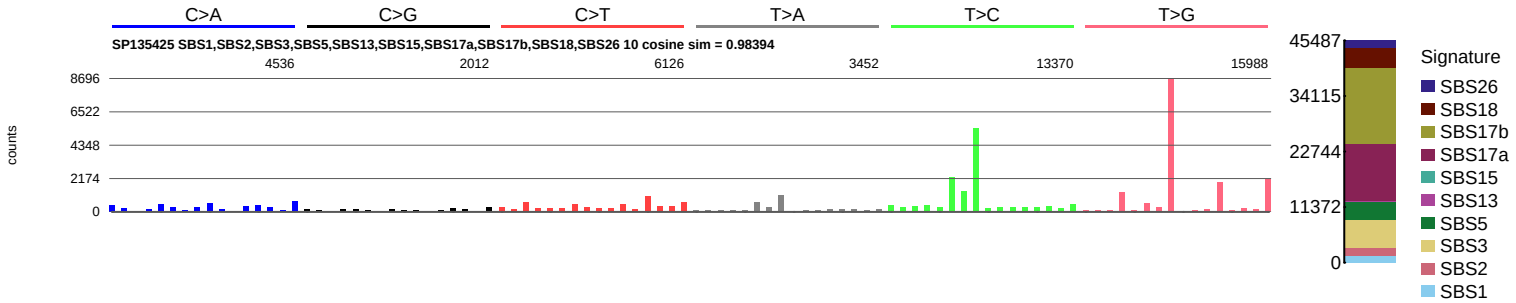
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98395
4905
2187
4809
3602
13373
16606
counts
8628
6471
4314
2157
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
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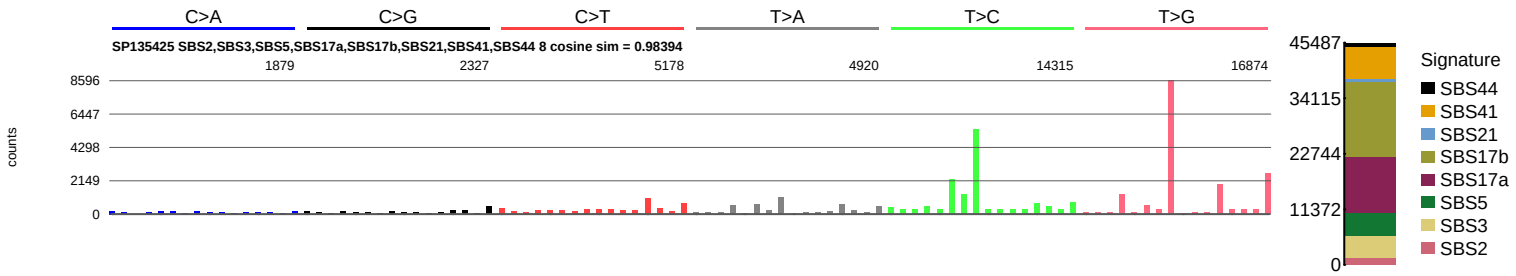
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98394
4536
2012
6126
3452
13370
15988
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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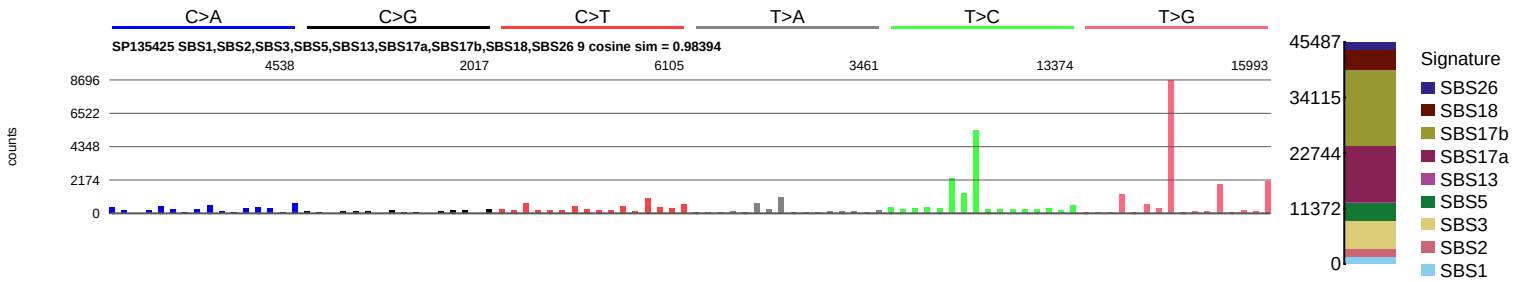
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98394
1879
2327
5178
4920
14315
16874
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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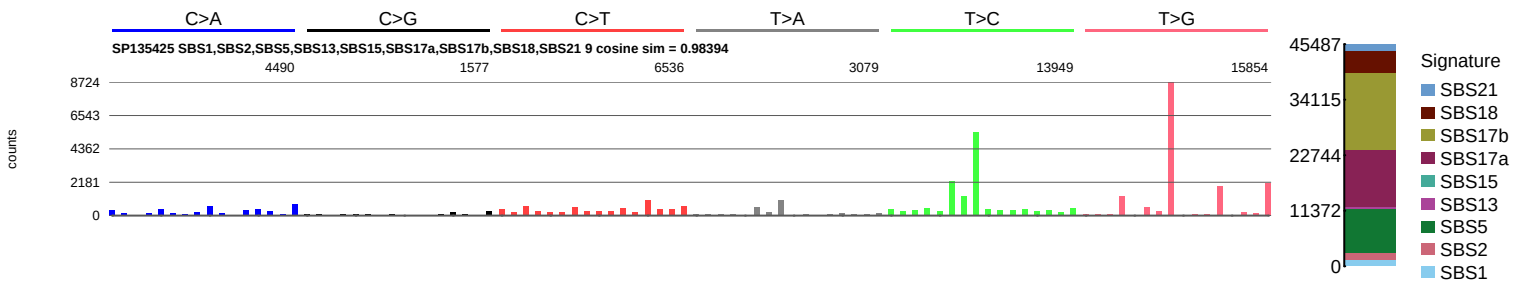
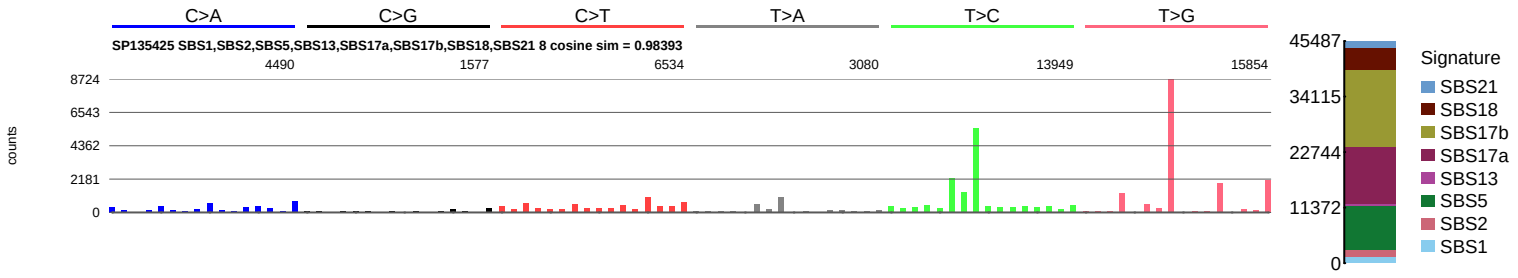
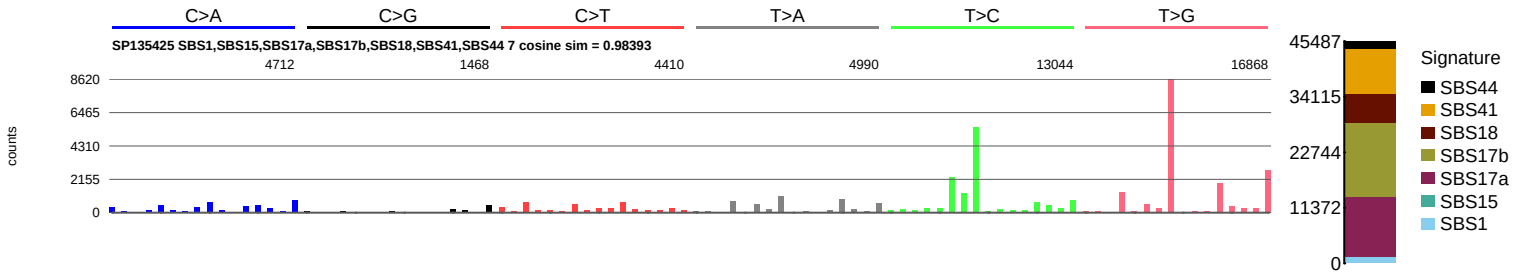
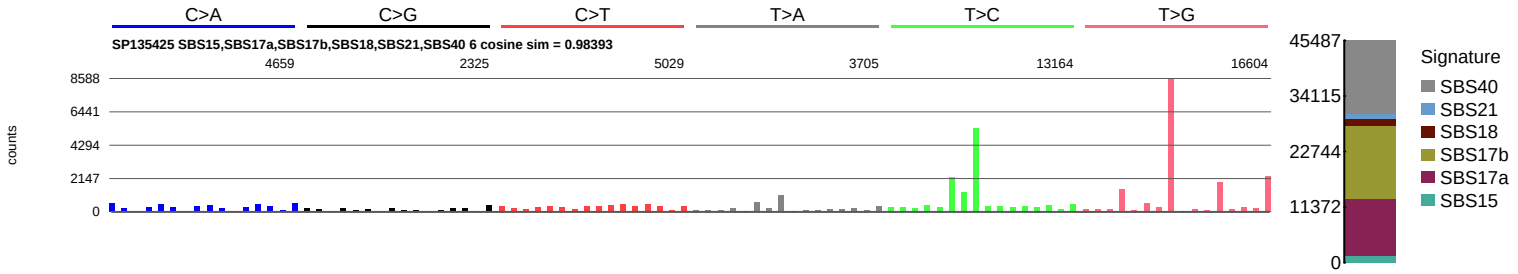
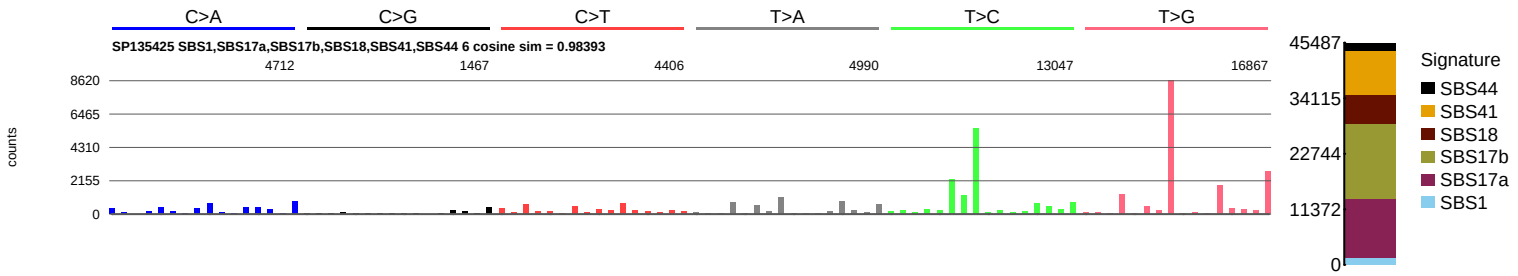
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98394
4538
2017
6105
3461
13374
15993
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

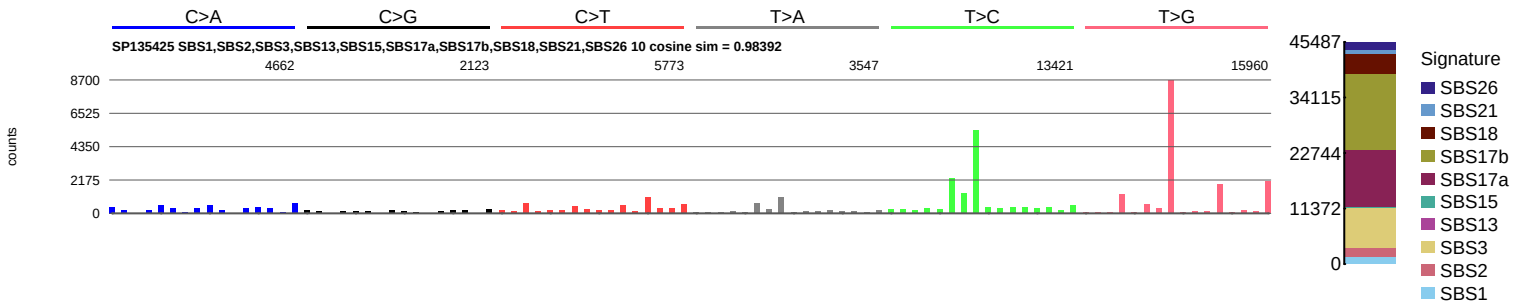

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98394
4490
1577
6536
3079
13949
15854
counts
8724
6543
4362
2181
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98393
4490
1577
6534
3080
13949
15854
counts
8724
6543
4362
2181
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98393
4712
1468
4410
4990
13044
16868
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98393
4659
2325
5029
3705
13164
16604
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98393
4712
1467
4406
4990
13047
16867
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1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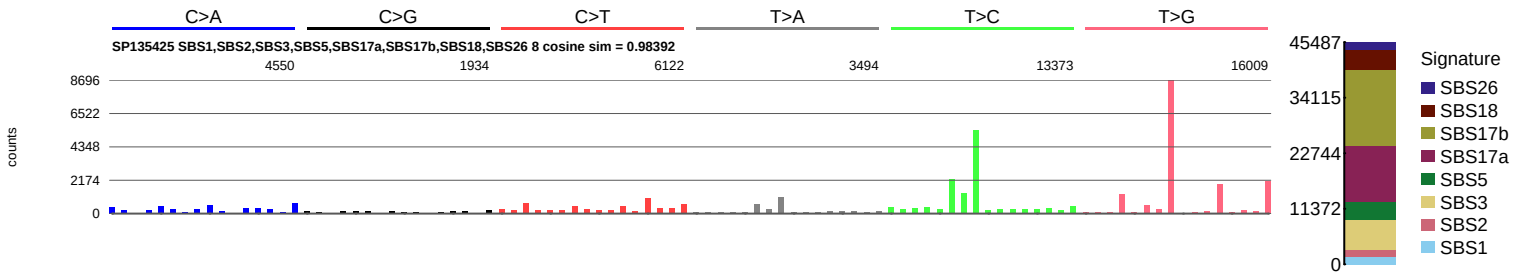
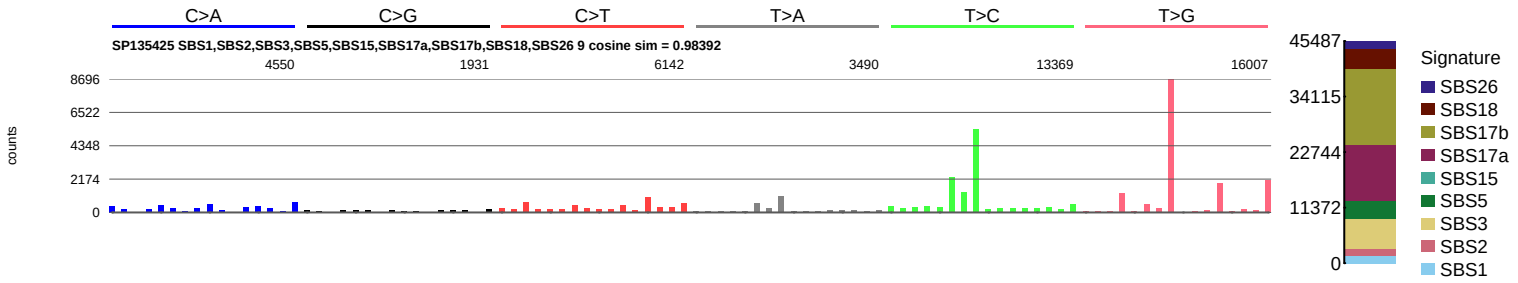
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98392
4662
2123
5773
3547
13421
15960
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98392
4550
1934
6122
3494
13373
16009
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98392
4550
1931
6142
3490
13369
16007
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
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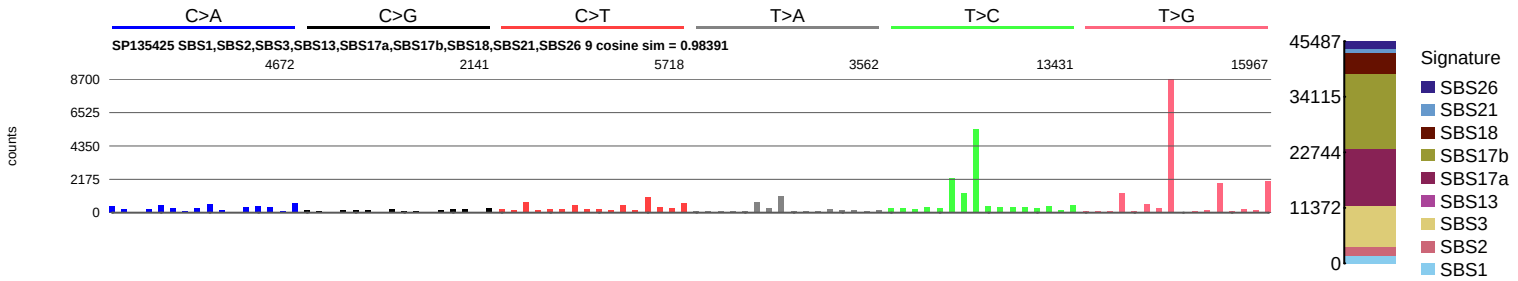
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98391
4672
2141
5718
3562
13431
15967
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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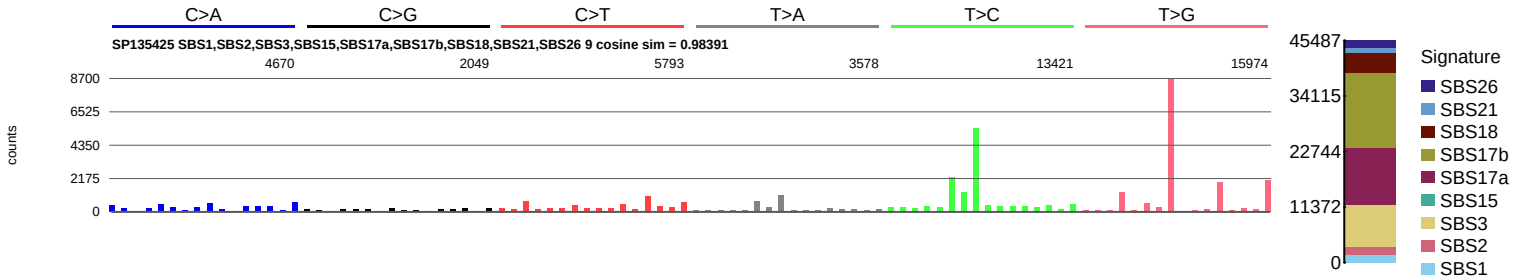
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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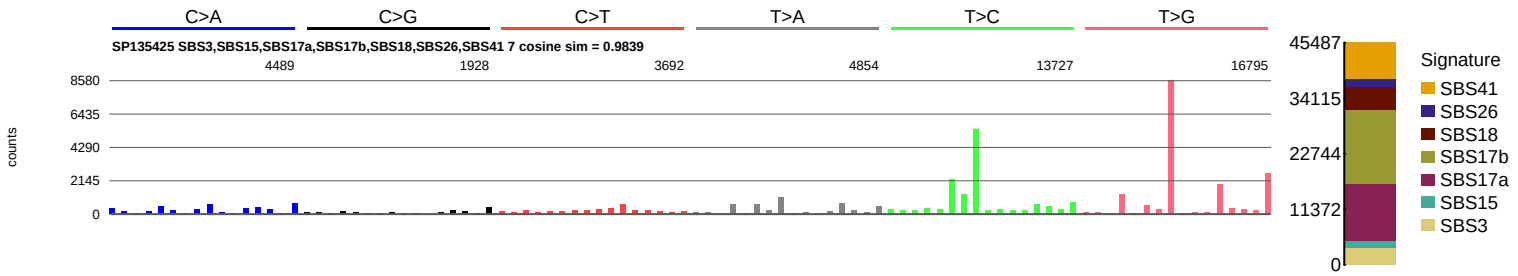
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98391
4670
2049
5793
3578
13421
15974
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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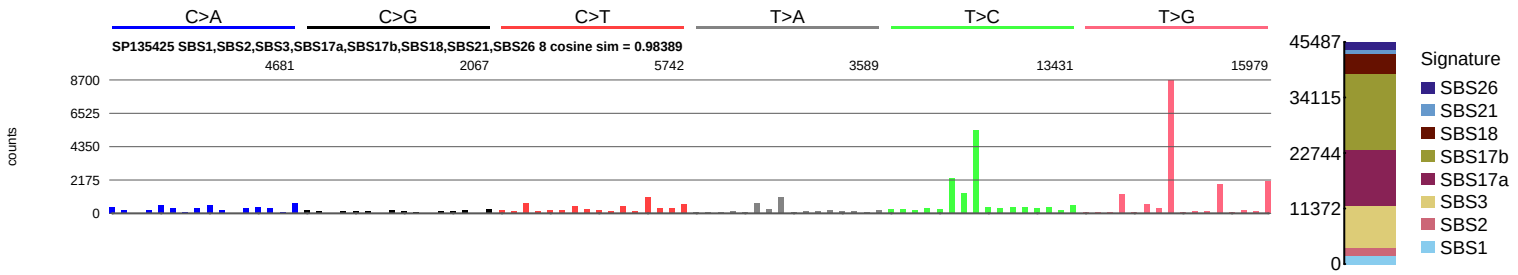
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.9839
4489
1928
3692
4854
13727
16795
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3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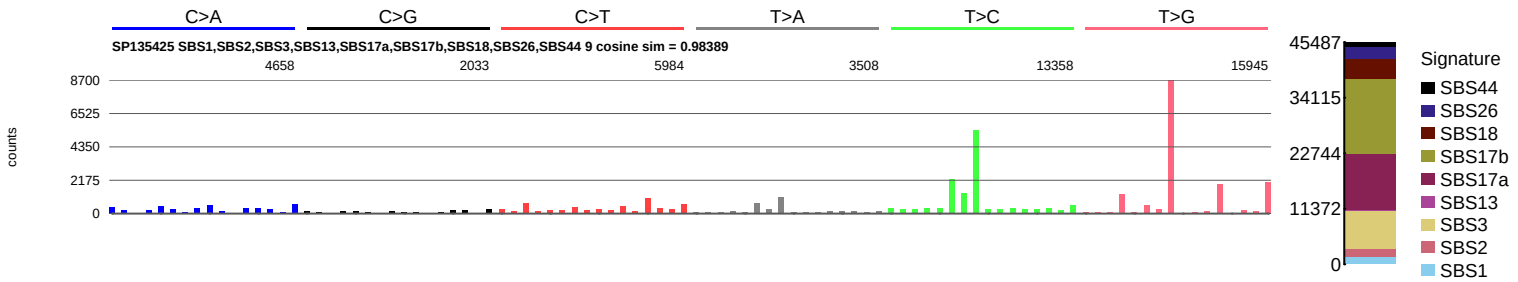
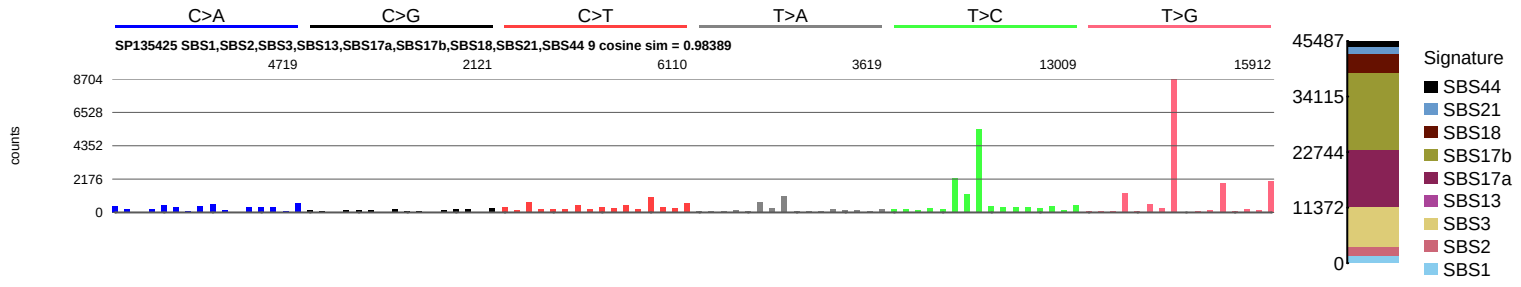
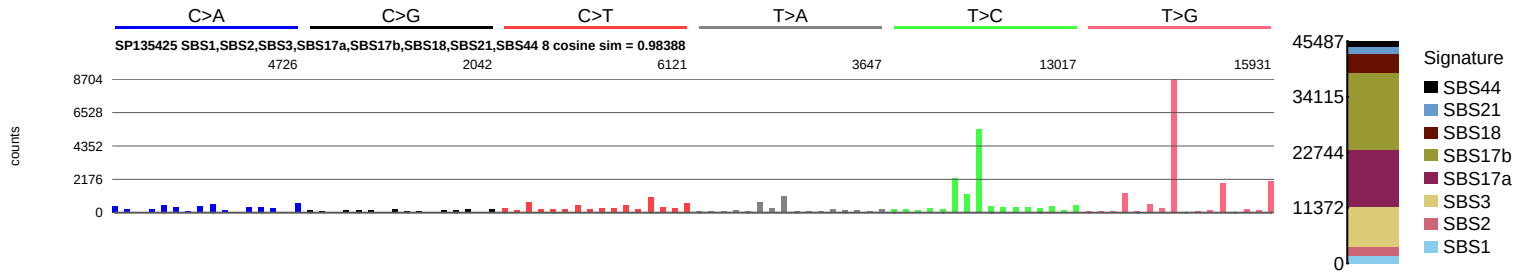
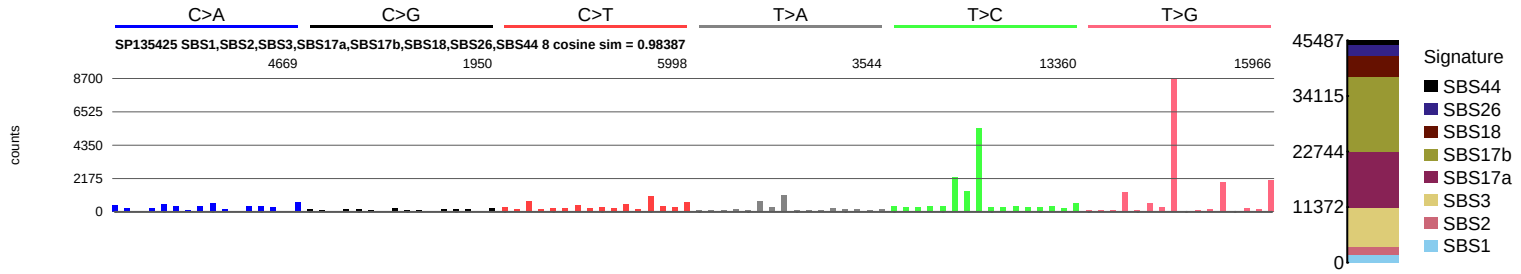
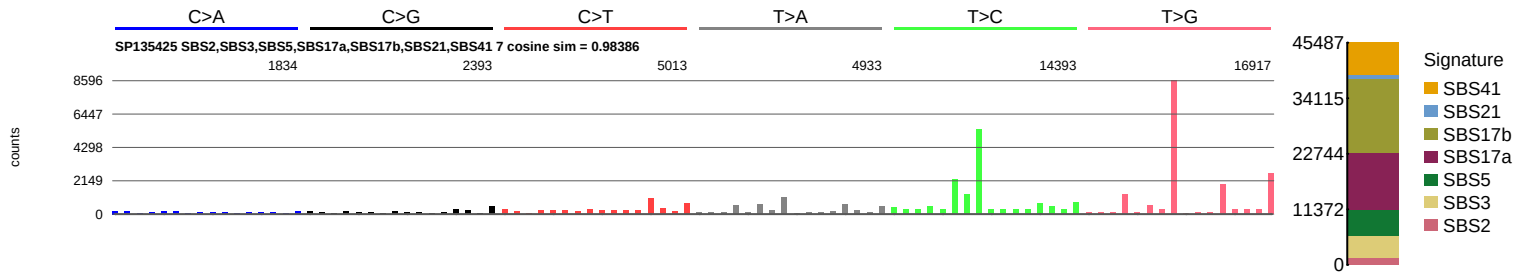
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98389
4681
2067
5742
3589
13431
15979
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

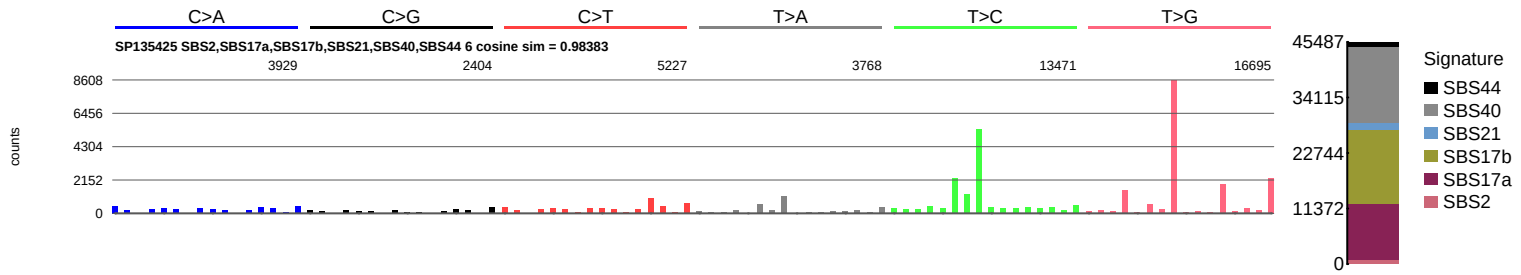

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98389
4658
2033
5984
3508
13358
15945
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98389
4719
2121
6110
3619
13009
15912
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98388
4726
2042
6121
3647
13017
15931
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98387
4669
1950
5998
3544
13360
15966
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98386
1834
2393
5013
4933
14393
16917
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2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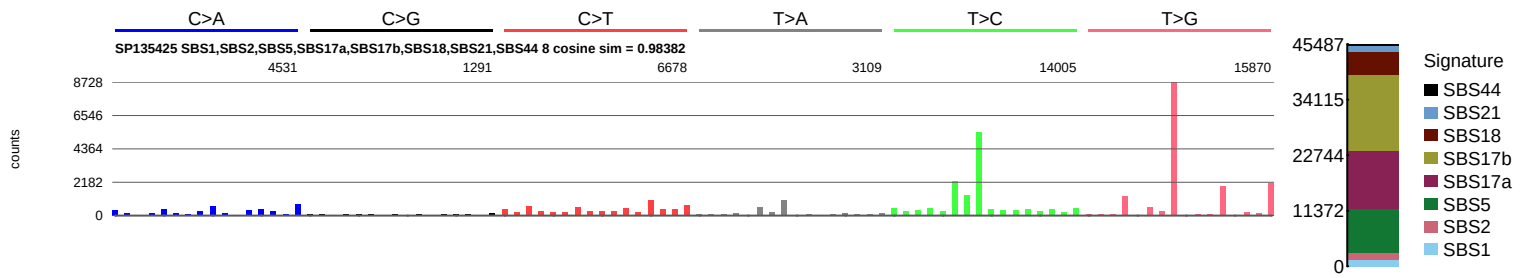
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98383
3929
2404
5227
3768
13471
16695
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2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C>T
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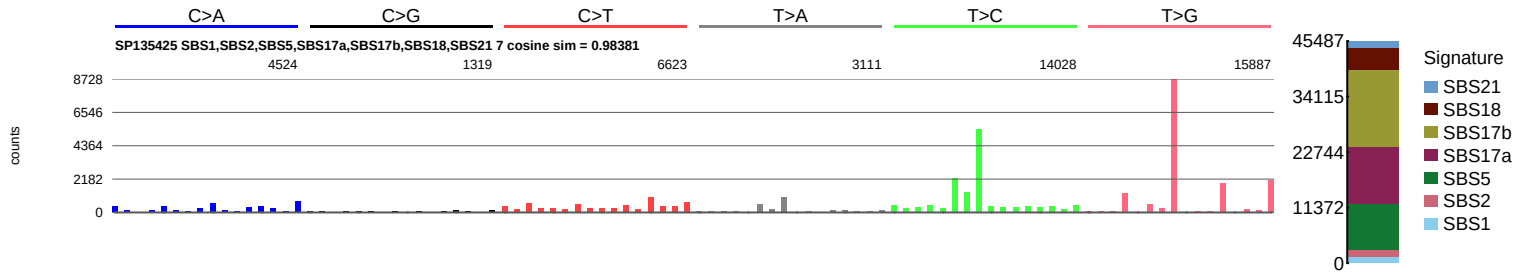
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98382
4531
1291
6678
3109
14005
15870
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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C>G
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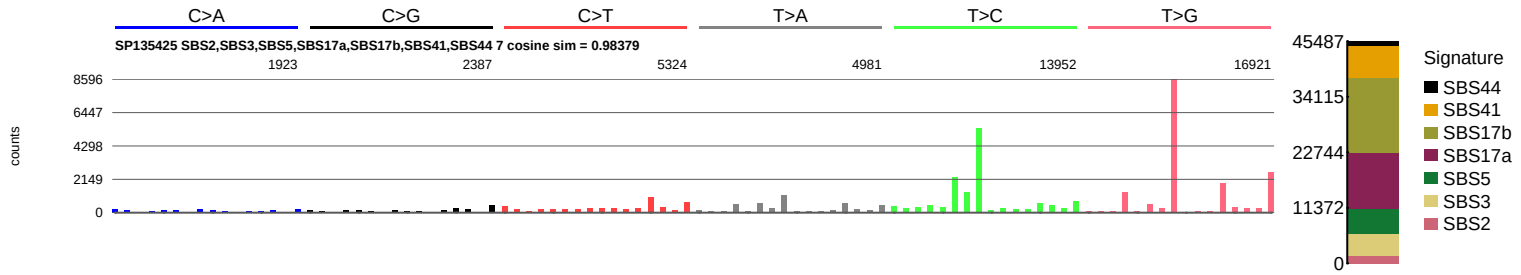
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98381
4524
1319
6623
3111
14028
15887
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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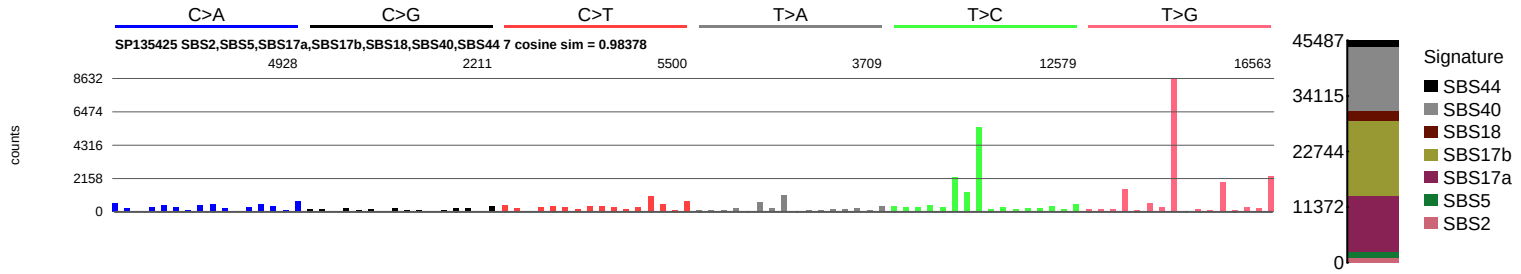
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98379
1923
2387
5324
4981
13952
16921
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2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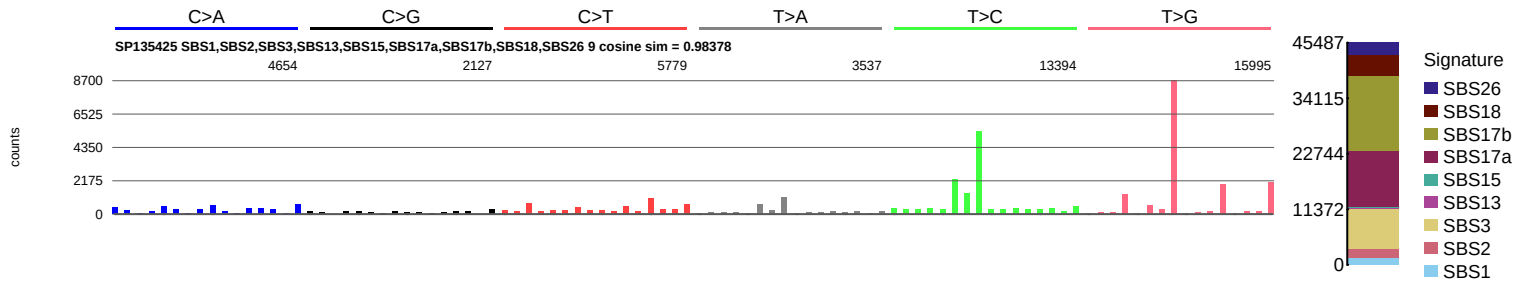
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98378
4928
2211
5500
3709
12579
16563
8632
6474
4316
2158
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2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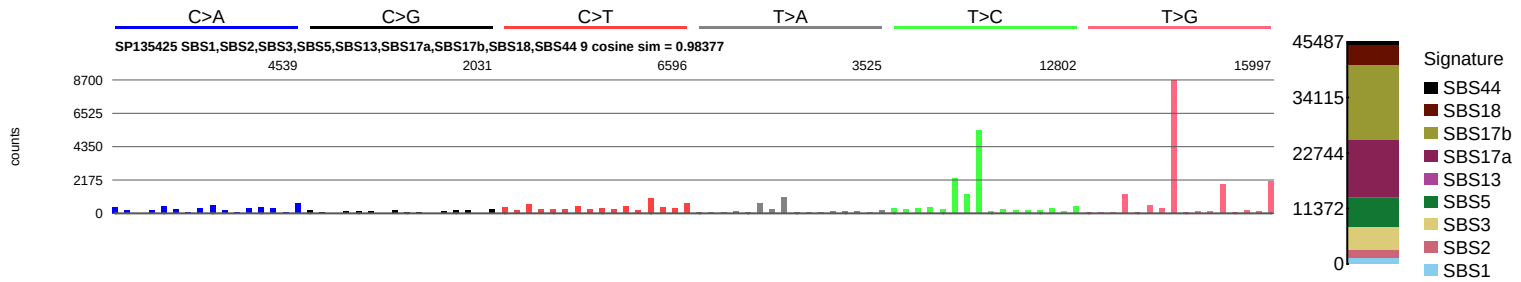
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98378
4654
2127
5779
3537
13394
15995
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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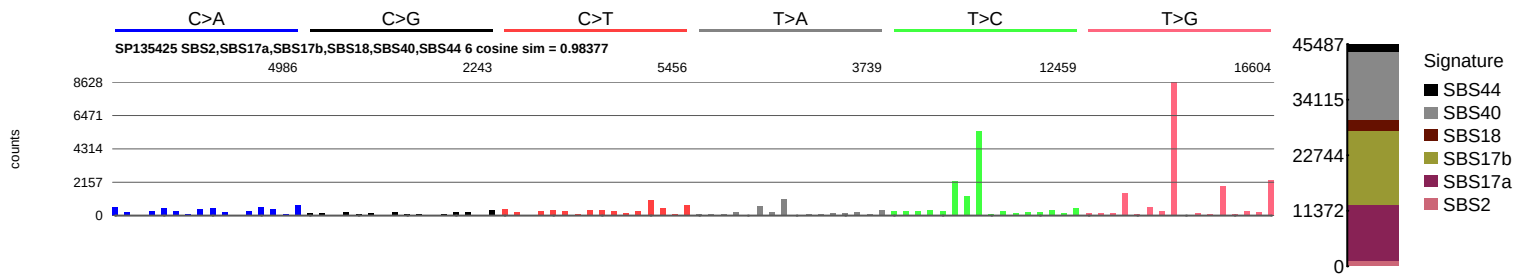
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98377
4539
2031
6596
3525
12802
15997
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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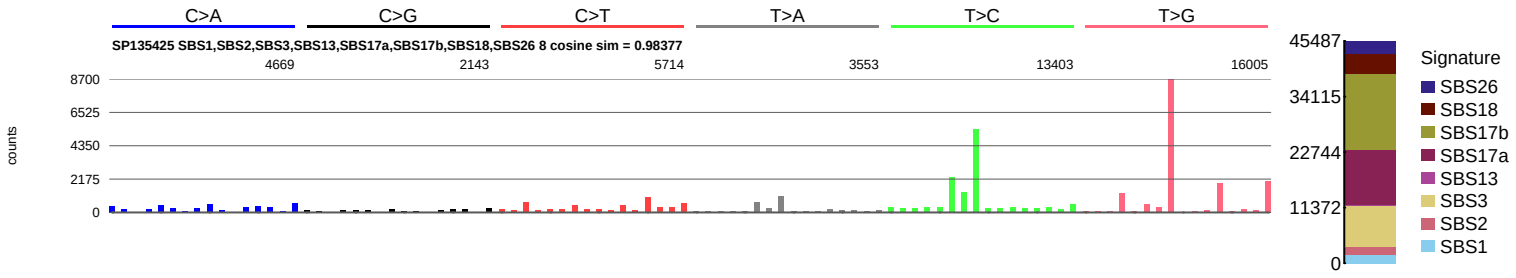
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98377
4986
2243
5456
3739
12459
16604
8628
6471
4314
2157
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98377
4669
2143
5714
3553
13403
16005
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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Signature
SBS26
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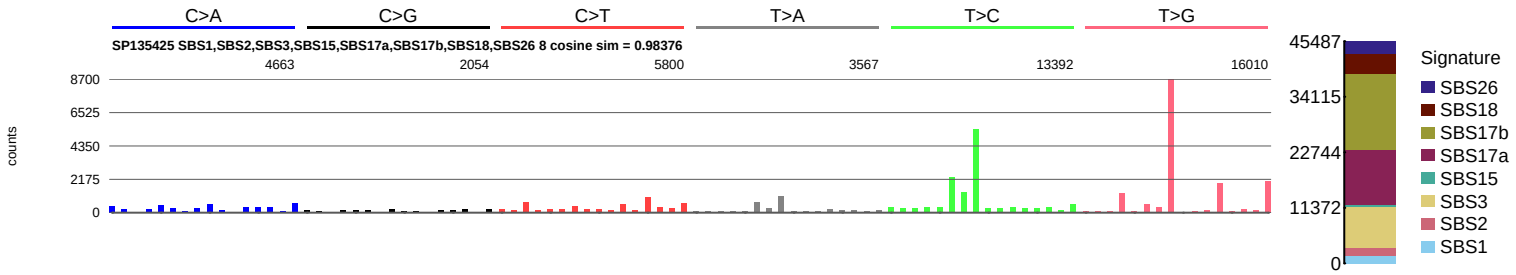
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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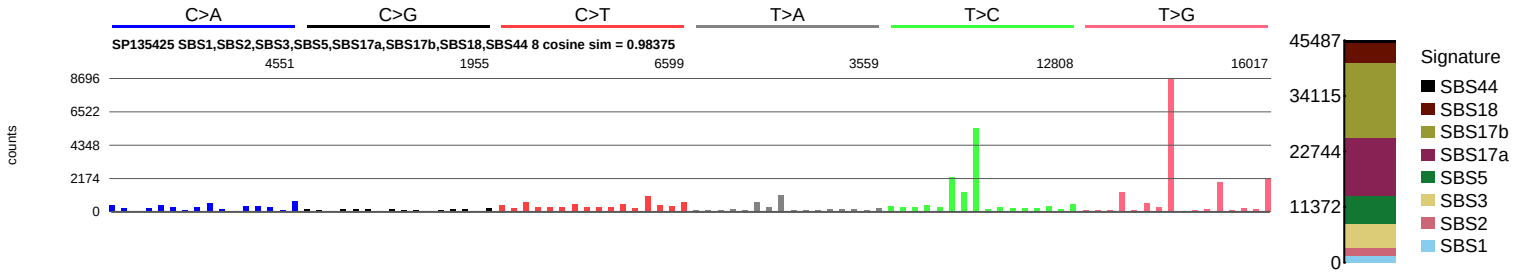
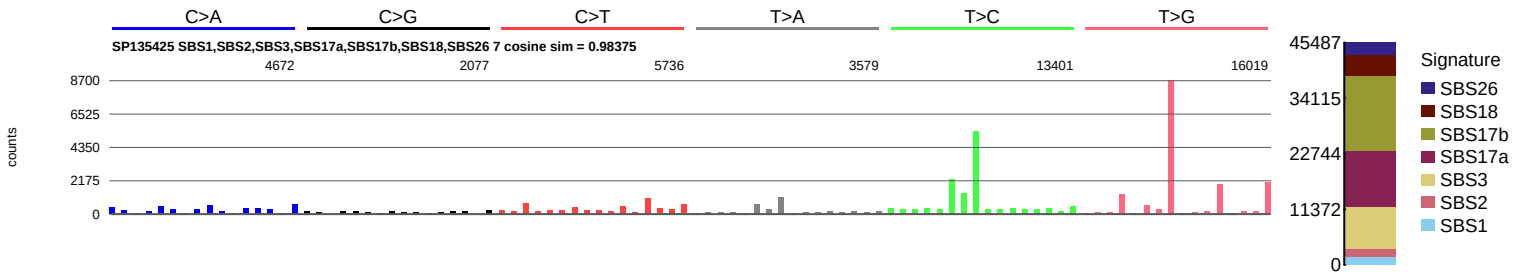
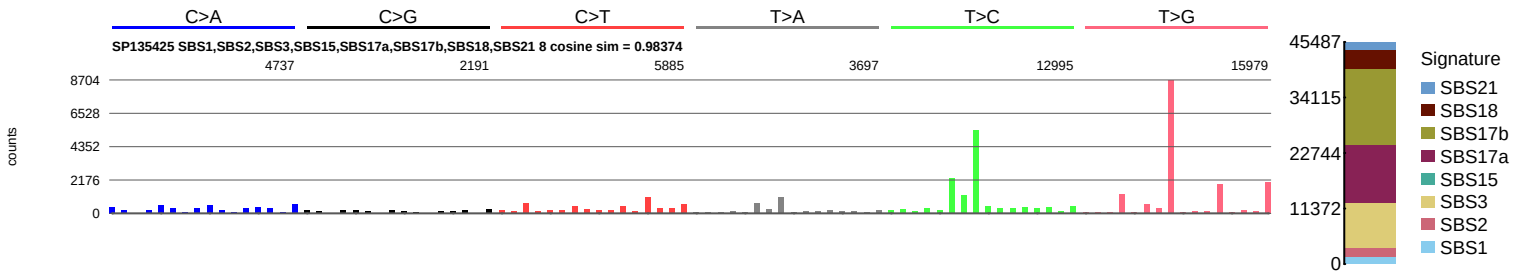
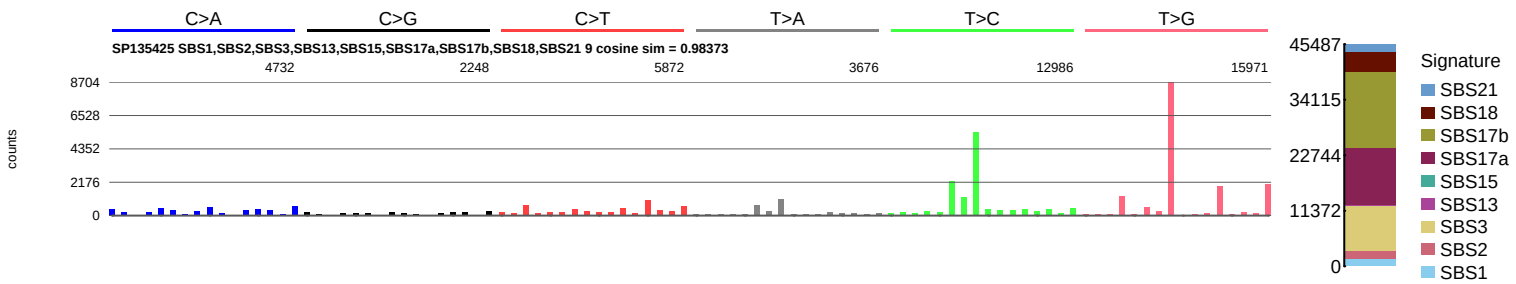
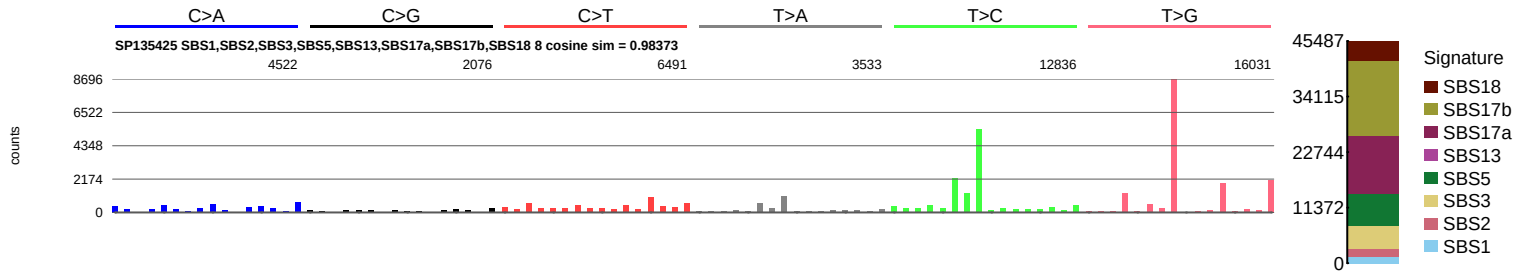
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98376
4663
2054
5800
3567
13392
16010
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

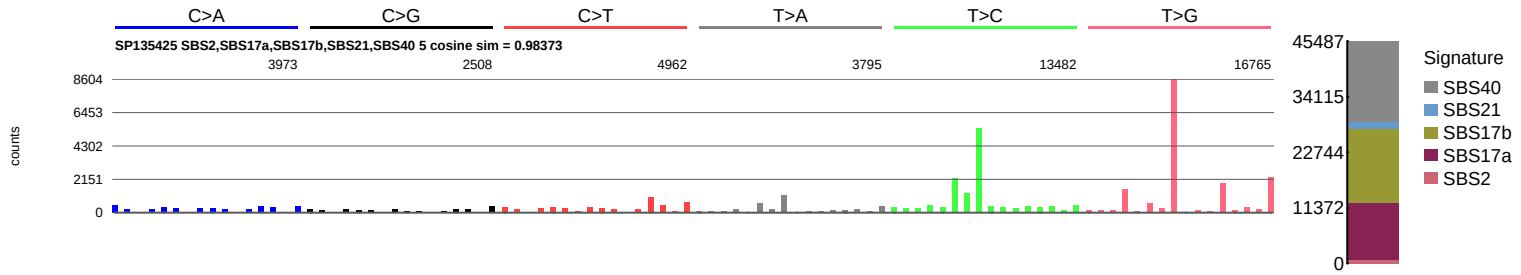

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98375
4551
1955
6599
3559
12808
16017
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98375
4672
2077
5736
3579
13401
16019
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98374
4737
2191
5885
3697
12995
15979
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98373
4732
2248
5872
3676
12986
15971
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98373
4522
2076
6491
3533
12836
16031
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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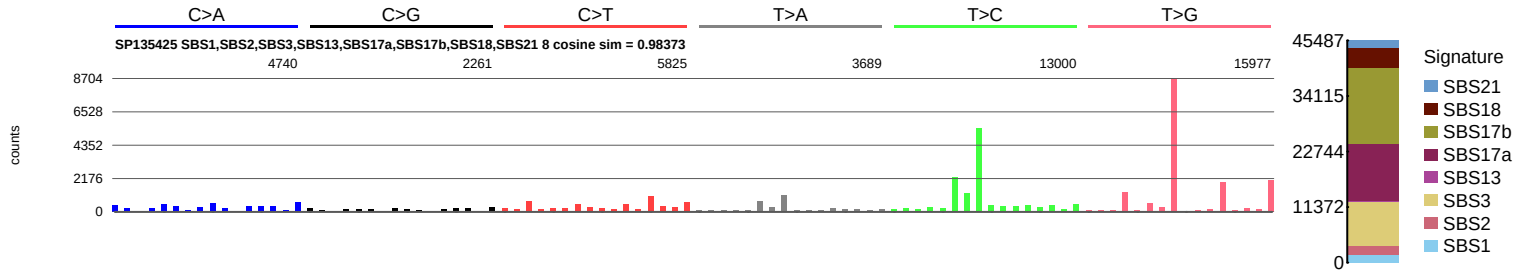
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98373
3973
2508
4962
3795
13482
16765
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2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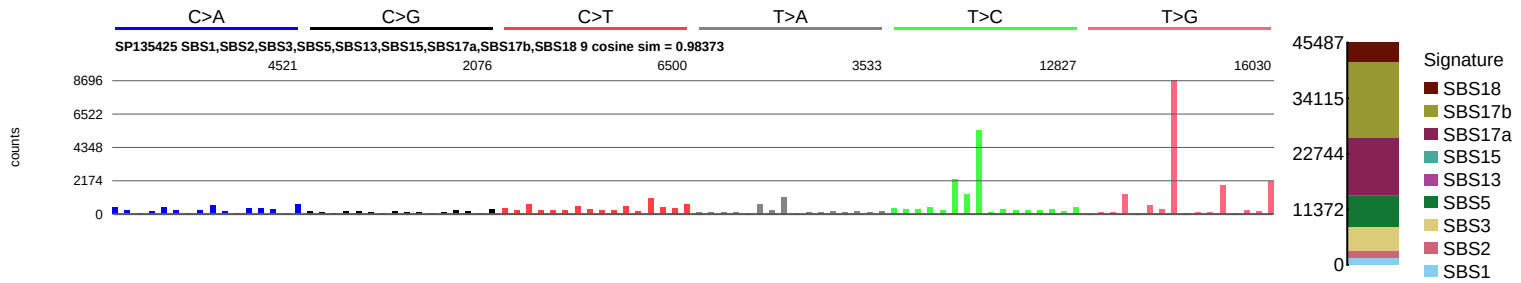
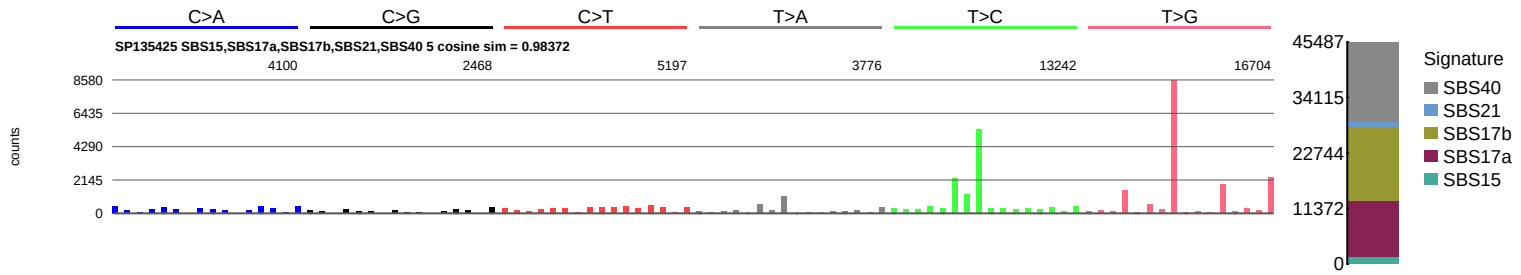
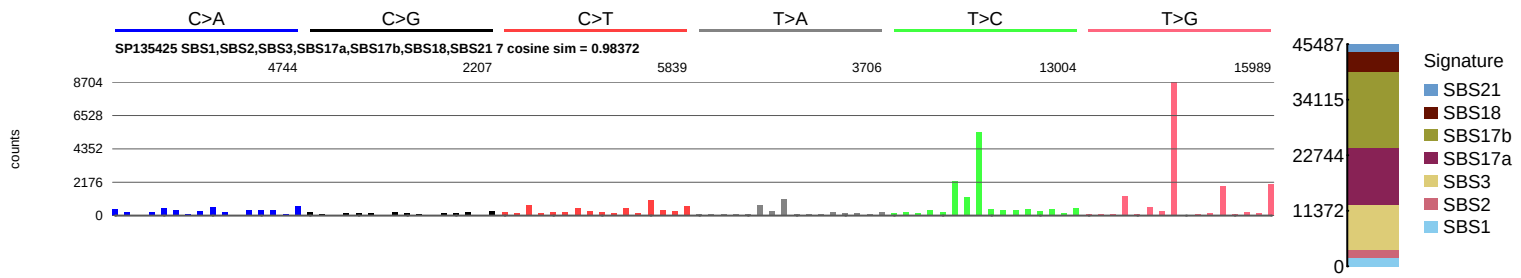
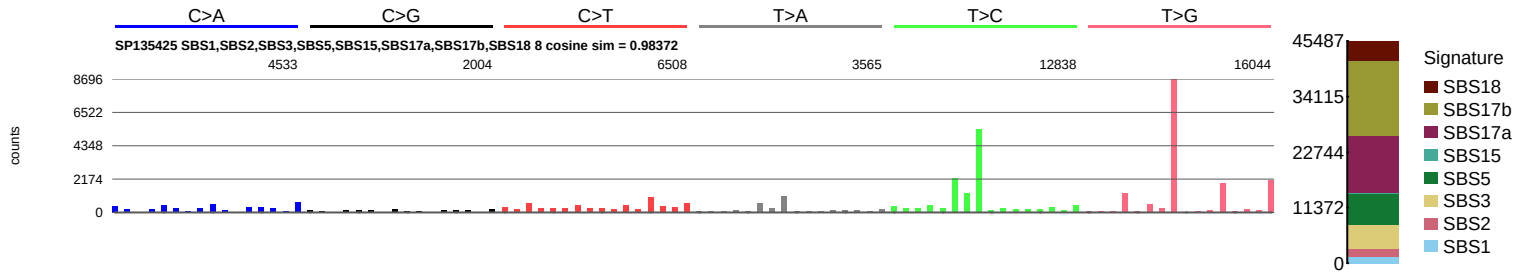
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98373
4740
2261
5825
3689
13000
15977
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98373
4521
2076
6500
3533
12827
16030
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.98372
4100
2468
5197
3776
13242
16704
counts
8580
6435
4290
2145
0
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98372
4744
2207
5839
3706
13004
15989
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98372
4533
2004
6508
3565
12838
16044
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
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Signature
SBS18
SBS17b
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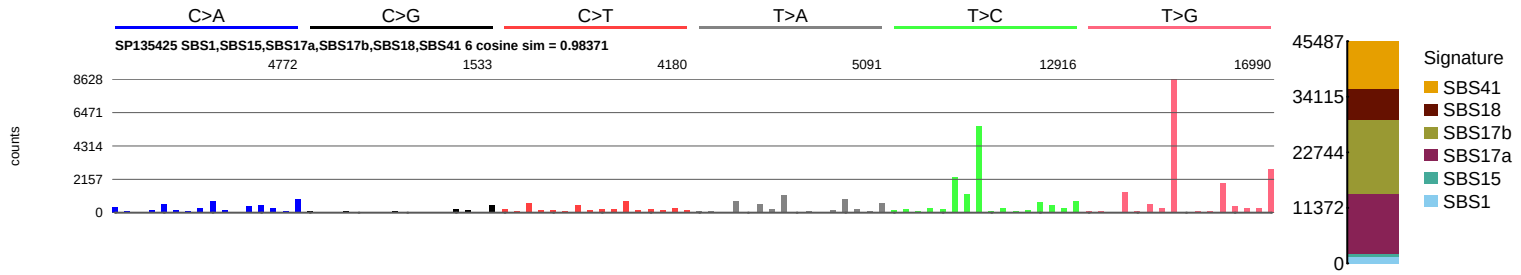
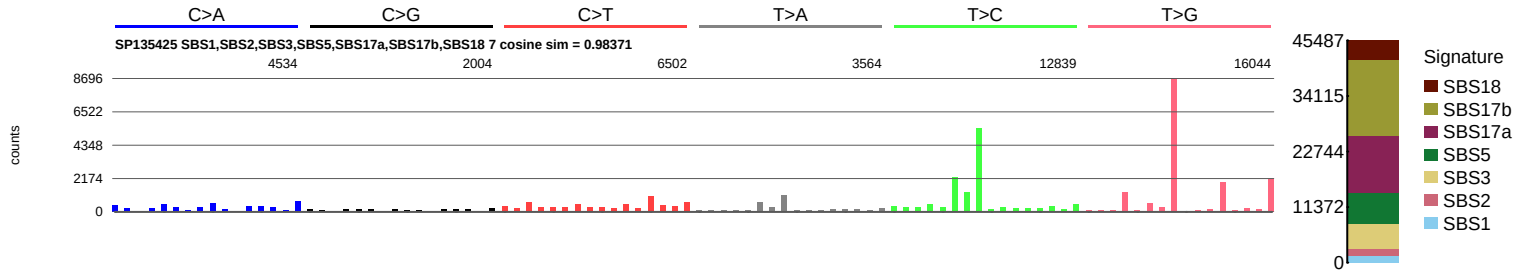
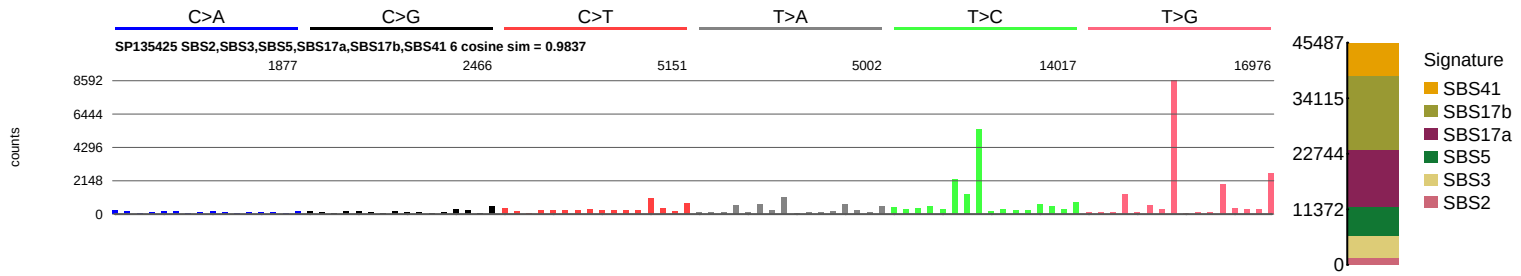
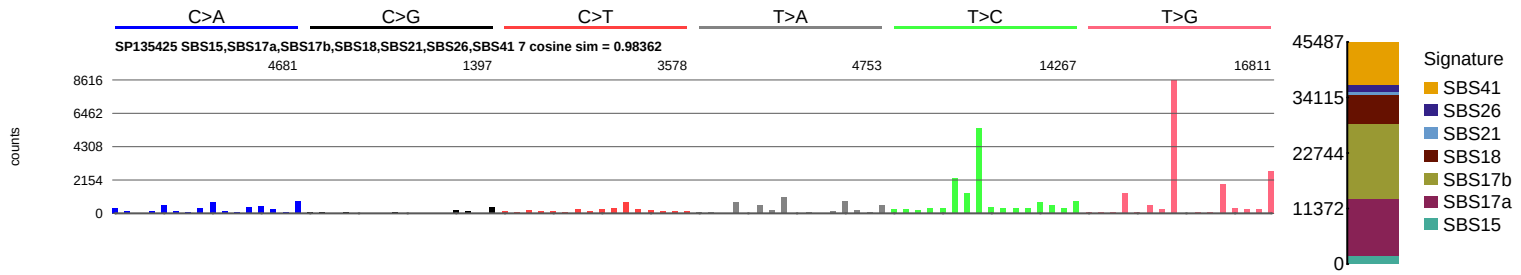
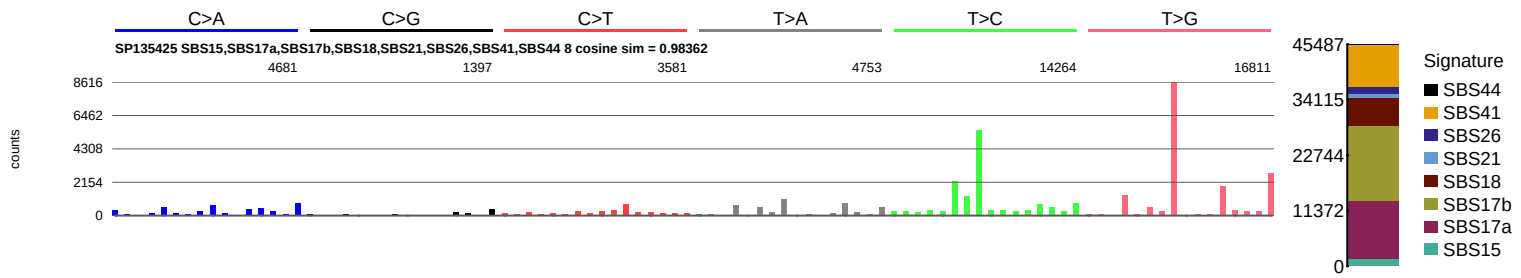
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98371
4772
1533
4180
5091
12916
16990
counts
8628
6471
4314
2157
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98371
4534
2004
6502
3564
12839
16044
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.9837
1877
2466
5151
5002
14017
16976
counts
8592
6444
4296
2148
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98362
4681
1397
3578
4753
14267
16811
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15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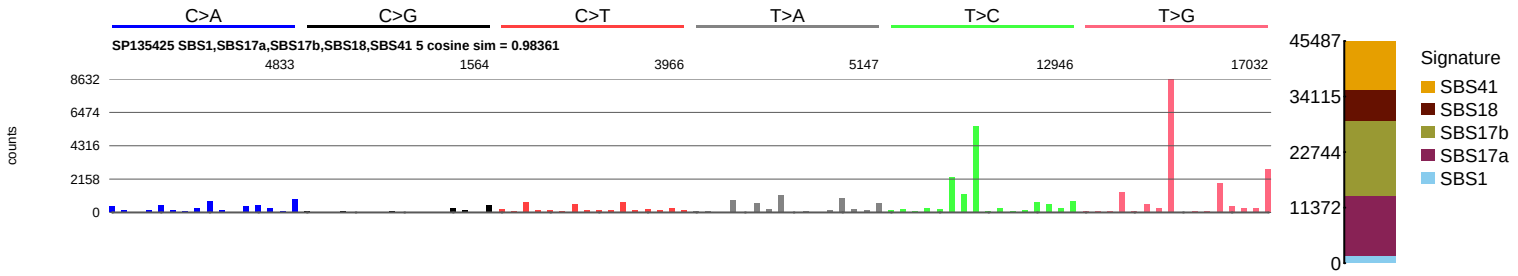
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98362
4681
1397
3581
4753
14264
16811
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15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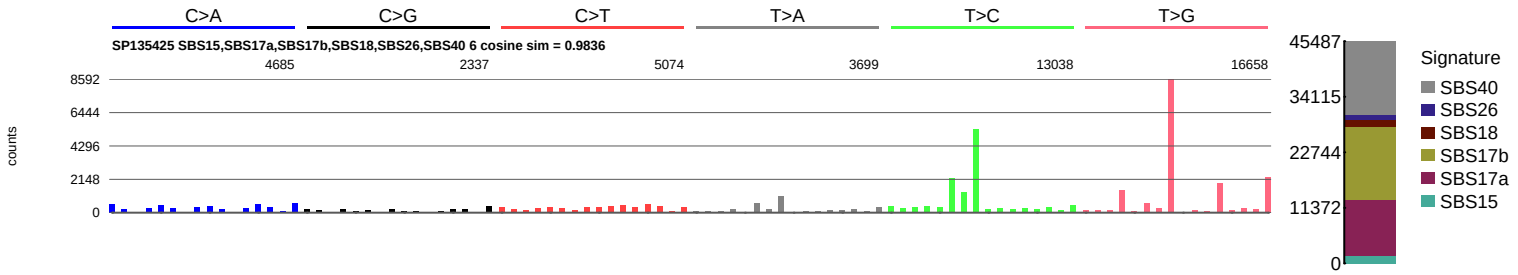
T>C
T>G
SP135425 SBS1,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98361
4833
1564
3966
5147
12946
17032
8632
6474
4316
2158
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
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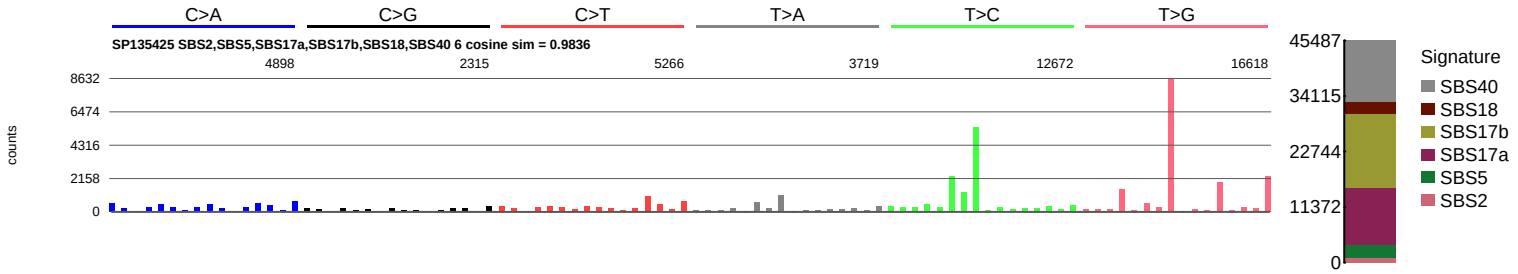
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.9836
4685
2337
5074
3699
13038
16658
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15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C>G
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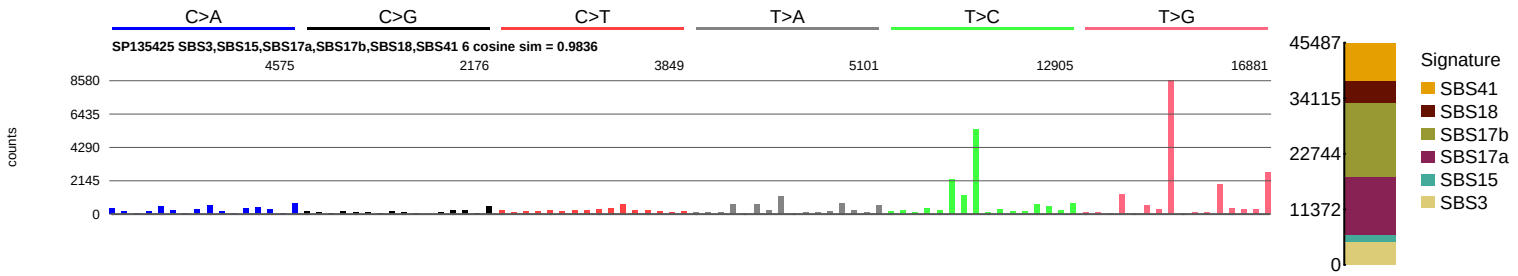
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.9836
4898
2315
5266
3719
12672
16618
8632
6474
4316
2158
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2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C>G
C>T
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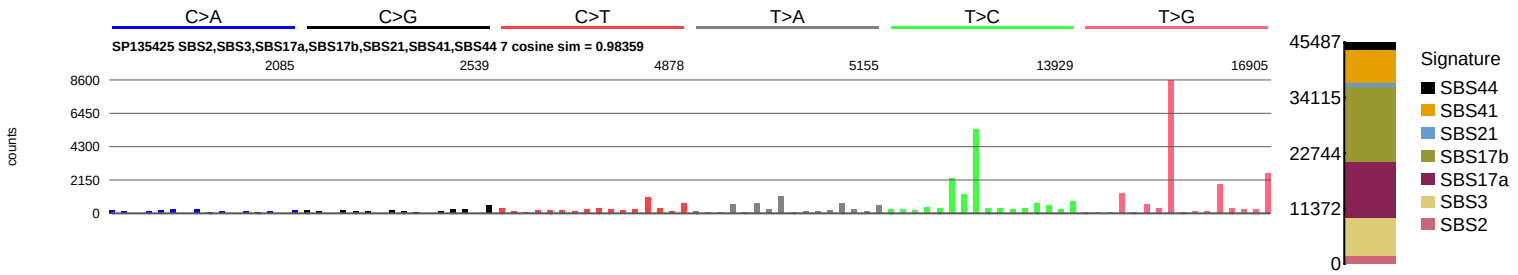
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.9836
4575
2176
3849
5101
12905
16881
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3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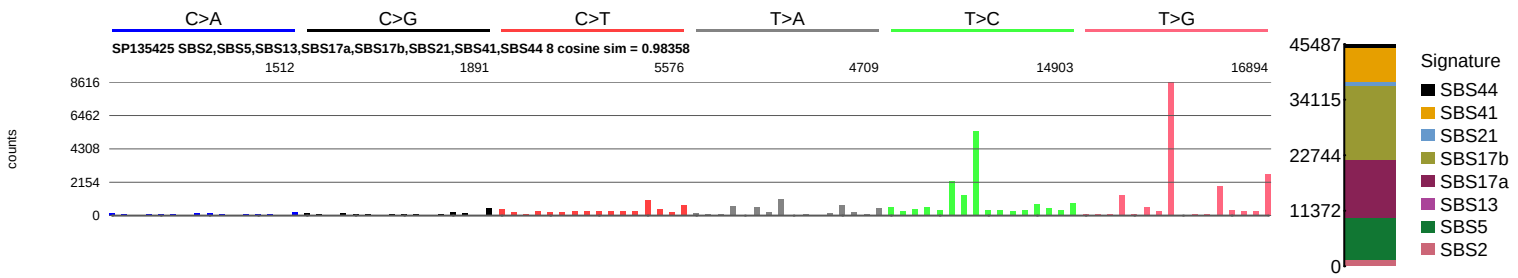
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98359
2085
2539
4878
5155
13929
16905
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2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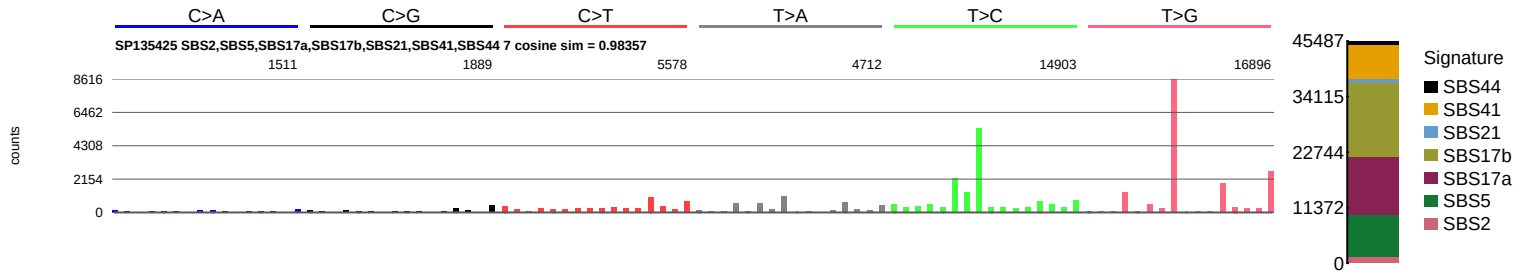
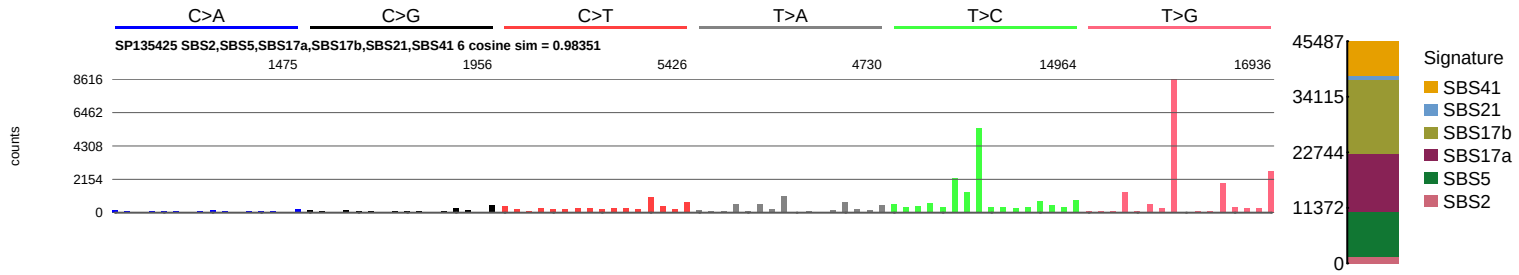
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98358
1512
1891
5576
4709
14903
16894
8616
6462
4308
2154
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98357
1511
1889
5578
4712
14903
16896
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2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C>G
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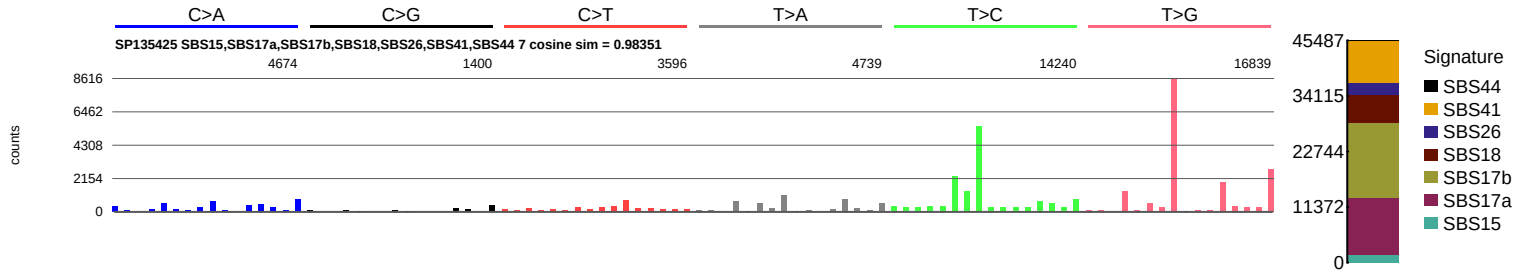
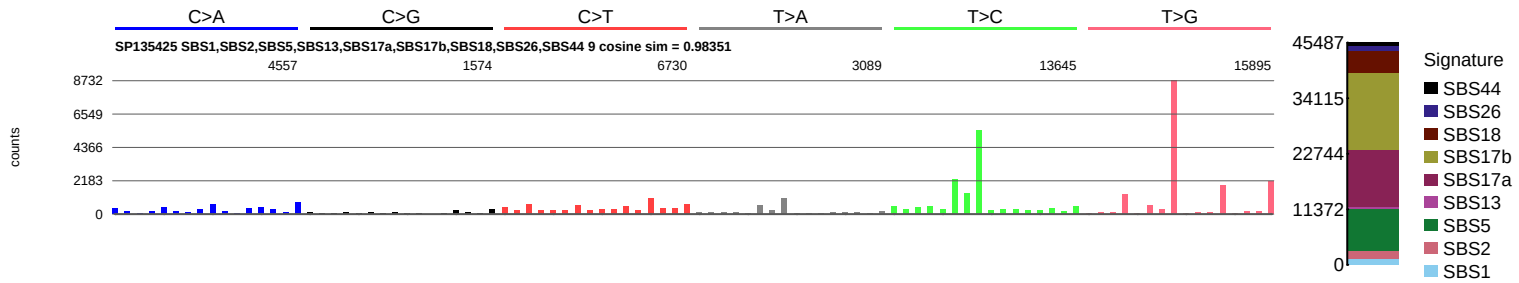
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98351
1475
1956
5426
4730
14964
16936
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98351
4674
1400
3596
4739
14240
16839
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15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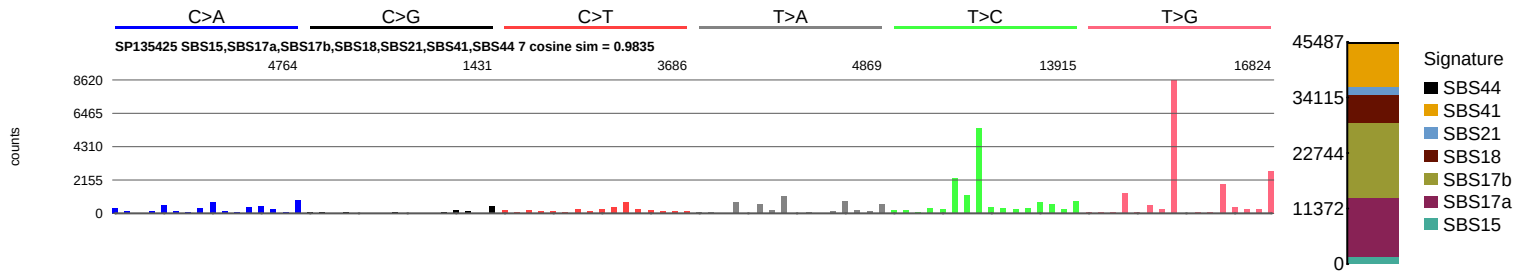
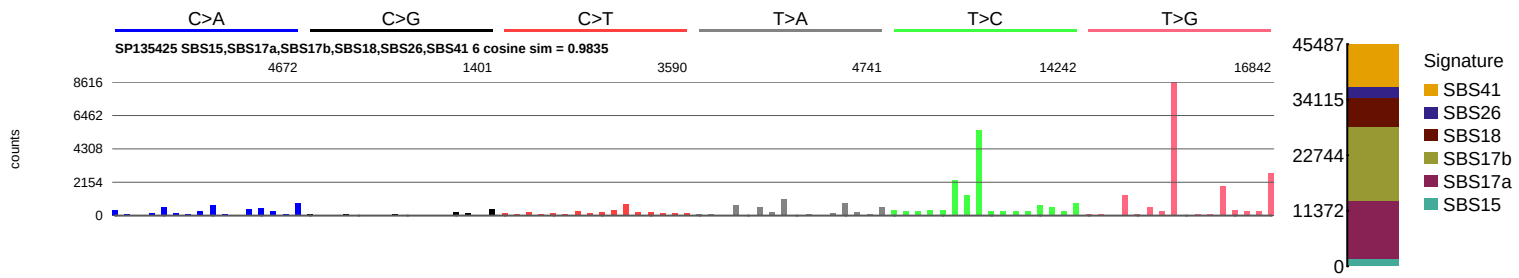
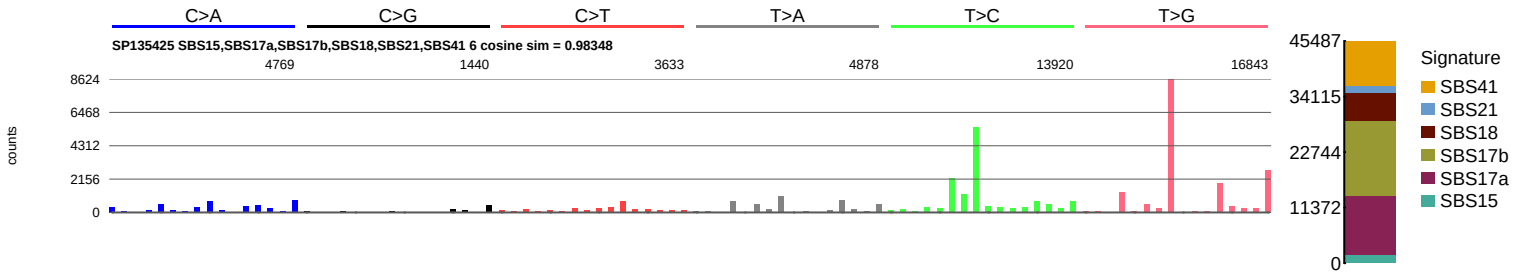
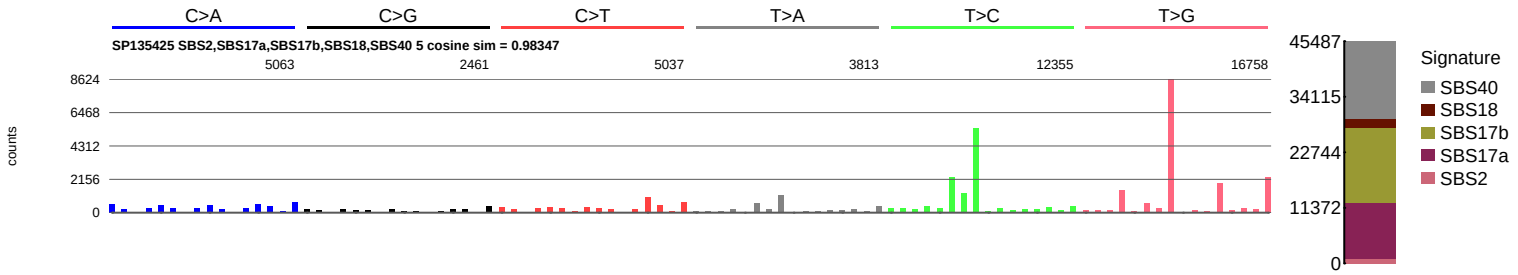
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98351
4557
1574
6730
3089
13645
15895
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.9835
4764
1431
3686
4869
13915
16824
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.9835
4672
1401
3590
4741
14242
16842
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98348
4769
1440
3633
4878
13920
16843
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15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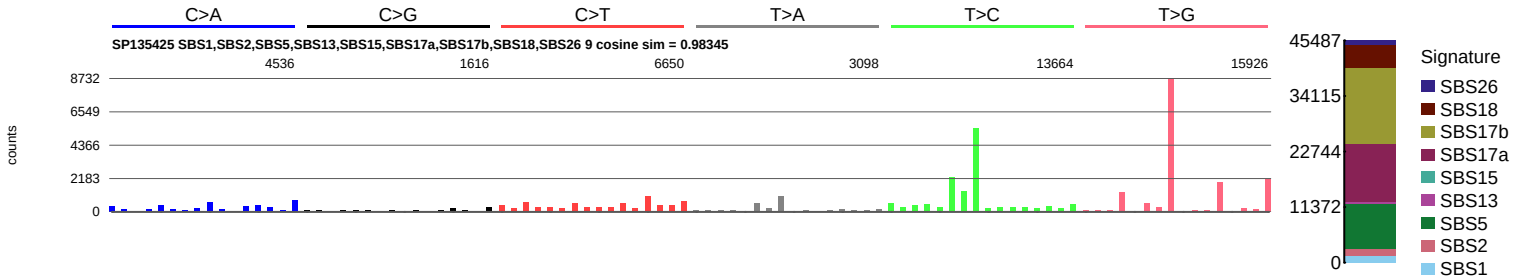
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98347
5063
2461
5037
3813
12355
16758
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2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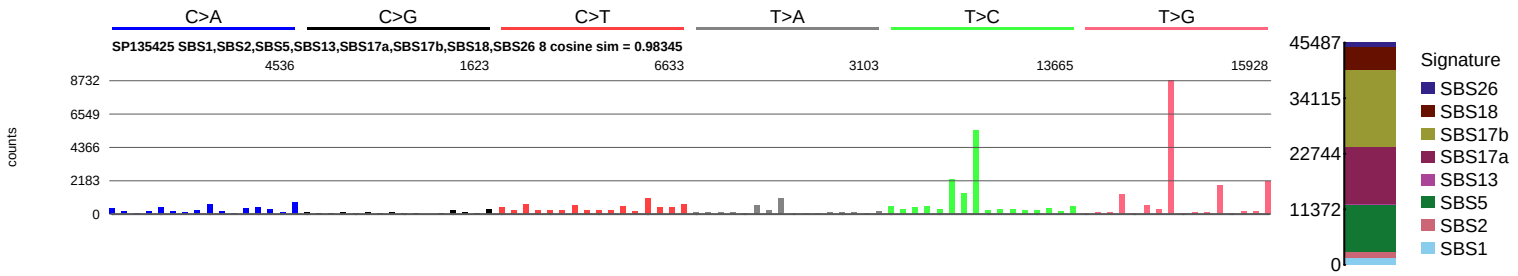
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98345
4536
1616
6650
3098
13664
15926
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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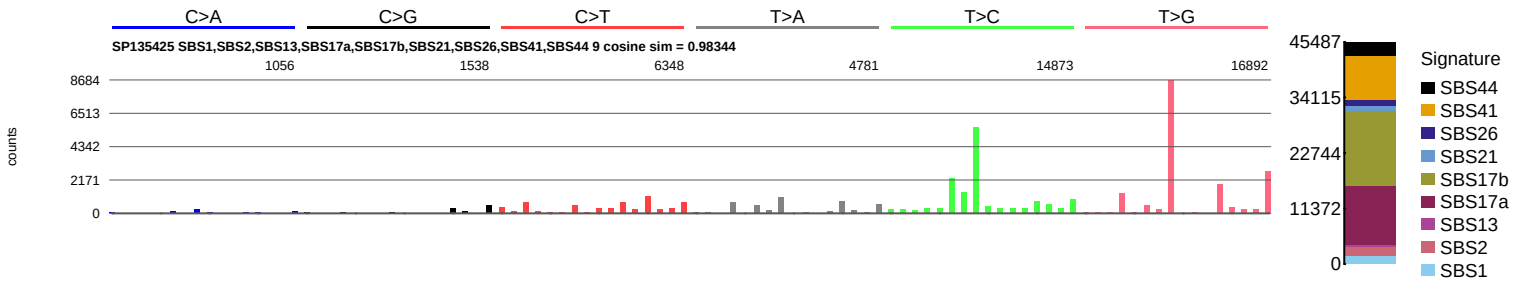
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98345
4536
1623
6633
3103
13665
15928
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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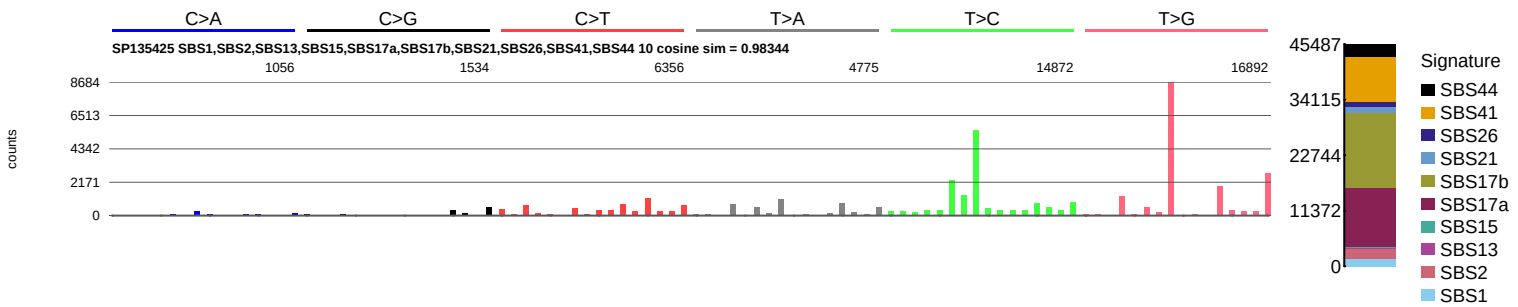
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98344
1056
1538
6348
4781
14873
16892
counts
8684
6513
4342
2171
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
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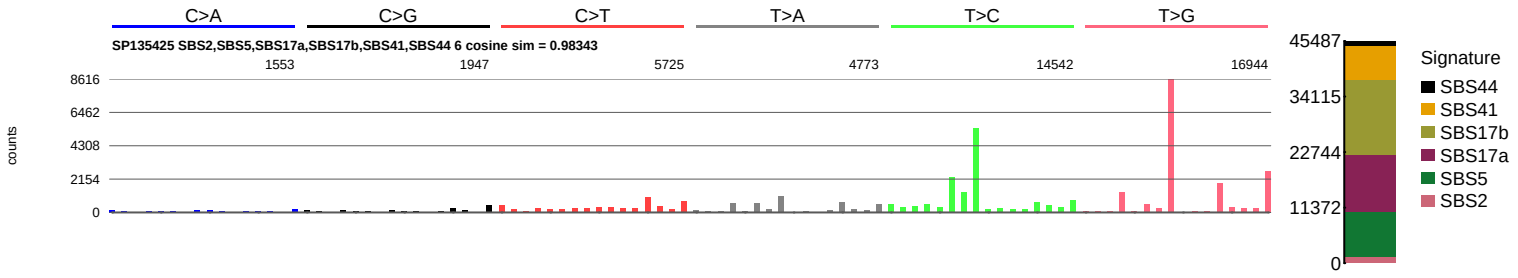
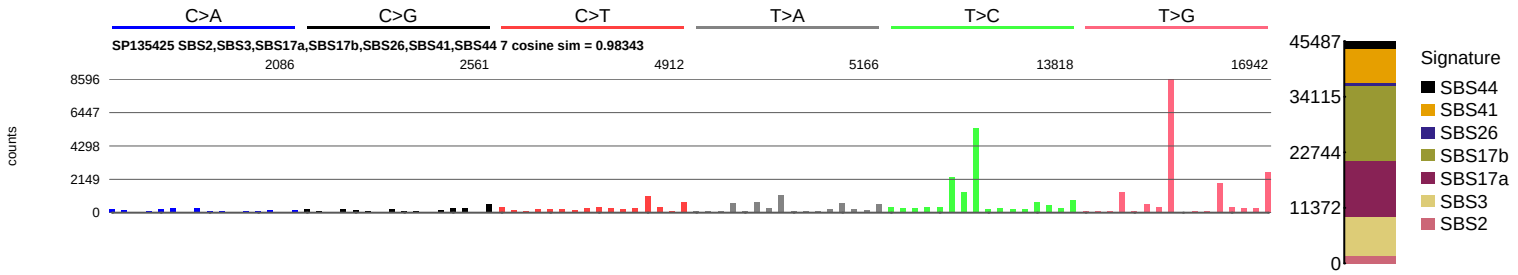
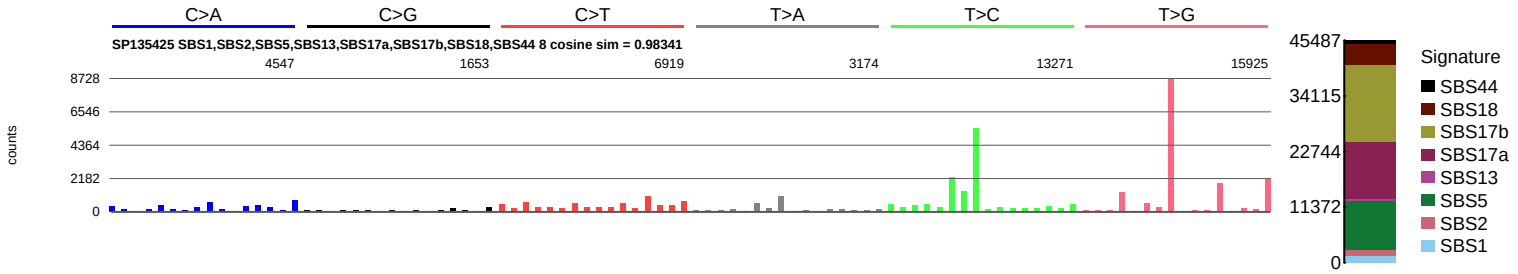
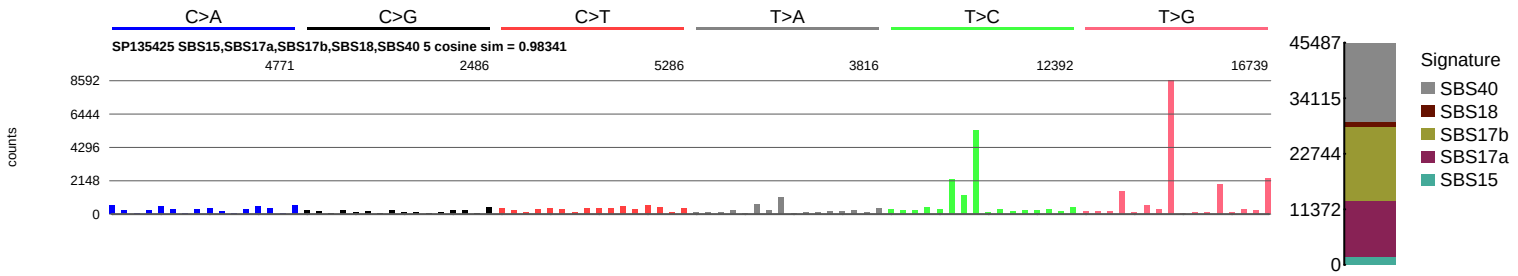
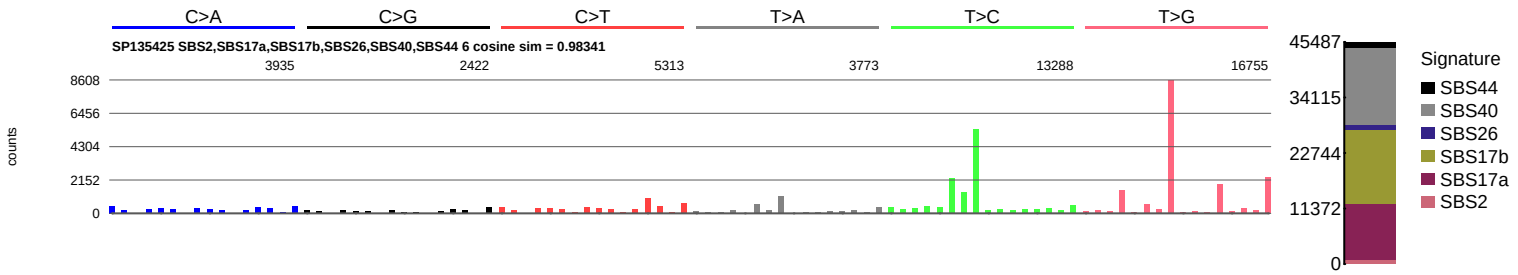
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98344
1056
1534
6356
4775
14872
16892
counts
8684
6513
4342
2171
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

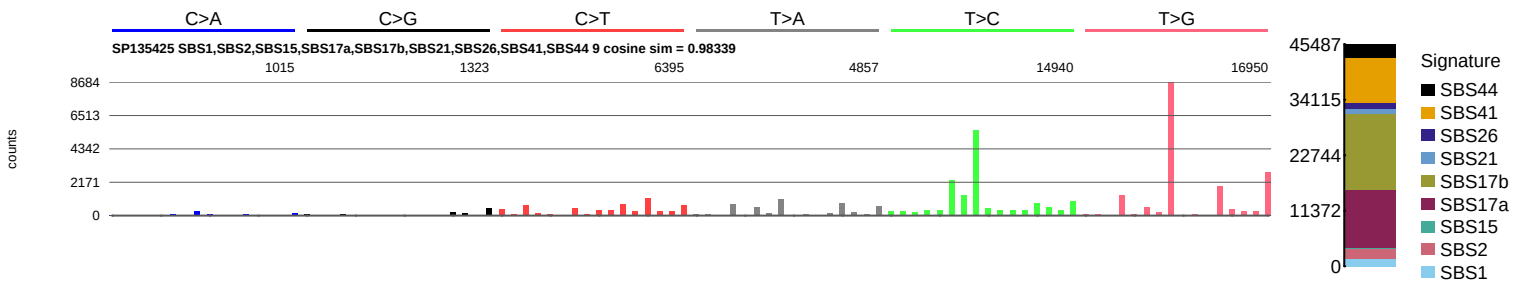

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98343
1553
1947
5725
4773
14542
16944
counts
8616
6462
4308
2154
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS5
SBS2

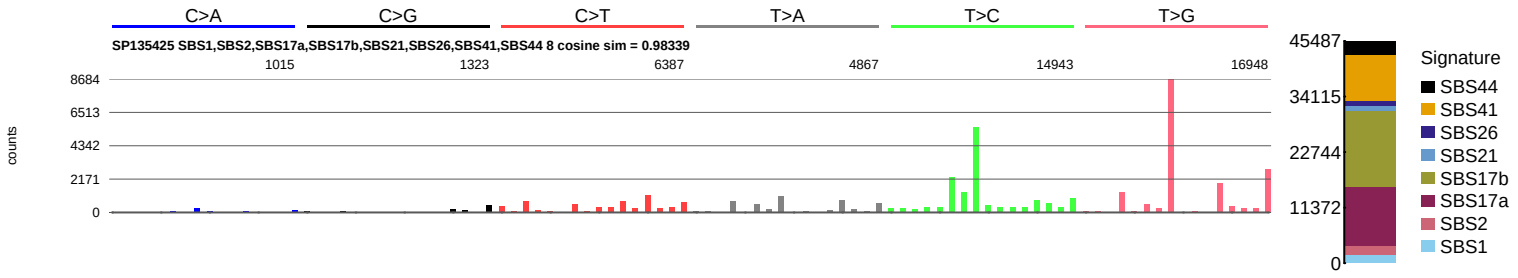

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98343
2086
2561
4912
5166
13818
16942
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98341
4547
1653
6919
3174
13271
15925
counts
8728
6546
4364
2182
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98341
4771
2486
5286
3816
12392
16739
counts
8592
6444
4296
2148
0
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98341
3935
2422
5313
3773
13288
16755
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98339
1015
1323
6395
4857
14940
16950
counts
8684
6513
4342
2171
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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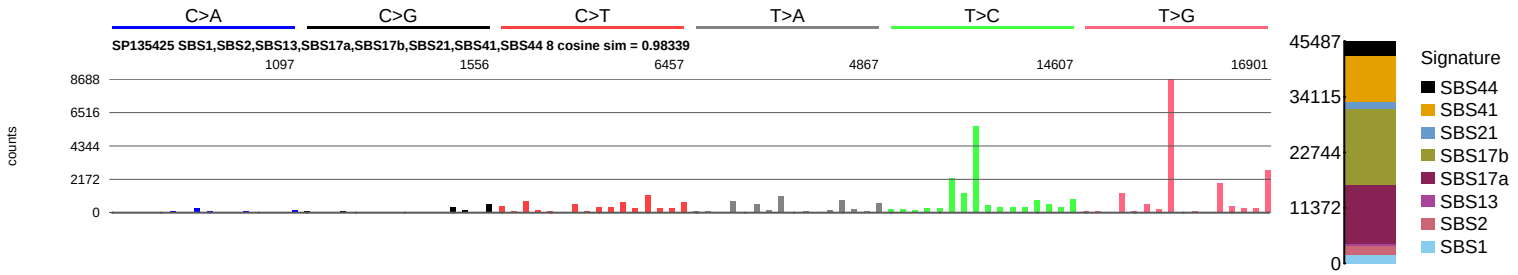
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98339
1015
1323
6387
4867
14943
16948
8684
6513
4342
2171
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1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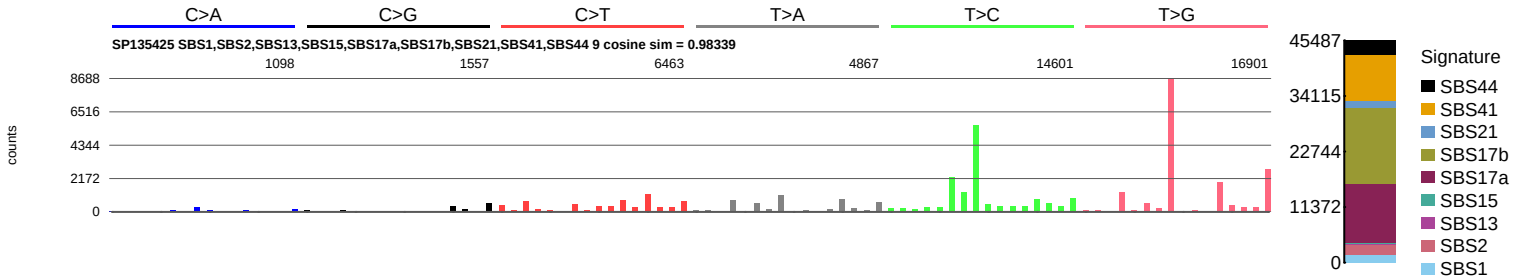
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98339
1097
1556
6457
4867
14607
16901
8688
6516
4344
2172
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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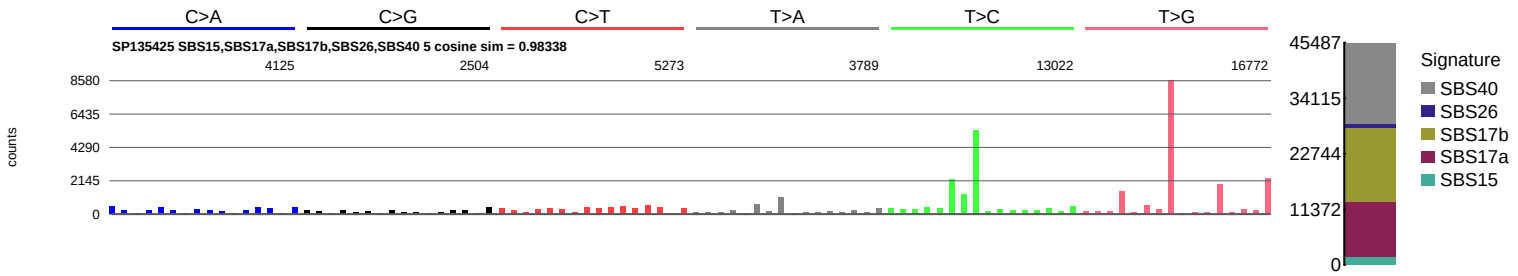
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.98339
1098
1557
6463
4867
14601
16901
8688
6516
4344
2172
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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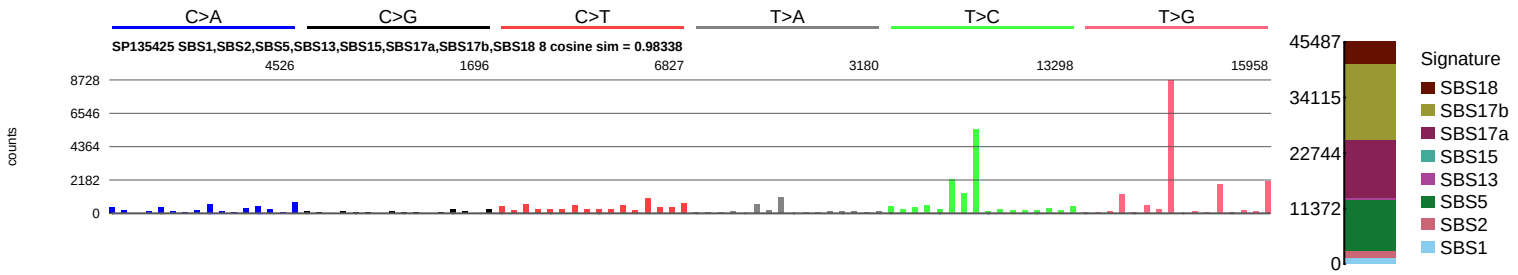
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98338
4125
2504
5273
3789
13022
16772
8580
6435
4290
2145
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15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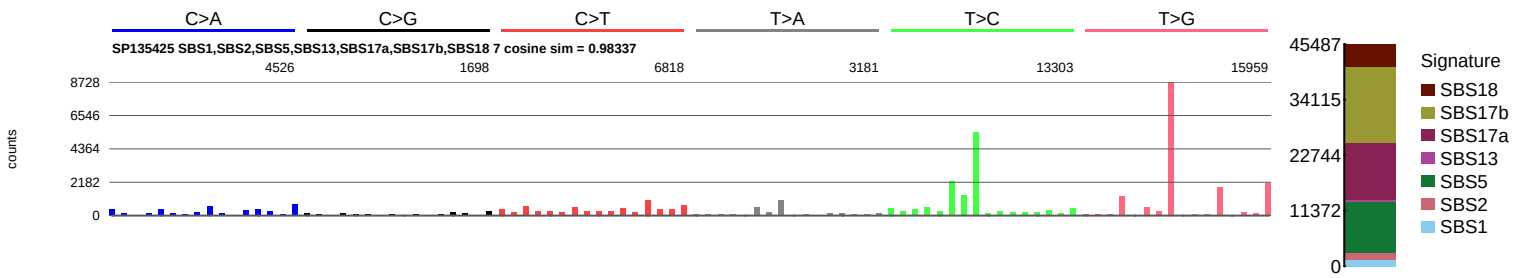
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98338
4526
1696
6827
3180
13298
15958
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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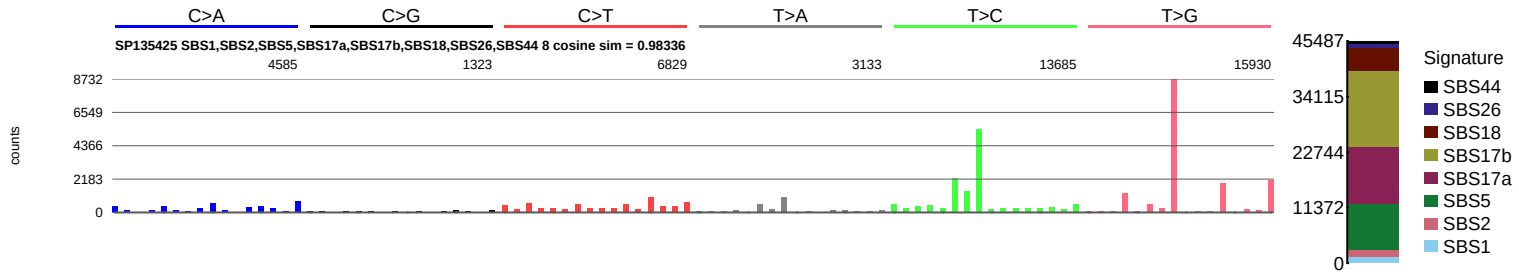
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98337
4526
1698
6818
3181
13303
15959
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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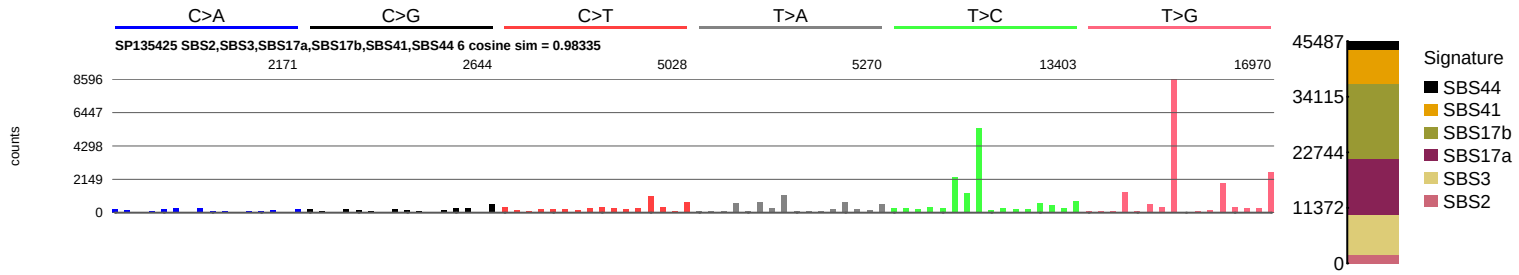
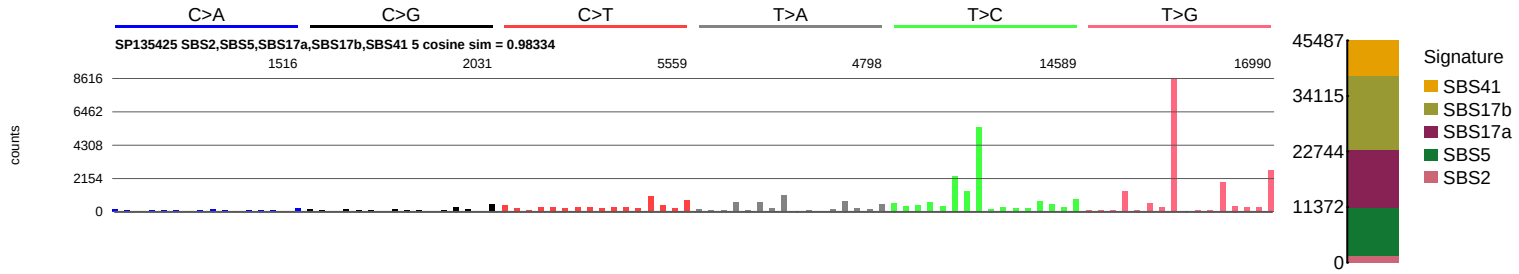
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98336
4585
1323
6829
3133
13685
15930
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98335
2171
2644
5028
5270
13403
16970
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2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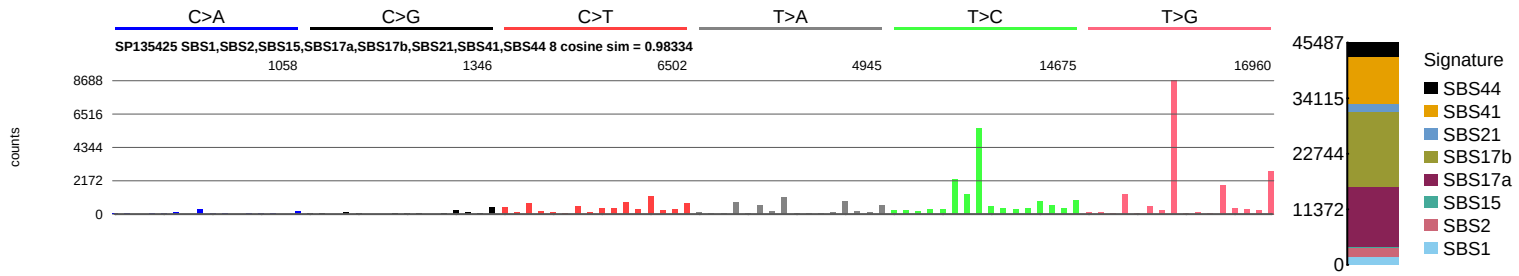
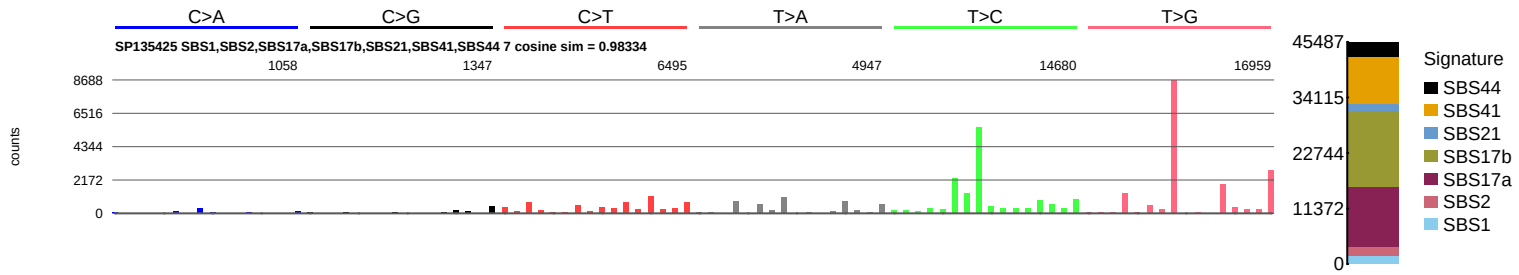
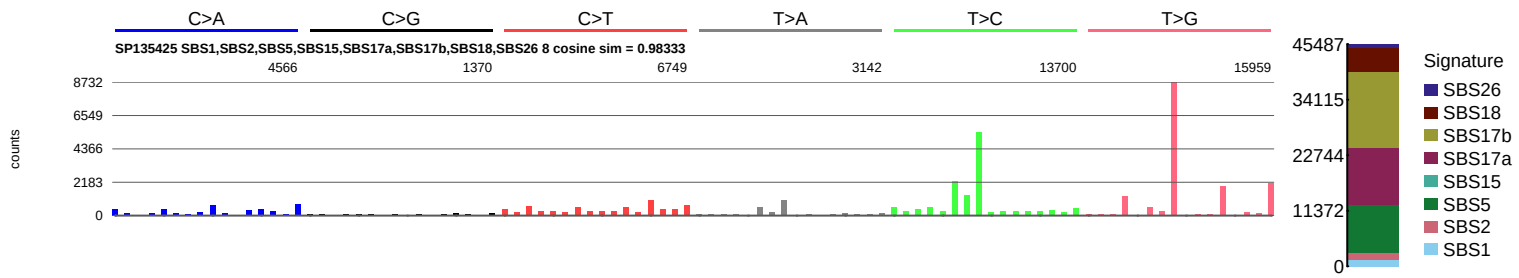
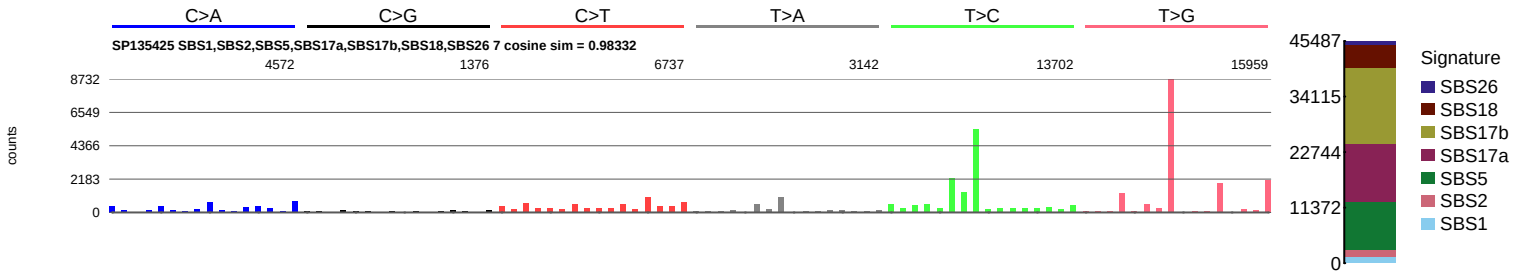
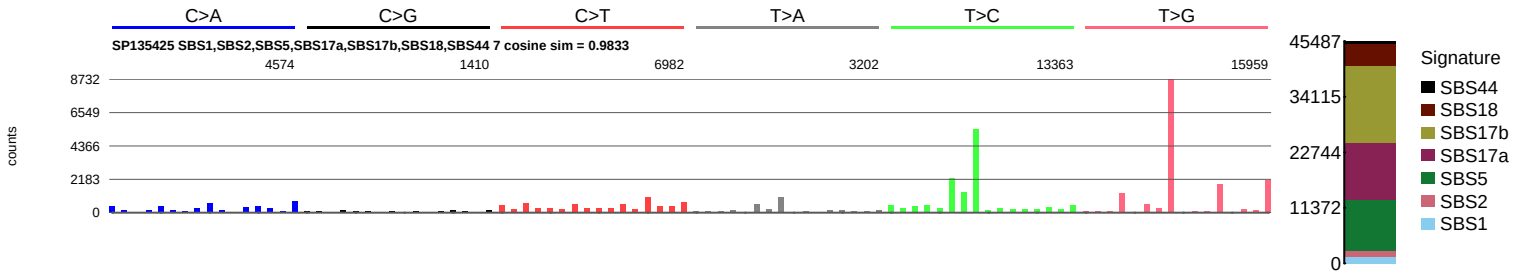
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98334
1516
2031
5559
4798
14589
16990
8616
6462
4308
2154
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2

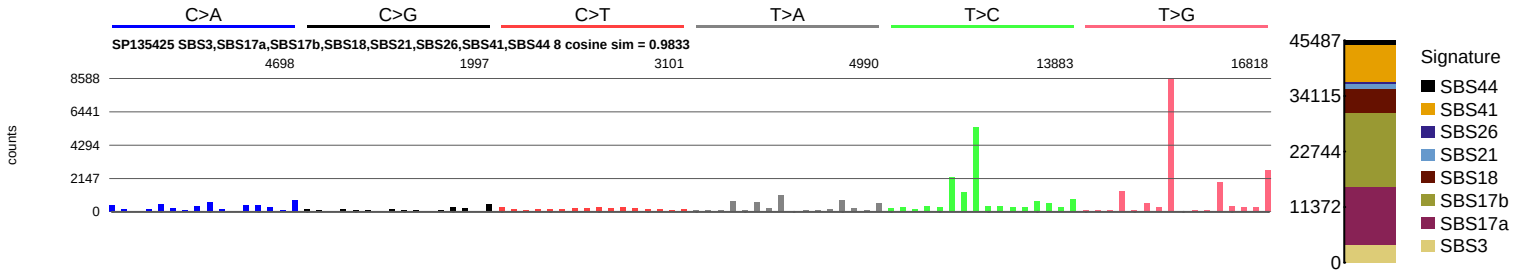

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98334
1058
1346
6502
4945
14675
16960
8688
6516
4344
2172
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98334
1058
1347
6495
4947
14680
16959
8688
6516
4344
2172
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98333
4566
1370
6749
3142
13700
15959
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98332
4572
1376
6737
3142
13702
15959
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.9833
4574
1410
6982
3202
13363
15959
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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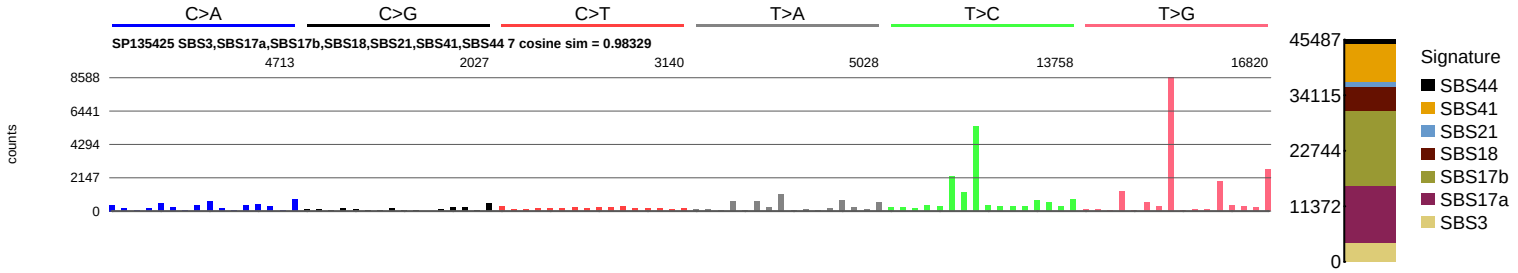
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.9833
4698
1997
3101
4990
13883
16818
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3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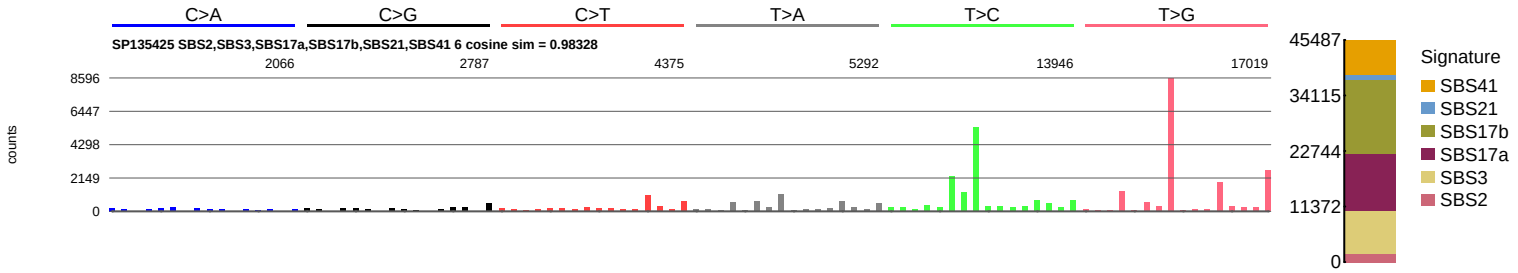
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98329
4713
2027
3140
5028
13758
16820
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3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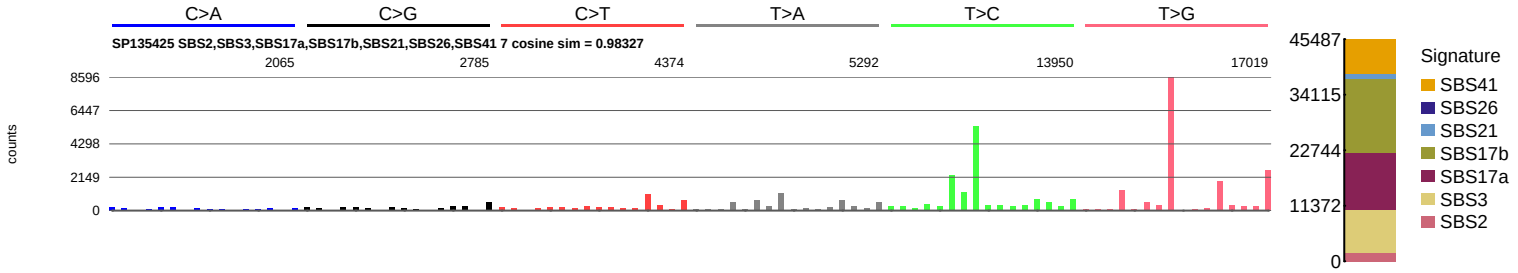
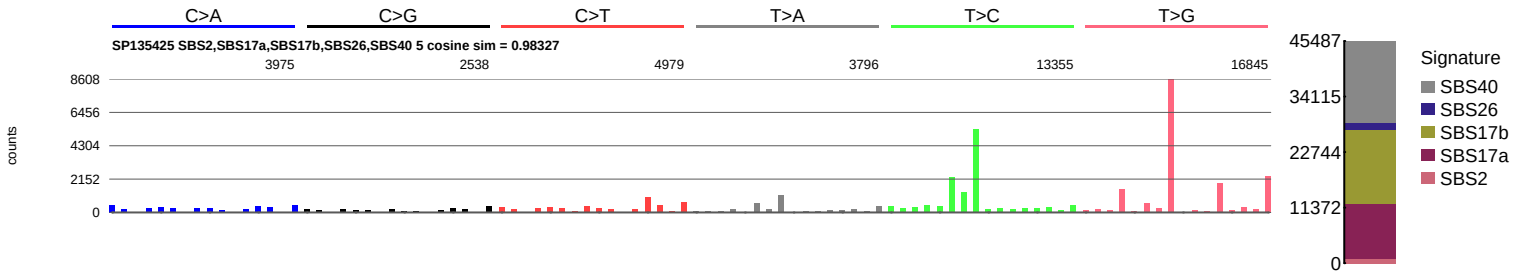
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98328
2066
2787
4375
5292
13946
17019
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98327
2065
2785
4374
5292
13950
17019
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2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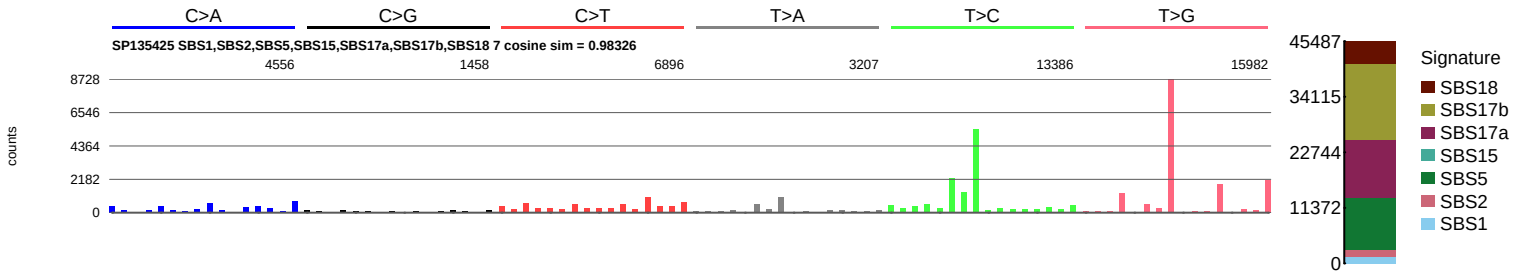
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98327
3975
2538
4979
3796
13355
16845
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98326
4556
1458
6896
3207
13386
15982
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
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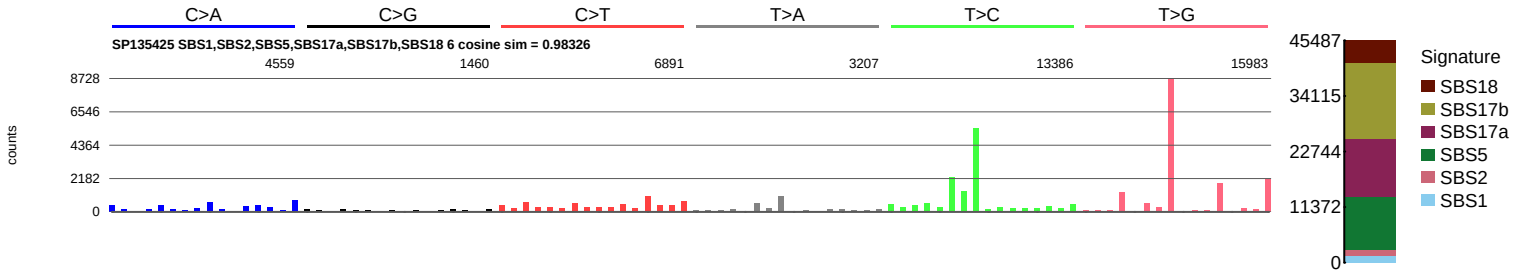
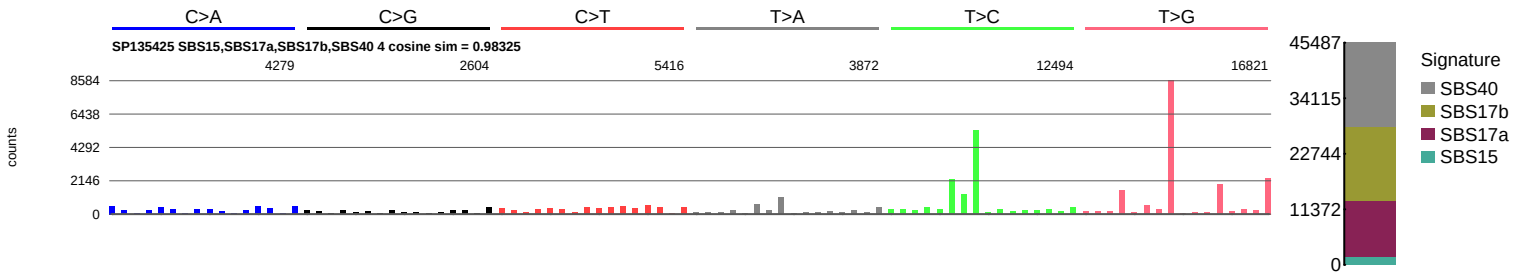
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98326
4559
1460
6891
3207
13386
15983
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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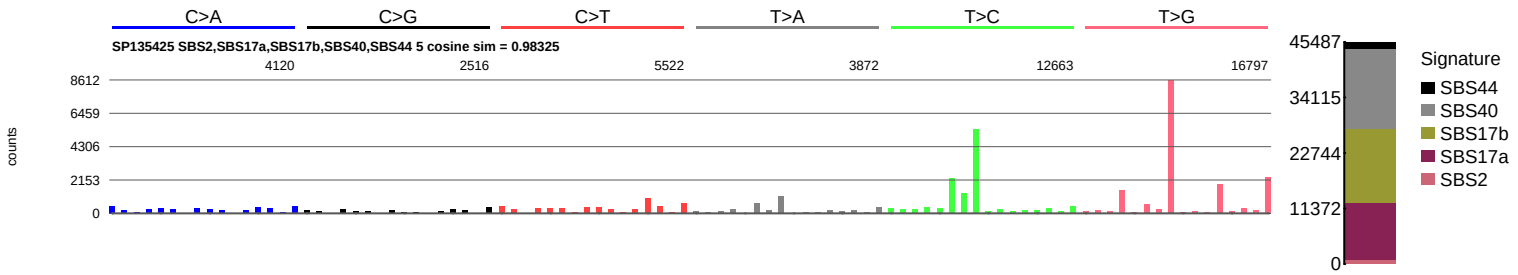
C>G
C>T
T>A
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98325
4279
2604
5416
3872
12494
16821
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS15


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.98325
4120
2516
5522
3872
12663
16797
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
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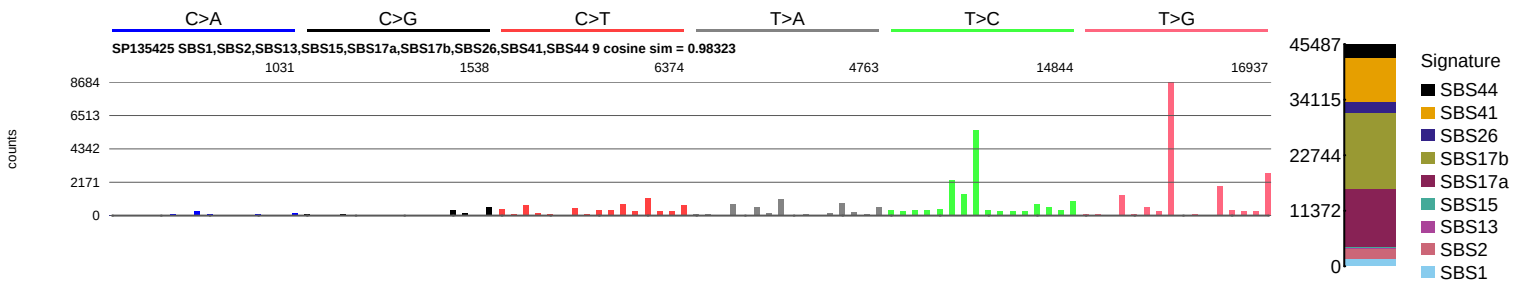
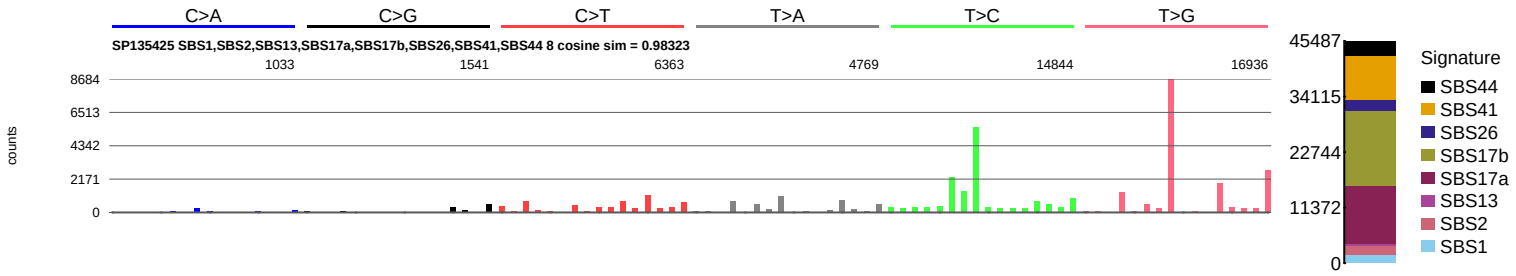
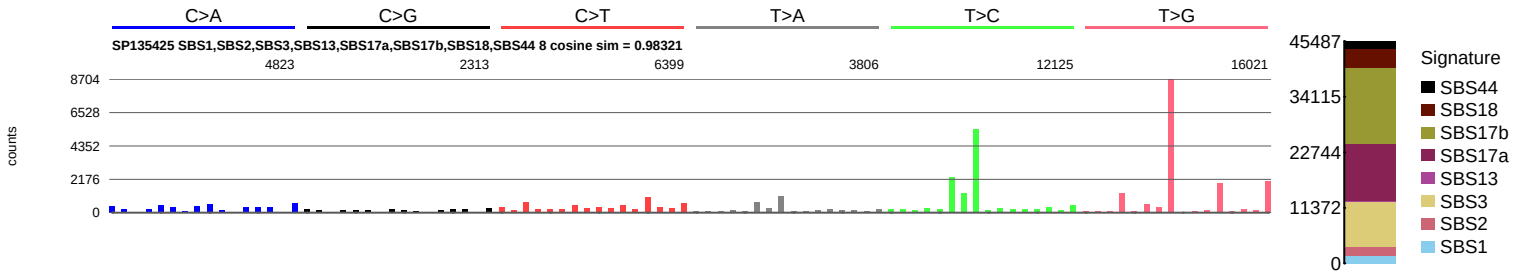
SBS44
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98323
1031
1538
6374
4763
14844
16937
8684
6513
4342
2171
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98323
1033
1541
6363
4769
14844
16936
8684
6513
4342
2171
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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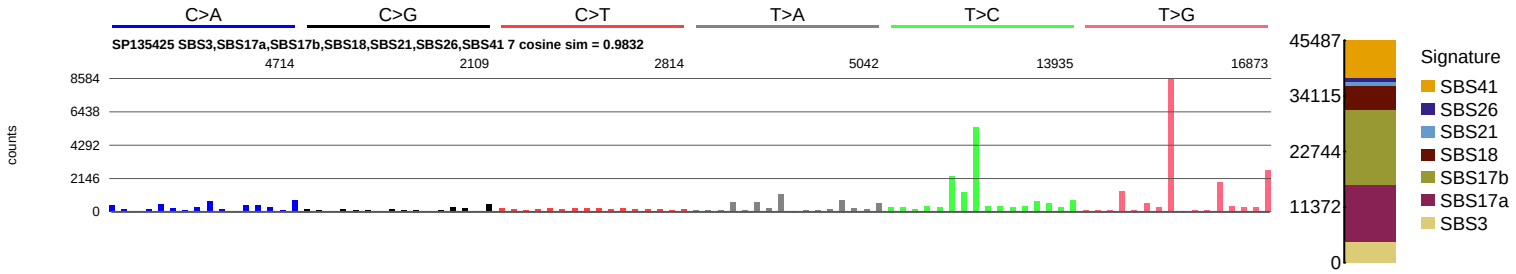
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98321
4823
2313
6399
3806
12125
16021
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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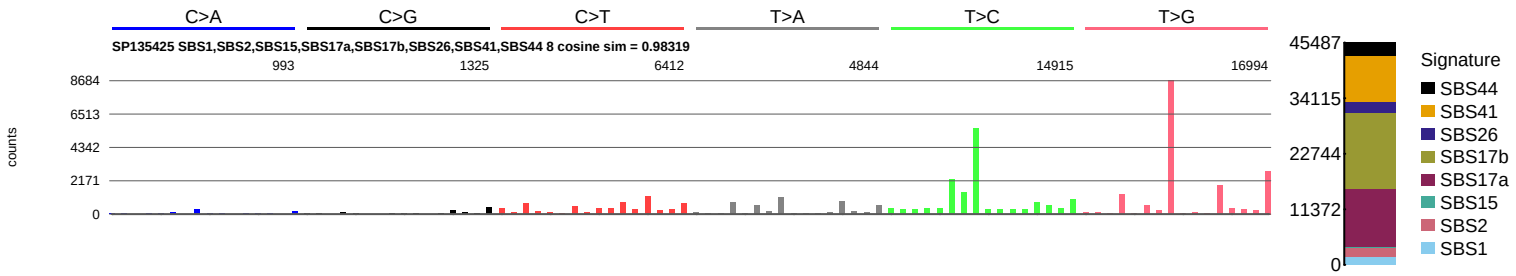
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.9832
4714
2109
2814
5042
13935
16873
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3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C>T
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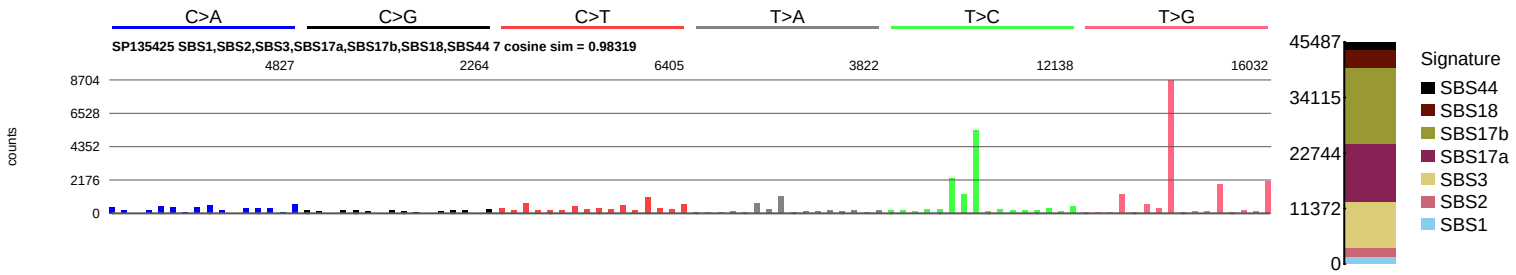
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98319
993
1325
6412
4844
14915
16994
8684
6513
4342
2171
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98319
4827
2264
6405
3822
12138
16032
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
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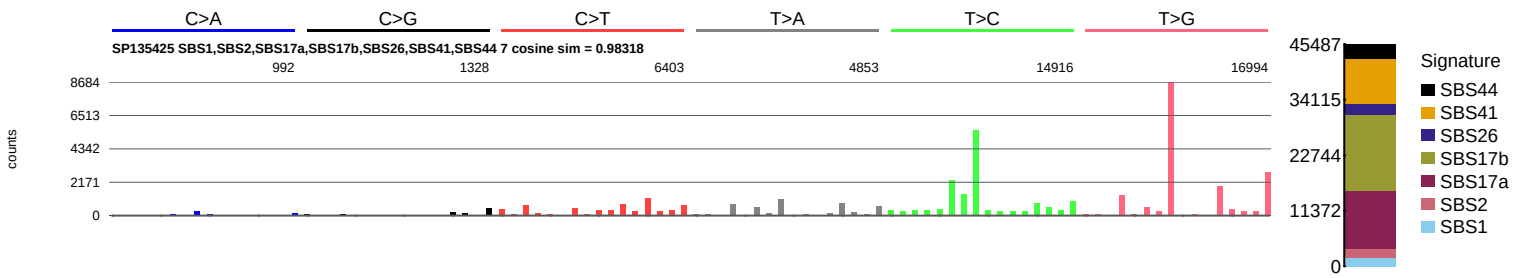
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
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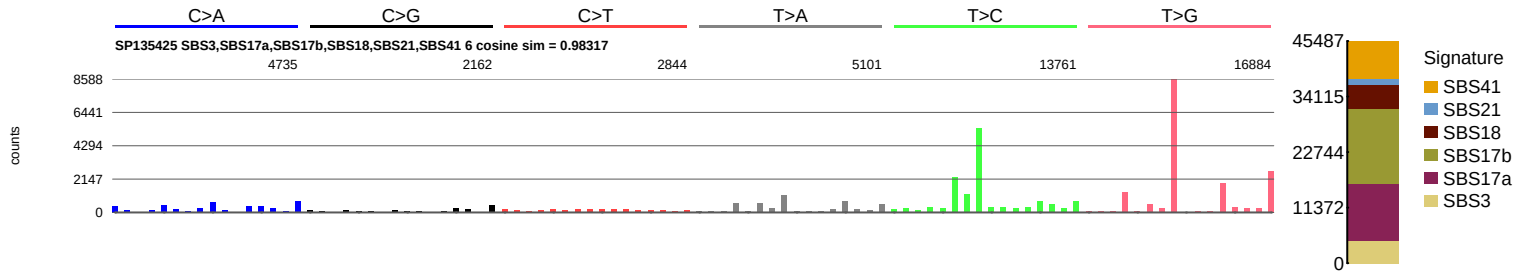
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98318
992
1328
6403
4853
14916
16994
8684
6513
4342
2171
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1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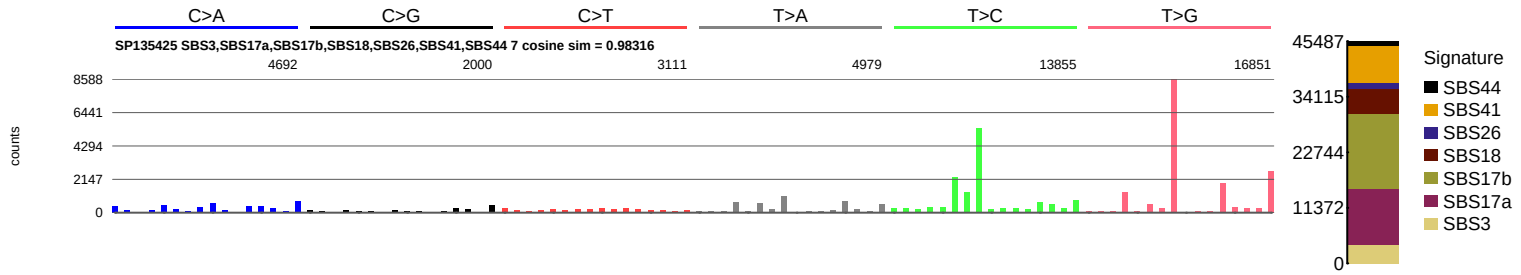
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98317
4735
2162
2844
5101
13761
16884
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3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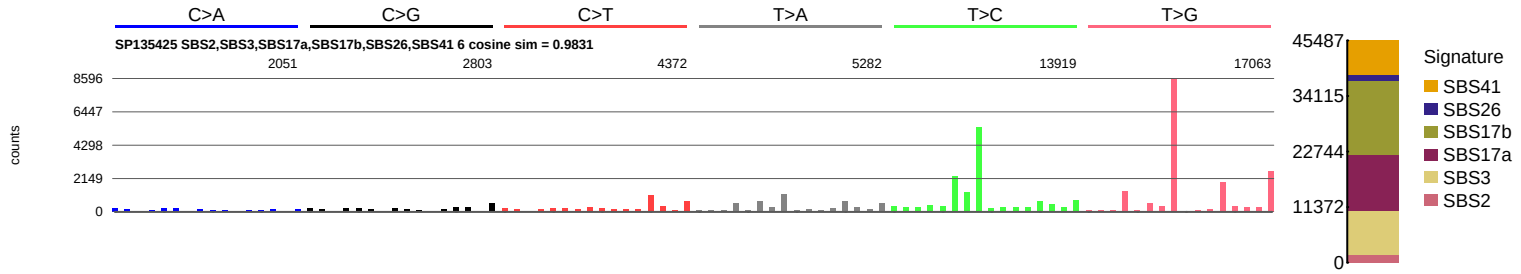
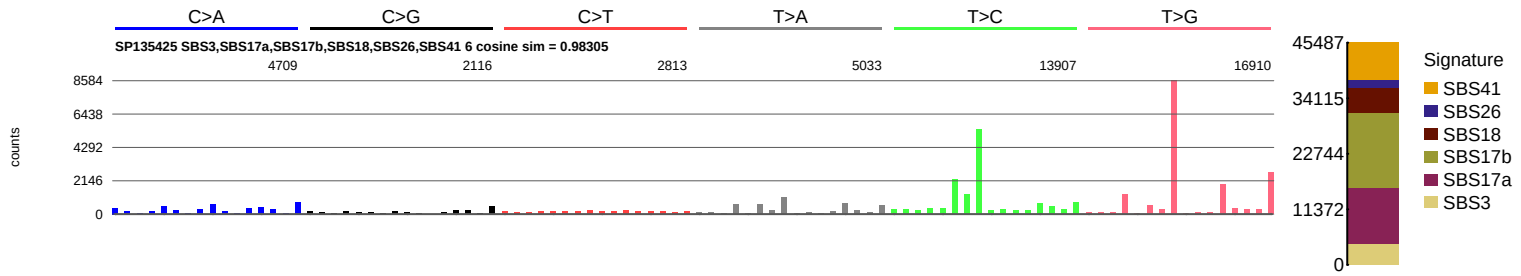
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98316
4692
2000
3111
4979
13855
16851
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.9831
2051
2803
4372
5282
13919
17063
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2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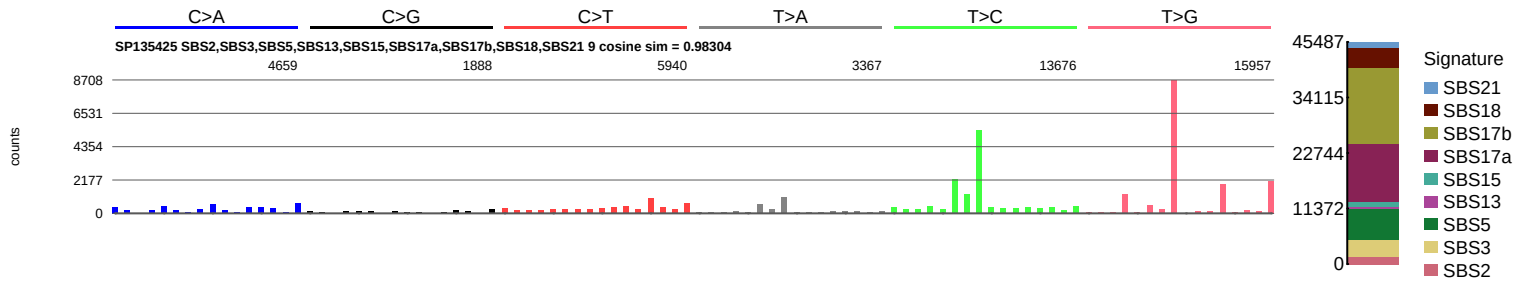
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98305
4709
2116
2813
5033
13907
16910
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3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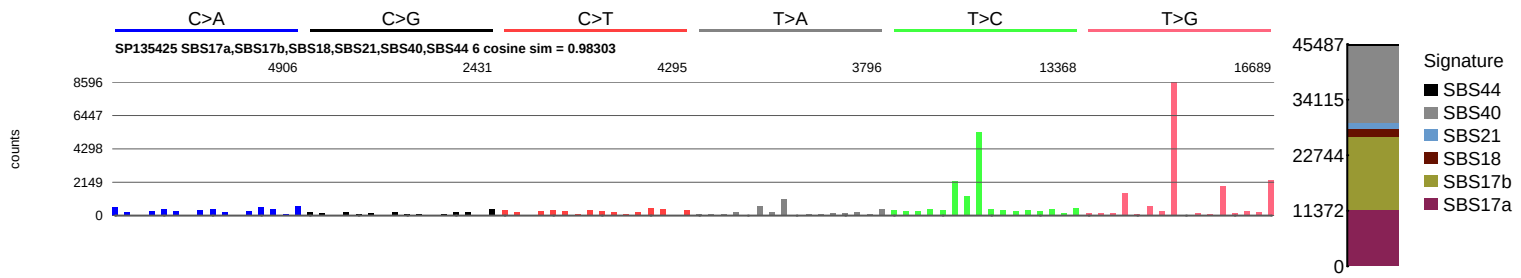
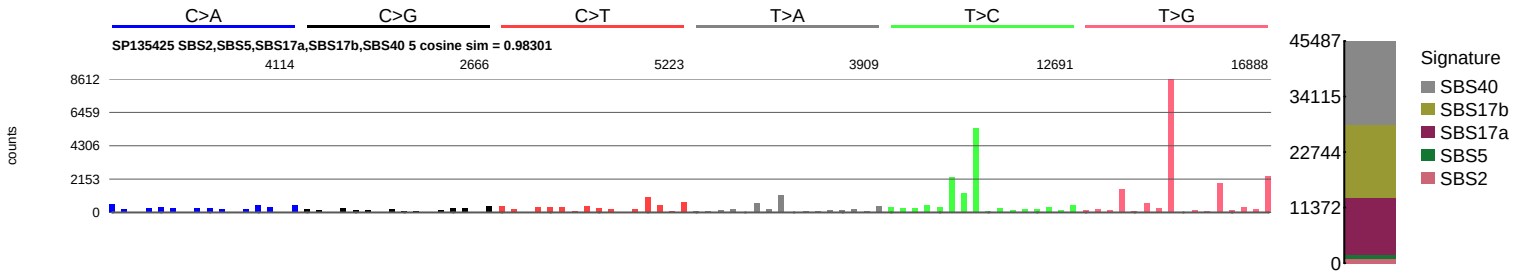
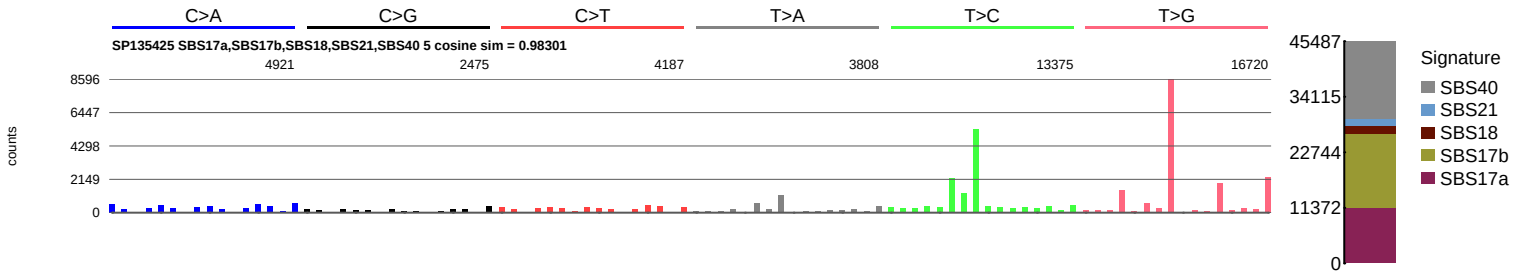
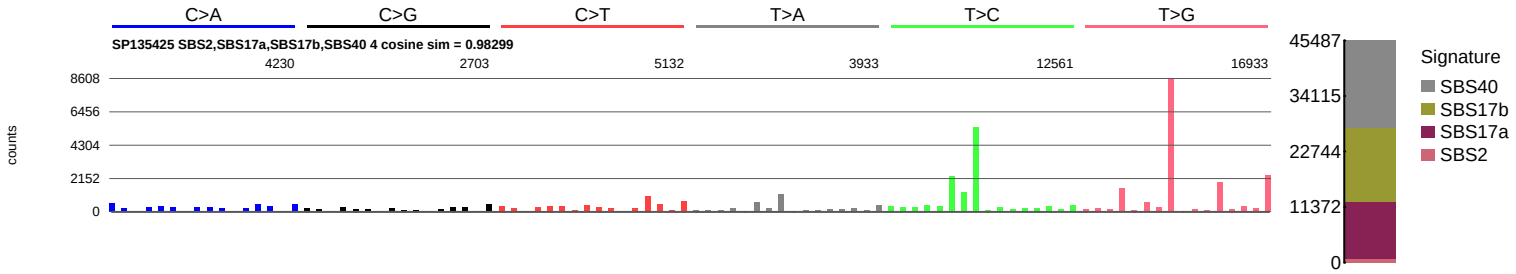
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98304
4659
1888
5940
3367
13676
15957
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 6 cosine sim = 0.98303
4906
2431
4295
3796
13368
16689
counts
8596
6447
4298
2149
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98301
4114
2666
5223
3909
12691
16888
8612
6459
4306
2153
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS40 5 cosine sim = 0.98301
4921
2475
4187
3808
13375
16720
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a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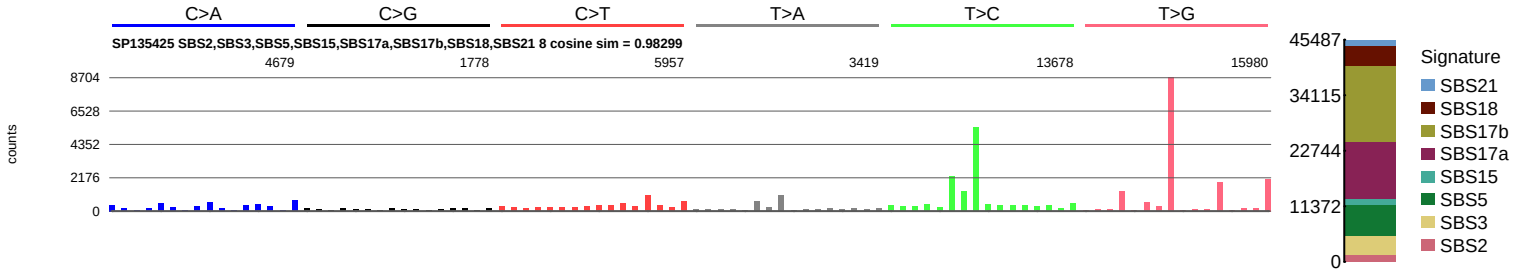
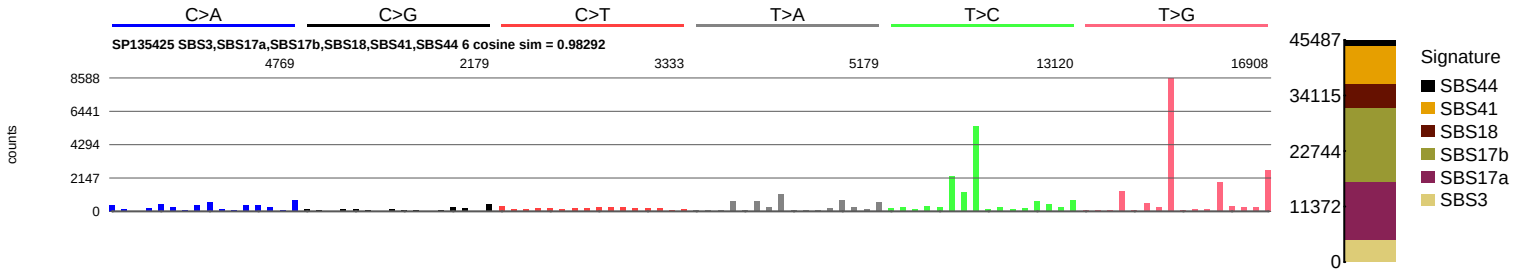
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98299
4230
2703
5132
3933
12561
16933
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98299
4679
1778
5957
3419
13678
15980
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2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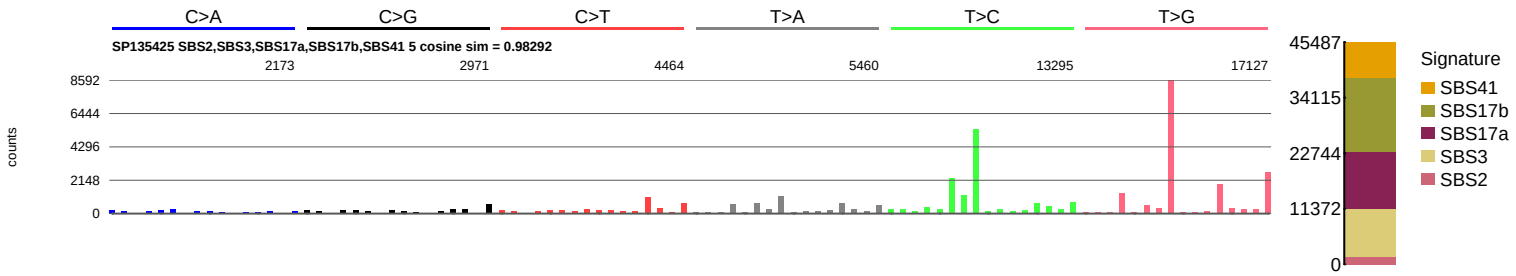
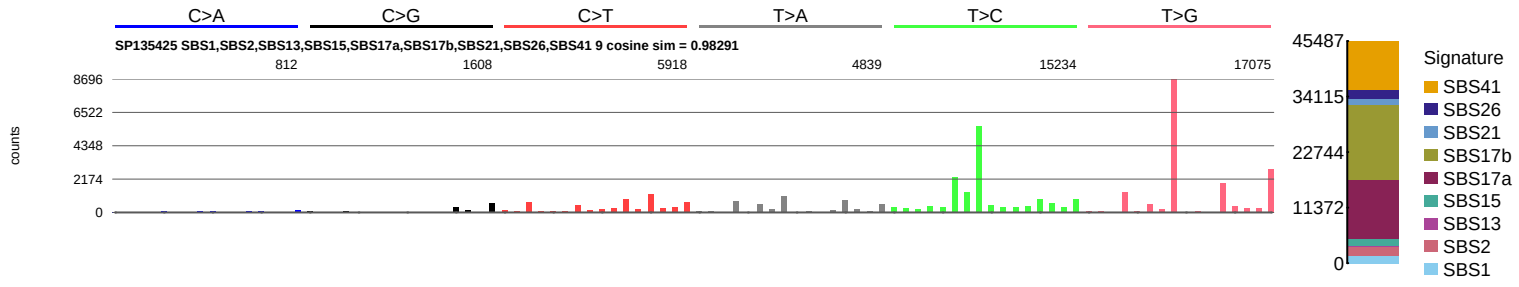
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98292
4769
2179
3333
5179
13120
16908
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98292
2173
2971
4464
5460
13295
17127
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2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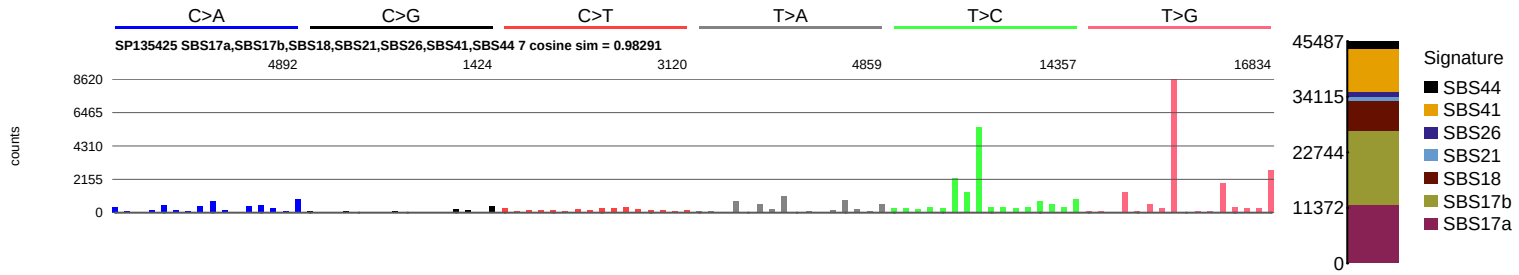
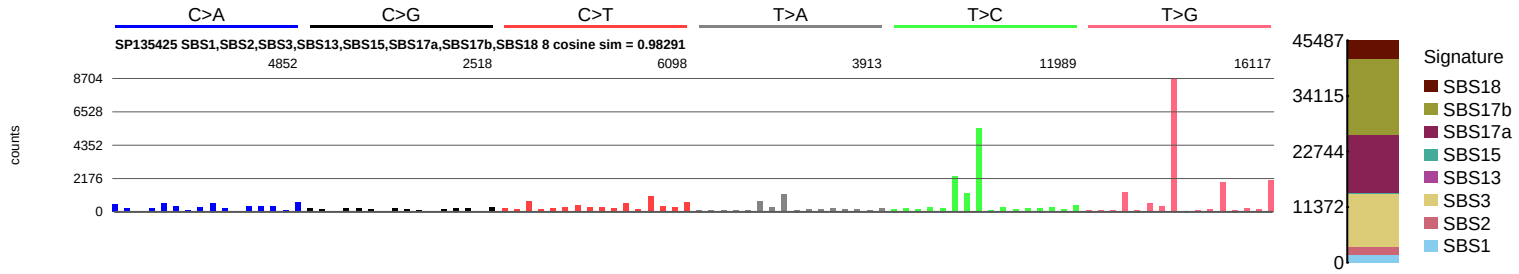
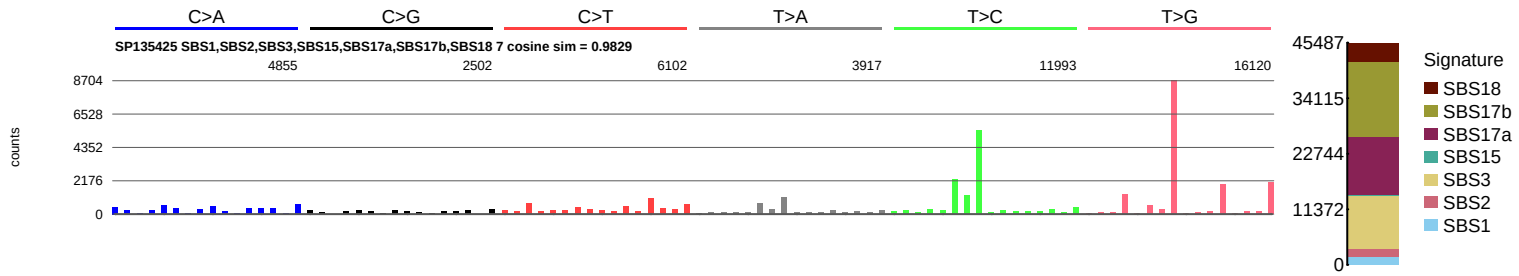
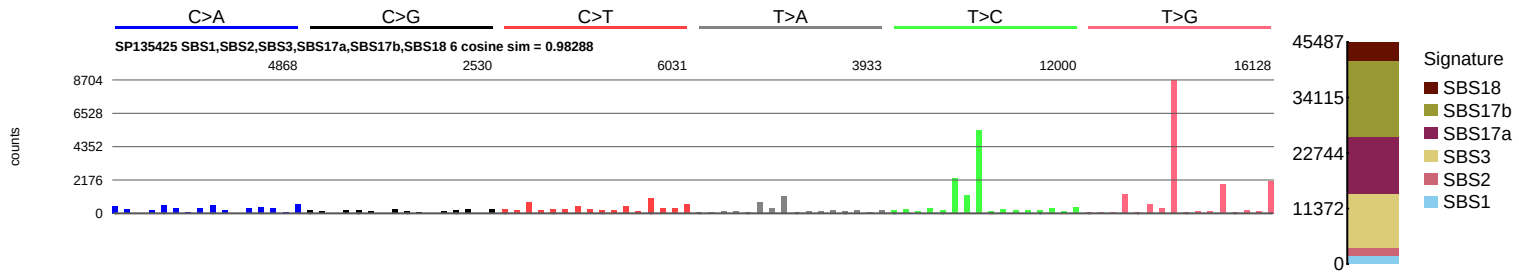
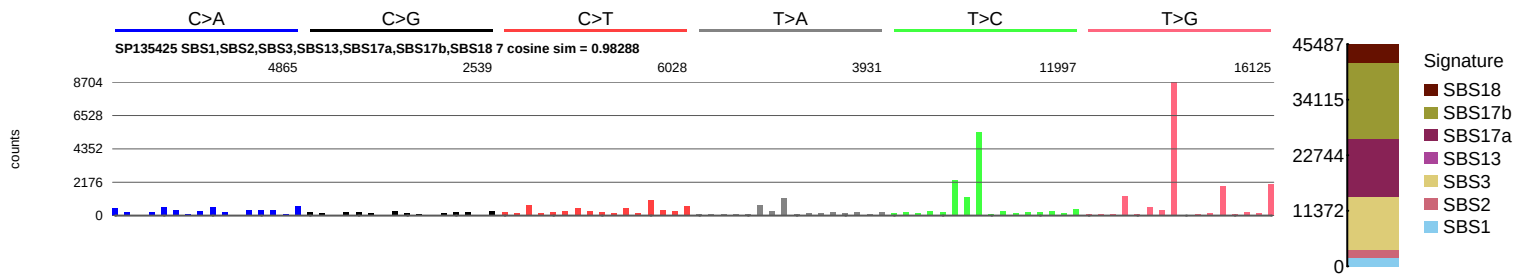
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98291
812
1608
5918
4839
15234
17075
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

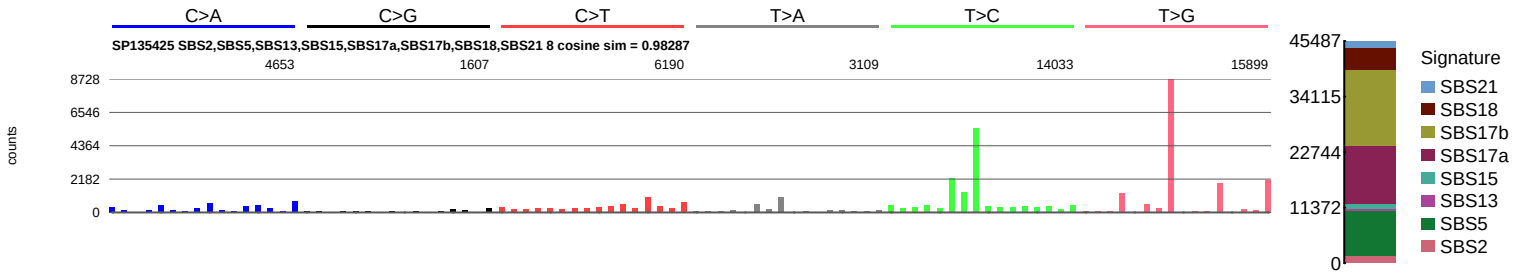

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98291
4892
1424
3120
4859
14357
16834
8620
6465
4310
2155
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a

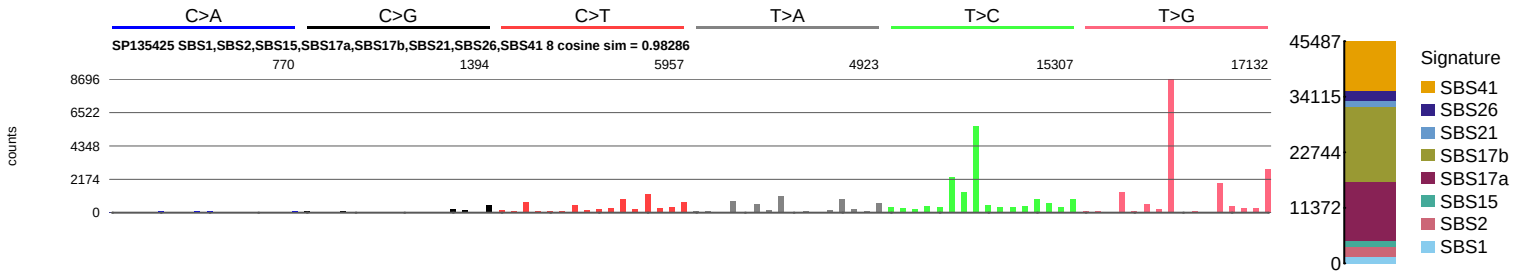

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98291
4852
2518
6098
3913
11989
16117
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

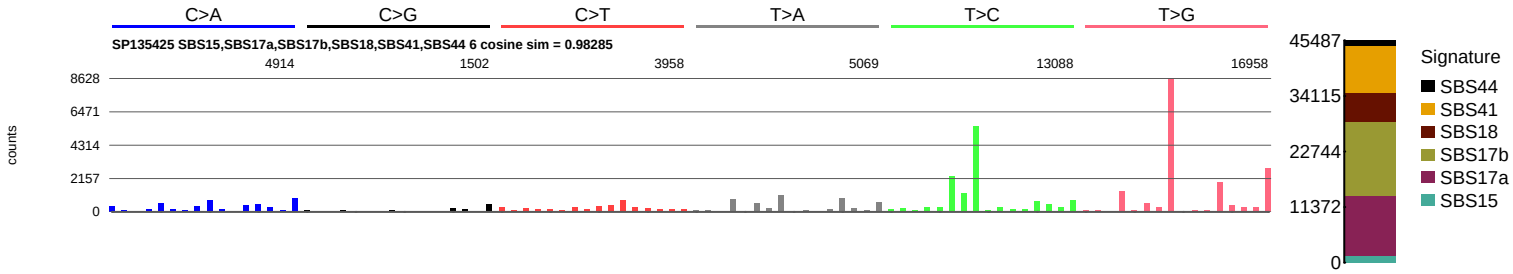

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.9829
4855
2502
6102
3917
11993
16120
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98288
4868
2530
6031
3933
12000
16128
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98288
4865
2539
6028
3931
11997
16125
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98287
4653
1607
6190
3109
14033
15899
counts
8728
6546
4364
2182
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98286
770
1394
5957
4923
15307
17132
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
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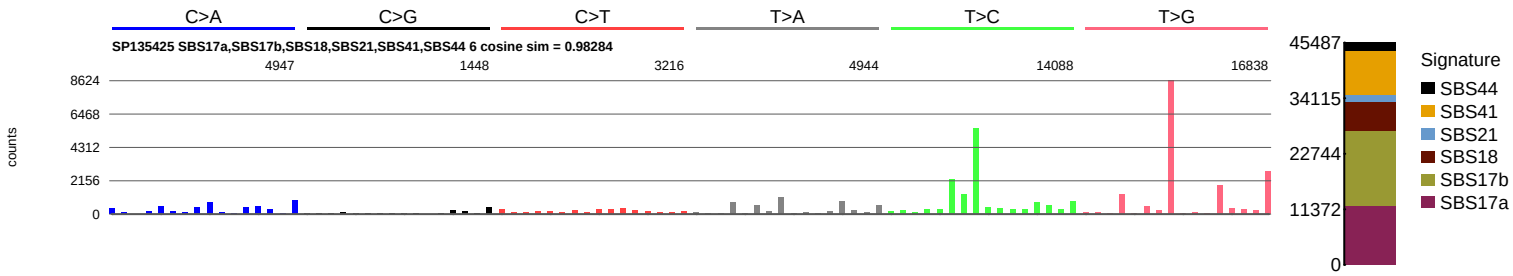
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98285
4914
1502
3958
5069
13088
16958
counts
8628
6471
4314
2157
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15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C>G
C>T
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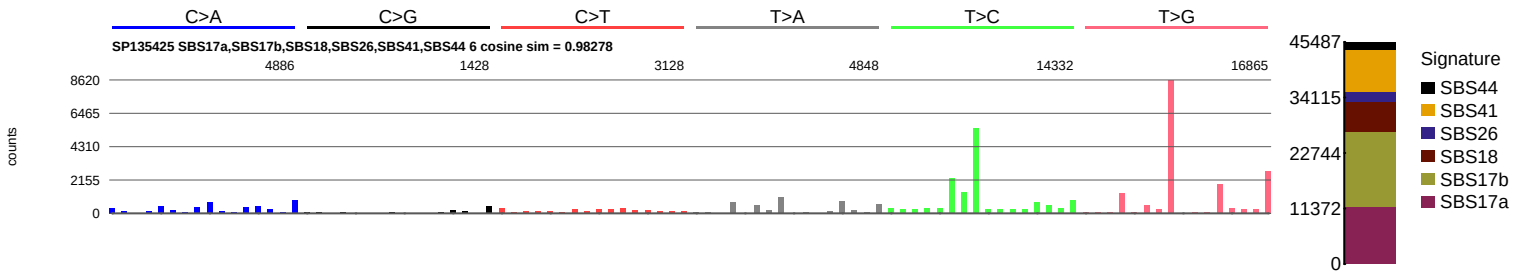
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 6 cosine sim = 0.98284
4947
1448
3216
4944
14088
16838
counts
8624
6468
4312
2156
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
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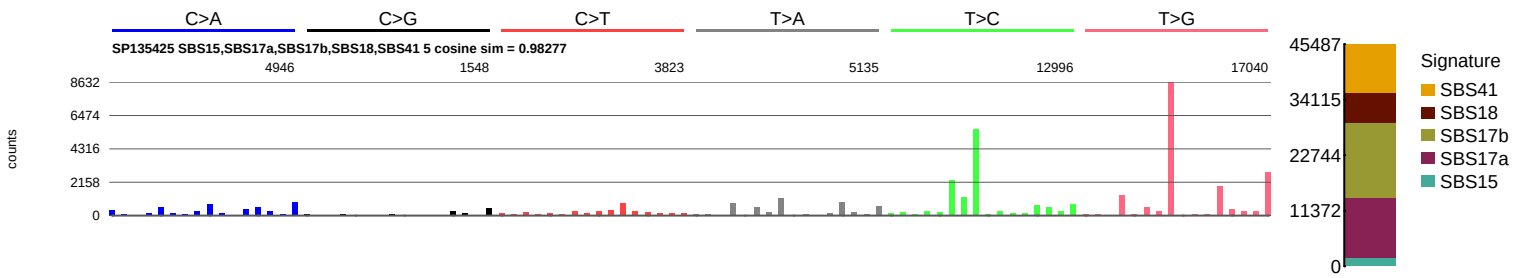
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 6 cosine sim = 0.98278
4886
1428
3128
4848
14332
16865
counts
8620
6465
4310
2155
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
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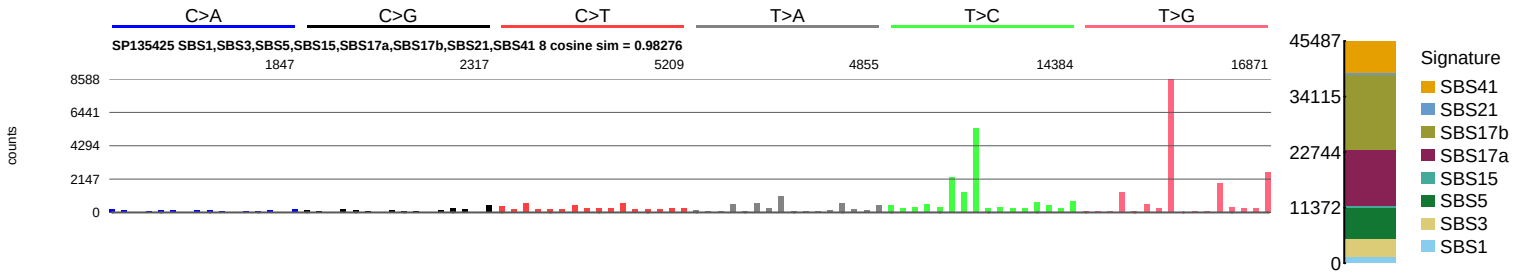
T>C
T>G
SP135425 SBS15,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98277
4946
1548
3823
5135
12996
17040
counts
8632
6474
4316
2158
0
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15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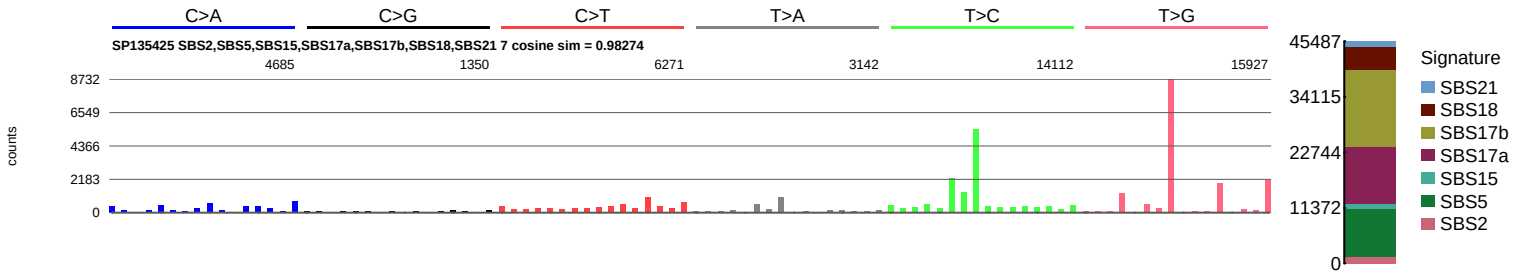
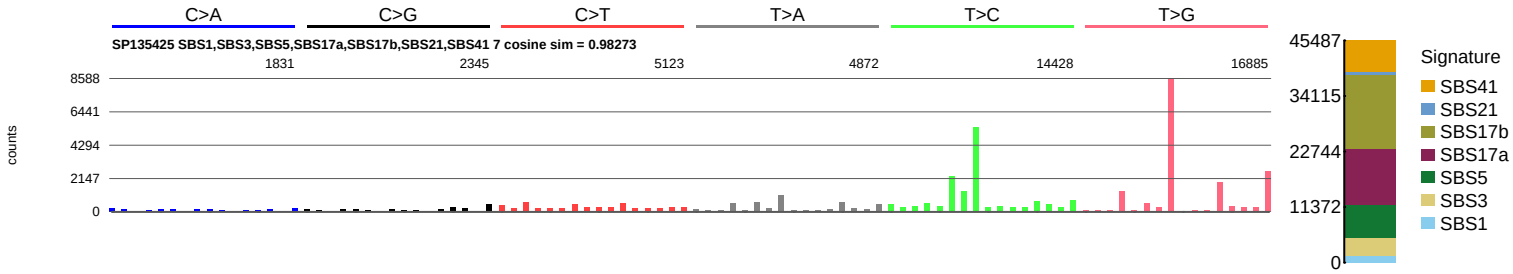
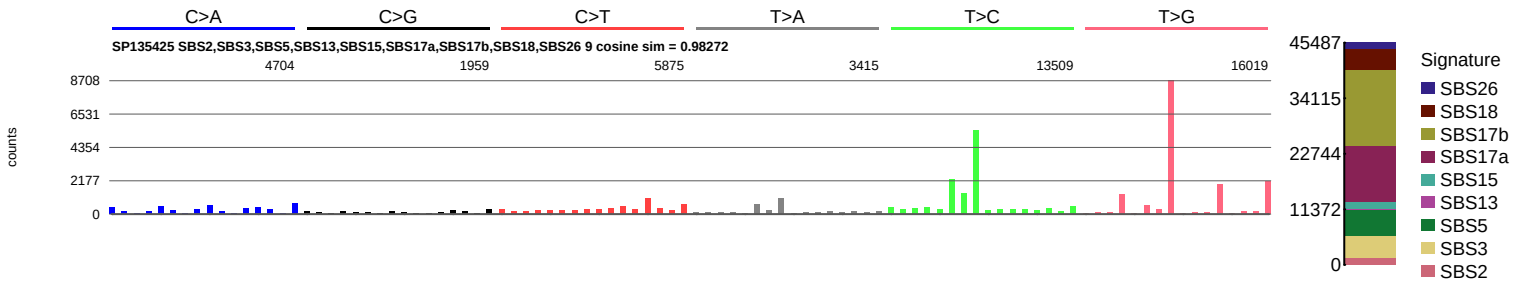
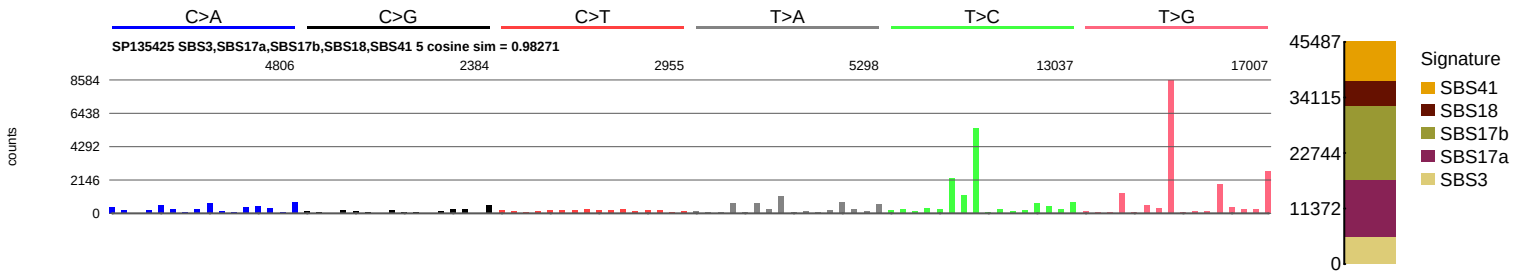
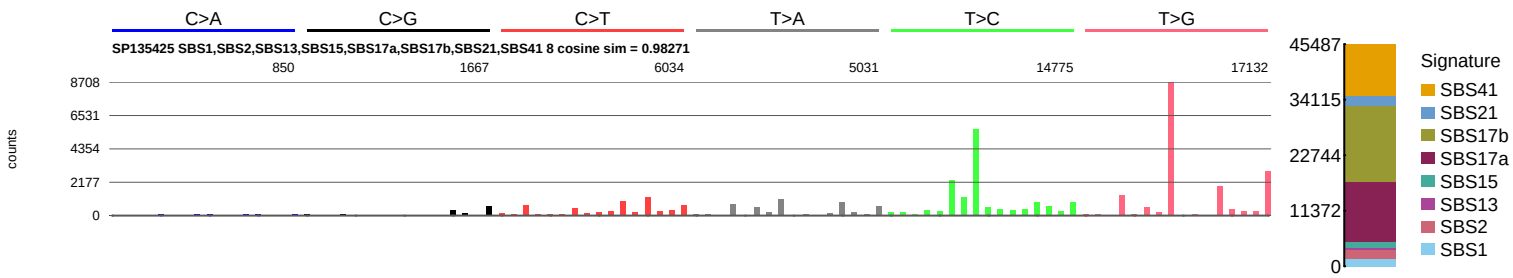
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98276
1847
2317
5209
4855
14384
16871
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

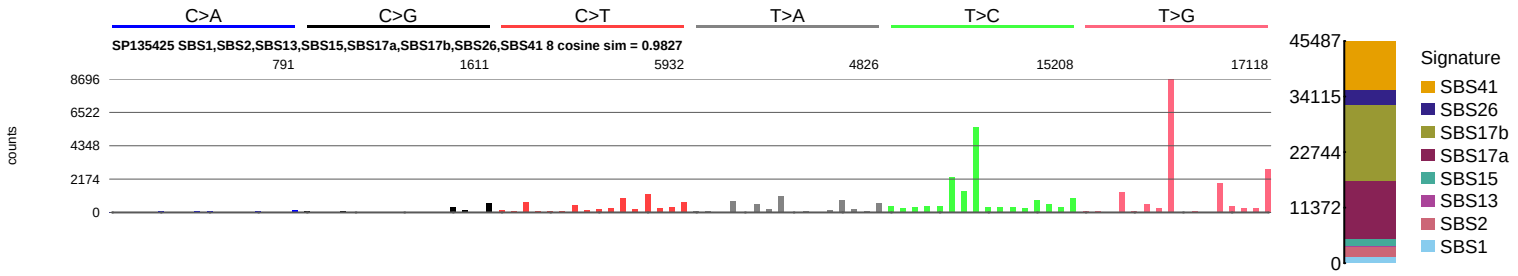

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98274
4685
1350
6271
3142
14112
15927
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

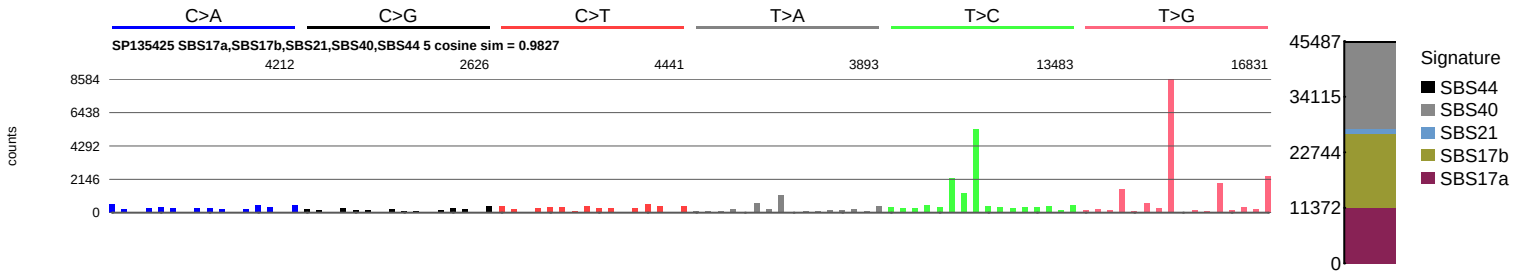

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98273
1831
2345
5123
4872
14428
16885
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1

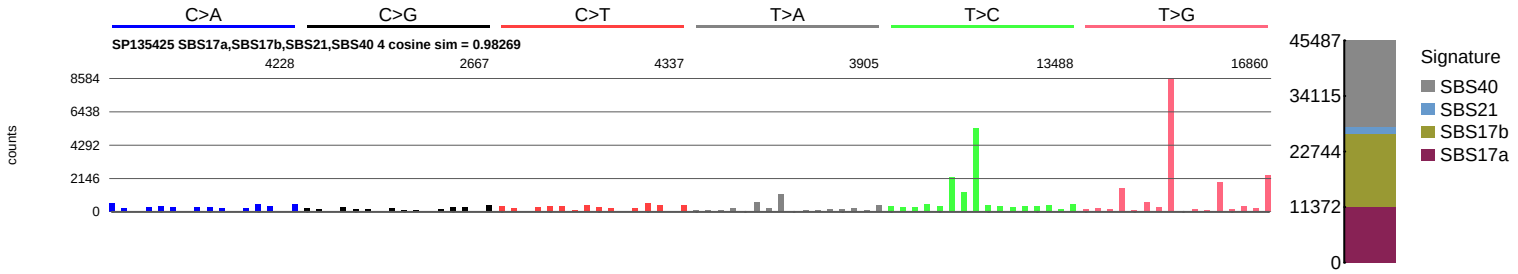

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98272
4704
1959
5875
3415
13509
16019
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

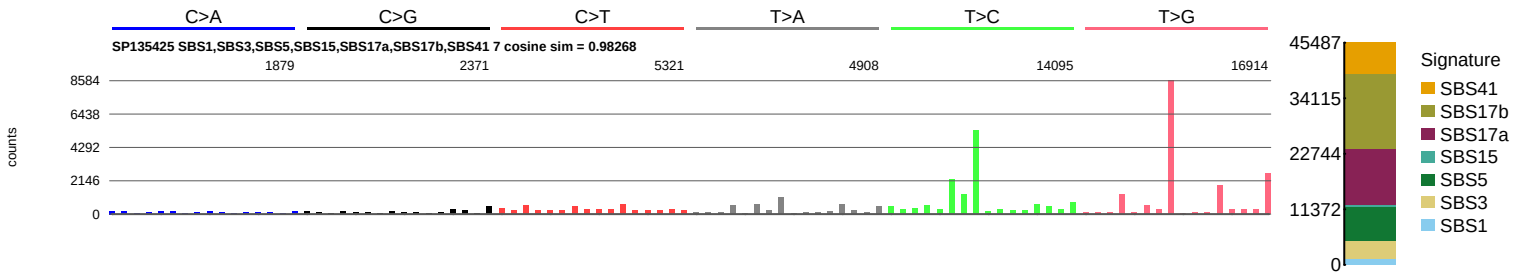

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS17a,SBS17b,SBS18,SBS41 5 cosine sim = 0.98271
4806
2384
2955
5298
13037
17007
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3

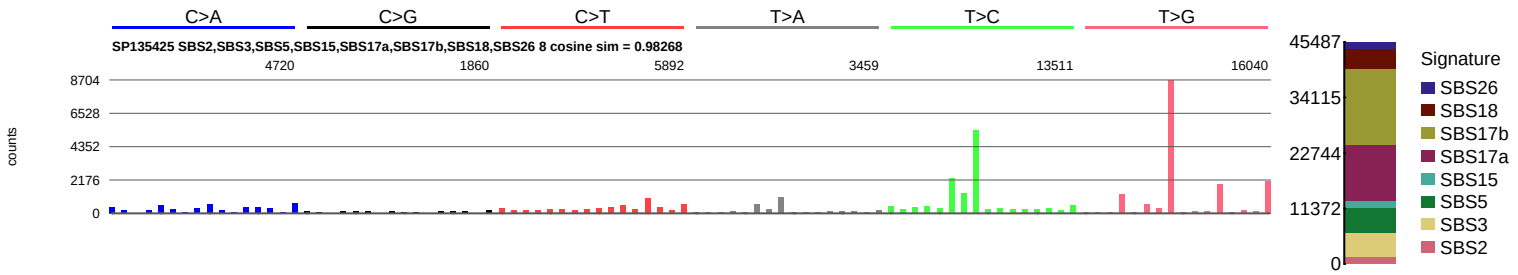

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98271
850
1667
6034
5031
14775
17132
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

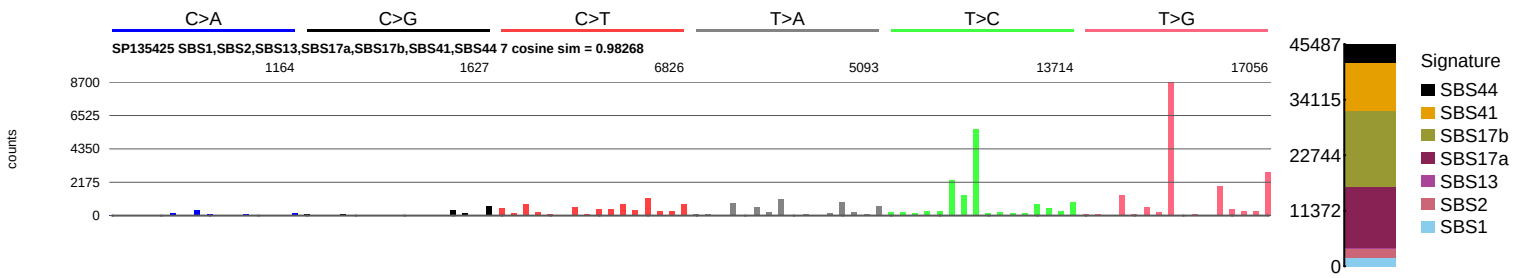

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.9827
791
1611
5932
4826
15208
17118
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

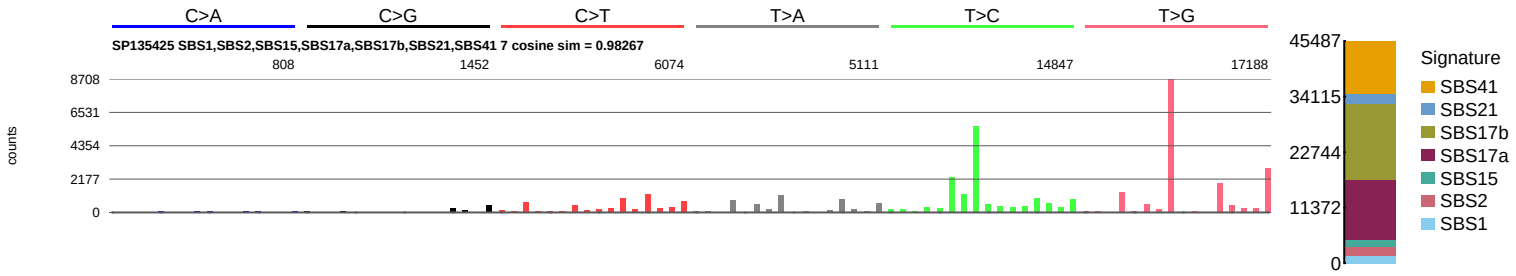

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS21,SBS40,SBS44 5 cosine sim = 0.9827
4212
2626
4441
3893
13483
16831
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a

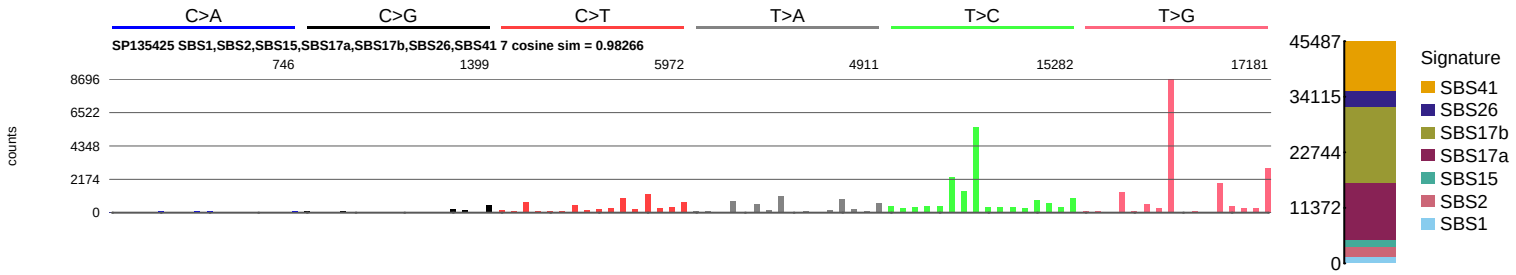

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS21,SBS40 4 cosine sim = 0.98269
4228
2667
4337
3905
13488
16860
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS21
SBS17b
SBS17a

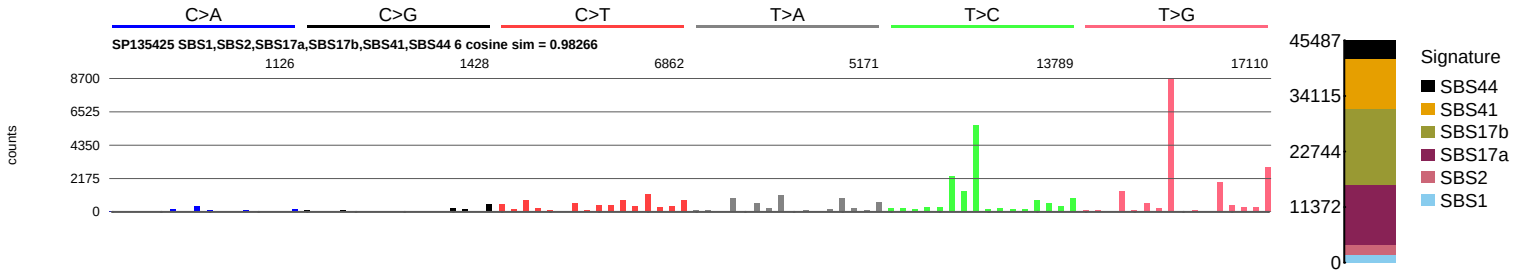

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98268
1879
2371
5321
4908
14095
16914
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

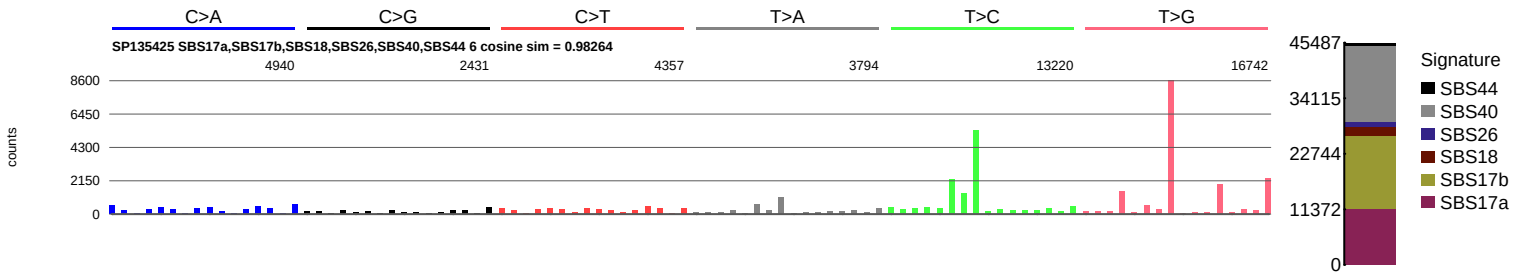

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98268
4720
1860
5892
3459
13511
16040
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

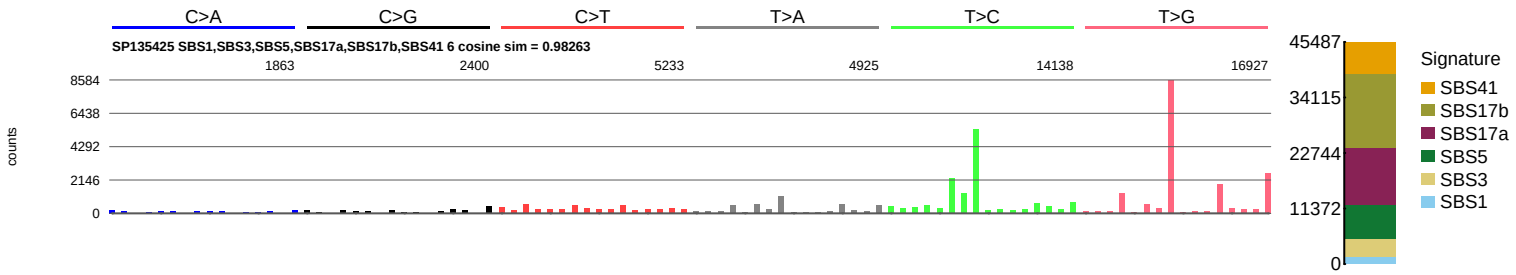

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98268
1164
1627
6826
5093
13714
17056
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1

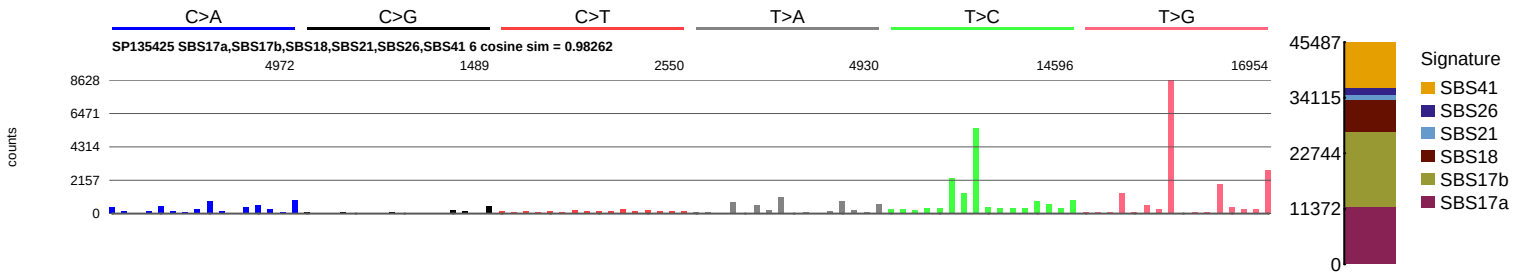

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98267
808
1452
6074
5111
14847
17188
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

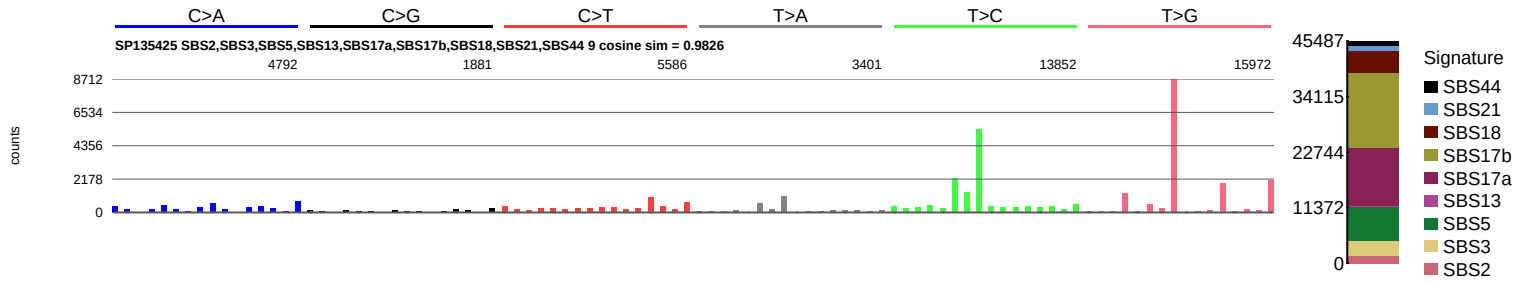

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98266
746
1399
5972
4911
15282
17181
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

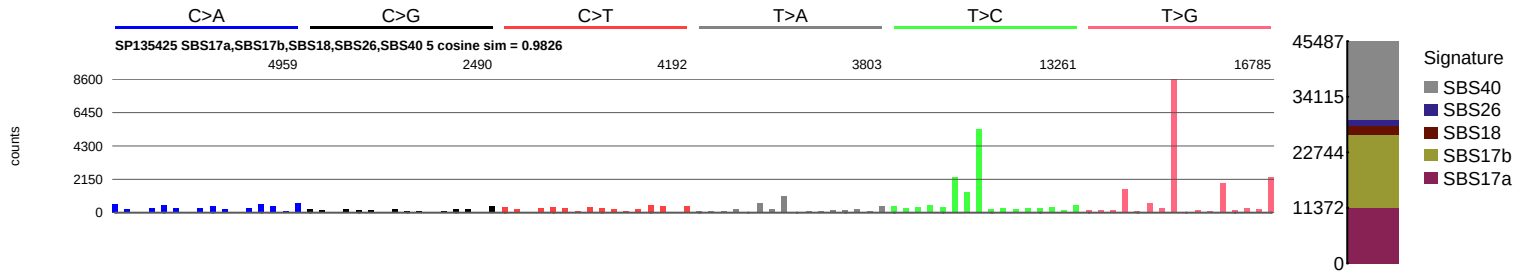

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98266
1126
1428
6862
5171
13789
17110
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2
SBS1

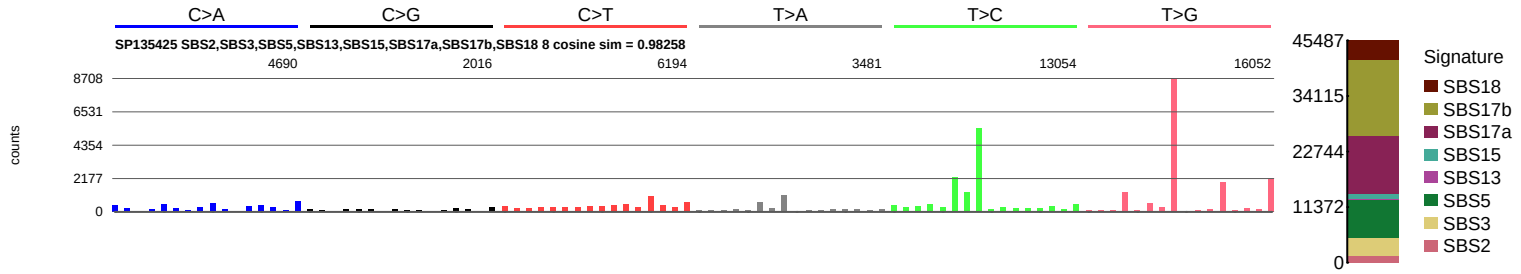

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 6 cosine sim = 0.98264
4940
2431
4357
3794
13220
16742
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98263
1863
2400
5233
4925
14138
16927
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 6 cosine sim = 0.98262
4972
1489
2550
4930
14596
16954
8628
6471
4314
2157
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.9826
4792
1881
5586
3401
13852
15972
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS40 5 cosine sim = 0.9826
4959
2490
4192
3803
13261
16785
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98258
4690
2016
6194
3481
13054
16052
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
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Signature
SBS18
SBS17b
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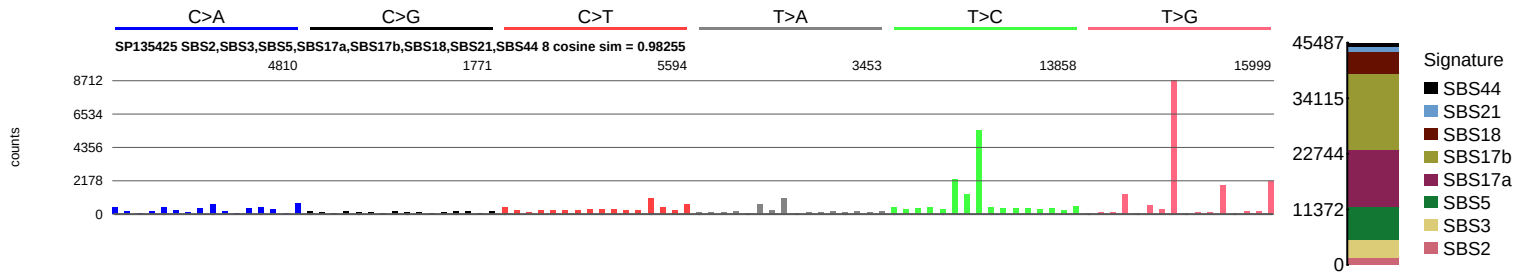
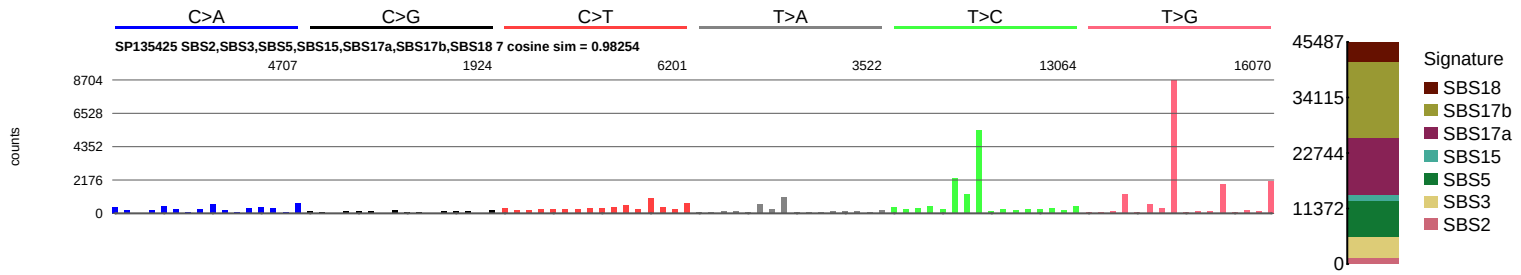
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98255
4810
1771
5594
3453
13858
15999
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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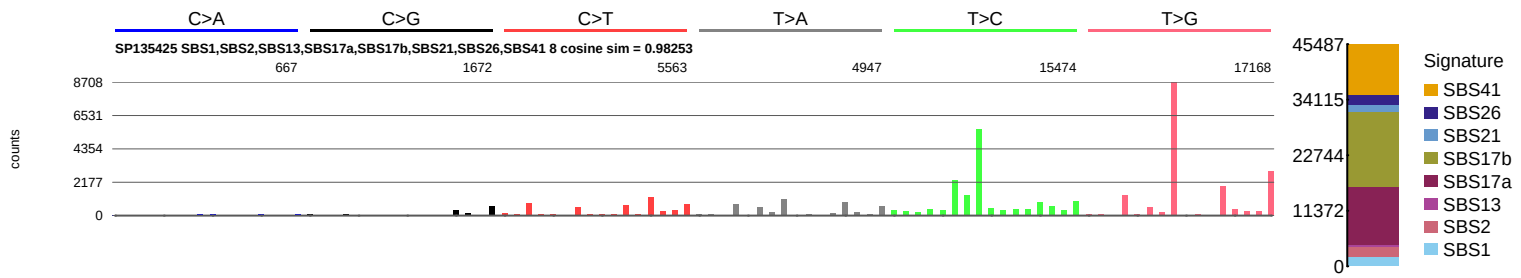
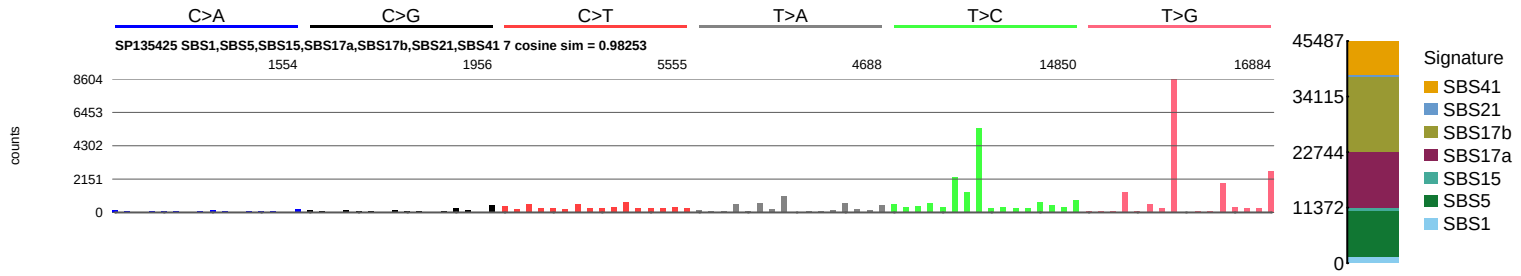
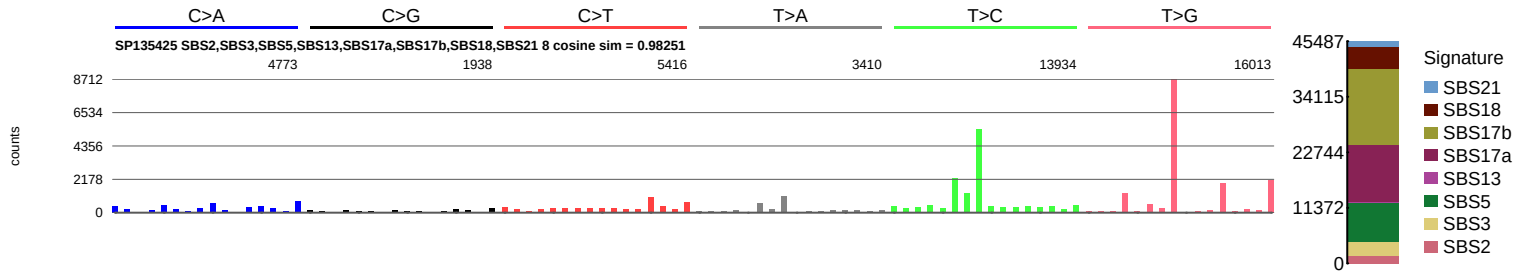
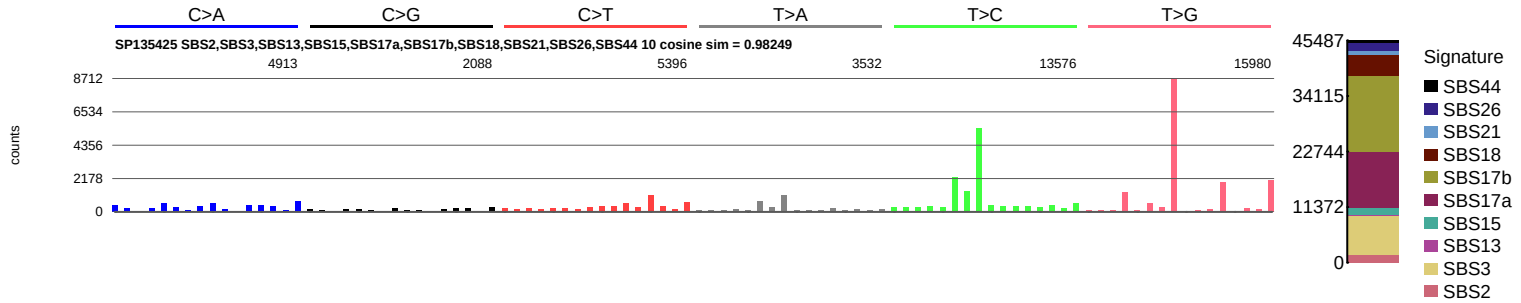
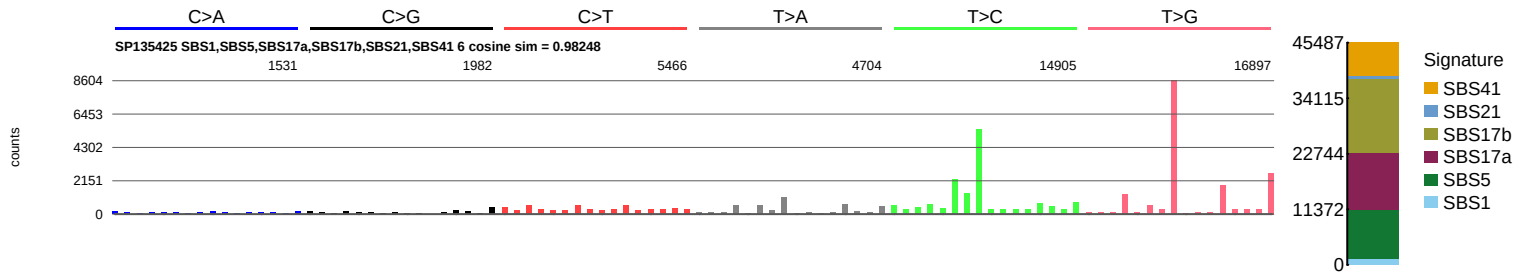
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98254
4707
1924
6201
3522
13064
16070
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

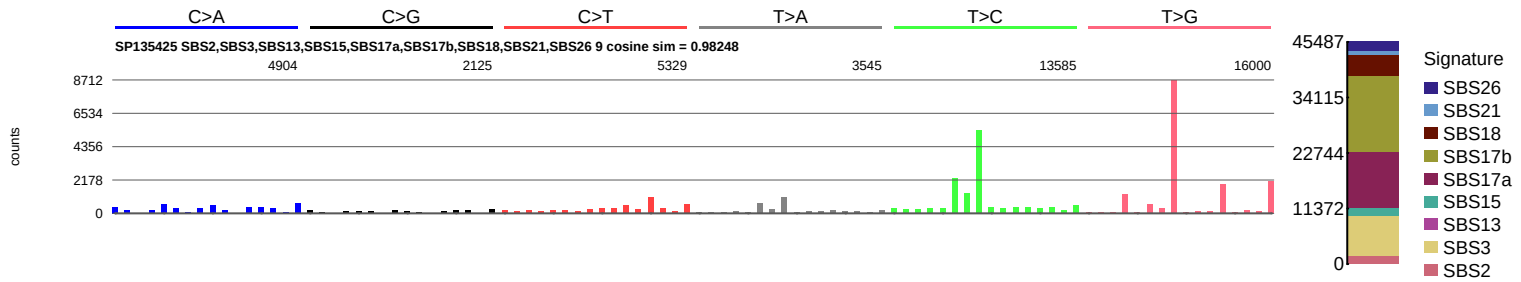

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98253
667
1672
5563
4947
15474
17168
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

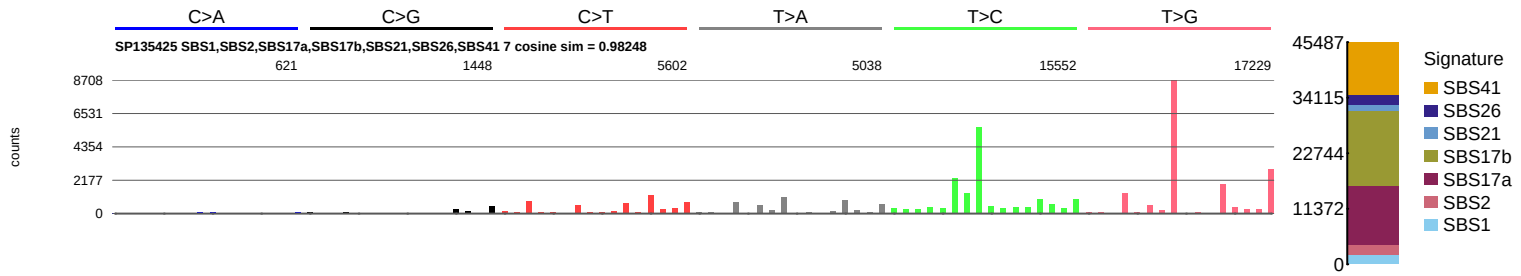

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98253
1554
1956
5555
4688
14850
16884
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

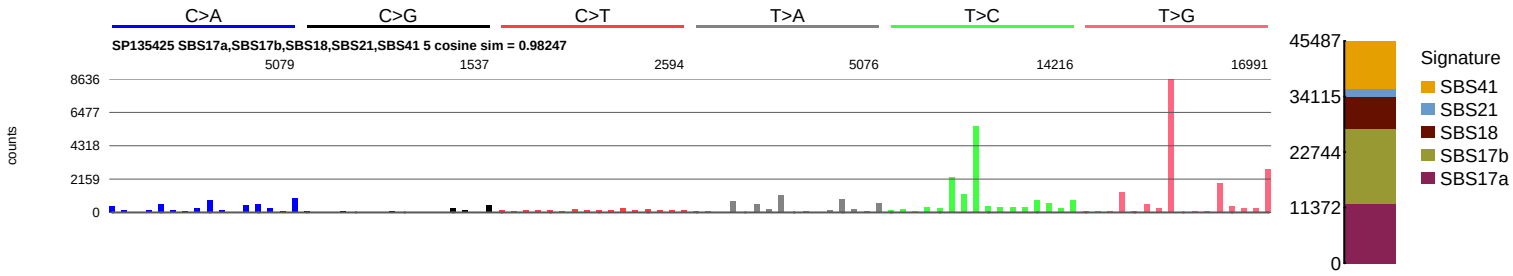

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98251
4773
1938
5416
3410
13934
16013
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2

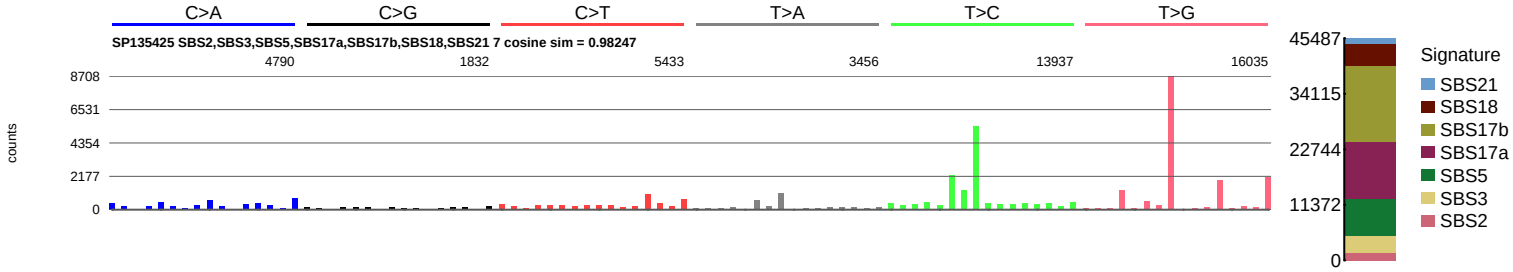

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98249
4913
2088
5396
3532
13576
15980
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

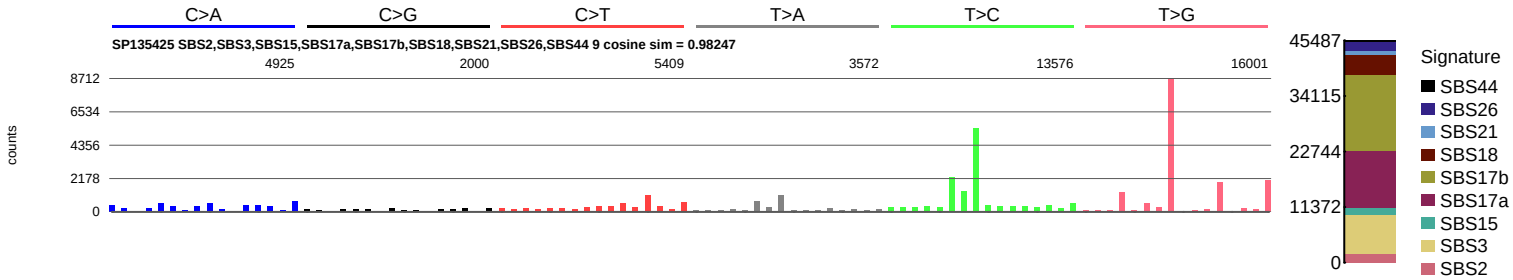

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98248
1531
1982
5466
4704
14905
16897
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS1

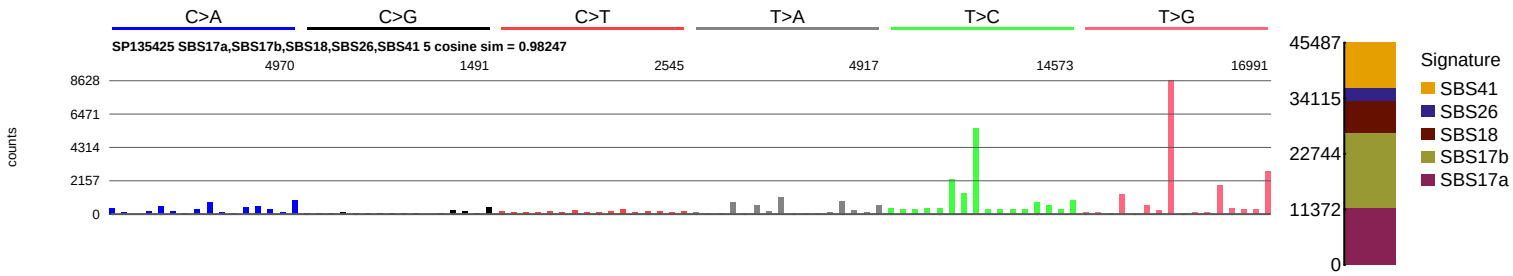

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98248
4904
2125
5329
3545
13585
16000
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

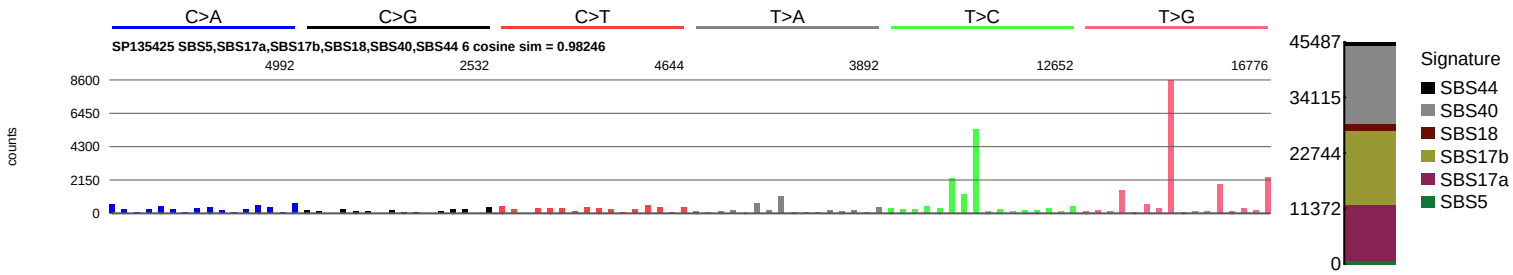

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98248
621
1448
5602
5038
15552
17229
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1

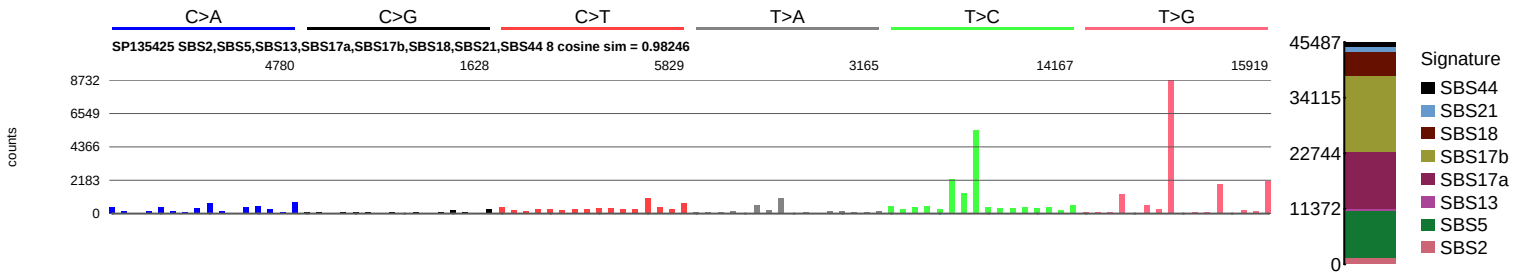

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS21,SBS41 5 cosine sim = 0.98247
5079
1537
2594
5076
14216
16991
8636
6477
4318
2159
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a

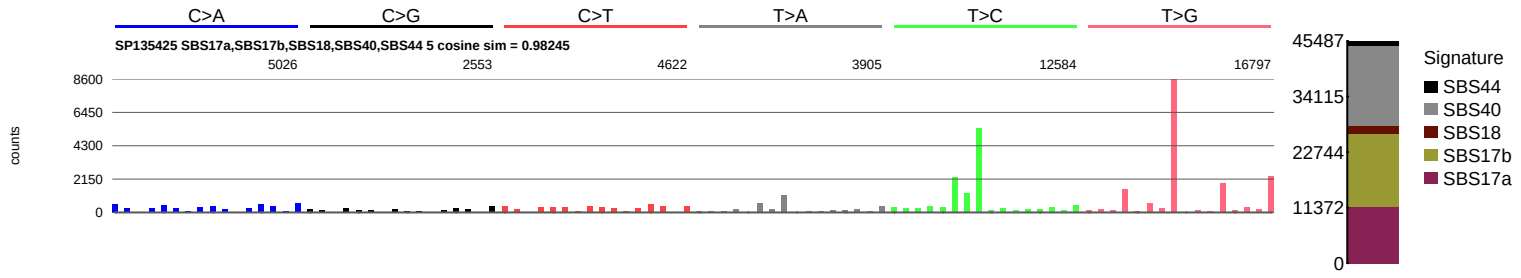

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98247
4790
1832
5433
3456
13937
16035
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

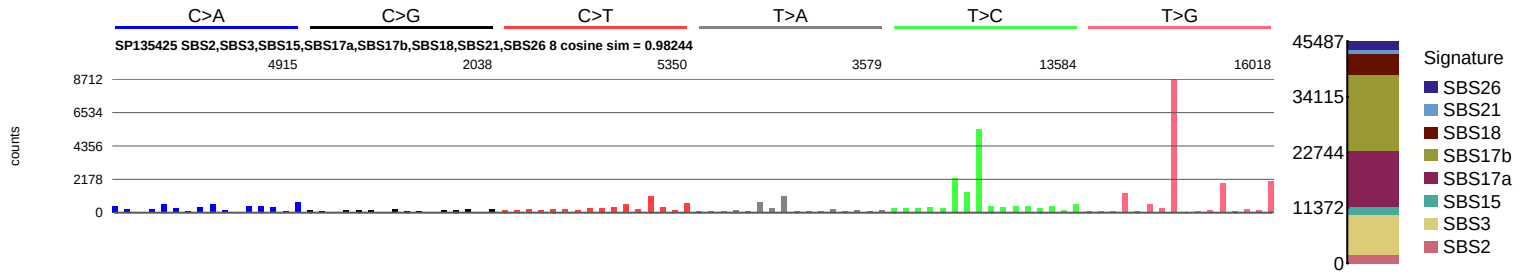

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98247
4925
2000
5409
3572
13576
16001
8712
6534
4356
2178
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

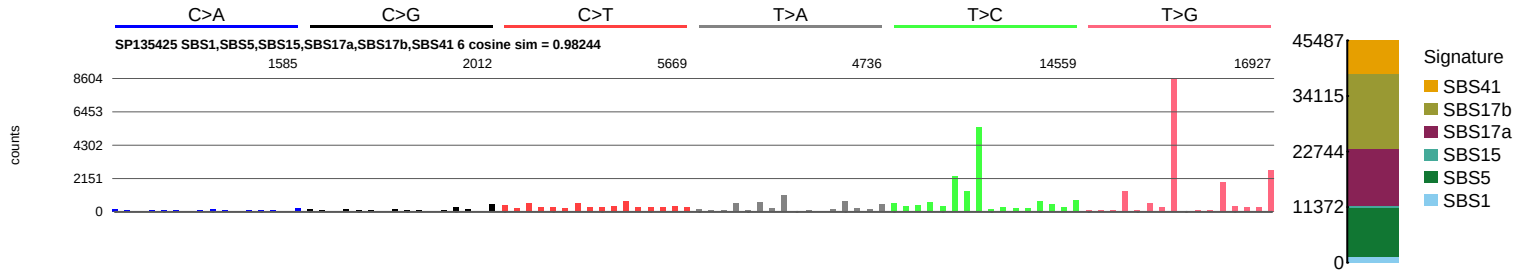

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS26,SBS41 5 cosine sim = 0.98247
4970
1491
2545
4917
14573
16991
8628
6471
4314
2157
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a

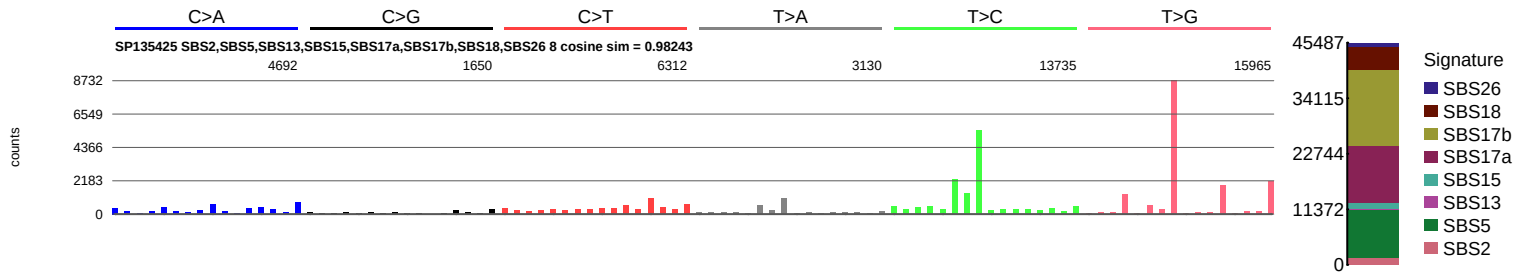

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98246
4992
2532
4644
3892
12652
16776
8600
6450
4300
2150
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5

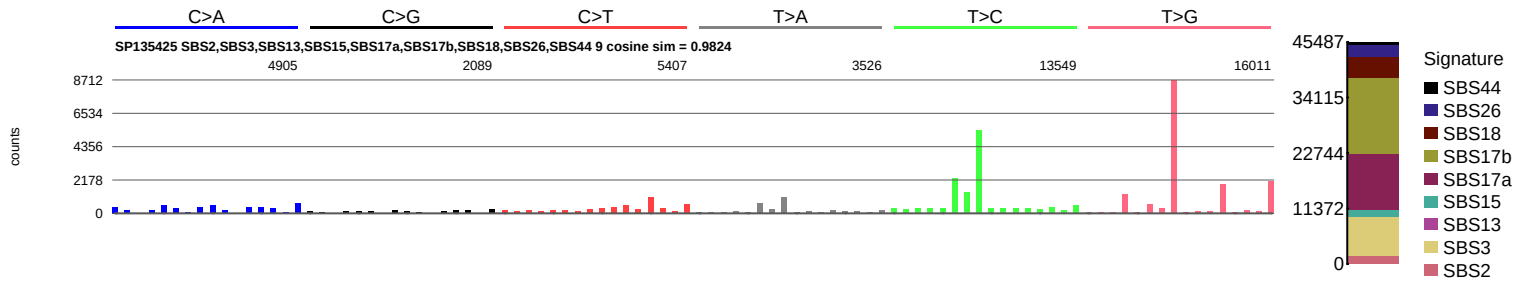

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98246
4780
1628
5829
3165
14167
15919
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2

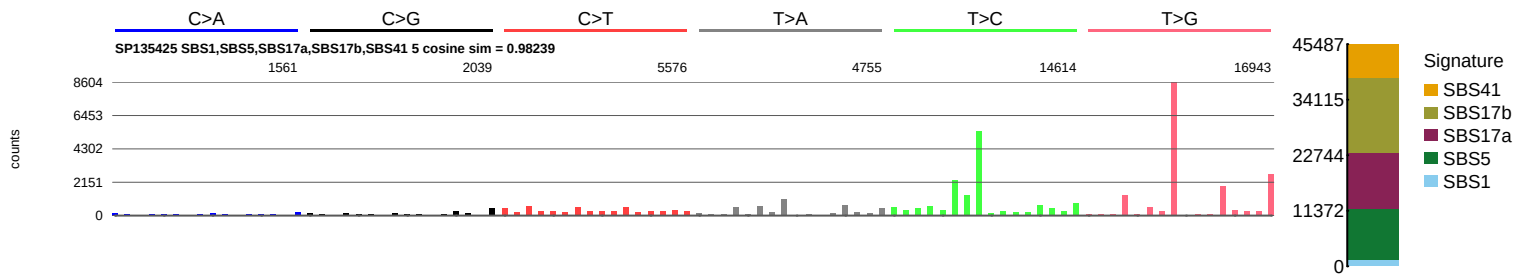

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS40,SBS44 5 cosine sim = 0.98245
5026
2553
4622
3905
12584
16797
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a

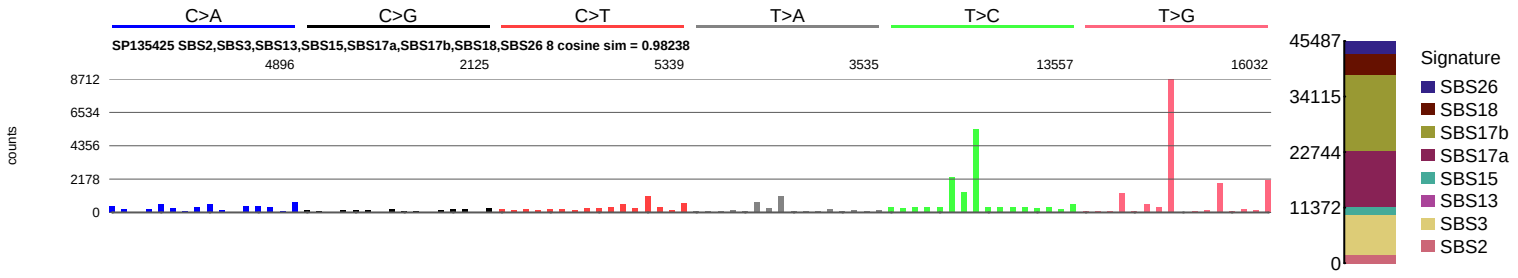

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98244
4915
2038
5350
3579
13584
16018
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

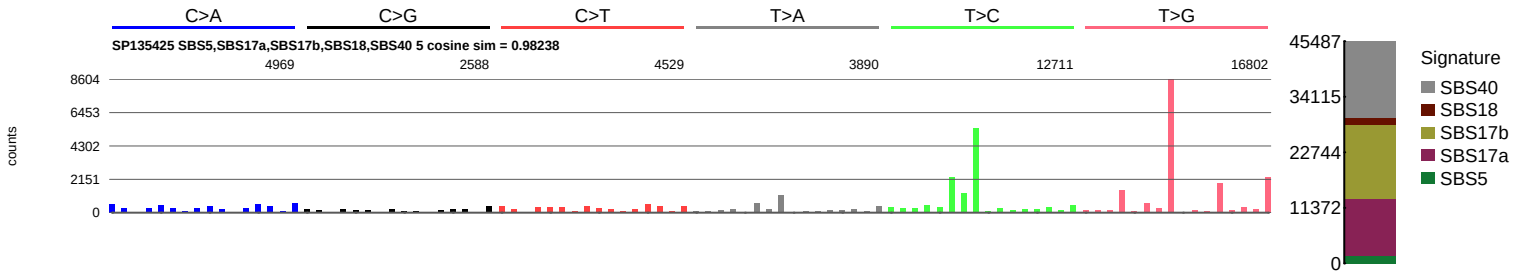

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98244
1585
2012
5669
4736
14559
16927
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1

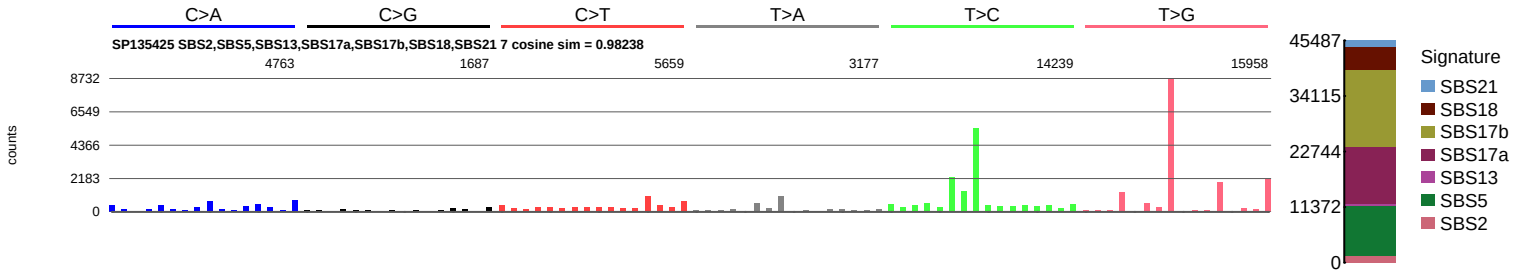

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98243
4692
1650
6312
3130
13735
15965
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2

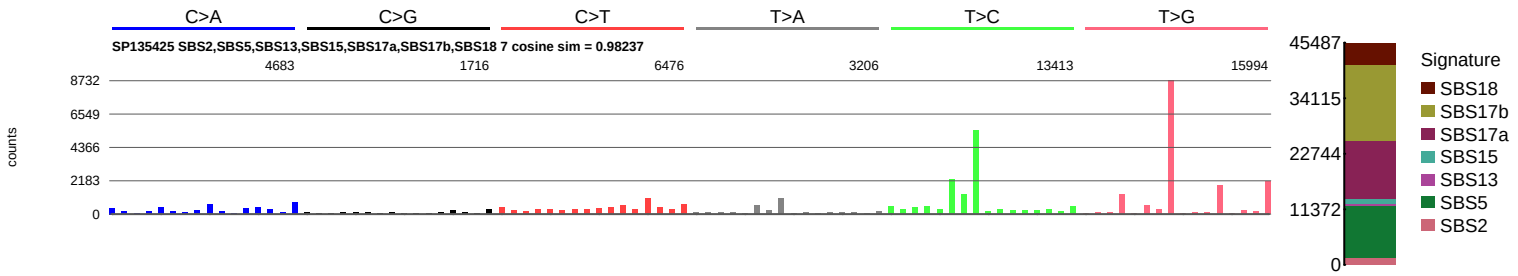

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.9824
4905
2089
5407
3526
13549
16011
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

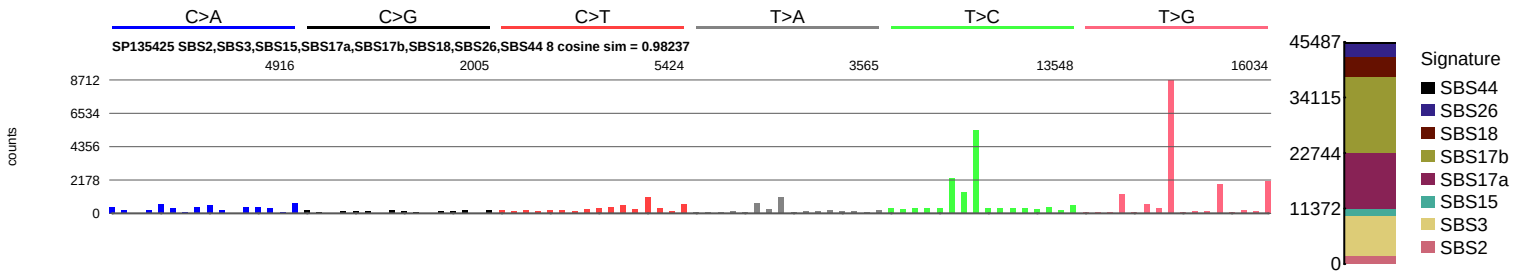

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98239
1561
2039
5576
4755
14614
16943
counts
8604
6453
4302
2151
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1

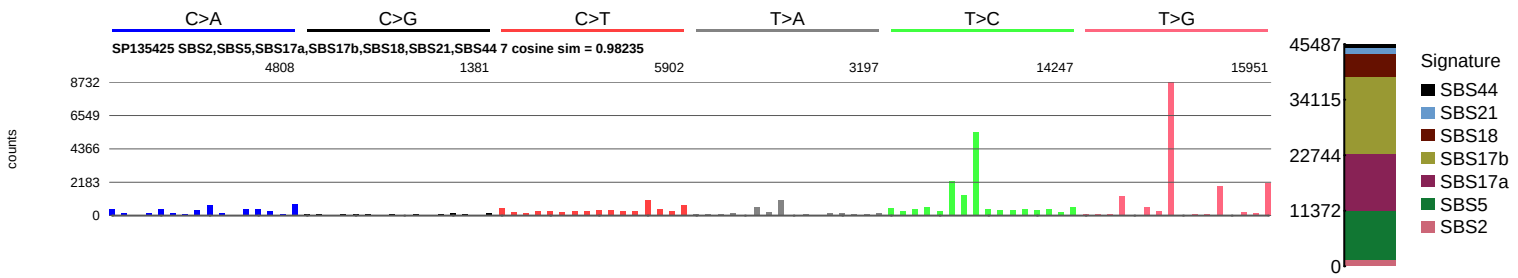

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98238
4896
2125
5339
3535
13557
16032
8712
6534
4356
2178
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

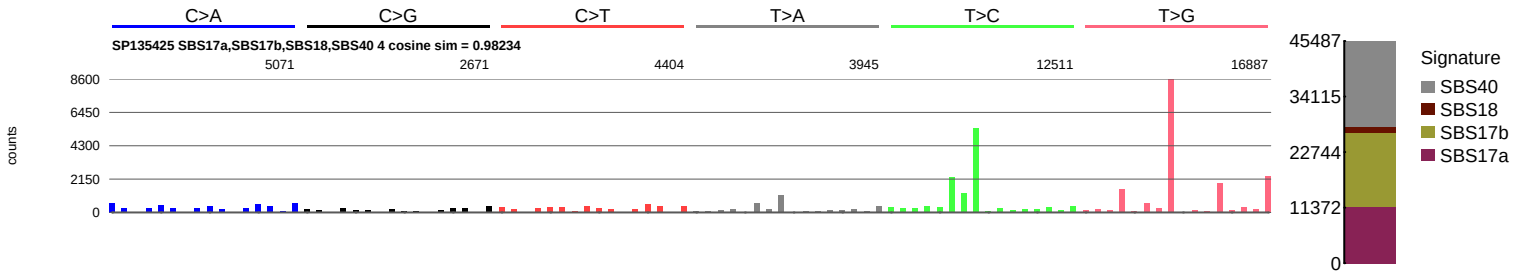

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98238
4969
2588
4529
3890
12711
16802
8604
6453
4302
2151
0
counts
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98238
4763
1687
5659
3177
14239
15958
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98237
4683
1716
6476
3206
13413
15994
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98237
4916
2005
5424
3565
13548
16034
8712
6534
4356
2178
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98235
4808
1381
5902
3197
14247
15951
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2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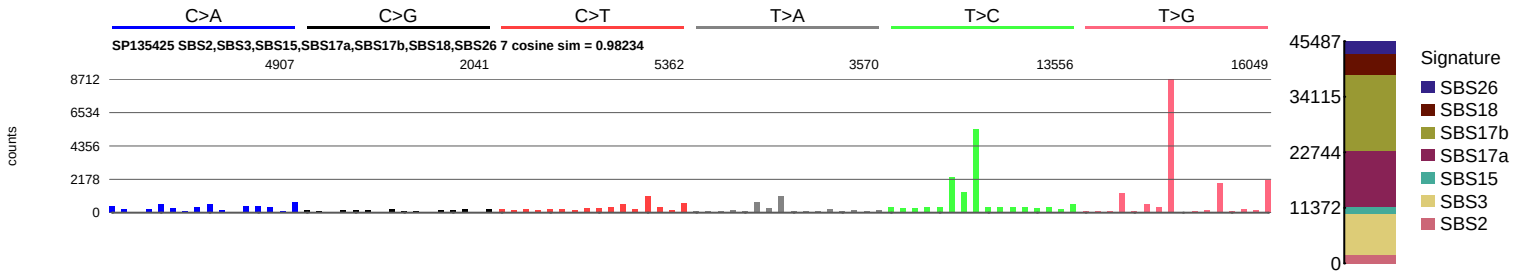
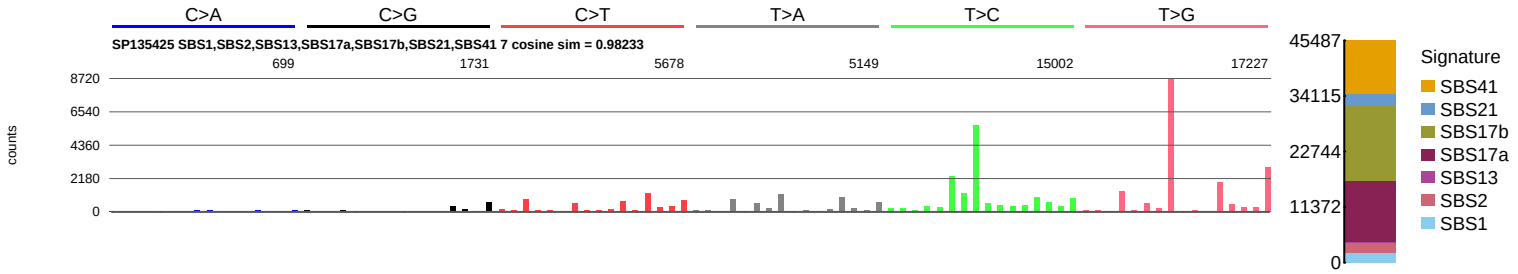
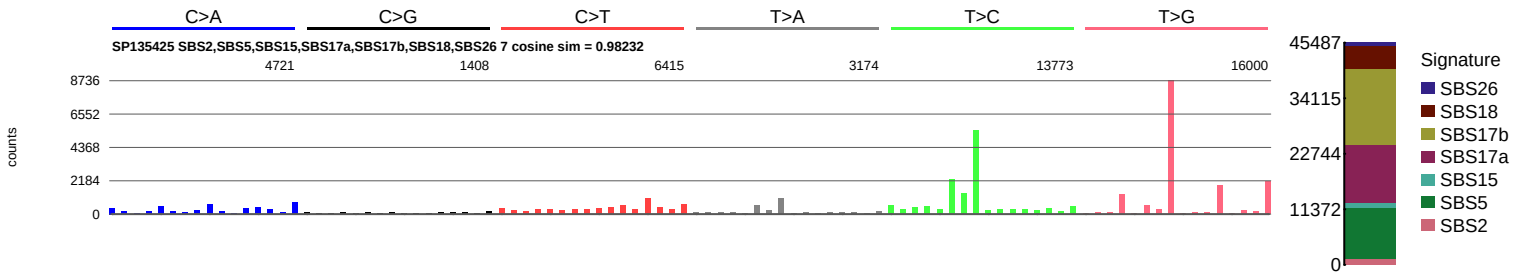
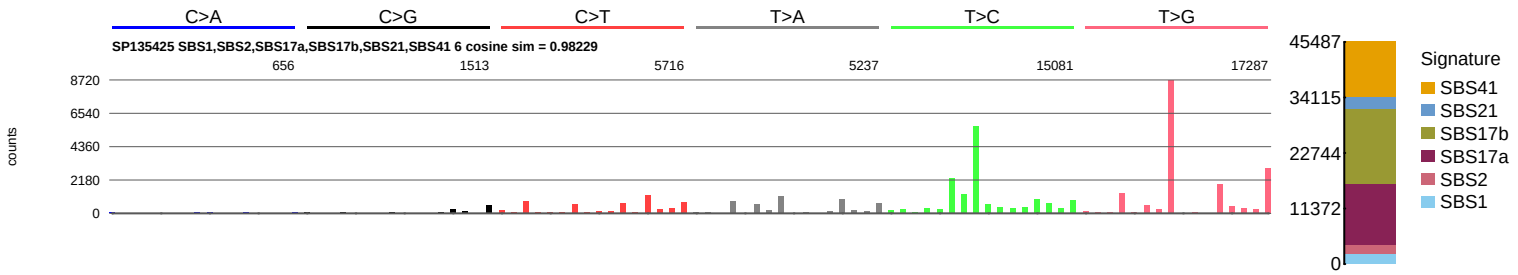
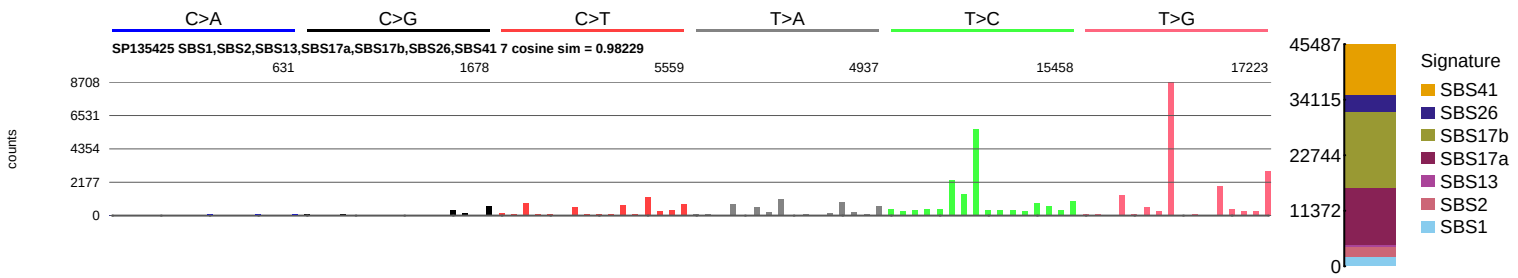
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS40 4 cosine sim = 0.98234
5071
2671
4404
3945
12511
16887
counts
8600
6450
4300
2150
0
45487
34115
22744
11372
0
Signature
SBS40
SBS18
SBS17b
SBS17a

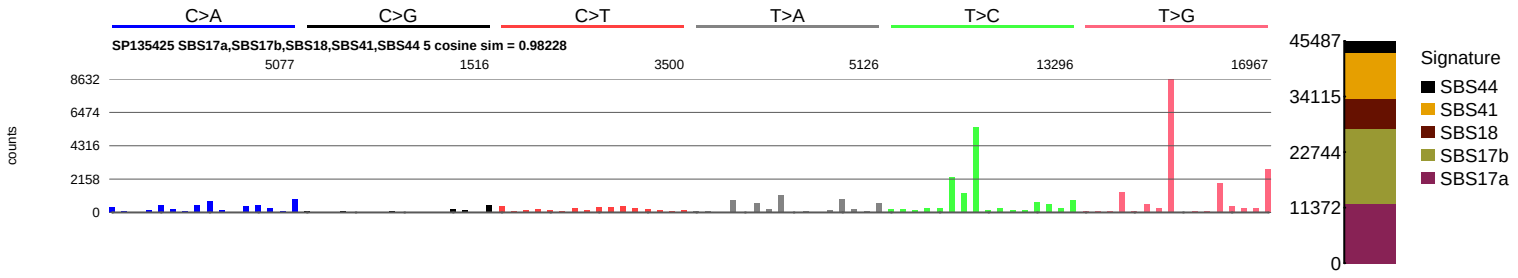

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98234
4907
2041
5362
3570
13556
16049
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

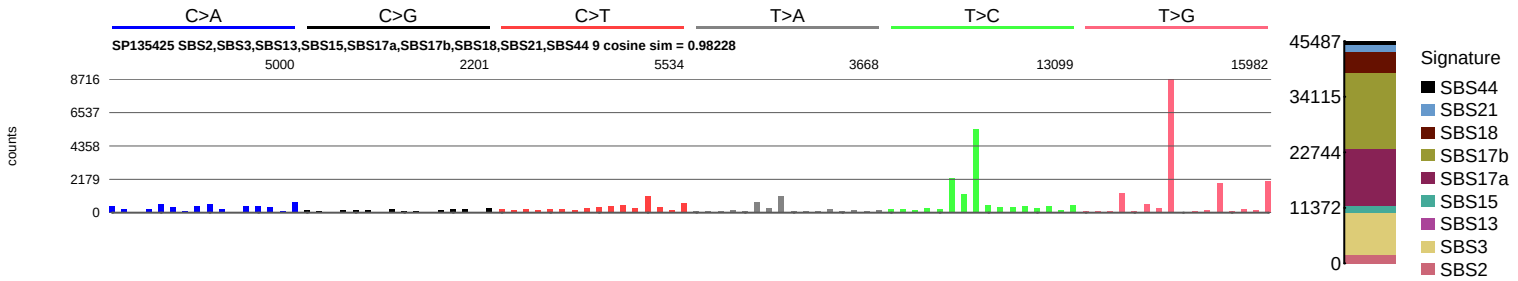

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98233
699
1731
5678
5149
15002
17227
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1

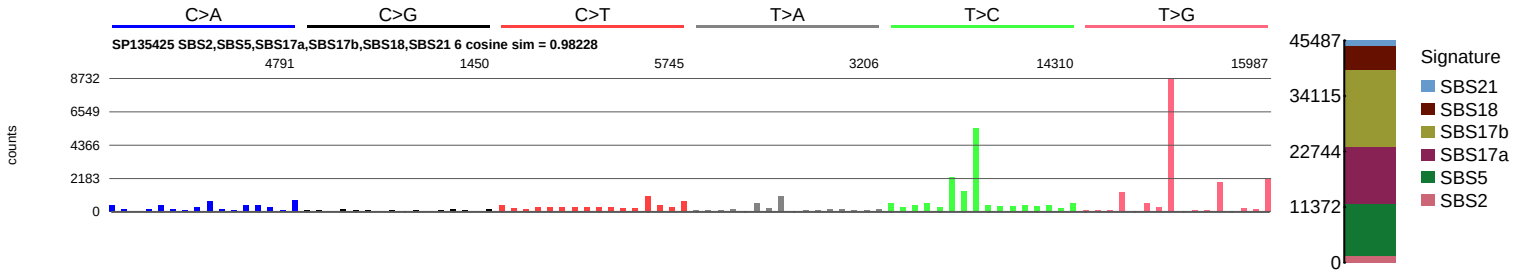

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98232
4721
1408
6415
3174
13773
16000
counts
8736
6552
4368
2184
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2

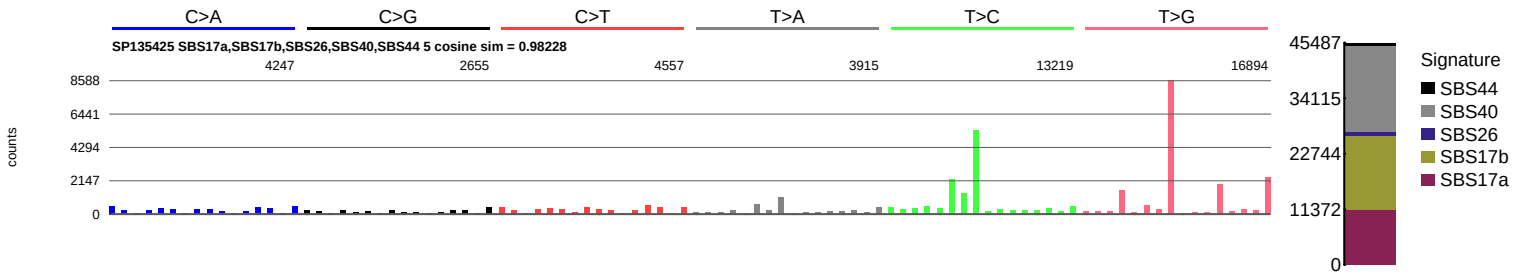

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98229
656
1513
5716
5237
15081
17287
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS2
SBS1

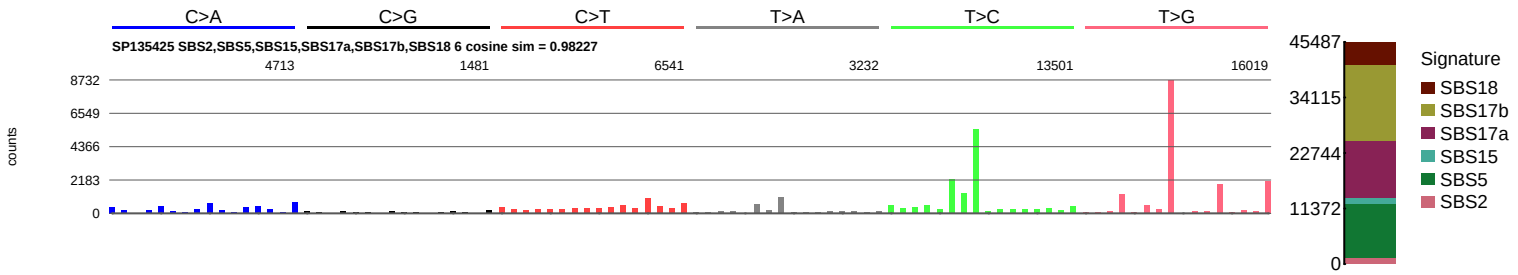

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98229
631
1678
5559
4937
15458
17223
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

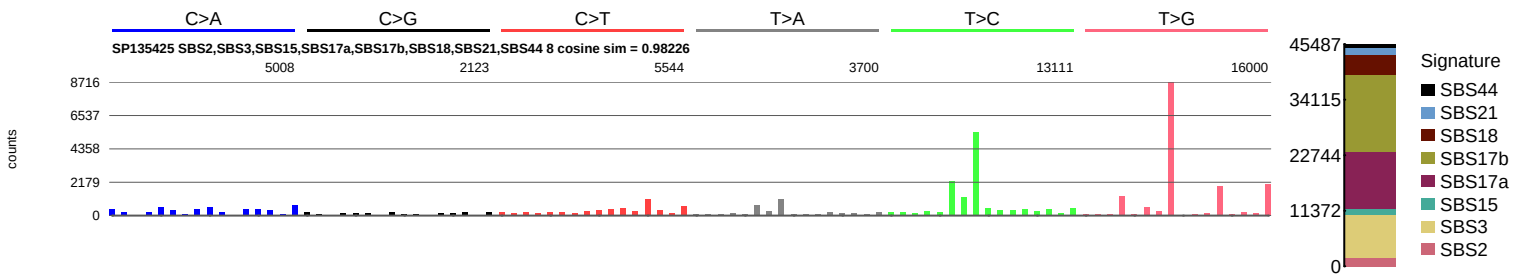

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS41,SBS44 5 cosine sim = 0.98228
5077
1516
3500
5126
13296
16967
8632
6474
4316
2158
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a

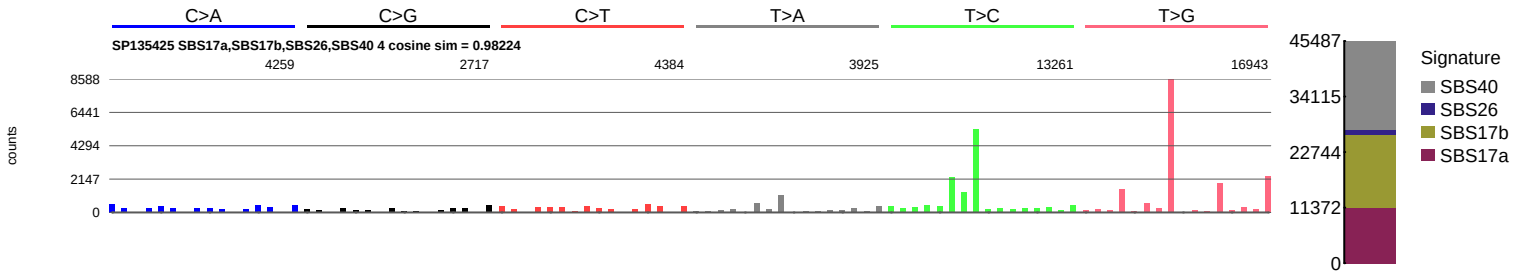

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98228
5000
2201
5534
3668
13099
15982
8716
6537
4358
2179
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

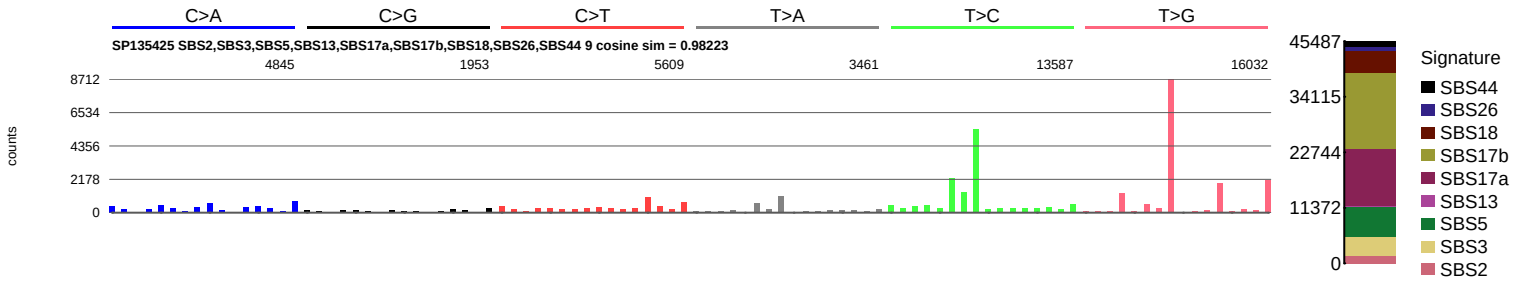

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98228
4791
1450
5745
3206
14310
15987
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS26,SBS40,SBS44 5 cosine sim = 0.98228
4247
2655
4557
3915
13219
16894
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98227
4713
1481
6541
3232
13501
16019
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98226
5008
2123
5544
3700
13111
16000
8716
6537
4358
2179
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS26,SBS40 4 cosine sim = 0.98224
4259
2717
4384
3925
13261
16943
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS40
SBS26
SBS17b
SBS17a


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98223
4845
1953
5609
3461
13587
16032
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
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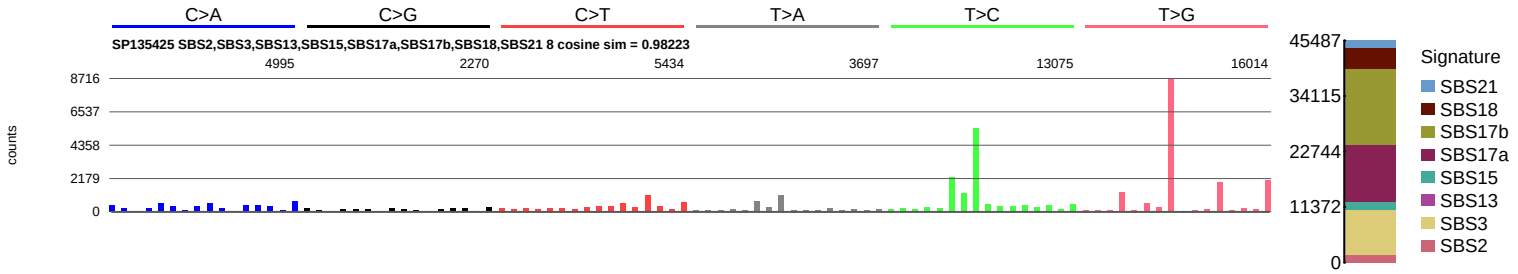
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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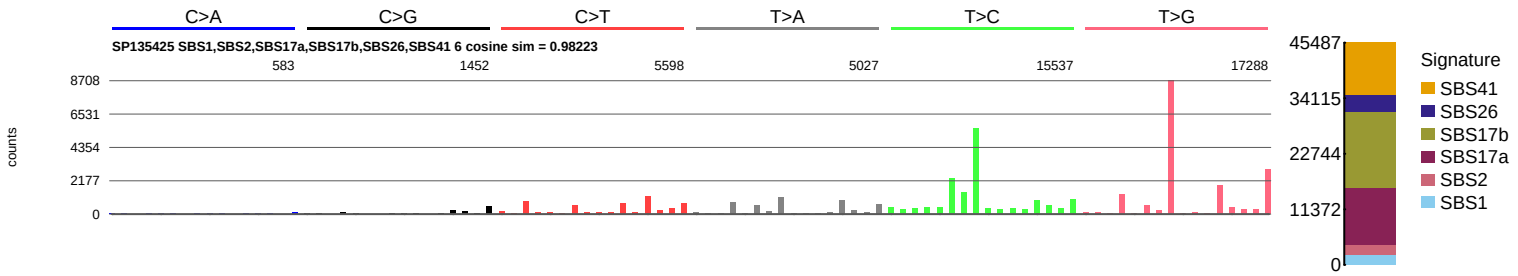
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98223
4995
2270
5434
3697
13075
16014
counts
8716
6537
4358
2179
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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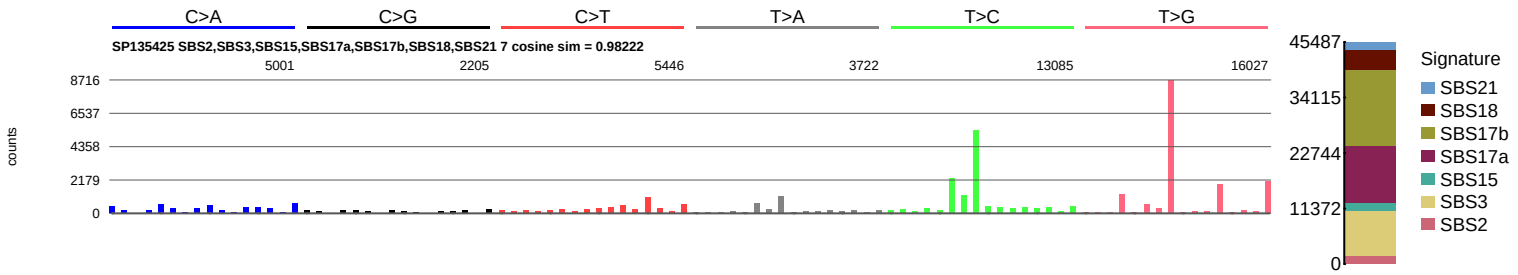
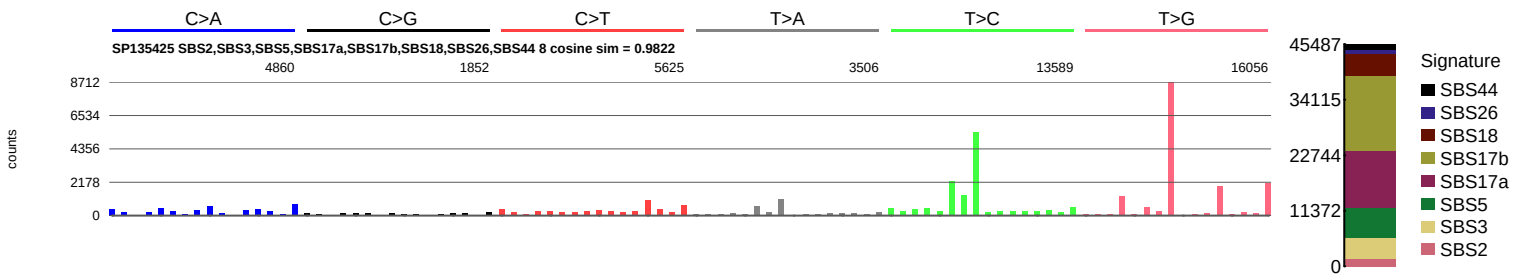
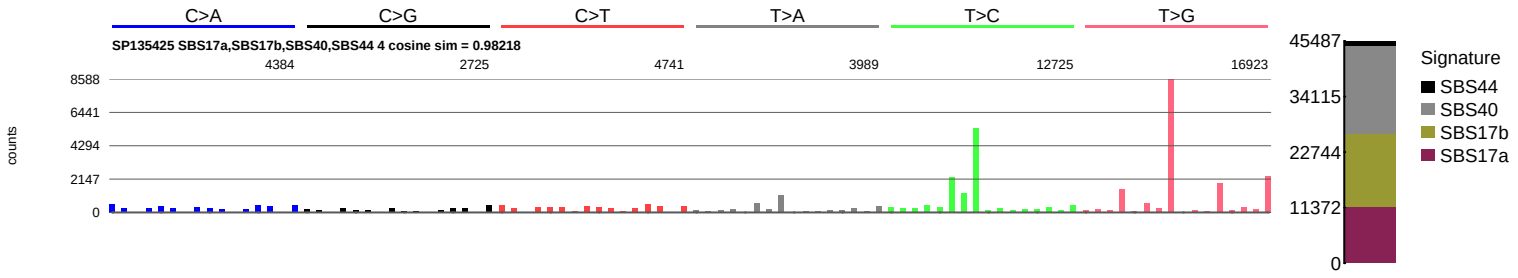
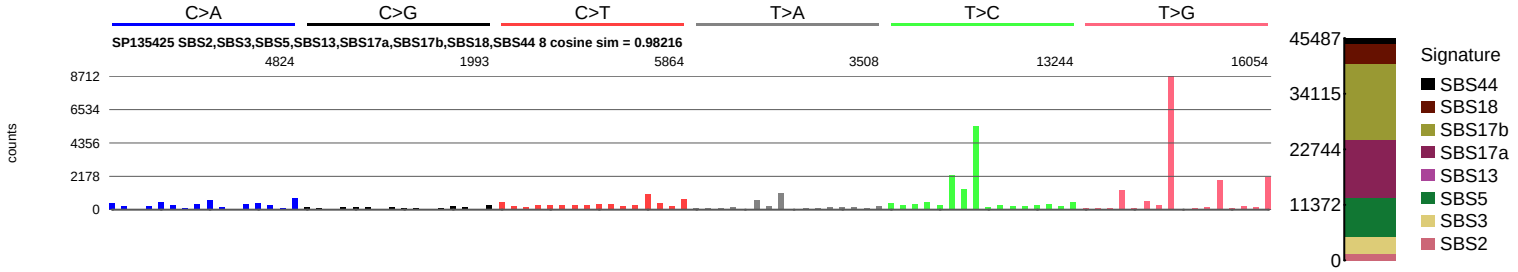
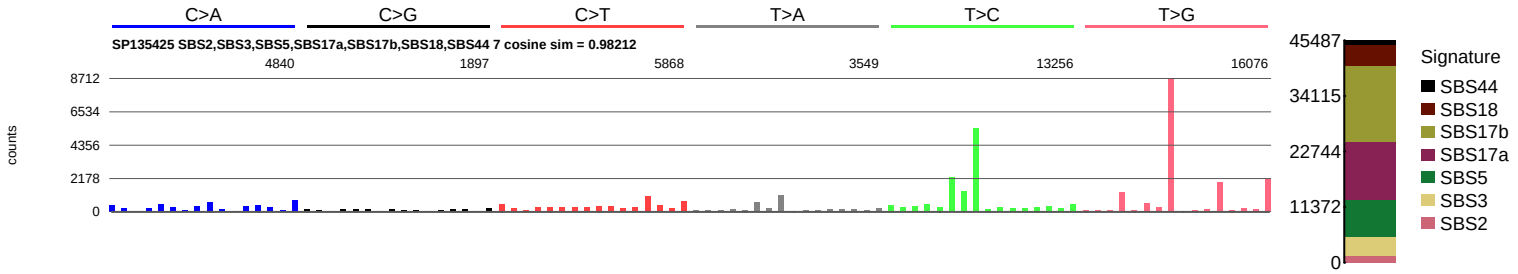
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98223
583
1452
5598
5027
15537
17288
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS2
SBS1

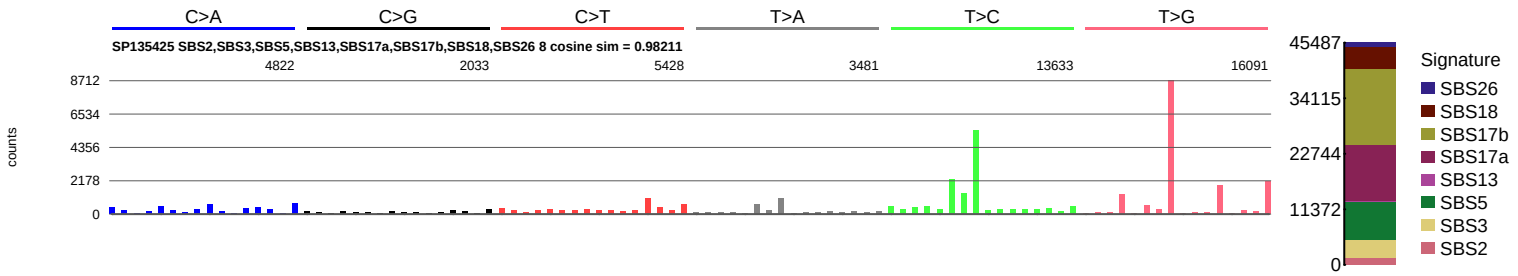

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98222
5001
2205
5446
3722
13085
16027
counts
8716
6537
4358
2179
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

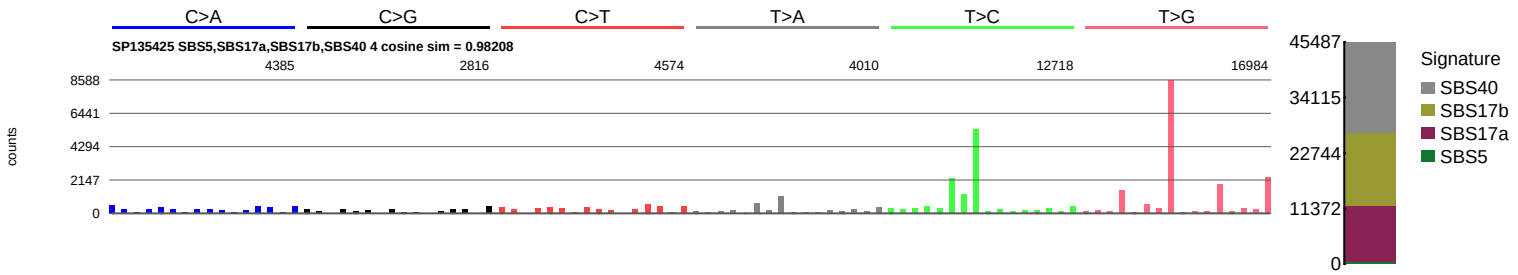

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9822
4860
1852
5625
3506
13589
16056
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2

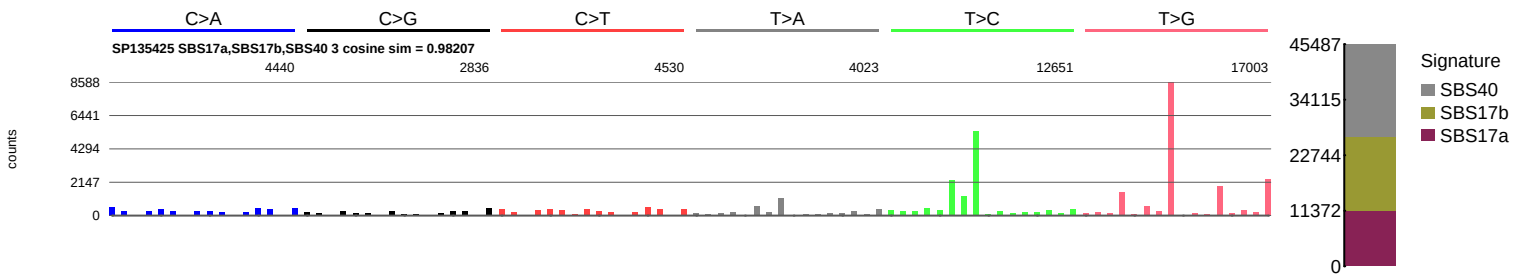

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS40,SBS44 4 cosine sim = 0.98218
4384
2725
4741
3989
12725
16923
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS44
SBS40
SBS17b
SBS17a

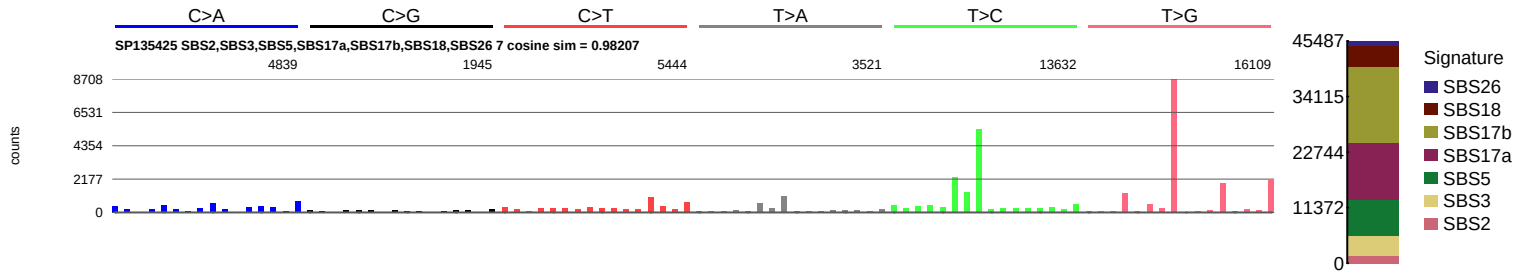

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98216
4824
1993
5864
3508
13244
16054
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98212
4840
1897
5868
3549
13256
16076
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98211
4822
2033
5428
3481
13633
16091
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98208
4385
2816
4574
4010
12718
16984
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS40 3 cosine sim = 0.98207
4440
2836
4530
4023
12651
17003
counts
8588
6441
4294
2147
0
45487
34115
22744
11372
0
Signature
SBS40
SBS17b
SBS17a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C>G
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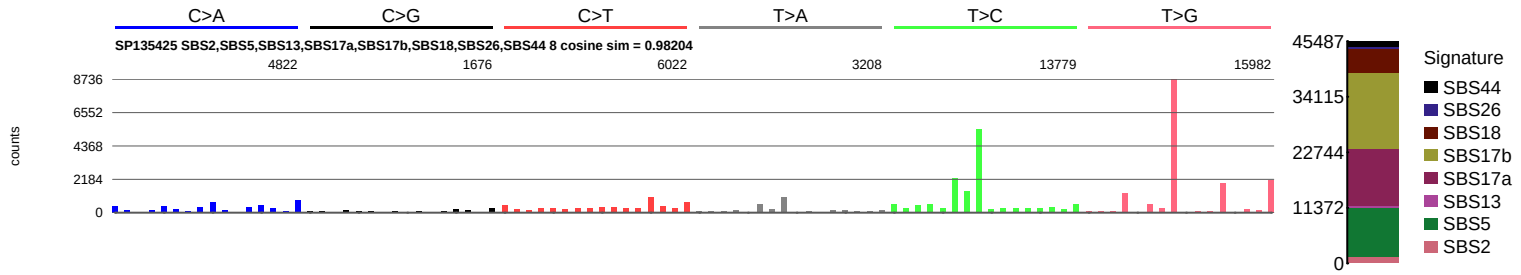
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98207
4839
1945
5444
3521
13632
16109
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2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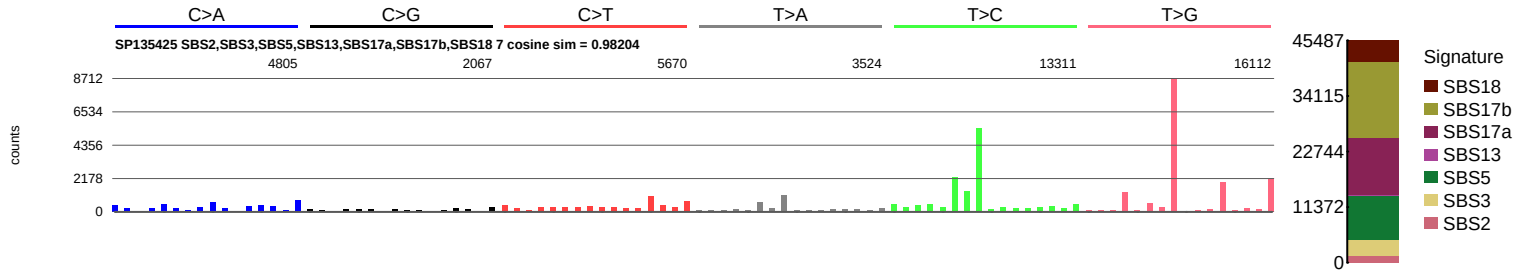
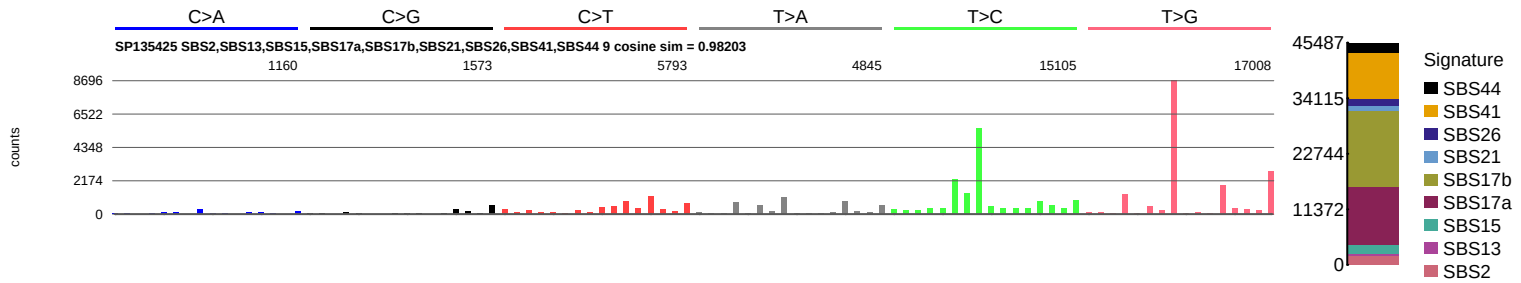
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98204
4822
1676
6022
3208
13779
15982
counts
8736
6552
4368
2184
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98204
4805
2067
5670
3524
13311
16112
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2


C>A
C>G
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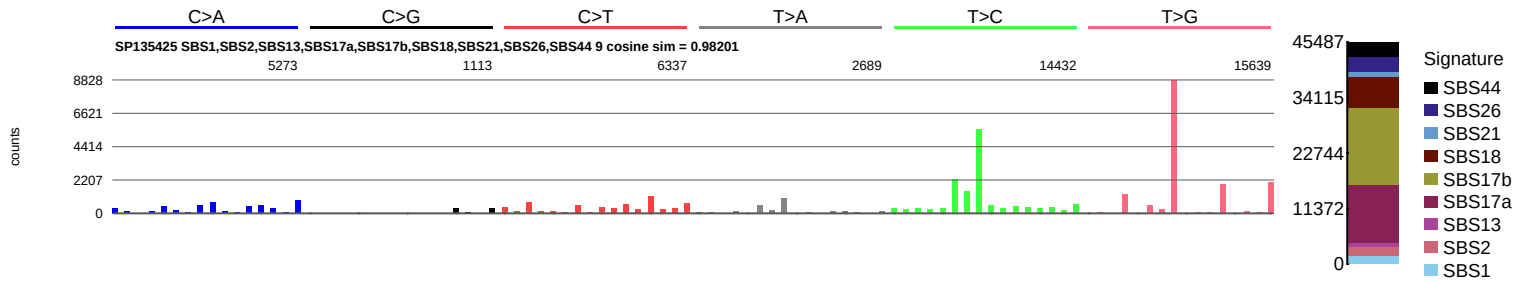
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98203
1160
1573
5793
4845
15105
17008
counts
8696
6522
4348
2174
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2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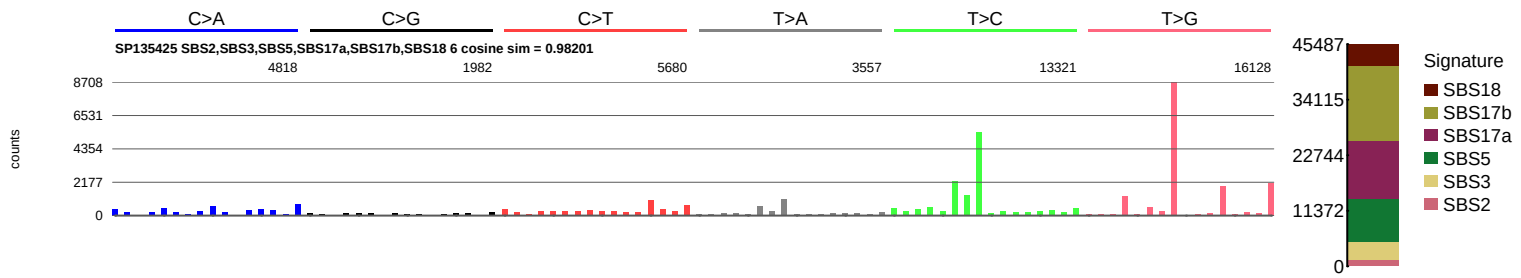
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98201
5273
1113
6337
2689
14432
15639
counts
8828
6621
4414
2207
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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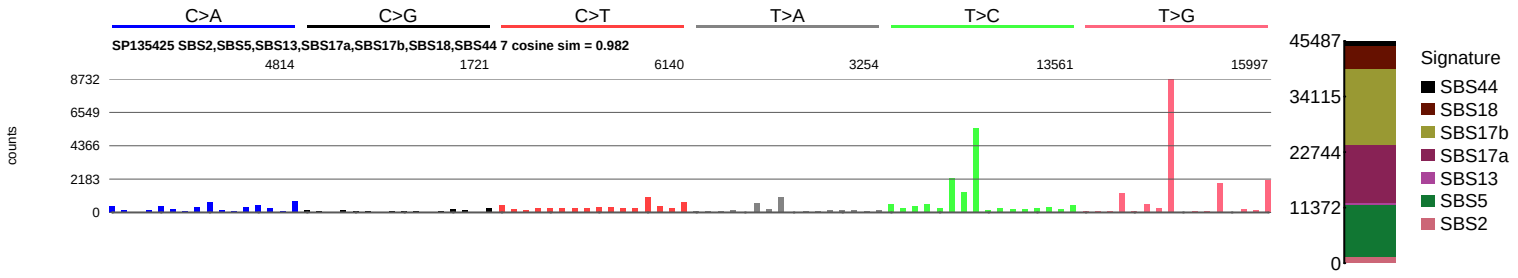
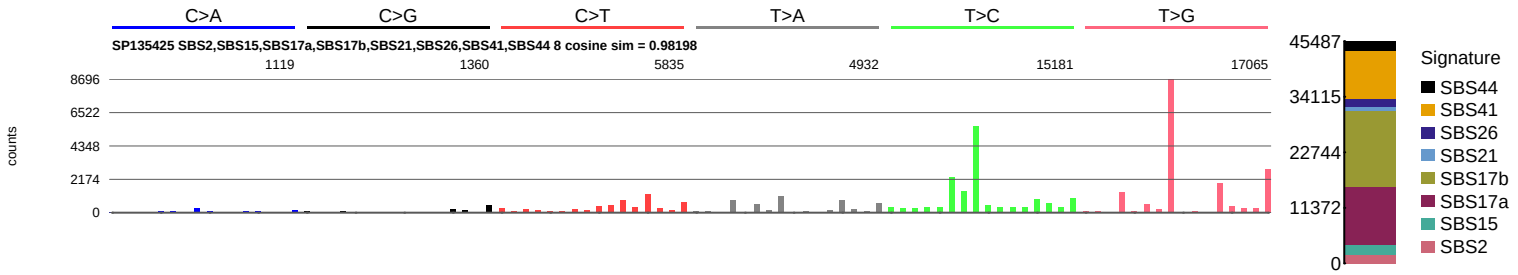
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98201
4818
1982
5680
3557
13321
16128
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.982
4814
1721
6140
3254
13561
15997
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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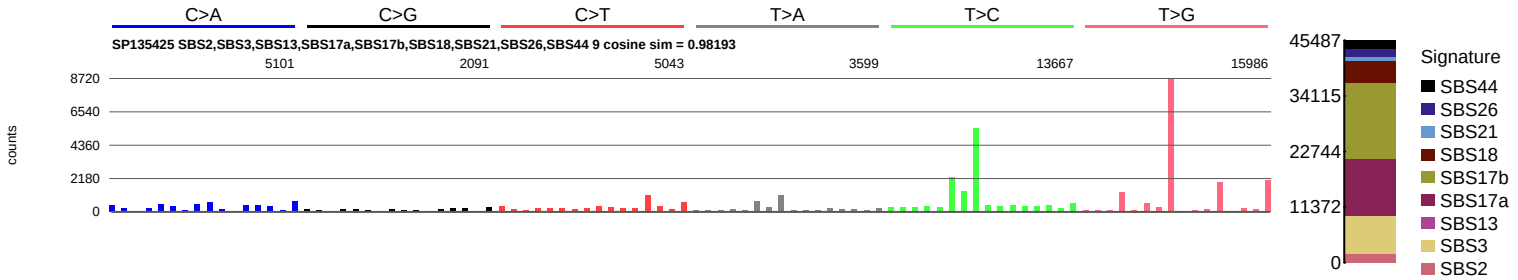
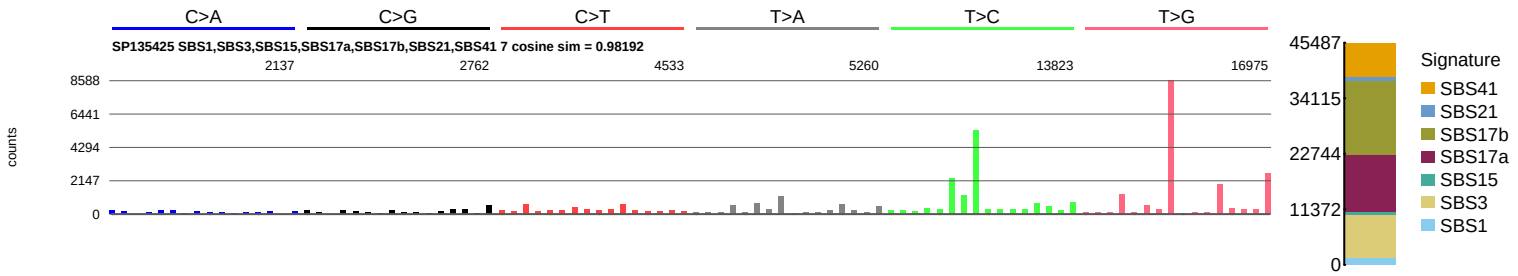
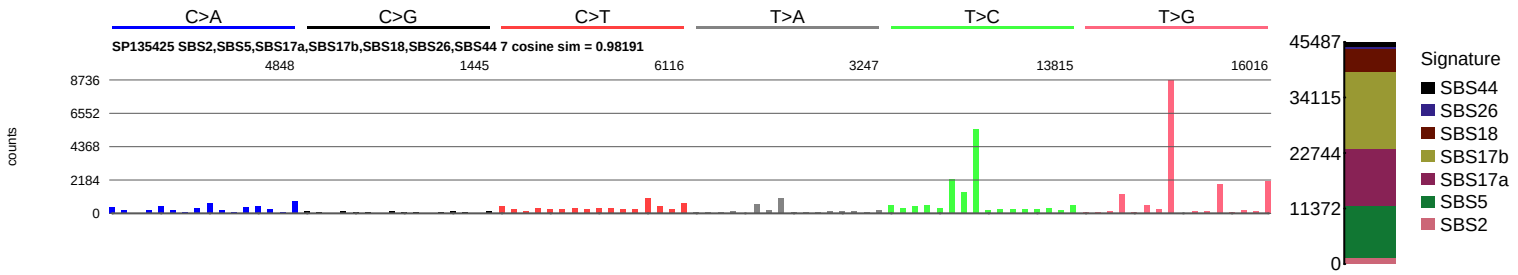
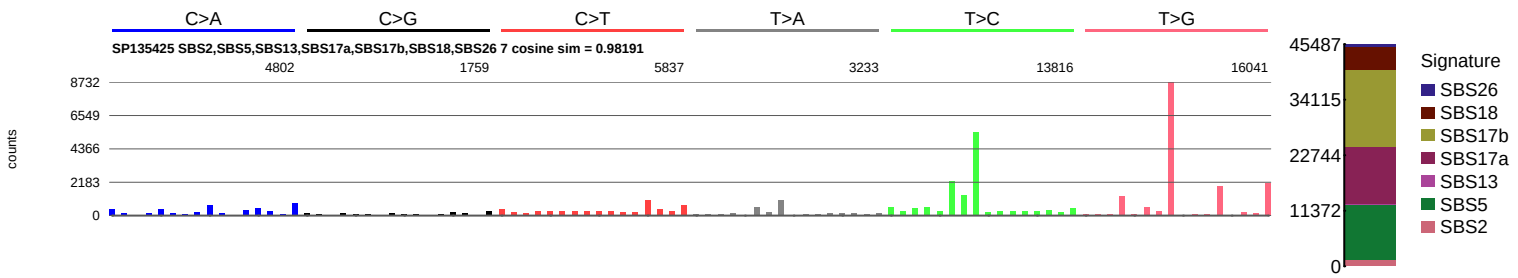
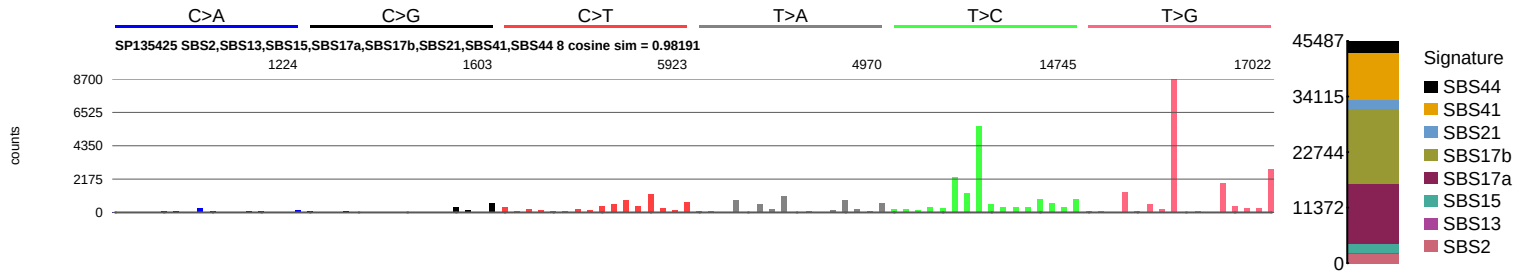
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98198
1119
1360
5835
4932
15181
17065
8696
6522
4348
2174
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2

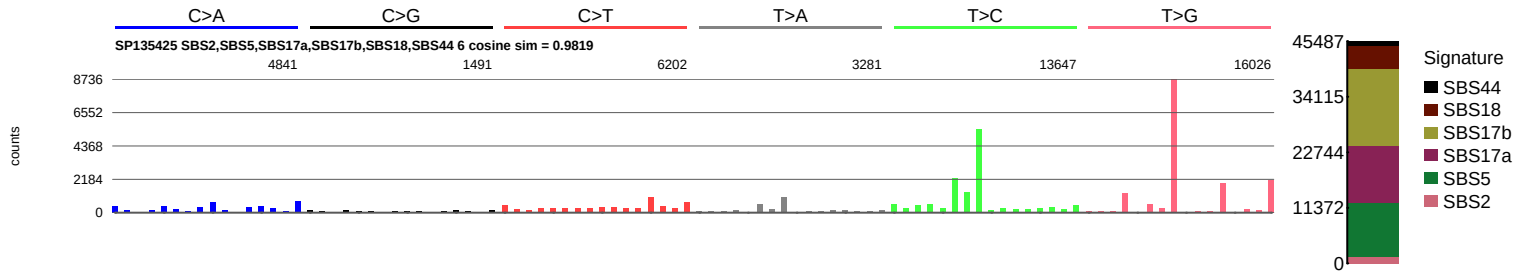

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98193
5101
2091
5043
3599
13667
15986
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

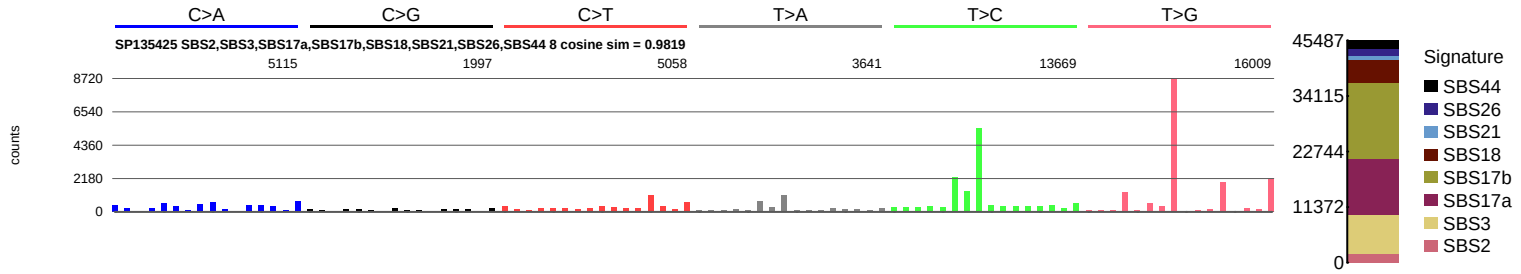

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98192
2137
2762
4533
5260
13823
16975
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1

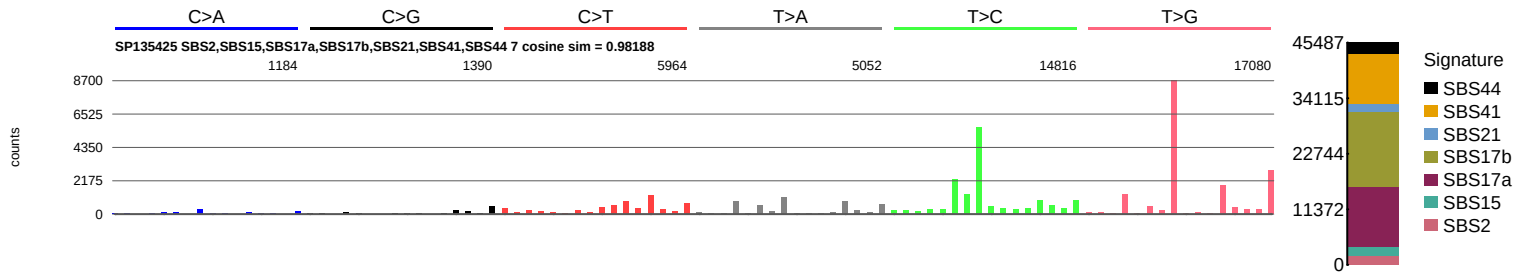

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98191
4848
1445
6116
3247
13815
16016
8736
6552
4368
2184
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98191
4802
1759
5837
3233
13816
16041
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98191
1224
1603
5923
4970
14745
17022
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.9819
4841
1491
6202
3281
13647
16026
counts
8736
6552
4368
2184
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.9819
5115
1997
5058
3641
13669
16009
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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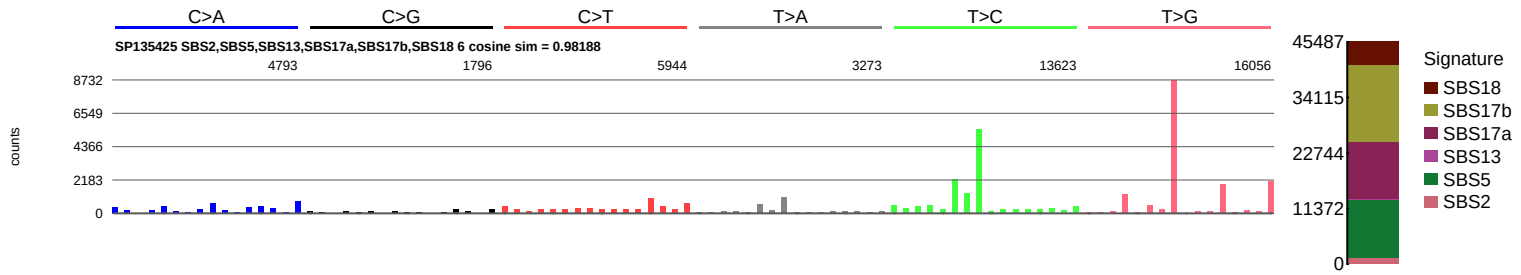
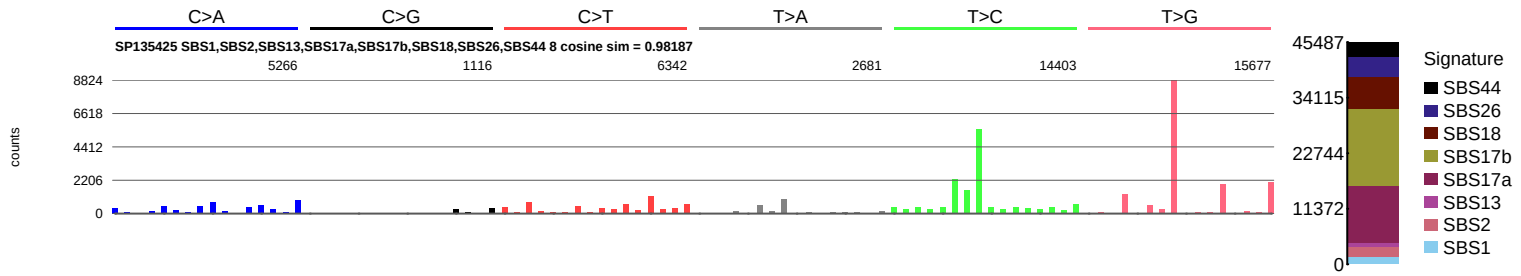
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98188
1184
1390
5964
5052
14816
17080
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98188
4793
1796
5944
3273
13623
16056
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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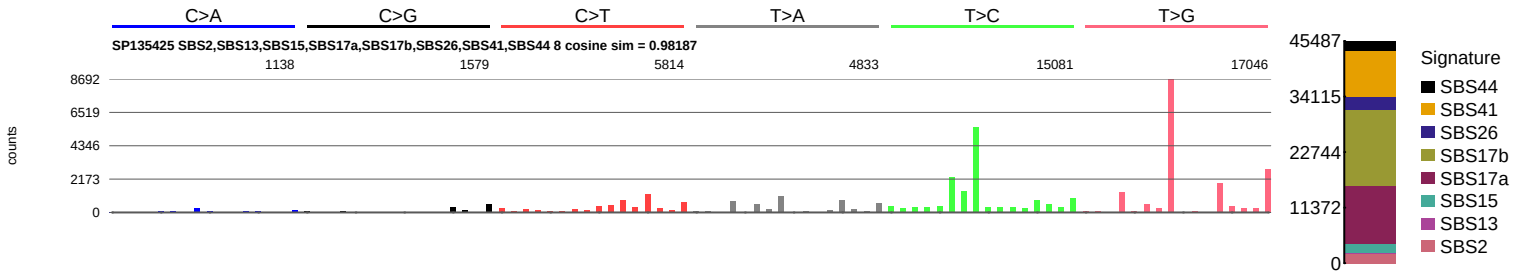
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98187
5266
1116
6342
2681
14403
15677
counts
8824
6618
4412
2206
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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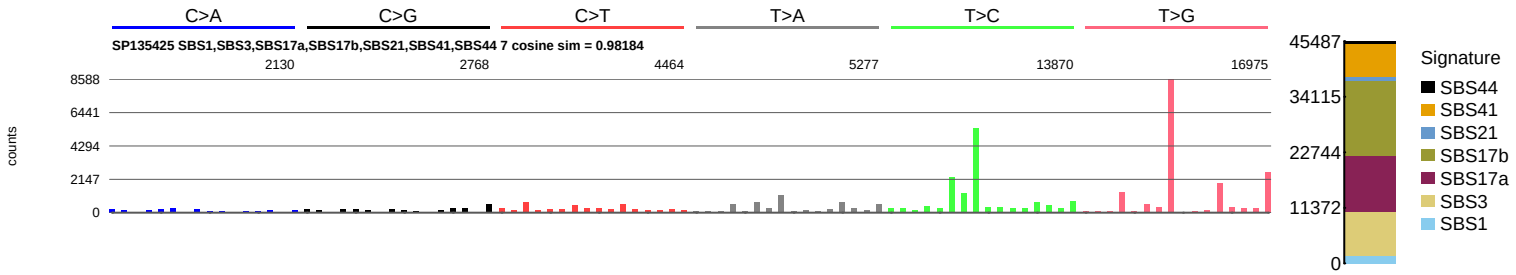
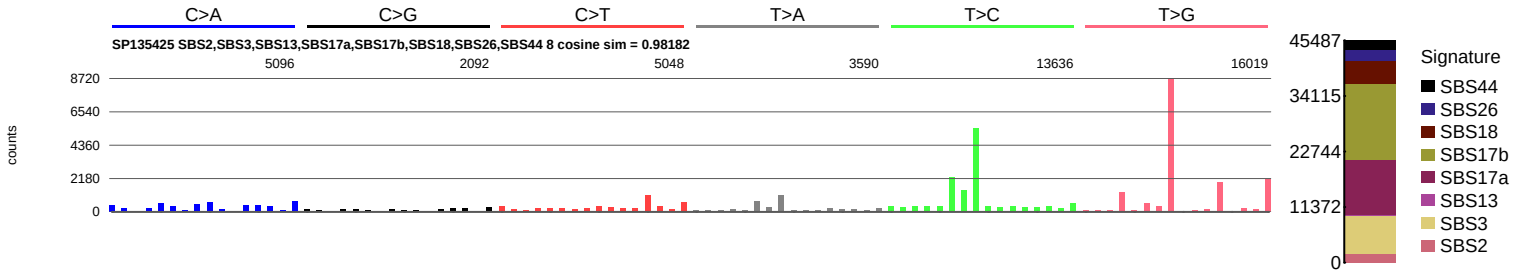
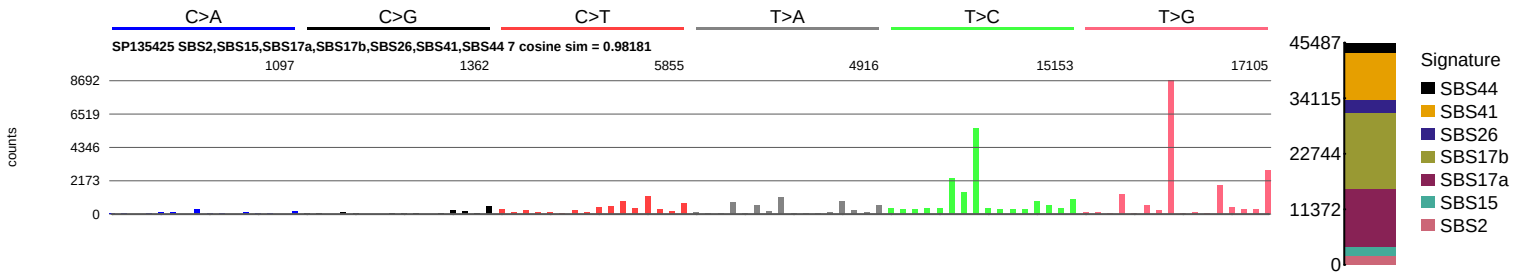
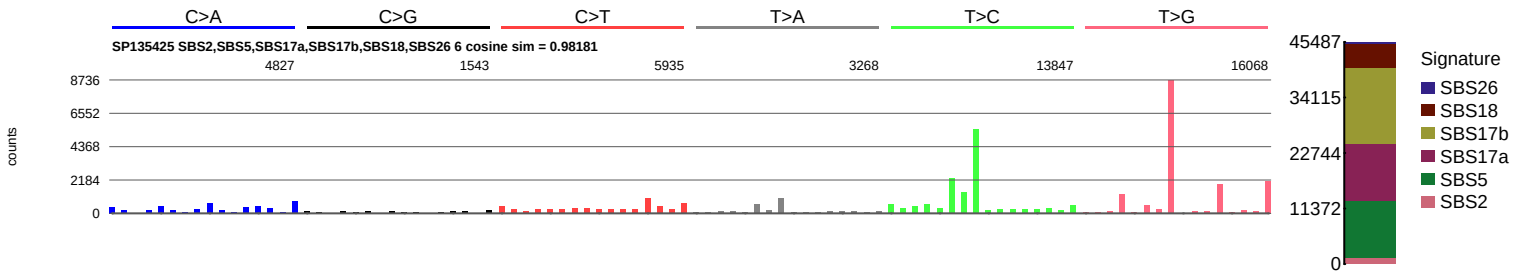
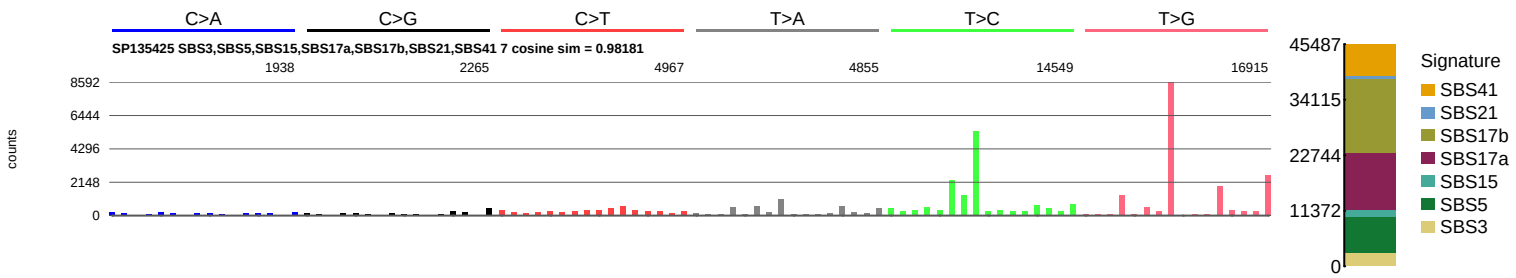
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98187
1138
1579
5814
4833
15081
17046
8692
6519
4346
2173
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2

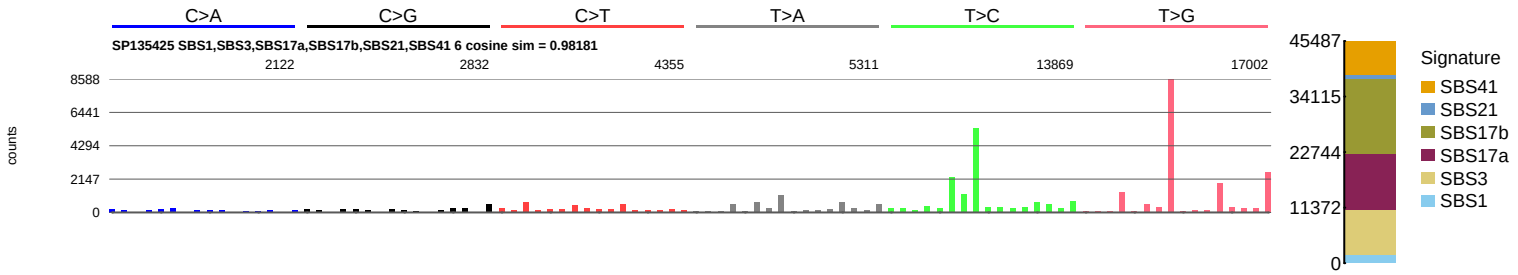

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98184
2130
2768
4464
5277
13870
16975
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1

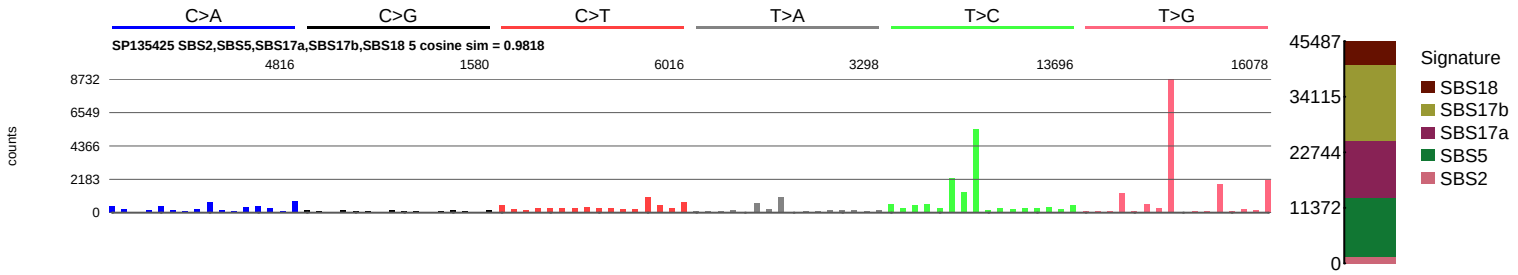

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98182
5096
2092
5048
3590
13636
16019
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98181
1097
1362
5855
4916
15153
17105
8692
6519
4346
2173
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98181
4827
1543
5935
3268
13847
16068
8736
6552
4368
2184
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98181
1938
2265
4967
4855
14549
16915
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98181
2122
2832
4355
5311
13869
17002
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1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C>G
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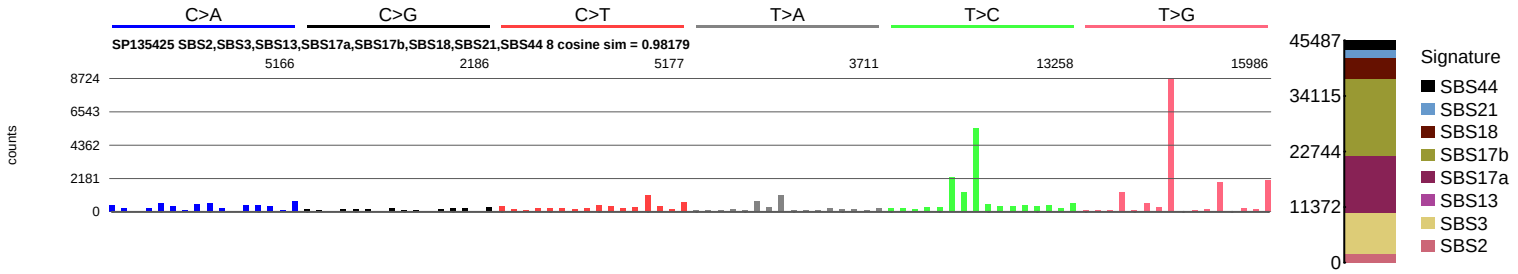
T>G
SP135425 SBS2,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9818
4816
1580
6016
3298
13696
16078
8732
6549
4366
2183
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2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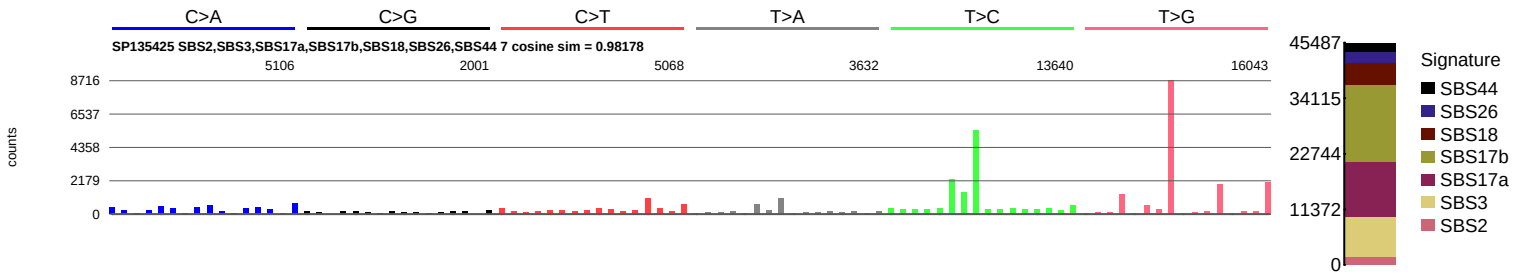
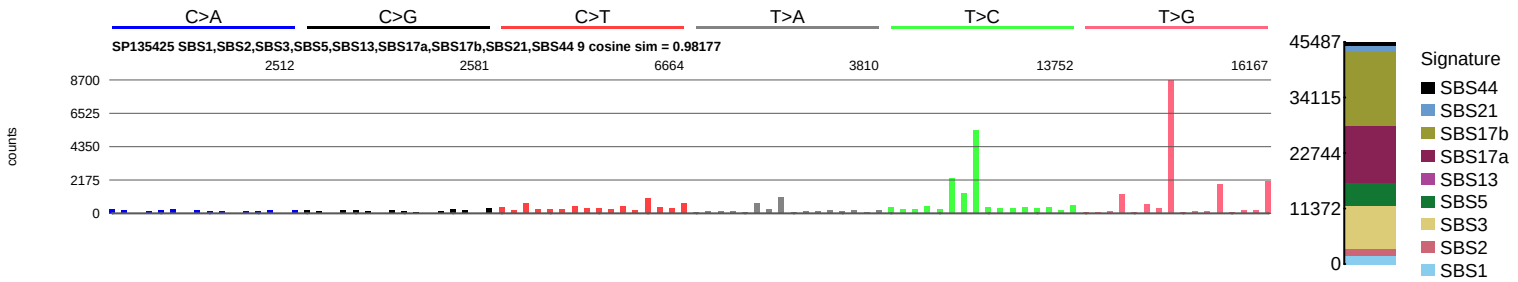
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98179
5166
2186
5177
3711
13258
15986
8724
6543
4362
2181
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98178
5106
2001
5068
3632
13640
16043
8716
6537
4358
2179
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2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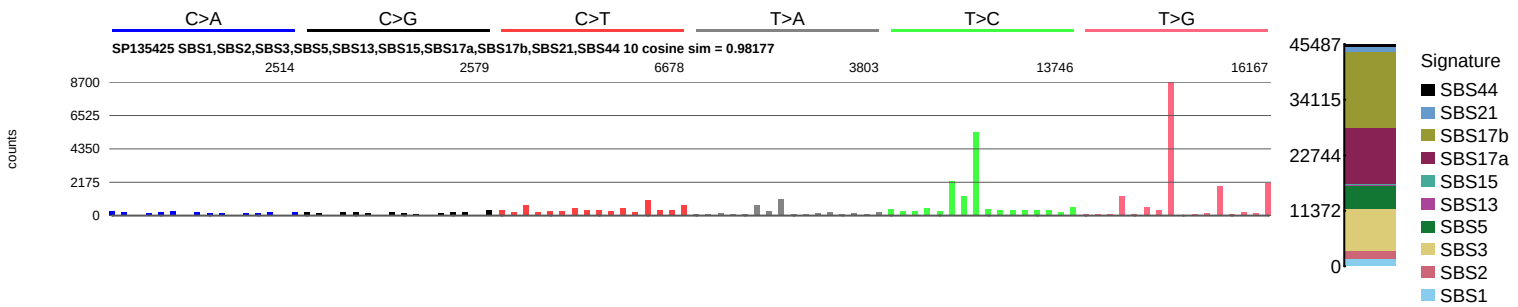
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98177
2512
2581
6664
3810
13752
16167
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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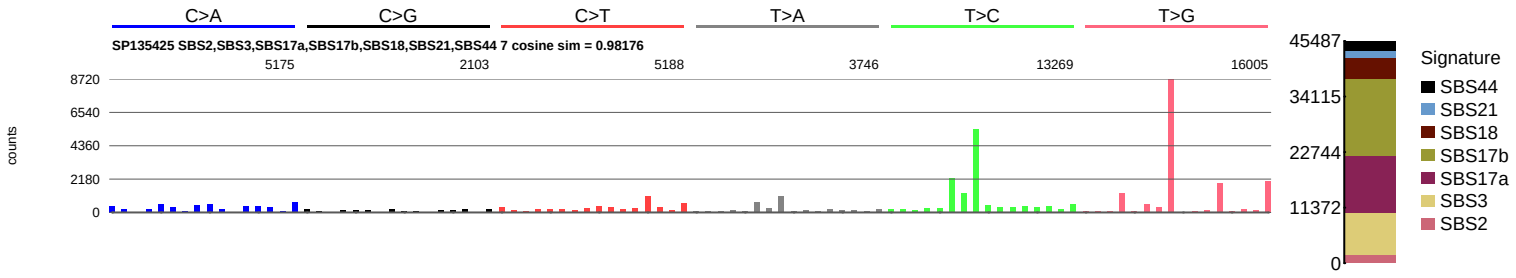
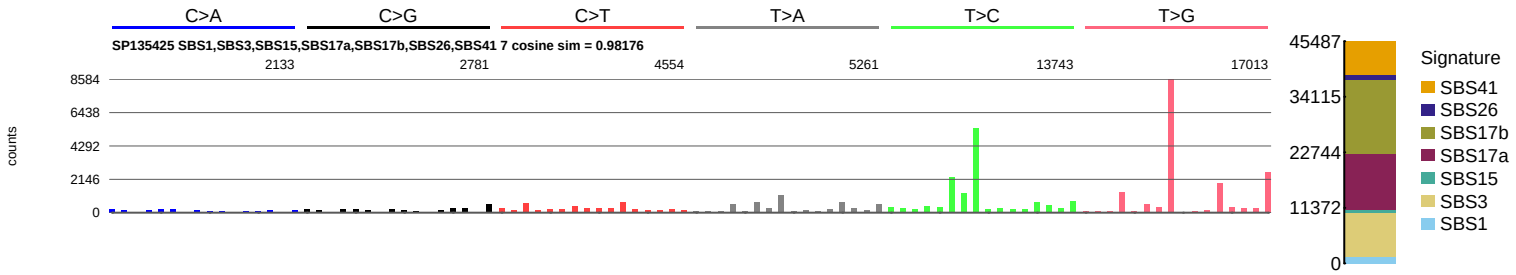
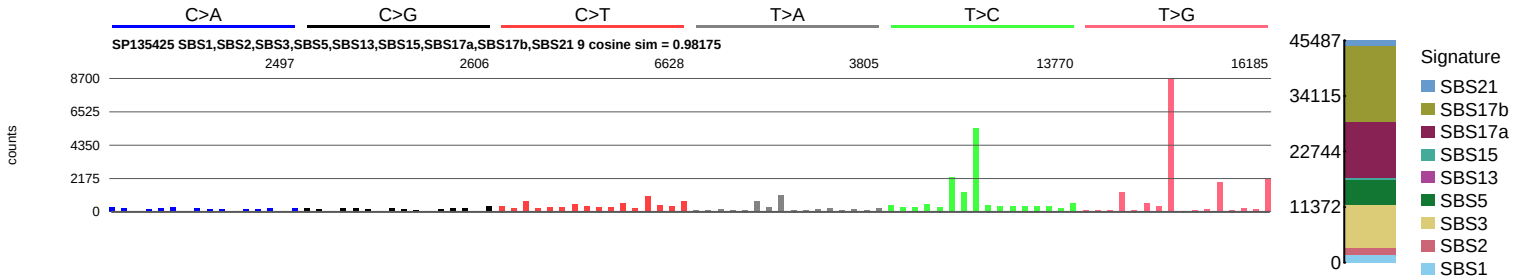
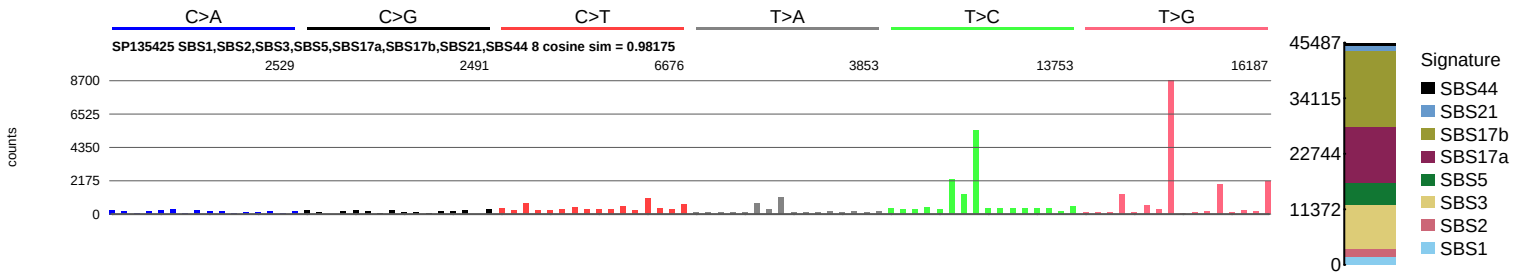
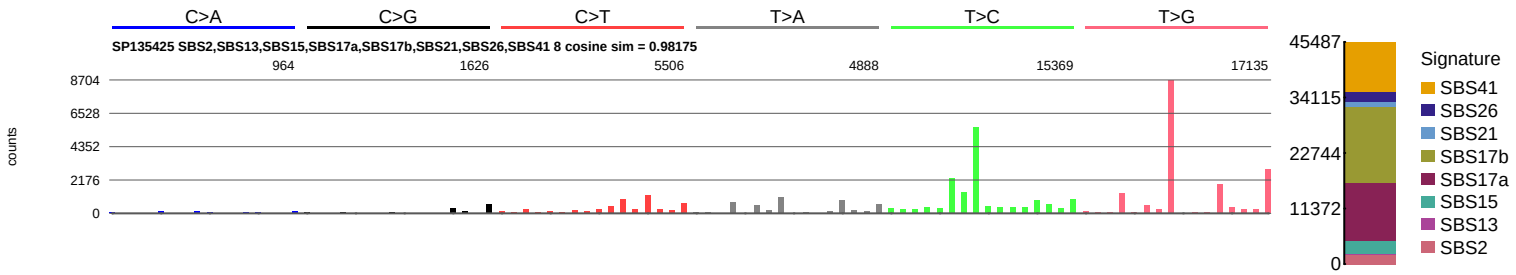
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 10 cosine sim = 0.98177
2514
2579
6678
3803
13746
16167
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

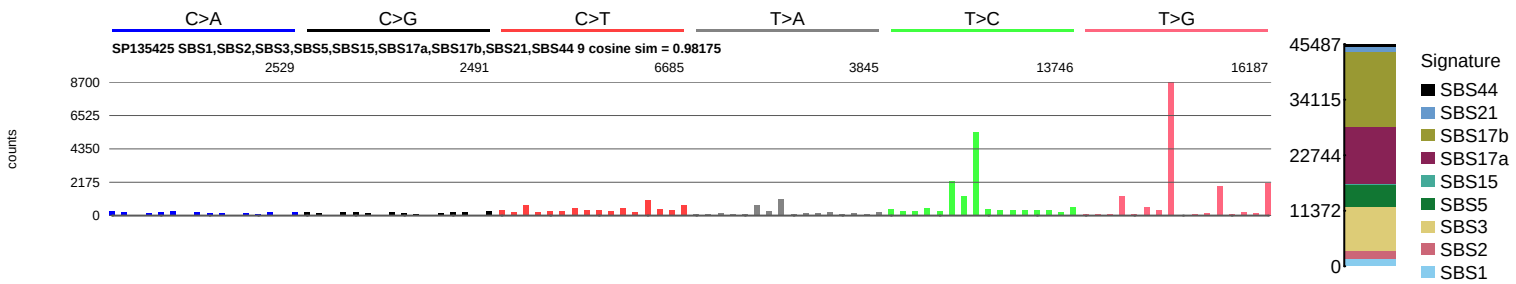

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98176
5175
2103
5188
3746
13269
16005
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2

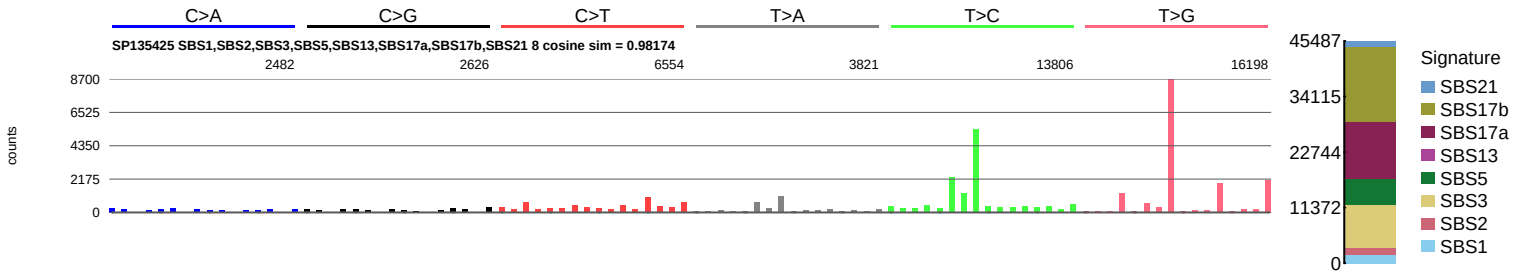

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98176
2133
2781
4554
5261
13743
17013
counts
8584
6438
4292
2146
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1

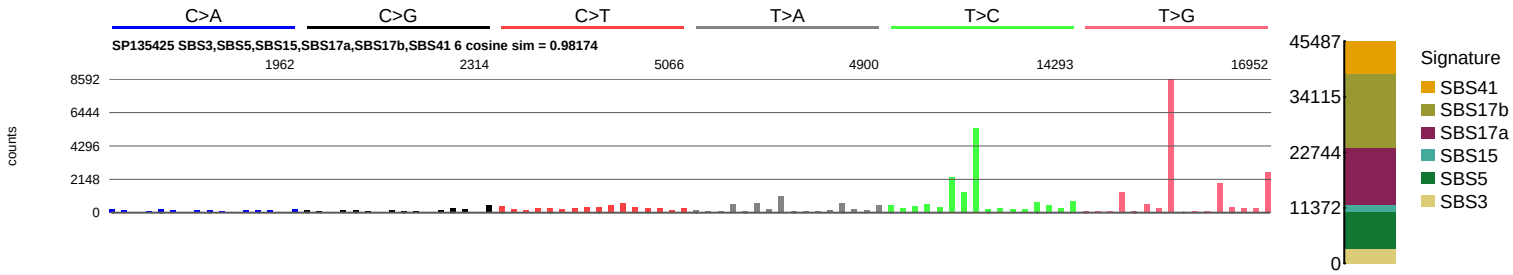

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.98175
2497
2606
6628
3805
13770
16185
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

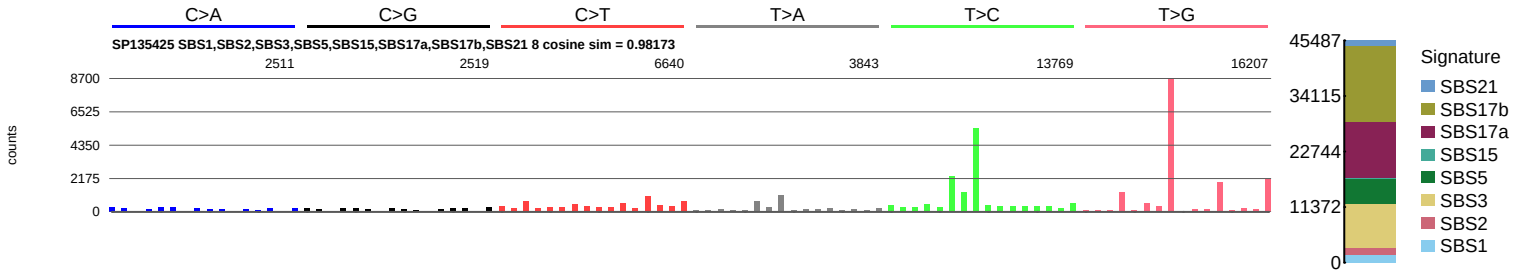

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98175
2529
2491
6676
3853
13753
16187
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98175
964
1626
5506
4888
15369
17135
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98175
2529
2491
6685
3845
13746
16187
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98174
2482
2626
6554
3821
13806
16198
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98174
1962
2314
5066
4900
14293
16952
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98173
2511
2519
6640
3843
13769
16207
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
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Signature
SBS21
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SBS17a
SBS15
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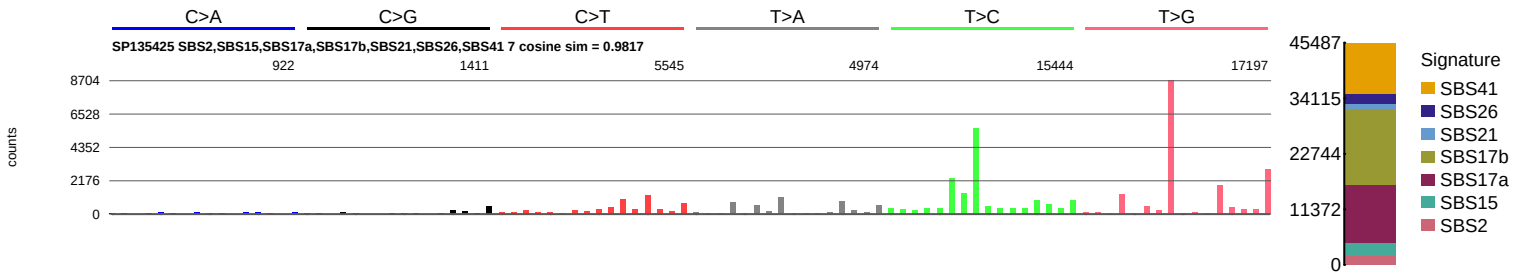
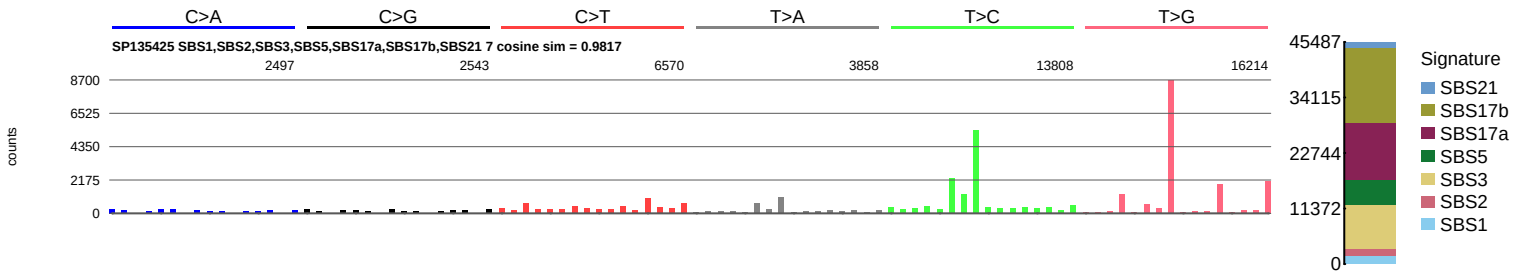
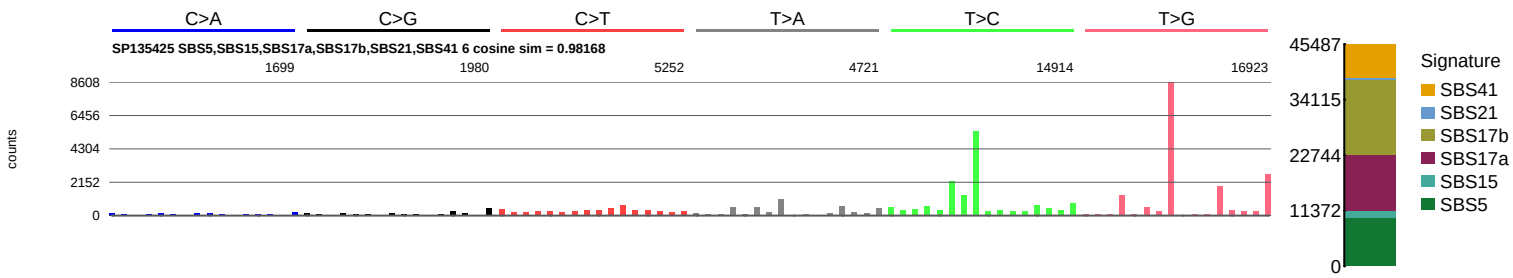
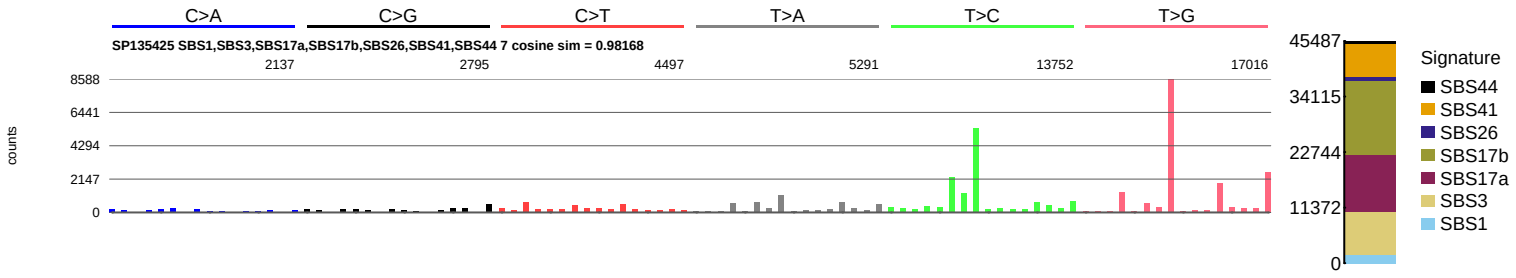
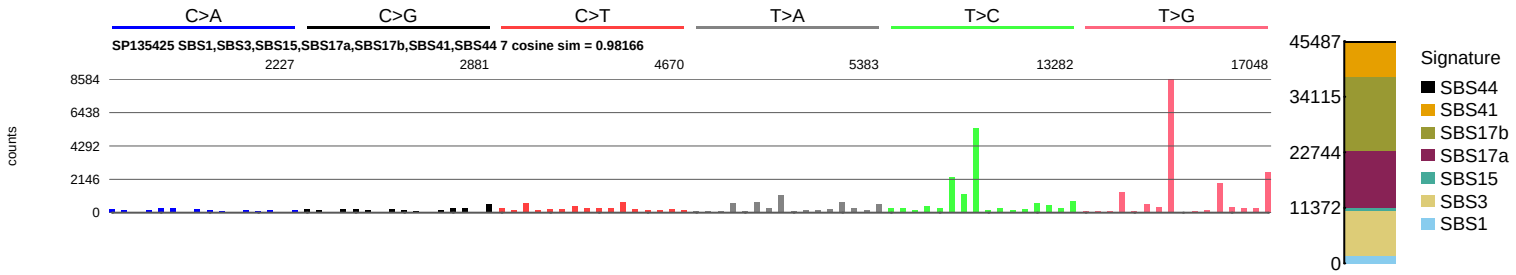
SBS3
SBS2
SBS1

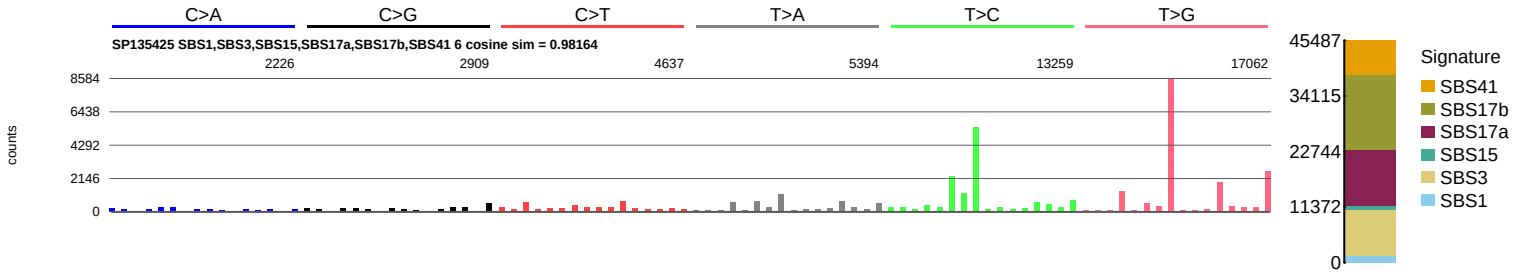

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.9817
922
1411
5545
4974
15444
17197
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2

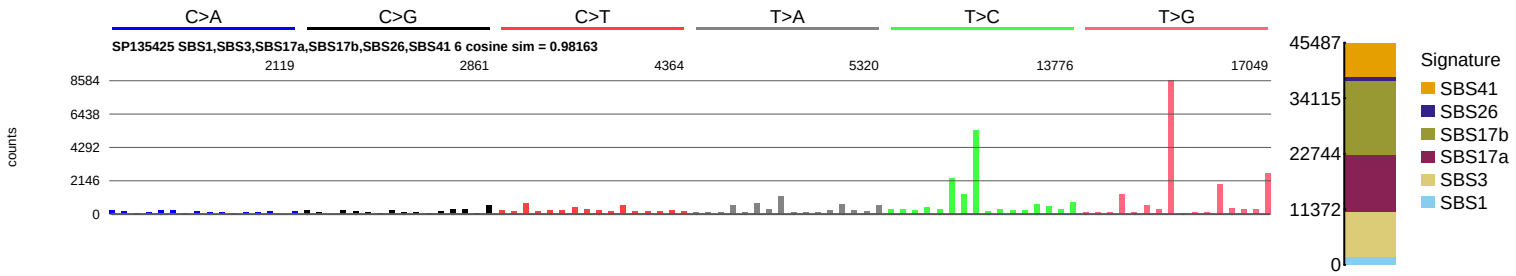

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9817
2497
2543
6570
3858
13808
16214
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

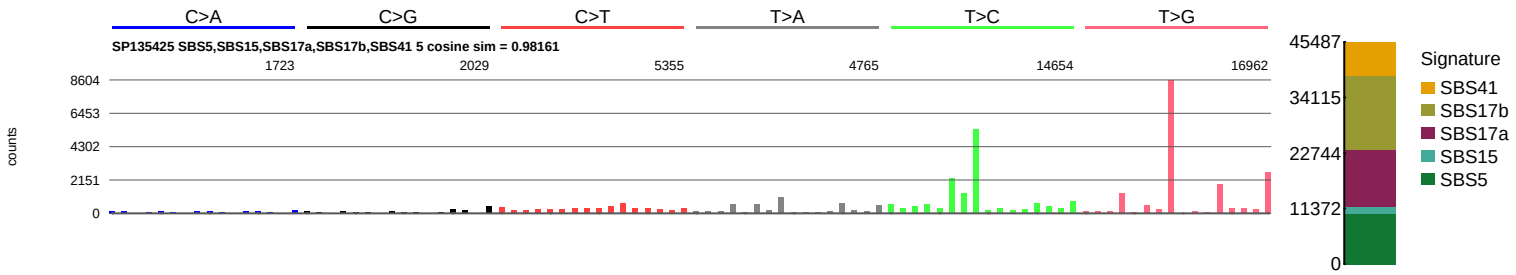

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98168
1699
1980
5252
4721
14914
16923
8608
6456
4304
2152
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5

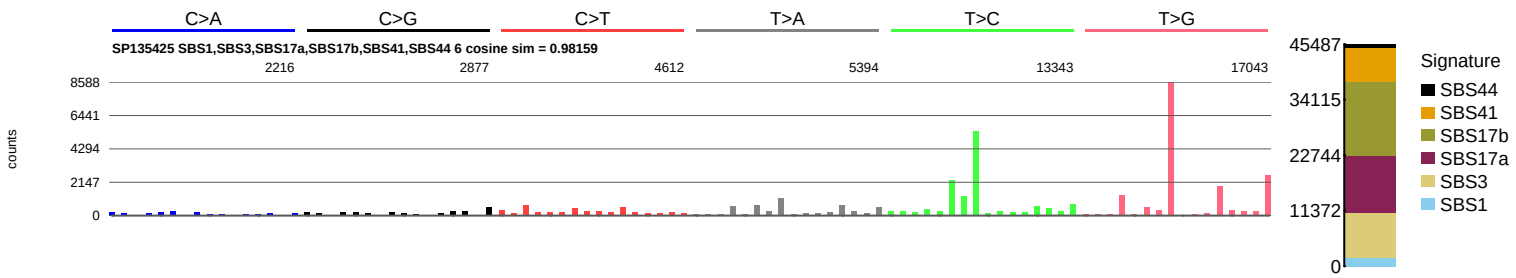

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98168
2137
2795
4497
5291
13752
17016
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1

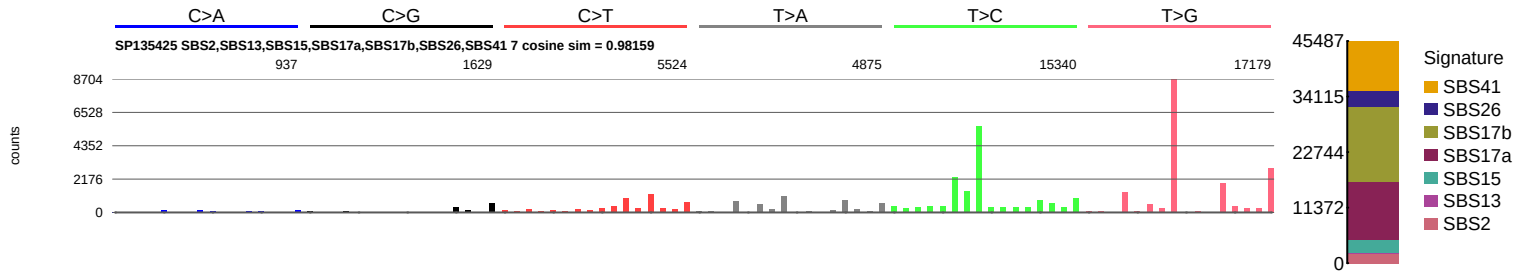

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98166
2227
2881
4670
5383
13282
17048
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

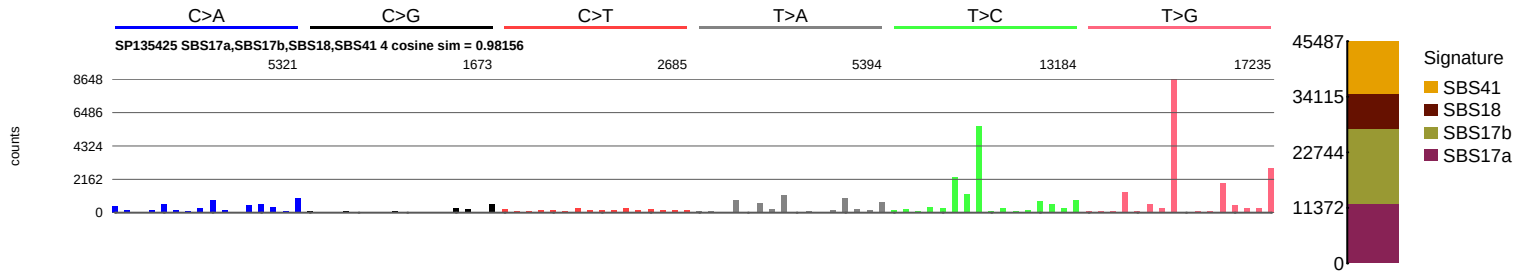

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98164
2226
2909
4637
5394
13259
17062
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1

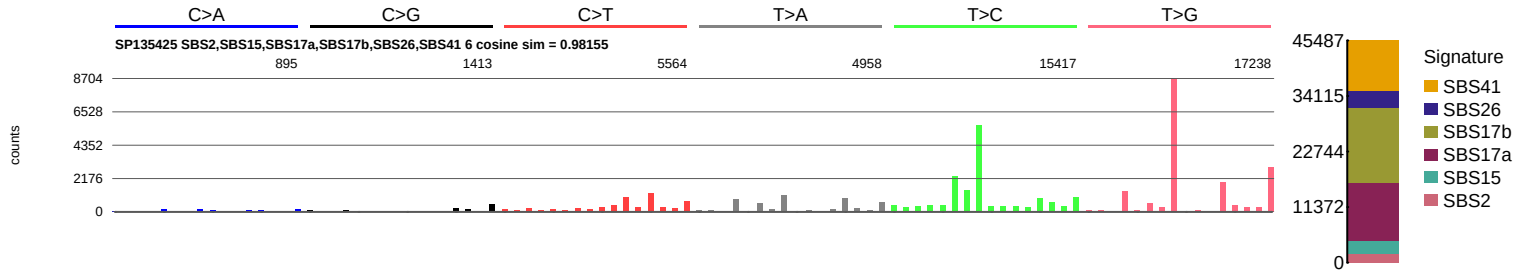

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98163
2119
2861
4364
5320
13776
17049
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98161
1723
2029
5355
4765
14654
16962
8604
6453
4302
2151
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98159
2216
2877
4612
5394
13343
17043
8588
6441
4294
2147
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98159
937
1629
5524
4875
15340
17179
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS17a,SBS17b,SBS18,SBS41 4 cosine sim = 0.98156
5321
1673
2685
5394
13184
17235
8648
6486
4324
2162
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS18
SBS17b
SBS17a


C>A
C>G
C>T
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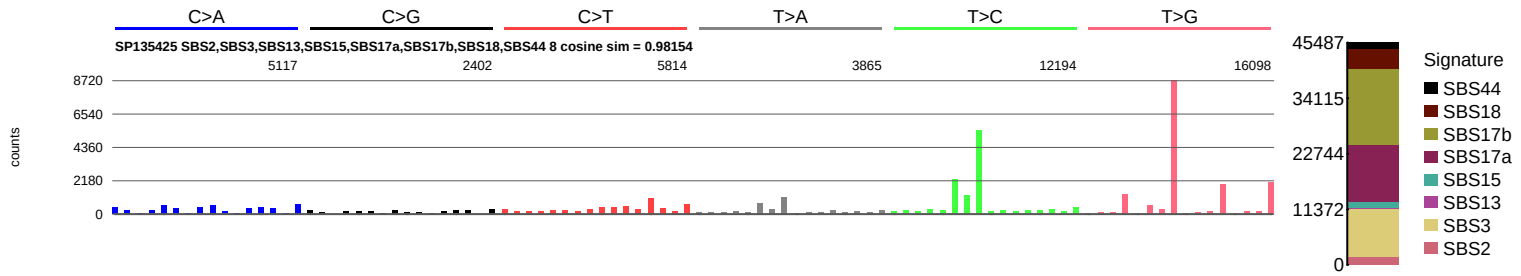
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98155
895
1413
5564
4958
15417
17238
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS2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C>G
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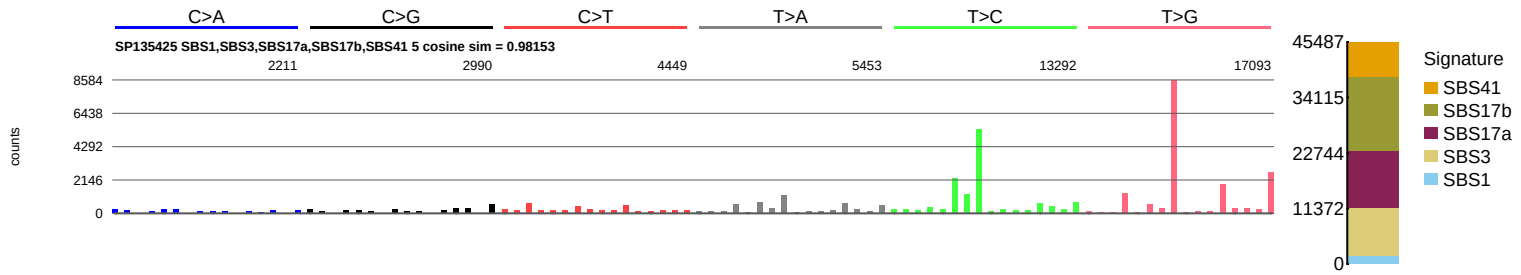
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98154
5117
2402
5814
3865
12194
16098
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
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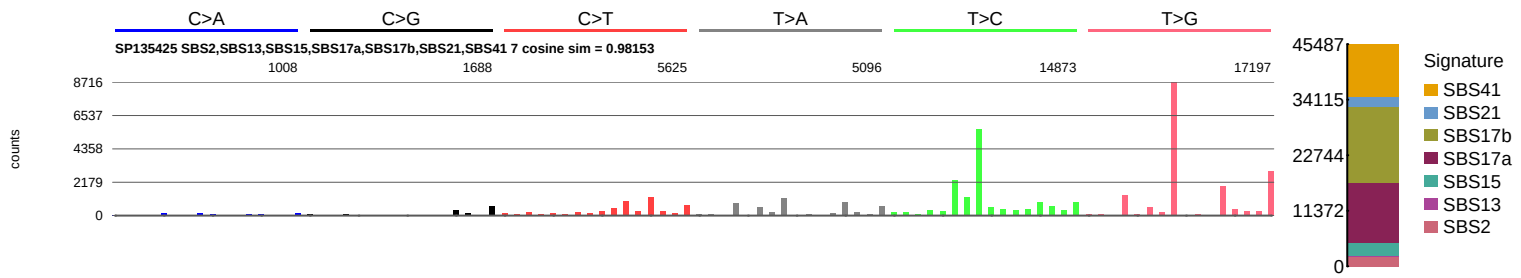
T>C
T>G
SP135425 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98153
2211
2990
4449
5453
13292
17093
8584
6438
4292
2146
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1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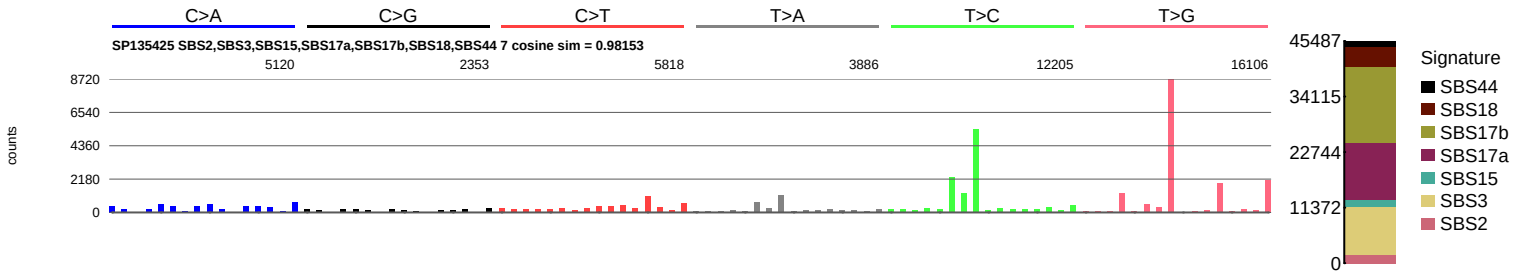
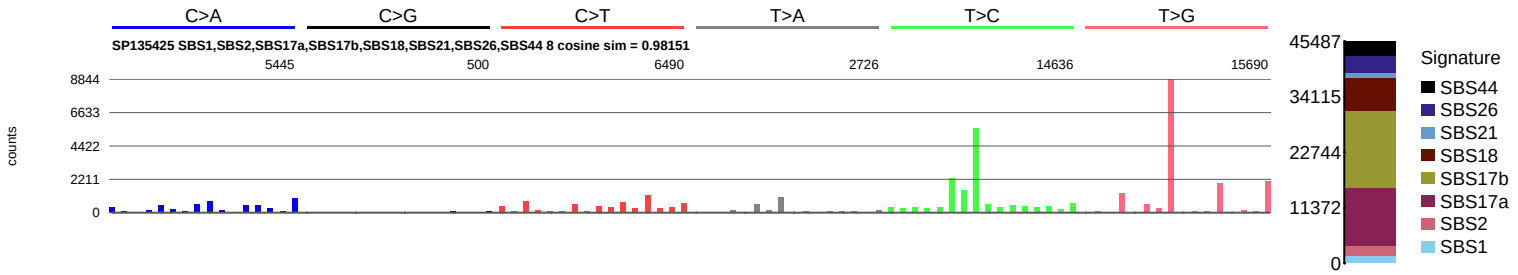
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98153
1008
1688
5625
5096
14873
17197
8716
6537
4358
2179
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98153
5120
2353
5818
3886
12205
16106
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2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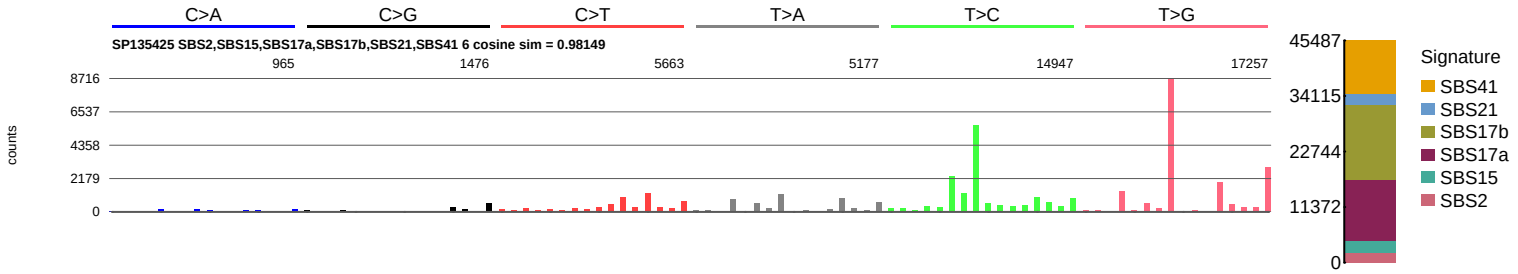
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98151
5445
500
6490
2726
14636
15690
counts
8844
6633
4422
2211
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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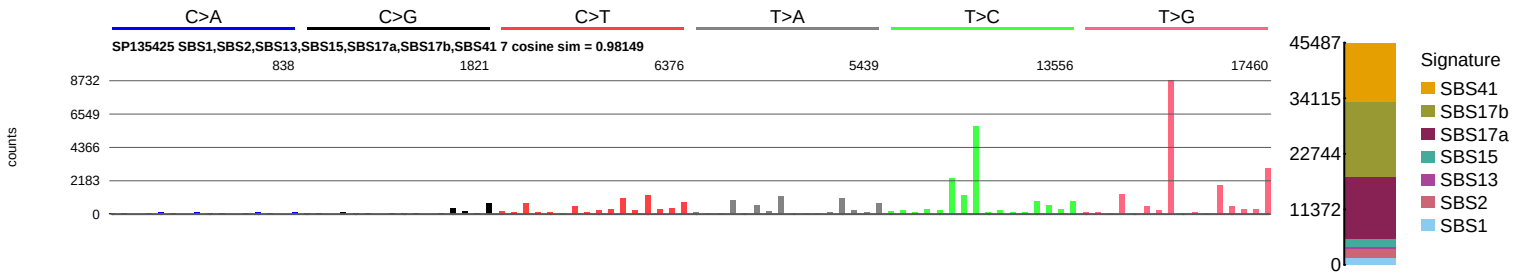
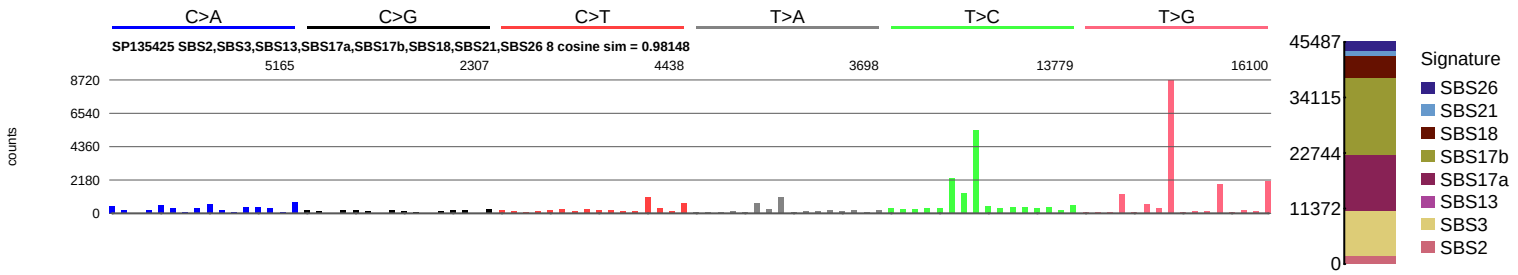
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98149
965
1476
5663
5177
14947
17257
counts
8716
6537
4358
2179
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98149
838
1821
6376
5439
13556
17460
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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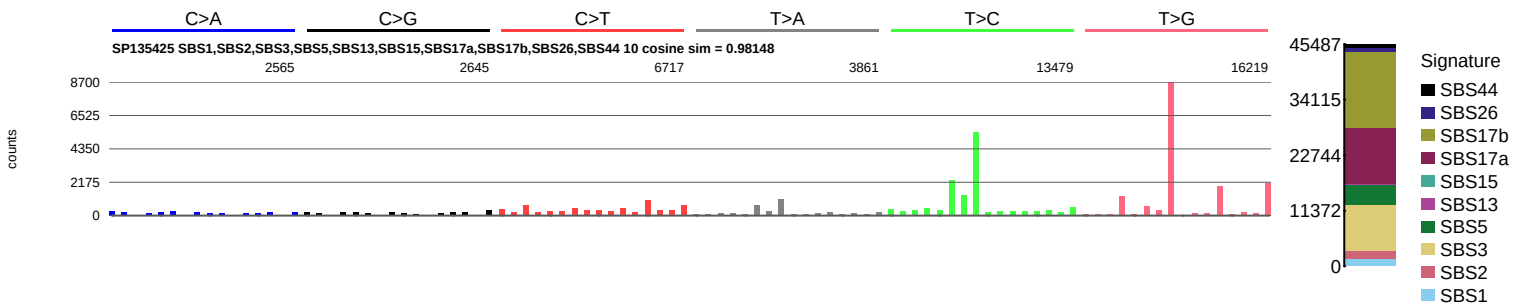
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98148
5165
2307
4438
3698
13779
16100
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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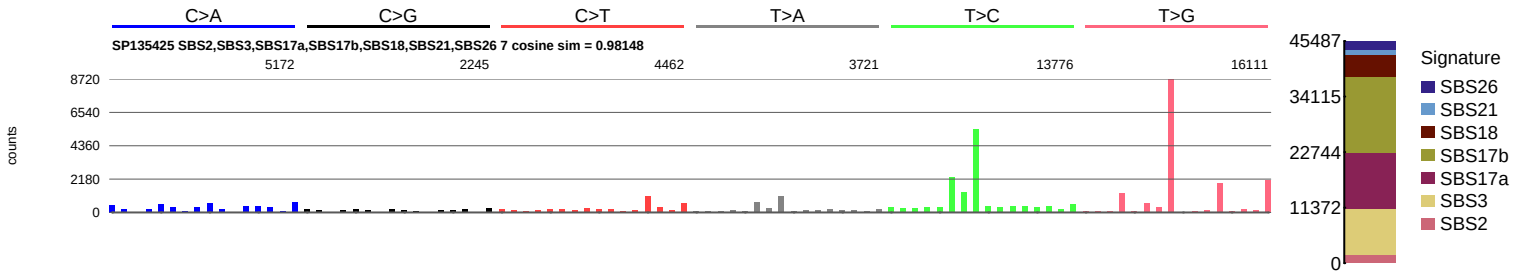
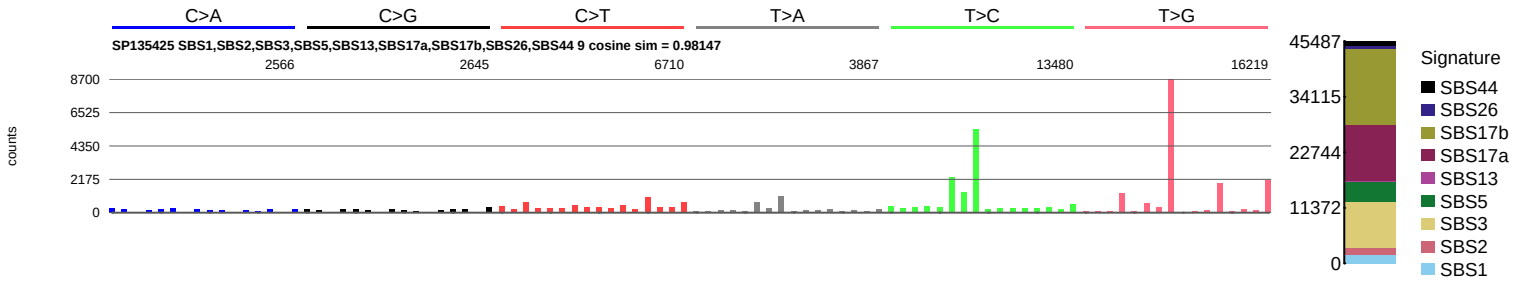
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 10 cosine sim = 0.98148
2565
2645
6717
3861
13479
16219
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98148
5172
2245
4462
3721
13776
16111
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2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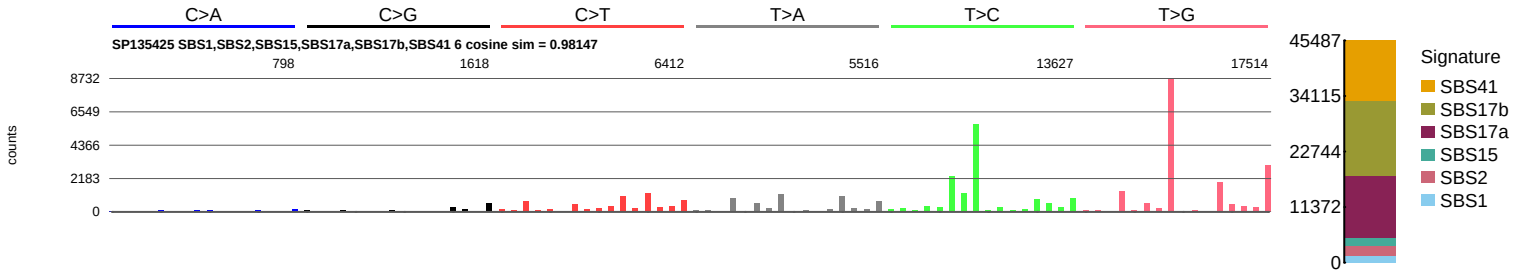
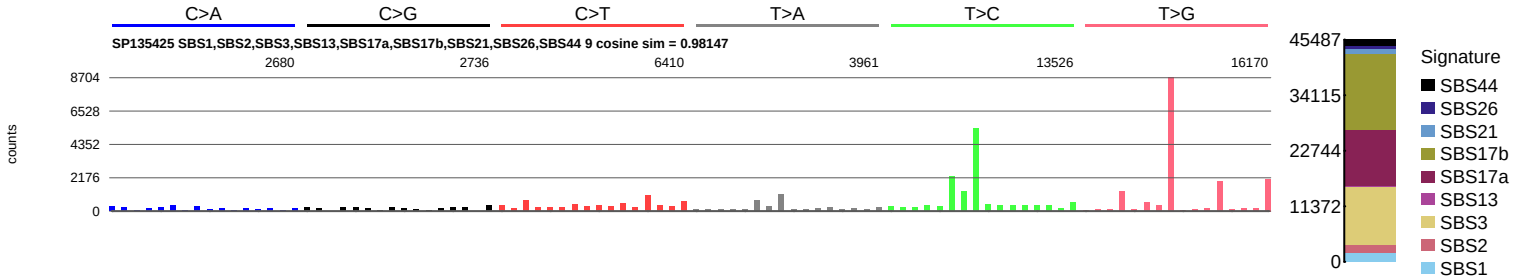
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98147
2566
2645
6710
3867
13480
16219
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98147
798
1618
6412
5516
13627
17514
counts
8732
6549
4366
2183
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2
SBS1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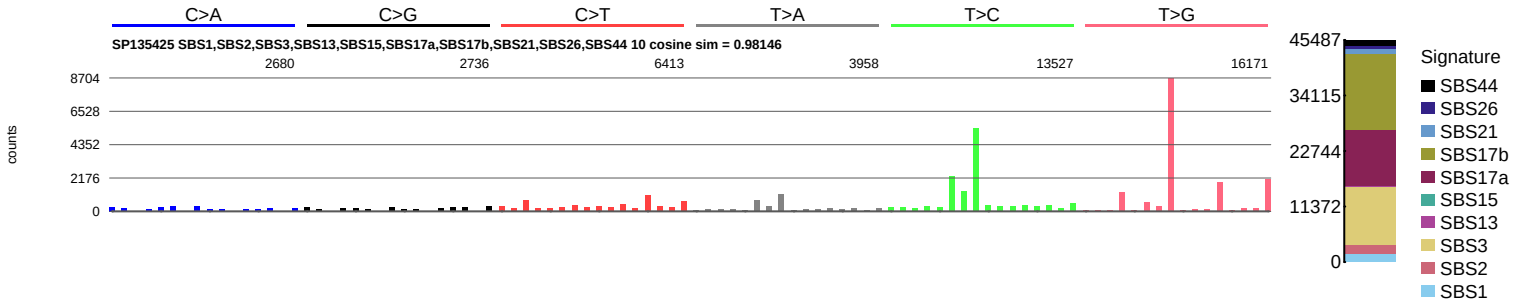
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98147
2680
2736
6410
3961
13526
16170
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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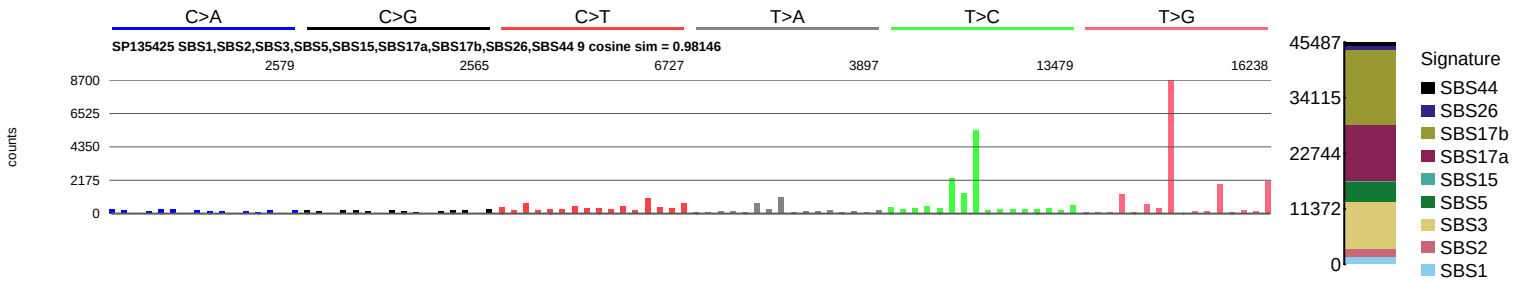
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 10 cosine sim = 0.98146
2680
2736
6413
3958
13527
16171
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98146
2579
2565
6727
3897
13479
16238
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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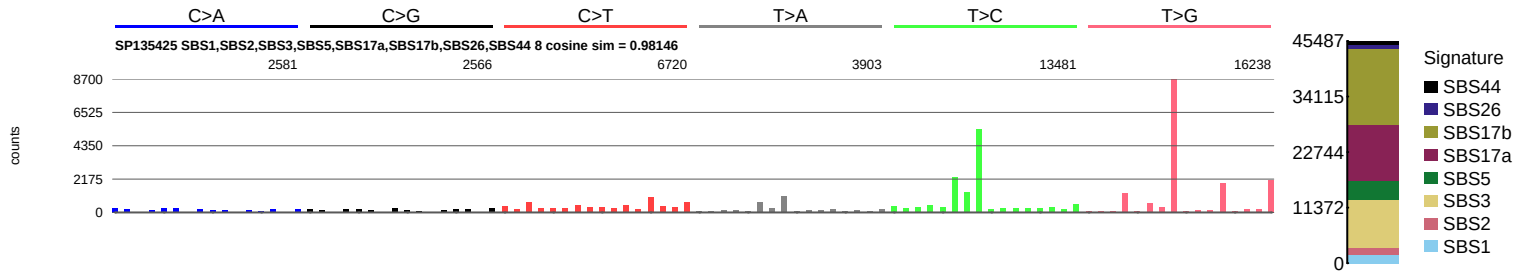
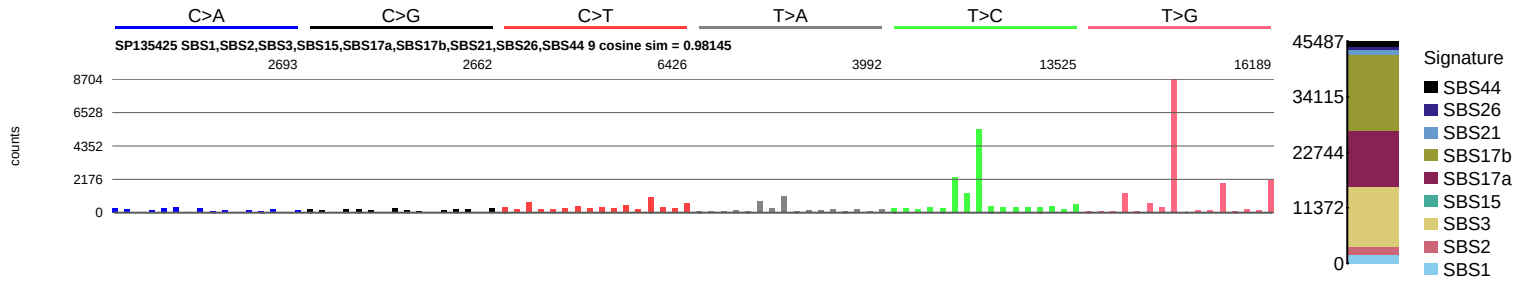
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98146
2581
2566
6720
3903
13481
16238
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS44 9 cosine sim = 0.98145
2693
2662
6426
3992
13525
16189
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
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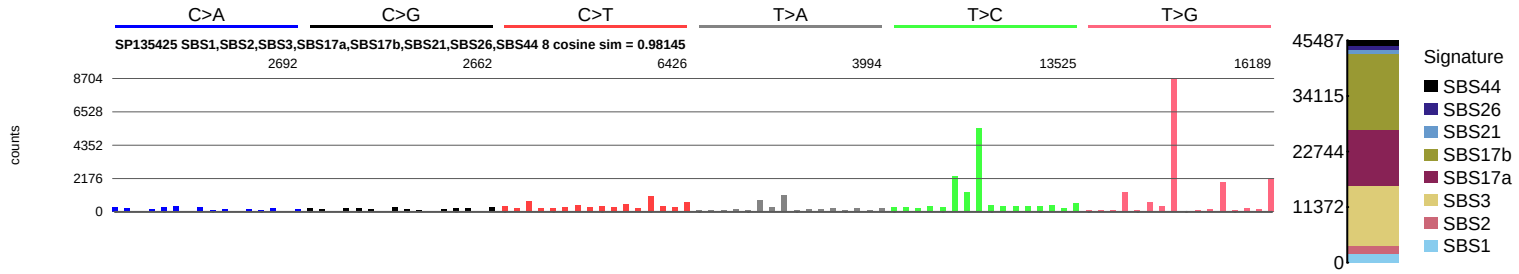
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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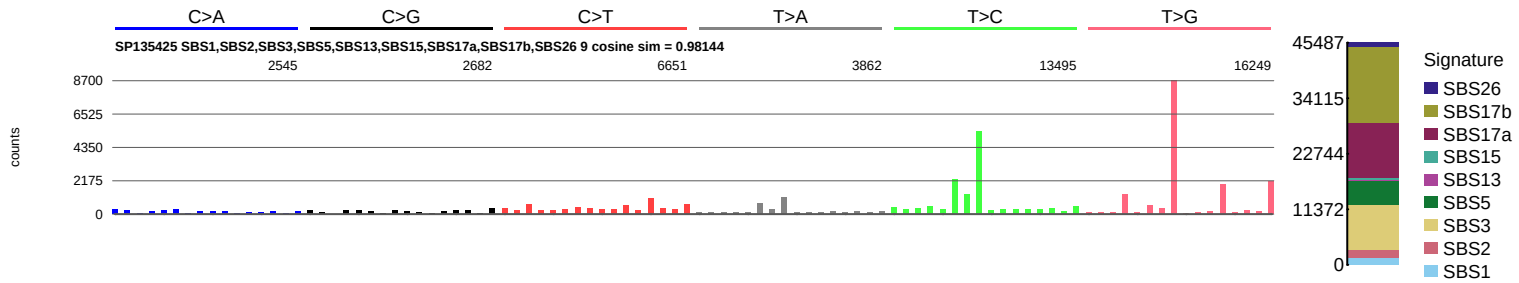
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS44 8 cosine sim = 0.98145
2692
2662
6426
3994
13525
16189
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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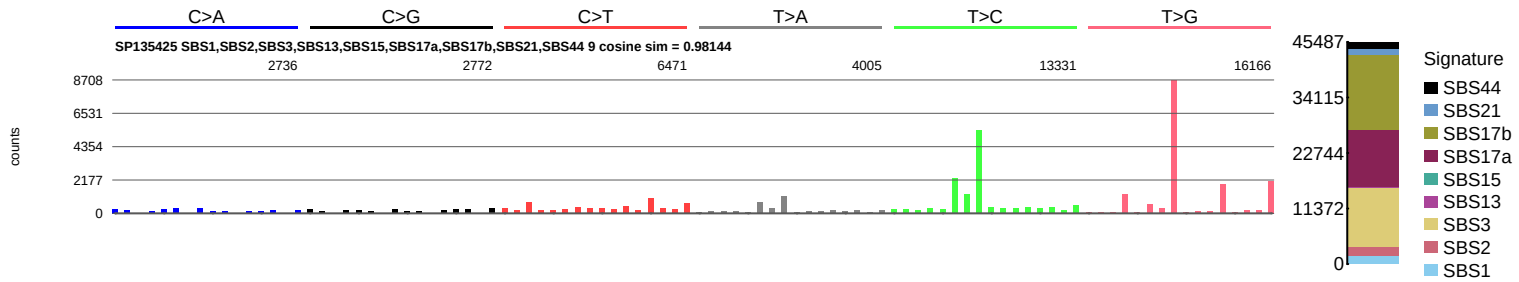
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 9 cosine sim = 0.98144
2545
2682
6651
3862
13495
16249
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98144
2736
2772
6471
4005
13331
16166
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
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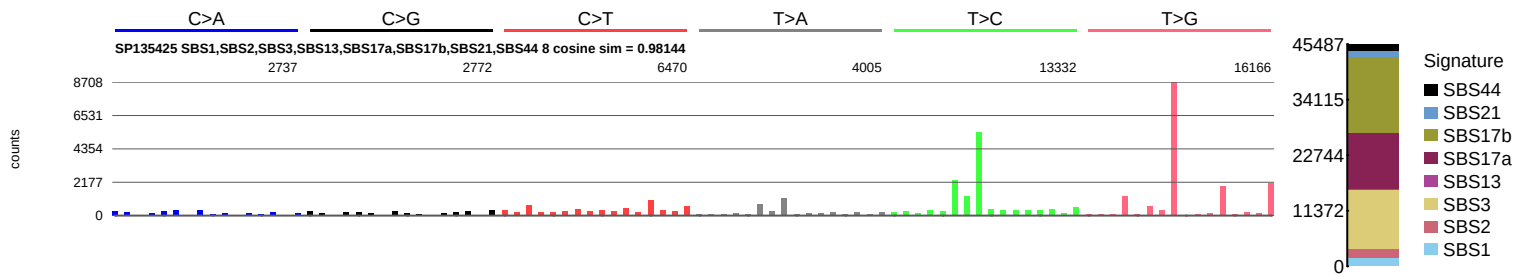
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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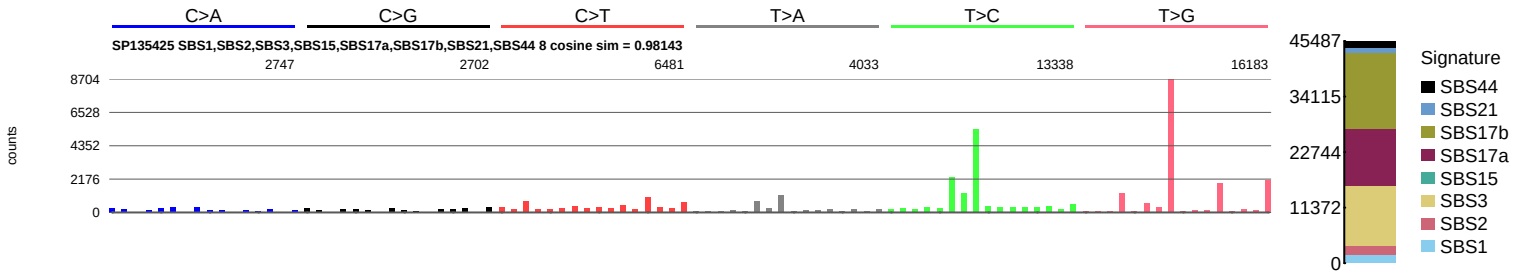
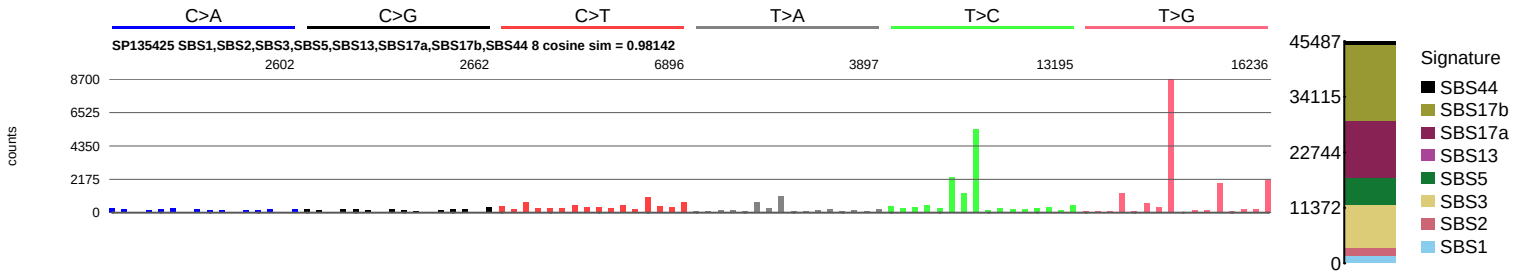
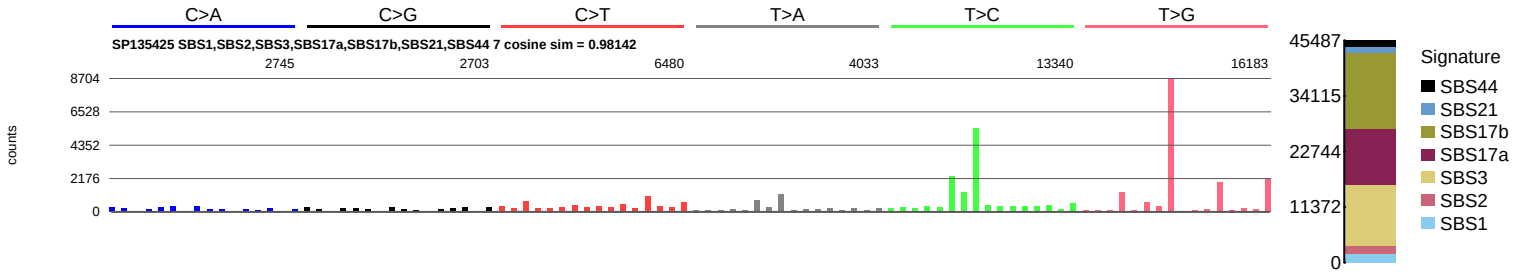
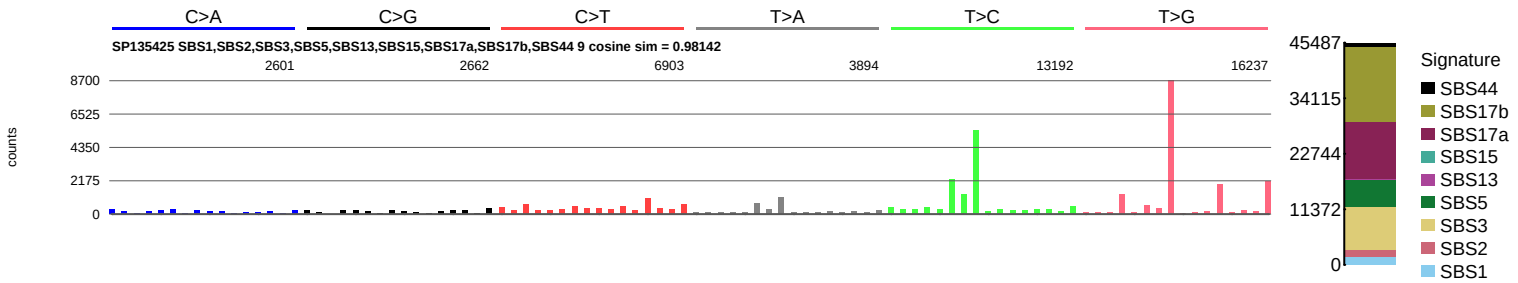
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98144
2737
2772
6470
4005
13332
16166
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98143
2747
2702
6481
4033
13338
16183
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98142
2602
2662
6896
3897
13195
16236
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98142
2745
2703
6480
4033
13340
16183
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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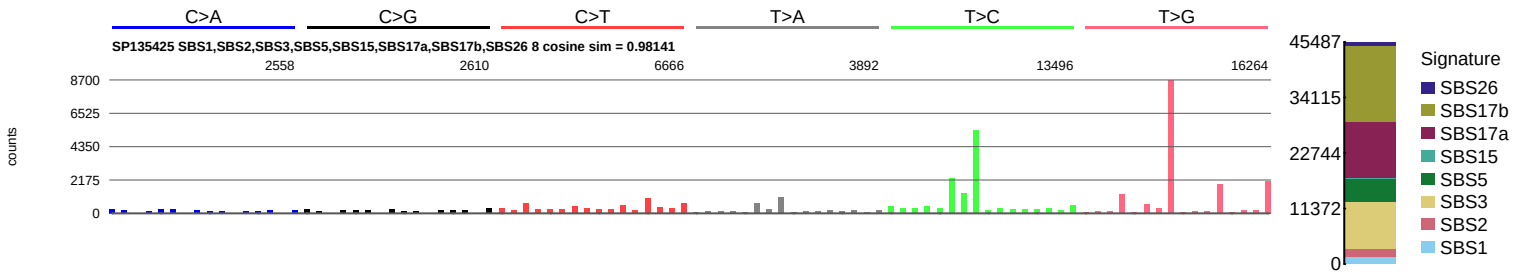
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS44 9 cosine sim = 0.98142
2601
2662
6903
3894
13192
16237
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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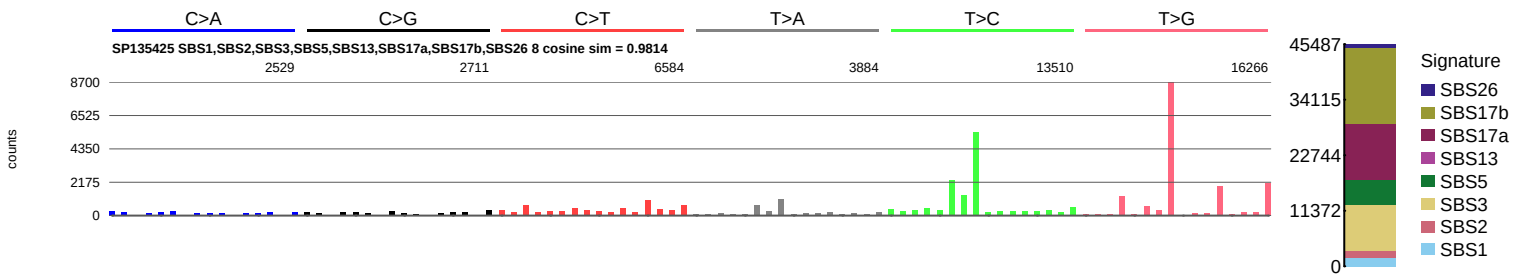
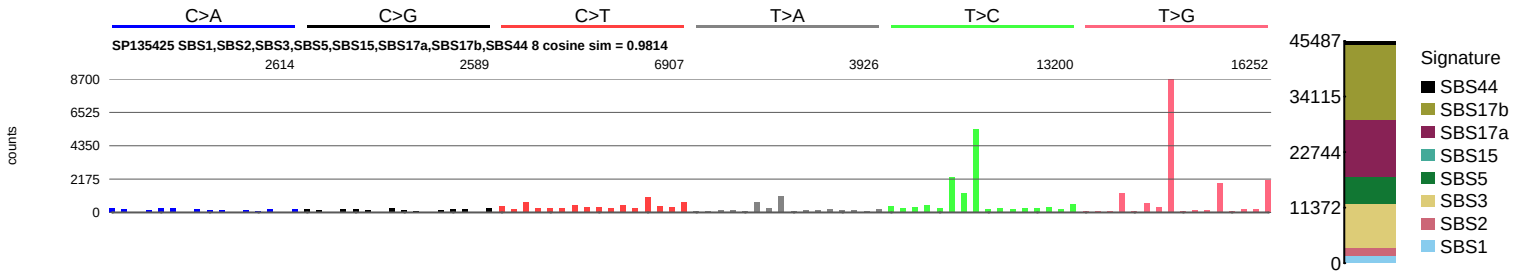
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98141
2558
2610
6666
3892
13496
16264
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26 8 cosine sim = 0.9814
2529
2711
6584
3884
13510
16266
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS44 8 cosine sim = 0.9814
2614
2589
6907
3926
13200
16252
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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Signature
SBS44
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SBS17a
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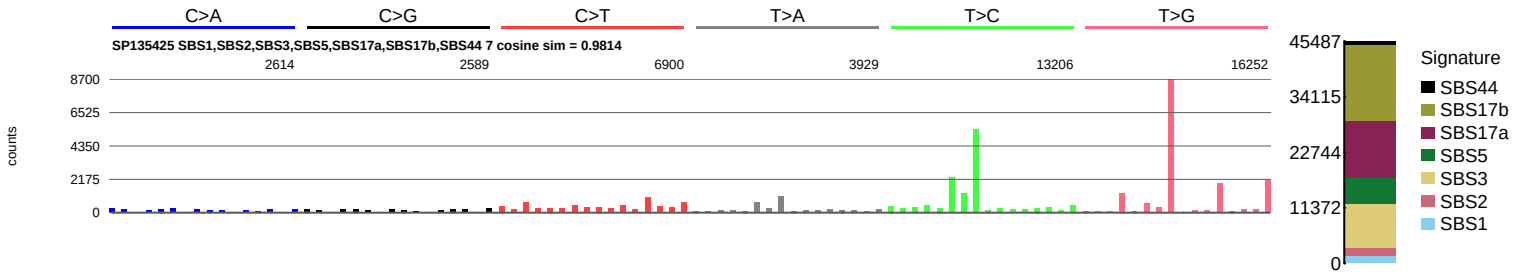
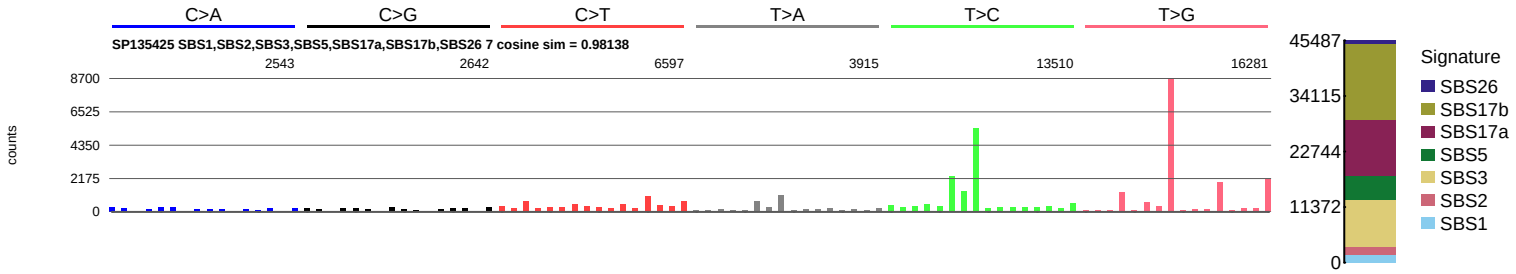
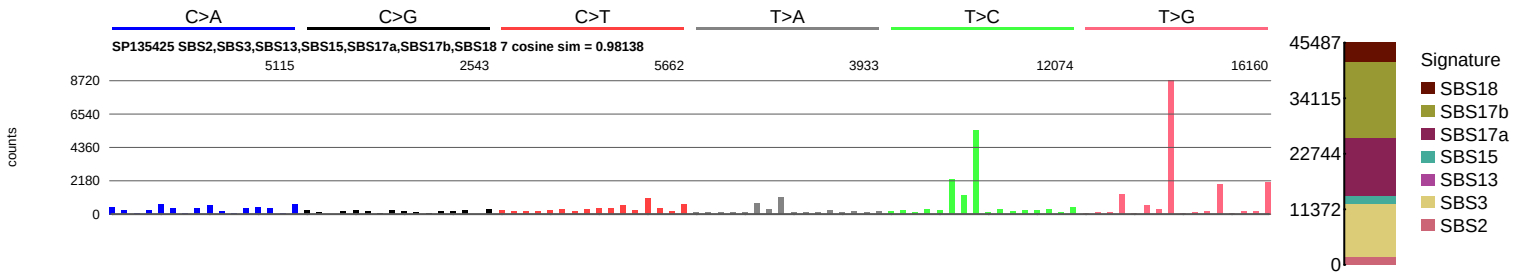
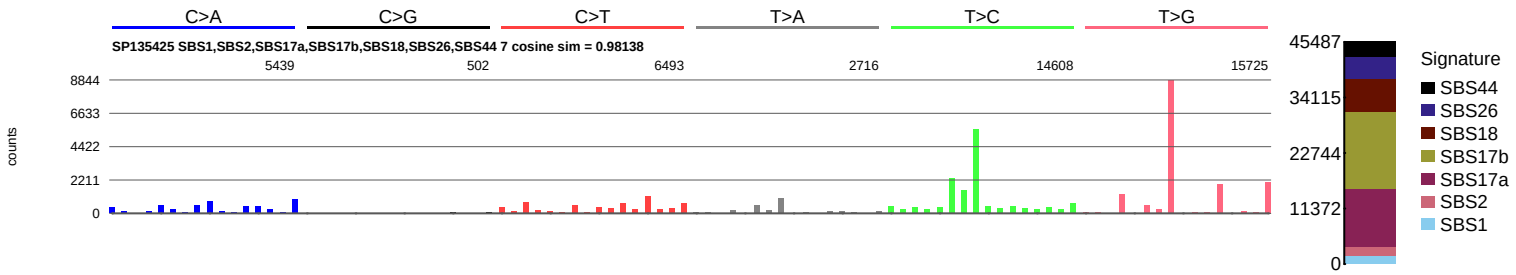
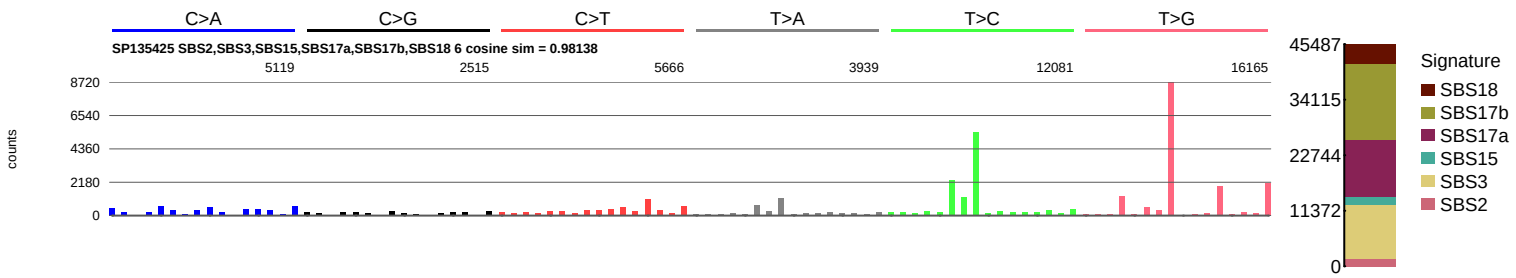
SBS3
SBS2
SBS1

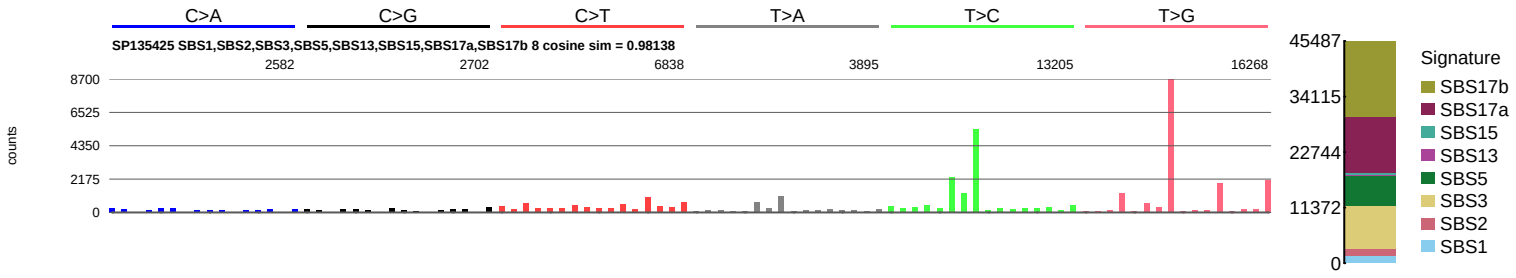

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS44 7 cosine sim = 0.9814
2614
2589
6900
3929
13206
16252
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS44
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

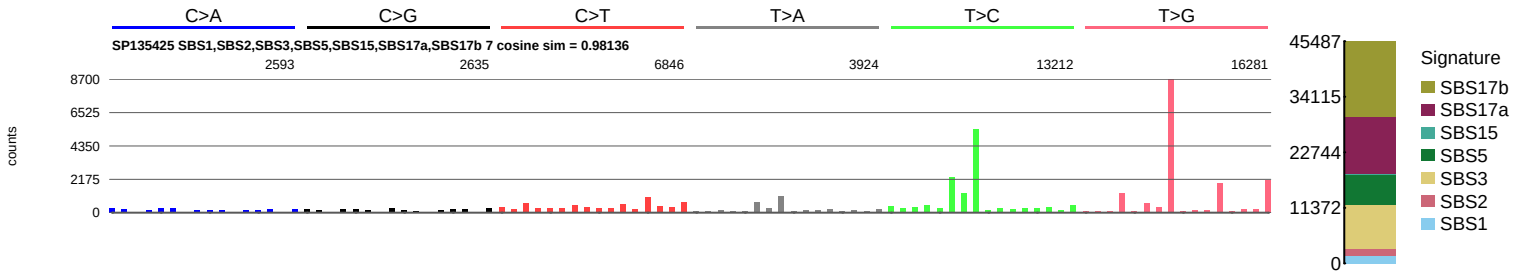

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98138
2543
2642
6597
3915
13510
16281
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

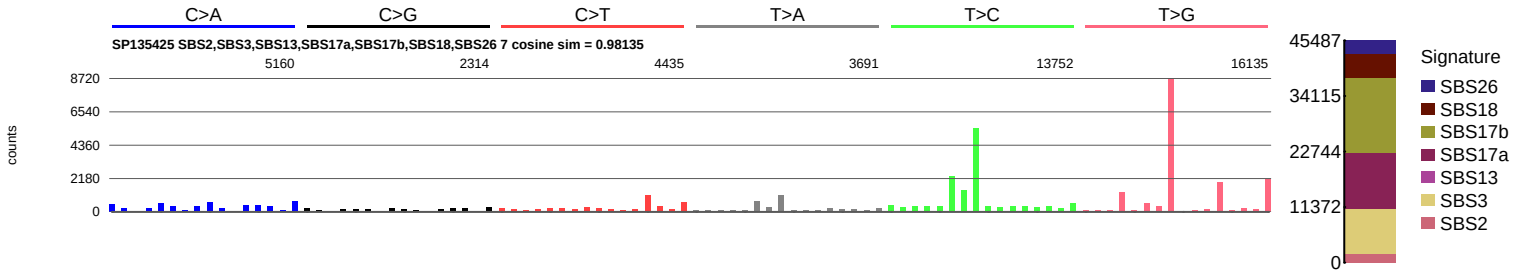

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98138
5115
2543
5662
3933
12074
16160
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2

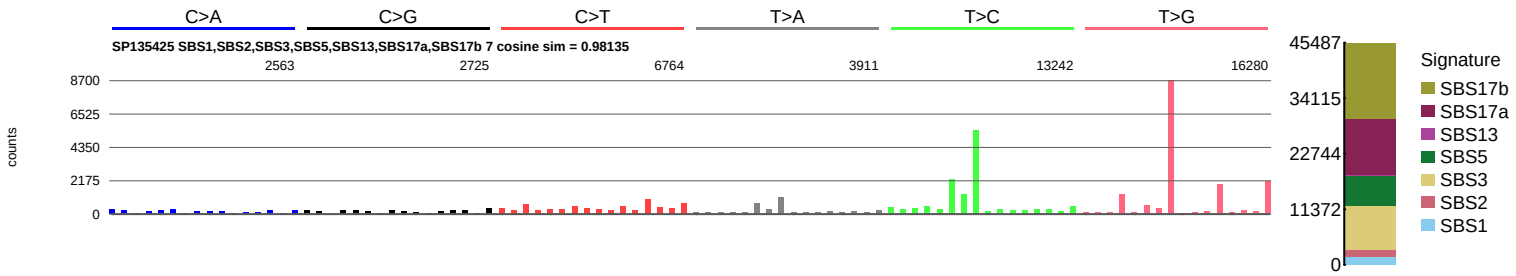

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98138
5439
502
6493
2716
14608
15725
counts
8844
6633
4422
2211
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

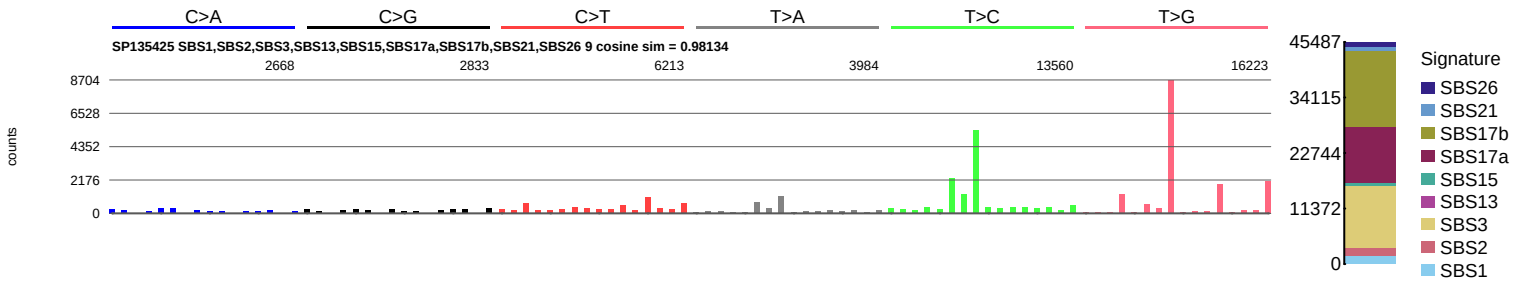

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98138
5119
2515
5666
3939
12081
16165
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2

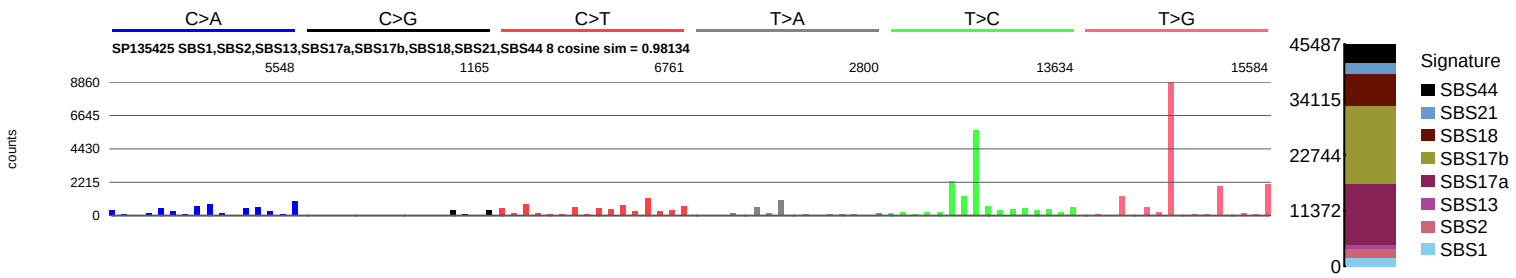

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98138
2582
2702
6838
3895
13205
16268
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

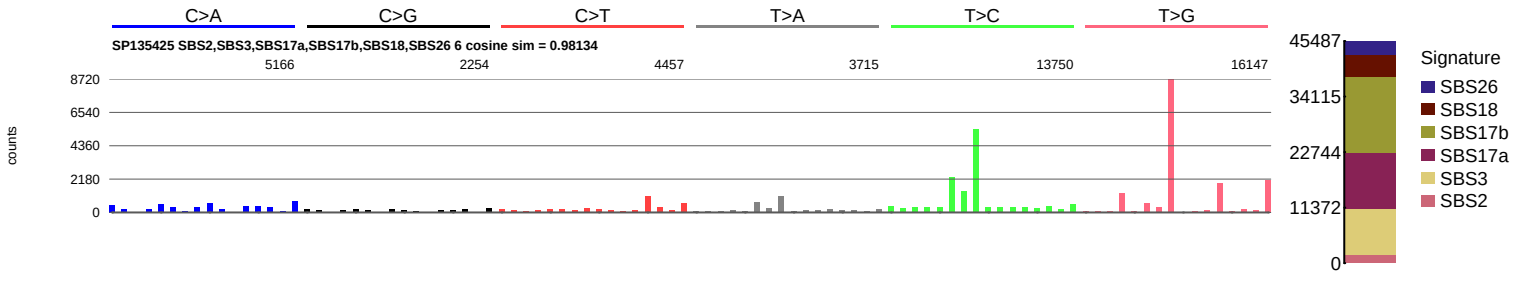

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98136
2593
2635
6846
3924
13212
16281
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

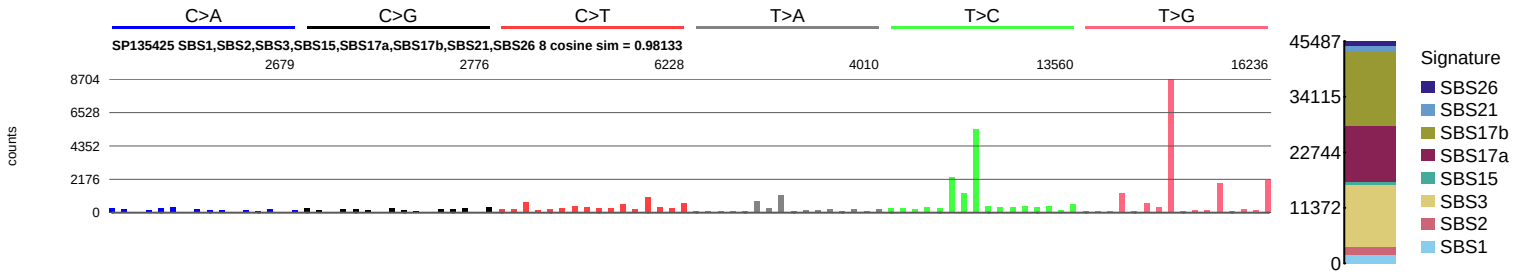

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98135
5160
2314
4435
3691
13752
16135
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2

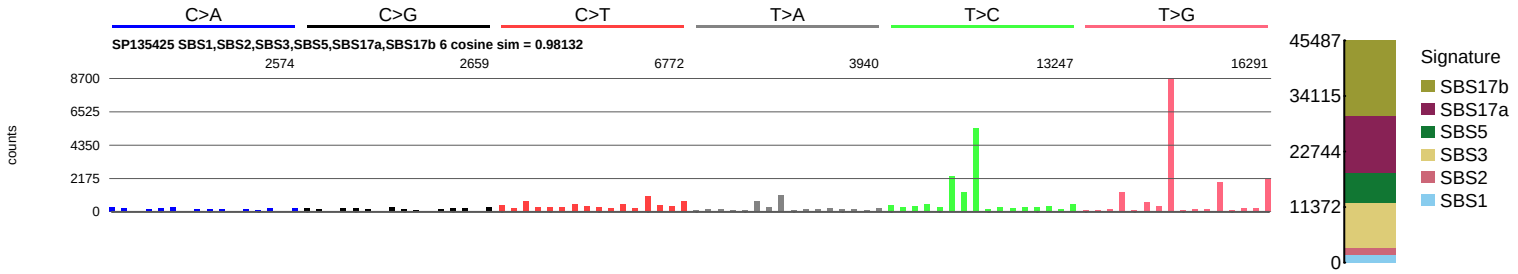

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 7 cosine sim = 0.98135
2563
2725
6764
3911
13242
16280
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

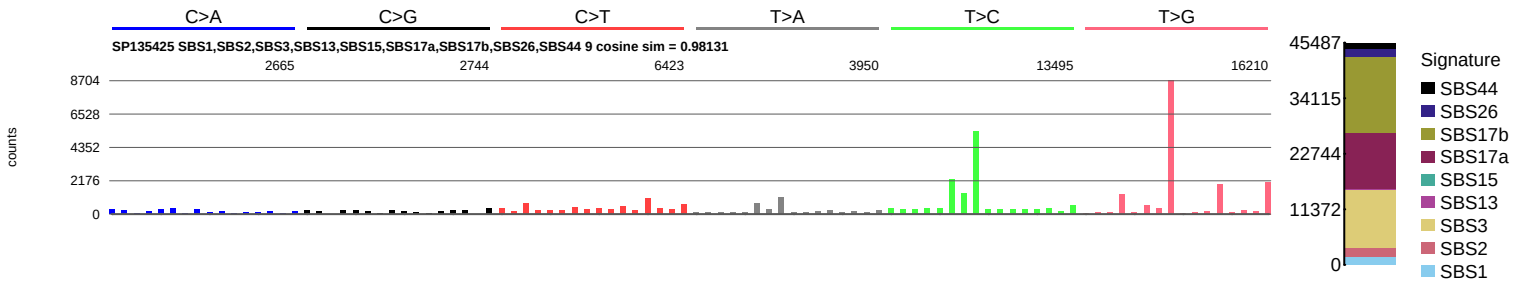

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26 9 cosine sim = 0.98134
2668
2833
6213
3984
13560
16223
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

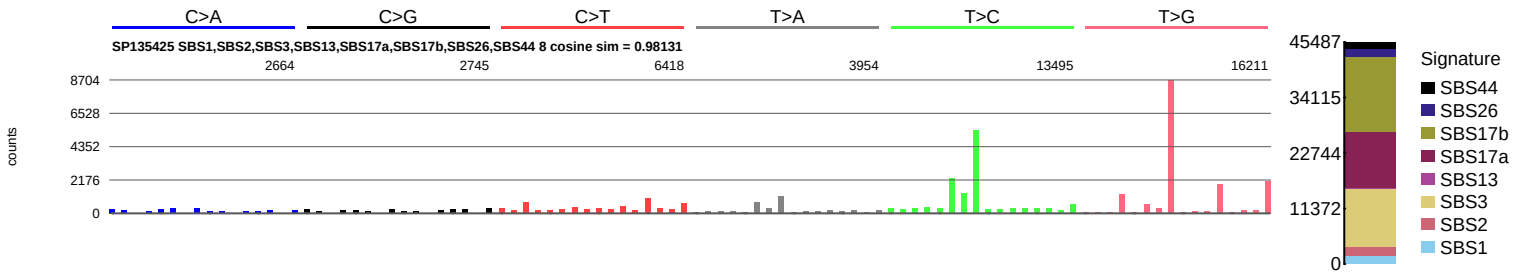

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98134
5548
1165
6761
2800
13634
15584
counts
8860
6645
4430
2215
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

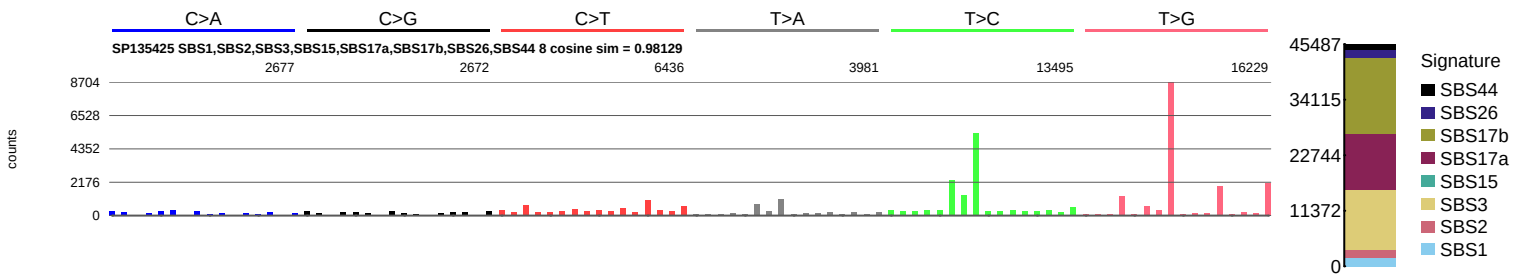

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98134
5166
2254
4457
3715
13750
16147
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2

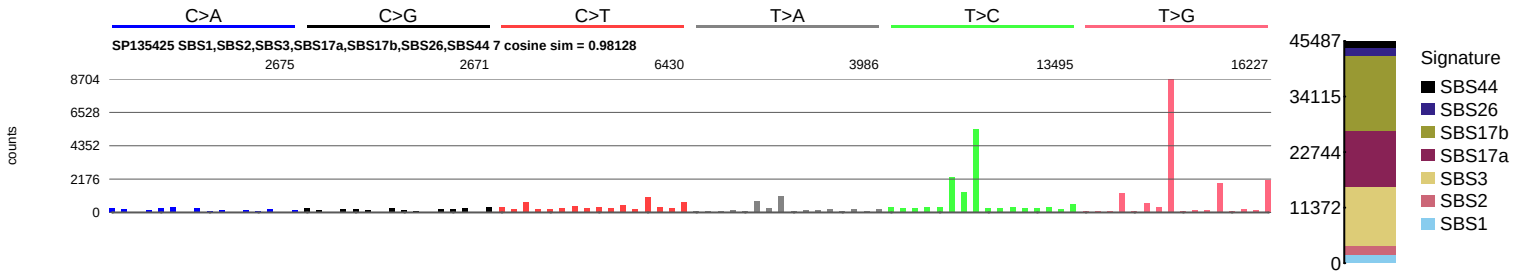

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98133
2679
2776
6228
4010
13560
16236
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

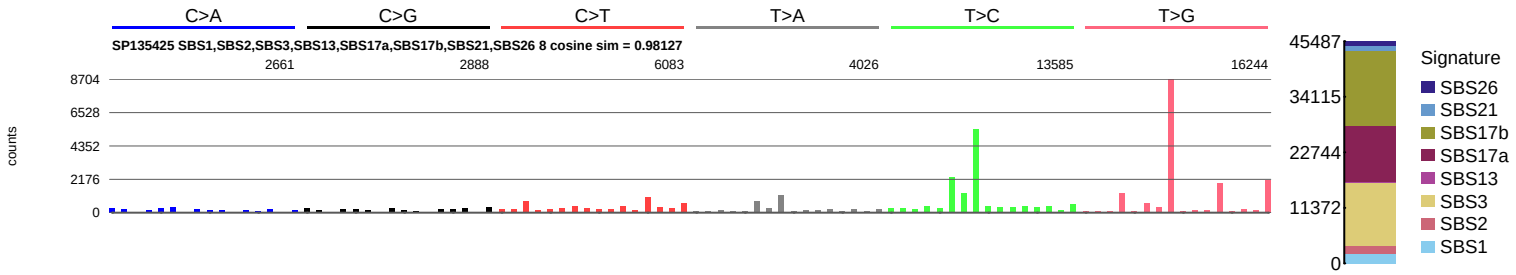

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b 6 cosine sim = 0.98132
2574
2659
6772
3940
13247
16291
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

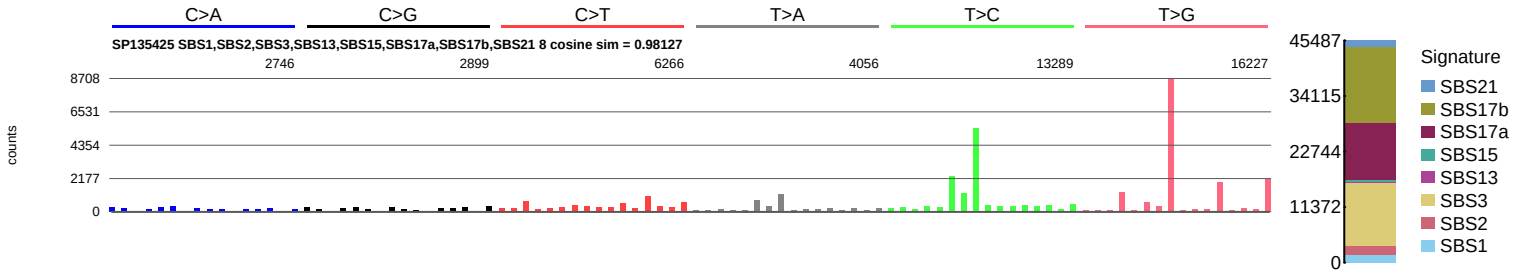

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS44 9 cosine sim = 0.98131
2665
2744
6423
3950
13495
16210
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98131
2664
2745
6418
3954
13495
16211
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS44 8 cosine sim = 0.98129
2677
2672
6436
3981
13495
16229
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS44 7 cosine sim = 0.98128
2675
2671
6430
3986
13495
16227
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26 8 cosine sim = 0.98127
2661
2888
6083
4026
13585
16244
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98127
2746
2899
6266
4056
13289
16227
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
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Signature
SBS21
SBS17b
SBS17a
SBS15
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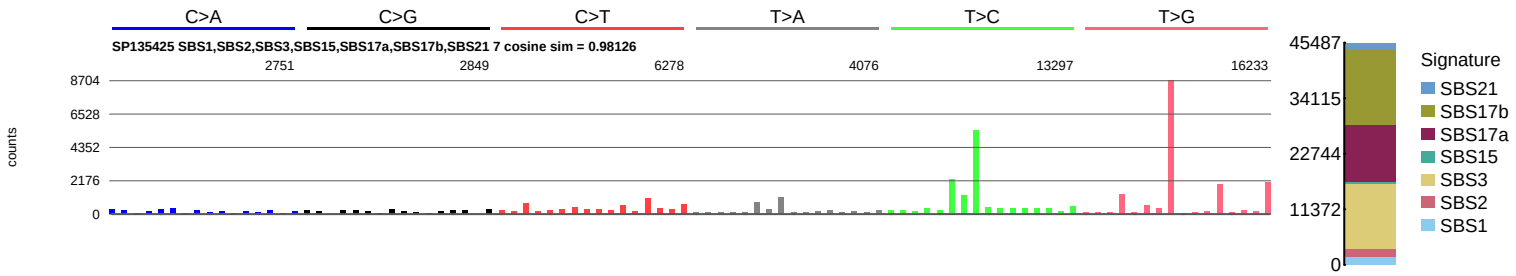
SBS3
SBS2
SBS1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SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98126
2751
2849
6278
4076
13297
16233
8704
6528
4352
2176
0
counts
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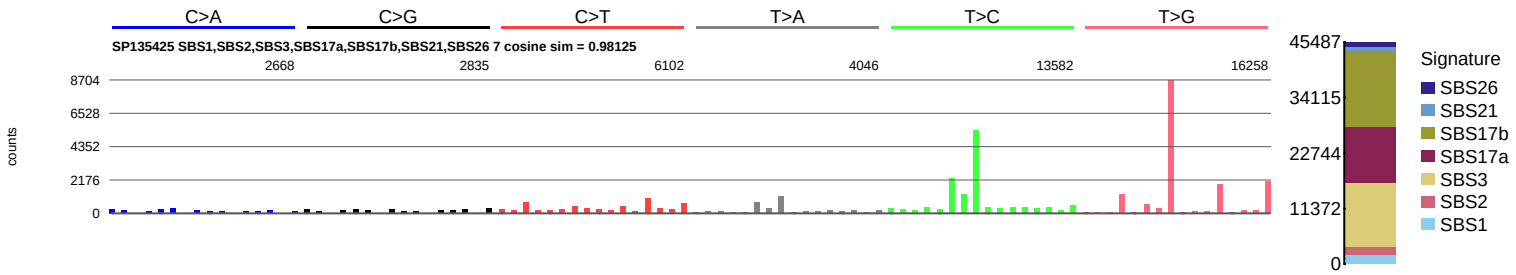
SBS2
SBS1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SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26 7 cosine sim = 0.98125
2668
2835
6102
4046
13582
16258
8704
6528
4352
2176
0
counts
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34115
22744
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0
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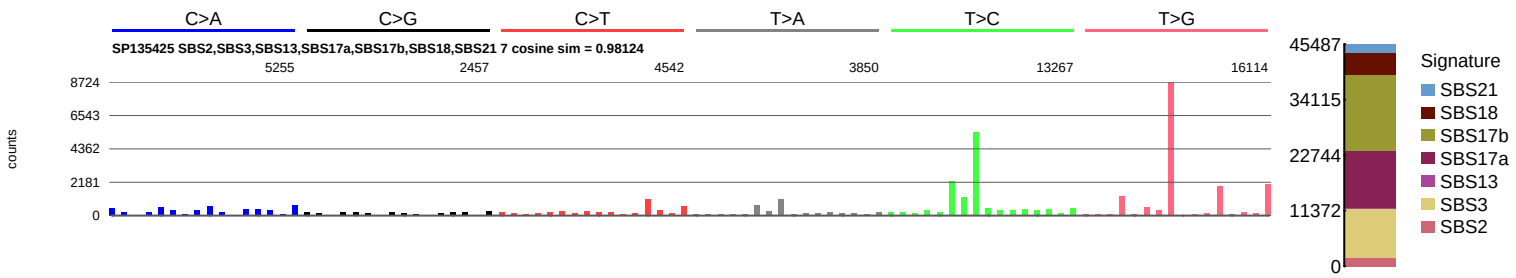
SBS3
SBS2
SBS1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SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98124
5255
2457
4542
3850
13267
16114
8724
6543
4362
2181
0
counts
45487
34115
22744
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SBS18
SBS17b
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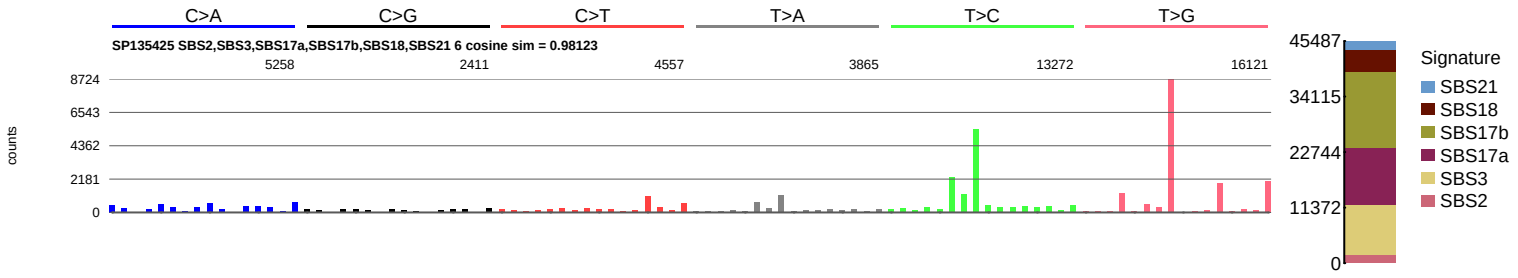
SBS13
SBS3
SBS2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SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98123
5258
2411
4557
3865
13272
16121
counts
8724
6543
4362
2181
0
45487
34115
22744
11372
0
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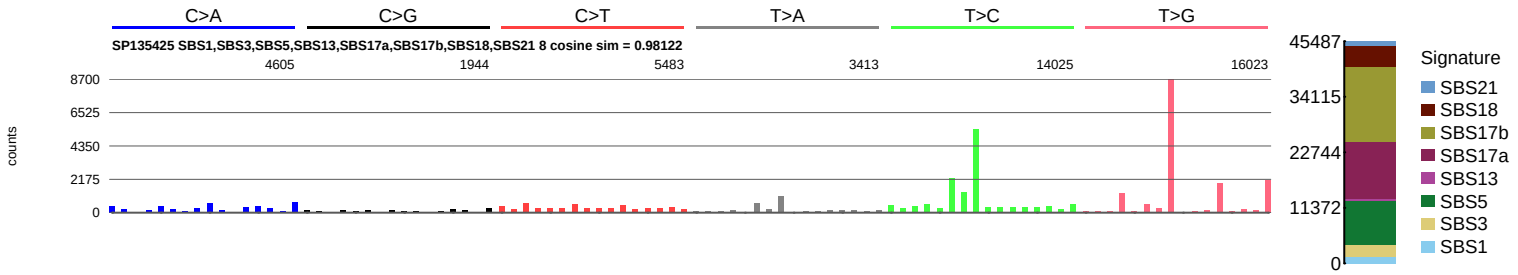
SBS18
SBS17b
SBS17a
SBS3
SBS2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SP135425 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98122
4605
1944
5483
3413
14025
16023
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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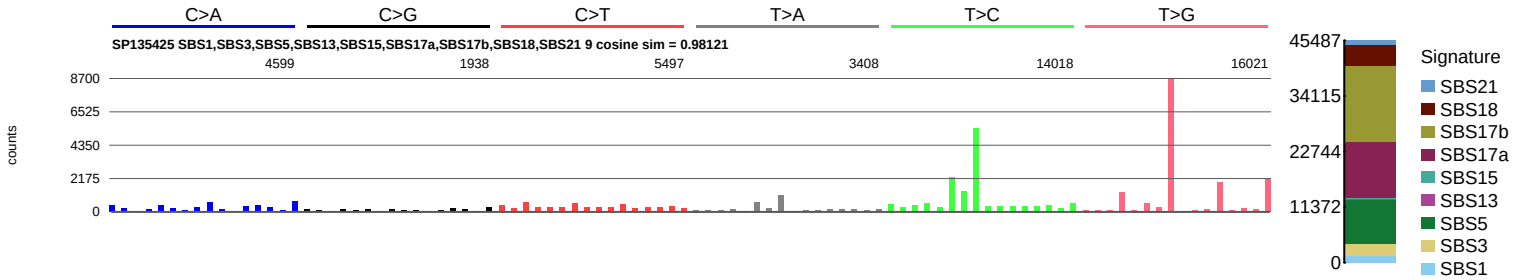
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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T>G
SP135425 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98121
4599
1938
5497
3408
14018
16021
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
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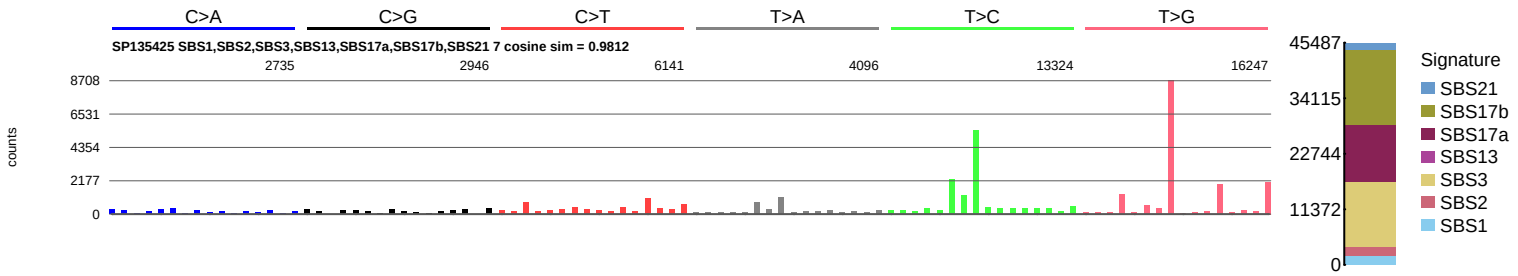
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
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T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.9812
2735
2946
6141
4096
13324
16247
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
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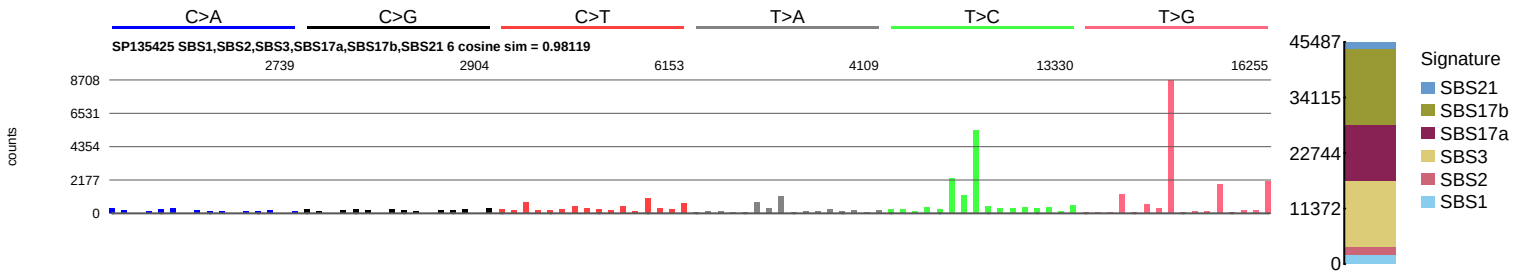
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98119
2739
2904
6153
4109
13330
16255
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
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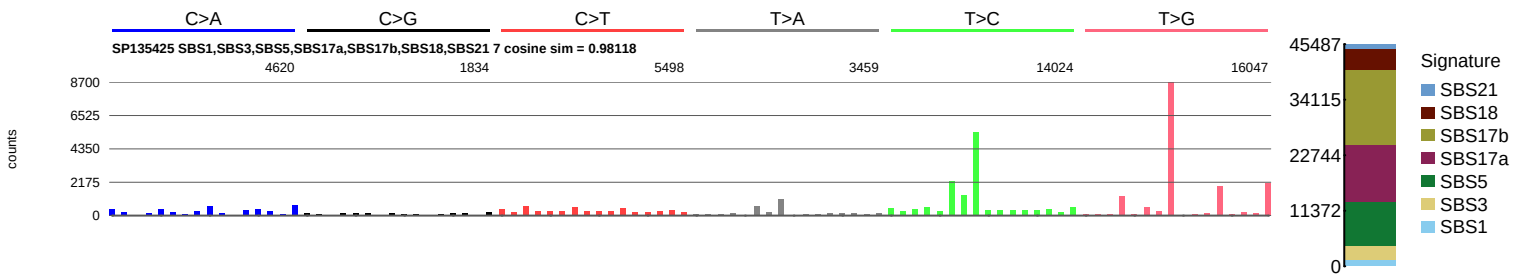
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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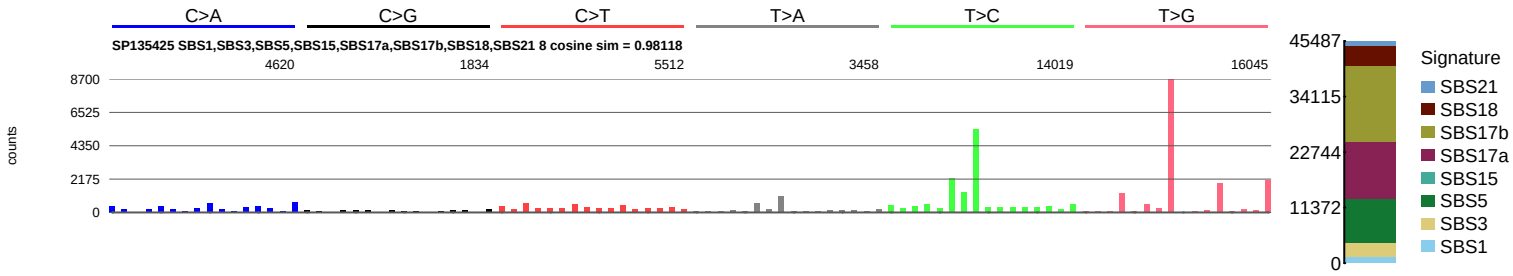
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98118
4620
1834
5498
3459
14024
16047
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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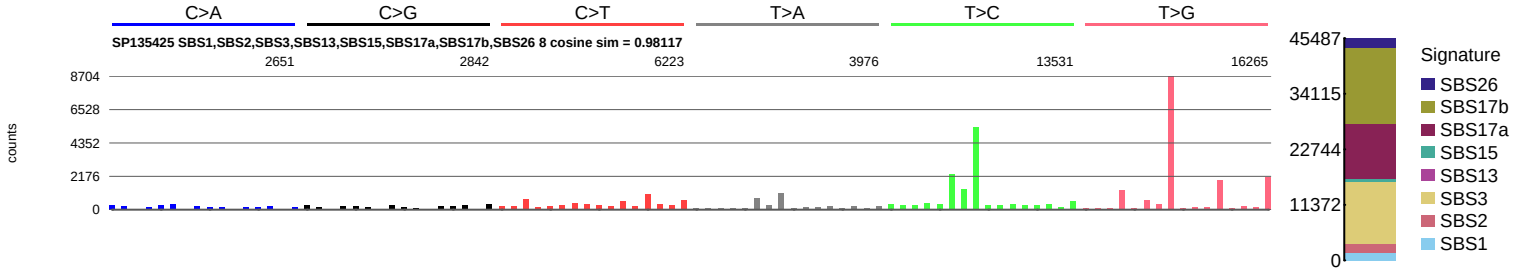
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98118
4620
1834
5512
3458
14019
16045
8700
6525
4350
2175
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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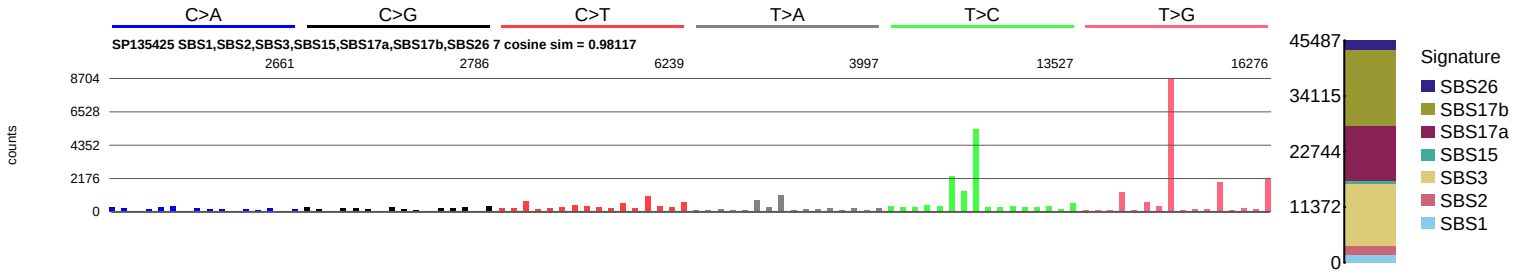
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.98117
2651
2842
6223
3976
13531
16265
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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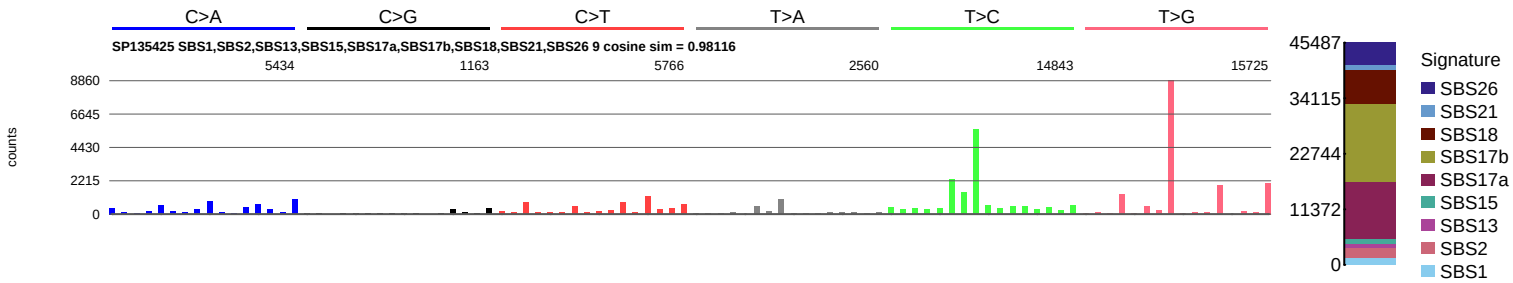
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98117
2661
2786
6239
3997
13527
16276
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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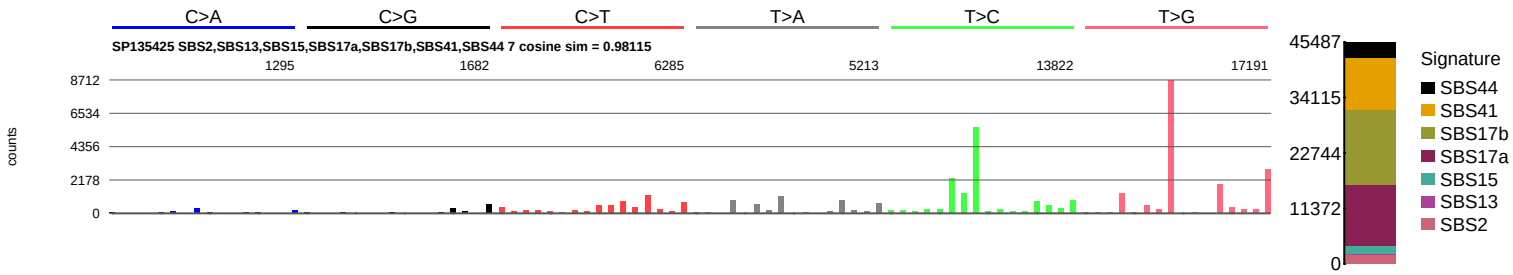
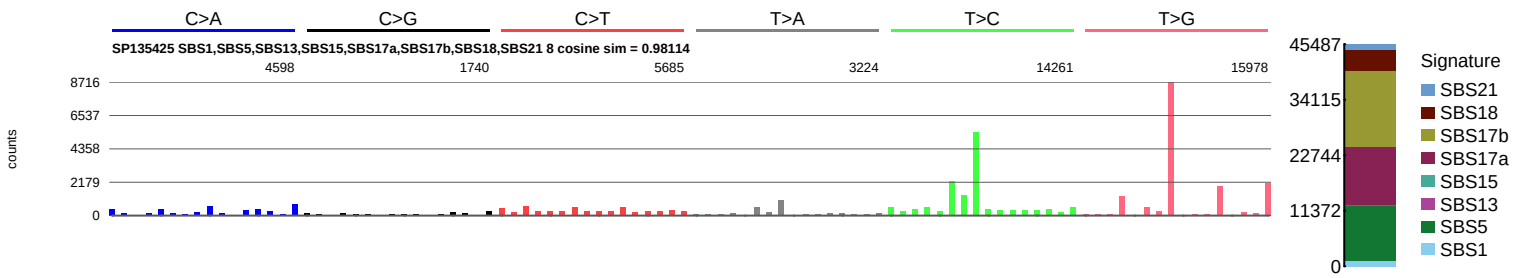
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98116
5434
1163
5766
2560
14843
15725
8860
6645
4430
2215
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.98115
1295
1682
6285
5213
13822
17191
8712
6534
4356
2178
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2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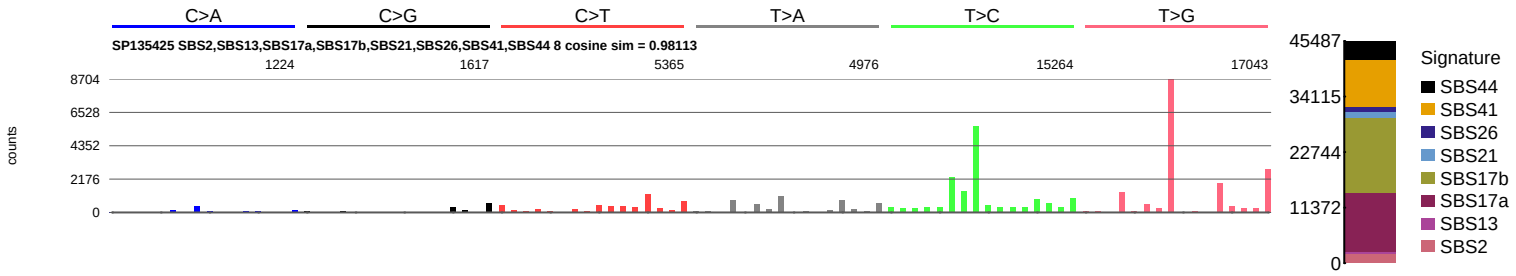
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98114
4598
1740
5685
3224
14261
15978
8716
6537
4358
2179
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98113
1224
1617
5365
4976
15264
17043
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
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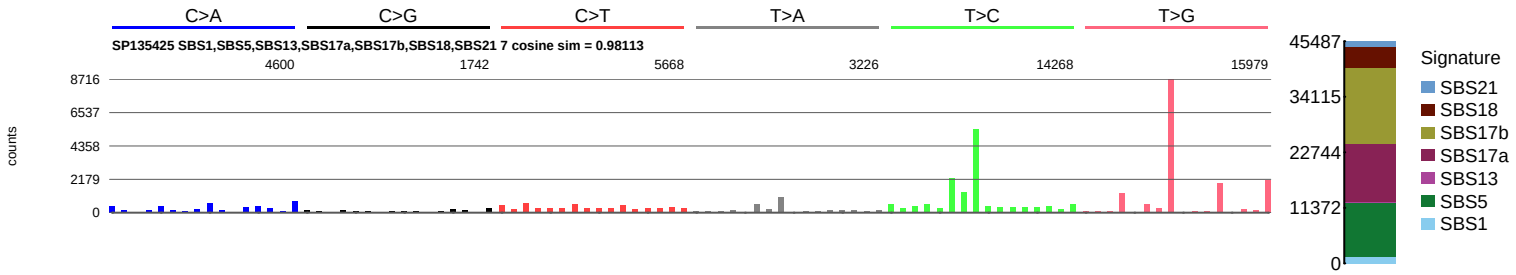
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2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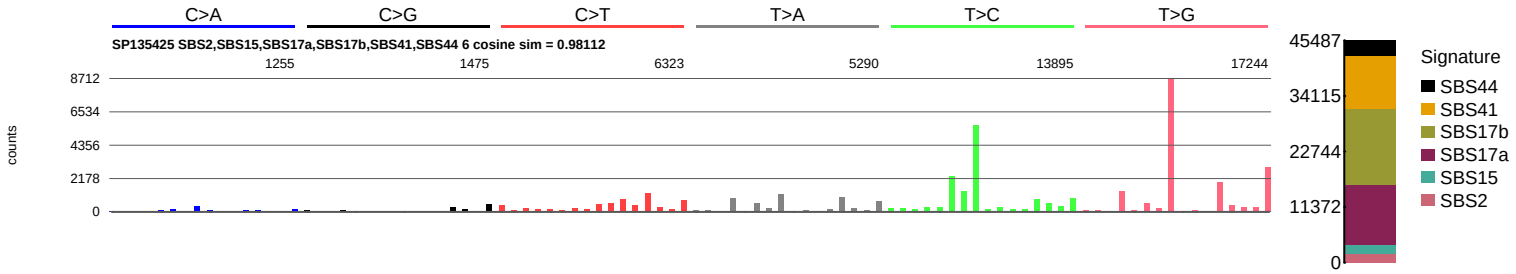
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98113
4600
1742
5668
3226
14268
15979
counts
8716
6537
4358
2179
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98112
1255
1475
6323
5290
13895
17244
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
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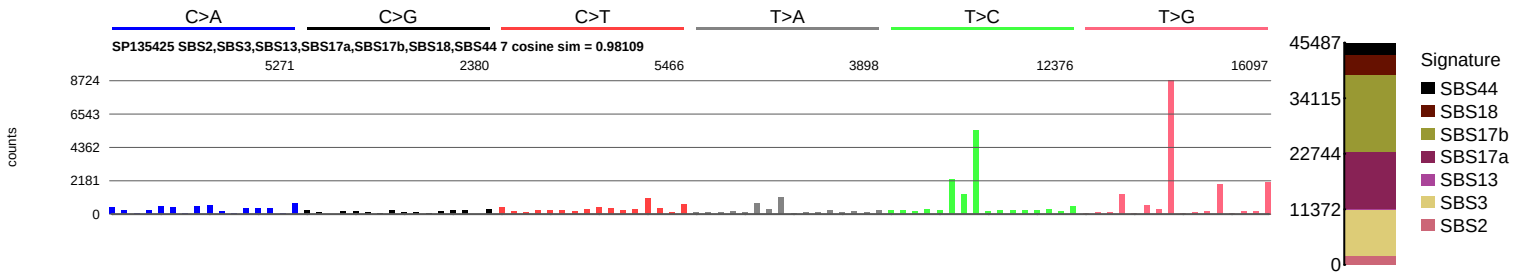
SBS41
SBS17b
SBS17a
SBS15
SBS2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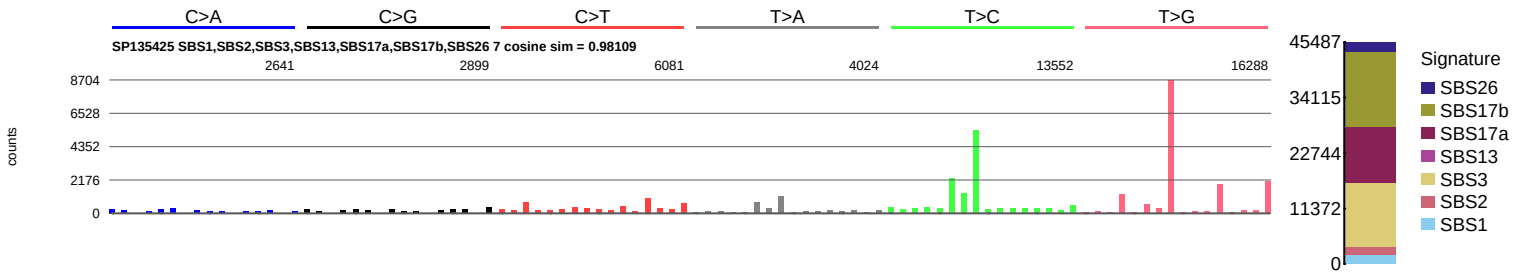
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98109
5271
2380
5466
3898
12376
16097
counts
8724
6543
4362
2181
0
45487
34115
22744
11372
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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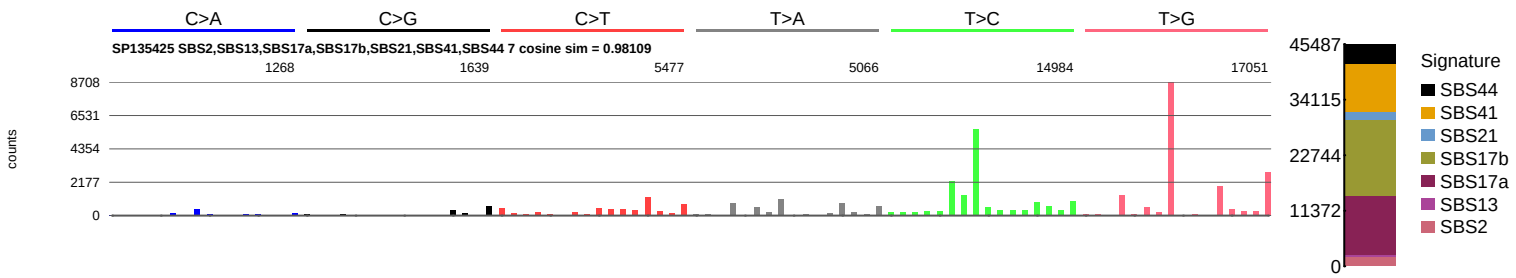
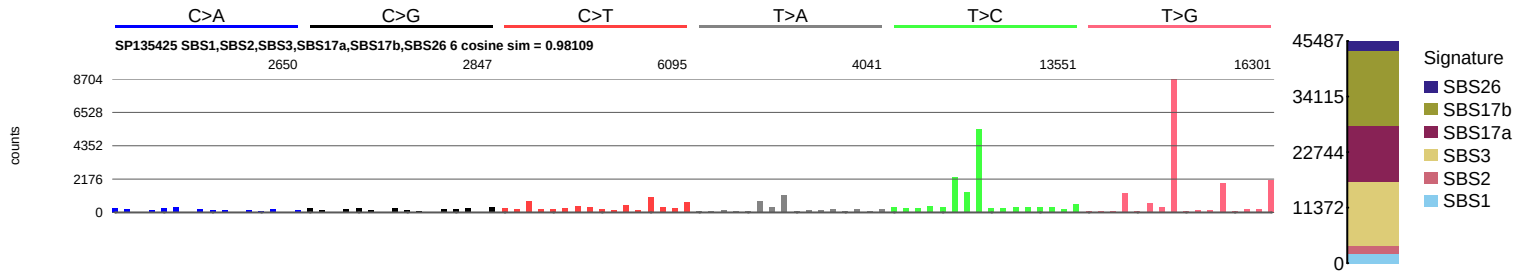
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98109
2641
2899
6081
4024
13552
16288
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98109
1268
1639
5477
5066
14984
17051
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS2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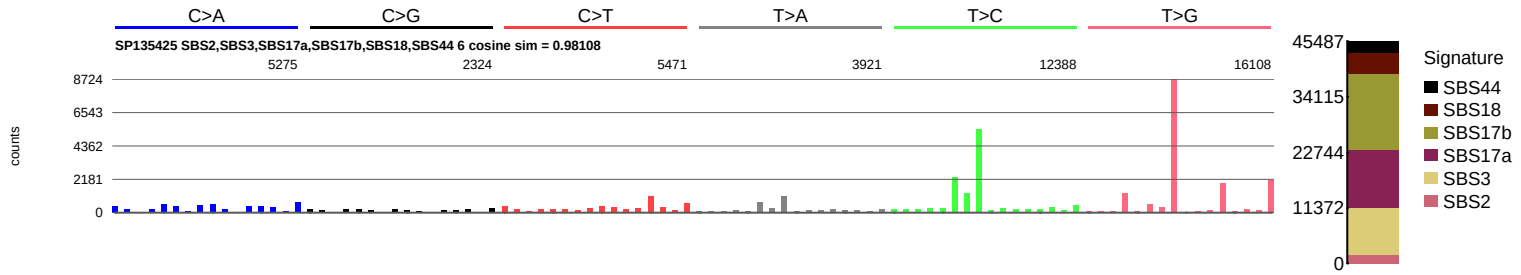
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26 6 cosine sim = 0.98109
2650
2847
6095
4041
13551
16301
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 6 cosine sim = 0.98108
5275
2324
5471
3921
12388
16108
counts
8724
6543
4362
2181
0
45487
34115
22744
11372
0
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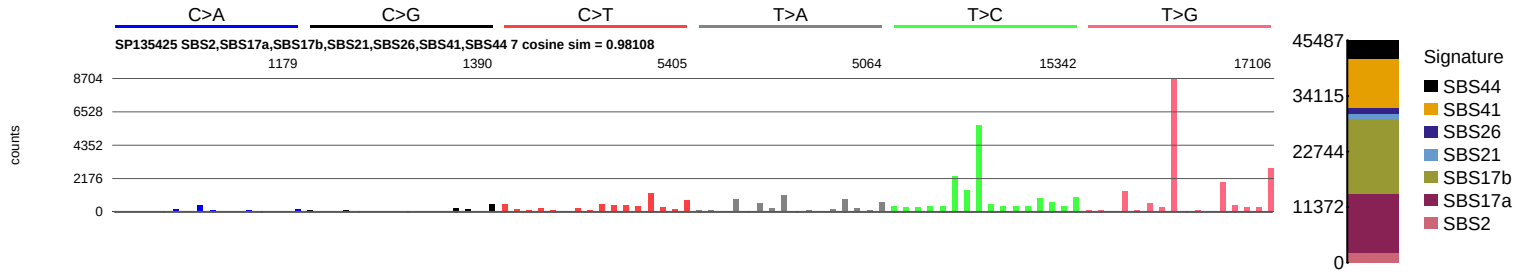
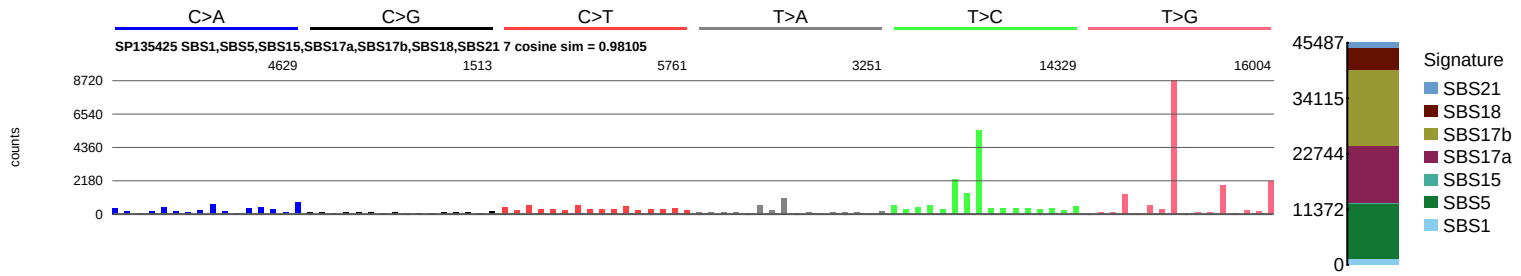
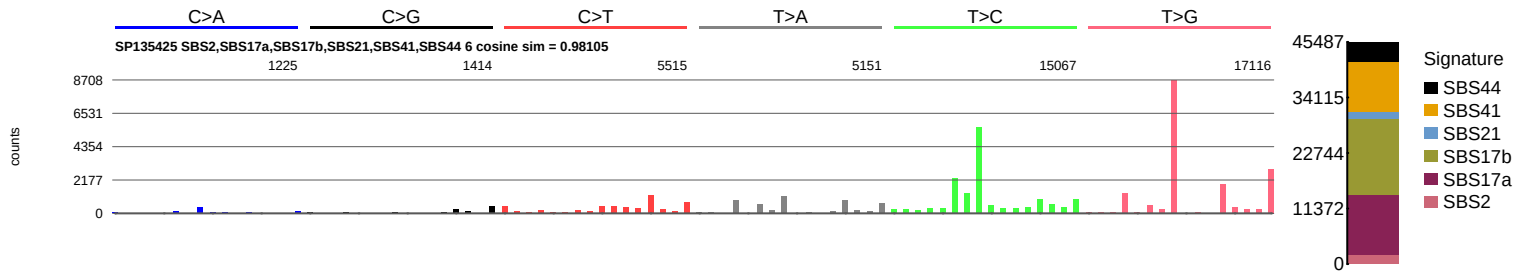
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 7 cosine sim = 0.98108
1179
1390
5405
5064
15342
17106
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98105
4629
1513
5761
3251
14329
16004
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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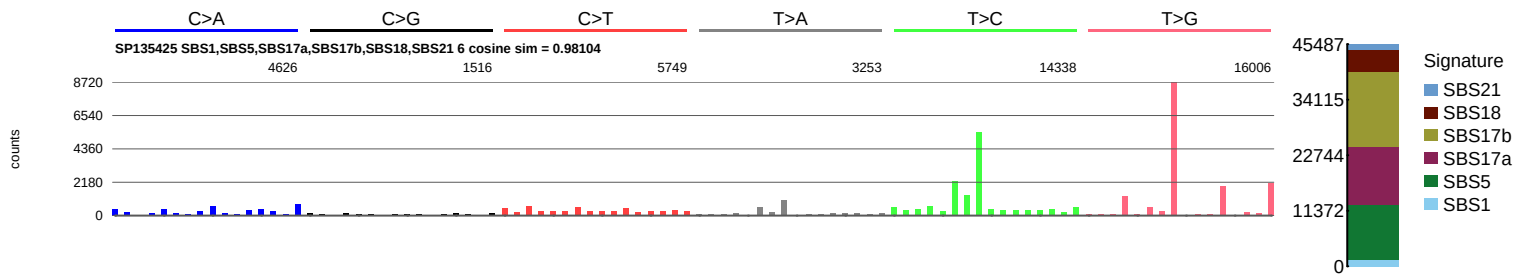
SP135425 SBS2,SBS17a,SBS17b,SBS21,SBS41,SBS44 6 cosine sim = 0.98105
1225
1414
5515
5151
15067
17116
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS2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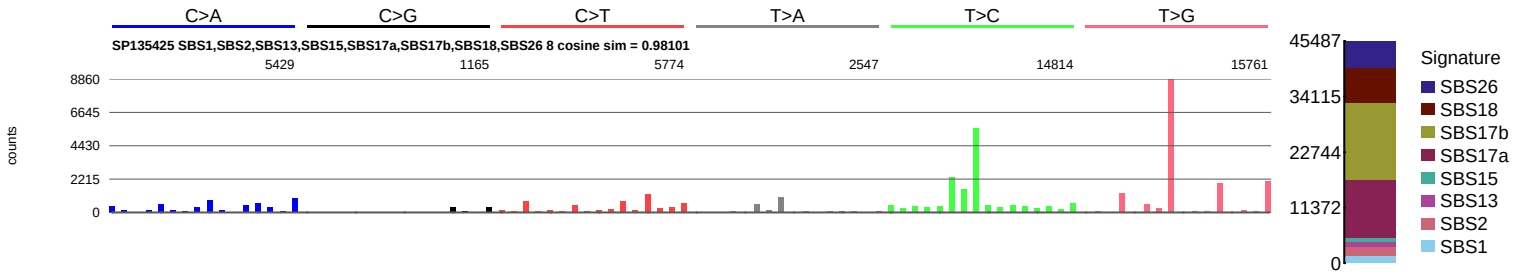
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98104
4626
1516
5749
3253
14338
16006
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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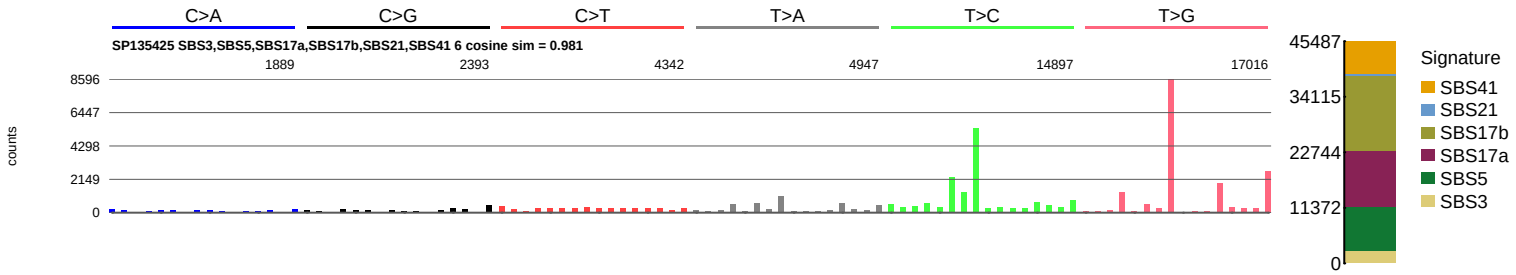
SP135425 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98101
5429
1165
5774
2547
14814
15761
8860
6645
4430
2215
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>T
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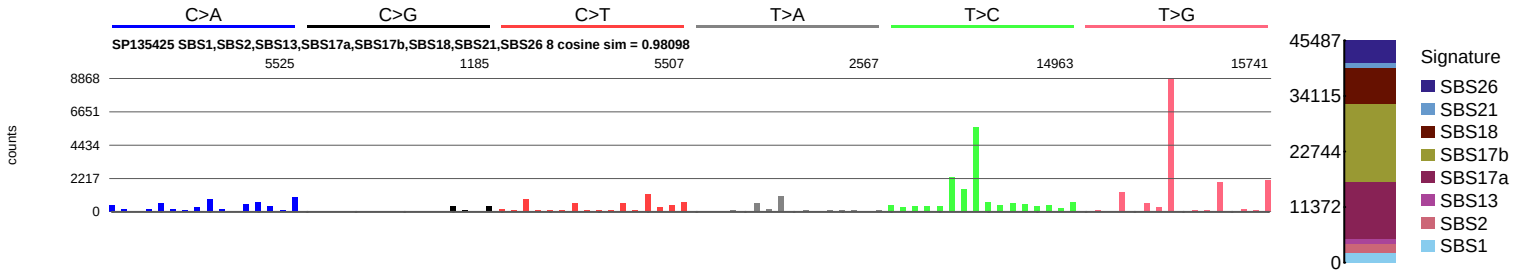
T>C
T>G
SP135425 SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.981
1889
2393
4342
4947
14897
17016
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3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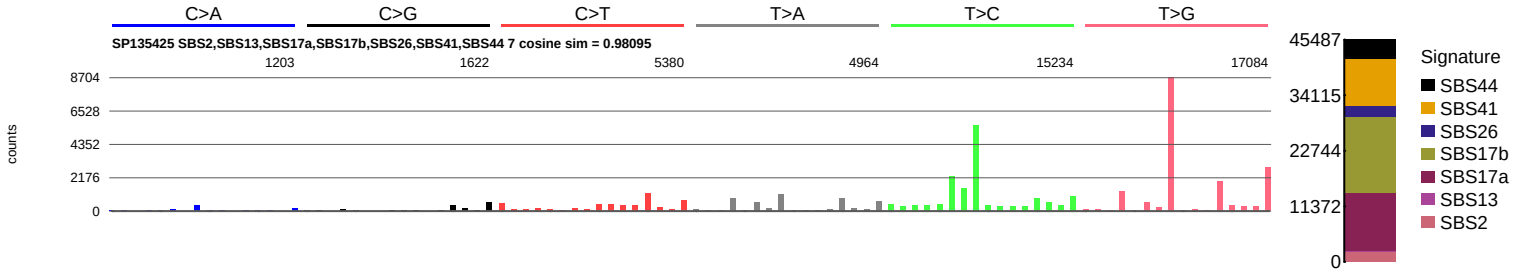
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98098
5525
1185
5507
2567
14963
15741
8868
6651
4434
2217
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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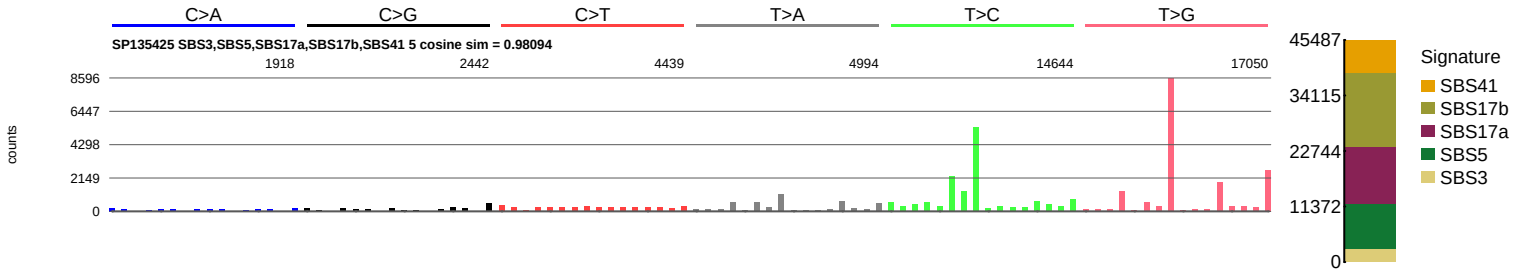
T>C
T>G
SP135425 SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98095
1203
1622
5380
4964
15234
17084
8704
6528
4352
2176
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS2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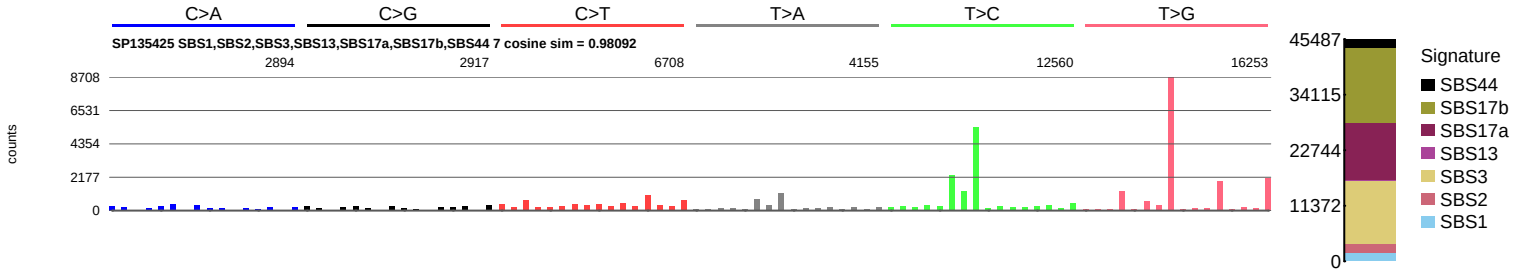
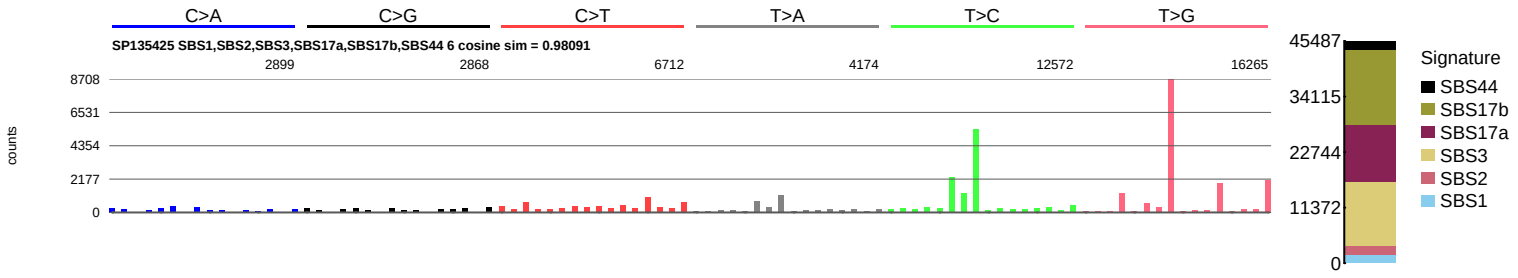
T>A
T>C
T>G
SP135425 SBS3,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98094
1918
2442
4439
4994
14644
17050
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS44 7 cosine sim = 0.98092
2894
2917
6708
4155
12560
16253
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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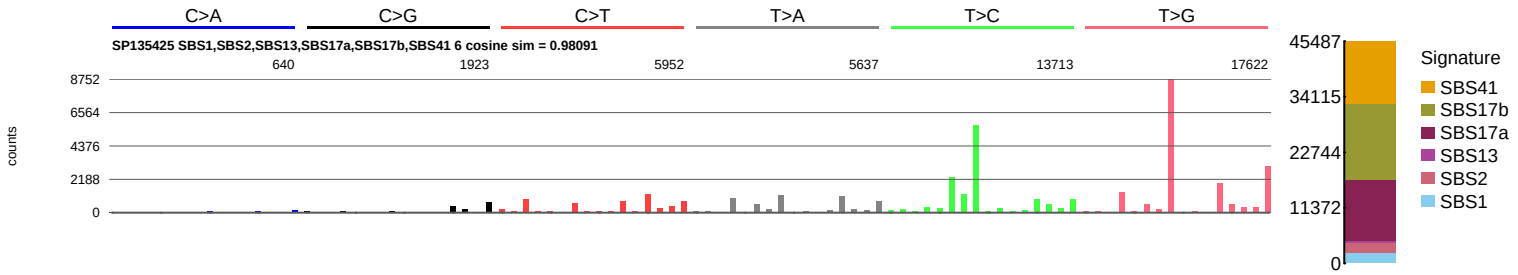
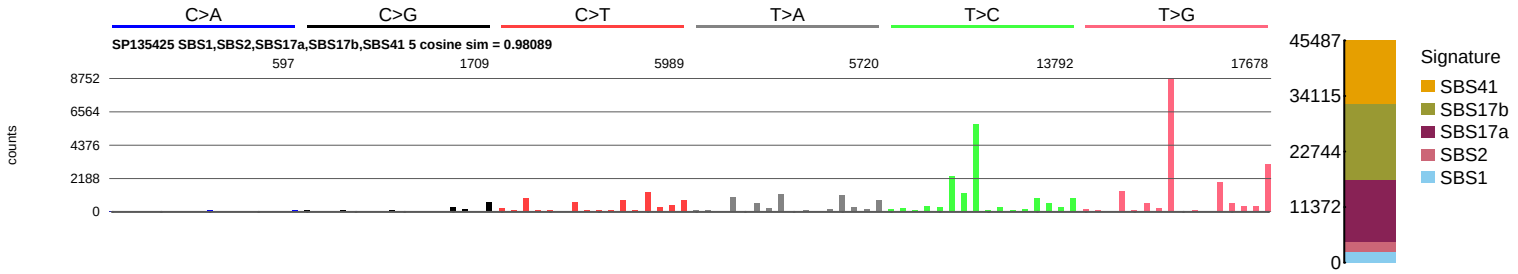
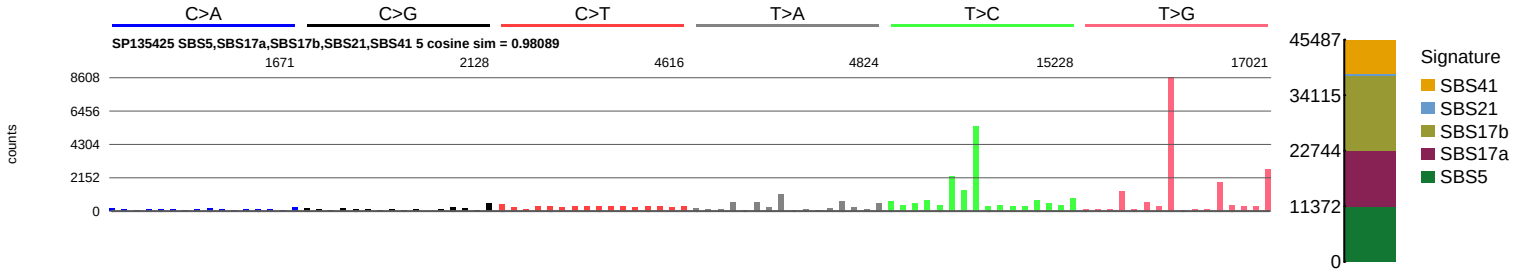
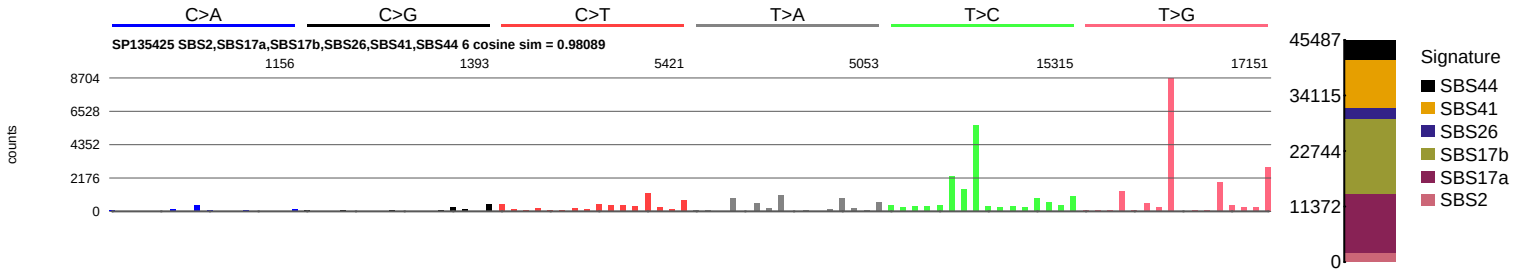
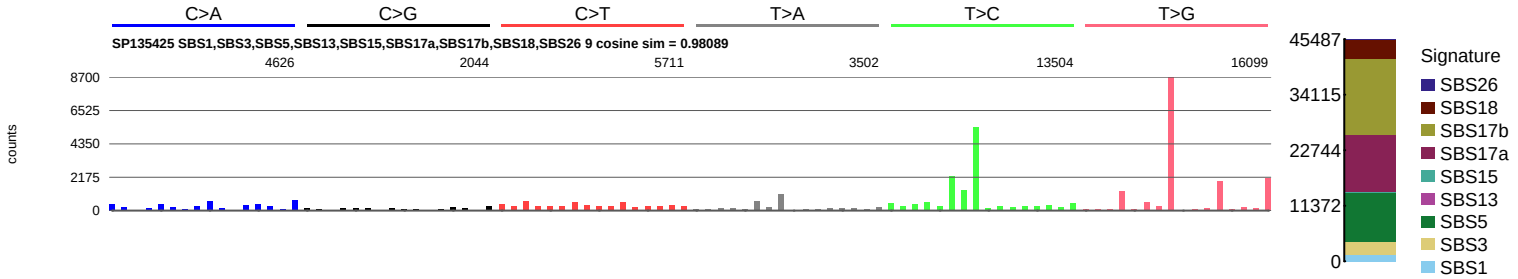
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS44 6 cosine sim = 0.98091
2899
2868
6712
4174
12572
16265
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS44
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98091
640
1923
5952
5637
13713
17622
counts
8752
6564
4376
2188
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98089
597
1709
5989
5720
13792
17678
counts
8752
6564
4376
2188
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS21,SBS41 5 cosine sim = 0.98089
1671
2128
4616
4824
15228
17021
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS26,SBS41,SBS44 6 cosine sim = 0.98089
1156
1393
5421
5053
15315
17151
counts
8704
6528
4352
2176
0
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS2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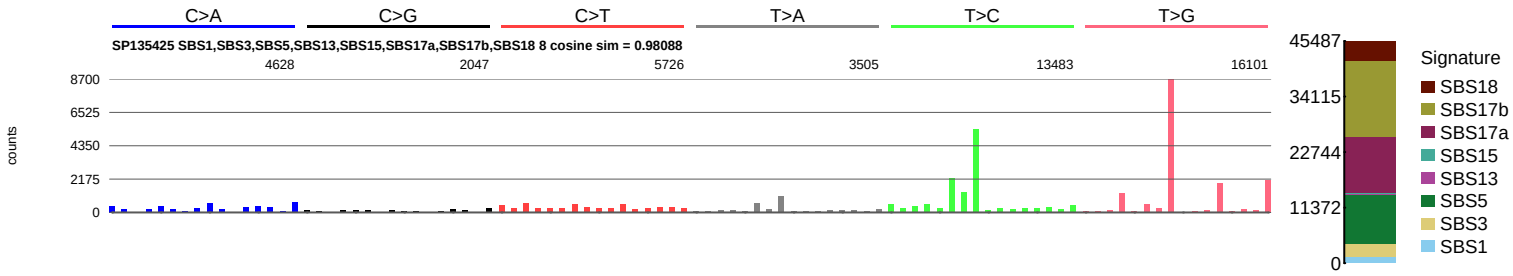
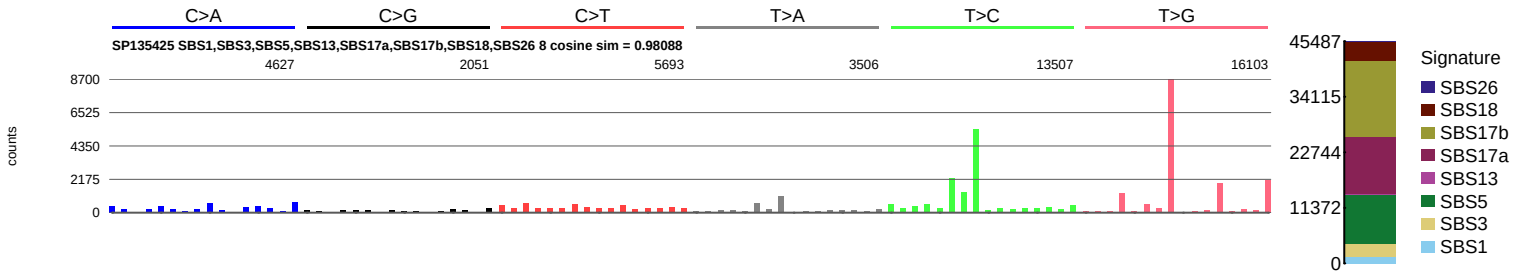
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98089
4626
2044
5711
3502
13504
16099
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98088
4628
2047
5726
3505
13483
16101
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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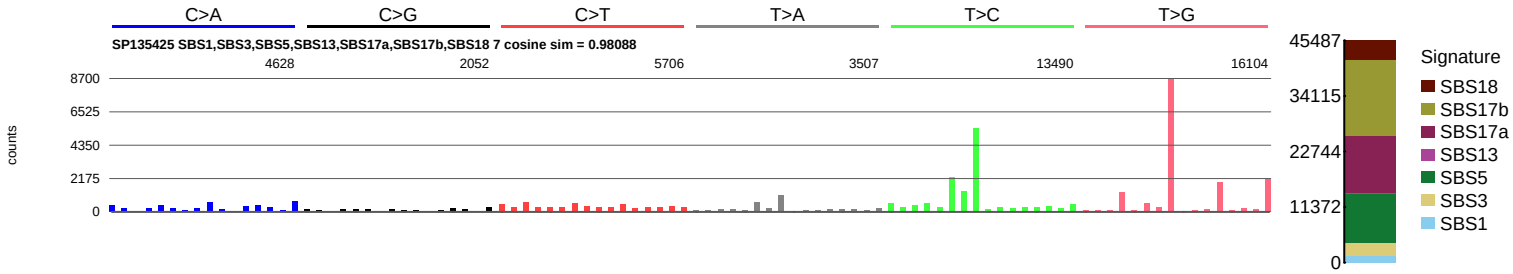
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98088
4627
2051
5693
3506
13507
16103
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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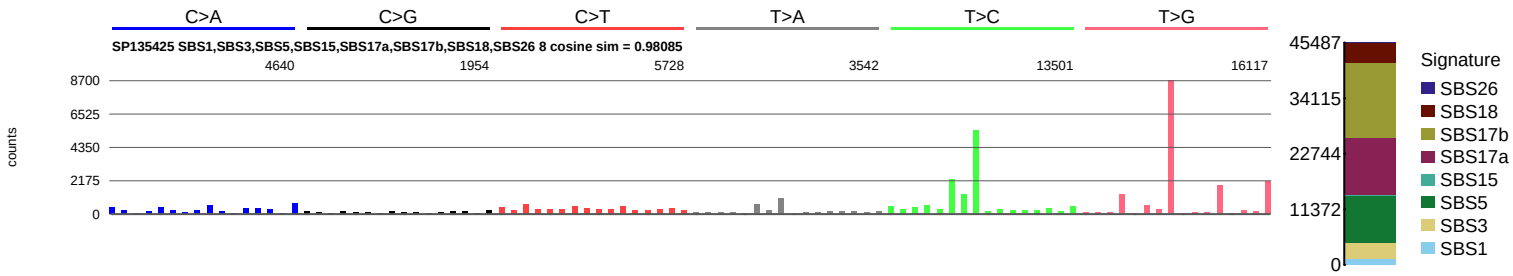
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98088
4628
2052
5706
3507
13490
16104
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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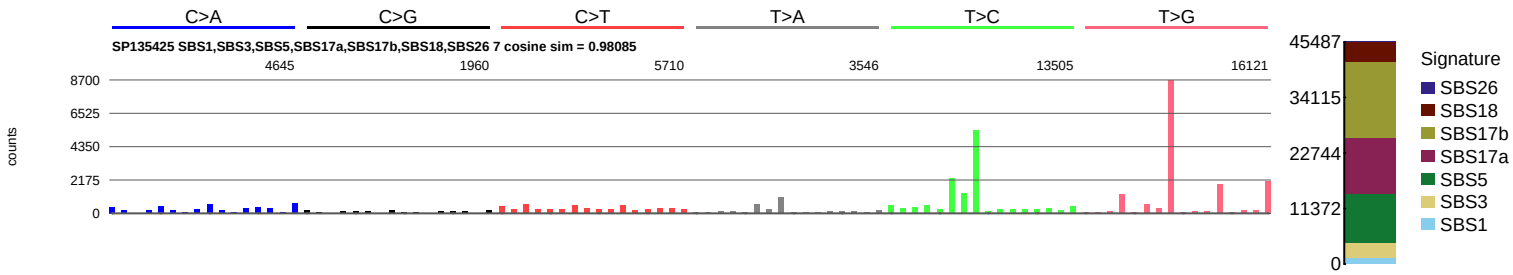
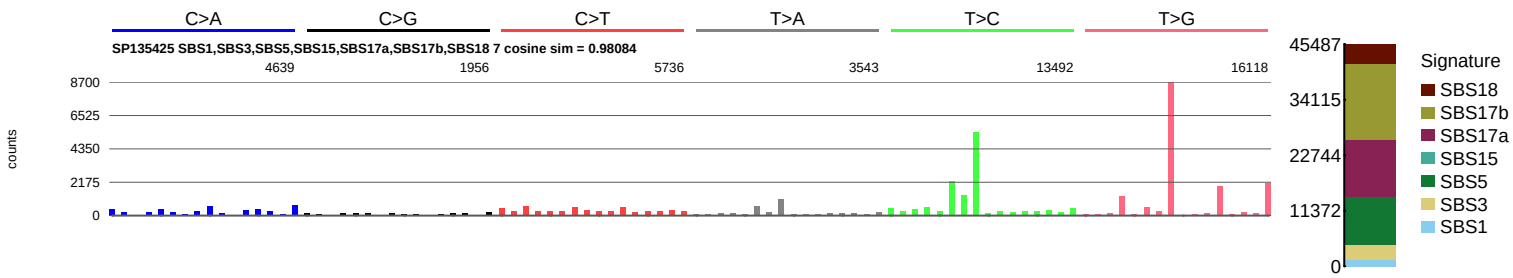
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98085
4640
1954
5728
3542
13501
16117
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98085
4645
1960
5710
3546
13505
16121
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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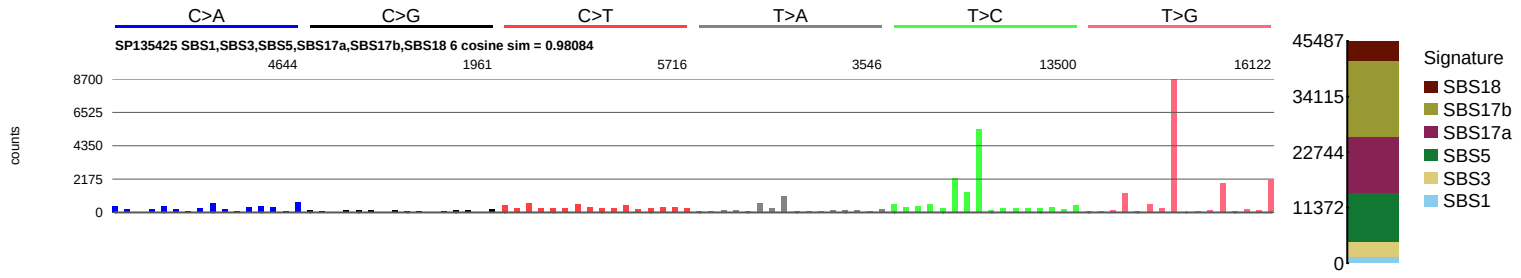
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98084
4639
1956
5736
3543
13492
16118
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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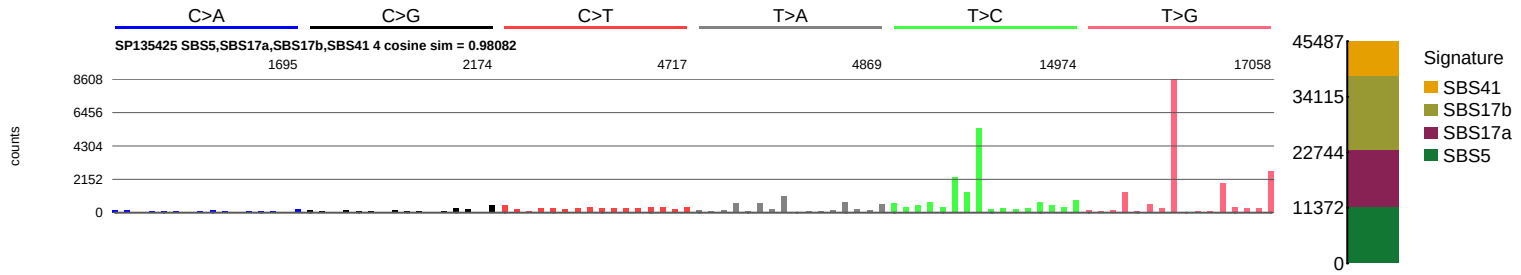
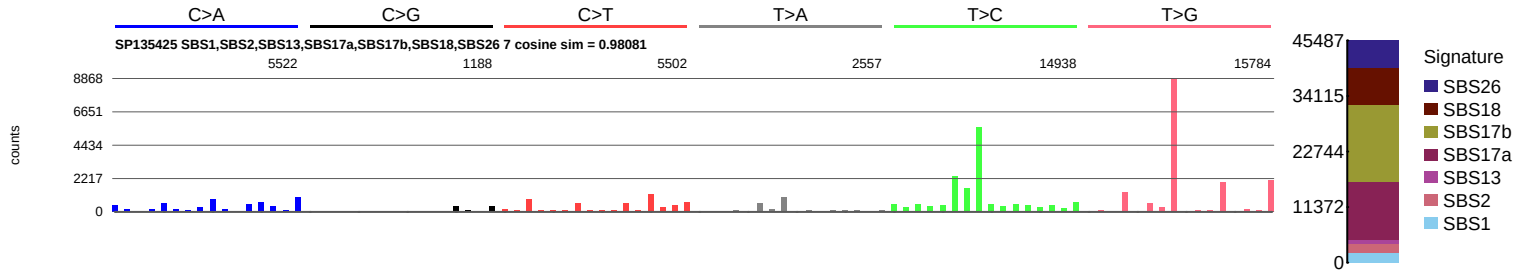
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98084
4644
1961
5716
3546
13500
16122
counts
8700
6525
4350
2175
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS17a,SBS17b,SBS41 4 cosine sim = 0.98082
1695
2174
4717
4869
14974
17058
counts
8608
6456
4304
2152
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS5


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98081
5522
1188
5502
2557
14938
15784
counts
8868
6651
4434
2217
0
45487
34115
22744
11372
0
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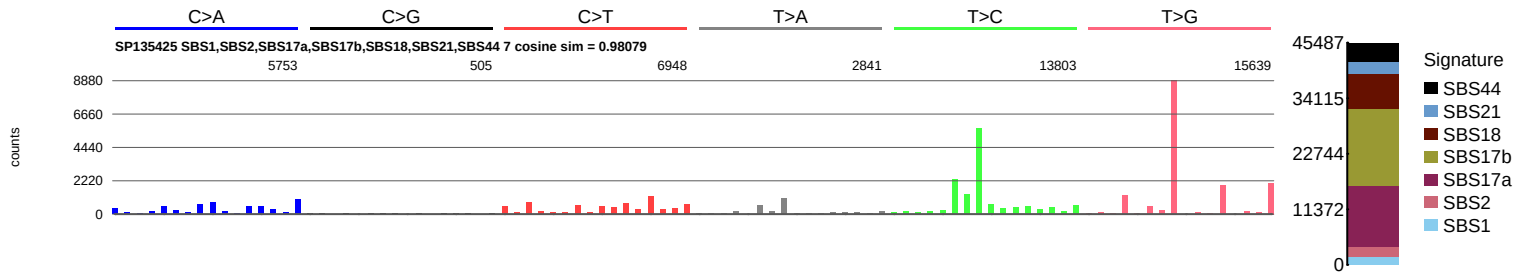
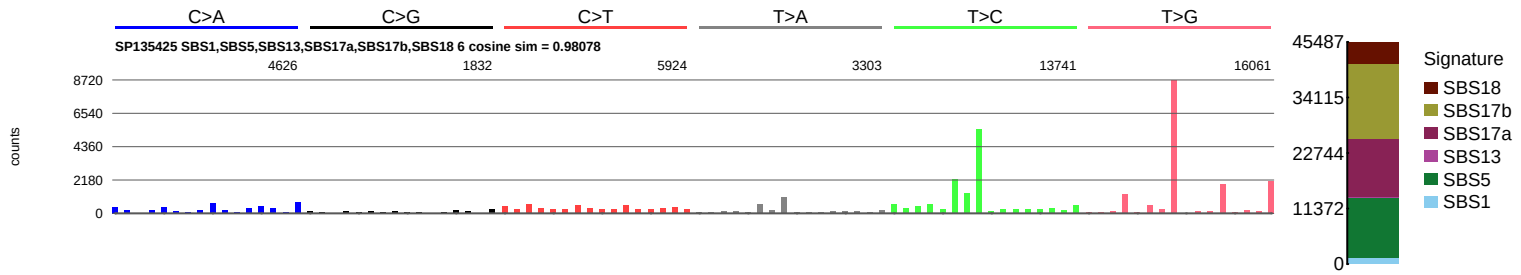
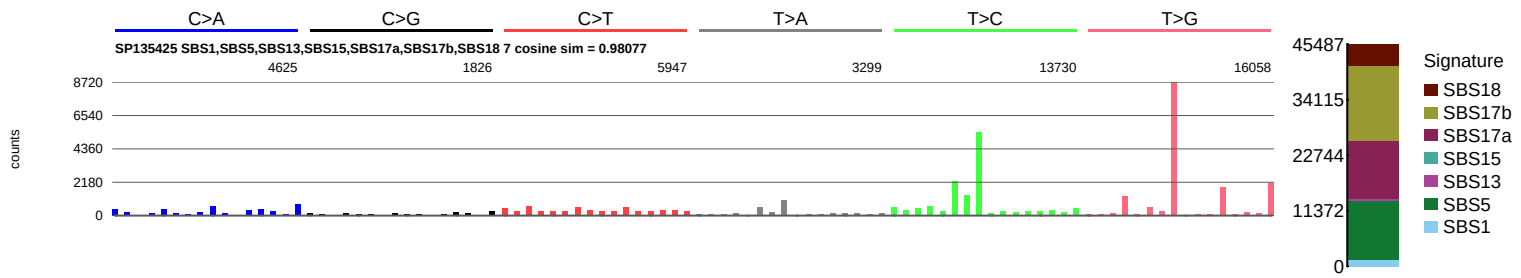
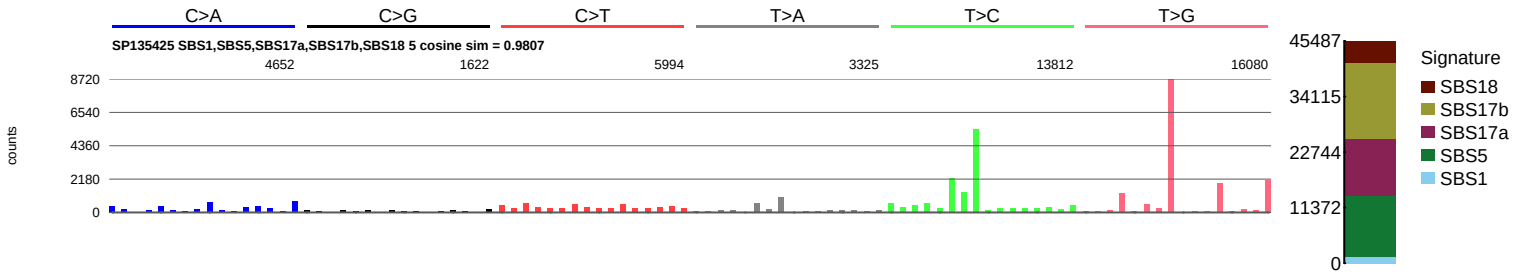
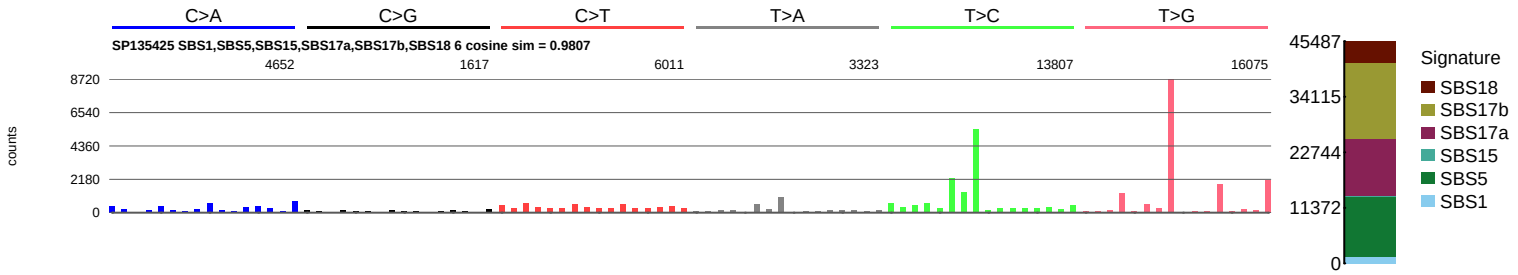
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

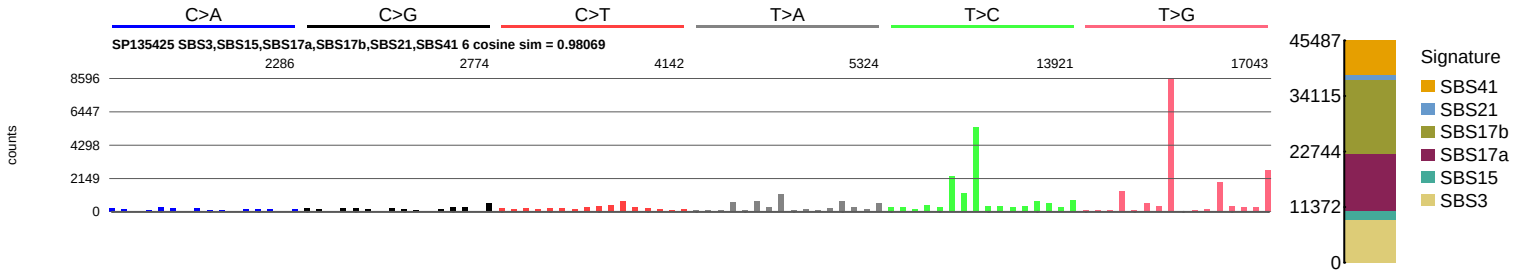

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.98079
5753
505
6948
2841
13803
15639
counts
8880
6660
4440
2220
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

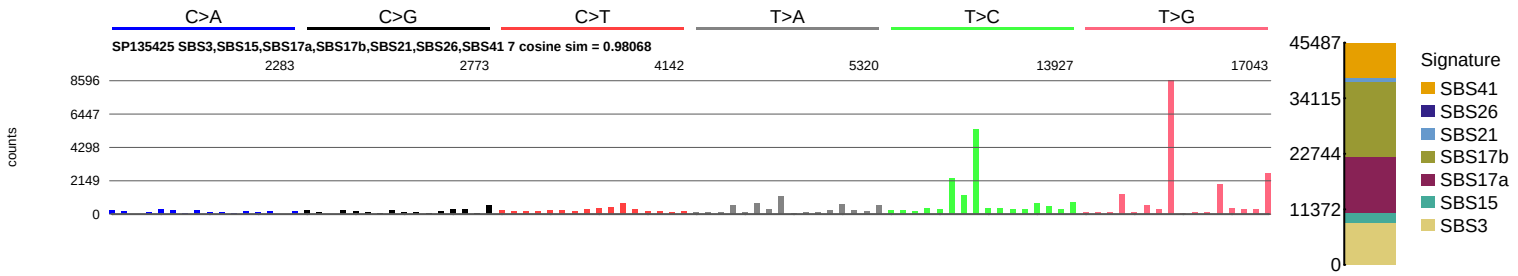

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98078
4626
1832
5924
3303
13741
16061
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

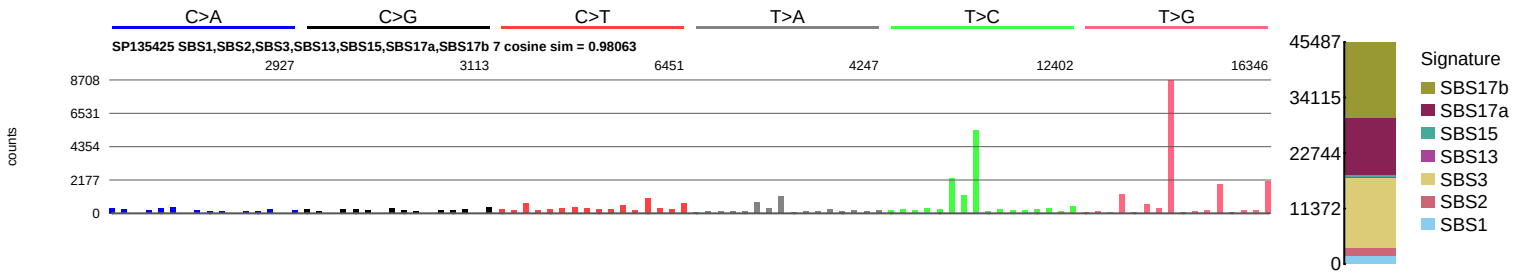

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98077
4625
1826
5947
3299
13730
16058
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

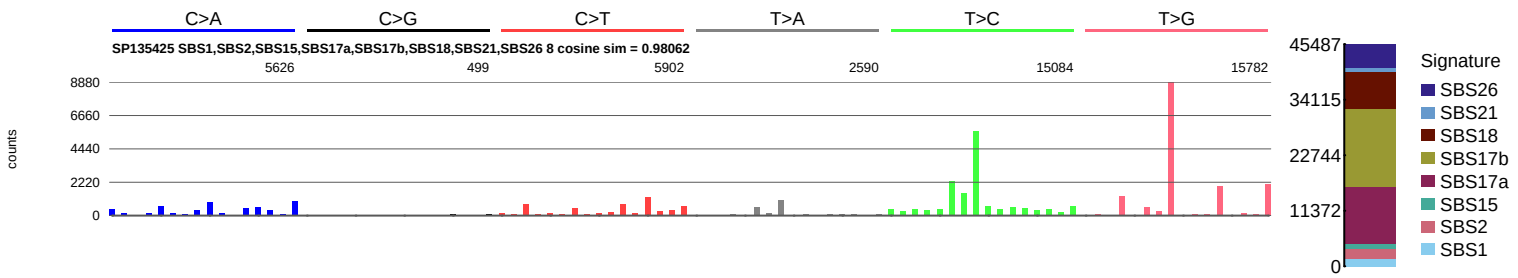

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.9807
4652
1622
5994
3325
13812
16080
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1

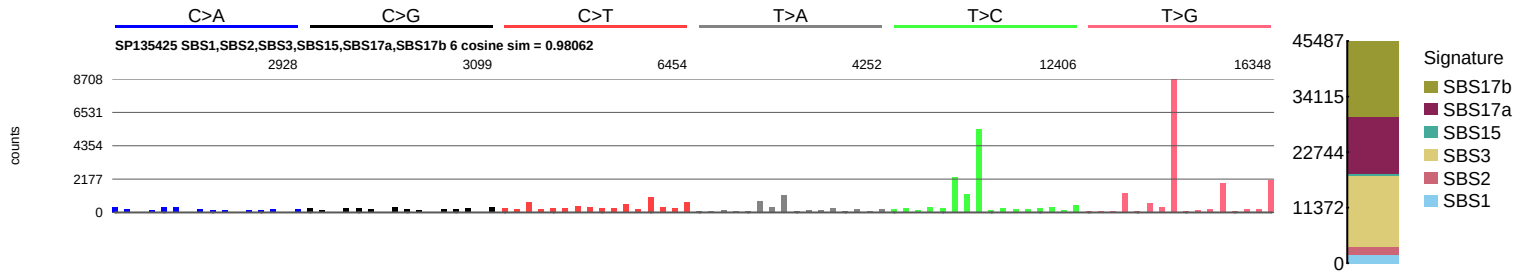

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9807
4652
1617
6011
3323
13807
16075
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

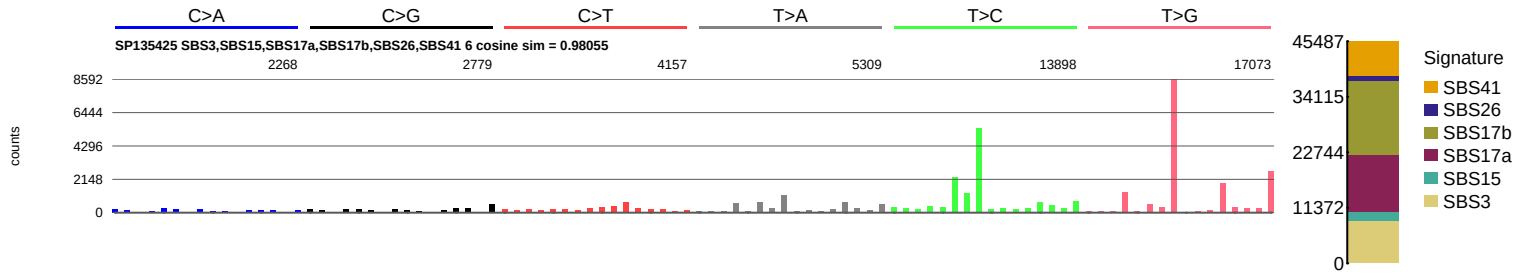

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98069
2286
2774
4142
5324
13921
17043
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3

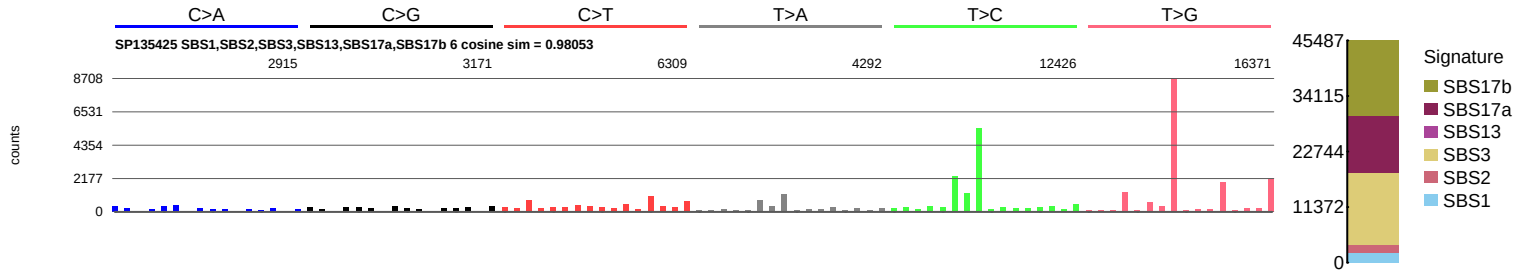

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98068
2283
2773
4142
5320
13927
17043
8596
6447
4298
2149
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3

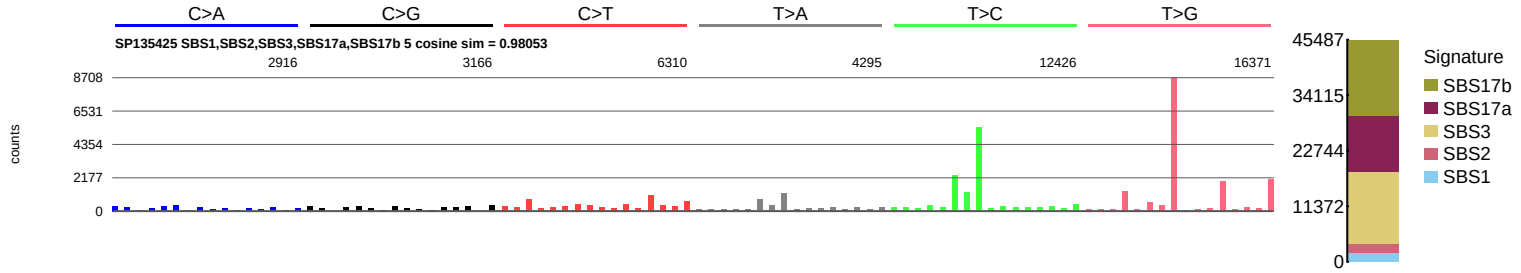

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98063
2927
3113
6451
4247
12402
16346
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

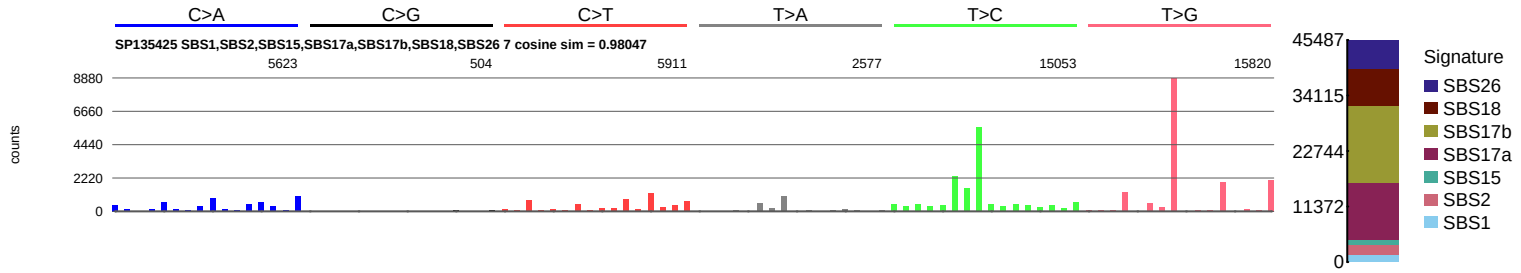

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98062
5626
499
5902
2590
15084
15782
8880
6660
4440
2220
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

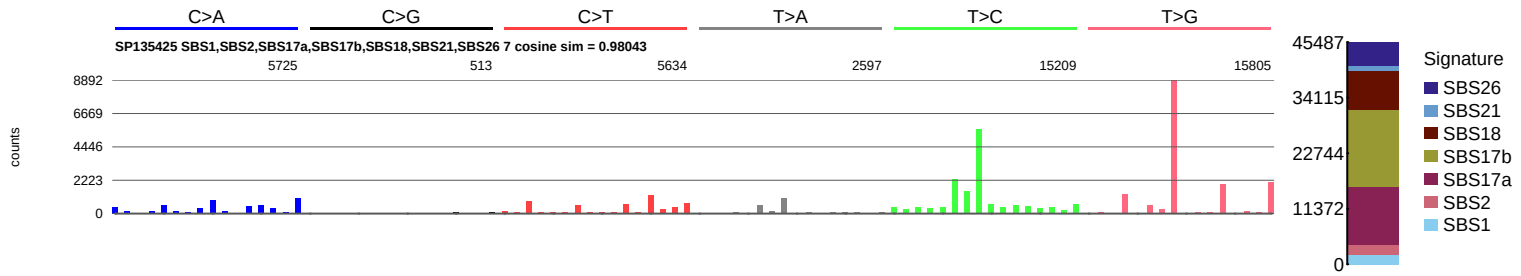

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98062
2928
3099
6454
4252
12406
16348
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

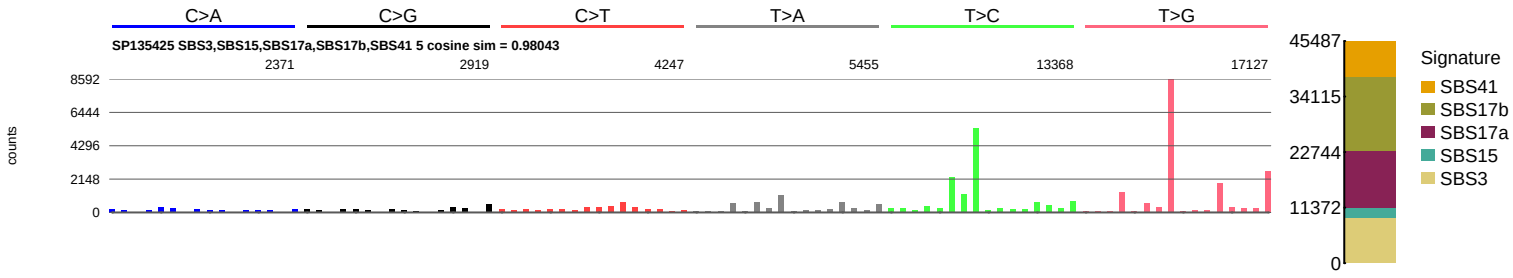

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98055
2268
2779
4157
5309
13898
17073
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3

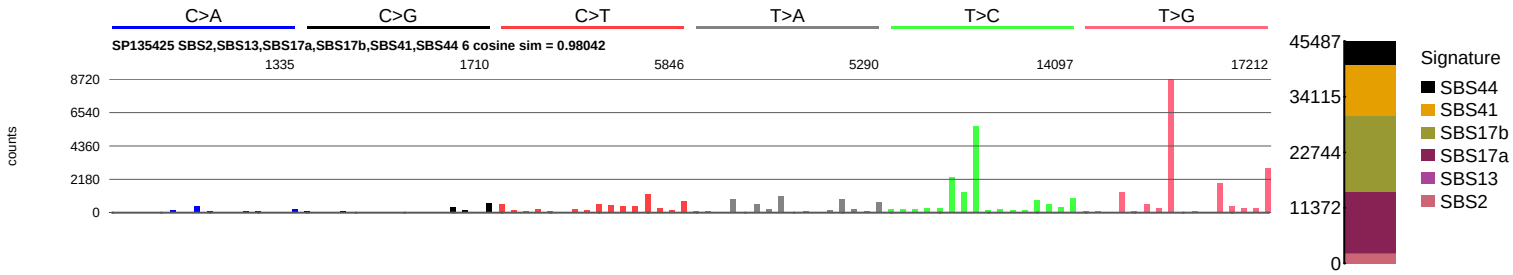

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b 6 cosine sim = 0.98053
2915
3171
6309
4292
12426
16371
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

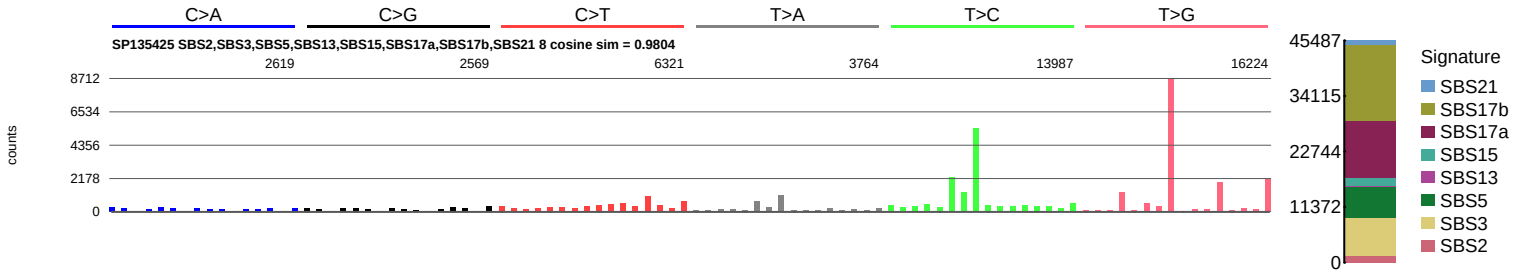

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS3,SBS17a,SBS17b 5 cosine sim = 0.98053
2916
3166
6310
4295
12426
16371
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS3
SBS2
SBS1

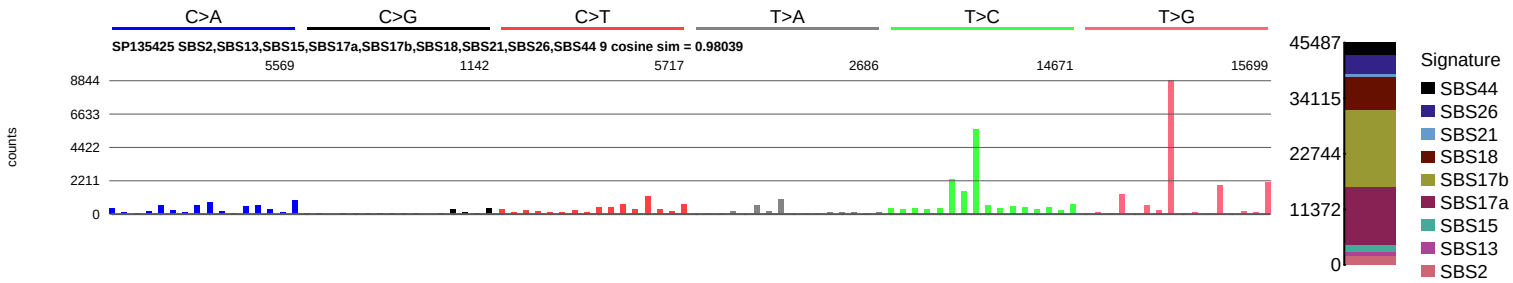

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98047
5623
504
5911
2577
15053
15820
8880
6660
4440
2220
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

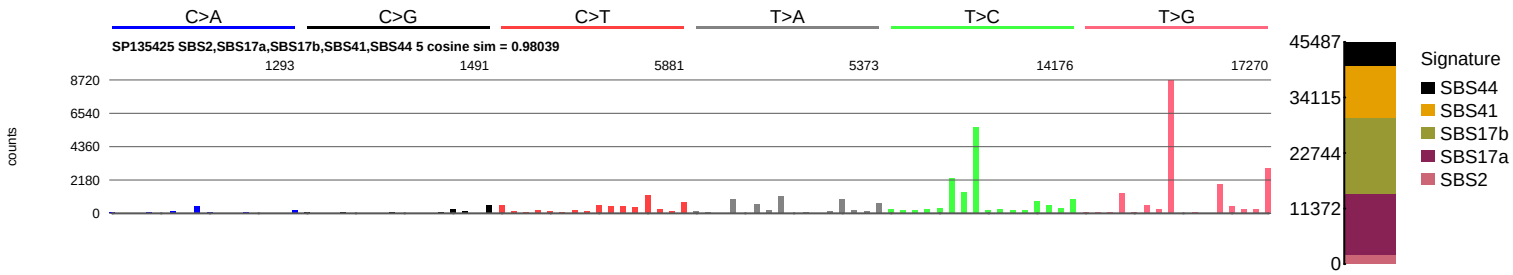

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98043
5725
513
5634
2597
15209
15805
8892
6669
4446
2223
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

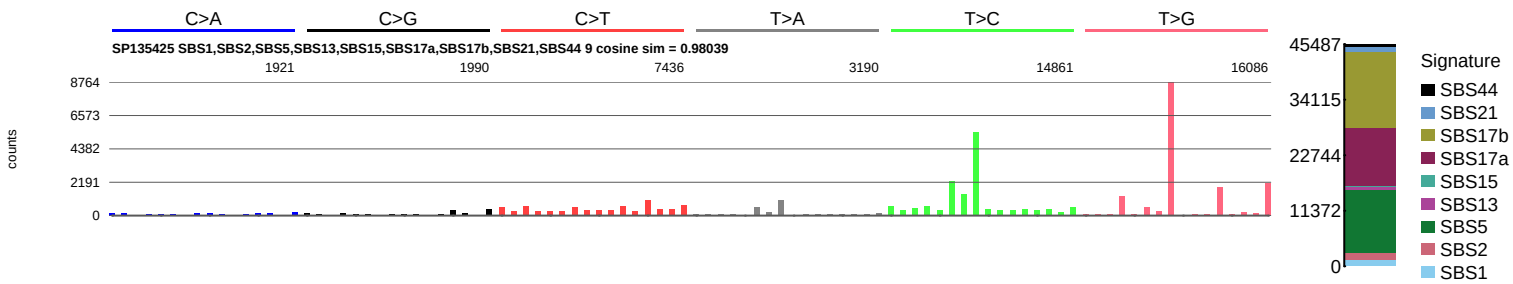

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98043
2371
2919
4247
5455
13368
17127
8592
6444
4296
2148
0
counts
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3

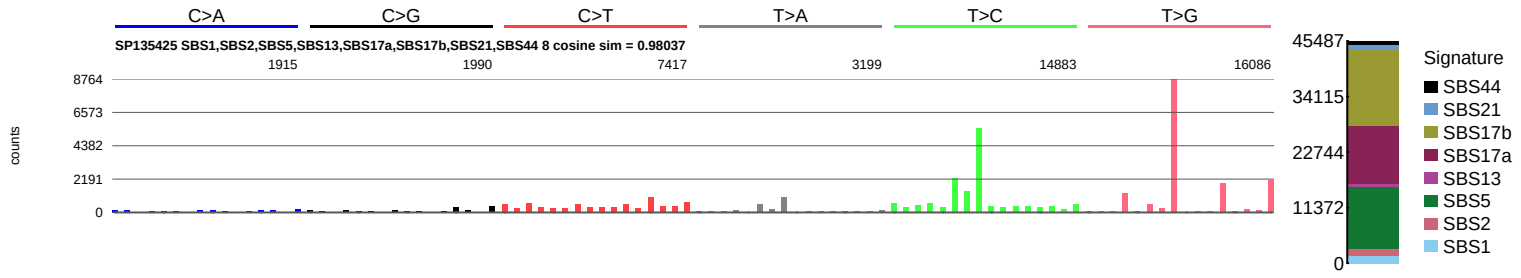

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.98042
1335
1710
5846
5290
14097
17212
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS2

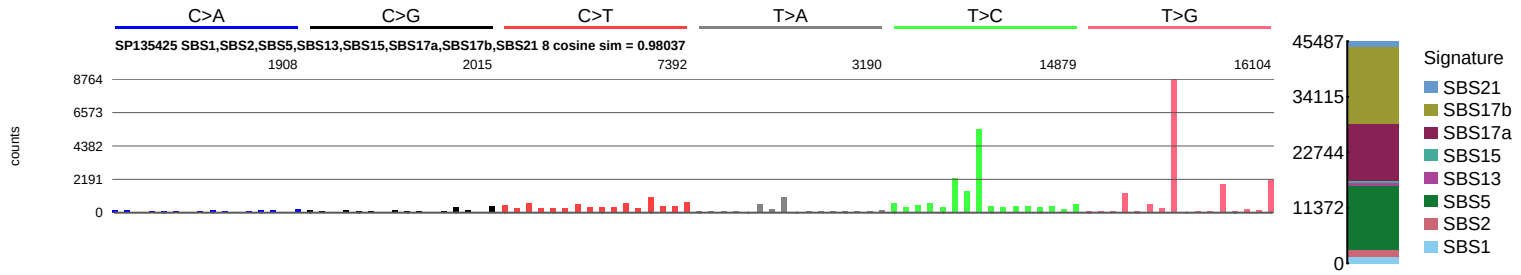

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.9804
2619
2569
6321
3764
13987
16224
8712
6534
4356
2178
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

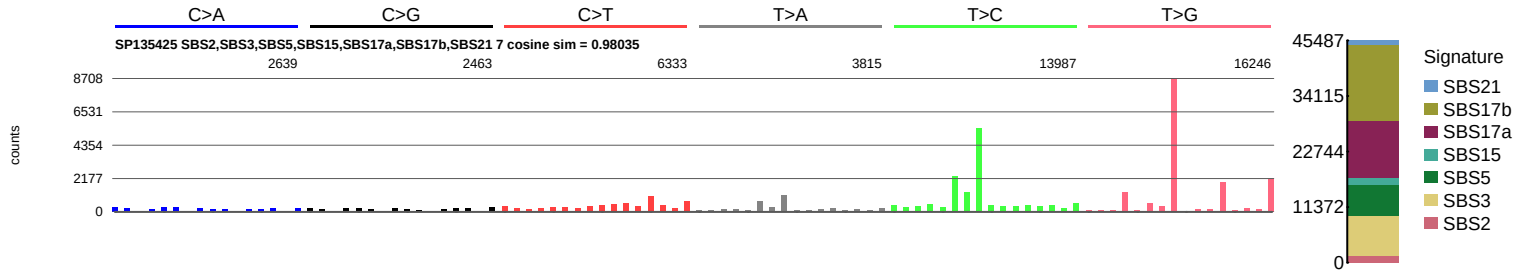

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98039
5569
1142
5717
2686
14671
15699
8844
6633
4422
2211
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2

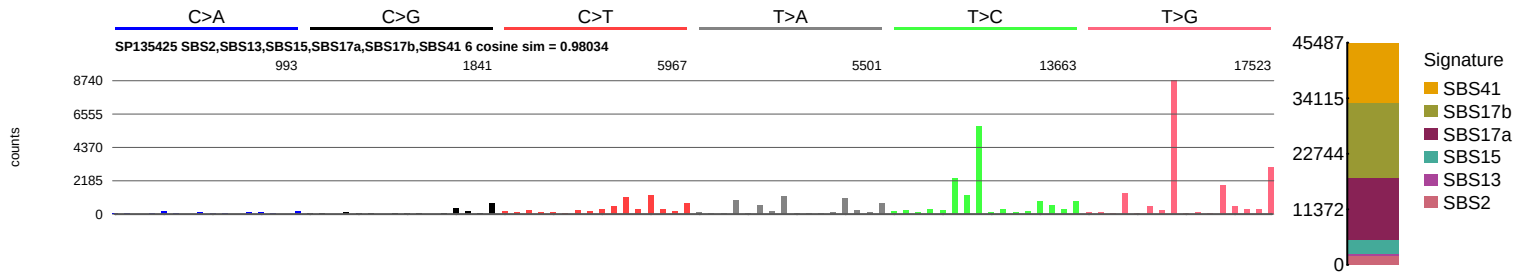

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS17a,SBS17b,SBS41,SBS44 5 cosine sim = 0.98039
1293
1491
5881
5373
14176
17270
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS2

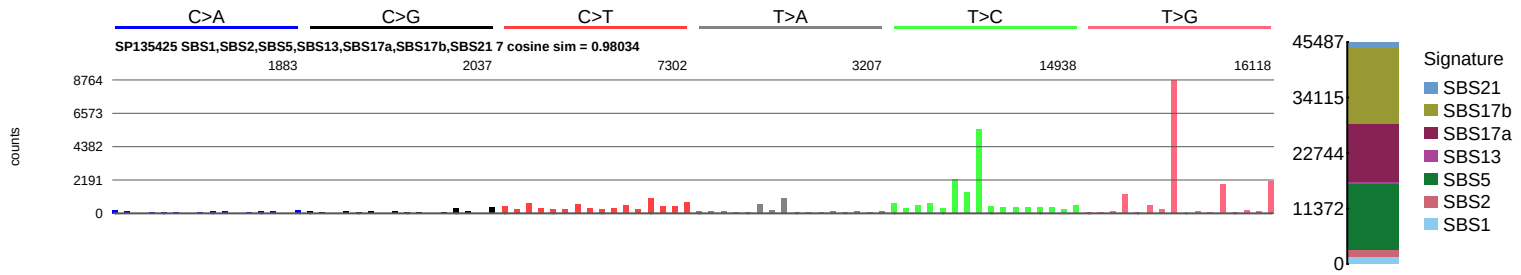

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98039
1921
1990
7436
3190
14861
16086
8764
6573
4382
2191
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

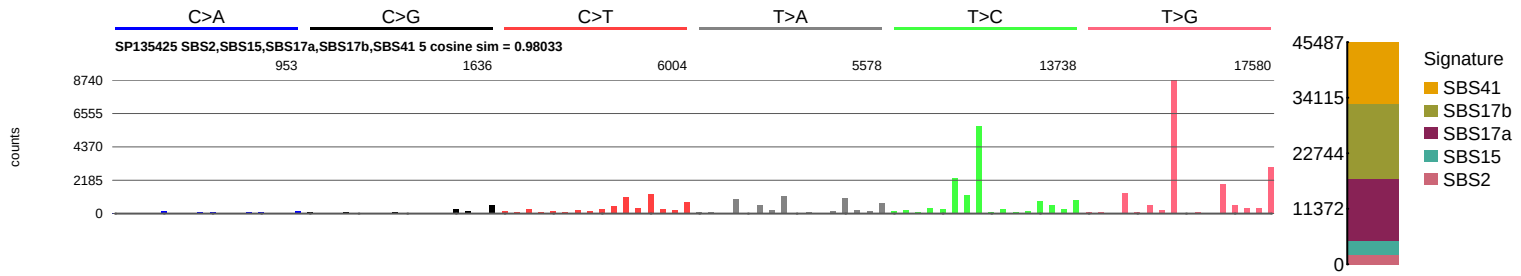

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98037
1915
1990
7417
3199
14883
16086
counts
8764
6573
4382
2191
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

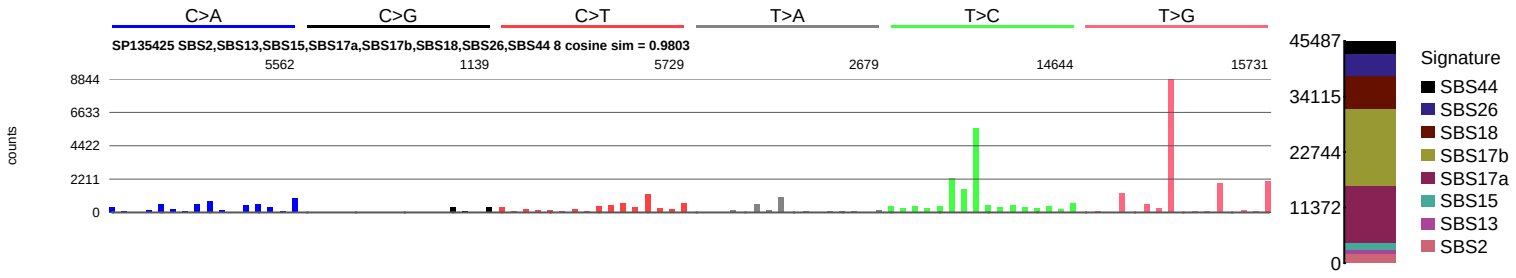

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98037
1908
2015
7392
3190
14879
16104
counts
8764
6573
4382
2191
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

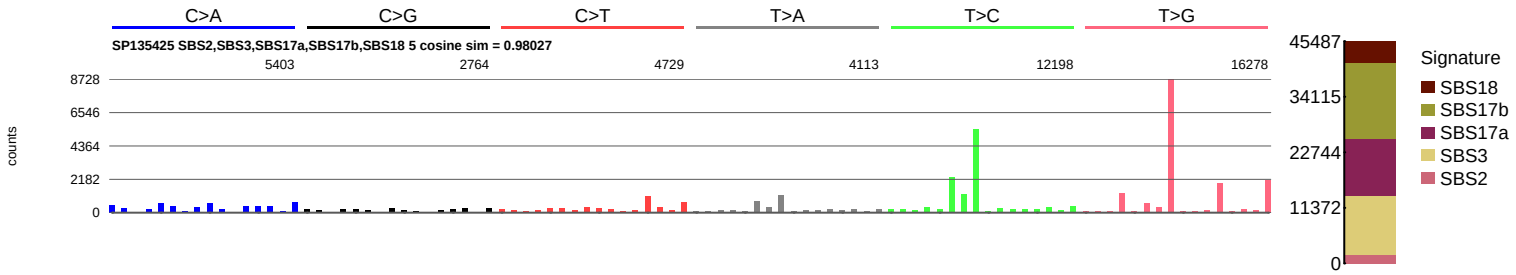

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98035
2639
2463
6333
3815
13987
16246
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2

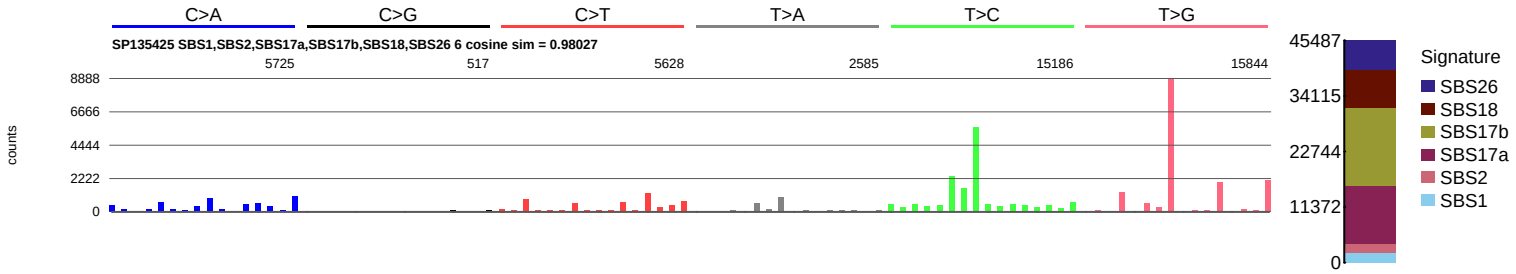

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98034
993
1841
5967
5501
13663
17523
counts
8740
6555
4370
2185
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98034
1883
2037
7302
3207
14938
16118
counts
8764
6573
4382
2191
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS15,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98033
953
1636
6004
5578
13738
17580
counts
8740
6555
4370
2185
0
45487
34115
22744
11372
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.9803
5562
1139
5729
2679
14644
15731
8844
6633
4422
2211
0
counts
45487
34115
22744
11372
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98027
5403
2764
4729
4113
12198
16278
8728
6546
4364
2182
0
counts
45487
34115
22744
11372
0
Signature
SBS18
SBS17b
SBS17a
SBS3
SBS2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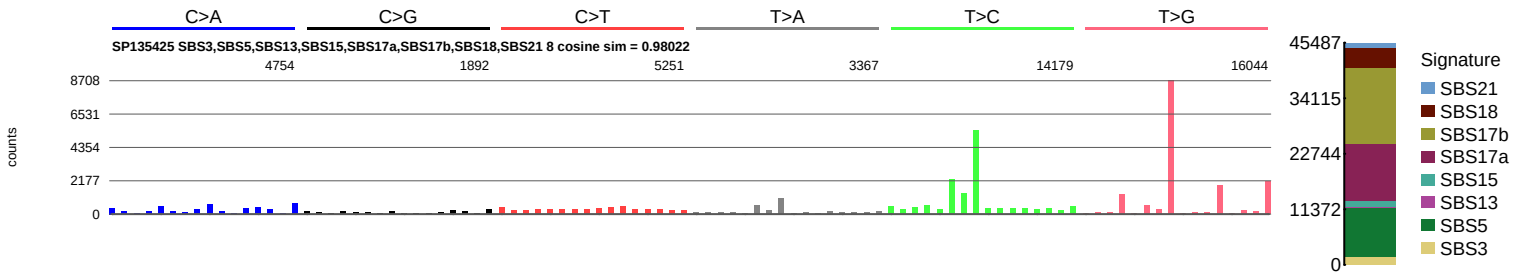
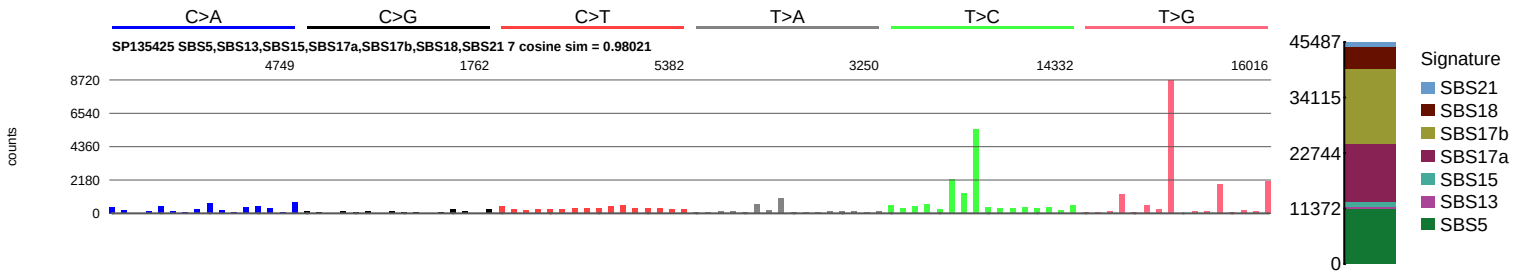
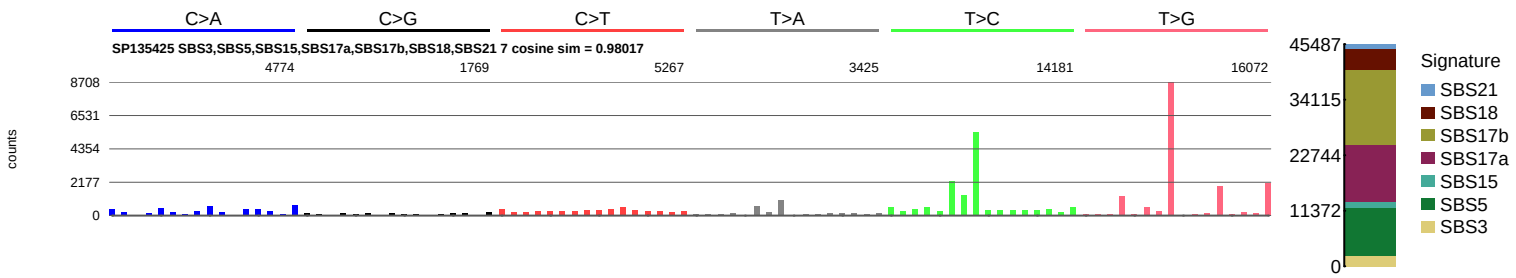
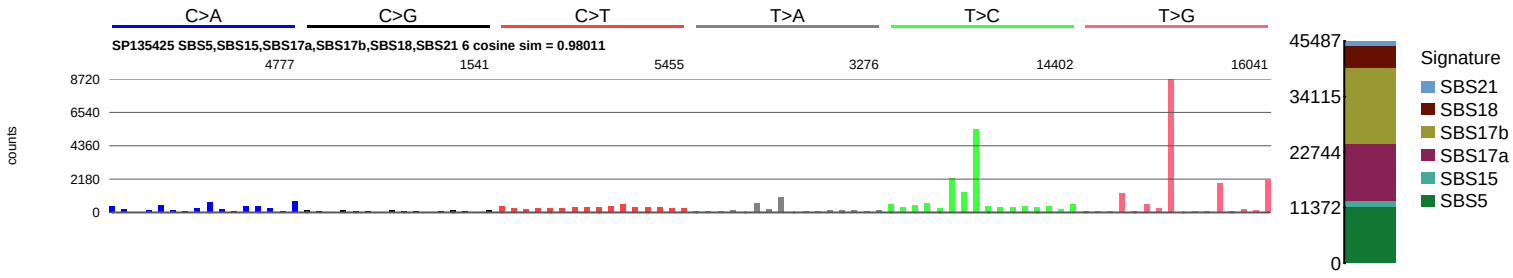
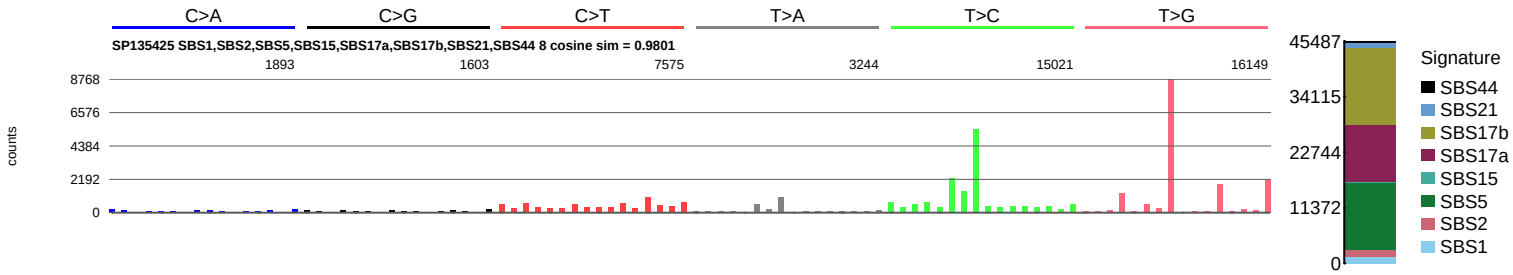
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98027
5725
517
5628
2585
15186
15844
8888
6666
4444
2222
0
counts
45487
34115
22744
11372
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

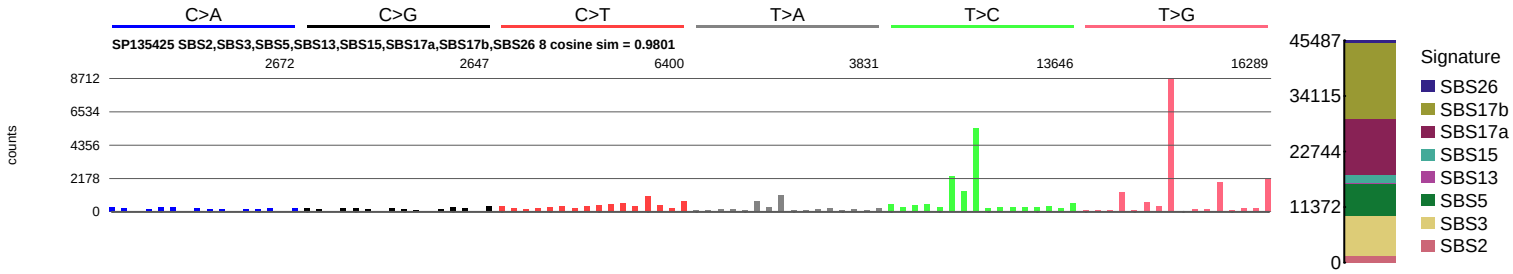

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98022
4754
1892
5251
3367
14179
16044
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

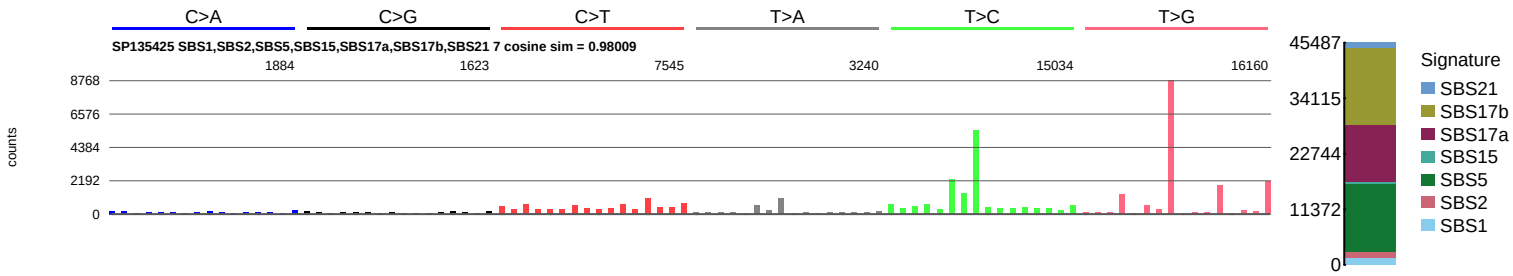

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98021
4749
1762
5382
3250
14332
16016
8720
6540
4360
2180
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5

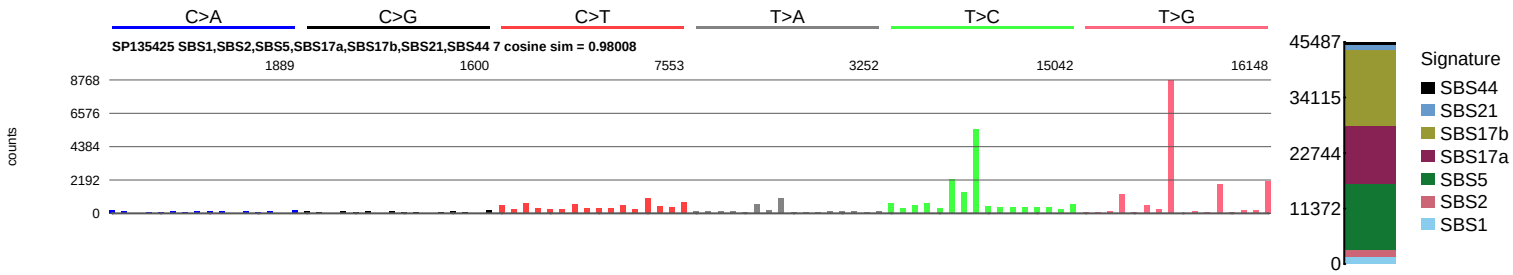

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98017
4774
1769
5267
3425
14181
16072
8708
6531
4354
2177
0
counts
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

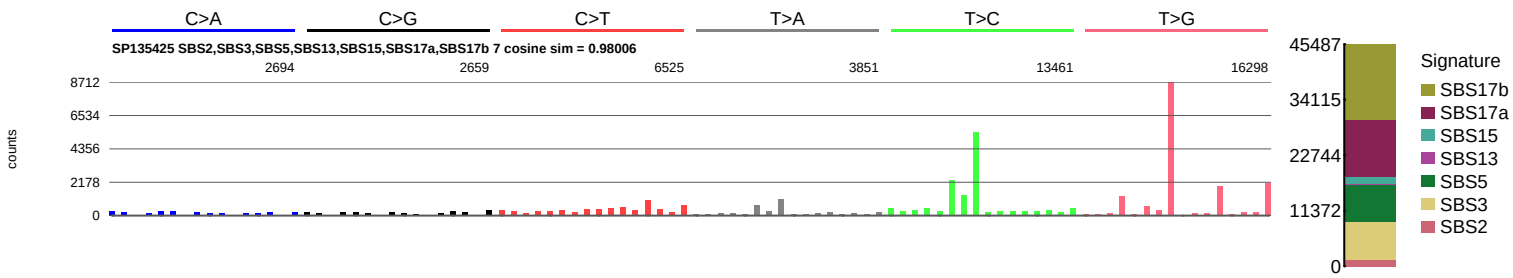

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.98011
4777
1541
5455
3276
14402
16041
counts
8720
6540
4360
2180
0
45487
34115
22744
11372
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5

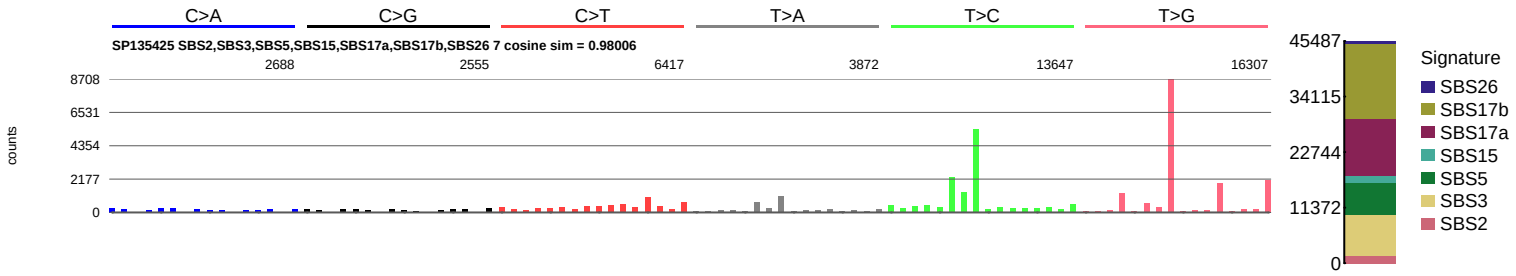

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.9801
1893
1603
7575
3244
15021
16149
counts
8768
6576
4384
2192
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

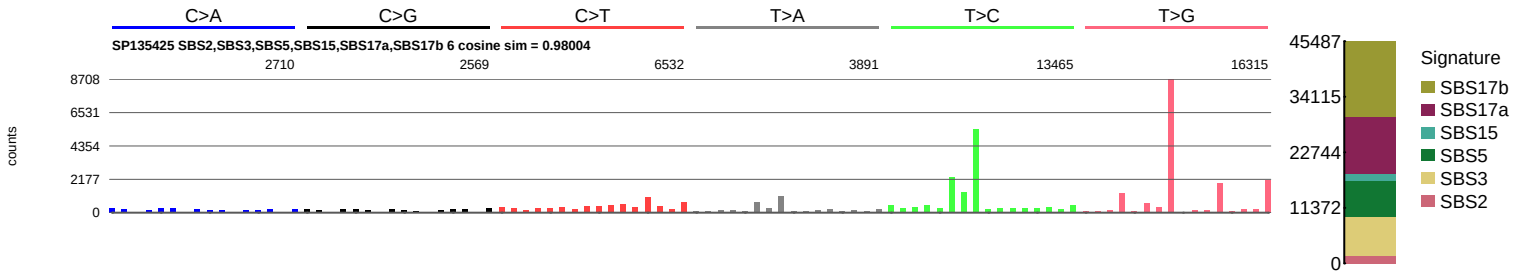

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26 8 cosine sim = 0.9801
2672
2647
6400
3831
13646
16289
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

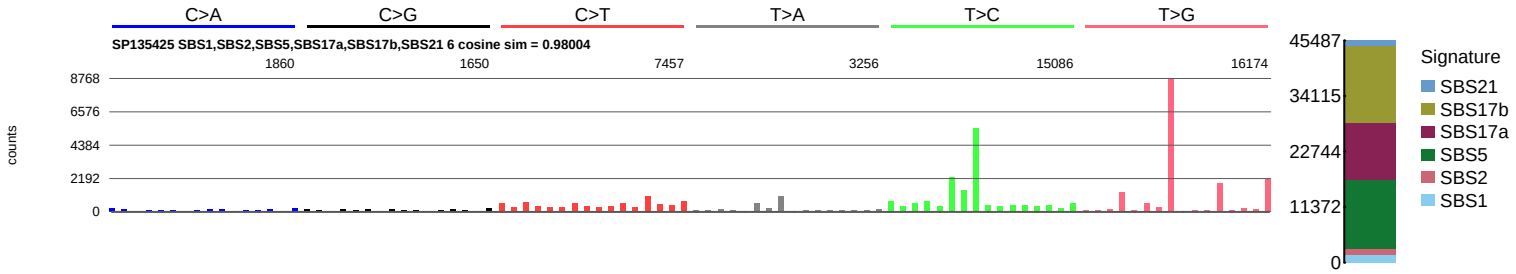

C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98009
1884
1623
7545
3240
15034
16160
counts
8768
6576
4384
2192
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS44 7 cosine sim = 0.98008
1889
1600
7553
3252
15042
16148
counts
8768
6576
4384
2192
0
45487
34115
22744
11372
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98006
2694
2659
6525
3851
13461
16298
counts
8712
6534
4356
2178
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26 7 cosine sim = 0.98006
2688
2555
6417
3872
13647
16307
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 6 cosine sim = 0.98004
2710
2569
6532
3891
13465
16315
counts
8708
6531
4354
2177
0
45487
34115
22744
11372
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP135425 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21 6 cosine sim = 0.98004
1860
1650
7457
3256
15086
16174
counts
8768
6576
4384
2192
0
45487
34115
22744
11372
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1